Prime-factor densities: gap spectra, zeta-zero criteria, and generalized Robin inequalities

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Abstract

We study reversals in the density of the kth smallest distinct prime factor and the finer arithmetic error in its consecutive-coefficient ratio. For fixed local laws with $\eta_p^{-1} = p/\nu + \beta + o(1)$, the finite maximal separated reversal count has exact rate $\limsup_k k^{-1} \log\log(3 + N_\eta(k)) = \sup_{H\geq 2} \gamma_H/(H+\nu)$, where γ_H records bounded-gap frequency. This spectrum has a finite phase profile and arithmetic realizations by affine prime sums. A uniform comparison with a full entire-function reference transfers the classical Nicolas error to the actual moving-rank coefficient ratio throughout a class of local laws. Its eventual sign gives RH criteria with exact forced-prime rank shifts and reference constants. Persistent excursions yield positive-area criteria; explicit short-interval sampling hypotheses also give localized reversal-count criteria. Hardy–Littlewood-shaped gap counts give separate conditional results on the last ascent. For consecutive-root polynomials plus a prime, we identify the exact complete correction and prove an unconditional negative sign when the degree is $1 + \lfloor N/(\log N)^j \rfloor$, for each fixed $j \geq 1$. If instead $d(N) \leq CN^\alpha$ for fixed $1/2 \leq \alpha < 1$, an eventual nonpositive complete profile excludes zeta zeros to the right of α; the signed premise remains unproved. A separate application to generalized divisor sums constructs explicit modified least common multiples with a positive normalized logarithm, uniformly for $1 < \kappa \leq 3/2 + 2/(5\log\log M_N)$ at unbounded integer cutoffs. Under RH the same window holds eventually at every integer cutoff; consequently, every fixed $1 < \kappa \leq 3/2$ has arbitrarily large counterexamples to the generalized upper inequality. An RH-conditional harmonic second moment measures excess zero multiplicity. We also prove prime-coordinate and residue-rational inverse theorems and construct distinct laws with identical entire profiles and sharp cubic finite-cutoff defects. The spectral and local compari-

son components have maintained Lean proofs, as does the full prime-coordinate inverse classification. The optimal cutoff distortion equals the least recurring prime gap; its adaptive characterization and full inverse are also Lean-formalized. The residue-rational classification and continuum-cardinality assertion remain ordinary proofs. Classical inputs and unproved arithmetic premises are distinguished; no new prime-gap bound or proof of RH is claimed.

1 Introduction and historical context

Let $d_k(p)$ be the natural density of positive integers whose kth smallest *distinct* prime factor is p. The question whether $p \mapsto d_k(p)$ first rises and then falls is now called Erdős Problem 690. Our subject is the finer structure left after unimodality fails: the number of separated reversals, their possible locations, and the arithmetic information encoded by their growth.

Erdős formulated the question in 1979 [1, p. 75]. The same paper discusses the typical scale $\log\log p_k(n) \sim k$ for prime factors of integers. This concerns where probability mass lies, not the mode of the individual-prime density. Erdős–Tenenbaum [2, Corollary 5] later corrected the earlier assertion about the mode: a maximizing prime satisfies $\log p \sim k/\log k$, rather than $\log p \sim k$. Neither statement counts reversals. The occurrence of a double exponential in classical prime-factor distribution does not by itself supply the theorem below.

Cambie [3, Theorem 5] proved unimodality for $k = 1, 2, 3$ and verified its failure for $4 \le k \le 20$. His Theorem 3 gives superpolynomially many local maxima for the different density $\delta_1(n, m)$ in Erdős Problem 692: the density of integers having exactly one divisor in $\{n+1, \dots, m-1\}$. Its moving variable is m, with n fixed. It is not a reversal count for $d_k(p)$.

Wang–Crapis [4, Theorem 1.1] proved non-unimodality for every $k \ge 4$. Their proof compares a symmetric-polynomial ratio with prime gaps and produces a descent followed by an ascent. We extend the comparison to local laws and make the construction quantitative. The question here is how the maximal number of witnesses grows with k. Its exact rate also requires a converse: a sufficiently late ascent must come from an appropriately short gap.

The first contribution is the exact reversal frequency spectrum, its arithmetic realization, and its inverse late-ascent interpretation. Quantitative bounded-gap results are existing analytic inputs, not new sieve results. The second is a uniform transfer of a classical RH-sensitive error into the actual moving-rank coefficient ratio, preserving its sign at a scale much smaller

than any fixed inverse power of $\log\log x$. The transfer holds throughout an explicit class of local probability laws, rather than only for the ordinary integers: each family contributes its full reference profile and the same centered arithmetic signal.

Nicolas's primorial criterion and its signed error [7, 8] supply the analytic source. The finite generating product and its saddle-point analysis already occur in Erdős–Tenenbaum [2]; the normalized Euler product also appears in mod-Poisson theory [9, 10]. We claim neither the generating function nor the Nicolas spectral expansion as new. The potentially new statements are the exact gap-frequency spectrum and the quantitative comparison with a full theta-centered coefficient reference, together with their stated applications. Priority remains provisional pending independent review.

There is also a separate divisor-sum application. Fan–Kobayashi–Molnar [13, Theorem 6.1 and §8] prove an eventual generalized Robin upper bound under RH for each fixed real exponent $\kappa > 3/2$ and ask about smaller exponents. Section 18 gives unconditional unbounded violations for every $1 < \kappa \leq 3/2$, and a shrinking window above the endpoint. The additional growing-family calculations in Section 16 retain complete arithmetic corrections. Their independent signs and their RH-sensitive conditional implications have different degree ranges. Section 17 states the exact N^{α} zero-free implication and its still-open signed hypothesis. The endpoint/window result and these family applications are potentially new; independent priority assessment remains outstanding.

Sections 4–8 prove the unconditional spectrum. Section 9 treats recurrence and the separate Hardy–Littlewood conditional location result. Sections 10–13 establish the coefficient transfer, pointwise RH criteria and reference capacities. Sections 14–15 treat persistent positive area and the localized signed-count criteria. Sections 16 and 18 give the growing arithmetic families and the generalized-divisor-sum application; Section 17 gives the power-degree criterion. The public repository contains maintained proofs and a provenance ledger. This is an AI-assisted research draft; independent expert review and assessment of novelty remain outstanding.

2 The argument in plain English

Choose a position in an integer's ordered list of prime factors, say the tenth distinct prime factor. For each prime, ask what fraction of sampled integers put that prime in the chosen position. These fractions form a sequence. A reversal means that the sequence drops and later rises again. We count

occurrences separated far enough that they do not reuse the same stretch of the sequence.

Two effects compete. Moving to a larger prime lowers its own chance of dividing the integer. At the same time, the additional smaller primes make it easier to have the required number of preceding factors. An exact formula measures this competition. On the relevant scale, a short gap between consecutive primes favors a rise, whereas a long gap favors a drop.

The useful measuring scale is the logarithm of the logarithm of the prime. On that scale the available rise threshold is approximately the rank divided by the scale. This explains the double-exponential locations in the argument. Eventually even the shortest possible prime gap is too large for a rise, so every fixed rank still has a final decreasing tail.

Many short gaps alone are not enough to count reversals. We must put a long gap before each selected short gap and keep the witnesses distinct. Periodic blocks of composite integers provide barriers that consecutive primes must jump. Grouping short gaps between the barriers and taking alternate occupied groups produces separated witnesses, losing only a fixed factor in their number.

Finally, different short-gap sizes can have different frequencies. Each size contributes its frequency divided by a threshold cost. The greatest contribution is exactly the reversal growth rate. For products of affine prime sums, the number of distinct factors changes that cost, while finitely many exceptional primes change only the initial details.

A second, much finer comparison uses the entire prime-product profile as a reference. The ratio that decides an ascent differs from this reference by a small signed error. We show that this error carries the same RH information as Nicolas's primorial inequality. Under RH it is eventually negative, with a bias from prime squares. If RH is false it has larger excursions of both signs. The rank moves outward with the prime cutoff in this statement; it does not contradict eventual decline at each fixed rank.

Changing the sampled family changes its local factor probabilities and its reference profile. We prove that a quadratic change in the local odds preserves the small RH-sensitive error after this exact reference has been subtracted. Forced primes shift the rank and also change the reference constant; both effects are retained. Thus one independently proved centered inequality in any qualifying family would imply RH.

This determines how the prefix changes the threshold for a possible ascent. An actual ascent also needs an actual prime gap below that threshold. RH controls the fine reference error, while a Hardy–Littlewood twin-count assumption supplies twins near the potential endpoint. Their roles and their

different precision are kept separate throughout.

Two further calculations test what arithmetic structure can actually supply. A polynomial with many prescribed roots makes divisibility by small primes less likely. We calculate its entire correction, rather than just its leading local probability, and prove a complete negative sign when the degree is only logarithmically smaller than the prime cutoff. This range must be distinguished from the much smaller power-sized degrees relevant to excluding zeta zeros.

For generalized divisor sums, increasing selected prime exponents in a least common multiple balances an extra height cost against a smaller local-factor loss. The full balance gives explicit counterexamples in a shrinking parameter window around 3/2. The proof works whether RH is true or false. It answers a separate inequality question; it does not resolve the signed premise needed for an independent proof of RH.

The finer error can also be detected by counting. We choose actual short prime gaps in many separate cells, place a long-gap descent before each, and test a finely spaced range of local laws. If RH fails, its positive error persists across a whole interval. Every selected cell then adds more actual ascents than the full reference predicts. A sufficiently accurate upper bound on this signed count would therefore imply RH. This uses known short-interval bounded gaps at some fixed width; it does not require twin primes, and it does not establish that upper bound.

3 Definitions and main results

All logarithms are natural. Index the primes by $p_0 = 2, p_1 = 3, \ldots$, and write $\mathcal{L}(x) = \log\log x$ for $x > 1$ and $\mathcal{E}(t) = \exp(\exp t)$. Ranks k are positive integers. A fixed *local law* consists of probabilities $0 \le \eta_p \le 1$. Define the finite products

$$M_r(p) = [z^r] \prod_{q<p} (1 - \eta_q + \eta_q z), \qquad D_{\eta,k}(p) = \eta_p M_{k-1}(p), \tag{1}$$

with $M_r = 0$ for $r < 0$. For the ordinary law $\eta_p = 1/p$, the Chinese remainder theorem identifies $D_{\eta,k}$ with d_k. The second canonical law is

$$\eta_2^{\mathrm{odd}} = 0, \qquad \eta_p^{\mathrm{odd}} = \frac{p-2}{(p-1)^2} \quad (p > 2). \tag{2}$$

Definition 3.1. A reversal at rank k consists of indices $d < a$ with

$$D_{\eta,k}(p_{d+1}) < D_{\eta,k}(p_d), \qquad D_{\eta,k}(p_a) < D_{\eta,k}(p_{a+1}).$$

A family of m such witnesses is separated if $a_i + 1 < d_j$ for $i < j$. Its maximal attainable size, when finite, is $N_\eta(k)$.

This counts raw-density descent–ascent witnesses, not oscillations of a centered error or a smoothed approximation.

For integers $H \geq 0$, set

$$\begin{aligned} G_H(X) &= \#\{i : p_i \leq X,\ p_{i+1} - p_i \leq H\}, \\ \gamma_H &= \limsup_{\substack{X\to\infty \\ X\in\mathbb{N}}} \frac{\log\log(3 + G_H(X))}{\mathcal{L}(X)}, \\ \Lambda(\nu) &= \sup_{H\geq 2} \frac{\gamma_H}{H+\nu} \quad (\nu > 0). \end{aligned} \tag{3}$$

The 3 keeps the logarithms defined when counts vanish. We have $0 \leq \gamma_H \leq 1$ and $\gamma_0 = \gamma_1 = 0$.

Theorem 3.2 (Exact reversal spectrum)**.** *Suppose $\nu > 0$, $\beta \in \mathbb{R}$, and, along sufficiently large primes,*

$$0 < \eta_p < 1, \qquad \eta_p^{-1} = p/\nu + \beta + \rho(p), \qquad \rho(p) \longrightarrow 0. \tag{4}$$

Every fixed-rank sequence $D_{\eta,k}$ is eventually nonincreasing, its reversal maximum is finite and attained, and

$$\limsup_{k\to\infty} \frac{\log\log(3 + N_\eta(k))}{k} = \Lambda(\nu) > 0. \tag{5}$$

The law, its parameters and its exceptional coordinates are fixed before k tends to infinity.

The weaker estimate $\eta_p = \nu/p + O(p^{-2})$ alone does not supply the vanishing difference of reciprocal errors used in this theorem.

Theorem 3.3 (Canonical bounds)**.** *For either canonical law and every $\varepsilon > 0$, for all sufficiently large k,*

$$\mathcal{E}(k/602) \leq N_\eta(k) \leq \mathcal{E}((1/3 + \varepsilon)k).$$

Each fixed-rank canonical sequence is eventually strictly decreasing.

The positive constant uses a conservative quantitative width-600 input; it is not a claim about the best known numerical prime-gap bound.

Theorem 3.4 (Affine arithmetic realization)**.** *Fix $m \geq 1$, positive integers c_j, and integers d_j, with $c_i d_j \neq c_j d_i$ for $i \neq j$. Put*

$$F(p,q) = \prod_{j=1}^{m} (c_j(p+q) + d_j).$$

Sample all ordered prime pairs $p, q \leq X$, including the diagonal, with denominator $\pi(X)^2$. At every fixed rank k and prime ℓ, the probability that ℓ is the kth smallest distinct prime factor of $|F(p,q)|$ has a limit $D_{F,k}(\ell)$; zero outputs are excluded from the event. The maximal separated reversal count $N_F(k)$ is finite and attained, and

$$\limsup_{k\to\infty} \frac{\log\log(3 + N_F(k))}{k} = \Lambda(m) > 0.$$

The size limit comes first, with family, rank and prime fixed. Outputs are weighted by their prime-pair representations.

The second main result, Theorem 11.1, concerns the ordinary coefficient ratio at rank $r = \lfloor \alpha(\gamma + \log\log\theta(x)) \rfloor$, for each fixed $\alpha > 0$. Its precise full-profile normalization is defined in Section 10. Eventual negativity of that centered ratio is equivalent to RH. This statement is separate from Theorem 3.2, whose hypotheses contain no RH assumption.

4 The exact comparison and uniform estimates

4.1 First differences

For consecutive primes $p < q$ with interior probabilities,

$$M_r(q) = (1 - \eta_p) M_r(p) + \eta_p M_{r-1}(p).$$

Direct subtraction gives

$$D_{\eta,k}(q) - D_{\eta,k}(p) = \eta_p \eta_q \{M_{k-2}(p) - T_\eta(p,q) M_{k-1}(p)\}, \tag{6}$$

where $T_\eta(p,q) = \eta_q^{-1} - \eta_p^{-1} + 1$. When $M_{k-1}(p) > 0$, an ascent is equivalent to $M_{k-2}(p)/M_{k-1}(p) > T_\eta(p,q)$. For the ordinary law, writing $R = M_{k-2}/M_{k-1}$ and $g = q - p$, the exact real-valued gap threshold is $R - 1$: a strict ascent occurs if and only if $g < R - 1$. Equality gives a flat step. This threshold does not constrain the actual gap to be small. The ordinary comparison level is $q - p + 1$; for the odd law at $p > 2$ it is

$$q - p + 1 - \frac{q-p}{(p-2)(q-2)}.$$

4.2 Finite exceptions and symmetric polynomials

Only finitely many coordinates in (4) can be forced ($\eta_p = 1$) or excluded ($\eta_p = 0$). Let f count the forced coordinates. Past them, each contributes a factor z to the generating polynomial; excluded coordinates contribute 1. Retain all interior coordinates, including the exceptional ones, and set

$$w_p = \frac{\eta_p}{1-\eta_p}, \qquad S(p) = \sum_{\substack{q<p \\ 0<\eta_q<1}} w_q.$$

For $k > f$, put $r = k-1-f$. The common positive product of $1-\eta_q$ cancels in the ratio, leaving $E_{r-1}(p)/E_r(p)$, where E_r is the elementary symmetric polynomial of degree r in the finite list of interior odds.

Lemma 4.1. *For at least $r \geq 1$ positive weights, let W_{r-1} be the largest sum of $r-1$ distinct weights. Then*

$$\frac{r}{S} \leq \frac{E_{r-1}}{E_r} \leq \frac{r}{S - W_{r-1}}, \tag{7}$$

and the right denominator is positive.

Proof. The terms of SE_{r-1} which choose a new index total rE_r. The remaining terms are

$$\sum_{|A|=r-1} \left(\sum_{j \in A} w_j \right) \prod_{j \in A} w_j,$$

between zero and $W_{r-1}E_{r-1}$. Rearrangement proves the bounds. At least one positive weight remains after any $r-1$ are removed. □

This specializes to Wang–Crapis [4, Lemmas 3.1–3.2]; the present use also retains the exact forced-rank shift.

Expanding (4) gives $w_p = \nu/p + O(p^{-2})$. The reciprocal-prime estimate and the convergent quadratic tail give

$$S(p) = \nu\mathcal{L}(p-1) + O_\eta(1), \qquad W_{r-1} = O_\eta(1 + \log(r+1)). \tag{8}$$

For the second estimate, dominate all but finitely many odds by C/p and use $p_j \geq j+2$. The finitely many exceptional odds contribute a fixed constant; their order need not be monotone.

Lemma 4.2 (Uniform sign margins)**.** *Fix $0 < u \le v$. Uniformly in the band $uk \le \mathcal{L}(p-1) \le vk$, as $k \to \infty$,*

$$\frac{E_{r-1}(p)}{E_r(p)} = \frac{k}{\nu \mathcal{L}(p-1)} + o(1), \qquad r = k - 1 - f.$$

Gaps at most H give strict ascents throughout the band if $(H+\nu)v < 1$. Gaps at least D give strict descents there if $Du > 2$, for sufficiently large rank and lower prime.

Proof. The prime number theorem ensures more than r interior primes below p uniformly in this band. Equations (7) and (8) have numerator $k + O_\eta(1)$ and denominator $\nu\mathcal{L}(p-1) + O_\eta(\log(k+1))$, bounded below by a positive multiple of k. This proves the uniform ratio statement.

Writing $g = q - p$, the threshold is $g/\nu + 1 + \rho(q) - \rho(p)$. Its error is bounded by twice the supremum of $|\rho|$ beyond p, even for large g. For $g \le H$ the limiting lower ratio is $1/(\nu v) > (H+\nu)/\nu$. For $g \ge D$ the upper ratio is $1/(\nu u) + o(1) < D/\nu$. Both margins are strict. □

Without an upper band endpoint, the same estimates give a one-sided bound: on $\mathcal{L}(p-1) \ge bk$ the ratio is at most $1/(\nu b) + o(1)$, uniformly for fixed $b > 0$.

4.3 Every fixed rank

For fixed $r \ge 1$, the ratio tends to zero by (7)–(8). Consecutive sufficiently large primes have gap at least two, so their threshold is uniformly positive. Equation (6) gives strict descent for supported degrees. If $k \le f$, the tail is zero. If $k = f + 1$, the remaining degree is zero, its predecessor coefficient vanishes, and the same identity again gives descent. Thus every fixed-rank tail is nonincreasing. For the ordinary law, eventual decrease was already noted by Cambie [3, §3]. The extension here retains the forced and excluded coordinates.

Once all ascent indices lie below an integer I, a separated family has size at most I. Attainable sizes form a nonempty bounded subset of $\mathbb{N}$ (zero is attainable), so their maximum is attained. The canonical laws have no forced coordinates and have positive supported tails, giving strict eventual decrease.

5 From short gaps to separated reversals

The following finite argument supplies the counting step and its loss. It turns actual small-gap supply into separated raw reversals. The later RH coefficient correction is not an input to this construction.

Lemma 5.1 (Periodic composite barriers)**.** *Fix integers $D \geq H+1$ and $M = D!$. Let $\mathcal{S}$ be a nonempty finite set of integer starts such that*

$$X \leq n, \quad n+H \leq Y, \quad \#([n, n+H] \cap \{primes\}) \geq 2, \quad X \geq M+2.$$

There are m separated pairs of consecutive-prime gaps, a gap at least D followed by one at most H, all endpoints in $[X, Y]$, with

$$|\mathcal{S}| \leq 2(M+H)(m+1). \tag{9}$$

Proof. Choose consecutive primes $p < q$ in each window, and assign them to $c = \lfloor (p-2)/M \rfloor$. Since $cM + j$ is composite for $2 \leq j \leq D$, and $c \geq 1$,

$$cM + D + 1 \leq p < q \leq (c+1)M + 1.$$

Indeed a prime q beyond the last endpoint but within $H < D$ of p would lie in the next composite block. A short pair cannot cross that barrier.

Starts assigned to cell c lie between $cM+D+1-H$ and $(c+1)M+1$, so there are at most $M+H$ of them. Order the occupied cells $c_0 < \cdots < c_{t-1}$ and choose one pair in each, with lower-prime index u_j. Between chosen primes in two successive occupied cells, a consecutive-prime gap crosses the destination cell's whole composite block. Its length is at least D, and it precedes the chosen short gap there.

Take destination cells $c_2, c_4, \ldots, c_{2m}$, where $m = \lfloor (t-1)/2 \rfloor$. Choose each crossing after the lower prime selected in the preceding occupied cell. The unused occupied cell between successive destinations ensures $a_i + 1 < d_{i+1}$. Every crossing lies between selected endpoints, so remains in $[X, Y]$. Finally $t \leq 2(m+1)$ and $|\mathcal{S}| \leq (M+H)t$, proving (9). □

No uniform distribution among cells is needed. Empty stretches are harmless, since a crossing still exists between selected occupied cells.

Fix $u, v > 0$ with $(H+\nu)v < 1$ and choose an integer $D \geq H+1$ with $D > 2/u$. If all primes from X through $Y+H$ lie in the sign band, the two lemmas imply, at sufficiently large k,

$$G_H(Y) \leq X + B(N_\eta(k)+1), \qquad B = 2(D!+H). \tag{10}$$

At most X lower primes are deleted below X; each retained short gap supplies a window starting at its lower prime. The upper endpoint is $Y + H$, since a counted gap may end above Y. All constants are fixed before k grows.

Corollary 5.2 (Positive-power transfer)**.** *Suppose that for fixed H and $\delta > 0$, for every sufficiently large X at least X^δ integer starts in $[X, 2X]$ have two primes in their width-H windows. For every $0 < a < 1/(H + \nu)$,*

$$N_\eta(k) \geq \mathcal{E}(ak) \quad \textit{for all sufficiently large } k.$$

Proof. Choose $a < b < 1/(H + \nu)$ and $X = \lceil \mathcal{E}(bk) \rceil$. Every endpoint in $[X, 2X + H]$ has $\mathcal{L}(p - 1) = bk + o(1)$. Fixed $u < b < v < 1/(H + \nu)$ therefore provide the sign band. The cell lemma gives $N_\eta(k) + 1 \geq X^\delta/B$. Since $\log(X^\delta/B) = \delta \exp(bk) + O(1)$ and $a < b$, the right side eventually exceeds $1 + \mathcal{E}(ak)$. □

We use the existing quantitative input

$$\#\{n \in [X, 2X] \cap \mathbb{N} : \#([n, n + 600] \cap \{\text{primes}\}) \geq 2\} \geq X^{1/3} \tag{11}$$

for all sufficiently large X. Its maintained declaration is `PrimeGaps.eventually_many_prime_windows_600`. Section 20 records its provenance. Quantitative prime-cluster theory, including Maynard's work [5], is the analytic background. We use this count, not merely infinitude of bounded gaps. With $\nu = 1$ the corollary applies at $a = 1/602 < 1/601$.

6 Proof of the exact spectrum

The two directions preserve the frequency subsequences and each individual gap's cutoff. This matching rate, rather than the presence of a double exponential alone, is the principal unconditional result.

6.1 Lower bound: arbitrary frequency subsequences

Fix H and $0 < a < \gamma_H/(H + \nu)$. Choose

$$0 < g < \gamma_H, \qquad 0 < v_0 < v < \frac{1}{H + \nu}, \qquad a < gv_0,$$

then $0 < u < a$ and $a < b < gv_0$. There are arbitrarily large integers Y with $3 + G_H(Y) > \mathcal{E}(g\mathcal{L}(Y))$. Assign $k = \lceil \mathcal{L}(Y)/v_0 \rceil$. The bounds

$v_0k - v_0 < \mathcal{L}(Y) \le v_0k$ imply $3 + G_H(Y) > \mathcal{E}(bk)$ at arbitrarily large assigned ranks.

Take $X = \lceil \mathcal{E}(uk) \rceil + 1$. The primes between X and $Y + H$ lie in a band with positive lower coefficient and upper coefficient v. Equation (10) applies. If $N_\eta(k) < \mathcal{E}(ak)$, then

$$3 + G_H(Y) \le 3 + X + B(1 + \mathcal{E}(ak)) < \mathcal{E}(bk)$$

eventually, a contradiction. Therefore $N_\eta(k) \ge \mathcal{E}(ak)$ for arbitrarily large k. Supremizing over H proves the lower bound $\Lambda(\nu)$ in (5). The argument keeps the actual frequency subsequence; it does not replace it by an all-large-X hypothesis.

6.2 Upper bound: each gap retains its own cutoff

For fixed integer $g \ge 2$ and $\varepsilon > 0$, the one-sided ratio bound implies that an ascent across a gap of size g must satisfy, apart from fixed initial exceptions,

$$\mathcal{L}(p - 1) < \left(\frac{1}{g + \nu} + \varepsilon \right) k. \tag{12}$$

The same estimate excludes all gaps greater than J above $\mathcal{L}(p - 1) = uk$ if $1/(J + 1 + \nu) < u$. In both statements, the reciprocal error is uniformly small beyond the lower prime.

Consequently

$$N_\eta(k) \le T_k + \sum_{2 \le g \le J} G_g(U_g(k)), \tag{13}$$

where we may take

$$T_k = \lceil \mathcal{E}(uk) \rceil + O(1), \qquad U_g(k) = \left\lceil \mathcal{E}\left(\left(\frac{1}{g + \nu} + \varepsilon \right) k \right) \right\rceil + 1.$$

Each witness has a distinct ascent index. There are at most T_k initial ones; all others have an actual gap $g \le J$ and are counted in that gap's term. Overcounting by the cumulative G_g only strengthens the upper bound.

Let $A > \Lambda(\nu)$. Choose $0 < u < A$ and then J as above. For each fixed g and $\delta > 0$, eventually

$$\log\log(3 + G_g(U)) \le (\gamma_g + \delta)\mathcal{L}(U).$$

Since $\gamma_g/(g + \nu) \le \Lambda(\nu)$, choose $\varepsilon, \delta > 0$ small enough that

$$(\gamma_g + \delta)\left(\frac{1}{g + \nu} + \varepsilon \right) < A \quad (2 \le g \le J).$$

There are only finitely many such comparisons. Rounding changes their double-logarithmic cutoffs by $o(1)$. Each term of (13), including T_k, has rate strictly below A. A fixed finite sum has rate at most the greatest term rate. Letting A decrease to $\Lambda(\nu)$ proves the matching bound.

The input (11) implies $\gamma_{600} = 1$. A given consecutive pair belongs to at most 601 integer-start windows of width 600, so the associated cumulative gap count has a positive-power lower bound along large cutoffs. The trivial bound $G_H(X) \le X$ gives the reverse inequality on its exponent. Hence $\Lambda(\nu) \ge 1/(600+\nu) > 0$. Together with the fixed-rank proof this completes Theorem 3.2.

For the canonical upper bound use the minimum eventual gap, two, and $\nu = 1$ in (12). Reducing ε slightly absorbs the rounding, leaving all ascents below $\mathcal{E}((1/3+\varepsilon)k)$. Their number bounds $N_\eta(k)$. Together with Corollary 5.2, this proves Theorem 3.3.

7 A finite phase profile

Theorem 7.1. *Let H_* be the least $H \ge 2$ for which $\gamma_H = 1$. This integer exists, and for every fixed $\nu > 0$,*

$$\Lambda(\nu) = \max_{2 \le H \le H_*} \frac{\gamma_H}{H+\nu}.$$

For all sufficiently large ν,

$$\Lambda(\nu) = \frac{1}{H_*+\nu}, \qquad \frac{1}{\Lambda(\nu)} - \nu = H_*. \tag{14}$$

Proof. Existence follows from $\gamma_{600} = 1$. If $H > H_*$, then $\gamma_H/(H+\nu) \le 1/(H+\nu) < 1/(H_*+\nu)$. For each $H < H_*$ put $\gamma = \gamma_H < 1$. With positive denominators, the inequality $\gamma/(H+\nu) < 1/(H_*+\nu)$ is equivalent to $(1-\gamma)\nu > \gamma H_* - H$. It holds beyond a finite threshold. Take the greatest threshold over the finitely many smaller classes. □

Here full frequency means only $\gamma_H = 1$. It need not imply positive relative density or a fixed power lower bound at every large cutoff. The theorem identifies a finite maximum without determining its entries numerically. The rate is an upper envelope of the gap-frequency data; the result recovers H_*, not every individual value of γ_H.

8 Arithmetic realization by affine prime sums

We prove Theorem 3.4 without discarding exceptional primes.

8.1 Fixed-modulus limits and independence

For fixed Q, the prime number theorem in arithmetic progressions gives

$$\frac{\#\{p \le X : p \text{ prime},\ p \equiv a \pmod{Q}\}}{\pi(X)} \longrightarrow \frac{1}{\varphi(Q)} \quad ((a,Q)=1).$$

Nonunit prime classes contain only primes dividing Q, so their normalized contribution vanishes. Full Cartesian sampling of the ordered pairs makes each fixed pair of unit classes have limiting probability $\varphi(Q)^{-2}$.

Take Q to be the product of any fixed finite set of primes. The Chinese remainder theorem identifies the unit-pair space with the product of the local unit-pair spaces. Divisibility of F at a prime depends only on the local pair, so the limiting divisibility indicators are independent, with exact law

$$\eta_\ell = \frac{\#\{(u,v) \in (\mathbb{F}_\ell^\times)^2 : \prod_j (c_j(u+v)+d_j) = 0\}}{(\ell-1)^2}. \tag{15}$$

For fixed k, ℓ the rank event uses only the primes at most ℓ. Summing their joint atoms gives (1).

If $F(p,q)=0$, then some $p+q = -d_j/c_j \le |d_j| \le \sum_j |d_j|$. Only finitely many such pairs exist, independently of X. Their normalized contribution is zero. The diagonal was included throughout; its $\pi(X)$ pairs are also negligible against $\pi(X)^2$. No modulus or rank was allowed to grow with X.

8.2 Exact local counts

Set $T_\ell = \{t \in \mathbb{F}_\ell : \prod_j (c_j t + d_j) = 0\}$. There are $\ell - 1$ unit pairs with sum zero, and $\ell - 2$ with any specified nonzero sum. Thus at every prime,

$$\eta_\ell = \frac{(\ell-2)|T_\ell| + \mathbf{1}_{0 \in T_\ell}}{(\ell-1)^2}. \tag{16}$$

If a factor is identically zero modulo ℓ, $T_\ell = \mathbb{F}_\ell$ and $\eta_\ell = 1$. An empty root set gives zero, and coincident roots are counted once. The formula covers all coefficient, shift, determinant, and small-prime exceptions.

Outside the finite set dividing the nonzero coefficients, nonzero shifts, or determinants $c_i d_j - c_j d_i$, there are m distinct roots. At most one shift is zero; let z_0 be one if it is and zero otherwise. Then

$$\eta_\ell = \frac{m(\ell-2)+z_0}{(\ell-1)^2}, \qquad \eta_\ell^{-1} = \frac{\ell}{m} - \frac{z_0}{m^2} + O_F(\ell^{-1}). \tag{17}$$

The law satisfies (4) with $\nu = m$. Theorem 3.2 proves the arithmetic theorem.

The simplest instances illustrate the exceptional coordinate:

$F(p,q)$	η_2	$\eta_\ell \quad (\ell > 2)$
$p+q$	1	$1/(\ell-1)$
$p+q+1$	0	$(\ell-2)/(\ell-1)^2$
$2(p+q)-1$	0	$(\ell-2)/(\ell-1)^2$.

The forced factor 2 in $p+q$ shifts subsequent ranks by one. Both odd examples have exactly the canonical odd law. All three have rate $\Lambda(1)$, despite different finite-rank behavior. Equations (5) and (14) also show how sufficiently large fixed numbers of distinct affine factors recover H_*. There is no uniformity claim when the family varies with the rank. The arithmetic realization uses fixed-modulus prime progression limits. Section 12 supplies the separate full-profile coefficient theorem that transfers the RH criterion to every one of these fixed families, with dimension $\nu = m$ and exact forced-coordinate rank shifts.

9 Late ascents and bounded prime gaps

For a canonical law, a *late ascent at coefficient* b is an ascent across consecutive primes $p < q$ with $\mathcal{L}(p-1) \geq bk$. Unbounded supply requires existence beyond every prescribed lower bound on both k and p.

Theorem 9.1 (Recurrence equivalence)**.** *Let $H \geq 1$ be an integer and $1/(H+2) < b < 1/(H+1)$. Gaps at most H recur arbitrarily far out if and only if for every K, Q there exist $k \geq K$ and consecutive primes $p < q$, with $p \geq Q$ and $\mathcal{L}(p-1) \geq bk$, at which both canonical densities ascend.*

Proof. Choose $b < a < 1/(H+1)$. To a sufficiently late gap at most H assign $k = \lfloor \mathcal{L}(p-1)/a \rfloor$. Then $k \to \infty$ with p, $\mathcal{L}(p-1) \geq ak \geq bk$, and $\mathcal{L}(p-1)/k \to a$. A small fixed band around a in Lemma 4.2 gives both ascents.

Conversely, $b > 1/(H+2)$ and the one-sided ratio bound exclude gaps at least $H+1$ from sufficiently late ascents. In the odd law the correction to the ordinary threshold has absolute value at most $1/(p-2)$, absorbed by the fixed strict margin. Unbounded lower primes therefore give recurring gaps at most H. □

Corollary 9.2 (Critical location)**.** *Suppose gaps at most $h \geq 1$ recur arbitrarily far out and every sufficiently late consecutive gap is at least h. The critical canonical late-ascent coefficient is $1/(h+1)$: every smaller positive coefficient has unbounded simultaneous ascent supply; every larger coefficient forces both densities to descend for all sufficiently large ranks and lower primes in that region. No assertion at equality is made.*

Proof. Use the forward rank assignment on gaps at most h and the one-sided upper ratio bound on gaps at least h. □

For every fixed $b > 1/5$, an ascent in either canonical law at sufficiently large k, p with $\mathcal{L}(p-1) \geq bk$ must cross a gap strictly below four. Gaps between odd primes are positive even integers, so it must be two. An independently proved unbounded supply would therefore imply infinitely many twin primes. That supply is not established here. A large finite ascent certifies a finite prime pair, not recurrence.

9.1 Infinitude, frequency and the last ascent

For either canonical family, let P_k denote the prime starting its last ascent, which exists for all sufficiently large k. If h is the least recurring prime gap, Corollary 9.2 and the upper cutoff imply

$$\limsup_{k\to\infty} \frac{\mathcal{L}(P_k)}{k} = \frac{1}{h+1}.$$

Indeed every smaller coefficient occurs at unbounded ranks and endpoints, whereas every larger coefficient is eventually excluded. Bounded-gap infinitude at width H therefore gives the lower location coefficient $1/(H+1)$. For example, taking width 186 as an input gives a coefficient at least $1/187$; it does not identify the least recurring gap. Twin-prime infinitude would give the location limsup exactly $1/3$. These implications concern location. A frequency hypothesis is needed for the corresponding reversal count.

Theorem 9.3 (Conditional last-ascent location)**.** *For either canonical law, let X_k be the largest supported prime p at which a hypothetical twin step satisfies the strict ascent criterion. Suppose that for some $C > 0$,*

$$\pi_2(X) \sim CX/(\log X)^2, \qquad \pi_2(X) = \#\{p \leq X : p,\ p+2 \text{ are prime}\}.$$

Then, for every sufficiently large k, P_k is exactly the last lower twin member at or below X_k, and

$$\frac{P_k}{X_k} \longrightarrow 1, \qquad \frac{\mathcal{L}(P_k)}{k} \longrightarrow \frac{1}{3}.$$

The same hypothesis and the count transfer give $\log\log(3+N_\eta(k))/k \to 1/3$ *for the canonical reversal counts.*

Proof. At a supported prefix, adding a positive weight decreases e_{r-1}/e_r by the elementary symmetric log-concavity inequality. The ordinary hypothetical twin threshold is 3; the odd threshold is $3-2/(p(p-2))$, which increases with $p > 2$. Thus the supported eligible primes form an initial segment. The uniform ratio estimates and Bertrand's theorem show that its last prime X_k exists for large k and $\mathcal{L}(X_k)/k \to 1/3$.

For every fixed $\varepsilon \in (0,1)$, subtracting the two assumed count asymptotics gives

$$\pi_2(X) - \pi_2((1-\varepsilon)X) \sim C\varepsilon X/(\log X)^2 > 0.$$

First use dyadic coverage in a fixed scale band strictly between 1/5 and 1/3. It supplies a twin ascent whose supported prefix serves as an anchor. Every later ascent must be a twin by the 1/5 exclusion, and lies at most at X_k. Conversely the last twin at or below X_k follows the anchor, has positive support and satisfies the strict criterion. This proves the exact identification. Now the displayed proportional coverage gives $(1-\varepsilon)X_k < P_k \le X_k$ eventually, for each fixed ε, and hence both location limits.

The global twin asymptotic also supplies a positive-power number of width-two windows in every sufficiently large dyadic interval. Corollary 5.2 for every $a < 1/3$, together with the canonical upper bound, proves the count limit. □

The usual Hardy–Littlewood twin conjecture supplies a particular positive constant C; the proof uses no value of that constant. The fixed-tuple asymptotic and the stronger uniform tuple assumptions used for short-interval distribution should be distinguished [14, (1) and Theorem 1]. Bare twin-prime infinitude does not supply the proportional coverage used here. Nor does the displayed asymptotic give a polylogarithmic or square-root bound for $X_k - P_k$ along these particular cutoffs. A limit on $\mathcal{L}(P_k)/k$ also does not imply $P_k/\mathcal{E}(k/3) \to 1$: a constant shift in the inner exponent already changes the original endpoint by a large multiplicative factor.

10 Prime-product errors and the full coefficient reference

We first develop the finer comparison for the ordinary integer law. Section 12 then extends the product and coefficient criteria to qualifying local laws. The

later arithmetic capacity, area and localized-count results retain their stated ordinary-law scope. We keep the arithmetic cutoff x and the rank r distinct: x tends to infinity, while r is comparable to $\log\log x$. All prime products in these sections include $p \le x$. For the prefix before an ascent prime p, the corresponding cutoff is $x = p - 1$.

10.1 Classical analytic quantities

Write

$$\theta(x) = \sum_{p \le x} \log p, \qquad \psi(x) = \sum_{p^a \le x} \log p, \qquad \kappa(t) = \frac{1/\log t + 1/(\log t)^2}{t^2}.$$

For sufficiently large x, set

$$\begin{aligned} J(x) &= \int_x^\infty (\psi(t) - t)\kappa(t)\, dt, \\ K(x) &= \int_x^\infty (\theta(t) - t)\kappa(t)\, dt, \\ P(x) &= J(x) - K(x) = \int_x^\infty (\psi(t) - \theta(t))\kappa(t)\, dt, \\ f(x) &= e^\gamma \log\theta(x) \prod_{p \le x} (1 - 1/p), \qquad F(x) = \log f(x). \end{aligned} \tag{18}$$

The quantitative prime number theorem gives convergence of these integrals. These are Nicolas's quantities [8, (1.9), (1.15)–(1.19)]; P retains every prime power. Let

$$A(x) = \sum_{p > x} \big(-\log(1 - 1/p) - 1/p\big),$$

and, when $\theta(x) > 1$, let

$$D(x) = \frac{\theta(x) - x}{x \log x} - \big(\log\log\theta(x) - \log\log x\big).$$

Partial summation and the convergent logarithmic prime-power series give the exact identity

$$F(x) = K(x) - D(x) + A(x) = J(x) - P(x) - D(x) + A(x). \tag{19}$$

Here $A(x) \ge 0$, $A(x) = O(1/x)$, and $D(x) \ge 0$ by concavity of $\log\log t$ on $t > 1$. No sign is assigned to $J(x)$.

We use the following classical facts, with their hypotheses retained. Nicolas's criterion is equivalent to eventual $F(x) < 0$. Under RH,

$$F(x) = -\frac{2 + W(x)}{\sqrt{x}\log x} + O\left(\frac{1}{\sqrt{x}(\log x)^2}\right), \qquad |W(x)| \le \beta_0, \tag{20}$$

where

$$W(x) = \operatorname{Re}\sum_{\rho} \frac{x^{i\operatorname{Im}\rho}}{\rho(1-\rho)}, \qquad \beta_0 = 2 + \gamma - \log(4\pi) < 2.$$

The zeros in this absolutely convergent series are counted with multiplicity. Defining the real projection explicitly matches the maintained formal statement. The expansion and its constant are classical [8, Proposition 2.1]. In the decomposition (19), the zero wave comes from J, while

$$P(x) \sim \frac{2}{\sqrt{x}\log x}$$

holds unconditionally and gives the leading prime-square contribution. Under RH the standard bounds $|\psi(x) - x| + |\theta(x) - x| = O(\sqrt{x}(\log x)^2)$ imply

$$D(x) = O((\log x)^3/x), \qquad |F(x) - K(x)| = O((\log x)^3/x). \tag{21}$$

Indeed the second derivative of $\log\log t$ is $-(\log t + 1)/(t^2(\log t)^2)$; Taylor's integral remainder, with $\theta(x)/x \to 1$, proves the first estimate, and (19) proves the second.

If RH fails, F has excursions of both signs of order x^{-b} for some $0 < b < 1/2$ [7, 8]. Section 11.1 records the zero-dependent version used in the coefficient transfer. In particular, the estimate for D in (21) is never used under the assumption that RH is false.

10.2 An entire profile and two different clocks

Define

$$\begin{aligned} G_x(z) &= \prod_{p \le x} \left(1 + \frac{z}{p-1}\right), & a_r(x) &= [z^r]G_x(z), \\ \mathcal{H}(z) &= \prod_{p} \left(1 + \frac{z}{p-1}\right)(1 - 1/p)^z, & C_r(u) &= [z^r]\mathcal{H}(z)e^{uz}. \end{aligned} \tag{22}$$

The power $(1-1/p)^z$ uses the real logarithm of its positive base. The product for $\mathcal{H}$ converges uniformly on compact subsets of $\mathbb{C}$: the logarithmic first-order terms cancel and the remaining tails are bounded by a constant times

$\sum_p p^{-2}$. It defines an entire function, positive on the nonnegative real axis, with $\mathcal{H}(0) = \mathcal{H}(1) = 1$.

The finite generating product is classical in this problem [2, (1.6)]; the normalized Euler factor also appears in the mod-Poisson description of prime divisibility [9, §4]. Our reference retains all its Taylor coefficients. It is not a finite-order truncation of a saddle-point expansion.

Let $\mathcal{H}_x$ be the finite product in (22), and put

$$B(x) = \log G_x(1), \qquad L(x) = \gamma + \log\log\theta(x). \tag{23}$$

Then $F(x) = L(x) - B(x)$ and, exactly,

$$G_x(z) = \mathcal{H}_x(z)e^{B(x)z}. \tag{24}$$

For positive supported degrees define

$$R_r(x) = \frac{a_{r-1}(x)}{a_r(x)}, \qquad \mathcal{R}_r(u) = \frac{C_{r-1}(u)}{C_r(u)}, \qquad \Delta_r(x) = R_r(x) - \mathcal{R}_r(L(x)).$$

The denominator conditions below ensure that these ratios have their ordinary real-valued meaning.

Theorem 10.1 (Uniform coefficient transfer)**.** *Fix $0 < a \le b < \infty$. There exist X and $K > 0$ such that, for every real $x \ge X$ and integer r with $a \le r/L(x) \le b$, both denominators are positive and*

$$\left|\Delta_r(x) - \frac{r}{L(x)^2}F(x)\right| \le K\left(\frac{|F(x)|}{L(x)^2} + \frac{1}{rx}\right). \tag{25}$$

The constants may depend on a, b, but not on r or x.

Proof. Write $\mathcal{H}(z) = \sum_{j\ge0} h_j z^j$ and

$$Q_r(u) = \frac{r!C_r(u)}{u^r} = \sum_{j=0}^{r} h_j \frac{(r)_j}{u^j}.$$

For $t = r/u$ in a fixed positive compact interval, the falling-factorial estimate

$$\left|\frac{(r)_j}{r^j} - 1 + \frac{j(j-1)}{2r}\right| \le \frac{j^4}{r^2}$$

holds also for $j > r$ if $(r)_j$ is extended by zero. For $j \le r$, apply the first two inclusion-exclusion bounds to $\prod_{i=0}^{j-1}(1 - i/r)$; the other case is immediate.

Cauchy coefficient bounds on a circle of radius larger than the compact interval justify summation with the factor j^4. Hence

$$Q_r(u) = \mathcal{H}(t) - \frac{t^2\mathcal{H}''(t)}{2r} + O(r^{-2}). \tag{26}$$

Since $\mathcal{H}(t) > 0$, the normalized denominators are bounded away from zero. Using $C_r' = C_{r-1}$ and the same expansion for $r-1, r-2$ gives

$$\mathcal{R}_r'(u) = \frac{C_{r-2}(u)}{C_r(u)} - \mathcal{R}_r(u)^2 = -\frac{r}{u^2} + O(r/u^3). \tag{27}$$

The first-order profile terms cancel in this paired difference.

On any fixed larger circle, $\mathcal{H}_x - \mathcal{H} = O(1/x)$ by the quadratic prime tail. To compare ratios without losing the cancellation, use

$$\frac{C_{r-1}(u)}{C_r(u)} = \frac{r}{u}\left(1 + \frac{Q_{r-1}(u) - Q_r(u)}{Q_r(u)}\right).$$

The identity $(r)_j - (r-1)_j = j(r-1)_{j-1}$ supplies a factor $1/u$ in the numerator difference. Replacing $\mathcal{H}$ by $\mathcal{H}_x$ therefore changes this ratio by $O(1/(rx))$, uniformly in the compact rank range. Equation (24) yields

$$R_r(x) = \mathcal{R}_r(B(x)) + O(1/(rx)).$$

Unconditionally $F(x) = o(1)$ by PNT and Mertens's theorem. Integrate (27) between L and $B = L - F$. The main term is rF/L^2; its error is $O(|F|/L^2)$ since r is comparable to L and $F = o(1)$. This proves (25). □

The coefficient expansion belongs to the saddle-point and mod-Poisson tradition [2, 9, 10]. The sign-preserving comparison (25), its theta centering, and its application to the actual ascent ratio are the potentially new statements here. Any fixed error $O(L^{-m})$ would be larger than the RH signal $1/(L\sqrt{x}\log x)$.

11 RH criteria and quantitative excursions

Theorem 11.1 (Moving-rank RH criterion)**.** *Fix $\alpha > 0$ and set $r(x) = \lfloor \alpha L(x) \rfloor$. Then RH is equivalent to*

$$R_{r(x)}(x) < \mathcal{R}_{r(x)}(L(x)) \quad \textit{for every sufficiently large } x.$$

Under RH, more precisely,

$$\frac{L(x)^2}{r(x)}\sqrt{x}\log x\,\Delta_{r(x)}(x) = -2 - W(x) + o(1). \tag{28}$$

If RH is false, there exists one $b \in (0, 1/2)$ such that, for every fixed $\alpha > 0$,

$$\limsup_{x\to\infty} x^b L(x)\Delta_{r(x)}(x) = +\infty, \qquad \liminf_{x\to\infty} x^b L(x)\Delta_{r(x)}(x) = -\infty. \tag{29}$$

The witnessing endpoints may depend on α and on the sign.

Proof. The floor rank is eventually in a fixed positive compact band. Under RH, substitute (20) into (25). After multiplication by $(L^2/r)\sqrt{x}\log x$, the two errors are respectively $O(1/L)$ and $O((\log x)/\sqrt{x})$. Both vanish, giving (28). Since $2 - \beta_0 > 0$, the sign is eventually negative.

Under false RH, Theorem 11.2 below supplies b and excursions of F with arbitrarily large amplitudes. In a band $a \le r/L \le b_1$, (25) gives, eventually,

$$\begin{aligned} F \ge 0 &\implies L\Delta_r \ge (a/2)F - K/(ax), \\ F \le 0 &\implies L\Delta_r \le (a/2)F + K/(ax). \end{aligned}$$

Here b_1 denotes the fixed band endpoint, not the excursion exponent. For the floor rank one may take $a = \alpha/2$. Since $x^{-1} = o(x^{-b})$, both excursions transfer with arbitrarily large amplitudes. This proves (29) and the reverse implication. □

The equality (28) is an ordinary corollary of the compiled linearization and spectral estimates. The maintained criterion and the quantitative false-RH consumer are separate named declarations; the ledger distinguishes these verification scopes.

11.1 The corrected logarithm at power-profile peaks

Theorem 11.2 (Zero-dependent excursions)**.** *Suppose $\zeta(\rho) = 0$ with $1/2 < \operatorname{Re}\rho < 1$. For every*

$$1 - \operatorname{Re}\rho < b \le 1/2$$

and each $U \in \{J, K, F\}$,

$$\limsup_{x\to\infty} x^b U(x) = +\infty, \qquad \liminf_{x\to\infty} x^b U(x) = -\infty.$$

Consequently, false RH supplies a common exponent $0 < b < 1/2$.

Proof. We give the analytic mechanism and the endpoint correction explicitly. For $a = \log 3$, initially in $\operatorname{Re} s > 0$, put

$$\widehat{J}(s) = \int_a^\infty J(e^u)e^{-su}\,du, \qquad T(z) = \int_3^\infty (\psi(t) - t)t^{-z-1}\,dt.$$

The Mellin identity is $T(z) = -\zeta'(z)/(z\zeta(z)) - 1/(z-1)$ minus the entire contribution of the finite interval from 1 to 3. Integration by parts gives

$$s\widehat{J}''(s) + 2\widehat{J}'(s) = T'(s+1) - T(s+1) + a^2 J(3)e^{-as}. \tag{30}$$

The cancellation at $z = 1$ and the quantitative PNT extend $\widehat{J}$ across $s = 0$ and along the real interval $s > -1/2$. At $s = \rho - 1$, the double pole from T' cannot cancel with the simple pole from T. Thus $\widehat{J}$ cannot be holomorphic there.

An eventual bound $J(x) \geq -Cx^{-b}$ would make $J(e^u) + Ce^{-bu}$ eventually nonnegative. Its Laplace transform has the preceding nonreal singularity, but is analytic on the real interval $s > -b$. Landau's singularity principle contradicts this: its abscissa of convergence lies between $\operatorname{Re}\rho - 1$ and 0, hence strictly above $-b$. Applying the same argument to $-J$ rules out the opposite bound. This is the classical Landau method; see [11, Theorem 3]. Since $P(x) = O(x^{-1/2}/\log x) = o(x^{-b})$, $K = J - P$ inherits both signs.

For negative excursions, $F \leq K + A$ and $A = O(1/x)$ suffice. For positive excursions one must control D without RH. Fix $A_0 > 0$, and choose an integer local maximum $x = n + 1$, $n \geq 2$, of the sequence $K(n) - A_0 n^{-b}$ between two lower values bracketing a high positive value. Such maxima exist arbitrarily far out because of the two unbounded excursions of K; rounding real endpoints to integers costs $O((\log x)/x) = o(x^{-b})$.

The two adjacent comparisons are

$$K(n) - A_0 n^{-b} \leq K(n+1) - A_0(n+1)^{-b},$$

$$K(n+2) - A_0(n+2)^{-b} \leq K(n+1) - A_0(n+1)^{-b}.$$

On each open unit interval θ is constant. Using $K(v) - K(w) = \int_v^w (\theta(t) - t)\kappa(t)\,dt$ and the possible endpoint jump of at most $\log x$ gives

$$0 < \theta(x) - x \leq (2A_0 + 1)x^{1-b}\log x.$$

Taylor's integral remainder for $\log\log$ and $A(x) \geq 0$ now yield

$$F(x) \geq K(x) - \frac{(2A_0+1)^2}{2}(\log x + 1)x^{-2b}.$$

The loss is $o(x^{-b})$. Choosing A_0 larger than any prescribed amplitude proves the positive excursions of F. Finally, if RH fails, symmetry of the zeros supplies such a ρ; choose any b strictly between $1 - \operatorname{Re}\rho$ and $1/2$. □

The existence of false-RH oscillations of F is classical [7, 8]. The displayed proof records the zero-dependent exponent and the exact nonlinear correction used by the formal coefficient consumer.

11.2 Integral, clock and positive-tilt criteria

Define

$$\mathcal{C}(x) = \exp\bigl(e^{-\gamma} G_x(1)\bigr).$$

The exact identity $\mathcal{C}(x) = \theta(x)^{1/f(x)}$ shows that eventual $\theta(x) < \mathcal{C}(x)$ is another form of Nicolas's criterion. Combining the preceding results gives

$$\text{RH} \iff K(x) < 0 \text{ eventually} \iff \theta(x) < \mathcal{C}(x) \text{ eventually}. \tag{31}$$

Under RH,

$$\frac{\mathcal{C}(x) - \theta(x)}{\sqrt{x}} = 2 + W(x) + o(1). \tag{32}$$

Indeed $\log \mathcal{C} - \log\theta = (1/f - 1)\log\theta$; (20), $\theta(x)/x \to 1$, and $e^t = 1 + t + O(t^2)$ prove (32). This is a change of variables in the classical Nicolas estimate.

For fixed $z > 0$, put

$$T_z(x) = \log\left(\frac{e^{\gamma z}\mathcal{H}(z)(\log\theta(x))^z}{G_x(z)}\right).$$

Then, exactly,

$$T_z(x) = zF(x) + \sum_{p>x}\left\{\log\left(1 + \frac{z}{p-1}\right) - z\log\left(1 + \frac{1}{p-1}\right)\right\}. \tag{33}$$

For fixed z the tail is $O_z(1/x)$. It is nonpositive for $z \geq 1$, and lies between 0 and Cz/x for $0 < z \leq 1$, with an absolute C. Concavity of $\log(1 + zw)$ proves the signs; the upper bound follows from $\log(1 + zw) \leq zw$ and $w - \log(1 + w) \leq w^2/2$. Therefore

$$\text{RH} \iff T_z(x) < 0 \text{ eventually}$$

for each fixed $z > 0$. Under RH a *single* sufficiently large cutoff works simultaneously for all $z > 0$, by splitting at $z = 1$ in (33); the converse follows from $T_1 = F$.

The corresponding positive kernel is

$$\kappa_z(t) = \frac{z}{(t-1)(t-1+z)\log t} + \frac{\log(1+z/(t-1))}{t(\log t)^2},$$
$$I_z(x) = \int_x^\infty (\theta(t)-t)\kappa_z(t)\,dt.$$

For $t \geq e$, $|\kappa_z(t) - z\kappa(t)| \leq 14(z+z^2)/(t^3 \log t)$. Chebyshev's bound gives $I_z = zK + O_z(1/(x\log x))$. It follows that eventual $I_z < 0$ is equivalent to RH and that

$$\sqrt{x}\log x\, I_z(x) = -z(2+W(x)) + o(1)$$

under RH. Under false RH every fixed positive tilt inherits the zero-dependent two-sided integral excursions. These tilt criteria transport the same leading error; they do not supply independent sign information by themselves.

12 RH transfer across qualifying local laws

12.1 Qualification and the exact family constants

Fix an independent local law with probabilities $0 \leq \eta_p \leq 1$. For an arithmetic ensemble, identifying this model with its densities also requires independence of its limiting finite-prime divisibility laws. Marginal probabilities alone do not establish that identification. Put

$$w_p = \begin{cases} \eta_p/(1-\eta_p), & \eta_p < 1, \\ 0, & \eta_p = 1. \end{cases}$$

A forced coordinate is removed from the odds product and retained in the rank shift below. The concrete qualification condition is

$$\nu > 0, \quad D \geq 0, \quad w_p \geq 0, \qquad \left| w_p - \frac{\nu}{p-1} \right| \leq \frac{D}{(p-1)^2} \quad \text{for every prime } p. \tag{34}$$

All parameters and the family are fixed before the cutoff tends to infinity. The condition follows from $\eta_p = \nu/p + O(p^{-2})$ along the primes, including arbitrary finite exceptions. Indeed, eventually $\eta_p \leq 1/2$, so

$$\frac{\eta_p}{1-\eta_p} = \eta_p + \frac{\eta_p^2}{1-\eta_p} = \frac{\nu}{p} + O(p^{-2}) = \frac{\nu}{p-1} + O(p^{-2}).$$

The finite head is absorbed into D. This argument is formalized for an explicit prime-tail bound as well as a bound at every prime. A limiting reciprocal offset is not required for this RH transfer; that stronger hypothesis remains part of the reversal-spectrum theorem.

For integer $N \geq 1$, define

$$G_{\eta,N}(z) = \prod_{p \leq N} (1 + w_p z), \qquad a_{\eta,r}(N) = [z^r] G_{\eta,N}(z),$$
$$\mathcal{H}_\eta(z) = \prod_{p} (1 + w_p z)(1 - 1/p)^{\nu z}, \qquad C_{\eta,r}(u) = [z^r] \mathcal{H}_\eta(z) e^{uz}.$$

The complex powers use the real logarithm of the positive base $1 - 1/p$. Write

$$\kappa_\eta(z) = \log \mathcal{H}_\eta(z), \qquad \Gamma_\eta(z) = e^{\nu z \gamma} \mathcal{H}_\eta(z) \quad (z \geq 0).$$

These are exact convergent constants; exceptional primes remain in the products. If $v_p = 1/(p-1)$, then $w_p \leq (\nu + D) v_p$. The elementary bounds $0 \leq t - \log(1+t) \leq t^2/2$ give

$$|\log(1 + w_p z) - \nu z \log(1 + v_p)| \leq K_\eta(z) v_p^2,$$
$$K_\eta(z) = \frac{z^2(\nu + D)^2}{2} + zD + \frac{\nu z}{2}.$$

Since $\sum_{p > N} v_p^2 \leq 2/N$, we have the explicit bound

$$\left| \kappa_\eta(z) - \sum_{p \leq N} \big(\log(1 + w_p z) - \nu z \log(1 + v_p) \big) \right| \leq \frac{2K_\eta(z)}{N}. \tag{35}$$

Theorem 12.1 (Family product criteria)**.** *Under* (34)*, fix $z > 0$ and set*

$$E_\eta(N, z) = \nu z \big(\gamma + \log\log \theta(N) \big) + \kappa_\eta(z) - \log G_{\eta,N}(z).$$

For $\theta(N) > 1$,

$$|E_\eta(N, z) - \nu z F(N)| \leq 2K_\eta(z)/N.$$

RH is equivalent to each of the following eventual statements:

$$E_\eta(N, z) < 0, \qquad E_\eta(N, z) \leq A/N, \qquad \Gamma_\eta(z) (\log \theta(N))^{\nu z} < G_{\eta,N}(z),$$

where A is any fixed real number.

Proof. The finite identity $F(N) = \gamma + \log\log\theta(N) - \sum_{p\le N}\log(1+v_p)$ and (35) give the error bound. Under RH, the Nicolas negative margin dominates every fixed multiple of $1/N$. If RH fails, its positive integer-cutoff excursions exceed every such multiple. Multiplication by the fixed positive number νz preserves both conclusions. Positivity of all factors turns the logarithmic criterion into the displayed product inequality. These are applications of Nicolas's criterion and error estimates [8, (1.10), Proposition 2.1]. □

There is a broader sufficient qualification for the product theorem. For fixed nonnegative weights and fixed $\nu, z > 0$, it is enough that some real κ and $C \ge 0$ satisfy

$$\left|\kappa - \sum_{p\le N}\big(\log(1+w_p z) - \nu z\log(1+v_p)\big)\right| \le C/N \quad \text{eventually.} \tag{36}$$

The same proof gives the product criterion with constant $\exp(\nu z\gamma+\kappa)$. This qualification contains neither θ nor an RH sign assumption. It is formalized separately. We do not assert that either sufficient condition is a necessary classification of every family that might admit an RH criterion.

12.2 The coefficient ratio and deterministic rank shifts

Let

$$U_\eta(N) = \nu L(N), \qquad R_{\eta,r}(N) = \frac{a_{\eta,r-1}(N)}{a_{\eta,r}(N)}, \qquad \overline{R}_{\eta,r}(u) = \frac{C_{\eta,r-1}(u)}{C_{\eta,r}(u)}.$$

Here $L(N) = \gamma + \log\log\theta(N)$ as before.

Theorem 12.2 (Uniform family coefficient transfer)**.** *Assume* (34). *On every fixed positive compact band for $r/U_\eta(N)$, both denominators are positive for large N, r, and for a constant K depending on the law and band,*

$$\left|R_{\eta,r}(N) - \overline{R}_{\eta,r}(U_\eta(N)) - \frac{r}{U_\eta(N)^2}\nu F(N)\right| \le K\left(\frac{|\nu F(N)|}{U_\eta(N)^2} + \frac{1}{rN}\right). \tag{37}$$

For every fixed $\alpha > 0$, put $r(N) = \lfloor \alpha U_\eta(N)\rfloor$. Each of the eventual inequalities

$$R_{\eta,r(N)}(N) < \overline{R}_{\eta,r(N)}(U_\eta(N)), \qquad R_{\eta,r(N)}(N) \le \overline{R}_{\eta,r(N)}(U_\eta(N))$$

is equivalent to RH.

Proof. For $|z| \le R$, the normalized local factor differs from one by at most $K_R^{\mathbb{C}} v_p^2$, where one may take

$$K_R^{\mathbb{C}} = \big((D+\nu/2)R + \nu^2 R^2 e^{\nu R}\big) e^{\nu R}.$$

The product therefore defines an entire function and its finite prefix approximates it uniformly on each closed disk with error $O_R(N^{-1})$. The proof uses product differences, so zeros of individual factors on a disk cause no division problem. On $0 \le t \le b$,

$$\mathcal{H}_\eta(t) \ge \exp\left(-K_\eta(b)\sum_p v_p^2\right) > 0.$$

Cauchy's estimate therefore bounds the coefficient of degree j in the prefix error by $O_R(N^{-1}R^{-j})$.

Apply the factorial-polynomial argument of Section 10 to this entire profile. If $h_j = [z^j]\mathcal{H}_\eta(z)$, its normalized coefficient polynomial is

$$P_r(t) = \sum_{j=0}^{r} h_j t^j \frac{(r)_j}{r^j}.$$

The entire coefficient series and its derivatives give uniform approximations to $\mathcal{H}_\eta$ and $\mathcal{H}_\eta'$ on a compact interval. The displayed positive lower bound controls the denominator. The exact identity

$$\overline{R}_{\eta,r}(u) = t - \frac{t^2}{r}\frac{P_r'(t)}{P_r(t)}, \qquad t = r/u,$$

then gives a paired prefix/reference error $O((rN)^{-1})$ and reference slope $-r/u^2 + O(u^{-2})$. These are the same general polynomial estimates used for Theorem 10.1; their hypotheses have now been proved for $\mathcal{H}_\eta$.

The actual finite product is restored at $B_\eta(N) = \nu\sum_{p\le N}\log(1+v_p)$, and $U_\eta(N) - B_\eta(N) = \nu F(N)$ exactly. Integrating the slope over the interval between these two clocks proves (37). The floor-rank band has positive lower and upper bounds. The negative RH margin and the positive false-RH excursions dominate its inverse-cutoff error, giving both asserted equivalences. □

To interpret the ratio arithmetically, put

$$f_N = \#\{p \le N : \eta_p = 1\}, \qquad M_{\eta,N}(j) = [z^j]\prod_{p\le N}(1-\eta_p+\eta_p z).$$

For $r \geq 1$ the exact identity, with total division at a zero denominator, is

$$\frac{M_{\eta,N}(f_N + r - 1)}{M_{\eta,N}(f_N + r)} = R_{\eta,r}(N). \tag{38}$$

Indeed, replace each forced probability by zero, shift by its exact count, and factor out the strictly positive survival product of the remaining coordinates. Excluded coordinates contribute zero odds and no rank shift. Both the identity and its two RH consumers are formalized. At a prime prefix $N = p - 1$, the corresponding distinct-factor rank is $k = f_N + r + 1$.

Every reciprocal-smooth law qualifies by its proved odds estimate. Consequently every fixed affine prime-pair family in Section 8 has these product and mass-ratio RH criteria, with $\nu = m$ and its exact residue probabilities. No bounded-gap or short-interval hypothesis is needed for these equivalences. Arithmetic interpretation still takes the input-size limit first, with the finite prime prefix fixed. Sparse support families require a separate qualification check; the affine constructor makes no assertion about them.

12.3 Reference-scale constants and combinations of families

The same entire-profile estimates give a locally unique positive-level reference crossing with

$$\overline{R}_{\eta,r}(u_{r,s}) = s, \qquad u_{r,s} = \frac{r}{s} - \frac{\mathcal{H}'_\eta(s)}{\mathcal{H}_\eta(s)} + O_{\eta,s}(r^{-1}).$$

For fixed $g \geq 0$, take $s = 1 + g/\nu$ and $T_{\eta,r,g} = \exp(\exp(u_{r,s}/\nu - \gamma))$. Then

$$\log\log T_{\eta,r,g} = \frac{r}{g+\nu} - \gamma - \frac{1}{\nu}\frac{\mathcal{H}'_\eta(1+g/\nu)}{\mathcal{H}_\eta(1+g/\nu)} + O_{\eta,g}(r^{-1}). \tag{39}$$

This generalizes the reference constant, including every finite exception in $\mathcal{H}_\eta$. In a density application, the free rank r must still be replaced by $k - f_N - 1$. The theorem describes the reference crossing. Identifying the actual last ascent additionally requires the stated arithmetic supply and sufficient precision of the actual local gap threshold.

Finally, for a fixed finite collection of qualifying laws and fixed real coefficients a_i, let $z_i \geq 0$ and $\lambda = \sum_i a_i \nu_i z_i$. The exact family bounds imply

$$\left| \sum_i a_i E_{\eta_i}(N, z_i) - \lambda F(N) \right| \leq \frac{1}{N} \sum_i |a_i|\, 2K_{\eta_i}(z_i).$$

If $\lambda > 0$, eventual negativity of this combination is another RH criterion. If $\lambda = 0$, only the inverse-cutoff remainder remains. Thus fixed finite family averaging retains one common arithmetic signal. An infinite or cutoff-dependent collection needs a separate uniform summability argument; no such transfer is asserted here.

13 Reference crossings and RH ascent capacities

The preceding centered criterion concerns $R_r - \mathcal{R}_r(L)$. A raw ascent instead concerns $R_r - (g+1)$ for an actual gap g. We now keep both quantities in the comparison.

Theorem 13.1 (Sharp reference crossing)**.** *For every fixed $s > 0$ there are $A, C > 0$ and r_0 such that, for $r \geq r_0$, the equation $\mathcal{R}_r(u) = s$ has a unique solution $u_{r,s}$ in $[r/s - A, r/s + A]$, and*

$$u_{r,s} = \frac{r}{s} - \frac{\mathcal{H}'(s)}{\mathcal{H}(s)} + O_s(1/r). \tag{40}$$

Proof. Expanding the adjacent normalized coefficients as in (26) gives, for $t = r/u$ in a compact neighborhood of s,

$$\mathcal{R}_r(u) = t - \frac{t^2}{r}\frac{\mathcal{H}'(t)}{\mathcal{H}(t)} + O(r^{-2}).$$

Its values on either side of r/s at a sufficiently large fixed distance straddle s. The derivative (27) is strictly negative throughout that interval, giving existence and uniqueness. Substitute $u = r/s + v$, with bounded v, into the last display to obtain (40). □

For $s = h + 1$, put $X_{r,h} = \mathcal{E}(u_{r,h+1} - \gamma)$. The reference crossing admits a local spatial interpretation. Fix a comparison constant $D_0 \geq 1$. In the window

$$B(x) \in [r/(h+1) - A, r/(h+1) + A], \qquad \mathcal{C}(x) \leq D_0 x, \qquad X_{r,h} \leq D_0 x,$$

the maintained coefficient comparison gives, for sufficiently large r, x and a constant K depending on the fixed parameters,

$$\begin{aligned} X_{r,h} - \mathcal{C}(x) > K \log x &\implies R_r(x) > h + 1, \\ \mathcal{C}(x) - X_{r,h} > K \log x &\implies R_r(x) < h + 1. \end{aligned} \tag{41}$$

Indeed $R_r(x) - \mathcal{R}_r(B(x)) = O(1/(rx))$, whereas $|\mathcal{R}_r'| \gg 1/r$ in the window. The uncertainty in B is $O(1/x)$; the derivative of $\mathcal{E}(u-\gamma)$ is $\mathcal{E}(u-\gamma)\log\mathcal{E}(u-\gamma)$, giving the stated spatial margin. For an actual gap h, these are ascent/descent tests. Equation (32) locates their RH transition in the θ coordinate, with displacement $(2+W(x))\sqrt{x}+o(\sqrt{x})$.

Theorem 13.2 (RH endpoint and reversal-count envelope)**.** *Assume RH. Let P_k be the lower prime of the last ordinary ascent and $N_k = N_{\mathrm{ordinary}}(k)$. For all sufficiently large k, both are attained, and with $r = k-1$,*

$$\theta(P_k - 1) < \mathcal{E}(u_{r,3} - \gamma). \tag{42}$$

For every $\varepsilon > 0$, eventually

$$(1-\varepsilon)N_k \log N_k \le \theta(P_k - 1), \qquad N_k \le (1+\varepsilon)\frac{\mathcal{E}(u_{r,3}-\gamma)}{\log\mathcal{E}(u_{r,3}-\gamma)}.$$

There is a constant C such that

$$\log\log N_k \le \frac{k-1}{3} - \gamma - \frac{\mathcal{H}'(3)}{\mathcal{H}(3)} + \frac{C}{k-1} \tag{43}$$

for all sufficiently large k.

Proof. The unconditional lower reversal bound supplies a last ascent whose endpoint is at least double-exponential in a positive multiple of k. The unconditional upper cutoff puts it below $\mathcal{E}((1/3+\varepsilon)k)$ for every fixed $\varepsilon > 0$. PNT therefore places $r/L(P_k - 1)$ in a fixed positive compact band. Under RH, Theorem 10.1 and (20) give $R_r(P_k - 1) < \mathcal{R}_r(L(P_k - 1))$ uniformly in that band. The actual ascent requires $R_r > g + 1 \ge 3$.

Near $u_{r,3}$, the reference is strictly decreasing. Away from its fixed neighborhood the uniform leading expansion separates its value from 3. These two comparisons imply $L(P_k - 1) < u_{r,3}$, which is (42). Every separated witness uses a distinct ascent index, so $N_k \le \pi(P_k)$. PNT yields the first two count inequalities. Equivalently one can count all primes with $\theta(p-1)$ below the reference endpoint; this monotone capacity has asymptotic $T/\log T$ as $T \to \infty$.

Taking two logarithms of that capacity gives $u_{r,3} - \gamma + o(1/r)$: the correction from division by $\log T$ is negligible on this scale, and the multiplicative PNT error is bounded eventually. Equation (40) now proves (43). □

The same comparison keeps the actual gap of each ascent: in a fixed positive compact rank band, RH implies

$$g_i + 1 < \mathcal{R}_{k-1}(L(p_i - 1)).$$

Thus, on any sufficiently late finite band of prime indices, the number of ascents is at most the number of indices satisfying this reference inequality with their *actual* g_i. Adding the number of earlier primes bounds every reversal family. Combining this upper count with the unconditional reversal supply forces many reference-eligible actual gaps, but does not lower their maximum width without an additional distribution argument.

Neither the square-root clock displacement nor the $O(1/k)$ count correction changes a leading coefficient measured by $\log\log X/k$. The rank is fixed in the raw reversal count; it moves with x in Theorem 11.1.

14 Persistent excursions and positive-area criteria

The pointwise criterion also admits a one-sided integral formulation. This retains information lost by the leading reversal spectrum. Write $v_+ = \max(v, 0)$ and define

$$\begin{aligned} Q_F(X) &= \int_X^{2X} F(t)_+ \, dt, \\ S_\alpha(x) &= \frac{L(x)^2}{\lfloor \alpha L(x) \rfloor} \Delta_{\lfloor \alpha L(x) \rfloor}(x), \\ Q_\alpha(X) &= \int_X^{2X} (S_\alpha(t))_+ \, dt. \end{aligned}$$

All statements below concern sufficiently large X; the rank and coefficient denominators are then positive. Arbitrary definitions on the bounded initial range do not change an eventual criterion.

Lemma 14.1 (Persistence at a controlled peak)**.** *If RH is false, there is $0 < b < 1/2$ such that for every fixed $A > 0$ and $d < 1 - b/2$, arbitrarily large integers n satisfy*

$$F(y) > An^{-b} \qquad (n \le y \le n + n^d).$$

More precisely, controlled positive peaks with $K(n) > A_0 n^{-b}$ give intervals of length $c_{A_0} n^{1-b/2}/\sqrt{\log n}$ on which $F(y) \ge (A_0/2)n^{-b}$, for some fixed $c_{A_0} > 0$.

Proof. The two-sided peak construction in Theorem 11.2 supplies arbitrarily late integer peaks x with $K(x) > A_0 x^{-b}$ and $0 < \theta(x) - x \le (2A_0 + 1)x^{1-b}\log x$. For $0 < h \le x$ and $x \le y \le x + h$, distinct integer prime positions give

$$|\theta(y) - y| \le M := (2A_0 + 1)x^{1-b}\log x + 3(h+1)\log x$$

eventually. The kernel satisfies $\kappa(t) \le 2/(x^2 \log x)$ on this interval. Taylor's integral remainder for $\log\log t$, on arguments at least $x/2$, gives

$$F(y) \ge K(x) - \frac{2Mh + 8M^2}{x^2 \log x} \quad \text{when } M \le x/2.$$

The nonnegative tail $A(y)$ was discarded only in this lower bound. Take $h = \eta x^{1-b/2}/\sqrt{\log x}$, with $\eta = \sqrt{A_0}/100$. The term $2Mh$, after the displayed normalization and division by x^{-b}, tends to zero. Using $M^2 \le 2(2A_0 + 1)^2 x^{2-2b} \log^2 x + 18(h+1)^2 \log^2 x$, the remaining loss has limsup at most $144\eta^2 < A_0/2$. Also $M = o(x)$. This proves the more precise assertion. Choose $A_0 > 2A$; every fixed x^d with $d < 1 - b/2$ is eventually shorter than h. □

The maintained formal proof establishes every fixed-power version and the finite loss envelope. The logarithmic critical-length refinement is the ordinary calculation displayed above.

Theorem 14.2 (Positive-area RH criteria)**.** *For every fixed $\alpha > 0$, each of the following is equivalent to RH:*

$$Q_F(X) = O\big(X^{1/4}(\log X)^B\big) \quad \textit{for some fixed } B \ge 0,$$

$$Q_\alpha(X) = O\big(X^{1/4}(\log X)^B\big) \quad \textit{for some fixed } B \ge 0.$$

The bounds are required at every sufficiently large real X. For $1/4 \le a < 1$, either bound $Q(X) = O_\varepsilon(X^{a+\varepsilon})$ for every $\varepsilon > 0$ implies that every nontrivial zero satisfies $\operatorname{Re}\rho \le (1+2a)/3$. These are conditional zero restrictions, not unconditional strips.

Proof. The coefficient transfer gives, with a fixed C_α,

$$|S_\alpha(x) - F(x)| \le \frac{C_\alpha}{L(x)}|F(x)| + \frac{C_\alpha}{x}.$$

Once $C_\alpha/L(x) \le 1/2$, taking positive parts and integrating yields

$$Q_\alpha(X) \le \tfrac{3}{2}Q_F(X) + C_\alpha \log 2, \qquad Q_F(X) \le 2Q_\alpha(X) + 2C_\alpha \log 2.$$

Under RH both positive parts vanish eventually.

Under false RH, Lemma 14.1 gives $Q_F(n) \ge An^{d-b}$ for every fixed $d < 1 - b/2$ at arbitrarily large n. Since $b < 1/2$, choose d with $d - b > 1/4$. This defeats every fixed logarithmic factor at exponent $1/4$. The two area

comparisons transfer the contradiction. For a zero of real part $\sigma > (1+2a)/3$, choose fixed exponents with

$$1-\sigma < b < \frac{2(1-a)}{3}, \qquad a+b < d < 1-b/2.$$

A sufficiently small fixed $\varepsilon > 0$ gives the same contradiction. □

For an explicit sufficient estimate, put

$$Z(x) = \sqrt{x}\log x\, J(x), \qquad q(X) = 2\left\lceil \frac{\log X}{8\log 2} \right\rceil.$$

The estimate is the following unproved assertion:

$$\frac{1}{X}\int_X^{2X} |Z(t)|^{q(X)}\, dt \le C\, 2^{-q(X)} \quad \text{eventually, with fixed } C. \tag{44}$$

Indeed $P(x) \sim 2/(\sqrt{x}\log x)$, $D \ge 0$ and $A = O(1/x)$ give $F(x) \le (Z(x) - 1)/(\sqrt{x}\log x)$ eventually. For $q \ge 1$, $(u-1)_+ \le |u|^q$. Thus (44) implies $Q_F(X) \le CX^{1/4}/\log X$ and hence RH. A fixed second-moment envelope instead gives only $O(\sqrt{X}/\log X)$ by $(v-m)_+ \le v^2/(4m)$. This identifies the strength needed by that particular estimate; it does not prove that every second-moment argument must fail. No unconditional bound (44) is asserted.

14.1 A growing positive tilt

The exact identity (33) also permits a growing parameter, provided its tail is estimated directly. For fixed $c > 0$, set $z = c\sqrt{x}$. PNT and partial summation give

$$\sum_{p>x} (p-1)^{-2} \sim \frac{1}{x\log x}, \qquad \sum_{p>x} (p-1)^{-3} = O\left(\frac{1}{x^2\log x}\right).$$

The estimate needed here is uniform in the growing parameter. Put $N = \lfloor x \rfloor$ and $S_2(N) = \sum_{p>N} (p-1)^{-2}$. The elementary bounds $0 \le \log(1+v) - v + v^2/2 \le v^3/3$ for $v \ge 0$ give

$$\left| \log \mathcal{H}(z) - \log \mathcal{H}_N(z) + \frac{z^2 - z}{2} S_2(N) \right| \le \frac{2(z^3+z)}{3N^2} \qquad (N \ge 1,\ z \ge 0).$$

This weaker cubic bound already suffices: after division by $z = c\sqrt{x}$, its error is $O_c(x^{-1})$, which is $o_c(1/(\sqrt{x}\log x))$. PNT supplies $N \log N\, S_2(N) \to 1$, and $N/x \to 1$ retains the real cutoff. Consequently

$$\frac{T_{c\sqrt{x}}(x)}{c\sqrt{x}} = F(x) - \frac{c}{2\sqrt{x}\log x} + o_c\left(\frac{1}{\sqrt{x}\log x}\right). \tag{45}$$

It follows that RH is equivalent to eventual $T_{c\sqrt{x}}(x) < 0$ for each fixed $c > 0$. The RH direction follows from the negative source margin; the converse uses the positive x^{-b} excursions with $b < 1/2$, which dominate every fixed square-root-scale allowance. This argument does not apply a fixed-tilt error term uniformly without proof. Nor does a larger c establish the resulting product inequality. For comparison, a finite fixed combination satisfies $\sum_j a_j T_{z_j} = (\sum_j a_j z_j)F + O(1/x)$: canceling its leading coefficient cancels this RH signal as well.

15 RH detected by localized reversal counts

This section uses a specified collection of witnesses and a growing finite tilt grid. It does not replace the original maximal count $N_\eta(k)$ by a monotone proxy. In fact, ascent inclusion alone does not imply monotonicity of maximal separated reversal counts: the step signs $(-,+,+,-,+,+)$ allow two separated witnesses, while $(-,+,+,+,+,+)$ allow only one.

15.1 Actual tilted laws and exact rounding

For each $t > 0$ put $\eta_p(t) = t/(p-1+t)$. Its odds are $t/(p-1)$, so its degree-r elementary coefficient is exactly $t^r a_r$. At a supported prefix $x = p-1$ and a consecutive gap $g = q-p$, the rank-$(r+1)$ density therefore ascends exactly when

$$g + t < R_r(x);$$

it descends exactly when $R_r(x) < g + t$. These are statements about actual local-law densities. No unweighted arithmetic realization for every real t is assumed.

For $t_j = 1 + j/M$, $0 \le j \le HM$, let $A_M(R,g) = \#\{j : g + t_j < R\}$. If $M \ge 1$ and $1 \le R - g \le H + 1$, then

$$A_M(R,g) = \lceil M(R-g-1)\rceil.$$

Thus, when both $R-g$ and $U-g$ lie in this interval,

$$|A_M(R,g) - A_M(U,g) - M(R-U)| < 1. \tag{46}$$

The weak lower and strict upper endpoints follow from the strict ascent test; a threshold exactly on a grid point is not counted. For a nonempty finite set of pairs, summing this identity loses less than one unit per pair.

15.2 The sampling input and the specified cells

Alweiss–Luo [6, Corollary 1.2] prove that for each fixed $\delta \in [0.525, 1]$, some fixed width H and constants $c > 0, m$ give at least $ch/(\log Y)^m$ consecutive prime pairs of gap at most H in every sufficiently late $[Y - h, Y]$, whenever $Y^\delta \le h \le Y$. We use $\delta = 3/5$ and enlarge H to at least two. This H is not supplied by a theorem about global infinitude at a numerical width such as 186.

For integer X sufficiently large, set

$$h_X = \lceil 3X^{3/5} \rceil, \quad m_X = \left\lfloor \frac{X^{7/10}}{10h_X} \right\rfloor, \quad M_X = \lceil \sqrt{X} \rceil, \quad r_X = \lfloor (H+2)L(X) \rfloor.$$

The cell indexed by $0 \le i < m_X$ starts at $z_i = X + 1 + 10h_X i$. Its three sampling windows are

$$[z_i, z_i + h_X], \quad [z_i + 3h_X, z_i + 4h_X], \quad [z_i + 6h_X, z_i + 7h_X].$$

The explicit $+1$ ensures that every relevant prefix $p - 1$ is at least X. All prefixes used below are at most $X + X^{7/10}$. The source theorem applies to all three windows simultaneously, since their right endpoints are at most $2X$ and $h_X \ge (2X)^{3/5}$ eventually.

Fix $\ell = H + 3$. Every multiple of $\ell!$ starts a composite block at offsets $2, \dots, \ell$. Since $h_X \ge \ell! + 2$ eventually, a block can be placed between the left and middle windows. Choose the first qualifying consecutive bounded pair in each window and use its lower prime in the left and middle windows. A consecutive-prime step crossing the intervening block has length at least ℓ. Fix a choice of these crossings once and for all, using only their prime geometry. Denote its lower-prime index by d_i and that of the selected right pair by a_i. These deterministic choices use only primes and the fixed schedule.

The crossing lies between the selected left and middle primes, so $d_i < a_i$. The right endpoint in one cell is strictly before the next cell's descent lower prime; therefore $a_i + 1 < d_j$ for $i < j$. Finally $m_X \sim X^{1/10}/30$. These are genuine separated witnesses, not repeatedly counted copies of one reversal.

15.3 Two finite integer counts

Use the same grid $t_j = 1 + j/M_X$, $0 \le j \le HM_X$, in every cell. Define

$$C_H(X) = \sum_{i=0}^{m_X-1} \sum_{j=0}^{HM_X} \mathbf{1}\{d_i < a_i \text{ and the tilted rank-}(r_X+1) \text{ density descends at } d_i \text{ and ascends at } a_i\},$$

$$\overline{C}_H(X) = \sum_{i=0}^{m_X-1} \sum_{j=0}^{HM_X} \mathbf{1}\{g_{a_i} + t_j < \mathcal{R}_{r_X}(L(p_{a_i} - 1))\}.$$

Use index zero when a specified selection is unavailable on the bounded initial range. The finite formulas then define total counts; their initial values do not enter either eventual criterion. The reference uses the same actual prime pairs and their actual gaps.

Uniformly on the window, both actual and reference ratios are $H + 2 + o(1)$, with positive denominators. This follows from $R_r, \mathcal{R}_r = r/L + O(1/L)$ on compact rank bands and $|L(y) - L(X)| \le 1$ on $[X, 2X]$ eventually. Every chosen long step is consequently a strict descent for every grid tilt. For the short step, $2 \le g_{a_i} \le H$ puts both ratio-minus-gap thresholds inside $[1, H+1]$. Writing $\Delta_i = \Delta_{r_X}(p_{a_i} - 1)$, we obtain

$$C_H(X) - \overline{C}_H(X) = M_X \sum_{i=0}^{m_X-1} \Delta_i + e_X, \qquad |e_X| < m_X \tag{47}$$

whenever $m_X > 0$.

Theorem 15.1 (Localized separated-reversal criteria)**.** *For the preceding fixed width and deterministic construction, RH is equivalent to each of the following:*

$$C_H(X) \le \overline{C}_H(X) \quad \textit{eventually,} \tag{48}$$

$$C_H(X) - \overline{C}_H(X) \le CX^{1/10} \quad \textit{eventually for some fixed } C > 0. \tag{49}$$

Under false RH the signed difference divided by $X^{1/10}$ is unbounded above along integer X.

Proof. Under RH, the uniform coefficient sign comparison applies to r_X at every selected prefix. Actual ascent tests are included in reference tests, proving (48) and hence (49).

If RH is false, Lemma 14.1, followed by the uniform coefficient transfer, supplies arbitrarily large integer n such that

$$\Delta_r(y) > Tn^{-1/2} \qquad (n \le y \le n + n^{7/10})$$

for every prescribed fixed $T > 0$ and every rank in a fixed positive compact $r/L(y)$ band. Indeed its retained main term is $\gg n^{-b}/L(y)$ for some $b < 1/2$, and $n^{1/2-b}/\log\log n \to \infty$. All selected prefixes at $X = n$ lie in this interval, and the single rank r_n lies in the required band throughout. Since $M_n \ge \sqrt{n}$, (47) gives

$$C_H(n) - \overline{C}_H(n) > m_n(T-1).$$

Now $m_n \ge c_0 n^{1/10}$ eventually, with fixed $c_0 > 0$. Choosing $T > 1 + C/c_0$ contradicts any proposed eventual bound (49). This proves both converses. □

For each individual tilt, the selected witnesses are bounded by $N_{\eta(t)}(r_X + 1)$. The sum in the theorem ranges over a growing finite grid, so it is not the maximal count for one fixed law. Its reference also retains θ and the entire profile. An independent proof of either eventual count inequality would prove RH; the equivalence itself supplies no such estimate.

An ascent-only variant uses every consecutive gap at most the source width in $[X+1, X+1+\lceil 2X^{7/10}\rceil]$ and the same grid and rank. Let its actual and reference counts be $A_H(X), B_H(X)$. The source at $\delta = 3/5$, applied to this longer window, supplies $\gg X^{7/10}/(\log X)^m$ pairs. At the persistent false-RH peaks, (46) then gives

$$A_H(X) - B_H(X) \gg \frac{X^{6/5-b}}{(\log X)^m \log\log X}, \qquad b < 1/2.$$

Thus RH is also equivalent to eventual $A_H \le B_H$, and to an eventual one-sided bound $A_H - B_H = O(X^{7/10})$. The persistence interval may be taken with any fixed exponent between $7/10$ and $3/4$, which absorbs the shift and rounding in this ascent-only window.

16 Growing families and signed bounds

A fixed-family RH criterion does not by itself justify varying the family with the prime cutoff. We first give a transfer with explicit error intervals, then two arithmetic calculations in which the complete correction can be retained. Throughout this section, a finite-prime limiting law is taken before the cutoff or polynomial degree varies. We make no assertion uniform in a finite sampling box.

16.1 Correction intervals

Theorem 16.1 (Transfer through a correction interval)**.** *Let $E, \ell, u : \mathbb{N} \to \mathbb{R}$ and let $c > 0$ be fixed. Suppose that*

$$\ell(N) \le E(N) - cF(N) \le u(N) \quad \textit{for all sufficiently large } N.$$

Assume that, for every fixed $0 < b < 1/2$, there is a constant $C_b > 0$ such that $u(N) - \ell(N) \le C_b N^{-b}$ eventually. Then RH is equivalent to eventual $E(N) \le u(N)$, and also to eventual $E(N) < u(N)$.

Proof. Under RH, $F(N) < 0$ eventually, so the upper interval bound gives $E(N) < u(N)$. If RH fails, the integer-cutoff Nicolas excursion theorem gives a fixed $0 < b < 1/2$ and arbitrarily large N with $cF(N) > (C_b + 1)N^{-b}$. At these N, the lower interval bound implies $E(N) > u(N)$, a contradiction. □

The interval center need not be small or positive. What matters is the width and its proved arithmetic relation to the same E and F. Defining an endpoint from the unknown error would not provide an independent estimate. The maintained theorem has a fixed coefficient c; a moving coefficient requires a separate uniform argument.

16.2 An explicit polynomial-plus-prime family

For a positive integer d, put

$$P_d(a) = \prod_{j=1}^{d} (a - j), \qquad V_d(a, q) = P_d(a) + q,$$

where a ranges over integers and q over primes, with ordered representations retained. For each fixed finite set of primes, uniform integer residues, prime progression limits and the Chinese remainder theorem give independent local divisibility laws. The polynomial has exactly $\min(d, p)$ roots modulo p. Consequently

$$\eta_d(p) = \frac{p - \min(d, p)}{p(p-1)}. \tag{50}$$

In particular, every prime $p \le d$ is excluded. Use this exact law, including those exclusions, to form $E_d(N) = E_{\eta_d}(N, 1)$ as in Theorem 12.1. Set

$$Q_2(N) = \sum_{p > N} \frac{1}{(p-1)^2}, \qquad T(N, b) = \sum_{p > N} \log\left(1 + \frac{b}{(p-1)^2}\right) \quad (b \ge 0).$$

Theorem 16.2 (Exact correction and finite bounds)**.** *For $N \geq \max(d, 3)$,*

$$E_d(N) = F(N) - T(N, d-1). \tag{51}$$

For $N \geq 1$ and $b \geq 0$,

$$\left(b - \frac{b^2}{2N^2}\right) Q_2(N) \leq T(N, b) \leq bQ_2(N), \qquad 0 \leq Q_2(N) \leq \frac{2}{N}. \tag{52}$$

Moreover $N \log N \, Q_2(N) \to 1$.

Proof. For $p > d$, direct substitution into the odds gives

$$(1 + w_d(p))(1 - 1/p) = \left(1 + \frac{d-1}{(p-1)^2}\right)^{-1}.$$

The exact profile identity expresses $E_d - F$ as the sum of these logarithms over $p > N$. All excluded primes already lie in the finite prefix; their profile factors cancel exactly, proving (51). For $t \geq 0$, $t - t^2/2 \leq \log(1+t) \leq t$. Since $p > N$ implies $p - 1 \geq N$, summing these inequalities proves (52). Comparison with $\sum_{n \geq N} n^{-2} \leq 2/N$ gives the elementary upper bound. Partial summation of the prime number theorem gives the stated asymptotic for Q_2. □

Theorem 16.3 (Independent signs at logarithmically reduced degree)**.** *For every fixed integer $j \geq 1$, let*

$$d_j(N) = 1 + \left\lfloor \frac{N}{(\log N)^j} \right\rfloor.$$

Then, unconditionally,

$$(\log N)^{j+1} E_{d_j(N)}(N) \longrightarrow -1.$$

In particular $E_{d_j(N)}(N) < 0$ for every sufficiently large integer N. The diagonal degree $d = N$ also satisfies $E_N(N) \log N \to -1$.

Proof. For large N, $d_j(N) \leq N$, and $b_N = d_j(N) - 1 = N/(\log N)^j + O(1)$. The two bounds in (52), together with $b_N \leq N$, imply

$$(\log N)^{j+1} T(N, b_N) \longrightarrow 1.$$

For every fixed positive integer A, the quantitative prime number theorem gives the unconditional limit

$$F(N)(\log N)^A \longrightarrow 0.$$

Subtracting in (51) proves the result. For the diagonal, use $b = N - 1$ in the same bounds to obtain

$$T(N, N-1)\log N \longrightarrow 1. \qquad \square$$

These are complete independent signed estimates, but the family degree is $N^{1-o(1)}$. For every fixed $\alpha < 1$ and fixed j, $d_j(N)/N^\alpha \to \infty$. The theorem therefore supplies no fixed-power zero-free conclusion. Neither its onset nor its estimates are asserted to be uniform when j grows with N.

16.3 The effect of adding two primes

Let a base residue family have zero probability $r_p \in [0, 1]$ at each prime p. Add two independent uniform nonzero residues. Their exact zero probability is

$$\eta(p) = \frac{p - 2 + r_p}{(p-1)^2}.$$

Indeed there are $p - 1$ ordered nonzero pairs with sum zero and $p - 2$ with any prescribed nonzero sum. Thus the laws of $q_1 + q_2 + 1$ and $q_1 + q_2$ attain the two endpoints, independently of the complexity of the base family.

Theorem 16.4 (Attained two-prime bracket)**.** *For every such finite residue family, every $N \geq 3$, and $z \geq 0$,*

$$E_{q_1+q_2+1}(N, z) \leq E_{\text{base}+q_1+q_2}(N, z) \leq E_{q_1+q_2}(N, z).$$

At $z = 1$, define the convergent tails

$$T_3(N) = \sum_{p>N} \log\left(1 + \frac{1}{(p-1)^3}\right), \qquad U_2(N) = \sum_{p>N} -\log\left(1 - \frac{1}{(p-1)^2}\right).$$

Then the exact bracket is

$$F(N) - T_3(N) \leq E_{\text{base}+q_1+q_2}(N, 1) \leq F(N) + U_2(N).$$

Proof. For tail primes $p > N \geq 3$, all three probabilities are less than one. Both $\eta/(1-\eta)$ and $\log(1 + z\eta/(1-\eta))$ increase with η. Compare the convergent tail corrections to $zF(N)$, including their exact normalizations. Substitution of $r_p = 0, 1$ gives the two displayed tails at $z = 1$. $\square$

The favorable endpoint correction is asymptotic to $1/(2N^2 \log N)$ by partial summation. Thus adding arbitrary base complexity cannot enlarge this correction within the stated two-prime-addition class. This controls a

correction, not the sign of the complete expression containing $F(N)$. Additional independent prime additions can be included in the base distribution; this ordinary specialization does not assert uniform finite-box limits or a result for changing nonunit coefficients.

17 Power-sized degrees and zero-free regions

The distinction between a logarithmic saving and a fixed power saving can be made precise for the actual polynomial family of Section 16. The result below is a conditional implication. It does not assume a zero-free region, but it does assume the eventual sign of the complete arithmetic profile.

Theorem 17.1 (Power-degree sign criterion)**.** *Let $d : \mathbb{N} \to \mathbb{N}$ satisfy $d(N) \geq 1$. Fix real numbers $C > 0$ and $1/2 \leq \alpha < 1$. Suppose that, for every sufficiently large integer N,*

$$d(N) \leq CN^{\alpha}, \qquad E_{d(N)}(N) \leq 0, \tag{53}$$

where E_d is the complete profile of the consecutive-root polynomial $P_d(a)+q$, with all small-prime exceptions included. Then

$$\zeta(\rho) \neq 0 \qquad \text{whenever } \alpha < \operatorname{Re}\rho < 1.$$

The degree bound and the signed inequality refer to the same chosen sequence of families and both hold eventually, not merely along possibly different unbounded subsequences.

Proof. Since $\alpha < 1$, the degree bound implies $d(N) \leq N$ eventually. The exact identity (51) and (52) therefore give

$$F(N) \leq T(N, d(N)-1) \leq (d(N)-1)Q_2(N) \leq 2CN^{\alpha-1} \quad \text{eventually.} \tag{54}$$

Suppose $\zeta(\rho) = 0$ with $\beta = \operatorname{Re}\rho > \alpha$ and $\beta < 1$. Choose

$$b = \frac{(1-\beta)+(1-\alpha)}{2}.$$

Then $0 < 1-\beta < b < 1-\alpha \leq 1/2$. The zero-dependent integer Nicolas excursion argument from Section 11.1 gives arbitrarily large integers N with $F(N) > 2CN^{-b}$. But $\alpha - 1 < -b$, so for $N \geq 1$,

$$F(N) \leq 2CN^{\alpha-1} \leq 2CN^{-b}$$

by (54), a contradiction. This uses the already established integer excursions, including the nonlinear logarithm correction. It neither differentiates an asymptotic error nor assumes a new real-to-integer interpolation. □

Corollary 17.2 (Square-root degree)**.** *If $d(N) \geq 1$, $C > 0$, and eventually*

$$d(N) \leq C\sqrt{N}, \qquad E_{d(N)}(N) \leq 0,$$

then RH holds.

Proof. Theorem 17.1 excludes every nontrivial zero to the right of $1/2$. The functional equation then places every nontrivial zero on the critical line. □

The new content here is the exact passage from this arithmetic degree budget to the classical zero-dependent Nicolas obstruction; the underlying oscillation mechanism is classical [7, 8]. External priority for this polynomial-family formulation is not established.

The complete sign in (53) remains unproved independently for power-sized degrees. In contrast, Theorem 16.3 proves that sign at $d(N) = 1 + \lfloor N/(\log N)^j \rfloor$ for every fixed $j \geq 1$. Such a degree exceeds CN^α eventually for every fixed $C > 0$ and $\alpha < 1$. Thus that unconditional theorem cannot be substituted into (53). Increasing a fixed logarithmic saving does not reach any fixed power saving.

This criterion concerns the specified polynomial-plus-prime law. No degree-only assertion is made for arbitrary changing coefficients or sampling rules: those can introduce additional excluded primes and a different correction. Nor does an arbitrary sequence of individually qualifying fixed laws automatically satisfy the uniform estimate used here.

18 A generalized Robin critical window

The same signed prime-product calculation has a separate application to generalized divisor sums. It gives actual counterexamples to a proposed extension of a generalized Robin inequality. It does not supply an independent RH criterion.

For a real parameter $\kappa > 1$, define

$$S_\kappa(n) = \sum_{d|n} \mu(n/d)\sigma(d)^\kappa, \qquad \mathcal{G}_\kappa(n) = \frac{\zeta(\kappa)S_\kappa(n)}{(e^\gamma n \log\log n)^\kappa} \quad (n \geq 3).$$

Here κ is a real divisor-sum exponent, not the prime-factor rank k. These are the LCM-power functions of Fan–Kobayashi–Molnar [13, Definition 1.2]. For an integer $N \geq 8$, put

$$M_N = \operatorname{lcm}(1, \ldots, N) \prod_{\substack{p \leq N \\ N < p^2 \leq 2N}} p, \qquad K_N = \frac{3}{2} + \frac{2}{5 \log\log M_N}. \tag{55}$$

Theorem 18.1 (Uniform critical window)**.** *For every $X \in \mathbb{N}$, there is an integer $N > X$, $N \geq 8$, such that simultaneously for every real $1 < \kappa \leq K_N$,*

$$\log \mathcal{G}_\kappa(M_N) > \frac{1}{100\sqrt{N}\log N}. \tag{56}$$

Under RH, this holds for every sufficiently large integer N, with a common cutoff for the entire parameter window.

Corollary 18.2. *For every fixed real $1 < \kappa \leq 3/2$, there are arbitrarily large integers n for which $\mathcal{G}_\kappa(n) > 1$.*

18.1 Exact arithmetic and the full signed comparison

Let $a_p = v_p(M_N)$ for $p \leq N$, and define

$$u_p = p^{-a_p-1}, \qquad U_N = \sum_{p\leq N} u_p,$$
$$R_N = 4\sum_{p\leq N}\frac{u_p}{p} + 32\sum_{p\leq N} u_p^2, \qquad D_N = \log\log\log M_N - \log\log\theta(N).$$

All finite sums have the same prime support $p \leq N$. Write

$$\mathcal{T}_\kappa(N) = \sum_{p>N} -\log(1-p^{-\kappa}).$$

For $\kappa > 1$ this converges and equals the logarithm of the omitted part of the zeta Euler product.

The defining divisor sum is multiplicative, and at a prime power

$$S_\kappa(p^a) = \sigma(p^a)^\kappa - \sigma(p^{a-1})^\kappa = \left(\frac{p^a}{1-1/p}\right)^\kappa (1-p^{-\kappa})\, C_\kappa(p,a),$$

where

$$C_\kappa(p,a) = \frac{(1-p^{-a-1})^\kappa - p^{-\kappa}(1-p^{-a})^\kappa}{1-p^{-\kappa}} > 0.$$

Therefore the exact normalized identity is

$$\log \mathcal{G}_\kappa(M_N) = \mathcal{T}_\kappa(N) - \kappa\big(F(N) + D_N\big) + \sum_{p\leq N} \log C_\kappa(p,a_p). \tag{57}$$

This retains the actual integer height $\log M_N$.

Lemma 18.3 (Uniform finite lower bound)**.** *For $N \geq 8$ and $1 < \kappa \leq 2$,*

$$\log \mathcal{G}_\kappa(M_N) \geq \mathcal{T}_\kappa(N) - \kappa(F(N) + D_N + U_N) - R_N. \tag{58}$$

If $1 < \kappa \leq K \leq 2$, its right side may be replaced by

$$\mathcal{T}_K(N) - K \max(F(N) + D_N + U_N, 0) - R_N.$$

Proof. Put $u = p^{-a-1}$, $v = p^{-\kappa}$ and $t = p^{-a}$. For $1 \leq \kappa \leq 2$, $u \leq 1/8$ and $v \leq 1/2$, Bernoulli's inequality gives

$$C_\kappa(p, a) \geq 1 - \frac{\kappa u}{1 - v}.$$

The number $w = \kappa u/(1 - v)$ satisfies $0 \leq w \leq 4u \leq 1/2$ and $w \leq \kappa u + 4uv$. The bound $\log(1 - w) \geq -w - 2w^2$ yields

$$\log C_\kappa(p, a) \geq -\kappa u - 4uv - 32u^2 \geq -\kappa u - 4u/p - 32u^2.$$

For the exponents of M_N, $p^{a_p+1} > N$; hence $u_p \leq 1/8$ when $N \geq 8$. Sum the bound in (57). Finally $\mathcal{T}_\kappa \geq \mathcal{T}_K$, and for every real A, $-\kappa A \geq -K \max(A, 0)$. □

18.2 The two arithmetic costs and the positive tail

The ordinary prime number theorem gives

$$\sqrt{N} \log N \, D_N \longrightarrow \sqrt{2}, \qquad \sqrt{N} \log N \, U_N \longrightarrow \sqrt{2}, \qquad \sqrt{N} \log N \, R_N \longrightarrow 0. \tag{59}$$

Here is a direct explanation of all three limits. The exact height is

$$\log M_N = \psi(N) + \theta(\sqrt{2N}) - \theta(\sqrt{N}).$$

Since $\psi(N) - \theta(N) = \theta(\sqrt{N}) + O(N^{1/3} \log N)$, we have $\log M_N - \theta(N) \sim \sqrt{2N}$. The derivative of $\log \log t$ is $1/(t \log t)$, giving the first limit.

For $p > \sqrt{2N}$, $a_p = 1$, so the upper-band contribution is

$$\sum_{\sqrt{2N} < p \leq N} p^{-2}.$$

Its normalized limit is $\sqrt{2}$. Below $\sqrt{2N}$, split at $N^{1/3}$ to bound the remaining contribution by

$$N^{-1} \pi(N^{1/3}) + \sum_{p > N^{1/3}} p^{-3}.$$

Both terms are negligible on the displayed scale. Finally, $u_p < 1/N$ gives $R_N = O((1+\log N)/N)$ and proves the third limit.

We need only a finite lower comparison for the moving zeta tail. For $q > 1$, $m \in \mathbb{N}$, and $1 < K \le 2$, put

$$\begin{aligned} A_2(x) &= \sum_{p>x} p^{-2}, \\ L_m(N,q,K) &= N^{2-K} A_2(N) \\ &\quad + \sum_{j=0}^{m-1} \big((Nq^{j+1})^{2-K} - (Nq^j)^{2-K}\big) A_2(Nq^{j+1}). \end{aligned} \tag{60}$$

Then $\mathcal{T}_K(N) \ge L_m(N,q,K)$. To see this, first use $-\log(1-p^{-K}) \ge p^{-K}$, and then bound the increasing function p^{2-K} from below by its step approximation at the displayed thresholds. Every increment is nonnegative, so summation preserves the bound.

Fix $s > 1$ and m before taking $N \to \infty$, and put $q = s^2$. Because $x \log x\, A_2(x) \to 1$ and $\log\log M_N / \log N \to 1$, the finitely many terms give

$$\begin{aligned} \sqrt{N} \log N\, L_m\left(N, s^2, \frac{3}{2} + \frac{c}{\log\log M_N}\right) &\longrightarrow C_m(s) e^{-c}, \\ C_m(s) &= 1 + \frac{1}{s} - \frac{1}{s^{m+1}}. \end{aligned} \tag{61}$$

Only fixed-dilation prime-square-tail limits are used. In particular, we have not assumed an asymptotic for a prime sum uniform in a moving exponent.

Choose $s = 1001/1000$ and $m = 10^6$. Bernoulli's inequality gives $C_m(s) > 999/500$. With $\beta_0 \le 1/20$ from the classical Nicolas constant, the limiting complete RH lower bound exceeds

$$C_m(s) e^{-2/5} - 3(\sqrt{2}-1) - \frac{3}{2}\beta_0 > \frac{7}{600}. \tag{62}$$

For example, $\log(3/2) > 2/5$ and $\sqrt{2} < 99/70$ show that the left side is greater than $333/250 - 87/70 - 3/40 = 99/7000 > 7/600$. The finite choice $m = 10^6$ is a symbolic parameter choice; no million-term numerical verification is used.

18.3 Proof in the two RH cases

Proof of Theorem 18.1. Under RH, (20) gives

$$\limsup_{N\to\infty} \sqrt{N} \log N\, F(N) \le \beta_0 - 2.$$

Insert this and (59)–(61) into Lemma 18.3, with $K = K_N$. For large N, $K_N < 2$. The limiting lower bound for the common scaled comparison is at least the quantity in (62). Since $1/100 < 7/600$, one cutoff works for every $1 < \kappa \le K_N$.

If RH fails, negative Nicolas excursions give arbitrarily large integer N with

$$F(N)\sqrt{N} < -1. \tag{63}$$

For completeness, take real excursions $F(x) < -2x^{-b}$ with $0 < b < 1/2$ and $x > 3$, and let $N = \lfloor x \rfloor$. The prime products and θ are constant on $[N, N+1)$, so $F(N) = F(x)$. Also $x < N+1 \le 4N$. Thus $F(N) < -2/\sqrt{x} \le -1/\sqrt{N}$. Only F is rounded in this argument; no floor invariance of M_N or W is used.

By (59), the positive arithmetic costs satisfy $(D_N + U_N)\sqrt{N}\log N \to 2\sqrt{2}$, whereas (63) gives $F(N)\sqrt{N}\log N < -\log N$. Hence $F(N) + D_N + U_N < 0$ at arbitrarily large integer cutoffs. At those cutoffs the maximum in Lemma 18.3 vanishes. Equations (59) and (61) then give the stronger positive limiting lower bound $C_m(s)e^{-2/5}$. Taking cutoffs beyond the common asymptotic thresholds proves (56). The same explicit integer family therefore works in both logical cases. □

Proof of Corollary 18.2. We have $\log M_N \ge \theta(N) \to \infty$, so $M_N \to \infty$. Also $K_N > 3/2$. Apply Theorem 18.1 beyond any prescribed integer-height threshold and use positivity of $\mathcal{G}_\kappa$ to exponentiate its positive logarithm. □

The theorem concerns integer cutoffs and supplies no effective onset or frequency of the exceptional cutoffs without RH. It establishes neither optimality of $2/5$ nor that M_N is colossally abundant. Every fixed $\kappa > 3/2$ eventually lies outside the shrinking window, so this result is consistent with the fixed-parameter upper theorem of [13]. In particular, the counterexamples do not violate ordinary Robin's inequality and do not prove or disprove RH.

19 Implications and scope

The spectrum gives a quantitative dictionary. Bounded-gap counts yield reversal counts through the finite selection lemma; the upper comparison shows that no other late ascents can improve the rate. Finite changes in an admissible law preserve that rate. Many different affine prime-pair expressions consequently share a spectrum despite different forced primes and small ranks.

Location and frequency encode different information. Corollary 9.2 depends on the least recurring gap. The rate $\Lambda(\nu)$ depends on how frequently bounded gap classes occur. Infinitude alone need not give $\gamma_H > 0$: a hypothetical count growing as a power of $\log X$ has $\gamma_H = 0$. This is why bare infinitude cannot replace the input to Corollary 5.2.

For Goldbach-type expressions, the arithmetic theorem describes factorization in the full representation-weighted ensemble. It gives no lower bound for representations of each fixed even integer. Fixed-modulus independence likewise does not supply uniform estimates when a prime cutoff grows with input size. Moving-tail distribution laws require additional uniformity.

Repeated raw reversals and centered moving-rank excursions have different quantifiers. At each fixed rank there is a final decreasing tail. The RH criterion lets the rank grow with the endpoint and compares its ratio with the full reference profile. It is the transfer in Theorem 10.1 that relates this second question to Nicolas's error. Theorem 12.2 extends the centered comparison to the qualifying local laws and the actual affine families. An independently proved centered inequality in any one such family would therefore imply RH. A coarse reversal rate or a numerical prime-gap bound does not itself provide that sign.

The exact raw comparison can be written as

$$R_r(x) - (g+1) = \big(\mathcal{R}_r(L(x)) - (g+1)\big) + \Delta_r(x).$$

RH controls the second term. The first term and the occurrence of the actual gap remain necessary for an ascent. In particular, for fixed $1/5 < b < c < 1/3$ and all sufficiently large k, an ascent at a prime with $bk \le \mathcal{L}(p-1) \le ck$ occurs exactly when the next gap is two. This follows directly from the uniform ratio margins. Proving positive ascent supply in this window independently would prove twins; the identity itself is not such a supply proof.

The Hardy–Littlewood input in Theorem 9.3 provides actual twins approaching the eligibility cutoff. RH instead refines that cutoff's reference error and gives Theorem 13.2. The global twin asymptotic allows a distance $o(X_k)$ from the cutoff, which can exceed $\sqrt{X_k}$; it cannot by itself transfer the RH square-root clock correction to the location of the last actual ascent. No implication from RH to twin-prime infinitude is established by these results.

The relation to Robin and Lagarias is through their established equivalences with RH [12]. Thus the eventual inequality in Theorem 11.1, if proved independently, would imply the corresponding universal divisor-sum inequalities. A valid violation of either criterion would imply the false-RH

coefficient excursions. These are logical consequences of known equivalences, not new bounds for colossally abundant numbers. Indeed, for $n = \prod p^{a_p}$,

$$\frac{\sigma(n)}{n} = \prod_{p|n} \frac{1 - p^{-(a_p+1)}}{1 - p^{-1}},$$

and optimizing the exponent-dependent numerator factors together with $\log\log n$ remains a separate problem.

The classical inputs include Nicolas's criterion and spectral formula, Landau oscillation, prime-factor generating products and saddle-point estimates. The potentially new contributions are the exact reversal-frequency spectrum, its affine realization and the sign-preserving moving-coefficient transfer with its applications. These attributions are provisional: proof verification does not establish priority. Speculative RH and independent small-gap-supply investigations are excluded from the theorem claims of this paper.

19.1 Finite smaller gaps and the complete spectrum

Suppose, for a fixed even $h \geq 2$, that all sufficiently late consecutive gaps are at least h. Every fixed initial prime prefix has finite support in the rank variable. Consequently, for all sufficiently large ranks, none of its steps can be an ascent. The RH argument in Theorem 13.2 then uses the threshold $h + 1$ throughout and gives

$$\theta(P_k - 1) < T_{k-1,h} := \mathcal{E}(u_{k-1,h+1} - \gamma), \qquad N_k \leq (1 + \varepsilon)\frac{T_{k-1,h}}{\log T_{k-1,h}}$$

eventually, and hence

$$\log\log N_k \leq \frac{k-1}{h+1} - \gamma - \frac{\mathcal{H}'(h+1)}{\mathcal{H}(h+1)} + \frac{C_h}{k-1}. \tag{64}$$

Constants and onsets depend on the fixed h. There is no fixed-width sign transition: $\mathcal{H}(s) > 0$ at every positive s, and the RH source multiplier $-(2 + W)$ stays negative in every fixed positive rank band.

Finite twins give $h = 4$ as an eventual lower bound and $\gamma_2 = \gamma_3 = 0$. If additionally $\Lambda(1) = 1/5$, the spectrum forces $\gamma_4 = 1$: every $H \geq 5$ contributes at most $1/6$, while the only remaining possible maximizing term is $\gamma_4/5$. It follows that

$$\Lambda(\nu) = \frac{1}{4 + \nu} \qquad (\nu > 0).$$

Indeed the full-frequency width-four term dominates every larger width, and smaller widths have zero exponent. The RH ceiling (64) now has the same leading coefficient $1/5$. The matching leading values yield no opposing signed inequality. Nor does $\gamma_H = 1$ specify an actual density: both $G_H(X) \sim X^{\delta}$, with any fixed $\delta > 0$, and $G_H(X) \sim cX/\log^2 X$ have that exponent.

19.2 When local frequency reveals the RH displacement

A stronger frequency hypothesis makes the reference displacement visible in actual last-ascent locations. Fix even $h \geq 2$, assume smaller gaps occur only finitely often, and assume every sufficiently late interval

$$[X, X + \ell(X)], \qquad \ell(X) = \frac{\sqrt{X}}{(\log X)^2},$$

contains the lower member of a consecutive gap exactly h. Let V_k be the last supported prime eligible for a hypothetical gap-h ascent, and P_k the last actual ascent prime. Then

$$P_k/V_k \longrightarrow 1, \qquad T_{k-1,h} - \mathcal{C}(P_k - 1) = o(\sqrt{P_k}). \tag{65}$$

Under RH this gives

$$\frac{T_{k-1,h} - \theta(P_k - 1)}{\sqrt{P_k}} = 2 + W(P_k - 1) + o(1).$$

Here is the precision argument. Monotonicity of the supported ordinary coefficient ratio makes eligible primes an initial segment. Ordinary prime existence provides a supported seed below $V_k/2$. Apply local coverage at $X = V_k - 2\ell(V_k)$. For large V_k, $X \geq V_k/2$ and $\ell(X) \leq 2\ell(V_k)$, so the supplied gap-h pair has lower prime in $(V_k - 2\ell(V_k), V_k]$. Support propagates from the seed, and monotonicity makes that step an actual ascent. Beyond V_k, the eventual lower gap bound h makes any ascent eligible for the hypothetical gap-h test, contradicting the definition of V_k. Thus all finite exceptions are accounted for and $V_k - 2\ell(V_k) < P_k \leq V_k$.

At $x = V_k - 1$ and $y = w - 1$, with w the next prime after V_k, the finite-product comparison gives $R_r(x) = \mathcal{R}_r(B(x)) + O_h(1/(rx))$, and likewise at y. Only the prime V_k is added between these prefixes, so $B(y) - B(x) = \log(V_k/(V_k - 1)) = O(1/V_k)$. Ordinary prime existence and the eventual cut-off bound first place $r/B(x)$, $r/B(y)$ and $r/u_{r,h+1}$ in a common fixed positive compact interval. The uniform signed coefficient comparison then applies on both connecting segments. The strict eligible sign at x, the noneligible sign

at y, and the displayed one-prime jump give $|B(x) - u_{r,h+1}| = O_h(1/V_k)$. No prior spatial closeness of the two clocks is assumed.

The quantitative PNT-strengthened Mertens estimate gives $e^{-\gamma}G_x(1) - \log x = o(1)$, and hence $\mathcal{C}(x)/x \to 1$. The weaker relation $\log \mathcal{C}(x) \sim \log x$ would not suffice. Exponentiating the proved clock difference now gives $|\mathcal{C}(V_k - 1) - T_{r,h}| = O_h(\log V_k)$. Each prime-clock jump in the remaining interval is $O(\log V_k)$. There are at most $2\ell(V_k) + 1$ integer sites, so their total change is $O(\sqrt{V_k}/\log V_k) = o(\sqrt{V_k})$. This proves (65); add (32) at the same actual endpoint for the RH assertion.

A global pair-count asymptotic alone gives only proportional coverage. For the exact-gap count $G_h^{=}(X)$, the maintained sufficient condition is

$$G_h^{=}(X) = c_h \operatorname{Li}_2(X) + E(X), \qquad c_h > 0,$$
$$E(X) = o(\sqrt{X}/\log^4 X), \qquad \operatorname{Li}_2(X) = \int_2^X \frac{dt}{(\log t)^2}.$$

Subtract this estimate at X and $X + \ell(X)$. The main increment is of order $\sqrt{X}/\log^4 X$ and dominates both endpoint errors. The full integral main term is essential at this precision; $c_h X/\log^2 X$ cannot replace it. A square-root-plus-epsilon global error is insufficient for this subtraction. This is a sufficient route to local coverage, not its necessary strength.

Finally, on any of the finite supports in Section 15, the signed excess is exactly the number of grid thresholds in $[U_i, R_i)$ minus the number in $[R_i, U_i)$, where $U_i = \mathcal{R}_r(L(p_i - 1))$. A positive excess arbitrarily far out contradicts RH. Neither finite twins nor a positive main term for another gap class has been shown to force that signed excess. These frequency statements therefore identify additional information needed by this route; they establish neither RH nor twin-prime infinitude.

20 Formalization and reproducibility

The principal maintained owners are:

- `Assembly/ReciprocalSmoothSpectrum.lean`: all-rank finite maxima and Theorem 3.2;
- `Assembly/Headline.lean`: Theorem 3.3;
- `Assembly/AffineSpectrum.lean`: Theorem 3.4 and arithmetic phase recovery;
- `Assembly/GapSpectrumPhase.lean`: Theorem 7.1;

- `Assembly/GapRecurrence.lean` and `Proof/Upper/GapScaleCutoff.lean`: Theorem 9.1, Corollary 9.2, and the conditional inverse twin-prime transfer;
- `Helpers/PrimeProfileCanonicalLinearization.lean` and `Helpers/PrimeProfileUniformSign.lean`: the full-profile coefficient comparison and uniform sign transfer;
- `Assembly/FamilyRHResults.lean`: Theorems 12.1 and 12.2, the general probability-law constructor, exact forced-rank mass criteria, actual affine-family consumers, and the generalized reference constant;
- `Helpers/FamilyIntervalTransfer.lean` and `Helpers/DoublePrimeAdditionCriterion.lean`: Theorems 16.1 and 16.4;
- `Helpers/ConsecutiveRootProbability.lean`, `Helpers/ConsecutiveRootCriterion.lean`, `Helpers/PrimeAddedRootBudget.lean` and `Helpers/LogarithmicRootCriterion.lean`: the exact root law, finite correction bounds, and Theorem 16.3;
- `Helpers/ConsecutiveRootZeroFree.lean`: Theorem 17.1 and Corollary 17.2, with the degree and complete-sign premises explicit;
- `Helpers/LcmDivisorArithmetic.lean`, `Helpers/LcmNormalizedComparison.lean` and the modified-LCM height, defect and finite-tail modules: the actual divisor sum and the complete finite and asymptotic bounds used in Section 18;
- `Helpers/ModifiedLcmRHWindow.lean`, `Helpers/ModifiedLcmNegativeSupply.lean` and `Helpers/ModifiedLcmCriticalWindow.lean`: Theorem 18.1 and Corollary 18.2, with separate RH and false-RH intermediate statements and unconditional maintained endpoints;
- `Helpers/NicolasPowerPeak.lean` and `Helpers/PrimeProfilePowerExcursions.lean`: the corrected logarithm and actual moving-rank power excursions;
- `Helpers/NicolasLogSpectral.lean`, `Helpers/NicolasSpatialClock.lean` and the positive-tilt modules: the classical spectral inputs and their clock/integral consumers;
- `Assembly/GeneralRHSharpCapacity.lean`: attained last ascents and sharp count capacities under any eventual lower gap, including every finite exception;

- `Helpers/NicolasPowerPersistence.lean`, `Helpers/NicolasCriticalPersistence.lean`, `Helpers/NicolasPositiveArea.lean` and `Helpers/PrimeProfilePositiveArea.lean`: power and critical logarithmic-length positive intervals and both positive-area RH criteria;

- `Helpers/NicolasMomentConsumer.lean` and `Helpers/MomentSecondArea.lean`: the growing-moment sufficient condition and the fixed-second-moment area envelope. Their proposed moment hypotheses remain unproved;

- `Helpers/TiltedDensity.lean`, `Helpers/TiltedReversalCount.lean`, `Helpers/TiltedCountRHUpper.lean`, `Helpers/TiltedCountExcursion.lean` and `Helpers/PrimeProfileWindowBands.lean`: actual tilted densities, finite signed counts, RH inclusion, false-RH interval margins and uniform fixed-window rank bands;

- `Helpers/LocalizedReversalCriterion.lean` and `Helpers/LocalReversalSampling.lean`: both localized reversal-count equivalences, explicit cell selection and the all-short-interval arithmetic input boundary;

- `Helpers/LocalizedAscentCriterion.lean` and `Helpers/LocalAscentSampling.lean`: both ascent-only criteria, using all actual pairs and an explicit quantitative sampling premise;

- `Helpers/TiltedSignedExcess.lean` and `Helpers/FixedTiltCombinations.lean`: exact signed counts, aggregate rounding and fixed-tilt cancellation;

- `Helpers/GapFrequencyCoverage.lean`: the global Li_2 error implication for actual local exact-gap occurrence;

- `Assembly/ConditionalLocation/FineLocationRH.lean`: the actual last-ascent sandwich, relative limit, square-root clock tracking, and full RH wave under the stated local coverage hypothesis;

- `Helpers/NicolasQuantitativeClock.lean`: unconditional $F(x)\log x \to 0$ and $\mathcal{C}(x)/x \to 1$, with sufficient precision before exponentiation;

- `Helpers/NicolasGrowingTilt.lean`, `Helpers/NicolasGrowingTiltAsymptotic.lean`, `Helpers/FiniteTwinsSpectrum.lean` and `Helpers/ReversalCountCounterexample.lean`: growing-tilt RH criteria and the sharp growing-tilt correction, the conditional finite-twins spectrum consequence, and the generic count-monotonicity counterexample.

Paths are relative to `PrimeFactorOscillations/`. The theorem ledger maps intermediate arguments to declarations. Ordinary prose sometimes combines several compiled lemmas; these combinations are distinguished from individually named declarations.

The analytic inputs are the reciprocal-prime estimate, fixed-modulus prime progression limits and (11). Narrow leaves of the pinned Erdős 690 development supply ordinary density and symmetric bounds; its conditional classification is not an input. The width-600 count is extracted from a separately pinned prime-cluster development. The RH part additionally reuses the Robin1984 analytic development and the verified explicit-formula and complex-analytic dependencies. Section 15 additionally uses the published Alweiss–Luo all-short-interval input. This input is not a discharged Lean dependency. The complete ordinary proof above must be distinguished from formal count theorems conditional on explicit arithmetic sampling hypotheses. Those hypotheses concern actual prime positions, gap widths, separation and the number of cells, and contain no density sign or RH conclusion. No project axiom asserts the source theorem. Classical analytic results in the maintained closed development are retained as proved dependencies, not research-target axioms. The paper provides the coefficient-transfer argument and its consumers, while citing the underlying classical prime-distribution and Nicolas theorems.

The relative-location implication in Theorem 9.3 is maintained in `Assembly/ConditionalLocation/TwinCountAsymptoticBridge.lean`. Its cutoff, order and last-ascent dependency closure has passed the maintained build and a public foundational-axiom audit. The positive twin-count asymptotic is explicit in its theorem type. The resulting logarithmic location and count limits are ordinary consequences of that proof and the maintained transfers. Likewise (28) combines verified estimates, but is not advertised as a separately compiled normalized-wave declaration. These qualifications concern formal integration, not an assumption that Hardy–Littlewood or RH has been proved.

The public Challenge states two headline theorems over Mathlib primitives. Solution contains their actual proofs. The two deliberate Challenge placeholders are absent from the Solution closures. The 5 October proof audit checked the two public compiled statement pairs and 539 selected theorem closures, finding only `propext`, `Classical.choice`, and `Quot.sound`. A fresh 6 October audit, importing the maintained family owners and Solution, checks 18 exact theorem statements and dependency closures for the additions in Sections 16– 18. Every axiom set is exactly the same three standard foundations. This includes all four power-degree declarations and the un-

conditional modified-LCM endpoints. The power-degree proof was already maintained before this editorial revision; it was independently rechecked, not replaced by an assumption wrapper. Its eventual signed premise remains explicit. The RH statements have maintained consumers in Solution with their RH, false-RH or zero hypotheses explicit. An axiom audit does not remove any of these hypotheses. The two Challenge statements concern the original reversal and affine-spectrum headlines; they are not claimed to be a Challenge comparison for every later RH declaration.

The dependency record and build guide specify Windows Lean 4.34.0, source revisions and artifact hashes. The project reuses the Erdős dependency's published cache without rebuilding or redistributing it. Additional analytic compatibility artifacts still require an existing matching view; a complete fresh-machine bootstrap is not claimed. The verification record separates the initial publication, the 4 October analytic development, the completed 5 October post-paper formalization, and the PDF checks. Branch links provide mutable navigation; the records retain checked source hashes.

No numerical experiment proves an asymptotic step here. The exact inequality $1/602 < 1/601$ supplies the canonical constant's strict margin; no effective onset rank is given. Official Comparator/NanoDa replay and independent expert review have not been claimed. Kernel checking verifies proof terms for the stated definitions under the reported foundations. Source fidelity, attribution, novelty and significance require scrutiny beyond the kernel. AI assistance was used for research, proof development and exposition.

21 Second moments, simplicity defects, and prime-profile rigidity

The statements in this section separate three kinds of information: a conditional spectral identity, an unconditional inverse theorem for local probabilities, and an unconditional limit on what the normalized profile can identify. No improved zero-free boundary is asserted.

21.1 The full second moment and a simplicity defect

Assume RH. Let Γ be the set of distinct ordinates of the nontrivial zeros, including both signs, write m_γ for their analytic multiplicities, and put

$$c_\gamma = (1/4 + \gamma^2)^{-1}, \qquad C = \gamma_{\mathrm{E}} + 2 - \log(4\pi).$$

The existing zero-wave formula is absolutely convergent and has the form $W(x) = \sum_{\gamma\in\Gamma} m_\gamma c_\gamma x^{i\gamma}$. Its reality follows from conjugate pairing, and $|W(x)| \le C$.

Theorem 21.1 (Universal second moment and excess multiplicity)**.** *For any two fixed qualifying quadratic prime laws F, G and any fixed positive rank parameters α, α', let W_F, W_G denote the normalized moving-rank observables already compared to W. Under RH,*

$$\lim_{M\to\infty} \frac{1}{\log M} \sum_{n=2}^{M} \frac{W_F(n^4)W_G(n^4)}{n} = V := \sum_{\gamma\in\Gamma} m_\gamma^2 c_\gamma^2.$$

Define

$$B = \sum_{\gamma\in\Gamma} m_\gamma c_\gamma^2, \qquad \mathcal{D} = V - B.$$

Then

$$0 \le \mathcal{D} = \sum_{\gamma\in\Gamma} m_\gamma(m_\gamma - 1)c_\gamma^2 \le C^2 - B.$$

In particular $\mathcal{D} = 0$ if and only if all these zeros are simple. For every finite set $S \subset \Gamma$,

$$\sum_{\gamma\in S} m_\gamma(m_\gamma - 1)c_\gamma^2 \le \mathcal{D}, \qquad \frac{m_\gamma(m_\gamma - 1)}{(1/4 + \gamma^2)^2} \le C^2 - B.$$

Proof. For every real u, $(\log M)^{-1} \sum_{n=2}^{M} n^{iu}/n$ tends to 1 for $u = 0$ and to 0 otherwise. Absolute summability permits termwise extraction from the product of two spectra. All pairs of frequencies whose sum is zero remain; conjugate symmetry therefore gives $\sum_\gamma m_\gamma^2 c_\gamma^2$, with no simplicity assumption. The established comparison gives $W_F(n^4) - W(n^4) \to 0$ and $W_G(n^4) - W(n^4) \to 0$. Since W is bounded, their product minus $W(n^4)^2$ tends to zero. The normalized harmonic average of that error tends to zero as well. Thus the same limit holds for both probability observables. The pointwise bound $W(n^4)^2 \le C^2$ gives $V \le C^2$. Subtracting B produces the nonnegative series displayed above. Every summand is zero precisely when its positive integer multiplicity is one. Finite subsums and a single complete frequency fiber give the remaining inequalities. □

Here n^4 is a divisor cutoff for each fixed family; it is not an argument substituted into a two-variable polynomial family. Finite exceptional primes and forced-rank shifts are included in the exact reference used to define each normalized observable. These bounds do not prove simplicity, RH, or a better unconditional multiplicity estimate than those already available.

21.2 What the entire Bernoulli profile determines

For a law $\eta(p) \in [0,1]$ with $\sum_p \eta(p)^2 < \infty$, define

$$\Phi_\eta(z) = \prod_p (1 + (z-1)\eta(p)) \exp(-(z-1)\eta(p)).$$

This is a locally uniformly convergent entire product. Its zeros, with multiplicity, are exactly $1 - \eta(p)^{-1}$ for $\eta(p) > 0$. Thus the profile recovers the unordered positive probabilities. This standard canonical-product observation alone does not label them by primes.

The analytic input now has a maintained Lean proof in `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileZeros.lean` under the project prefix. For $w = \eta(p)(z-1)$,

$$(1+w)e^{-w} - 1 = -(e^w - 1 - w)e^{-w}.$$

Consequently, on $|z-1| \le R$, each factor's defect from one is at most $R^2 e^{2R} \eta(p)^2$. Summing this bound gives locally uniform product convergence and entire continuation. At a point where no linear factor vanishes, the summable-defect nonvanishing theorem for infinite products makes the product nonzero. Conversely a zero factor makes the product zero. Thus the compiled zero-set statement is exactly

$$\Phi_\eta(z) = 0 \quad \Longleftrightarrow \quad \exists p : \ \eta(p) > 0, \quad z = 1 - \eta(p)^{-1}.$$

The same module proves invariance under every bijective relabelling. The companion maintained module `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileMultiplicity.lean` also proves the full multiplicity identity

$$\operatorname{ord}_{z=1-1/c} \Phi_\eta = m_c, \qquad m_c = \#\{p : \eta(p) = c\}, \quad c > 0.$$

Indeed square summability makes this fibre finite. Splitting off the *entire* fibre writes the profile as the m_c-th power of $(1 + c(z-1))e^{-c(z-1)}$ times the profile of the remaining coordinates. The displayed factor has a simple zero at $1 - 1/c$; the remaining product is entire and nonzero there by the proved zero-set theorem. Orders add, giving the identity. It follows that equal profiles recover every positive probability with its complete multiplicity. The case $c = 1$, giving a zero at $z = 0$, is included. Zero probabilities contribute the unit factor and are invisible to the profile itself. The maintained companion `Mathlib/Analysis/SpecialFunctions/`

`Log/BernoulliProfileReconstruction.lean` now proves the complete unlabelled inverse statement: two admissible, square-summable arrays have equal entire profiles if and only if there is a bijection between their positive coordinates preserving every probability value. To construct it, match each finite positive-value fibre using the proved multiplicity identity, then assemble these fibre bijections over all positive values. Conversely, delete the zero coordinates, whose factors are one, and reindex the remaining product by the value-preserving bijection. No hypothesis about the Riemann hypothesis or prime distributions enters this unlabelled reconstruction. The next maintained module, `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileTailMatching.lean`, also proves the ordered-tail matching step. If a_n, b_n have eventually strictly decreasing positive tails, $a_n \to 0$, and the same set of positive values, then there are fixed indices A, B such that

$$a_{A+n} = b_{B+n} \qquad (n \geq 0).$$

To exclude finite exceptions, take a tail value a_A smaller than every positive exceptional value in either sequence. Its matching value b_B lies in the ordered tail. Every positive value at most a_A then lies after A and after B, respectively. These two tails have the same range. Their mutual value-matching maps are strictly increasing maps of the natural numbers and are inverse to one another; each strictly increasing self-map satisfies $n \leq f(n)$, so both maps must be the identity. The Lean equal-profile corollary consumes the positive-coordinate bijection above. The maintained assembly `Assembly/PrimeProfileInverse.lean` now proves the prime-labelled tail conclusion as well: equal profiles and reciprocal asymptotics $a_n^{-1} = p_n/\nu_1 + b_1 + o(1)$ and $b_n^{-1} = p_n/\nu_2 + b_2 + o(1)$, with $\nu_i > 0$, force $\nu_1 = \nu_2$, $b_1 = b_2$, and $a_n = b_n$ eventually. Here p_n is the increasing enumeration of primes. Its proof derives tail positivity, convergence to zero and strict ordering from the reciprocal remainder, then consumes the compiled matching theorem. The existing unconditional theta PNT and exact prime-prefix identity give $p_{n+b}/p_{n+a} \to 1$ for every fixed a, b. This identifies the leading scales. The remaining integer prime difference converges to $\nu(b_1 - b_2)$, hence is eventually constant. A positive constant would translate every sufficiently large prime to a prime, contradicting the maintained prime-translation theorem. These complete consumers are implemented in `Helpers/PrimeProfilePrimeRatio.lean`, `Helpers/PrimeProfileInverseAsymptotics.lean`, and `Helpers/PrimeProfileShiftRigidity.lean`. The finite-exception conclusion is now compiled as well. The main-

tained modules `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileIdentity.lean` and `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileFinitePermutation.lean` split off the common tail and cancel it near $z = 1$, where it has value one. The analytic identity theorem makes the remaining finite profiles equal everywhere. Their positive coordinates admit a value-preserving bijection; the equal total number of coordinates then gives equally many zeros. Joining the two bijections gives a permutation of the exceptional prefix, extended by the identity on its entire complement. Probabilities equal to one are included. The assembled theorem `prime_profile_eq_iff_finite_permutation` proves both directions of this classification together with equality of the scale and intercept. The general finite-accumulation version is also compiled, with equality expressed as frequent agreement in a punctured neighborhood of the accumulation point.

Theorem 21.2 (Prime-coordinate rigidity)**.** *Suppose two such laws satisfy, outside finite sets,*

$$0 < \eta_i(p) < 1, \qquad \eta_i(p)^{-1} = p/\nu_i + b_i + o(1), \quad \nu_i > 0.$$

If $\Phi_{\eta_1} = \Phi_{\eta_2}$, then $\nu_1 = \nu_2$, $b_1 = b_2$, and $\eta_1(p) = \eta_2(p)$ for every sufficiently large prime. The complete laws differ only by a permutation of finitely many prime coordinates. Conversely every such finite permutation preserves the profile. Equality on a real interval, or on any set with a finite accumulation point, suffices.

Proof. Write the reciprocal remainders as $r_i(p) \to 0$. For sufficiently large consecutive primes the reciprocal increment is at least $1/\nu_i - 2\sup|r_i| > 0$. Each tail is therefore strictly ordered. Equality of entire functions gives equality of their zero multisets. After removing finitely many exceptional probabilities, the two ordered tails consequently match with one fixed prime-index shift: $\eta_1(p_{j+a}) = \eta_2(p_{j+b})$ for all sufficiently large j. The prime number theorem gives $p_{j+a}/p_{j+b} \to 1$, so comparison of the leading reciprocal terms forces $\nu_1 = \nu_2 = \nu$. Now

$$p_{j+b} - p_{j+a} = \nu(b_1 - b_2) + o(1).$$

An integer sequence with a real limit is eventually constant. If this constant were nonzero, one direction of the matching would give a fixed positive integer c such that every sufficiently large prime p has $p + c$ prime. Iteration would then make $p + pc = p(1+c)$ prime, a contradiction. Hence the constant and the shift are zero, and $b_1 = b_2$. The prime-labelled tails coincide.

Cancel the common tail product in a neighborhood of $z = 1$. The remaining finite products have the same positive-probability multiset. Enlarging to one common finite prime set gives the same number of zero probabilities too, hence a finite permutation. This also retains probabilities equal to one, whose zeros lie at $z = 0$. The converse follows by reordering finitely many factors. The identity theorem gives the last assertion. □

21.3 Sharp cubic camouflage outside the rigid class

Theorem 21.3 (Identical profiles with optimally small finite-cutoff defects)**.** *There are continuum many admissible laws $\eta(p) = p^{-1} + O(p^{-2})$, pairwise distinct on infinitely many primes, whose entire profiles are identical. They can be chosen so that for each fixed $T \geq 0$, every $N \geq 1200$ and $0 \leq t \leq T$,*

$$0 \leq L_\eta(N,t) - L_0(N,t) \leq 4800\, T^2 N^{-3},$$
$$L_\eta(N,t) = \sum_{p \leq N} \{\log(1 + t\eta(p)) - t\eta(p)\}.$$

For every nontrivial construction with infinitely many swaps and every fixed $t > 0$, the difference is not $o(N^{-3})$. These laws need not come from a fixed polynomial arithmetic family.

Proof. The established bounded-gap theorem supplies disjoint pairs $p_j < q_j \leq p_j + H$, where $H = 600$, $p_j > 2H$, and $q_j < p_{j+1}$. Swap the probabilities $1/p_j, 1/q_j$ on any chosen subset of these pairs. The resulting law obeys $|\eta(p) - 1/p| \leq 2H/p^2$. The entire product is unchanged, because its locally uniform absolute convergence permits permutation.

Every cutoff meets at most one incomplete pair. Completed pairs cancel exactly. If $p \leq N < q$ is an incomplete selected pair, the complete signed defect is

$$d_t(p,q) = \log(1 + t/q) - \log(1 + t/p) + t(1/p - 1/q).$$

Applying $\log u \leq u - 1$ to both reciprocal ratios gives

$$\frac{t^2(q-p)}{pq(q+t)} \leq d_t(p,q) \leq \frac{t^2(q-p)}{pq(p+t)} \leq \frac{t^2(q-p)}{p^3}.$$

All denominators are positive. Since $p > N - H \geq N/2$, this proves the asserted bound. At the selected cutoffs $N = p_j$ the two-sided bounds imply $p_j^3 d_t(p_j, q_j)/(q_j - p_j) \to t^2$. As $q_j - p_j \geq 1$, the error cannot be $o(N^{-3})$. Partition the pair indices into countably many infinite blocks and swap all

indices in a block according to one binary choice. This gives continuum many laws, any two differing on infinitely many coordinates. Finally their reciprocal deviations $\eta(p)^{-1} - p$ take positive and negative integer values at infinitely many swapped pairs; they have no limiting constant. Thus they lie outside the preceding rigidity hypothesis, demonstrating the significance of its $o(1)$ remainder. □

The canonical product is classical mod-Poisson machinery; the prime-coordinate rigidity and arithmetic swap comparisons are the specific conclusions considered here, without a claim of literature priority. The prime-coordinate inverse, an infinite swap construction and its exact optimal constant are now Lean-formalized. The continuum-cardinality assertion retains its ordinary proof. Additional maintained owners for completed surrounding inputs, relative to `PrimeFactorOscillations/`, are:

- `Helpers/XiWaveMoments.lean` and `Helpers/XiMultiplicityBudget.lean`;
- `Mathlib/Analysis/Fourier/LogarithmicBilinear.lean`;
- `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocal.lean`;
- `Mathlib/NumberTheory/Primes/Translation.lean`;
- `Assembly/PrimeProfileRigidity.lean`.

The latest promotion audit is separate from the older verification totals elsewhere in this manuscript; those historical totals do not certify this new batch or an improved analytic exponent.

21.4 Universal cutoff rigidity and recurring prime gaps

The swap construction admits a converse without a reciprocal-intercept assumption. Write $p_0 = 2 < p_1 < \cdots$, and suppose $0 \le a_j \le 1$, $\sum_j a_j^2 < \infty$, and $\Psi_a = \Psi_{(1/p_j)}$. Define the complete signed defect

$$\Delta_{a,n}(t) = \sum_{j=0}^{n} \{\log(1 + ta_j) - ta_j - \log(1 + t/p_j) + t/p_j\}, \qquad t > 0.$$

The term $t/p_j - ta_j$ is part of the observable and is retained throughout.

Theorem 21.4 (Universal adaptive cutoff inverse)**.** *For every positive probe sequence* (t_n)*,*

$$\frac{p_n^2(p_n+t_n)}{t_n^2}\Delta_{a,n}(t_n)\longrightarrow 0$$

if and only if $a_j = 1/p_{\sigma(j)}$ *for a permutation* σ *fixing every sufficiently large index. In particular, for each fixed* $t>0$*,*

$$\Delta_{a,n}(t)=o(p_n^{-3}) \quad\Longleftrightarrow\quad a \text{ is a finite permutation of } (1/p_j).$$

Proof. Multiplicity recovery shows that the positive coordinates of a are exactly the distinct numbers $1/p_j$. First suppose no coordinate is zero, so $a_j = 1/p_{\sigma(j)}$ for a permutation σ. Put $H_t(j)=\log(1+t/p_j)-t/p_j$, which is strictly increasing and negative.

For $B_n=\{0,\dots,n\}$ and $S_n=\sigma(B_n)$, cancel the common coordinates in the signed sum. If $m_n=|S_n\setminus B_n|=|B_n\setminus S_n|$, every entering weight is at least $H_t(n+1)$ and every leaving weight is at most $H_t(n)$. Consequently

$$\Delta_{a,n}(t)\ge m_n\{H_t(n+1)-H_t(n)\}\ge 0. \tag{21.1}$$

Our signed scalar comparison gives, with $p=p_n$ and $q=p_{n+1}$,

$$H_t(n+1)-H_t(n)\ge\frac{t^2(q-p)}{pq(q+t)}.$$

Eventually $q\le 2p$, by the unconditional adjacent-prime ratio limit. Since $q-p\ge 1$, uniformly for every $t>0$,

$$\frac{p^2(p+t)}{t^2}\{H_t(n+1)-H_t(n)\}\ge\frac{p(p+t)}{q(q+t)}\ge\frac14. \tag{21.2}$$

A vanishing normalized defect therefore forces $m_n=0$ at every sufficiently large index. Two successive invariant prefixes imply $\sigma(n)=n$: the image of n belongs to B_n but cannot belong to the invariant prefix B_{n-1}. Conversely, a permutation fixing a tail preserves every sufficiently large prefix, whose defect is exactly zero.

It remains to rule out a zero coordinate. After the first zero has entered B_n, the positive probabilities in that prefix correspond to a set S_n of at most n distinct prime indices. If $r=|S_n\setminus B_n|$ and $s=|B_n\setminus S_n|$, then $s\ge r+1$. The same common-coordinate cancellation, now using the boundary weight $H_t(n)$ on both sides, yields

$$\Delta_{a,n}(t)\ge(r-s)H_t(n)\ge -H_t(n).$$

For $y > 0$,

$$y - \log(1+y) \geq \frac{y^2}{2(1+y)}.$$

Indeed the difference vanishes at zero and its derivative is $y^2/(2(1+y)^2) \geq 0$. Taking $y = t/p_n$ gives

$$\frac{p_n^2(p_n+t)}{t^2}\Delta_{a,n}(t) \geq \frac{p_n}{2} \longrightarrow \infty, \tag{21.3}$$

uniformly over all positive probes. Thus a zero coordinate cannot satisfy the proposed accuracy condition. For fixed t, the normalizing factor is asymptotic to p_n^3/t^2, proving the final equivalence. □

Theorem 21.5 (Optimal distortion records the least recurring prime gap)**.** *Let G be the least positive integer occurring infinitely often as $p_{n+1} - p_n$. This integer exists by bounded-gap supply and satisfies $2 \leq G \leq 600$. Every same-profile law a that is not a finite permutation satisfies, for every positive probe sequence (t_n),*

$$\limsup_{n\to\infty} \frac{p_n^2(p_n+t_n)}{t_n^2}\Delta_{a,n}(t_n) \geq G.$$

There is one infinitely supported permutation law attaining equality for every such probe sequence. It satisfies $a(p) = 1/p + O(p^{-2})$.

Proof. Every gap smaller than G occurs only finitely often; there are only finitely many such integer gaps. Hence eventually $q - p \geq G$. An infinite permutation moves infinitely many prefixes, and at each such sufficiently large prefix (21.1) gives normalized defect at least

$$G\frac{p(p+t)}{q(q+t)} \geq G\left(\frac{p}{q}\right)^2 \longrightarrow G,$$

uniformly in $t > 0$. Zero coordinates instead give the divergent lower bound (21.3).

For attainment, choose disjoint adjacent prime pairs $p < q = p + G$ at arbitrarily large heights, and swap each pair. At any cutoff at most one pair is incomplete; at its lower endpoint the exact defect is $d_t(p,q)$. At every other prime cutoff it is zero. The two-sided signed comparison gives

$$G\left(\frac{p}{q}\right)^2 \leq \frac{p^2(p+t)}{t^2}d_t(p,q) \leq G\frac{p}{q} \leq G.$$

Both endpoints tend to G uniformly in $t > 0$, proving the claimed limsup for every probe sequence, including probes tending to zero or growing without bound. The bounded-displacement estimate proves the stated local-law error. □

Thus the optimal normalized cost among all infinite rearrangements is exactly G. In particular, cost 2 is equivalent to infinitely many twin primes. This is an inverse characterization, not a new bound on G or a zero-free region.

Corollary 21.6 (The same constant governs late ascents)**.** *Let C denote the optimal distortion constant in the preceding theorem. The critical late-ascent scale for the ordinary and generic odd laws is*

$$b_* = \frac{1}{1+C} = \frac{1}{1+G}.$$

Every $0 < b < b_*$ admits arbitrarily late simultaneous ascents in both laws; every $b > b_*$ has eventual strict descent beyond the corresponding moving-rank scale. The endpoint $b = b_*$ is left open.

Proof. Apply the maintained recurring-gap/late-ascent critical-scale theorem to the actual least recurring gap G. The preceding optimality theorem identifies $C = G$. □

The full adaptive inverse is maintained in `Assembly/PrimeProfileZeroHoles.lean`. The exact optimality theorem, including the definition and bounds $2 \le G \le 600$, its universal lower threshold for every law *not a finite permutation*, and a single witness for all positive probe sequences, is maintained in `Assembly/PrimeProfileMinimax.lean`. The formal statements use the equivalent threshold formulation: every witness defect is at most G, and for every $\varepsilon > 0$ and N some $n \ge N$ has defect at least $G - \varepsilon$. The witness also satisfies the explicit bound $|a_n - 1/p_n| \le 2G/p_n^2$ whenever $p_n \ge 2G$, retaining all small-prime exceptions. Its signed construction is in `Assembly/PrimeProfileAttainment.lean` and `Mathlib/Combinatorics/AdjacentSwapPrefix.lean`. The late-ascent corollary has its own compiled declaration. The stronger zero-hole inequality (21.3) retains its ordinary proof; the formal inverse uses the sufficient uniform bound $p_n/4$. Finite permutations have eventual zero defect and are excluded from every positive lower-distortion assertion. No literature priority is claimed.

21.5 Two-coordinate cutoff transport

For a prime-coordinate permutation σ, define the crossing width

$$m_n = \big|\sigma(\{0,\dots,n\}) \setminus \{0,\dots,n\}\big|,$$
$$D_n(t) = \frac{p_n^2(p_n+t)}{t^2}\Delta_{a,n}(t), \qquad a_j = 1/p_{\sigma(j)}.$$

Put $g_j = p_{j+1} - p_j$. The full signed defect detects more than whether a prefix moves: it distinguishes simultaneous crossings.

Lemma 21.7 (Complete two-crossing cost)**.** *For every $n \geq 1$ with $m_n \geq 2$, and every $t > 0$,*

$$D_n(t) \geq T_n \left(\frac{p_n}{p_{n+2}}\right)^2, \qquad T_n = g_{n-1} + 2g_n + g_{n+1}.$$

Proof. Write $W_t(j) = \log(1 + t/p_j) - t/p_j$, an increasing weight. Cancel common coordinates from the two prefixes. There are m_n entering and m_n leaving coordinates. Every entering coordinate has index at least $n+1$, and at least one has index at least $n+2$. Every leaving coordinate has index at most n, and at least one has index at most $n-1$. Consequently,

$$\begin{aligned}\Delta_{a,n}(t) &\geq m_n(W_t(n+1) - W_t(n)) \\ &\quad + W_t(n+2) - W_t(n+1) + W_t(n) - W_t(n-1) \\ &\geq W_t(n+1) + W_t(n+2) - W_t(n) - W_t(n-1).\end{aligned}$$

The remaining crossing contributions have been retained in this inequality; no unsigned coefficient bound is used.

The maintained scalar comparison, with a middle reference denominator, states that for $t > 0$, $0 < p \leq q$ and $p \leq r$,

$$\frac{r^2(r+t)}{t^2}\left\{\log(1+t/q) - t/q - \log(1+t/p) + t/p\right\} \geq (q-p)\left(\frac{r}{q}\right)^2.$$

Apply it to $(p,q,r) = (p_{n-1}, p_{n+1}, p_n)$ and (p_n, p_{n+2}, p_n). Replace the first ratio by the smaller common ratio p_n/p_{n+2} and add the two signed costs. Their denominator displacements sum to T_n. □

Lemma 21.8 (The second prime-geometry threshold)**.** *For $n \geq 3$, $T_n \geq 12$. Equality holds exactly when*

$$(g_{n-1}, g_n, g_{n+1}) = (2,4,2) \quad \text{or} \quad (4,2,4).$$

The exceptional initial values are $T_1 = 7$ and $T_2 = 10$; a double crossing is impossible at $n = 0$.

Proof. The four primes start beyond 3, so all three gaps are even and at least 2. Two consecutive gaps cannot both be 2: among $p, p+2, p+4$, one is divisible by 3 and exceeds 3. If the central gap is at least 4, the weighted sum is at least $2 + 2 \cdot 4 + 2 = 12$, with equality only for $(2, 4, 2)$. If the central gap is 2, both neighbours are at least 4, giving $4+2\cdot 2+4 = 12$, with equality only for $(4, 2, 4)$. The initial values use $2, 3, 5, 7, 11$ directly. □

Corollary 21.9 (Double-crossing rigidity)**.** *Fix any positive probe sequence* (t_n)*. For a law that is* not a finite permutation *and has* $m_n \geq 2$ *at arbitrarily large indices,*

$$\limsup_{n\to\infty} D_n(t_n) \geq 12.$$

Equivalently, an upper limit strictly below 12 *forces* $m_n \leq 1$ *eventually. If the upper limit is strictly below* 14 *and double crossings persist, one of the two displayed four-prime patterns occurs arbitrarily far out.*

Proof. The fixed-shift prime ratio p_n/p_{n+2} tends to 1. The first two lemmas therefore give the lower threshold uniformly over every positive probe sequence. If the integer T_n is not 12 at a sufficiently large double crossing, its parity gives $T_n \geq 14$. Such crossings would force an upper limit at least 14, contradicting the stated ceiling. Thus every sufficiently late double crossing has one of the two equality patterns. Persistent double crossings give their recurrence. □

Theorem 21.10 (A sharp probability-profile equivalence)**.** *The following statements are equivalent:*

1. *Arbitrarily late consecutive four-prime windows, starting beyond* 3*, have offsets* $(0, 2, 6, 8)$ *or* $(0, 4, 6, 10)$*.*

2. *There is one permutation* σ *with* $m_n \geq 2$ *at arbitrarily large cutoffs and*
$$\forall \varepsilon > 0 \ \exists N \ \forall n \geq N \ \forall t > 0, \qquad D_n(t) \leq 12 + \varepsilon.$$

For the law $a_j = 1/p_{\sigma(j)}$ *in the construction,* $0 < a_j \leq 1$*,* $\sum_j a_j^2 < \infty$*, and the entire canonical profile equals that of* $u_j = 1/p_j$*. It is* not a finite permutation*. For every positive probe sequence, chosen after the law,*

$$\limsup_{n\to\infty} D_n(t_n) = 12.$$

The cutoff in the upper estimate is independent of the positive probe.

Proof. Suppose the windows recur. Greedily choose an unbounded set A of their starting indices with $a + 4 \le b$ whenever $a < b$ in A. This is possible by selecting the next start beyond the previous start plus four. Set $B = A \cup (A + 1)$. No j and $j + 2$ both belong to B: such a pair would put two selected starts at distance less than four. Define

$$\sigma(j) = \begin{cases} j + 2, & j \in B, \\ j - 2, & j \notin B,\ j \ge 2,\ j - 2 \in B, \\ j, & \text{otherwise.} \end{cases}$$

The absence of two-step chains makes this an involution. In each selected block it swaps the first with the third coordinate and the second with the fourth. Both prime displacements equal six, for either offset pattern. At cutoff $n = a+1$ the distinct exterior coordinates $a+2, a+3$ enter, so double crossings recur.

For an arbitrary real weight W, completed pairs cancel exactly:

$$\sum_{j=0}^{k+1} \big(W(\sigma(j)) - W(j)\big) = \mathbf{1}_B(k+1)\big(W(k+3) - W(k+1)\big) + \mathbf{1}_B(k)\big(W(k+2) - W(k)\big).$$

This identity includes every finite-prefix term. It follows either by pairing completed swaps or by induction, retaining the two pairs that can cross the boundary.

We also need the upper comparison at a middle reference. Write $\delta_t(p, q) = \log(1 + t/q) - t/q - \log(1 + t/p) + t/p$. For $t > 0$ and $0 < p \le r \le q$,

$$\frac{r^2(r+t)}{t^2}\delta_t(p, q) \le (q - p)\left(\frac{r}{p}\right)^2.$$

Indeed, $\delta_t(p, q) = \int_p^q t^2/(x^2(x+t))\,dx \le t^2(q-p)/(pq(p+t))$. After normalization, the desired comparison follows from the nonnegative identity

$$\left\{(q - p)(r/p)^2 - \frac{(q - p)r^2(r + t)}{pq(p + t)}\right\} p^2q(p + t) = (q - p)r^2\{p(q - r) + t(q - p)\} \ge 0.$$

All denominators are positive.

Apply the exact prefix identity to $W = W_t$ and this upper comparison with reference $r = p_{k+1}$. Each present pair has displacement six. The pair

starting at $k+1$ costs at most 6, and the pair starting at k costs at most $6(p_{k+1}/p_k)^2$. Absent pairs contribute zero. Thus, for every $k \geq 0$ and $t > 0$,

$$D_{k+1}(t) \leq 6(p_{k+1}/p_k)^2 + 6.$$

The right side tends to 12, proving the upper estimate with a cutoff uniform in t. The previous lower threshold at recurring double crossings supplies arbitrarily late values at least $12 - \varepsilon$ for every positive probe sequence. This proves the exact upper limit, including the rank shift $n = k + 1$.

Reindexing preserves square summability and the canonical entire product, so the constructed sequence has the asserted probability properties and profile. A finite permutation fixes all sufficiently late coordinates and preserves all later prefixes; it cannot have recurring double crossings. The selected blocks start beyond 3, so the exceptional prime coordinates $2, 3$ are fixed. The first cutoff $n = 0$ plays no role in the eventual estimate.

Conversely, use the constant probe $t = 1$ and $\varepsilon = 1$ in statement (ii). The eventual ceiling is $13 < 14$. The compiled inverse criterion therefore forces an equality pattern at every sufficiently late double crossing. Their recurrence gives statement (i). □

The complete signed finite-set inequality is maintained in `Mathlib/Combinatorics/TwoBoundaryTransport.lean`; the actual prime-profile specialization and its middle-prime normalization are maintained in `Helpers/PrimeProfileDoubleCrossing.lean`. The prime-congruence classification is maintained in `Mathlib/NumberTheory/Primes/WeightedQuadruplet.lean`. The quantified uniform lower threshold and arbitrarily late witnesses are compiled there, as is the eventual width bound under a ceiling $C < 12$. Their limsup formulations are mathematical consequences of these quantified statements. The stronger inverse under an eventual ceiling $C < 14$ is maintained in `Helpers/PrimeProfileQuadrupletInverse.lean`. The exceptional initial values $7, 10$ retain their elementary checks.

The converse is maintained in `Assembly/PrimeProfileQuadrupletAttainment.lean`. It formally proves the actual probability-profile equivalence, with one law before every positive probe, as well as positivity, the unit bound, square summability, full profile equality, arbitrarily late lower witnesses and exclusion of finite permutation representations. Its compiled upper formulation has one cutoff for all positive probes. The sparse block selection and the complete uniform upper estimate are maintained in `Helpers/PrimeQuadrupletSupport.lean` and `Helpers/PrimeProfileTwoStepTransport.lean`; the distance-two involution, exact

signed prefix identity and double-crossing witness are reusable owners in `Mathlib/Combinatorics/TwoStepSwapPrefix.lean`. The matching middle-reference upper comparison is maintained in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalNormalization.lean`. No infinitude of either four-prime pattern is asserted.

21.6 Quantized distortion and probe independence

The whole finite transport controls the distortion of every bounded profile. For a permutation σ, put

$$S_n = \sum_{j=0}^{n} p_{\sigma(j)} - \sum_{j=0}^{n} p_j.$$

This is the exact signed prime-denominator transport. It is nonnegative, because the first $n+1$ primes have the least possible sum.

Lemma 21.11 (Localization and a uniform transport error)**.** *Let $r = p_n$, $C \geq 0$, $C < r/4$, and suppose that $D_n(t) \leq C$ for one $t > 0$. Then every entering prime is at most $r + 4C$, and every leaving prime is at least $r - C$. Moreover,*

$$0 \leq S_n \leq 4C, \qquad S_n \left(\frac{r}{r+4C}\right)^2 \leq D_n(v) \leq S_n \left(\frac{r}{r-C}\right)^2 \quad (v > 0),$$

and, uniformly over every new positive probe,

$$|D_n(v) - S_n| \leq \frac{32C^2}{p_n}.$$

Proof. Let E and L be the finite sets of entering and leaving prime values, after common coordinates cancel. They have equal cardinalities. For $\delta_t(p,q) = \log(1 + t/q) - t/q - \log(1 + t/p) + t/p$ and $N_r(t) = r^2(r+t)/t^2$, exact insertion of the common reference gives

$$D_n(t) = N_r(t) \left\{ \sum_{q \in E} \delta_t(r,q) + \sum_{p \in L} \delta_t(p,r) \right\},$$
$$S_n = \sum_{q \in E} (q - r) + \sum_{p \in L} (r - p).$$

Every retained summand is nonnegative. Thus each reference cost is at most the full cost C. The middle-reference lower comparison gives $N_r(t)\delta_t(p,r) \geq$

$r-p$ for a leaving prime. If an entering prime satisfies $q \geq 2r$, monotonicity in q gives

$$N_r(t)\delta_t(r,q) \geq N_r(t)\delta_t(r,2r) \geq r/4,$$

contrary to $C < r/4$. Hence $q < 2r$, and the lower comparison gives $N_r(t)\delta_t(r,q) \geq (q-r)(r/q)^2 \geq (q-r)/4$. This proves both localization bounds for every transported coordinate.

For any $v > 0$ and $0 < \ell \leq p \leq r \leq q \leq u$, the two scalar comparisons already proved give

$$(q-p)(r/u)^2 \leq N_r(v)\delta_v(p,q) \leq (q-p)(r/\ell)^2.$$

Use $\ell = r - C > 0$ and $u = r + 4C$, and sum this inequality on the *same* two reference transports. This proves the displayed two-sided comparison for $D_n(v)$ and S_n. Applying its lower bound at the original probe t, with $r/(r+4C) > 1/2$, gives $S_n \leq 4C$.

Write $\lambda = (r/(r+4C))^2$ and $\mu = (r/(r-C))^2$. The following nonnegative identities justify the remaining constants:

$$\{\lambda - (1 - 8C/r)\}\, r(r+4C)^2 = 48rC^2 + 128C^3,$$
$$\{1 + 4C/r - \mu\}\, r(r-C)^2 = C\{2r(r-4C) + rC + 4C^2\}.$$

Thus $1 - 8C/r \leq \lambda$ and $\mu \leq 1 + 4C/r$. Retaining the signed two-sided comparison and $0 \leq S_n \leq 4C$ gives

$$S_n - D_n(v) \leq 8CS_n/r \leq 32C^2/r,$$
$$D_n(v) - S_n \leq 4CS_n/r \leq 16C^2/r.$$

The asserted absolute bound concerns this isolated error after the complete signed transport has been compared. □

Theorem 21.12 (A probe-independent even distortion level)**.** *Fix a permutation σ. Suppose that one positive probe sequence (t_n) has eventually bounded normalized distortion: $D_n(t_n) \leq C$ for all sufficiently large n, with one fixed $C \geq 0$. Then there is an integer $K \geq 0$, independent of subsequent probes, such that*

$$\limsup_{n\to\infty} D_n(v_n) = 2K$$

for every positive sequence (v_n). More strongly, for every $\varepsilon > 0$ one cutoff gives $D_n(v) \leq 2K + \varepsilon$ for all later n and all $v > 0$; arbitrarily late cutoffs satisfy $D_n(v) \geq 2K - \varepsilon$ for all $v > 0$ *simultaneously. The level is zero exactly for a finite permutation. Consequently a law that is* not a finite permutation *has a positive even level whenever one probe has eventually bounded cost.*

Proof. Apply the lemma at every sufficiently large cutoff using the original probe t_n. Since $p_n \to \infty$, it gives a single eventual bound $0 \le S_n \le 4C$ and an error tending to zero uniformly in every new positive probe. This step only uses $p_n \ge n$, not a quantitative prime number theorem.

Once $p_n - C > 2$, every entering and leaving prime is odd. The two transported sets have equal cardinalities, so their sums have the same parity. Hence S_n is a nonnegative even integer. Only finitely many such values lie in $[0, 4C]$. At least one occurs arbitrarily late; let $2K$ be the largest value that does. Every larger value then disappears after one cutoff, so $S_n \le 2K$ eventually, whereas $S_n = 2K$ at arbitrarily late cutoffs. The error estimate in the lemma gives both uniform assertions, with the same cutoffs for all positive probes, and therefore every displayed limsup.

A finite permutation has zero defect at every sufficiently late prefix, so its level is zero. Conversely, level zero and nonnegativity give $D_n(1) \to 0$. The maintained adaptive cutoff inverse then forces the permutation to be finite. □

The localization theorem is maintained in `Helpers/PrimeProfileCostLocalization.lean`. The complete cross-probe comparison, nonnegative transport, budget $S_n \le 4C$ and explicit error $32C^2/p_n$ are maintained in `Helpers/PrimeProfileTransportComparison.lean`. The even-lattice and uniform-tail owners are `Helpers/PrimeProfileTransportParity.lean` and `Helpers/PrimeProfileTransportTail.lean`. The quantified uniform even-level theorem is maintained in `Helpers/PrimeProfileQuantization.lean`. Its single cutoff is selected before every new positive probe, and its lower witnesses are shared by all positive probes. The limsup wording and zero-level characterization above are mathematical consequences of that compiled formulation and the previously maintained cutoff inverse; they are not advertised as separate named Lean declarations. The generic finite reference transport and parity lemmas are maintained in `Mathlib/Combinatorics/FinitePrefixDisplacement.lean`; the bounded natural-sequence recurrence lemma is maintained in `Mathlib/Order/BoundedNatSequence.lean`. No independent arithmetic bound on an RH detector is inferred from this profile classification.

Theorem 21.13 (Complete shared-profile distortion alternatives)**.** *Let $a_j \in [0,1]$, $\sum_j a_j^2 < \infty$, and suppose its entire canonical profile equals the*

reciprocal-prime profile. Write

$$D_{a,n}(t) = \frac{p_n^2(p_n + t)}{t^2} \sum_{j=0}^{n} \{\log(1 + ta_j) - ta_j - \log(1 + t/p_j) + t/p_j\}.$$

Exactly one of the following alternatives holds:

1. *There is an integer $K \geq 0$ such that, for every $\varepsilon > 0$, one cutoff gives $D_{a,n}(t) \leq 2K + \varepsilon$ for all later n and every $t > 0$, while arbitrarily late cutoffs satisfy $D_{a,n}(t) \geq 2K - \varepsilon$ for every $t > 0$ simultaneously.*

2. *For every real C and every N, some $n \geq N$ satisfies $D_{a,n}(t) > C$ for every $t > 0$ simultaneously.*

Thus every positive adaptive probe sequence has the same upper limit, belonging to $\{0, 2, 4, \ldots\} \cup \{+\infty\}$. The upper limit is zero exactly when a is a finite coordinate permutation of the reciprocal-prime law. If even one coordinate of a is zero, the stronger conclusion

$$\forall C \in \mathbb{R}\ \exists M\ \forall n \geq M\ \forall t > 0 : \quad D_{a,n}(t) > C$$

holds.

Proof. First suppose that all coordinates are positive. The maintained entire-profile reconstruction gives one permutation σ with $a_j = 1/p_{\sigma(j)}$. Substituting this equality into the complete signed prefix sum gives $D_{a,n}(t) = D_n(t)$ exactly.

If the constant unit probe $D_n(1)$ is eventually bounded, the preceding theorem gives the first alternative. Otherwise, fix $C \in \mathbb{R}$ and N, and put $A = \max(C, 0)$. There are arbitrarily late cutoffs with $D_n(1) > 12A$; choose one beyond N with $p_n > 4A$. If any positive probe v had $D_n(v) \leq A$, the localization lemma, applied with v as the original probe, would give

$$D_n(1) \leq S_n + \frac{32A^2}{p_n} \leq 4A + 8A = 12A,$$

a contradiction. Thus every positive probe exceeds $A \geq C$ at this same cutoff. This proves the second alternative without selecting different cutoffs for different probes.

If $a_i = 0$, the maintained zero-hole estimate instead gives $D_{a,n}(t) \geq p_n/4$ for every $n \geq i$ and every $t > 0$. Since $p_n \geq n$, it gives the asserted eventual uniform divergence. Nonnegativity of the coordinates makes these two cases exhaustive.

The alternatives are disjoint: in the first, use $\varepsilon = 1$; in the second, use $C = 2K + 2$ and a cutoff beyond that upper envelope. Their quantified bounds give the stated common upper limit. A finite permutation has eventual zero defect. Conversely, the zero level is incompatible with a zero coordinate, so reconstruction reduces it to the positive permutation case, where the preceding cutoff inverse applies. In particular, a law that is *not a finite permutation* has a positive even level or infinite upper limit. □

The quantitative implication $D_n(v) \le C \Rightarrow D_n(1) \le 12C$ for $p_n > 4C$, the shared unbounded witnesses, and the permutation dichotomy are compiled in `Helpers/PrimeProfileUniformDichotomy.lean`. The general-law dichotomy and eventual uniform zero-hole divergence are compiled in `Assembly/PrimeProfileDistortionClassification.lean`. The exact coordinate-substitution identity is maintained in `Assembly/PrimeProfileCutoffInverse.lean`. The formal alternatives use the displayed quantified upper and lower envelopes; their upper-limit wording, exclusivity and zero-level characterization are mathematical consequences, not additional named Lean declarations. The prime 2 is retained in every finite prefix; evenness is used only after the localization window excludes it. No new prime-pattern infinitude or improvement of the certified 7/8 RH boundary follows from this classification.

Theorem 21.14 (Exact finite-accuracy transport readout)**.** *Fix a prime-coordinate permutation σ and a cutoff with $r = p_n \ge 3$. Suppose $C \ge 0$, $D_n(t) \le C$ for one $t > 0$, and $32C^2 < r$. Then there is one integer $K \ge 0$ such that*

$$S_n = 2K, \qquad |D_n(v) - 2K| < 1 \quad \textit{for every } v > 0.$$

For each positive probe v separately, K is the unique nonnegative integer j with $|D_n(v) - 2j| < 1$.

Proof. The precision hypothesis implies $C < r/4$: otherwise $32C^2 \ge 2r^2 \ge 6r > r$. Hence $r - C > 3r/4 \ge 9/4 > 2$. Localization excludes the exceptional prime 2 from the transported sets, so their exact signed sum is $S_n = 2K$ for some $K \ge 0$. The uniform error estimate gives $|D_n(v) - 2K| \le 32C^2/r < 1$ for every $v > 0$. Two distinct even integers have distance at least 2; they cannot both lie at distance strictly less than 1 from the same observation. This proves uniqueness for each probe, with the same recovered value for every probe. □

The theorem has no global boundedness assumption. For example, suppose one positive adaptive probe has eventual cost at most $C_n \geq 0$, where $C_n^2/p_n \to 0$. Then, at every sufficiently late cutoff, *every* positive probe determines the exact moving transport S_n by its unique nearest even integer. Moreover the uniform readout error is at most $32C_n^2/p_n \to 0$. The levels may grow with n; no constant distortion level is assumed. The finite theorem, including per-probe uniqueness, is compiled in `Helpers/PrimeProfileTransportReadout.lean`. The growing-budget statement is a mathematical consequence of that finite certificate and the maintained uniform error estimate, not a separately named limit declaration.

Theorem 21.15 (Pairwise recovery and discrete separation)**.** *Let σ, τ be two permutations of the prime coordinates. Suppose one positive adaptive probe sequence (t_n) and one $C \geq 0$ satisfy*

$$D_n^\sigma(t_n) \leq C, \qquad D_n^\tau(t_n) \leq C$$

for all sufficiently large n. Then

$$|D_n^\sigma(t_n) - D_n^\tau(t_n)| \longrightarrow 0 \quad \Longleftrightarrow \quad \sigma(n) = \tau(n) \text{ for all sufficiently large } n.$$

More quantitatively, if the two permutations do not agree eventually, then for every $\varepsilon > 0$ and every N there is $n \geq N$ such that

$$|D_n^\sigma(v) - D_n^\tau(w)| \geq 2 - \varepsilon \qquad \text{for every } v, w > 0$$

simultaneously. The cutoff is shared by all pairs of probes. The hypothesis of nonagreement excludes pairs differing only finitely; in particular, at least one permutation is not a finite permutation*.*

Proof. At a cutoff with $p_n > 4C$ and $p_n - C > 2$, let $S_n^\sigma = 2K$ and $S_n^\tau = 2L$ be the exact signed transports. For any two positive probes, the two previously proved error bounds give

$$|S_n^\sigma - S_n^\tau| \leq |D_n^\sigma(v) - D_n^\tau(w)| + \frac{64C^2}{p_n}.$$

Consequently

$$|D_n^\sigma(v) - D_n^\tau(w)| + \frac{64C^2}{p_n} < 2 \quad \Longrightarrow \quad S_n^\sigma = S_n^\tau,$$

because distinct even integers have distance at least 2.

Exact consecutive-prefix subtraction gives

$$S^{\sigma}_{n+1} - S^{\sigma}_{n} = p_{\sigma(n+1)} - p_{n+1},$$

and the same identity holds for τ. Thus eventual equality of the signed transport sequences forces eventual equality of the prime coordinates, hence of their indices. If σ and τ do not agree eventually, the transports therefore differ at arbitrarily late cutoffs. Choose one beyond N, both cost cutoffs, and a threshold where

$$p_n > \max(4C,\, 2 + C,\, 64C^2/\varepsilon).$$

Its two even transports differ by at least 2. The displayed error comparison then proves the claimed $2 - \varepsilon$ separation for every pair of positive probes at this one cutoff.

In particular, asymptotically equal observations are incompatible with eventual nonagreement: use separation with $\varepsilon = 1$ and the original probes. Conversely, if the permutations agree past a cutoff, their images of every later finite prefix are identical. Indeed those images have the same complement, namely the common image of the remaining tail. All weighted prefix sums, and therefore the normalized defects at each common probe, then agree exactly. □

The finite comparison, shared separation witnesses, and pairwise inverse are compiled in `Helpers/PrimeProfilePairwiseRigidity.lean`. The formal inverse states convergence by its explicit ε–cutoff condition. The reusable consecutive-defect identity, two-prefix coordinate recovery, equal-tail prefix-image identity and eventual exact-defect equivalence are compiled in `Mathlib/Combinatorics/FinitePrefixDisplacement.lean`. This recovery compares two possibly infinite rearrangements; it does not assume either is the reference order. It identifies their tails even though their entire canonical profiles are equal. The separation constant 2 is proved, with no claim here that it is attained by two distinct classes.

21.7 Two-moment compensation and a larger exact-readout range

The preceding readout uses one logarithmic probe. Two finite probability moments cancel its first nonlinear transport error. Write $I_n = \{0, \dots, n\}$, $A_n = \sigma(I_n)$, $r = p_n$, and

$$E_n = A_n \setminus I_n, \qquad L_n = I_n \setminus A_n.$$

Both sets are finite and have equal cardinality. Define

$$\Delta_j = \sum_{i=0}^{n} \left(p_i^{-j} - p_{\sigma(i)}^{-j} \right), \qquad Q_n = 3r^2\Delta_1 - r^3\Delta_2.$$

Thus Q_n uses only the first two finite power sums of the actual Bernoulli probabilities and their reference probabilities. These are finite-prefix observations; equality of the entire canonical profiles does not assert equality of these labelled finite sums.

Theorem 21.16 (Cubic compensated transport and exact readout)**.** *Suppose $D_n(t) \le C$ for one $t > 0$, where $C \ge 0$ and $C < r/4$. Then the full signed transport satisfies*

$$0 \le S_n - Q_n \le \frac{64C^3}{r^2}.$$

If instead one assumes $r \ge 3$ and $32C^3 < r^2$, the condition $C < r/4$ follows automatically, and $S_n = 2K$ is the unique nonnegative even integer in $[Q_n, Q_n + 2)$. In particular, K is determined by the compensated observation without knowing S_n.

Proof. Set

$$\Phi_r(p) = -\frac{3r^2}{p} + \frac{r^3}{p^2}.$$

For positive p, r, direct algebra gives the exact identity

$$\Phi_r(p) - \Phi_r(r) = (p - r) - \frac{(p - r)^3}{p^2}.$$

By the definition of the finite moments, $Q_n = \sum_{i=0}^{n} \Phi_r(p_{\sigma(i)}) - \sum_{i=0}^{n} \Phi_r(p_i)$. Cancel the common coordinates and insert the common reference $\Phi_r(r)$. Equal cardinalities cancel its constant contribution. The analogous reference decomposition of S_n therefore gives

$$S_n - Q_n = \sum_{i \in E_n} \frac{(p_i - r)^3}{p_i^2} + \sum_{i \in L_n} \frac{(r - p_i)^3}{p_i^2}.$$

This retains the compensation and the unit coefficient of S_n; both displayed remainders are nonnegative.

The proved one-probe localization gives $0 \le p_i - r \le 4C$ for $i \in E_n$ and $0 \le r - p_i \le C$ for $i \in L_n$. On the entering set, $p_i \ge r$. On the leaving set,

$p_i \geq r - C > 3r/4$. Hence, on either set, the displacement divided by p_i is at most $4C/r$. It follows that

$$\begin{aligned} S_n - Q_n &\leq \frac{16C^2}{r^2}\left(\sum_{i\in E_n}(p_i - r) + \sum_{i\in L_n}(r - p_i)\right) \\ &= \frac{16C^2}{r^2}S_n \leq \frac{64C^3}{r^2}, \end{aligned}$$

using the maintained bound $S_n \leq 4C$.

For the readout statement, if $C \geq r/4$ then $32C^3 \geq r^3/2 \geq 3r^2/2 > r^2$, a contradiction. Thus $r - C > 3r/4 \geq 9/4 > 2$. Localization excludes the exceptional prime 2 from both transported sets, but retains it in the original prefixes. Their equal cardinalities force $S_n = 2K$, $K \geq 0$. The error bound and the precision hypothesis give $0 \leq 2K - Q_n < 2$. An interval of the form $[Q_n, Q_n + 2)$ cannot contain two distinct even integers. This proves the asserted recovery. □

For growing budgets, $C_n^3/p_n^2 \to 0$ gives eventual exact recovery with error tending to zero. In particular, the sufficient scale is now $C_n = o(p_n^{2/3})$, compared with $o(p_n^{1/2})$ for the preceding single-probe readout. This is a stronger observation theorem. It supplies no new arithmetic upper bound for the probe cost itself, so it changes no RH boundary.

The exact scalar identity and signed entering/leaving comparisons are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CubicReciprocalTransport.lean`. The two-moment identity, complete signed comparison, cubic error and unique finite even readout are compiled in `Helpers/PrimeProfileCubicReadout.lean`. The growing-budget statement is a mathematical consequence of the finite certificates, rather than a separately named limit declaration. The coefficient cancellation is elementary; no literature-priority claim is made for this quantitative application to prime-profile transport.

21.8 Geometric compensation and stable logarithmic-moment readout

For an integer $m \geq 0$, define the centered reciprocal polynomial

$$\Psi_{m,r}(p) = r\sum_{j=1}^{m}(1-r/p)^j, \qquad P_{m,n} = \sum_{i=0}^{n}\big(\Psi_{m,r}(p_{\sigma(i)}) - \Psi_{m,r}(p_i)\big), \quad r = p_n.$$

The empty sum is zero. Equivalently,

$$P_{m,n} = r\sum_{j=1}^{m} M_{j,n}, \qquad M_{j,n} = \sum_{i=0}^{n}\big((1-r/p_{\sigma(i)})^j - (1-r/p_i)^j\big).$$

These are finite centered power-sum differences of the probabilities. Binomial expansion expresses $M_{j,n}$ using their ordinary power sums of degrees at most j; the constant terms cancel. For $m = 2$, the polynomial is $\Phi_r(p) + 2r$, so $P_{2,n} = Q_n$.

Theorem 21.17 (All even orders preserve the exact transport)**.** *Suppose one positive probe satisfies $D_n(t) \le C$, with $0 \le C < r/4$. For every even integer $m = 2k \ge 0$,*

$$0 \le S_n - P_{m,n} \le 4C(4C/r)^m.$$

If $r \ge 3$ and $4C(4C/r)^m < 2$, the exact transport $S_n = 2K$ is the unique nonnegative even integer in $[P_{m,n}, P_{m,n} + 2)$.

Proof. The finite geometric identity gives, for every $p > 0$,

$$\Psi_{m,r}(p) = (p-r) - \frac{(p-r)^{m+1}}{p^m}, \qquad \Psi_{m,r}(r) = 0.$$

One may verify the identity by induction on m; the successor adds $r(1 - r/p)^{m+1}$ and cancels the preceding remainder. After the same common-coordinate cancellation and equal-cardinality reference decomposition as before, evenness of m gives exactly

$$S_n - P_{m,n} = \sum_{i\in E_n} \frac{(p_i - r)^{m+1}}{p_i^m} + \sum_{i\in L_n} \frac{(r-p_i)^{m+1}}{p_i^m}.$$

Every summand is nonnegative. The proved localization implies that the displacement divided by p_i is at most $4C/r$ on both sets. Each summand is therefore at most $(4C/r)^m$ times its displacement. Their displacement sum is $S_n \le 4C$, proving the bound. This argument includes $m = 0$: both sides reduce to the existing transport bound. For the readout statement, $r - C > 2$ excludes the exceptional prime from the transported sets, and the error interval of width less than two identifies the even integer uniquely. □

Theorem 21.18 (Stable recovery from logarithmically many centered moments)**.** *Suppose $r = p_n \ge 3$ and one positive probe satisfies $D_n(t) \le r/8$. Put*

$$k = \lfloor \log_4 r \rfloor + 1, \qquad m = 2k.$$

There is an integer $K \geq 0$ with $S_n = 2K$ such that, simultaneously for every real observation y,

$$|y - P_{m,n}| < \tfrac{1}{2} \quad \Longrightarrow \quad |y - 2K| < 1.$$

For each such y, K is the unique nonnegative integer j with $|y - 2j| < 1$. Thus nearest-even rounding recovers the exact transport despite an observation error of almost one half.

Proof. Since $r < 4^k$, the preceding signed estimate with $C = r/8$ gives

$$0 \leq S_n - P_{m,n} \leq \frac{r}{2}(1/2)^{2k} = \frac{r}{2 \cdot 4^k} < \frac{1}{2}.$$

The localization window satisfies $r - C = 7r/8 \geq 21/8 > 2$, so $S_n = 2K$ for some $K \geq 0$. If $|y - P_{m,n}| < 1/2$, the retained one-sided error gives $-1 < y - 2K < 1/2$, hence $|y - 2K| < 1$. Two distinct even integers cannot both be at distance less than one from y, proving uniqueness with the same K for every observation. □

The number of centered moments is at most $\log_2 r + 2$. The noiseless readout needs only $m = 2\lfloor \log_4 r \rfloor$, using the interval of width two; the two extra moments above leave a fixed observation-error allowance. For example, if each centered difference $M_{j,n}$ is supplied with absolute error at most $1/(4rm)$, the combined observation $y = r \sum_{j=1}^{m} \widehat{M}_{j,n}$ has error at most $1/4$, so the stable readout applies. This last statement follows by the finite sum inequality. Its precision concerns the centered differences; expanding them into raw powers requires retaining the binomial coefficient amplification.

The finite geometric identity, reference value and arbitrary-order remainder are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalMomentCore.lean`. The signed coordinate comparisons are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalMomentTransport.lean`. The complete signed estimate, finite readout, explicit degree 2 $\mathrm{Nat.log}_4(p_n)$ and uniform noisy readout with degree $2(\mathrm{Nat.log}_4(p_n) + 1)$ are compiled in `Helpers/PrimeProfileMomentReadout.lean`. Here the natural-number logarithm is the integer base-four logarithm. The real-logarithm moment-count inequality and the per-moment measurement-error consequence are ordinary mathematical consequences, not separate named Lean declarations. The error theorem retains $r = 2$; the stated even-integer

recovery certificates explicitly require $r \geq 3$. The construction improves finite observation accuracy under an assumed arithmetic cost bound, and leaves the certified RH boundary unchanged.

21.9 Uniform recovery on the full localization window

The actual coordinate denominators sharpen the preceding geometric estimate. No separation from the endpoint $C = r/4$ is needed.

Theorem 21.19 (Full-window signed accuracy and stable readout)**.** *Suppose one positive probe satisfies $D_n(t) \leq C$, where $0 \leq C < r/4$ and $r = p_n$. For every even order $m = 2k \geq 0$,*

$$0 \leq S_n - P_{m,n} \leq 2^{-m} S_n \leq 4C\, 2^{-m}.$$

If $r \geq 3$, put

$$k = \lfloor \log_4(2r) \rfloor + 1, \qquad m = 2k.$$

Then there is one integer $K \geq 0$, with $S_n = 2K$, such that every observation y satisfying $|y - P_{m,n}| < 1/2$ has $2K$ as its unique nonnegative even integer at distance less than one. The degree is independent of C throughout $0 \leq C < r/4$.

Proof. For an entering coordinate q, localization gives $q - r \leq 4C < r$, hence $q < 2r$ and

$$\frac{q-r}{q} < \frac{1}{2}.$$

For a leaving coordinate p, it gives $r - p \leq C < r/4$, hence $p > 3r/4$ and

$$\frac{r-p}{p} < \frac{1}{3} < \frac{1}{2}.$$

The exact even-order remainder from the preceding theorem therefore satisfies, with its two signs retained,

$$\begin{aligned} S_n - P_{m,n} &= \sum_{i \in E_n} (p_i - r) \left(\frac{p_i - r}{p_i} \right)^m + \sum_{i \in L_n} (r - p_i) \left(\frac{r - p_i}{p_i} \right)^m \\ &\leq 2^{-m} \left(\sum_{i \in E_n} (p_i - r) + \sum_{i \in L_n} (r - p_i) \right) = 2^{-m} S_n. \end{aligned}$$

The maintained bound $S_n \leq 4C$ completes the uniform error estimate, including $m = 0$.

For the chosen degree, $2r < 4^k$, so

$$0 \le S_n - P_{m,n} \le \frac{4C}{4^k} < \frac{r}{4^k} < \frac{1}{2}.$$

Since $r - C > 2$, the same exceptional-prime argument gives $S_n = 2K$. Combining the retained one-sided error with $|y - P_{m,n}| < 1/2$ yields $-1 < y - 2K < 1/2$. Thus $|y - 2K| < 1$, and distinct even integers again prove uniqueness with the same K before all possible observations are chosen. □

The degree is at most $\log_2 r + 3$. This covers every budget in the full localization window, including budgets approaching its upper endpoint, with no deterioration of the moment count or the observation noise allowance. The previously stated per-centered-moment accuracy condition applies with this new value of m.

The arbitrary relative-ratio remainder and the two half-window coordinate comparisons are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalMomentWindow.lean`. The complete signed sandwich, retained $2^{-m}S_n$ error, and uniform noisy readout are compiled in `Helpers/PrimeProfileFullWindowReadout.lean`. The formal degree uses $\mathrm{Nat.}\log_4(2p_n)$; the real-logarithm upper bound on the degree is an ordinary consequence. As before, the error estimate includes the first prime and the even-integer readout explicitly assumes $p_n \ge 3$. This is a stronger observation certificate under the same localization hypothesis; it supplies no new arithmetic estimate of that hypothesis.

21.10 Polynomial square filters preserve the full transport

The geometric polynomial is one choice in a larger class of finite probability observables. This gives freedom to suppress the complete remainder while retaining the principal transport coefficient exactly.

Theorem 21.20 (Arbitrary normalized polynomial filters)**.** *Let $H \in \mathbb{R}[X]$ satisfy $H(0) = 1$, and suppose*

$$|H(x)| \le \eta \qquad (1/2 \le x \le 4/3).$$

There is a polynomial G with $\deg G \le 2 \deg H$ such that, for every $p > 0$,

$$\Phi_{H,r}(p) := (p - r)\big(1 - H(r/p)^2\big) = rG(r/p).$$

Define the complete signed observable

$$A_{H,n} = \sum_{i=0}^{n} \Phi_{H,r}(p_{e(i)}) - \sum_{i=0}^{n} \Phi_{H,r}(p_i), \qquad r = p_n.$$

Under the same hypothesis $D_n(t) \le C$ at one positive probe, with $0 \le C < r/4$,

$$0 \le S_n - A_{H,n} \le \eta^2 S_n \le 4C\eta^2.$$

Thus $A_{H,n}$ is a finite linear combination of probability moments of orders at most $2 \deg H$, with principal coefficient one.

Proof. Put $R = 1 - H^2$. Since $R(0) = 0$, write $R = XQ$ and set $G = (1 - X)Q$. Then $\deg G \le 2 \deg H$; the zero polynomial case is immediate. For $p \neq 0$,

$$rG(r/p) = r(1 - r/p)Q(r/p) = (p - r)R(r/p) = \Phi_{H,r}(p).$$

The constant coefficient of G cancels between the two equal-size prefixes, so this is genuinely a finite probability-moment observable.

Cancel the common coordinates before estimating anything. On E_n, localization gives $r \le p_i < 2r$; on L_n, it gives $3r/4 < p_i \le r$. Consequently $r/p_i \in [1/2, 4/3]$ on both retained supports. No such assertion is needed for a common small-prime coordinate. Equal cardinalities give the exact signed identity

$$\begin{aligned} S_n - A_{H,n} = &\sum_{i \in E_n} (p_i - r) H(r/p_i)^2 \\ &+ \sum_{i \in L_n} (r - p_i) H(r/p_i)^2. \end{aligned}$$

Every displayed displacement is nonnegative. Applying the filter bound to this complete remainder gives

$$0 \le S_n - A_{H,n} \le \eta^2 \left(\sum_{i \in E_n} (p_i - r) + \sum_{i \in L_n} (r - p_i) \right) = \eta^2 S_n.$$

Finally use the already proved $S_n \le 4C$. The principal term has not been discarded or bounded separately. □

The polynomial realization and degree bound are compiled in
`Mathlib/Analysis/SpecialFunctions/Log/`

`PolynomialReciprocalTransport.lean`. The actual profile polynomial representation, complete signed comparison and retained $\eta^2 S_n$ error are compiled in `Helpers/PrimeProfileSquareFilter.lean`. These statements allow the identity profile and finite permutations; the recovered transport may be zero.

21.11 An accelerated quadratic filter and fewer moments

Consider the explicit polynomial

$$B(x) = 1 - \frac{528}{217}x + \frac{288}{217}x^2 = \frac{2(12x-11)^2 - 25}{217}.$$

It satisfies $B(0) = 1$. This is the normalized shifted quadratic Chebyshev polynomial; the construction below uses its explicit formula.

Theorem 21.21 (Accelerated stable moment readout)**.** *Set $H_k = B^k$ and $A_{k,n} = A_{H_k,n}$, for each integer $k \geq 0$. Under $D_n(t) \leq C$ at one positive probe, $0 \leq C < r/4$, the observable uses probability moments of degree at most $4k$, and*

$$0 \leq S_n - A_{k,n} \leq \left(\frac{25}{217}\right)^{2k} S_n \leq 64^{-k} S_n \leq 4C\, 64^{-k}.$$

If $r \geq 3$, choose

$$k = \lfloor \log_{64}(2r) \rfloor + 1.$$

There is a single $K \geq 0$ with $S_n = 2K$ such that every observation y with $|y - A_{k,n}| < 1/2$ satisfies $|y - 2K| < 1$, and no other nonnegative even integer has distance less than one from y.

Proof. For $1/2 \leq x \leq 4/3$, the quantity $12x - 11$ lies in $[-5, 5]$. Hence

$$-25 \leq 2(12x-11)^2 - 25 \leq 25, \qquad |B(x)| \leq \frac{25}{217}.$$

It follows that $H_k(0) = 1$, $\deg H_k \leq 2k$, and $|H_k(x)| \leq (25/217)^k$ throughout the retained window. The preceding theorem therefore gives the exact-rate error. The rational inequality $(25/217)^2 < 1/64$ gives the stated convenient geometric bound, also valid at $k = 0$.

For the specified k, we have $2r < 64^k$, and consequently

$$0 \leq S_n - A_{k,n} \leq \frac{4C}{64^k} < \frac{r}{64^k} < \frac{1}{2}.$$

The localization inequality $r - C > 2$ excludes the prime 2 from the transported coordinates. Equal entering and leaving cardinalities then imply $S_n = 2K$ for some $K \geq 0$, as in the earlier parity proof. Combining the one-sided error with $|y - A_{k,n}| < 1/2$ gives $-1 < y - 2K < 1/2$. Distinct even integers are separated by two, which proves uniqueness uniformly over all observations. □

At the same degree $m = 4k$, the previous geometric certificate gave $2^{-m}S_n = 16^{-k}S_n$; the quadratic filter gives $64^{-k}S_n$. Thus the certified remainder is smaller by a factor 4^{-k}. The explicit degree is at most

$$4\big(\lfloor \log_{64}(2r) \rfloor + 1\big) \leq \frac{2}{3}\log_2 r + \frac{14}{3},$$

compared with the earlier $\log_2 r + 3$ bound. The leading coefficient in the logarithmic moment count is reduced by one third. This real-logarithm comparison is an ordinary consequence of the compiled integer-logarithm certificate; it is not a separate Lean theorem.

The fixed noise allowance concerns the combined observable. It does not assert that all raw probability moments can be measured with the same accuracy. If $G(x) = \sum_{j=0}^{m} a_j x^j$ and the measured raw moment differences have errors at most ϵ_j, their induced observable error is bounded by

$$r \sum_{j=1}^{m} |a_j| r^j \epsilon_j.$$

Requiring this quantity to be less than $1/2$ suffices. The constant moment cancels exactly.

The explicit quadratic normalization, degree and all-power window bounds are compiled in `Mathlib/Analysis/SpecialFunctions/Log/QuadraticReciprocalFilter.lean`. The degree-$4k$ profile representation, full $64^{-k}S_n$ comparison, and explicit noisy even-integer recovery are compiled in `Helpers/PrimeProfileQuadraticFilterReadout.lean`. The sharper $(25/217)^{2k}$ error is a consequence of the compiled general filter theorem and compiled norm bound. The error statements retain the first prime; the even-integer readout assumes $p_n \geq 3$ explicitly. No minimax optimality is asserted. This improves observation accuracy under the same arithmetic cost hypothesis. It does not establish a stronger bound on that cost or change the certified RH boundary.

21.12 All-order filters and a halved logarithmic moment count

The quadratic can be replaced by a polynomial of every order. The required uniform bound and nonzero denominator follow from an elementary coupled recurrence; no trigonometric continuation is needed.

Define polynomials T_k, U_k by

$$T_0 = 1, \quad U_0 = 0, \qquad \begin{cases} T_{k+1} = XT_k - (1 - X^2)U_k, \\ U_{k+1} = T_k + XU_k. \end{cases}$$

Here U_k is the shifted second-kind coordinate, so the initial zero polynomial is part of the definition. Induction gives $\deg T_k \le k$ and $\deg U_k \le k - 1$ for $k \ge 1$, with $U_0 = 0$ handled separately.

Lemma 21.22 (Unit-window invariant and positive normalizer)**.** *For every $k \ge 0$ and real x,*

$$T_k(x)^2 + (1 - x^2)U_k(x)^2 = 1.$$

Consequently $|T_k(x)| \le 1$ on $[-1, 1]$. At the exterior point $11/5$, the normalizer $a_k = T_k(11/5)$ satisfies

$$a_k > 0, \qquad 4^k \le 2a_k.$$

Proof. Writing $d = 1 - x^2$, the recursion preserves the quadratic form because

$$(xT - dU)^2 + d(T + xU)^2 = T^2 + dU^2.$$

The initial value is one. For $|x| \le 1$ its second term is nonnegative, so $T_k(x)^2 \le 1$, including both endpoints.

For the exterior estimate, put $b_k = U_k(11/5)$. The exact recursion is

$$a_{k+1} = \frac{11}{5}a_k + \frac{96}{25}b_k, \qquad b_{k+1} = a_k + \frac{11}{5}b_k.$$

We retain the coupled inequalities

$$b_k \ge 0, \qquad a_k \ge \frac{9}{5}b_k, \qquad V_k := a_k + \frac{9}{5}b_k \ge 4^k.$$

They hold at $k = 0$. Assuming them at k, the next b is nonnegative, and

$$a_{k+1} - \frac{9}{5}b_{k+1} = \frac{2}{5}a_k - \frac{3}{25}b_k \ge \frac{3}{5}b_k \ge 0.$$

Moreover

$$V_{k+1} = 4V_k + \frac{3}{5}b_k \geq 4^{k+1}.$$

Thus all three inequalities hold by induction. Since $V_k \leq 2a_k$, we obtain $4^k \leq 2a_k$ and $a_k > 0$. The normalizing denominator has therefore been controlled before division. □

Define

$$H_k(x) = \frac{T_k((11-12x)/5)}{a_k}, \qquad \mathcal{A}_{k,n} = A_{H_k,n}.$$

Then $H_k(0) = 1$, $\deg H_k \leq k$, and the affine map sends $[1/2, 4/3]$ into $[-1, 1]$. The preceding lemma gives

$$|H_k(x)| \leq \frac{1}{a_k} \leq 2\,4^{-k} \qquad (1/2 \leq x \leq 4/3).$$

Theorem 21.23 (All-order stable readout)**.** *Under the same one-positive-probe hypothesis $D_n(t) \leq C$, where $0 \leq C < r/4$ and $r = p_n$, the observable $\mathcal{A}_{k,n}$ uses probability moments of degree at most $2k$, and*

$$0 \leq S_n - \mathcal{A}_{k,n} \leq 4\,16^{-k} S_n \leq 16C\,16^{-k} \qquad (k \geq 0).$$

If $r \geq 3$, choose

$$k = \lfloor \log_{16}(8r) \rfloor + 1.$$

There is a single $K \geq 0$ with $S_n = 2K$ such that every observation y satisfying $|y - \mathcal{A}_{k,n}| < 1/2$ has $2K$ as its unique nonnegative even integer at distance less than one.

Proof. Apply the complete signed square-filter theorem with $\eta = 2\,4^{-k}$. The principal coefficient remains one, and the actual error is still

$$\sum_{i \in E_n} (p_i - r) H_k(r/p_i)^2 + \sum_{i \in L_n} (r - p_i) H_k(r/p_i)^2.$$

The squared norm is $4\,16^{-k}$, giving the stated error bound. The polynomial representation has degree at most $2 \deg H_k \leq 2k$.

For the displayed choice of k, $8r < 16^k$. Hence

$$0 \leq S_n - \mathcal{A}_{k,n} \leq \frac{16C}{16^k} < \frac{4r}{16^k} < \frac{1}{2}.$$

As before $r - C > 2$, so the retained entering and leaving primes are odd and equal cardinalities imply $S_n = 2K$ with $K \geq 0$. The combined error bounds give $-1 < y - 2K < 1/2$. Separation of even integers proves uniqueness for every observation with this same K. □

The explicit degree now satisfies

$$m \le 2\big(\lfloor \log_{16}(8r) \rfloor + 1\big) \le \frac{1}{2}\log_2 r + \frac{7}{2}.$$

Thus its leading logarithmic coefficient is half the original full-window geometric certificate's coefficient. At a matched even degree $m = 2k$, the new complete error bound is $4\,4^{-m}S_n$, compared with $2^{-m}S_n$ previously; it is strictly smaller for every $m > 2$. These real-logarithm and matched-degree comparisons are ordinary consequences of the compiled formulas. The observation-noise allowance remains a condition on the combined observable. The coefficient-weighted raw-moment accuracy requirement from the preceding subsection still applies.

The recurrence, degree bounds, quadratic invariant, unit-window norm and rational growth packet are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalChebyshevCore.lean`. The positive-denominator normalization, affine degree bound and uniform filter norm are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalChebyshevFilter.lean`. The actual probability-moment representation, complete signed error and explicit integer-logarithm noisy recovery are compiled in `Helpers/PrimeProfileChebyshevReadout.lean`. All readout certificates are exported together by `Assembly/PrimeProfileReadouts.lean`. The error theorem includes $p_n = 2$; even-integer recovery assumes $p_n \ge 3$. Identity profiles and finite permutations remain allowed. No minimax optimality or external-priority claim is made. The gain concerns finite observation accuracy under an assumed arithmetic cost bound, and supplies no new RH boundary.

21.13 Exact spectral error and cost-adaptive recovery

The coupled argument admits an exact refinement. This keeps both characteristic branches of the normalizer and gives a sufficient recovery condition in terms of the available cost budget.

For a real exterior point $a \ge 1$, choose $0 \le c \le a$ with $c^2 \le a^2 - 1$, and put $\delta = a^2 - 1 - c^2 \ge 0$. Writing $A = T_k(a)$ and $B = U_k(a)$, the exact step identities are

$$T_{k+1}(a) - cU_{k+1}(a) = (a - c)(A - cB) + \delta B,$$
$$T_{k+1}(a) + cU_{k+1}(a) = (a + c)(A + cB) + \delta B.$$

They preserve the cone $B \ge 0$, $A - cB \ge 0$ and imply $(a + c)^k \le A + cB \le 2A$. In particular the exterior normalizer is positive. On the characteristic

boundary $\delta = 0$, both linear forms have exact powers:

$$T_k(a) + cU_k(a) = (a + c)^k, \qquad T_k(a) - cU_k(a) = (a - c)^k.$$

Indeed the displayed step identities multiply the two forms by $a + c$ and $a - c$, respectively, and both start at one.

Take $a = 11/5$ and $c = 4\sqrt{6}/5$, and define

$$\lambda = \frac{11 + 4\sqrt{6}}{5}, \qquad \mu = \frac{11 - 4\sqrt{6}}{5}, \qquad W_k = \lambda^k + \mu^k.$$

Here $\lambda > 4$, $\mu > 0$, and $\lambda\mu = 1$, so

$$W_k = \lambda^k + \lambda^{-k} = 2T_k(11/5) > 0.$$

These sign and nonvanishing conditions hold for every $k \geq 0$, including $W_0 = 2$. The preceding normalized filter therefore satisfies

$$|H_k(x)| \leq \frac{2}{W_k} \qquad (1/2 \leq x \leq 4/3).$$

Theorem 21.24 (Exact spectral certificate)**.** *For the preceding observable $\mathcal{A}_{k,n}$, suppose $D_n(t) \leq C$ at one positive probe, $0 \leq C < r/4$, $r = p_n$. Then, for every $k \geq 0$,*

$$0 \leq S_n - \mathcal{A}_{k,n} \leq \frac{4S_n}{W_k^2} \leq \frac{16C}{W_k^2}.$$

The degree remains at most $2k$. If $r \geq 3$ and

$$32C < W_k^2,$$

there is a single $K \geq 0$ with $S_n = 2K$ such that every observation y with $|y - \mathcal{A}_{k,n}| < 1/2$ has $2K$ as its unique nonnegative even integer at distance less than one.

Proof. The exact characteristic identities prove the formula for W_k. The numerator of H_k has absolute value at most one on the retained window, so division by its positive normalizer $W_k/2$ gives the stated filter norm. Apply the same complete signed comparison, with its unchanged remainder

$$\sum_{i \in E_n} (p_i - r) H_k(r/p_i)^2 + \sum_{i \in L_n} (r - p_i) H_k(r/p_i)^2.$$

This proves $0 \leq S_n - \mathcal{A}_{k,n} \leq 4S_n/W_k^2$. The maintained estimate $S_n \leq 4C$ gives the last inequality.

Under $32C < W_k^2$, positivity of W_k^2 gives $16C/W_k^2 < 1/2$. The exceptional-prime argument is unchanged: $r - C > 2$, so $S_n = 2K$ for some $K \geq 0$. Combining the one-sided error with the observation noise gives $-1 < y - 2K < 1/2$, and separation of even integers proves uniqueness. The same K works before any observation is chosen. □

This is a condition on C and the chosen order, rather than a requirement to use the earlier worst-case order depending only on r. Since $\mu > 0$ and $\lambda > 4$, we have $W_k > 4^k$ for every $k \geq 0$. Thus the exact error coefficient $4/W_k^2$ is strictly smaller than the previously used $4\,16^{-k}$. Also $W_k^2 \geq \lambda^{2k}$, so a sufficient condition at an even degree $m = 2k$ is

$$\lambda^m > 32C.$$

For $C > 0$, any nonnegative even $m > \log(32C)/\log\lambda$ suffices. Uniformly over $C < r/4$, one can instead choose an even degree with

$$m > \frac{\log(8r)}{\log\lambda}, \qquad m \leq \frac{\log(8r)}{\log\lambda} + 2.$$

Its leading coefficient relative to $\log_2 r$ is $\log 2/\log\lambda < 1/2$. These real-logarithm choices and coefficient comparisons are ordinary consequences of the compiled exact formula and spectral signs. No minimum-degree or minimax assertion is made. The noise condition still concerns the combined observable; raw-moment measurement errors retain their coefficient weights.

The quantified exterior cone and both characteristic identities are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalChebyshevCore.lean`. The exact two-branch denominator, reciprocal-root identities, positivity, and exact filter norm are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalChebyshevFilter.lean`. The full signed spectral bound and the cost-adaptive noisy readout are compiled in `Helpers/PrimeProfileChebyshevReadout.lean`. The error estimate includes the first prime, whereas even-integer recovery explicitly assumes $p_n \geq 3$. Finite permutations and zero transport remain included. This proves a stronger finite observation certificate under the same arithmetic hypotheses, without a new arithmetic cost estimate or an improved RH boundary.

21.14 Nonlinear coercivity and the half-reference window

The preceding localization used a tangent estimate for each entering coordinate. Integrating the exact compensated kernel gives a stronger bound on the whole signed transport. Fix a cutoff $n \geq 0$ and put $r = p_n$, with the prefix containing exactly $n+1$ primes. The probe $t > 0$ is arbitrary and may be chosen independently at each cutoff. Write

$$B_t(x) = \log(1 + t/x) - t/x,$$
$$N_t(r; p, q) = \frac{r^2(r+t)}{t^2}\big(B_t(q) - B_t(p)\big),$$
$$f_r(d) = \frac{r}{2}\left(1 - \frac{r^2}{(r+d)^2}\right).$$

In the gauge f_r we require $r > 0$ and $d \geq 0$.

Theorem 21.25 (Full signed nonlinear coercivity)**.** *For every prime-coordinate permutation, every cutoff, and every positive probe, the complete displacement and normalized cost satisfy*

$$0 \leq S_n, \qquad f_r(S_n) \leq D_n(t).$$

Consequently, if $D_n(t) \leq C < r/2$, then

$$S_n \leq r\left(\frac{1}{\sqrt{1 - 2C/r}} - 1\right).$$

The hypotheses imply $C \geq 0$. Every denominator is positive on the stated domain.

Proof. The exact kernel is

$$B_t'(x) = \frac{t^2}{x^2(x+t)} \qquad (x > 0).$$

For an entering coordinate $q \geq r$, the function

$$F(x) = \frac{r^2(r+t)}{t^2}B_t(x) + \frac{r^3}{2x^2}$$

has derivative

$$F'(x) = \frac{r^2 t(x-r)}{x^3(x+t)} \geq 0 \qquad (r \leq x \leq q).$$

Comparing its endpoint values gives the probe-uniform inequality

$$N_t(r; r, q) \geq \frac{r}{2}\left(1 - \frac{r^2}{q^2}\right) = f_r(q - r).$$

For a leaving coordinate $0 < p \leq r$, the maintained exact lower comparison gives

$$N_t(r; p, r) \geq r\left(\frac{r}{p} - 1\right) \geq r - p \geq f_r(r - p).$$

The middle inequality follows from $r(r/p - 1) - (r - p) = (r - p)^2/p \geq 0$. For the last inequality, the exact identity

$$d - f_r(d) = \frac{d^2(3r + 2d)}{2(r + d)^2} \geq 0$$

holds for every $d \geq 0$; also $f_r(d) \geq 0$.

The gauge is subadditive. To see this without changing any signed main term, set

$$a = \frac{r}{r + x}, \quad b = \frac{r}{r + y}, \quad c = \frac{r}{r + x + y} \qquad (x, y \geq 0).$$

Then $0 \leq a, b, c \leq 1$ and

$$c - ab = \frac{rxy}{(r + x)(r + y)(r + x + y)} \geq 0.$$

Hence $c^2 \geq a^2b^2$, and

$$1 - c^2 \leq 1 - a^2b^2 = (1 - a^2) + (1 - b^2) - (1 - a^2)(1 - b^2) \leq (1 - a^2) + (1 - b^2).$$

Multiplying by $r/2$ proves $f_r(x + y) \leq f_r(x) + f_r(y)$; induction gives its finite-sum form.

Let

$$E_n = e(\{0, \ldots, n\}) \setminus \{0, \ldots, n\}, \qquad L_n = \{0, \ldots, n\} \setminus e(\{0, \ldots, n\}).$$

They have equal cardinality. Cancelling common coordinates and retaining the boundary reference gives the exact identities

$$S_n = \sum_{j \in E_n} (p_j - r) + \sum_{j \in L_n} (r - p_j),$$
$$D_n(t) = \sum_{j \in E_n} N_t(r; r, p_j) + \sum_{j \in L_n} N_t(r; p_j, r).$$

Each displacement is nonnegative. Apply finite subadditivity to the first identity and the scalar bounds to each reference cost in the second. This proves $0 \le f_r(S_n) \le D_n(t)$ and $S_n \ge 0$. No estimate of the magnitudes of the original signed terms is used.

Finally, $f_r(S_n) \le C < r/2$ implies

$$0 < 1 - \frac{2C}{r} \le \left(\frac{r}{r + S_n}\right)^2.$$

Both square roots are nonnegative, and $r + S_n > 0$, so

$$\sqrt{1 - 2C/r}\,(r + S_n) \le r.$$

Division by the strictly positive square root proves the displayed localization. Nonnegativity of the gauge also gives $C \ge 0$. □

The finite-set theorem is more general than the prime specialization: it applies to any two equal-cardinality finite sets whose retained coordinates lie on the stated sides of a positive reference. Only the leaving coordinates need an explicit positivity hypothesis; the entering coordinates are automatically positive.

This extends the available localization domain from $C < r/4$ to $C < r/2$. Within the former domain the new upper bound approaches $r(\sqrt{2} - 1)$ as C increases to $r/4$, whereas the old bound $S_n \le 4C$ approaches r. The new theorem includes the first prime $r = 2$, every finite permutation, and zero displacement. It makes no even-integer readout assertion at the exceptional first prime and does not assume an eventual prime-ratio estimate. The boundary prime is p_n throughout, while both finite prefixes have length $n + 1$; there is no shift to p_{n+1} in the cost.

The exact derivative, integrated reference costs, subadditive gauge, finite signed assembly and inverse bound are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalCoercivity.lean`. The prime-profile coercivity and half-reference localization are compiled in `Helpers/PrimeProfileNonlinearCoercivity.lean`. The public export is `Assembly/ProbabilityProfiles.lean`. This is a stronger estimate of the finite probability-profile cost. It does not estimate the outstanding signed Möbius packet, and the certified unconditional RH boundary remains 7/8.

21.15 Rational characteristic acceleration from the integrated window

The integrated reference costs also sharpen the retained coordinate window. Keep $D_n(t) \le C$, $0 \le C < r/4$, and $r = p_n$. For each entering prime q, its individual reference cost is at most the full cost, so

$$\frac{r}{2}\left(1 - \frac{r^2}{q^2}\right) \le C < \frac{r}{4}, \qquad \left(\frac{r}{q}\right)^2 > \frac{1}{2}.$$

For each leaving prime p,

$$r\left(\frac{r}{p} - 1\right) \le C < \frac{r}{4}, \qquad \frac{r}{p} < \frac{5}{4}.$$

The entering ratios are at most one, and the leaving ratios are at least one. Since $(45/64)^2 = 2025/4096 < 1/2$, every retained ratio therefore belongs to

$$J = [45/64, 5/4].$$

This assertion concerns precisely $E_n \cup L_n$ after cancellation. It makes no claim about a common small prime.

The rational left endpoint is chosen to give exact characteristic roots. The affine map and its exterior parameters are

$$z(x) = \frac{125 - 128x}{35}, \qquad a = z(0) = \frac{25}{7}, \qquad c = \frac{24}{7}.$$

The map sends J to $[-1, 1]$, while $c^2 = a^2 - 1$, $a + c = 7$, and $a - c = 1/7$. Define

$$V_k = 7^k + 7^{-k}, \qquad H_k(x) = \frac{2T_k(z(x))}{V_k} \qquad (k \ge 0).$$

The characteristic identities already proved above give $V_k = 2T_k(25/7) > 0$. In particular $H_k(0) = 1$ and $\deg H_k \le k$. The retained-window bound is

$$|H_k(x)| \le \frac{2}{V_k} \le 2\,7^{-k} \qquad (x \in J).$$

Both branches are retained; $V_0 = 2$ and $H_0 = 1$.

Use the same signed observable

$$A_{k,n} = \mathrm{Def}_{e,n+1}\big((p - r)(1 - H_k(r/p)^2)\big),$$

where $\mathrm{Def}_{e,n+1}$ denotes the permuted finite prefix sum minus the original finite prefix sum.

Theorem 21.26 (Rational characteristic readout)**.** *Under the stated quarter-budget hypotheses, for every $k \geq 0$ the observable $A_{k,n}$ is a probability-moment combination of degree at most $2k$, and*

$$0 \leq S_n - A_{k,n} \leq \frac{4S_n}{V_k^2} \leq \frac{16C}{V_k^2},$$
$$S_n - A_{k,n} \leq 4\,49^{-k} S_n \leq 16C\,49^{-k}.$$

If $r \geq 3$ and $32C < V_k^2$, there is a single $K \geq 0$ with $S_n = 2K$ such that every observation satisfying $|y - A_{k,n}| < 1/2$ has $2K$ as its unique nonnegative even integer at distance less than one. A uniform sufficient integer order is

$$k = \lfloor \log_{49}(8p_n) \rfloor + 1.$$

The Lean statement uses the natural-number logarithm at the positive integer argument $8p_n$.

Proof. The exact kernel comparisons prove the window as just shown. The coupled rotation invariant bounds $|T_k(z)|$ by one for $-1 \leq z \leq 1$, and the exterior characteristic identity gives the strictly positive denominator V_k. This proves the stated polynomial norm and normalization at zero.

The normalized square-filter representation supplies a polynomial G_k of degree at most $2k$ with

$$(p - r)(1 - H_k(r/p)^2) = rG_k(r/p).$$

Thus $A_{k,n} = r\,\mathrm{Def}_{e,n+1}(G_k(r/p))$. Only the retained coordinates are used in estimating its error:

$$S_n - A_{k,n} = \sum_{j \in E_n} (p_j - r) H_k(r/p_j)^2 + \sum_{j \in L_n} (r - p_j) H_k(r/p_j)^2.$$

Every displayed displacement is nonnegative, and the principal coefficient remains one. The support-only comparison therefore gives $0 \leq S_n - A_{k,n} \leq 4S_n/V_k^2$. The previously compiled full signed estimate $S_n \leq 4C$ proves the first cost bound. Since $V_k \geq 7^k$, squaring gives $V_k^2 \geq 49^k$ and proves both geometric bounds.

For recovery, $r \geq 3$ and $C < r/4$ give $r - C > 2$. All retained primes are odd, so the maintained parity theorem gives $S_n = 2K$ for some nonnegative integer K. The condition $32C < V_k^2$ makes the full signed observation error less than $1/2$. Combining it with $|y - A_{k,n}| < 1/2$ puts y at distance less than one from $2K$. Two different even integers cannot both be at distance

less than one from the same y, proving uniqueness with the same K for all admissible observations.

Finally the chosen natural logarithm gives $49^k > 8r > 32C$, hence $V_k^2 > 32C$. This proves the sufficient order without a finite sampling or growing-cutoff assumption. □

At matched degree $m = 2k$, the geometric error factor changes from $4\,16^{-k}$ to $4\,49^{-k}$. The explicit sufficient moment degree has the ordinary real-logarithm bound

$$m \le \frac{\log r}{\log 7} + \frac{3\log 2}{\log 7} + 2.$$

Its leading coefficient relative to $\log_2 r$ is $\log 2/\log 7$, compared with $1/2$ for the earlier geometric certificate. These degree comparisons are ordinary consequences of the compiled natural-logarithm choice; no minimum-degree or minimax claim is made. The exact formula retains the contracting branch and can require less than this uniform sufficient order.

The integrated scalar window is compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalWindow.lean`; its two-support prime specialization is in `Helpers/PrimeProfileReciprocalWindow.lean`. The exact rational filter and all normalization and norm certificates are in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalSevenFilter.lean`. The reusable signed comparison requiring a norm only on the retained supports is in `Helpers/PrimeProfileSupportedFilter.lean`. The degree, exact and geometric full error, cost-adaptive recovery, and explicit logarithmic recovery are compiled in `Helpers/PrimeProfileSevenReadout.lean`, publicly exported by `Assembly/PrimeProfileReadouts.lean`. The error theorem includes $r = 2$; the even-integer recovery explicitly requires $r \ge 3$. Every finite permutation and zero transport remain included. The noise allowance concerns the combined observable, with the previously stated coefficient weights on raw moment errors. This proves a stronger finite observation theorem under the same cost hypothesis. It supplies no independent bound on the open signed Möbius packet; the unconditional RH boundary remains 7/8.

21.16 The exact locality threshold for one fixed probe

The universal half-reference estimate can be complemented by an exact threshold when the positive probe itself is fixed. Fix $n \geq 0$, $r = p_n$, and $t > 0$. Define

$$c_t(r) = \frac{r^2(r+t)}{t^2},$$
$$K_t(r) = c_t(r)\left(\frac{t}{r} - \log(1 + t/r)\right).$$

The threshold is strictly positive. Throughout this subsection n and t are held fixed while the coordinate permutation varies.

Theorem 21.27 (Exact fixed-probe locality threshold)**.** *For every real budget C, the following are equivalent:*

$$\textit{There is an integer } M \textit{ such that, for every permutation } e,$$
$$D_n(t) \leq C \implies e(\{0, \dots, n\}) \subset \{0, \dots, M-1\};$$
$$C < K_t(r).$$

More quantitatively, if $D_n(t) \leq C < K_t(r)$, every entering prime $q = p_j$ satisfies

$$(K_t(r) - C)\, q(q+t) \leq r^2(r+t).$$

At every budget $C \geq K_t(r)$ the failure is already witnessed by finite transpositions: for every real B and integer M, a swap of n with some $j > \max(n, M)$ has

$$S_n > B, \qquad D_n(t) < K_t(r) \leq C.$$

The same fixed t works for all these witnesses.

Proof. For $u > 0$, the strict logarithmic tangent inequality and its reciprocal application give

$$0 < u - \log(1+u) \leq \frac{u^2}{1+u}.$$

Indeed $\log(1+u) < u$. Also $\log(1/(1+u)) \leq 1/(1+u) - 1$, hence $\log(1+u) \geq u/(1+u)$, which proves the upper bound. Taking $u = t/r$ proves $K_t(r) > 0$.

For any positive coordinate q, exact cancellation gives

$$K_t(r) - N_t(r; r, q) = c_t(r)\left(\frac{t}{q} - \log(1 + t/q)\right),$$
$$0 < K_t(r) - N_t(r; r, q) \leq \frac{r^2(r+t)}{q(q+t)}.$$

This keeps the full compensated logarithmic expression. For an entering coordinate, the maintained signed reference decomposition gives $N_t(r; r, q) \leq D_n(t) \leq C$. Put $\delta = K_t(r) - C > 0$. Then

$$\delta \leq K_t(r) - N_t(r; r, q) \leq \frac{r^2(r+t)}{q(q+t)}.$$

Multiplying by the strictly positive denominator proves the quantitative coordinate constraint.

Let $A = r^2(r+t)$. Since a prime $q \geq 2$ and $t > 0$ have $q + t > 1$, the same constraint gives $\delta q \leq A$ and thus $q \leq A/\delta$. The zero-based prime enumeration satisfies $j \leq p_j = q$. Choose an integer

$$M > \max\left(n+1, \frac{A}{\delta}\right).$$

An image coordinate common to the original prefix has index at most $n < M$. Every other image coordinate is entering, and its index satisfies $j \leq q \leq A/\delta < M$. This proves the finite ambient prefix assertion, uniformly over all permutations satisfying the same budget. The argument also covers negative C, when applicable cost constraints may be vacuous.

For the converse, fix any $B \in \mathbb{R}$ and $M \geq 0$. Choose an integer $Q > B + r$ and put $j = Q + n + M + 1$. Then $j > \max(n, M)$ and $p_j \geq j \geq Q$. The permutation exchanging exactly n and j has the exact crossing-prefix identities

$$S_n = p_j - r > B, \qquad D_n(t) = N_t(r; r, p_j).$$

The positive residual already proved gives $N_t(r; r, p_j) < K_t(r)$. This constructs arbitrarily large displacement and entering index with the same fixed probe. If $C \geq K_t(r)$, each such swap satisfies $D_n(t) \leq C$. Its image prefix contains $j > M$, contradicting any proposed uniform ambient prefix of length M. Thus finite uniform localization implies $C < K_t(r)$. □

The residual estimate also shows, as an ordinary consequence, that the swap costs approach $K_t(r)$ from below with error at most $r^2(r+t)/(p_j(p_j + t))$, while their displacements tend to infinity. The sharpness assertion concerns the budget value for one fixed probe and cutoff. It does not change the probe with the escaping coordinate. In the converse, the counterexamples are finite permutations; no assertion restricted to permutations that are not finite is used.

This gives an exact criterion for reducing reconstruction from a single cost inequality to a finite search over possible prefix images. The complete

probability profile and its existing inverse theorems supply additional information beyond that one inequality. All statements here include the first prime $r = 2$, use a prefix of length $n + 1$, and keep the boundary reference p_n.

The strict logarithmic deficit, positive saturation, complete rational residual and coordinate constraint are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalSaturation.lean`. The crossing transposition identity is compiled in `Mathlib/Combinatorics/FinitePrefixTransposition.lean`. The full-cost incoming bound, uniform ambient-prefix theorem, unbounded finite-swap witnesses and exact equivalence are compiled in `Helpers/PrimeProfileSaturationThreshold.lean`, exported by `Assembly/ProbabilityProfiles.lean`. The displayed asymptotic consequence follows from the compiled residual and unbounded prime enumeration; it is not a separately compiled limit declaration. No assertion of external priority or optimal moment degree is made. The outstanding signed arithmetic estimate and certified RH boundary 7/8 are unchanged.

21.17 The sharp locality threshold uniformly over positive probes

The order of quantifiers changes when a single ambient prefix must work for every positive probe. The preceding fixed-probe theorem keeps t fixed; the next theorem requires the same M for all $t > 0$. Again $n \geq 0$ is fixed, $r = p_n$, and the observed prefix has length $n + 1$.

Theorem 21.28 (Exact probe-uniform locality threshold)**.** *For every real C, the following are equivalent:*

There is an integer M such that, for every $t > 0$ and permutation e,

$$D_n(t) \leq C \implies e(\{0, \dots, n\}) \subset \{0, \dots, M-1\};$$

$$C < \frac{r}{2}.$$

In the subcritical case, every entering prime q satisfies the probe-independent quadratic constraint

$$(r - 2C)q^2 \leq r^3.$$

At the critical budget itself, finite transpositions already give unbounded displacement. More precisely, for every real B and integer M, choose $j > \max(n, M)$ with $q = p_j$ and $q - r > B$. The positive probe

$$t_{r,q} = \frac{r^3}{2(q^2 - r^2)}$$

and the swap of n with j satisfy

$$S_n = q - r > B,$$

$$D_n(t_{r,q}) \leq \frac{r}{2} - \frac{r^3}{4q^2} < \frac{r}{2}.$$

Thus the failure at every $C \geq r/2$ uses finite permutations and strictly positive probes, with no limiting probe admitted as an observation.

Proof. We first prove the upper bound complementary to the maintained integrated lower bound. For $t > 0$ and $q \geq r > 0$, put

$$c = \frac{r^2(r+t)}{t^2}, \qquad a = \frac{r^2(r+t)}{2}, \qquad F(x) = -cB_t(x) - \frac{a}{x^2}.$$

The exact derivative $B_t'(x) = t^2/(x^2(x+t))$ gives

$$F'(x) = \frac{r^2(r+t)t}{x^3(x+t)} > 0 \qquad (x > 0).$$

Consequently $F(r) \le F(q)$, which rearranges to

$$\begin{aligned} N_t(r;r,q) &\le \frac{a}{r^2} - \frac{a}{q^2} \\ &= \frac{r+t}{2}\left(1 - \frac{r^2}{q^2}\right). \end{aligned}$$

Together with the integrated lower bound this gives the two-sided, fully compensated estimate

$$\frac{r}{2}\left(1 - \frac{r^2}{q^2}\right) \le N_t(r;r,q) \le \frac{r+t}{2}\left(1 - \frac{r^2}{q^2}\right).$$

Suppose first that $D_n(t) \le C < r/2$. In the exact signed reference decomposition,

$$D_n(t) = \sum_{j \in E} N_t(r;r,p_j) + \sum_{i \in L} N_t(r;p_i,r),$$

the common coordinates have canceled, and every retained reference cost is nonnegative. Thus an entering q has $N_t(r;r,q) \le C$. The lower integrated bound yields

$$\frac{r}{2}\left(1 - \frac{r^2}{q^2}\right) \le C, \qquad (r-2C)q^2 \le r^3.$$

All denominators multiplied here are positive. Write $\delta = r - 2C > 0$. Since $q \ge 2$, we have $q \le q^2$, so $q \le r^3/\delta$. Choose once and for all an integer

$$M > \max\left(n+1, \frac{r^3}{r-2C}\right).$$

A coordinate retained from the original prefix has index at most $n < M$. Every entering coordinate has $j \le p_j = q \le r^3/\delta < M$. This M depends only on n and C, and therefore works simultaneously for every $t > 0$ and every admissible permutation.

For the converse, let B and M be arbitrary. The unbounded prime enumeration supplies $j > \max(n, M)$ with $q = p_j$ and $q - r > B$. In particular $q > r$, so $t_{r,q} > 0$. Its exact substitution into the upper bound gives

$$\begin{aligned} N_{t_{r,q}}(r;r,q) &\le \frac{r + t_{r,q}}{2}\left(1 - \frac{r^2}{q^2}\right) \\ &= \frac{r}{2} - \frac{r^3}{4q^2} < \frac{r}{2}. \end{aligned}$$

The crossing transposition has exactly this normalized cost and displacement $q - r$. For $C \geq r/2$ it is budget-admissible, but its image prefix contains $j > M$. Since M was arbitrary, no common finite ambient prefix exists. This also treats the critical equality $C = r/2$. □

The theorem proves that the earlier half-reference condition for localization is sharp when the positive probe is allowed to vary. In a fixed-probe experiment the exact threshold remains $K_t(r)$ from the preceding subsection. Both results concern the information supplied by one cost-budget inequality; the complete profile inverse theorem has additional information. The small prime $r = 2$, cutoff $n = 0$, and finite permutations are included without parity restrictions or a rank shift.

The integrated upper bound, positivity of the explicit escaping probe, and quantitative half-reference margin are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalUpperCost.lean`. The probe-independent coordinate bound, common finite-prefix theorem, unbounded finite-swap witnesses and exact equivalence are compiled in `Helpers/PrimeProfileUniformProbeLocality.lean`, exported by `Assembly/ProbabilityProfiles.lean`. The earlier integrated lower bound and exact fixed-probe theorem are consumed directly. No external-priority claim is made. The unresolved signed arithmetic estimate and certified RH boundary 7/8 are unchanged.

21.18 The complete escape spectrum of a fixed profile probe

The one-coordinate locality threshold is the first member of a complete hierarchy. The underlying statement holds for any increasing sequence with a finite ceiling. This also separates a finite search for the entire prefix from a finite search for all but a prescribed number of its coordinates.

Let $w : \mathbb{N} \to \mathbb{R}$ be nondecreasing, let ℓ be a finite upper bound, and assume it is approached from below:

$$w_i \le \ell \quad (i \ge 0), \qquad \forall a < \ell \; \exists M : \; a < w_M.$$

Write

$$A_m = \sum_{i=0}^{m-1} w_i, \qquad \Delta_N(e) = \sum_{i=0}^{N-1} w_{e(i)} - A_N,$$

and, for $1 \le k \le N$, define

$$T_k = k\ell - (A_N - A_{N-k}).$$

Theorem 21.29 (Exact finite-prefix escape spectrum)**.** *For every $N \ge 1$, $1 \le k \le N$, and real C, the following are equivalent:*

$$\exists M \; \forall e : \quad \Delta_N(e) \le C \implies$$
$$\#\big(e(\{0,\dots,N-1\}) \setminus \{0,\dots,M-1\}\big) < k;$$

$$C < T_k.$$

The subcritical implication comes from the complete signed bound

$$\Delta_N(e) \ge k w_M + A_{N-k} - A_N$$

whenever at least k image coordinates lie outside the prefix of length M. For every $C \ge T_k$ and every proposed M, a permutation fixed outside a finite interval has at least k image coordinates outside that prefix and satisfies $\Delta_N(e) \le T_k \le C$.

Proof. First, any finite set $S \subset \mathbb{N}$ of cardinality m satisfies

$$\sum_{i \in S} w_i \ge A_m.$$

To see this without discarding signed terms, cancel $S \cap \{0,\dots,m-1\}$. The remaining entering and leaving sets have equal cardinalities. Every entering weight is at least w_m, and every leaving weight is at most w_m.

Their complete difference is therefore nonnegative. This is an application of the maintained signed equal-cardinality reference comparison.

Suppose now that $S = e(\{0, \dots, N-1\})$ contains at least k indices outside $\{0, \dots, M-1\}$. Choose exactly k such indices, forming U. The remainder $S \setminus U$ has cardinality $N-k$, so

$$\begin{aligned}\sum_{i \in S} w_i &= \sum_{i \in U} w_i + \sum_{i \in S \setminus U} w_i \\ &\geq k w_M + A_{N-k}.\end{aligned}$$

Subtracting A_N gives the stated signed bound. In particular, if $C < T_k$, choose M with

$$w_M > \ell - \frac{T_k - C}{k};$$

the denominator is positive. The ceiling-approximation hypothesis supplies such an M. A budget-admissible prefix with k outside indices would then have

$$\Delta_N(e) \geq T_k - k(\ell - w_M) > C,$$

a contradiction. The same M works for every admissible permutation.

For the converse put $a = N - k$ and choose $b \geq \max(N, M)$. Exchange the two disjoint half-open integer blocks $[a, a+k)$ and $[b, b+k)$ pointwise, fixing all other indices. This is an involution and fixes every index at least $b+k$. Its first N images consist of the unchanged first $N-k$ indices and the complete far block. Thus at least k images are outside the prefix of length M, and its exact cost is

$$\begin{aligned}\Delta_N(e) &= \sum_{i=0}^{k-1} w_{b+i} - \sum_{i=0}^{k-1} w_{a+i} \\ &\leq k\ell - (A_N - A_{N-k}) = T_k.\end{aligned}$$

This proves failure at the critical equality as well as at every larger budget. The entire leaving block remains in the formula. □

For the probability-profile application fix $t > 0$ and $n \geq 0$, put $N = n+1$, $r = p_n$, and use the actual scaled weights

$$w_i = c_t(r) B_t(p_i), \qquad \ell = 0.$$

They are strictly increasing and strictly negative. The proved logarithmic deficit bound gives

$$0 < -B_t(p_i) \leq \frac{t^2}{p_i(p_i + t)}.$$

Since $p_i \geq i$, these weights approach zero from below. This verifies the ceiling hypothesis on the entire prime enumeration without a prime number theorem. Moving the fixed positive factor $c_t(r)$ inside the finite sums gives the exact identity $\Delta_{n+1}(e) = D_n(t; e)$.

Corollary 21.30 (Every fixed-probe escape threshold)**.** *For $1 \leq k \leq n+1$, set*

$$T_k(t, n) = -c_t(r) \sum_{i=n+1-k}^{n} B_t(p_i).$$

For every real C,

$$\exists M \ \forall e : \quad D_n(t; e) \leq C \implies$$
$$\#\big(e(\{0, \ldots, n\}) \setminus \{0, \ldots, M-1\}\big) < k$$

holds if and only if $C < T_k(t, n)$. At $C = T_k(t, n)$, finite block permutations already put at least k coordinates beyond every proposed ambient prefix. The thresholds obey

$$T_0(t, n) = 0, \qquad T_1(t, n) = K_t(r),$$
$$T_{k+1}(t, n) - T_k(t, n) = -c_t(r) B_t(p_{n-k}) > 0 \qquad (0 \leq k \leq n).$$

Hence they form a strictly ordered finite spectrum.

Proof. Apply the general theorem to the scaled prime weights just verified. Canceling the first $n + 1 - k$ terms of their prefix sum gives the displayed threshold. The first increment is precisely $c_t(r)(-B_t(r)) = K_t(r)$. Each later increment is the positive logarithmic deficit of the next earlier prime, with the same fixed normalizer. These cancellations also give $T_0 = 0$ and the exact successor difference. □

Consequently, for $0 \leq j < n + 1$ and a nonnegative budget in $[T_j, T_{j+1})$, all but at most j coordinates have one common finite ambient prefix, while j coordinates can escape it in finite-permutation examples. At a budget at least T_{n+1}, all $n + 1$ observed coordinates can be sent beyond every proposed cutoff. This interval interpretation is a mathematical consequence of the compiled threshold family. It quantifies precisely how much positional information one cost inequality retains. The probe is fixed throughout this spectrum; the preceding probe-uniform theorem has a different order of quantifiers. All prime-profile statements include $p_n = 2$ and $n = 0$, retain prefix length $n + 1$, and require no parity hypothesis.

The explicit finite block involution, fixed-tail statement, complete signed transport identity and escaping cardinality are compiled in `Mathlib/Combinatorics/SeparatedBlockPrefix.lean`. The minimal-prefix comparison, retained tail-count lower bound, subcritical finite-prefix theorem, finite-permutation witnesses and exact general equivalence are compiled in `Mathlib/Combinatorics/FinitePrefixTailSpectrum.lean`. The prime-weight ceiling verification, scaled-cost identity, complete profile spectrum, finite witnesses, first-threshold consistency and strictly positive threshold increments are compiled in `Helpers/PrimeProfileEscapeSpectrum.lean`, exported by `Assembly/ProbabilityProfiles.lean`. The interval interpretation is not a separately compiled combined declaration. No external-priority claim is made. These results concern probability-profile information; the unresolved signed arithmetic estimate and certified RH boundary 7/8 are unchanged.

21.19 Simultaneous locality for arbitrary probe collections

The block witness can be chosen independently of the weight sequence. This gives an exact statement about simultaneous budgets, including collections with infinitely many probes.

Let I be any index set. For each $\alpha \in I$, let $w^{(\alpha)}$ be nondecreasing, bounded above by a finite ℓ_α, and cofinal below that ceiling as in the preceding theorem. At one common prefix length N put

$$\Delta_{\alpha,N}(e) = \sum_{i=0}^{N-1} w^{(\alpha)}_{e(i)} - \sum_{i=0}^{N-1} w^{(\alpha)}_i,$$
$$T_{\alpha,k} = k\ell_\alpha - \sum_{i=N-k}^{N-1} w^{(\alpha)}_i.$$

The budget C_α is any real number for each α.

Theorem 21.31 (Exact joint-budget locality)**.** *For $N \geq 1$ and $1 \leq k \leq N$, the following are equivalent:*

$$\exists M\ \forall e: \quad \big(\forall \alpha \in I,\ \Delta_{\alpha,N}(e) \leq C_\alpha\big) \implies$$
$$\#\big(e(\{0,\dots,N-1\}) \setminus \{0,\dots,M-1\}\big) < k;$$

$$\exists \alpha \in I: \quad C_\alpha < T_{\alpha,k}.$$

More strongly, for every M there is one finite permutation e with at least k image coordinates outside that prefix and

$$\Delta_{\alpha,N}(e) \leq T_{\alpha,k} \qquad \text{for every } \alpha \in I.$$

The same permutation works for the entire index set.

Proof. Suppose an index α has $C_\alpha < T_{\alpha,k}$. Apply the preceding one-weight theorem to obtain a finite ambient prefix for its budget. Every permutation satisfying all the budgets satisfies this particular budget, so that same ambient prefix proves the joint assertion.

Conversely, suppose $C_\alpha \geq T_{\alpha,k}$ for every α. Given any proposed M, take $a = N - k$ and $b = \max(N, M)$, and exchange $[a, a+k)$ with $[b, b+k)$, fixing every other coordinate. This single permutation has at least k images outside the prefix of length M and fixes every coordinate at least $b+k$. For

each α, the complete signed block identity gives

$$\begin{aligned}\Delta_{\alpha,N}(e) &= \sum_{i=0}^{k-1} w_{b+i}^{(\alpha)} - \sum_{i=0}^{k-1} w_{a+i}^{(\alpha)} \\ &\le k\ell_\alpha - \sum_{i=0}^{k-1} w_{a+i}^{(\alpha)} \\ &= T_{\alpha,k} \le C_\alpha.\end{aligned}$$

The construction of e uses only N, k, M and is therefore independent of α. It proves all the inequalities pointwise, so a uniform convergence rate over the index set is unnecessary. This disproves every proposed joint ambient prefix. The same calculation proves the stronger common-witness assertion at the saturation budgets themselves. □

The empty index set is included. In that case the constraints on the left are vacuous, and the finite block exchange disproves localization; the existential condition on the right is false as well.

Corollary 21.32 (Arbitrarily many simultaneous positive probes)**.** *Fix $n \ge 0$ and $1 \le k \le n+1$. Let $t, C : I \to \mathbb{R}$ with $t_\alpha > 0$ for every α. Then*

$$\begin{gathered}\exists M\ \forall e: \quad \big(\forall \alpha \in I,\ D_n(t_\alpha; e) \le C_\alpha\big) \implies \\ \#\big(e(\{0,\dots,n\}) \setminus \{0,\dots,M-1\}\big) < k\end{gathered}$$

holds if and only if

$$\exists \alpha \in I: \quad C_\alpha < T_k(t_\alpha, n).$$

For every proposed M, one finite block permutation has at least k image coordinates outside it while satisfying $D_n(t_\alpha; e) \le T_k(t_\alpha, n)$ for every α simultaneously.

Proof. Use

$$w_i^{(\alpha)} = c_{t_\alpha}(p_n) B_{t_\alpha}(p_i), \qquad \ell_\alpha = 0.$$

All increasing-weight and ceiling hypotheses were proved in the preceding specialization for every positive probe. The exact scaled-cost and threshold identities identify the general joint theorem with the displayed statement. □

Here a single permutation must satisfy all probe budgets simultaneously. The earlier probe-uniform theorem asks for a common ambient prefix for

each positive probe's admissible permutations. These are different quantifiers. The present theorem classifies the information supplied by the conjunction of these upper cost bounds on the same ordered prime coordinates and the same cutoff. Exact profile values and additional signed arithmetic relations supply other information. The first prime 2, the cutoff $n = 0$, and finite permutations remain included.

The canonical signed block-cost bound, common finite witness for arbitrary weight families, and exact general joint equivalence are compiled in `Mathlib/Combinatorics/FinitePrefixTailSpectrum.lean`. The positive-probe common witness and exact joint-budget criterion are compiled in `Helpers/PrimeProfileJointProbeLocality.lean`. The locality results are grouped in `Assembly/PrimeProfileLocality.lean` and exported through `Assembly/ProbabilityProfiles.lean`. No finiteness or nonemptiness assumption on the probe index type is used. No external-priority claim is made. The unresolved signed arithmetic estimate and certified RH boundary 7/8 are unchanged.

21.20 The rational endpoint envelope of all positive probes

The preceding joint criterion can be evaluated without optimizing a transcendental function. The exact answer involves two rational sums. Throughout this subsection the observed prefix is fixed; only the positive real probe varies.

For $r, p, t > 0$ write

$$v_t(r,p) = \frac{r^2(r+t)}{t^2}\left(\frac{t}{p} - \log(1+t/p)\right).$$

For a nonempty finite set S and positive coordinates $(p_i)_{i\in S}$, put

$$Q_t = \sum_{i\in S} v_t(r,p_i), \quad A = \sum_{i\in S} \frac{r^3}{2p_i^2},$$
$$B = \sum_{i\in S} \frac{r^2}{p_i}, \qquad H = \sum_{i\in S} \frac{r^3}{2p_i^3}.$$

In particular $A, B, H > 0$ and $m = \#S \geq 1$.

Theorem 21.33 (Exact rational envelope and explicit probe witnesses)**.** *For every such $S, r, (p_i)$,*

$$\sup_{t>0} Q_t = \max(A, B).$$

More precisely, for every $t > 0$ and $a \geq 1$,

$$Q_t \leq \frac{r}{r+t}A + \frac{t}{r+t}B \leq \max(A,B),$$
$$Q_t \geq A - tH,$$
$$Q_t \geq (1-1/a)B - \frac{mr^2a}{t}.$$

For every real C,

$$(\exists t > 0 :\ C < Q_t) \quad \Longleftrightarrow \quad C < \max(A,B).$$

If $C < A$, the explicit choice $t = (A-C)/(2H)$ works. If $C < B$, put $\delta = B - C > 0$; the explicit choices

$$a = 1 + \frac{4B}{\delta}, \qquad t = \frac{4mr^2a}{\delta}$$

work. Thus the supremum assertion includes its least-upper-bound property and actual finite positive witnesses below it.

Proof. We first prove the upper bound with its complete dependence on the probe. For any real u and $z \geq 0$ define

$$F_u(z) = (1+u)^2 \log(1+z) - (1+2u)z + \frac{z^2}{2}.$$

Direct differentiation gives

$$F_u(0) = 0, \qquad F_u'(z) = \frac{(z-u)^2}{1+z} \geq 0.$$

The denominator is positive on the stated domain. Consequently

$$(1+2u)z - \frac{z^2}{2} \leq (1+u)^2 \log(1+z).$$

Substitute $u = t/r$ and $z = t/p$ and multiply by $r^4/[t^2(r+t)] > 0$. Expanding both sides yields exactly

$$v_t(r,p) \leq \frac{r}{r+t}\frac{r^3}{2p^2} + \frac{t}{r+t}\frac{r^2}{p}.$$

Summation gives the first inequality. Both coefficients are nonnegative and sum to one, so it also gives the asserted upper bound by $\max(A, B)$.

For the small-probe lower bound use

$$G(z) = z - \log(1+z) - \frac{z^2}{2(1+z)}, \qquad G'(z) = \frac{z^2}{2(1+z)^2} \geq 0, \qquad G(0) = 0.$$

Thus

$$v_t(r,p) \geq \frac{r^2(r+t)}{2p(p+t)}.$$

The exact difference between this rational expression and $r^3/(2p^2) - tr^3/(2p^3)$ is

$$\frac{r^2 t(p^2 + rt)}{2p^3(p+t)} \geq 0.$$

All denominators are positive. Summing proves $Q_t \geq A - tH$.

For the large-probe lower bound, the logarithmic tangent inequality $\log x \leq x - 1$ for $x > 0$ implies, for $a \geq 1$,

$$\begin{aligned} \log(1 + t/p) &= \log((1 + t/p)/a) + \log a \\ &\leq (1 + t/p)/a - 1 + a - 1 \\ &\leq a + \frac{t}{ap}. \end{aligned}$$

Here the last step uses $1/a \le 1$. The exact identity

$$v_t(r,p) = \frac{r^2}{p} - \frac{r^2}{t}\log(1+t/p) + \frac{r^3}{t^2}\left(\frac{t}{p} - \log(1+t/p)\right)$$

has a nonnegative final term, again by the logarithmic tangent inequality. Therefore

$$v_t(r,p) \ge (1-1/a)\frac{r^2}{p} - \frac{r^2 a}{t}.$$

Summing gives the third estimate.

If $C < A$, the displayed small probe is positive and satisfies

$$Q_t \ge A - \frac{A-C}{2} = \frac{A+C}{2} > C.$$

If $C < B$, the displayed large probe is positive, $a \ge 1$, and

$$a\delta = \delta + 4B, \qquad \frac{B}{a} < \frac{\delta}{4}, \qquad \frac{mr^2 a}{t} = \frac{\delta}{4}.$$

Hence $Q_t > B - \delta/2 > C$. Since $C < \max(A,B)$ means $C < A$ or $C < B$, this proves the witness implication. Its converse follows from the upper bound. Finally, an upper bound strictly below $\max(A,B)$ would itself be exceeded by a positive-probe witness. This proves the exact least-upper-bound assertion without assuming that a maximizing finite probe exists. □

Corollary 21.34 (Exact all-positive-probe escape criterion)**.** *Fix $n \ge 0$, put $r = p_n$, and let $1 \le k \le n+1$. Define*

$$A_{n,k} = \sum_{i=n+1-k}^{n} \frac{r^3}{2p_i^2}, \qquad B_{n,k} = \sum_{i=n+1-k}^{n} \frac{r^2}{p_i}.$$

Then

$$\sup_{t>0} T_k(t,n) = \max(A_{n,k}, B_{n,k}).$$

For every real C, the assertion

$$\begin{aligned} &\exists M\ \forall e: \quad \big(\forall t>0,\ D_n(t;e) \le C\big) \implies \\ &\#\big(e(\{0,\dots,n\}) \setminus \{0,\dots,M-1\}\big) < k \end{aligned}$$

holds if and only if $C < \max(A_{n,k}, B_{n,k})$. For $k = 1$ this cutoff is exactly $C < p_n$. At or above the cutoff, every proposed M has a finite block-permutation counterexample satisfying all the probe budgets simultaneously.

Proof. Cancel the initial $n+1-k$ terms in the complete signed threshold:

$$T_k(t,n) = \sum_{j=0}^{k-1} v_t(r, p_{n+1-k+j}).$$

The index identity $(n+1-k)+k = n+1$ uses $k \le n+1$. The endpoint theorem applies to this nonempty finite block, proving the supremum and the strict-witness criterion. Apply the preceding joint-budget locality theorem to the index set of all positive reals, with probe t and constant budget C. Its existential condition is precisely $\exists t > 0 : \ C < T_k(t,n)$, which the endpoint theorem has evaluated.

When $k = 1$, the block consists of $p_n = r$ alone, giving $A_{n,1} = r/2$ and $B_{n,1} = r$. Since $r > 0$, their maximum is r. If C is at least the maximum, every individual threshold is at most C, so the common finite block witness from the joint theorem satisfies every positive-probe budget at once. □

The quantifier order matters. The cutoff p_n here concerns a single permutation satisfying every positive-probe bound. The earlier cutoff $p_n/2$ concerns the union of the admissible permutations as the probe varies. These classify different information. Every statement above includes $n = 0$, $p_n = 2$, the full prefix length $n+1$, and finite permutations; no odd-prime or large-rank exception is needed.

The scalar square-derivative envelope and the two endpoint lower estimates are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalEndpoint.lean`. The finite-sum bounds, explicit witnesses, strict-sublevel equivalence and actual least-upper-bound theorem are compiled in `Mathlib/Analysis/SpecialFunctions/Log/FiniteLogDeficitEndpoint.lean`. The exact prime-block identity, rational supremum, all-positive-probe locality equivalence and single-coordinate specialization are compiled in `Helpers/PrimeProfileAllProbeThreshold.lean`, exported through `Assembly/PrimeProfileLocality.lean` and `Assembly/ProbabilityProfiles.lean`. These theorems measure finite probability-profile information. No external-priority claim or zero-free exponent gain is made; the certified RH boundary remains 7/8.

21.21 RH readout on the same prime-quadruplet transport

The short-quadruplet theorem and the moving-coefficient RH criterion apply to the same probability law. The bridge is a quantitative bound on its actual local odds. It also clarifies which existential statement includes an additional prime-pattern conjecture.

Lemma 21.35 (Bounded prime reindexing preserves the RH law class)**.** *Let f send every prime p to a prime $q = f(p)$, and suppose $|q-p| \le H$ for every prime, where $H \ge 0$ is fixed. Then*

$$\left|\frac{1}{q-1} - \frac{1}{p-1}\right| \le \frac{H(H+1)}{(p-1)^2}.$$

Thus the weights $w(p) = 1/(f(p)-1)$ define a qualifying quadratic prime law with $\nu = 1$ and error constant $H(H+1)$. Bijectivity is unnecessary for this qualification. For each fixed such law and every fixed $\alpha > 0$, its existing moving-coefficient reference-sign criterion is equivalent to RH. All primes, including 2, belong to the stated domain.

Proof. Put $x = p - 1 \ge 1$ and $y = q - 1 \ge 1$. The displacement hypothesis gives $|x - y| \le H$ and

$$x \le y + H \le (H+1)y.$$

Consequently, keeping the exact reciprocal difference,

$$\begin{aligned}\left|\frac{1}{y} - \frac{1}{x}\right| x^2 &= |x-y|\frac{x}{y} \\ &\le H(H+1).\end{aligned}$$

Division by the positive x^2 proves the estimate. The weights are positive, $\nu = 1 > 0$, and $H(H+1) \ge 0$, verifying every field of the maintained quadratic-law definition. The established quadratic-law RH equivalence now applies to this fixed law. No RH hypothesis is used to establish its qualification. □

To state the specialization precisely, let $B \subset \mathbb{N}$ satisfy

$$n \in B \implies n+2 \notin B, \qquad n \in B \implies p_{n+2} = p_n + 6.$$

Let e_B be the existing permutation exchanging n with $n+2$ for each $n \in B$, fixing all other coordinates. Selected starts may be adjacent; the first

condition prevents conflicting edges. The associated same-profile probability law is

$$a_i = \frac{1}{p_{e_B(i)}}.$$

For a prime-labelled coordinate $p = p_i$ its actual local odds are

$$w_B(p_i) = \frac{a_i}{1 - a_i} = \frac{1}{p_{e_B(i)} - 1}.$$

Write $\mathcal{L}_B$ for the quadratic law with these weights. Its finite coefficients and ratios are

$$\begin{aligned} E_{B,N,r} &= [z^r] \prod_{p_i \le N} (1 + w_B(p_i) z), \\ R_{B,N,r} &= \frac{E_{B,N,r-1}}{E_{B,N,r}}, \\ L(N) &= \gamma + \log\log \vartheta(N), \qquad r_N = \lfloor \alpha L(N) \rfloor. \end{aligned}$$

The reference ratio $\mathcal{R}_{B,r}(u)$ is the maintained factorial-convolution ratio formed from the entire reference profile of $\mathcal{L}_B$. The subscript records its defining law; the following invariance theorem identifies it with one canonical reference. The natural number N is an inclusive prime-value cutoff; i remains the prime index. On the finite observed probability prefix the earlier notation is $n + 1$, with endpoint p_n. No scalar is substituted into the permutation map.

Theorem 21.36 (RH on a fixed constellation witness)**.** *For every fixed B satisfying the two conditions above, $\mathcal{L}_B$ has $\nu = 1$ and error constant 42. For every fixed $\alpha > 0$,*

$$\text{RH} \quad \Longleftrightarrow \quad R_{B,N,r_N} < \mathcal{R}_{B,r_N}(L(N)) \quad \textit{for every sufficiently large } N.$$

The same fixed B is used throughout the limit. The coefficient denominators and moving ranks are positive for all sufficiently large N, as provided by the existing quadratic-law comparison. Recurrence of the selected blocks is unnecessary for this fixed-witness equivalence.

Proof. A selected start $i \in B$ is sent to $i + 2$, so its prime value increases by exactly 6. If i is the upper end of a selected edge, then $i \ge 2$, $i - 2 \in B$, and its prime value decreases by exactly 6. Every other coordinate is fixed. Thus

$$|p_{e_B(i)} - p_i| \le 6 \qquad (i \ge 0).$$

This covers every coordinate, including the indices below two in the backward-edge test. Apply the preceding lemma with $H = 6$:

$$|w_B(p) - 1/(p-1)| \le 42/(p-1)^2.$$

The odds identity is exact since all moved values are primes at least 2. The existing quadratic-law RH theorem, with $\nu = 1$, now gives precisely the displayed moving-rank equivalence for $\mathcal{L}_B$. □

Here is how the established sign comparison preserves the zero information. For this fixed B, α, there is $C \ge 0$ such that eventually, with the actual Nicolas log oscillation $F(N)$,

$$\begin{aligned} F(N) > C/N &\implies \mathcal{R}_{B,r_N}(L(N)) < R_{B,N,r_N}, \\ F(N) < -C/N &\implies R_{B,N,r_N} < \mathcal{R}_{B,r_N}(L(N)). \end{aligned}$$

Under RH the installed Nicolas estimate eventually gives the second inequality for every fixed C. If RH fails, its installed unbounded positive excursions give $NF(N) > C+1$ arbitrarily far out, forcing the opposite coefficient sign. This contradicts an eventual upper-reference sign. The proof uses an unconditional comparison and the appropriate RH or non-RH Nicolas input in each direction. The cutoff and the whole signed oscillation are retained. No uniform cutoff over B or α is asserted.

Corollary 21.37 (The exact existential constellation statement)**.** *Fix $\alpha > 0$. The following are equivalent:*

1. *RH holds and short prime quadruplets recur arbitrarily far out.*
2. *There is a set B satisfying the two support conditions above, with arbitrarily large k for which $k, k+1 \in B$, and whose actual law $\mathcal{L}_B$ has $R_{B,N,r_N} < \mathcal{R}_{B,r_N}(L(N))$ eventually.*

The short quadruplets are the two already defined consecutive-prime families

$$(p, p+2, p+6, p+8), \qquad (p, p+4, p+6, p+10), \qquad p > 3.$$

No disjointness assumption is imposed on their arithmetic occurrences. The support construction selects separated four-coordinate blocks when constructing its one permutation.

Proof. If RH and recurrence hold, the maintained support-selection theorem supplies B with unbounded adjacent selected starts. The fixed-law RH equivalence gives its eventual coefficient sign.

Conversely, the coefficient sign gives RH by that same equivalence. An adjacent selected pair $k, k+1$ forces two coordinates across the prefix cutoff $k+2$. These double crossings recur arbitrarily far out. The maintained signed transport estimate gives, for every $\varepsilon > 0$, eventually and uniformly over all positive probes,

$$D_{k+1}(t; e_B) \leq 12 + \varepsilon.$$

Take $\varepsilon = 1$ and the constant probe $t = 1$. The resulting eventual ceiling $13 < 14$, together with recurrent double crossings, satisfies every hypothesis of the existing quadruplet inverse theorem. It yields recurrent short quadruplets. In particular the critical cost and the coefficient-sign readout concern the same transport, rather than independently chosen probability laws. □

This establishes an RH connection for the constellation witness. It does not identify RH with quadruplet recurrence alone: the existential criterion above detects their conjunction. The cost threshold 12 records local four-prime geometry; the moving coefficient sign records the Nicolas oscillation. The full Bernoulli profile alone does not distinguish these transports, since it is invariant under coordinate permutations.

The reciprocal displacement bound, complete quadratic-law adapter, prime-index conversion and bounded-reindexing RH criterion are compiled in `Helpers/PrimeProfileBoundedReindexing.lean`. The complete distance-six verification, actual probability-odds identity, explicit law constants, fixed-witness RH equivalence and existential conjunction criterion are compiled in `Assembly/PrimeProfileQuadrupletRH.lean`, exported through `Assembly/ProbabilityProfiles.lean`. The existing short-quadruplet attainment, support selection and inverse theorem are consumed without a new prime-pattern supply assumption. No external-priority claim is made. The certified zero-free boundary remains 7/8.

21.22 One canonical RH reference for every constellation transport

The entire reference in the preceding criterion is independent of the chosen block support. This follows from an exact infinite signed pairing, using the quadratic bounds to justify the complete series.

Theorem 21.38 (Involutive invariance of the entire reference)**.** *Let $\mathcal{A}, \mathcal{B}$ be qualifying quadratic prime laws with the same parameter ν. Suppose σ is an involutive permutation of the primes and*

$$w_{\mathcal{A}}(p) = w_{\mathcal{B}}(\sigma(p)) \qquad \text{for every prime } p.$$

Then their entire reference profiles are equal. Consequently all reference Taylor coefficients are equal, including degree zero. Their factorial-convolution reference ratios agree wherever the common denominator is nonzero. The two laws may have different quadratic error constants.

Proof. For a fixed real $t \geq 0$, define the complete signed difference

$$\begin{aligned} g_t(p) &= [\log(1 + w_{\mathcal{A}}(p)t) - \nu t \log(1 + 1/(p-1))] \\ &\quad - [\log(1 + w_{\mathcal{B}}(p)t) - \nu t \log(1 + 1/(p-1))] \\ &= \log(1 + w_{\mathcal{B}}(\sigma(p))t) - \log(1 + w_{\mathcal{B}}(p)t). \end{aligned}$$

Each bracketed profile series is absolutely convergent by its proved quadratic-law logarithmic bound. Their difference is therefore summable. The normalizing terms have been canceled exactly. Involutivity gives

$$g_t(\sigma(p)) = -g_t(p).$$

Reindex only this summable difference:

$$\sum_p g_t(p) = \sum_p g_t(\sigma(p)) = -\sum_p g_t(p).$$

Its sum is zero. The two logarithmic reference profiles are thus equal for every $t \geq 0$. Exponentiating and applying the existing positive-real product identity shows equality of their entire profiles on the nonnegative real axis.

Both reference profiles are entire by the established quadratic-law construction. Their coincidence set has 0 as an accumulation point: for every $\varepsilon > 0$, the nonzero real point $\varepsilon/2$ lies within distance ε of zero and has equal profile values. The analytic identity principle therefore gives equality on the whole complex plane. Every Taylor coefficient at zero is equal. Substitution in the finite factorial convolutions proves the reference-ratio equality on its stated nonzero-denominator domain. □

Only the signed difference is reindexed in this proof. In particular, the argument does not rearrange either divergent positive sum $\sum_p \log(1+w_{\mathcal{A}}(p)t)$ or $\sum_p \log(1+w_{\mathcal{B}}(p)t)$ separately.

Let $\mathcal{L}_0$ be the fixed canonical law with

$$w_0(p) = \frac{1}{p-1}, \qquad \nu = 1, \qquad \text{quadratic error } 0,$$

and denote its reference ratio by $\mathcal{R}_{0,r}(u)$.

Corollary 21.39 (A common canonical constellation RH criterion)**.** *For every fixed support B from the preceding subsection,*

$$\Phi_{\mathcal{L}_B} = \Phi_{\mathcal{L}_0}, \qquad \mathcal{R}_{B,r}(u) = \mathcal{R}_{0,r}(u)$$

on the common nonzero-denominator domain of the ratios. The reference clock is exactly the same at every cutoff. For each fixed $\alpha > 0$,

$$\text{RH} \quad \Longleftrightarrow \quad R_{B,N,r_N} < \mathcal{R}_{0,r_N}(L(N)) \quad \textit{eventually in } N.$$

Thus the reference profile, its coefficients, its clock and its ratio are fixed independently of B. The actual finite coefficient ratio R_{B,N,r_N} still belongs to the chosen constellation transport. The existential criterion of the preceding subsection remains valid with $\mathcal{R}_0$ replacing every $\mathcal{R}_B$.

Proof. The exact correspondence $i \mapsto p_i$ identifies the permutation e_B with an involutive permutation of the primes. Its inverse coordinate is the number of primes strictly below the prime label; the two inverse identities hold on the entire enumeration, beginning at $p_0 = 2$. The actual odds satisfy

$$w_B(p_i) = w_0(p_{e_B(i)}).$$

Both laws qualify: $\mathcal{L}_B$ has the already proved error bound 42, and $\mathcal{L}_0$ has error zero; both have $\nu = 1$. Apply the theorem to obtain equality of the entire profiles and every reference coefficient and ratio. The clock identity follows directly from their common $\nu = 1$. Substitution into the already compiled fixed-witness and existential RH equivalences gives the assertions. Their eventual positive-denominator domain is retained. □

The infinite signed cancellation here removes the change in the reference caused by reindexing. The finite-cutoff coefficient ratio, the complete compensated cutoff cost, and the Nicolas clock oscillation remain in their respective statements. In particular the RH sign response is preserved; no zero signal is set to zero by this pairing. The cost threshold continues to distinguish the local prime geometry even though these entire references coincide.

The summable logarithmic pairing, entire-function equality, equality of all reference coefficients and ratios, and exact prime-coordinate conversion are compiled in `Helpers/PrimeProfileReferenceInvariance.lean`. Their application to the actual quadruplet law, the canonical clock, the fixed-reference RH criterion and the fixed-reference existential criterion are compiled in `Assembly/PrimeProfileQuadrupletCanonicalRH.lean`, exported through `Assembly/ProbabilityProfiles.lean`. The preceding corollary concerns each fixed support and rank parameter. The next theorem makes the cutoff uniform over all supports, and more generally over every prime permutation of a fixed displacement radius. No uniformity in a growing rank parameter is asserted.

21.23 A uniform RH criterion over bounded prime permutations

Write $P_N = \{p \text{ prime} : p \le N\}$ and $w_0(p) = 1/(p-1)$. For a permutation σ of the primes put

$$S_{\sigma,N} = \sigma(P_N), \qquad E_{\sigma,N,r} = e_r\big((w_0(p))_{p\in S_{\sigma,N}}\big), \qquad R_{\sigma,N,r} = \frac{E_{\sigma,N,r-1}}{E_{\sigma,N,r}}.$$

Thus these are exactly the coefficients obtained by assigning odds $w_0(\sigma(p))$ to the sampled prime coordinate $p \le N$. For the identity permutation write $E_{0,N,r}$ and $R_{0,N,r}$. The benchmark remains the fixed canonical reference $\mathcal{R}_{0,r}(u)$, rather than a reference chosen after σ. Set

$$A(N) = \sum_{p\in P_N} \log(1 + w_0(p)),$$
$$L(N) = \gamma + \log\log\vartheta(N), \qquad r_N = \lfloor \alpha L(N) \rfloor.$$

Theorem 21.40 (Uniform bounded-displacement RH criterion)**.** *Fix $H \in \mathbb{N}$ and $\alpha > 0$. Let $\mathcal{S}_H$ be the class of all permutations of the primes satisfying $|\sigma(p) - p| \le H$ at every prime. Then*

$$\mathrm{RH} \quad\Longleftrightarrow\quad \exists N_0\ \forall N \ge N_0\ \forall \sigma \in \mathcal{S}_H : \quad R_{\sigma,N,r_N} < \mathcal{R}_{0,r_N}(L(N)).$$

Under RH one may choose this same N_0 so that $r_N > 0$, all the displayed actual denominators, and the canonical reference denominator are positive. The cutoff depends on H and α, and is independent of σ. The theorem includes $H = 0$ and permutations with cycles of arbitrary length; no involution or prime-gap recurrence premise is required.

Proof. We first establish two exact finite comparisons.

If $H \le N$, bounded displacement applied to σ and its inverse gives

$$P_{N-H} \subseteq S_{\sigma,N} \subseteq P_{N+H}.$$

Indeed, $p \le N$ implies $\sigma(p) \le N + H$. If $q \le N - H$ and $p = \sigma^{-1}(q)$, then $p \le q + H \le N$, so $q \in S_{\sigma,N}$. This accounts for every crossing coordinate, including those at the endpoints.

Adding a nonnegative coordinate to a finite nonnegative vector can only decrease e_{r-1}/e_r wherever its initial denominator is positive. To see the sign, write $a \ge 0$ for the new coordinate and use $e_j^{\text{new}} = e_j + ae_{j-1}$, with $e_{-1} = 0$. After multiplying the positive denominators, the assertion reduces to

$$a(e_{r-1}^2 - e_{r-2}e_r) \ge 0.$$

The established elementary-symmetric log-concavity inequality gives this for $r \ge 2$; for $r = 1$ the numerator is $e_0 = 1$ and the denominator increases. Iteration over the finite complement proves

$$R_{0,N+H,r} \le R_{\sigma,N,r} \le R_{0,N-H,r} \tag{U1}$$

whenever $r \ge 1$ and $E_{0,N-H,r} > 0$. All intermediate and larger-prefix denominators remain positive. The maintained proof first works over the nonnegative rational odds and then transports the exact symmetric coefficients and inequalities to $\mathbb{R}$.

Next, for every $M \ge 1$ the complete clock correction satisfies

$$\begin{aligned} 0 \le A(M+H) - A(M) &= \sum_{\substack{p \text{ prime} \\ M < p \le M+H}} \log\left(1 + \frac{1}{p-1}\right) \\ &\le \frac{H}{M}. \end{aligned} \tag{U2}$$

There are at most H integers in the interval, and each summand is nonnegative and at most $1/(p-1) \le 1/M$. No prime, logarithmic term, or positive correction is discarded. The argument includes $H = 0$ and the prime 2. In particular, for fixed H this increment tends to zero.

Apply the existing moving-clock coefficient-sign theorem to the single canonical law, at prefix M, with arithmetic clock $A(M)$ and reference clock $L(M+H)$. Its hypotheses hold: $L(M+H) \to \infty$, and

$$A(M) - L(M+H) = [A(M+H) - L(M+H)] - [A(M+H) - A(M)] \longrightarrow 0.$$

It supplies a constant $C \geq 0$, depending only on H, α and the canonical law, and an eventual domain with positive rank and denominators, on which

$$L(M+H) - A(M) < -\frac{C}{M} \quad \Longrightarrow \quad R_{0,M,r_{M+H}} < \mathcal{R}_{0,r_{M+H}}(L(M+H)). \tag{U3}$$

Its finite factorial convolutions are the actual elementary symmetric coefficients by the proved generating-product identity.

Assume RH. Write $F(N) = L(N) - A(N)$ on the eventual domain $\vartheta(N) > 1$. The established Nicolas consequence of RH gives $F(N) < -K/N$ eventually for every fixed real K. Choose $K = 2(C + H + 1)$ and take $M \geq \max(1, H)$, $N = M + H$. Then $N \leq 2M$, and (U2) gives the fully signed estimate

$$\begin{aligned} L(N) - A(M) &= F(N) + [A(N) - A(M)] \\ &< -\frac{2(C+H+1)}{N} + \frac{H}{M} \\ &\leq -\frac{C+H+1}{M} + \frac{H}{M} = -\frac{C}{M} - \frac{1}{M} < -\frac{C}{M}. \end{aligned}$$

Thus (U3) applies. Combining it with the upper inequality in (U1) proves the desired strict inequality for every $\sigma \in \mathcal{S}_H$. Every threshold used here concerns only M, H, α and the canonical law. Replacing M by $N - H$ therefore yields one eventual cutoff for the entire class, with all positivity conclusions retained.

Conversely, the identity permutation belongs to $\mathcal{S}_H$ for every $H \geq 0$. Selecting it in the asserted eventual inequality gives the canonical moving-rank coefficient criterion, whose already proved converse implies RH. This also proves nonvacuity at $H = 0$. □

Corollary 21.41 (One cutoff for all constellation supports)**.** *For each fixed $\alpha > 0$, RH is equivalent to the eventual inequality*

$$R_{B,N,r_N} < \mathcal{R}_{0,r_N}(L(N))$$

holding simultaneously for every support B satisfying the separation and distance-six hypotheses of the quadruplet construction. Under RH the actual denominators are positive on this common domain.

Proof. The proved all-coordinate displacement theorem bounds the associated prime permutation by 6, in both swap directions and at the small indices. Apply the uniform theorem with $H = 6$; its actual finite odds are

exactly those of $\mathcal{L}_B$. For the converse select the empty support, which satisfies both support hypotheses and gives the canonical criterion. Equivalently, apply the already proved fixed-support RH converse to that support. □

The quantifiers are stronger than $\forall B\,\exists N_0(B)\,\forall N \geq N_0(B)$: here $\exists N_0\,\forall N \geq N_0\,\forall B$. Consequently the same sign test also permits an admissible support or permutation chosen separately at each cutoff. The theorem does not assert uniformity for $H = H(N)$ or $\alpha = \alpha(N)$. It supplies no constellation occurrence by itself: adding a recurrent-witness requirement retains the earlier conjunction of RH with constellation recurrence. The canonical zero signal is retained through the signed clock correction. No zero-free exponent gain is claimed; the certified boundary remains $7/8$.

The finite rational and real ratio comparisons and the exact prime-prefix sandwich are compiled in `Assembly/PrimeProfilePrefixSandwich.lean`. The H/M clock bound and shifted-clock limits are compiled in `Helpers/PrimeProfileClockShift.lean`. The uniform RH equivalence, its positive-denominator forward theorem and the simultaneous quadruplet specialization are compiled in `Assembly/PrimeProfileUniformConstellationRH.lean`, exported through `Assembly/ProbabilityProfiles.lean`.

21.24 Shared excursions and every prescribed constellation shape

The uniform criterion has a stronger converse than selecting the identity. A false-RH excursion is visible simultaneously through every bounded prime permutation. This will supply an RH converse for a single shape-specific transport, including cycles longer than two.

Theorem 21.42 (Shared positive excursions)**.** *Fix $H \in \mathbb{N}$ and $\alpha > 0$. If RH is false, then arbitrarily large integer cutoffs N satisfy*

$$\mathcal{R}_{0,r_N}(L(N)) < R_{\sigma,N,r_N} \qquad \text{for every } \sigma \in \mathcal{S}_H$$

simultaneously. At these cutoffs $r_N > 0$ and all actual and reference denominators are positive.

Proof. The past reference clock $L(M-H)$ tends to infinity. Moreover

$$A(M) - L(M-H) = [A(M-H) - L(M-H)] + [A(M) - A(M-H)] \longrightarrow 0.$$

For the second term apply (U2) at $M - H$, using $(M - H) + H = M$ eventually. Both assertions include $H = 0$.

Apply the canonical moving-clock sign theorem at prefix M with reference clock $L(M - H)$. It supplies $C \geq 0$ and an eventual domain on which a positive clock difference greater than C/M forces the actual ratio above the reference. Put $M = N + H$. The fully retained signed difference is

$$L(N) - A(N + H) = F(N) - [A(N + H) - A(N)].$$

Set $K = C + H + 1$. If $N \geq 1$ and $F(N) > K/N$, then

$$\begin{aligned} L(N) - A(N + H) &> \frac{C + H + 1}{N} - \frac{H}{N} \\ &= \frac{C}{N} + \frac{1}{N} > \frac{C}{N + H}. \end{aligned}$$

Thus the canonical ratio at $N + H$, with rank r_N, exceeds the reference. The lower inequality in (U1) transfers the same strict inequality to every $\sigma \in \mathcal{S}_H$.

For completeness, the denominator required by (U1) is available unconditionally: the canonical moving-clock theorem with prefix $N - H$ and reference clock $L(N)$ supplies positivity of $E_{0,N-H,r_N}$ and the reference denominator eventually, by the forward-shift clock limits already proved. Their finite superset comparisons supply every actual denominator.

If RH is false, the proved Nicolas excursion theorem gives, beyond every threshold, an integer N with $NF(N) > K$. Choose N beyond all the preceding eventual domains. Dividing by the positive N gives the required hypothesis. Neither C, K, nor the selected cutoff depends on σ. □

Corollary 21.43 (Every individual bounded permutation reads RH)**.** *For any fixed $\sigma \in \mathcal{S}_H$ and $\alpha > 0$,*

$$\text{RH} \quad \Longleftrightarrow \quad R_{\sigma,N,r_N} < \mathcal{R}_{0,r_N}(L(N)) \quad \textit{eventually in } N.$$

Even the nonstrict eventual upper inequality implies RH.

Proof. The uniform RH theorem gives the forward implication. If RH were false, the shared excursions would violate the asserted eventual upper inequality for this same σ. The argument uses its actual coefficients against the canonical reference, without assuming that σ is an involution. □

We now retain a complete finite constellation shape. Let $k \geq 2$, let $h_0, \ldots, h_{k-1}$ be distinct nonnegative integers, and suppose $h_j \leq W$ for every

j. The usual normalized description is $0 = h_0 < h_1 < \cdots < h_{k-1} = W$; the formal theorem only needs injectivity of the offset list and its bound by W. Write $\mathrm{Rec}(\mathcal{H})$ for

$$\forall X \in \mathbb{N}\ \exists q \geq X : \quad q + h_j \text{ is prime for every } j < k.$$

A *full-shape cycle at q* in a prime permutation σ means that there are prime labels $a_0, \ldots, a_{k-1}$ satisfying

$$a_j = q + h_j, \qquad \sigma(a_j) = a_{j+1 \bmod k} \qquad (0 \leq j < k).$$

Distinctness of the offsets gives distinct labels. This definition records the whole shape, including its cyclic order.

Theorem 21.44 (Arbitrary-shape constellation and RH equivalence)**.** *Fix the finite shape above and $\alpha > 0$. Then the following are equivalent:*

1. *RH holds and* $\mathrm{Rec}(\mathcal{H})$ *holds.*
2. *There is a permutation σ of the primes with $|\sigma(p) - p| \leq W$ for every prime, with full-shape cycles at arbitrarily large anchors q, such that*

$$R_{\sigma,N,r_N} < \mathcal{R}_{0,r_N}(L(N)) \quad \textit{eventually in } N.$$

The construction in the forward direction can use disjoint intervals $[q, q+W]$ and fix every prime outside the selected copies.

Proof. Assume recurrence of the shape. Apply the proved separated-subset selection theorem to its set of anchors, with separation $W + 1$. This yields an unbounded set A of valid anchors such that

$$a < b,\ a, b \in A \quad \Longrightarrow \quad a + W < b.$$

For $a \in A$ and $j < k$, define the prime label $\iota(a, j) = a + h_j$. This map from $A \times \{0, \ldots, k-1\}$ to the primes is injective. Indeed, if $a < b$, then $a + h_i \leq a + W < b \leq b + h_j$, excluding equality. The reverse ordering is excluded in the same way. Equal anchors then force equal offsets, and injectivity of the offset list forces equal indices.

Rotate the second coordinate of this labelled set and extend the resulting permutation along ι, fixing its complement. The established embedding-extension API gives a genuine permutation of all primes, with

$$\sigma(\iota(a, j)) = \iota(a, j + 1 \bmod k).$$

On a selected copy the anchor cancels from the displacement:

$$|\sigma(\iota(a,j)) - \iota(a,j)| = |h_{j+1 \bmod k} - h_j| \le W.$$

On the complement the displacement is zero. Every selected anchor gives its full-shape cycle, and the anchors are unbounded. Under RH, the preceding individual-permutation criterion supplies its eventual canonical coefficient sign.

Conversely, a full-shape cycle supplies prime labels equal to every $q + h_j$. Arbitrarily large cycle anchors therefore imply $\mathrm{Rec}(\mathcal{H})$, with every offset retained. The eventual coefficient sign for this one bounded permutation implies RH by the preceding corollary. This proves both required conclusions. □

The construction starts with all occurrences and selects disjoint copies; it never assumes that the original occurrences do not overlap. No consecutive-prime condition is imposed: intervening primes may exist. If consecutive patterns are desired, their absence-of-extra-primes condition is additional geometric information. The analytic sign cutoff, when RH holds, depends only on W and α, and works for every permitted shape and cycle transport inside that displacement class. No uniformity for a growing W or a growing α is claimed.

For a compact comparison, let $s_\sigma(N)$ denote the canonical coefficient inequality at cutoff N, let E_σ denote its eventual validity, and let $C_{\mathcal{H}}(\sigma)$ denote recurrence of full-shape cycles. For every fixed $\sigma \in \mathcal{S}_H$ we have

$$\mathrm{RH} \iff E_{\mathrm{id}} \iff E_\sigma.$$

The uniform and shape-specific statements are

$$\begin{aligned} \mathrm{RH} &\iff \exists N_0\ \forall N \ge N_0\ \forall \sigma \in \mathcal{S}_H : s_\sigma(N), \\ \mathrm{RH} \wedge \mathrm{Rec}(\mathcal{H}) &\iff \exists \sigma \in \mathcal{S}_W : C_{\mathcal{H}}(\sigma) \wedge E_\sigma. \end{aligned}$$

The smallest-recurring-gap location theorem records one coarse geometric parameter. The full-shape cycle word records an arbitrary finite geometric pattern. The centered sign tests the common global Nicolas bias, which cannot be hidden by choosing a different bounded transport: false RH produces shared positive excursions. These are separate requirements in the equivalence; no logical independence theorem is asserted. An independently proved eventual sign for one such witness would imply RH. Shape recurrence alone does not supply that sign, and RH alone does not supply shape recurrence. The certified zero-free boundary remains 7/8.

The past-clock limits are compiled in `Helpers/PrimeProfileExcursionClock.lean`. The shared positive response, simultaneous false-RH excursions and the individual bounded-permutation RH criterion are compiled in `Assembly/PrimeProfileSharedExcursions.lean`. The full-shape construction and decoding are compiled in `Helpers/PrimeConstellationCycles.lean`. The arbitrary-shape RH-and-recurrence equivalence is compiled in `Assembly/PrimeConstellationRH.lean`, exported through `Assembly/ProbabilityProfiles.lean`. The formal shape framework also permits the degenerate sizes zero and one; the full nontrivial-cycle interpretation above concerns $k \geq 2$.

21.25 Prime progressions with unbounded additive diameter

The fixed-shape theorem requires recurrence of one prescribed offset vector. Prime arithmetic progressions offer a different, established supply: the length is prescribed, while the common difference may vary. We retain this distinction throughout.

Consider finite blocks of primes

$$B_i = \{p_{i,0} < \cdots < p_{i,k_i-1}\}, \qquad k_i \geq 2,$$
$$p_{i,k_i-1} < p_{i+1,0}, \qquad p_{i,k_i-1} \leq 2p_{i,0}.$$

Rotate each block in increasing order, sending its last element to its first, and fix every other prime. This defines a permutation σ; the block lengths and additive displacements may be unbounded. Put

$$P_{\sigma,N}(z) = \prod_{p \leq N} \left(1 + \frac{z}{\sigma(p) - 1}\right), \qquad Q_{\sigma,N}(z) = P_{\sigma,N}(z) e^{-A(N)z},$$

where $A(N)$ is the canonical arithmetic clock. Write $P_{0,N}$ and $Q_{0,N}$ for the identity permutation, and Φ_0 for the canonical entire profile.

Theorem 21.45 (One-boundary stability)**.** *For every fixed $R \geq 0$, the normalized products satisfy*

$$\sup_{|z| \leq R} |Q_{\sigma,N}(z) - \Phi_0(z)| \leq \frac{C_R}{N}$$

for all sufficiently large N. No upper bound on k_i or on the additive diameter is needed. Consequently, for each fixed $b \geq 0$ there are r_0, N_0, K such that,

for $r \geq r_0$, $N \geq N_0$, $u > 0$ and $r/u \leq b$, the finite and reference factorial-convolution denominators are positive and their adjacent coefficient ratios differ by at most $K/(rN)$.

Proof. All complete blocks cancel exactly. If a block is incomplete at N, let p_0 be its first prime and p_m its first prime exceeding N. The included primes form an initial segment, whose shifted product obeys

$$P_{\sigma,N}(z)(1 + z/(p_0 - 1)) = P_{0,N}(z)(1 + z/(p_m - 1)).$$

This identity is valid at zeros of either full product. Define instead the boundary multiplier

$$D_N(z) = \frac{1 + z/(p_m - 1)}{1 + z/(p_0 - 1)}.$$

If no block is incomplete, set $D_N = 1$. In the incomplete case, $p_0 \leq N < p_m \leq 2p_0$, so

$$0 \leq \frac{1}{p_m - 1} \leq \frac{1}{p_0 - 1} \leq \frac{4}{N}.$$

For $N \geq \max(1, 8R)$ and $|z| \leq R$, the boundary denominator has norm at least $1/2$. Thus

$$|D_N(z) - 1| = \left| \frac{(1/(p_m - 1) - 1/(p_0 - 1))z}{1 + z/(p_0 - 1)} \right| \leq \frac{8R}{N}.$$

The cross-multiplied identity now gives $Q_{\sigma,N} = Q_{0,N} D_N$ without dividing by $P_{0,N}$. If $|Q_{0,N}| \leq B_R$ and $|Q_{0,N} - \Phi_0| \leq C_{0,R}/N$, then

$$|Q_{\sigma,N} - \Phi_0| \leq \frac{C_{0,R} + 8RB_R}{N}.$$

The canonical prefix bound and profile approximation supply these constants.

Cauchy's estimate on a disc of radius $b+1$ bounds every Taylor coefficient error by $C/[N(b+1)^j]$. The geometric descending-factorial estimates give $O(1/N)$ errors for both the normalized coefficient polynomial and its first derivative. The positive canonical lower bound controls their denominators. In the exact adjacent-ratio identity the common leading term cancels, leaving $(r/u)^2/r$ times a difference of logarithmic derivatives. This retains the bound $K/(rN)$, rather than $O(1/N)$. The initial cutoff is allowed to depend on the chosen radius throughout. □

Corollary 21.46 (Progression transports retain the RH criterion)**.** *Fix $\alpha > 0$ and the above permutation σ. Let $r_N = \lfloor \alpha L(N) \rfloor$, let $R_\sigma(N, r)$ denote the adjacent coefficient ratio of $P_{\sigma,N}$, and use the canonical reference ratio $\overline{R}_0$. Then*

$$\text{RH} \quad \Longleftrightarrow \quad R_\sigma(N, r_N) < \overline{R}_0(r_N, L(N)) \quad \textit{for every sufficiently large } N.$$

The ranks and both denominators are positive eventually. There are such permutations whose blocks are prime arithmetic progressions of lengths tending to infinity.

Proof. The preceding inverse-rank comparison and the canonical reference secant preserve both clock signs whenever $|L(N) - A(N)| > C/N$. Under RH the proved Nicolas estimate gives $L(N) - A(N) < -C/N$ eventually, for every fixed C. If RH fails, $N(L(N) - A(N))$ has arbitrarily late positive excursions exceeding every fixed constant. These force the reverse strict coefficient inequality and contradict even an eventual non-strict upper sign. The factorial-convolution coefficients at $A(N)$ are the coefficients of $P_{\sigma,N}$, since $P_{\sigma,N} = Q_{\sigma,N} e^{A(N)z}$.

For the progression supply, use Green–Tao's theorem on arbitrarily long prime progressions beyond any prescribed threshold.[1] From a progression of length $2k$, retain its latter k terms. If its first retained prime is $p_0 = a + kd$, its last is $a + (2k-1)d < 2p_0$. Choose these blocks successively beyond the preceding last prime and with k tending to infinity. Their disjoint cycles give the required permutation. This does not produce recurrence at a fixed common difference. □

The same ordinary construction works inside a fixed reduced residue class: its prime subset has positive relative density, so Green–Tao's Theorem 1.2 applies there. Every prime outside the chosen blocks remains fixed. The observable still uses all primes and the canonical RH reference.

The boundary bound, exact shifted-product identity and normalized-prefix transfer are compiled in `Helpers/PrimeProfileBoundaryMultiplier.lean`. The arbitrary-prefix Taylor-to-ratio transfer is compiled in `Helpers/PrimeProfileApproximationTransfer.lean`. The analytic interface is `Definitions/PrimeCyclePrefix.lean`; its uniform profile,

[1] B. Green and T. Tao, *The primes contain arbitrarily long arithmetic progressions*, Annals of Mathematics 167 (2008), 481–547, Theorems 1.1 and 1.2, `https://doi.org/10.4007/annals.2008.167.481`.

Cauchy and inverse-rank estimates are compiled in `Helpers/PrimeCyclePrefixApproximation.lean`. The moving-clock sign transfer is compiled in `Helpers/PrimeCyclePrefixClockSign.lean`. The per-model RH criterion is compiled in `Assembly/PrimeCyclePrefixRH.lean` and exported through `Assembly/ProbabilityProfiles.lean`. The concrete progression construction above is an ordinary argument using Green–Tao as an explicit external supply input; its complete Lean transport construction remains separate.

21.26 One residue class and the complete Dirichlet zero signal

An intrinsic class-only observable has a different reference from the full-prime transport above. There is a particularly useful choice: the identity class 1 mod q, with a fixed integer $q \geq 2$. All Dirichlet characters take value one at that class.

Theorem 21.47 (Positive character folding)**.** *Initially for $\Re s > 1$, put*

$$G_q(s) = \sum_{\substack{p \text{ prime} \\ p \equiv 1 \ (q)}} \frac{\log p}{p^s}.$$

Then

$$G_q(s) = -\frac{1}{\varphi(q)} \sum_{\chi \bmod q} \frac{L'}{L}(s,\chi) - H_q(s), \qquad H_q(s) = \sum_{\substack{p \text{ prime}, \ k \geq 2 \\ p^k \equiv 1 \ (q)}} \frac{\log p}{p^{ks}}.$$

The remainder is holomorphic on $\Re s > 1/2$. A zero ρ with $1/2 < \Re\rho < 1$ of any character in this family is a pole of G_q, with

$$\operatorname{Res}_{s=\rho} G_q(s) = -\frac{1}{\varphi(q)} \sum_{\chi \bmod q} m_\chi(\rho) \neq 0.$$

Here $m_\chi(\rho) = 0$ when that character has no zero at ρ.

Proof. In the absolutely convergent Euler region, expand

$$-\frac{L'}{L}(s,\chi) = \sum_{\substack{p \text{ prime} \\ p \nmid q}} \sum_{k \geq 1} \chi(p)^k \frac{\log p}{p^{ks}}.$$

Character orthogonality gives $\varphi(q)^{-1}\sum_\chi \chi(p)^k = \mathbf{1}_{p^k\equiv 1\ (q)}$. Separate $k=1$ to obtain the asserted identity. Primes dividing q contribute to neither side. For every fixed $\sigma_0 > 1/2$ and $\Re s \geq \sigma_0$,

$$|H_q(s)| \leq \frac{1}{1-2^{-\sigma_0}} \sum_{n\geq 2} \frac{\log n}{n^{2\sigma_0}} < \infty.$$

This modulus-independent majorant gives normal convergence and holomorphy. Continue by the finite sum of logarithmic derivatives. At a zero in the stated strip, each contributing logarithmic derivative has residue $m_\chi(\rho)$; no L-function has a pole there. All multiplicities have the same sign in their sum, while H_q is holomorphic. The residue is therefore nonzero, including when several characters share the zero. □

This identity has a direct probability interpretation. For each prime $p \equiv 1 \bmod q$, take a Bernoulli variable B_p of success probability $1/p$, independently over each finite prefix. Its odds are precisely $1/(p-1)$, as in the coefficient law. The weighted mean is

$$M_q(x) = \mathbb{E} \sum_{\substack{p\leq x \\ p\equiv 1\ (q)}} (\log p) B_p = \sum_{\substack{p\leq x \\ p\equiv 1\ (q)}} \frac{\log p}{p},$$

$$\int_{1^-}^{\infty} x^{-s}\, dM_q(x) = G_q(s+1) \qquad (\Re s > 0).$$

Consequently the centered transform

$$\mathcal{D}_q(s) = G_q(s+1) - \frac{1}{\varphi(q)s}$$

is holomorphic on $\Re s > -1/2$ if and only if GRH holds for the characters modulo q (equivalently, their associated primitive L-functions). The subtraction removes the principal-character pole. Nonprincipal characters are regular and nonzero at one; finite imprimitive Euler factors introduce no zeros in $\Re s > 1/2$. The functional equations pair χ with $\overline{\chi}$, both present in the family, so absence of right-half-strip zeros gives the full critical-line assertion.

For an explicit sufficient arithmetic estimate, fix $1/2 \leq c < 1$. If there is a constant C_q such that, for every $\varepsilon > 0$,

$$M_q(x) - \frac{\log x}{\varphi(q)} - C_q = O_{q,\varepsilon}(x^{c-1+\varepsilon}),$$

then every nontrivial zero in this Dirichlet family has real part at most c. Indeed, integration by parts gives, initially for $\Re s > 0$,

$$\mathcal{D}_q(s) = C_q + s \int_1^\infty \left(M_q(x) - \frac{\log x}{\varphi(q)} - C_q \right) x^{-s-1}\, dx.$$

The assumed estimate continues this expression holomorphically to $\Re s > c - 1$, contradicting the nonzero residue of any zero with $\Re \rho > c$.

Corollary 21.48 (Cyclotomic polynomial realization)**.** *Fix $q \geq 2$ and put $F_q(u) = \Phi_q(u)$, the qth cyclotomic polynomial. For a divisor prime $p \nmid q$, its local root count and probability odds are*

$$\rho_q(p) = \varphi(q) \mathbf{1}_{p \equiv 1 \ (q)}, \qquad \eta_q(p) = \frac{\rho_q(p)}{p - \rho_q(p)}.$$

For each fixed cutoff N, sample u uniformly modulo $\prod_{p \leq N,\ p \nmid q} p$. The weighted distinct-prime-divisor count has mean

$$\mathbb{E} \sum_{\substack{p \leq N,\ p \nmid q \\ p \mid F_q(u)}} \log p = \varphi(q) M_q(N).$$

Its Mellin–Stieltjes transform has residue $-\sum_{\chi \bmod q} m_\chi(\rho)$ at $s = \rho - 1$ for every zero $1/2 < \Re \rho < 1$ in the Dirichlet family. All characters contribute with the same sign.

Proof. Since $p \nmid q$, the polynomial $X^q - 1$ is separable over $\mathbb{F}_p$. The roots of Φ_q are exactly the elements of multiplicative order q. The cyclic group $\mathbb{F}_p^\times$ has $\varphi(q)$ such elements when $q \mid p - 1$, and none otherwise. In the nonzero case $p \geq q + 1 > \varphi(q)$, so the local success probability $\rho_q(p)/p$ is strictly below one. The Chinese remainder theorem makes these finite-prime divisibility events independent under the stated sampling law. Their odds give the displayed coefficient law, and linearity of expectation gives its weighted mean. The preceding character identity supplies the Mellin residue after multiplication by $\varphi(q)$. □

Theorem 21.49 (Exact finite cyclotomic sampling)**.** *Fix $q \geq 2$ and a finite set S of primes. Sample U uniformly modulo $M_S = \prod_{p \in S} p$, and define*

$$B_p(U) = \mathbf{1}_{p \nmid q,\ p \mid \Phi_q(U)}, \qquad \theta_{q,p} = \begin{cases} \varphi(q)/p, & p \nmid q,\ p \equiv 1 \pmod q, \\ 0, & \text{otherwise.} \end{cases}$$

For every prescribed pattern $\varepsilon : S \to \{0,1\}$,

$$\Pr(B_p(U) = \varepsilon_p \text{ for all } p \in S) = \prod_{\substack{p \in S \\ \varepsilon_p = 1}} \theta_{q,p} \prod_{\substack{p \in S \\ \varepsilon_p = 0}} (1 - \theta_{q,p}).$$

All factors satisfy $0 \le \theta_{q,p} < 1$. In particular, every $p \le q$ has $\theta_{q,p} = 0$. The statement includes $S = \varnothing$, with $M_S = 1$ and both products equal to one.

Proof. Chinese remaindering bijects scalar residues modulo M_S with tuples $(u_p)_{p \in S} \in \prod_{p \in S} \mathbb{F}_p$. A success coordinate has $\rho_q(p)$ permitted values, and a failure coordinate has $p - \rho_q(p)$, where $\rho_q(p) = p\theta_{q,p}$ includes the depleted case. The number of tuples giving the pattern is therefore exactly

$$\prod_{\substack{p \in S \\ \varepsilon_p = 1}} \rho_q(p) \prod_{\substack{p \in S \\ \varepsilon_p = 0}} (p - \rho_q(p)).$$

Dividing by M_S proves the formula. The preceding cyclotomic root calculation gives the local probabilities. If a nondepleted success is possible, $q \mid p - 1$, so $q \le p - 1$ and $\varphi(q) \le q < p$. This also excludes every $p \le q$. □

This is an exact law for each fixed finite divisor support, without a sampling-limit error. It asserts independence across divisor primes for fixed q. Different cyclotomic families evaluated at the same scalar need not be independent.

Thus polynomial inputs realize the intrinsic progression observable, including quadratic and quartic examples:

$$\Phi_3(u) = u^2 + u + 1, \qquad \Phi_4(u) = u^2 + 1,$$
$$\Phi_5(u) = u^4 + u^3 + u^2 + u + 1,$$
$$\Phi_8(u) = u^4 + 1, \qquad \Phi_{12}(u) = u^4 - u^2 + 1.$$

The class-1 mod 5 example includes characters of order four, beyond the quadratic characters modulo eight. Here u is the sampled polynomial input and N is the divisor cutoff. Primes dividing q are explicitly removed; no assertion about their root counts is used. For fixed q they form a finite exceptional set. The rescaled remainder is $\varphi(q)H_q$, so its bound carries this factor if q varies. The local first-order term is character dependent; this argument does not silently qualify the law as a pointwise quadratic perturbation of the canonical prime law.

The finite-field root counts, their prime-modulus specialization, strict local probabilities, the actual `PrimeClockLaw`, its general coefficient comparison, and the full finite sampling law are checked in Lean. The maintained owners, under the project namespace, are

```
Mathlib/FieldTheory/Finite/Cyclotomic.lean
Helpers/CyclotomicProbability.lean
Helpers/CyclotomicSampling.lean
```

All nineteen declarations have only the standard foundational axioms in their audits. The weighted expectation formula is an ordinary consequence of the checked finite distribution and linearity of expectation; it has no separately checked declaration here. Character folding and the analytic continuation argument remain ordinary written proofs, with no priority claim. They identify a concrete class-only probability observable and an explicit remaining arithmetic estimate. Existence of primes or progressions in the class does not itself provide that estimate.

21.27 An entire profile that remembers every prime

Each proper nonzero fraction ρ_p/p has a prime denominator in lowest terms. This permits exact reconstruction at every prime, including inactive primes. For the forward compensated Bernoulli product, see [9, Proposition 3.1]. Here the prime denominators supply an inverse conclusion without a common leading asymptotic.

Theorem 21.50 (Prime-labelled reconstruction from the zero set)**.** *For each prime p, let a_p, b_p be integers with $0 \le a_p < p$ and $0 \le b_p < p$, and suppose*

$$\sum_p (a_p/p)^2 < \infty, \qquad \sum_p (b_p/p)^2 < \infty.$$

The canonical probability-centred profiles

$$\mathcal{P}_a(z) = \prod_p \big(1 + (a_p/p)(z-1)\big) \exp\big(-(a_p/p)(z-1)\big),$$

and $\mathcal{P}_b$ are entire. Their zero sets recover every coordinate:

$$\begin{aligned} \mathcal{Z}(\mathcal{P}_a) = \mathcal{Z}(\mathcal{P}_b) &\iff a_p = b_p \quad \text{for every prime } p \\ &\iff \mathcal{P}_a = \mathcal{P}_b. \end{aligned}$$

Zero-set inclusion already implies $a_p = b_p$ wherever $a_p > 0$. Every zero is negative and simple.

Proof. On each compact set the compensated factor is $1 + O_R((a_p/p)^2)$, giving locally uniform convergence. A sufficiently late tail has an absolutely

convergent logarithm wherever its factors are nonzero. Consequently the zeros are exactly

$$z = 1 - \frac{p}{a_p} \qquad (a_p > 0).$$

If a zero contributed at p is also contributed at a prime q by $b_q > 0$, then $p\, b_q = q\, a_p$. Hence $p \mid q\, a_p$. Since $0 < a_p < p$, we have $p \nmid a_p$, so $p \mid q$, whence $p = q$. Cancellation gives $a_p = b_p$. This proves the inclusion assertion.

Apply this argument in both directions when the zero sets agree. If $a_p = 0 < b_p$, reverse inclusion would force $b_p = a_p$. Thus all coordinates agree, giving equality of the products as well. Conversely, equality of the entire functions gives equality of their zeros.

No two positive coordinates of one profile can contribute the same zero. At its unique vanishing factor, the remaining convergent product is nonzero. The linear factor has nonzero derivative, so the zero is simple. Finally $a_p < p$ implies $1 - p/a_p < 0$. □

The strict upper bound is essential: a deterministic success has factor $z \exp(-(z-1))$, and moving it to another prime preserves the profile. All probabilities may vanish, giving the constant profile one. The theorem recovers local root counts; different polynomials may have the same such counts.

Corollary 21.51 (Complete cyclotomic local-law fingerprint)**.** *Fix $q \geq 2$, use the depleted root counts $\rho_q(p)$ above, and put $\theta_{q,p} = \rho_q(p)/p$. Then*

$$\mathcal{C}_q(z) = \prod_p (1 + \theta_{q,p}(z-1)) \exp(-\theta_{q,p}(z-1))$$

is entire, with simple zeros exactly at

$$\left\{ 1 - \frac{p}{\varphi(q)} : p \text{ prime},\ p \equiv 1 \pmod{q} \right\}.$$

For $q, r \geq 2$, equality of these zero sets, equality of the entire profiles, and $\rho_q(p) = \rho_r(p)$ at every prime are equivalent.

Proof. The explicit root count gives

$$0 \leq \theta_{q,p} \leq \frac{\varphi(q)}{p} \leq \frac{\varphi(q)}{p-1}, \qquad \theta_{q,p} < 1.$$

The squared probabilities are therefore summable. Apply Theorem 21.50. At an active prime the numerator is precisely $\varphi(q)$; $q \mid p-1$ already implies $p \nmid q$, giving the displayed zero set. □

For the actual scalar sampling experiment, the exact finite law gives

$$\mathcal{C}_q(z) = \lim_{N\to\infty} \exp\left(-(z-1)\sum_{p\le N}\theta_{q,p}\right)\mathbb{E}_N z^{\sum_{p\le N} B_p(U_N)}.$$

Each finite expectation uses its complete Chinese remainder period. These profile zeros lie in the generating variable z. The Dirichlet zeros in the preceding subsection occur in the Mellin continuation of the corresponding weighted prime count.

A profile determines the local law rather than a unique modulus: $q = 3$ and $q = 6$ give the same depleted counts, since the relevant primes are odd. In contrast, the laws for $q = 3$ and $q = 4$ differ at $p = 7$. Both facts are consistent with exact every-prime reconstruction.

Theorem 21.52 (Exact intersection of cyclotomic profile zero sets)**.** *Fix $q, r \ge 2$, and let $L = \operatorname{lcm}(q, r)$. If $\varphi(q) \ne \varphi(r)$, the zero sets of $\mathcal{C}_q$ and $\mathcal{C}_r$ are disjoint. If $\varphi(q) = \varphi(r) = D$, their intersection is exactly*

$$\mathcal{Z}(\mathcal{C}_q) \cap \mathcal{Z}(\mathcal{C}_r) = \left\{1 - \frac{p}{D} : p \text{ prime},\ p \equiv 1 \pmod{L}\right\}.$$

Proof. For general proper prime-denominator laws, suppose a zero contributed by $a_p > 0$ is also contributed by $b_\ell > 0$. Then $p/a_p = \ell/b_\ell$, hence $pb_\ell = \ell a_p$. As p is prime and $0 < a_p < p$, it follows that $p = \ell$ and then $a_p = b_p$. Conversely, an agreeing positive coordinate contributes the same zero to both profiles. Thus a common zero is equivalent to an agreeing positive count at the same prime.

At an active cyclotomic coordinate the counts are $\varphi(q)$ and $\varphi(r)$. They must therefore be equal. When both equal D, a prime contributes to both profiles exactly when $q \mid p-1$ and $r \mid p-1$, or equivalently $L \mid p-1$. Its zero is $1 - p/D$. Every prime satisfying this condition contributes to both, which proves the reverse inclusion as well. □

Theorem 21.53 (Exact shared-support factorization)**.** *Let $q, r \ge 2$, $L = \operatorname{lcm}(q, r)$ and $d_m = \varphi(m)$. Define the residual entire function*

$$R_{q\setminus r}(z) = \prod_{\substack{p \text{ prime}\\ q\mid p-1,\ r\nmid p-1}} \left(1 + \frac{d_q}{p}(z-1)\right)\exp\left(-\frac{d_q}{p}(z-1)\right).$$

Then, with the displayed compensation retained exactly,

$$\mathcal{C}_q(z) = \mathcal{C}_L\left(1 + \frac{d_q}{d_L}(z-1)\right)R_{q\setminus r}(z).$$

The analogous identity holds with q, r interchanged. Moreover,

$$R_{q\setminus r}(z) = 0 \quad \Longleftrightarrow \quad \exists\, p \text{ prime}: \quad q \mid p-1, \quad r \nmid p-1, \quad z = 1 - \frac{p}{d_q}.$$

The two residual functions have disjoint zero sets. If $d_q = d_r = D$, the common entire factor is

$$A_{q,r}(z) = \mathcal{C}_L\left(1 + \frac{D}{\varphi(L)}(z-1)\right),$$

and $\mathcal{C}_q = A_{q,r}R_{q\setminus r}$, $\mathcal{C}_r = A_{q,r}R_{r\setminus q}$.

Proof. The prime support $q \mid p-1$ splits into $L \mid p-1$ and $q \mid p-1,\ r \nmid p-1$. On the first part,

$$\eta_q(p) = \frac{d_q}{p} = \frac{d_q}{d_L}\,\eta_L(p).$$

For $B_a(z) = (1+a(z-1))\exp(-a(z-1))$ the exact identity $B_{ca}(z) = B_a(1+c(z-1))$ includes both the linear factor and the exponential. Squared local probabilities are summable, so the products of B_a converge locally uniformly. Equivalently, $\sum_p |B_{\eta_q(p)}(z) - 1|$ converges for every fixed z. Splitting this convergent product over the two complementary supports therefore proves the factorization, including at its zeros.

The residuals inherit square summability, take local probabilities in $[0, 1)$, and are entire. The zero theorem for compensated Bernoulli products gives the displayed residual zero set. If the two residuals shared a zero, their full profiles would share it as well. The preceding common-zero theorem forces $d_q = d_r$. Their residual prime labels p, t would then satisfy $1 - p/d_q = 1 - t/d_q$, hence $p = t$. One residual requires $r \nmid p - 1$, whereas the other requires $r \mid p - 1$, a contradiction. □

The support-partition identity is proved for arbitrary square-summable Bernoulli laws and arbitrary, possibly infinite supports. The scaling identity itself needs no convergence hypothesis, since its local factors agree pointwise. The actual cyclotomic factorization, residual entireness and residual zero statements above are kernel checked; no division by a profile at a zero is used.

The infinitude in these formulas is unconditional, by Dirichlet's theorem in the identity class. More quantitatively, write $\pi(X; m, 1)$ for the number of primes at most X that are 1 modulo m, and count profile zeros in $[-T, 0]$.

The exact formulas and the prime number theorem in a fixed progression give

$$N_q(T) = \pi(\varphi(q)(T+1); q, 1) \sim \frac{T}{\log T},$$
$$N_{q,r}(T) = \pi(D(T+1); L, 1) \sim \frac{D}{\varphi(L)} \frac{T}{\log T} \quad \big(\varphi(q) = \varphi(r) = D\big).$$

Here q, r are fixed and $T \to \infty$; these prime-counting consequences are ordinary deductions, not additional checked Lean declarations. Every cyclotomic profile has the same leading zero density, while the overlap records the intersection of progressions. For example, $q = 5$ and $r = 8$ have $D = 4$ and $L = 40$, so their shared zeros have relative density $1/4$. For $q = 3, r = 4$ it is $1/2$. Different totients instead give no shared profile zeros. These statements concern zeros in z, and introduce no new Riemann or Dirichlet zero-free estimate.

Theorem 21.54 (Uniform separation and stable prime-label recovery)**.** *Under the hypotheses of Theorem 21.50, suppose additionally that $a_p, b_p \le D$ for every prime, where $D \ge 1$ is an integer. Any two distinct zeros z, w drawn from the two profiles satisfy*

$$|z - w| \ge \frac{1}{D^2}.$$

More precisely, distinct proper prime/count labels (p, r) and (ℓ, s), with $1 \le r < p$, $1 \le s < \ell$ and $r, s \le D$, satisfy

$$\left|\left(1 - \frac{p}{r}\right) - \left(1 - \frac{\ell}{s}\right)\right| \ge \frac{1}{rs} \ge \frac{1}{D^2}.$$

Consequently, an observed complex location within strictly less than $1/(2D^2)$ of a zero can match at most one such prime/count label. The constants are independent of the height of the zero.

Proof. The prime-label argument shows that $ps - \ell r$ is a nonzero integer for distinct admissible labels. Its absolute value is at least one. Since $r, s > 0$, clearing denominators gives

$$\left|\left(1 - \frac{p}{r}\right) - \left(1 - \frac{\ell}{s}\right)\right| = \frac{|ps - \ell r|}{rs} \ge \frac{1}{rs} \ge \frac{1}{D^2}.$$

Every profile zero has one of these labels by the exact zero formula, which proves separation between any two distinct zeros, including zeros taken from

different profiles. If an observation were within the stated strict tolerance of two different labels, the triangle inequality would put their locations strictly less than $1/D^2$ apart, contradicting the bound. □

For the actual progression profiles, take $D = \max(\varphi(q), \varphi(r))$ for fixed $q, r \geq 2$. Then every distinct pair of zeros from $\mathcal{C}_q$ and $\mathcal{C}_r$ has distance at least D^{-2}. This is a uniform lower bound, with no optimality claim. It concerns accuracy of zero locations: a bound on the error in function values alone still needs a separate analytic estimate to control the displacement of zeros. The determinant inequality used in the proof is elementary; the conclusion here is its application to exact and approximate prime-labelled profile recovery.

The reconstruction, support-partition and actual cyclotomic results have thirty-three checked declarations in the maintained owners

```
Mathlib/Analysis/SpecialFunctions/Log/
  BernoulliProfilePartition.lean
Mathlib/NumberTheory/Primes/BernoulliProfile.lean
Helpers/CyclotomicProfileRigidity.lean
```

under the project namespace. Their audits contain only the standard foundational axioms. The simple-zero assertion follows from the zero-factor argument above and is not a separately named declaration in this batch. All comparisons here use the displayed probability-centred compensation.

21.28 Special integers: reciprocal-divisor fingerprints

Directly sampling a special sequence and sampling divisors of one of its members are different experiments. The latter gives a single exact probability model for every positive integer, without an infinitude assumption on the class from which that integer is selected. For $n \geq 1$, choose $d \mid n$ with probability

$$\mathbb{P}_n(d) = \frac{d^{-1}}{\sum_{e|n} e^{-1}}, \qquad P_n(z) = \mathbb{E}_n z^{\omega(d)}.$$

Here $\omega(d)$ counts distinct prime factors and $\omega(1) = 0$. The divisor involution $d \mapsto n/d$ gives $\sum_{d|n} d^{-1} = \sigma(n)/n$, so $P_n(0) = n/\sigma(n)$.

Theorem 21.55 (Complete prime-power recovery and unitary divisibility)**.** *For positive integers n, m, the zero set of P_n determines n. More precisely, put*

$$S_{p,a} = 1 + p + \cdots + p^{a-1}, \qquad a \geq 1.$$

The zeros of P_n are exactly the distinct simple numbers

$$-\frac{p^a}{S_{p,a}} \qquad (p^a \| n).$$

Consequently,

$$\begin{aligned} Z(P_n) = Z(P_m) \quad &\iff \quad n = m, \\ Z(P_n) \subseteq Z(P_m) \quad &\iff \quad \exists\, k \geq 1 :\ m = nk,\ \gcd(n,k) = 1. \end{aligned}$$

For $n = 1$ the zero set is empty. Both conclusions also hold for the canonically compensated profiles

$$\mathcal{D}_n(z) = e^{-\mu_n(z-1)} P_n(z), \qquad \mu_n = \sum_{p^a \| n} \frac{S_{p,a}}{p^a + S_{p,a}}.$$

Proof. Under reciprocal divisor weight the exponent choices at distinct primes are independent, since both the support and the weight factor:

$$P_n(z) = \prod_{p^a \| n} \frac{1 + z\sum_{j=1}^{a} p^{-j}}{1 + \sum_{j=1}^{a} p^{-j}} = \prod_{p^a \| n} \frac{p^a + S_{p,a} z}{p^a + S_{p,a}}.$$

Each linear factor has the displayed zero. Since $S_{p,a} \equiv 1 \pmod{p}$, the fraction $p^a / S_{p,a}$ is reduced. Its positive numerator therefore identifies the prime p and the exponent a. In particular, distinct prime powers from different prime coordinates cannot contribute the same zero. The product has exactly one factor for each positive prime valuation, so all its zeros are simple. Its zero set recovers every positive valuation and hence the integer.

Inclusion of the zero sets says that every prime dividing n occurs in m with exactly the same exponent. This is equivalent to $m = nk$ with no prime common to n and k. Conversely, such a coprime factorization retains precisely those prime-power factors of P_n in P_m. Finally the exponential compensation never vanishes, so it changes neither zero set. □

The profile itself is a finite compensated Bernoulli product with success probabilities $S_{p,a}/(p^a + S_{p,a})$. The maintained Lean results prove its exact zero formula, recover arbitrary square-summable prime-indexed exponent functions, discharge summability for every integer factorization, and prove the two integer equivalences for $\mathcal{D}_n$, including unitary divisibility. Their transfer to P_n uses the ordinary product identity above. There are fourteen checked declarations in

`Mathlib/NumberTheory/Primes/BernoulliDivisorProfile.lean`
`Mathlib/NumberTheory/Primes/ReciprocalDivisorInteger.lean`

under the project namespace, with only standard foundational axioms. The finite weighted-divisor sampling identity in the displayed product is proved ordinarily here; it is not yet a separately named Lean sampling adapter. No priority claim is made for this inverse statement.

This fingerprint retains substantially more information than abundancy alone. For example, both 6 and 28 are perfect, so both have $P_n(0) = 1/2$, but their zero sets are respectively $\{-2, -3\}$ and $\{-4/3, -7\}$. Zero-set inclusion also differs from ordinary divisibility: $2 \mid 4$, but the zero -2 of P_2 is absent from P_4, whose zero is $-4/3$.

The classical RH criterion in the same model. Robin's criterion, together with its colossally abundant counterexample reduction, gives

$$\text{RH} \quad \Longleftrightarrow \quad P_n(0) > \frac{e^{-\gamma}}{\log\log n} \quad \text{for every integer } n \geq 5041.$$

One may equivalently quantify over just the colossally abundant integers $n \geq 5041$, or over just the superabundant integers in that range: a counterexample exists among the former if RH fails, and they are contained in the latter. These are translations of the classical criterion, not new independent estimates [12, Section 2 and equation (1.2)]. The denominator is positive on the stated range. Compensation must be undone before reading this probability: $P_n(0) = e^{-\mu_n}\mathcal{D}_n(0)$, rather than $\mathcal{D}_n(0)$ itself.

What the named classes contribute. The common divisor model above applies to every existing member of each class. Its relation to direct sampling is as follows.

Sophie Germain primes. For primes p with $2p + 1$ prime, direct counts are $\omega(p) = 1$ and $\omega(p(2p + 1)) = 2$. They have no growing distinct-factor rank regime. No infinitude assumption is made. The integer fingerprint is valid for each such p or product; conditioning a polynomial sieve family on both outputs being prime needs its own sampling theorem.

Fermat and Mersenne primes. Use the precise values $2^{2^j} + 1$ with $j \geq 0$, or $2^j - 1$ with $j \geq 2$, only when prime. Their individual divisor profiles each have one zero, at the negative prime itself. This neither

assumes infinitely many such primes nor transfers an index cutoff to a divisor cutoff. Uniform residue sampling of a cyclotomic polynomial is a separate experiment from evaluating it at the fixed base 2.

Even and odd perfect numbers. Every hypothetical or actual perfect integer has $P_n(0) = 1/2$. For even perfect numbers the Euclid–Euler form specifies the two prime coordinates and their exponents; the full profile retains them. For odd perfect numbers the statement is conditional on a witness, without assuming one exists. At $n \geq 5041$ the value $1/2$ already satisfies Robin's probability inequality, so restricting that inequality to perfect numbers supplies no stronger RH criterion.

Superabundant and colossally abundant numbers. Under the strict record convention, superabundant integers are exactly the strict lower records of $P_n(0)$. For a fixed $\epsilon > 0$, a colossally abundant optimizer minimizes $n^\epsilon P_n(0)$; all maximizing ties in the usual non-strict definition are retained. These classes lie in the actual extremal regime of the RH criterion. Exact reconstruction does not estimate their probability relative to the critical threshold.

Fibonacci values. Take $F_0 = 0, F_1 = 1$ and values F_j for $j \geq 2$, distinguishing all Fibonacci values from Fibonacci primes. Each positive value has the exact divisor fingerprint. Uniform index sampling instead gives local divisibility through ranks of apparition; simultaneous events are controlled by the least common multiple of those ranks, so independence requires a separate justification. The index cutoff, the value F_j, and a prime-divisor cutoff are distinct parameters.

Primorials, factorials and highly composite numbers. They also fit the finite model. A primorial uses exponent one at each supported prime, whereas factorials and the extremal classes retain nontrivial exponents. Direct sampling of these sequences can force every fixed small prime in the limit, so a finite rank shift does not justify applying a positive-rank prime clock to that direct measure.

The zero-set theorem is uniform over positive integers and requires no class-specific existence theorem. An RH implication still needs the appropriate independent inequality or a proved transfer of its probability law; injective encoding alone supplies neither.

21.29 Fibonacci carriers and the golden-ratio clock

The Fibonacci sequence supplies an exact prime clock through its *indices*. Set $F_0 = 0, F_1 = 1$, and, for each odd prime ℓ, let P_ℓ be the least prime factor of F_ℓ. This exists because $F_\ell > 1$ for $\ell \geq 3$. The sampled variable is the integer index n; the event is $P_\ell \mid F_n$. A label cutoff X means $\ell \leq X$. It is neither a bound on P_ℓ nor a bound on the value F_n.

Theorem 21.56 (A prime-index carrier of a strongly divisible sequence)**.** *Suppose $a_1 = 1$ and $\gcd(a_m, a_n) = a_{\gcd(m,n)}$ for a sequence of nonnegative integers. If ℓ is prime and $c \neq 1$ divides a_ℓ, then*

$$c \mid a_n \quad \Longleftrightarrow \quad \ell \mid n \qquad (n \geq 0).$$

In particular, for every odd prime ℓ and every prime divisor c of F_ℓ this equivalence holds with $a_n = F_n$. The selected primes P_ℓ are pairwise distinct. For every finite set S of odd primes and arbitrary marks z_ℓ,

$$\prod_{\ell \in S} z_\ell^{\mathbf{1}_{P_\ell \mid F_n}} = \prod_{\ell \in S} z_\ell^{\mathbf{1}_{\ell \mid n}}.$$

The product convention means a factor 1 at an unsuccessful event, including when a mark is zero.

Proof. If $\ell \nmid n$, primality gives $\gcd(\ell, n) = 1$. A common divisor c of a_ℓ and a_n would then divide $a_1 = 1$, contradicting $c \neq 1$. If $\ell \mid n$, the gcd identity instead gives $\gcd(a_\ell, a_n) = a_\ell$, so $a_\ell \mid a_n$ and hence $c \mid a_n$. For Fibonacci numbers use their strong divisibility identity $\gcd(F_m, F_n) = F_{\gcd(m,n)}$. If $P_\ell = P_q$ at two prime labels, the equivalence at $n = q$ gives $\ell \mid q$, hence $\ell = q$. Finally, each individual indicator in the two products is equal. □

The real closed form and its alternating correction. Write

$$\varphi = \frac{1 + \sqrt{5}}{2}, \qquad \psi = \frac{1 - \sqrt{5}}{2} = -\varphi^{-1}.$$

Binet's formula and its exact relative error are

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}} = \frac{\varphi^n}{\sqrt{5}}\big(1 - (-1)^n \varphi^{-2n}\big),$$
$$F_n - \frac{\varphi^n}{\sqrt{5}} = \frac{(-1)^{n+1}\varphi^{-n}}{\sqrt{5}}.$$

Both powers satisfy the Fibonacci recurrence, and their difference divided by $\varphi - \psi = \sqrt{5}$ has initial values $0, 1$; this proves the formula. Its absolute error is at most $1/\sqrt{5} < 1/2$, so F_n is the nearest integer to $\varphi^n/\sqrt{5}$ for every $n \geq 0$. The sign alternates with parity and the relative error is φ^{-2n}. This real approximation explains the exponential value scale. The divisibility clock uses the same quadratic equation over finite fields, as follows.

Theorem 21.57 (Golden-root order and Fibonacci divisibility)**.** *For roots u, v of $T^2 - T - 1$ in any commutative ring,*

$$(u - v)F_n = u^n - v^n.$$

In a field, if $v = 1 - u$ and $u \neq v$, then

$$F_n = 0 \quad \Longleftrightarrow \quad \left(\frac{u}{v}\right)^n = 1 \quad \Longleftrightarrow \quad (-u^2)^n = 1.$$

Here F_n is interpreted by the natural-number map into the field.

For an odd prime $r \neq 5$, choose $u = (1 + \sqrt{5})/2$ in $\mathbb{F}_{r^2}$ and put $\chi_5(r) = (5/r)$. The Fibonacci entry point $\mathfrak{z}(r) = \min\{n \geq 1 : r \mid F_n\}$ satisfies

$$\mathfrak{z}(r) = \operatorname{ord}(-u^2), \qquad \mathfrak{z}(r) \mid r - \chi_5(r).$$

The exceptional primes have the exact rules

$$2 \mid F_n \Longleftrightarrow 3 \mid n, \qquad 5 \mid F_n \Longleftrightarrow 5 \mid n.$$

Thus $\mathfrak{z}(2) = 3$, $\mathfrak{z}(5) = 5$, $P_3 = 2$ and $P_5 = 5$.

Proof. The ring identity follows from the initial values at $n = 0, 1$ and the recurrence: $u^2 = u + 1$ and $v^2 = v + 1$ ensure that both sides satisfy the same second-order recursion. For $v = 1 - u$, these identities give $uv = -1$, so $v \neq 0$ and $u/v = -u^2$. With $u - v \neq 0$, the ring identity can be divided by $u - v$ and v^n, proving the power test.

For $r \neq 2, 5$, the discriminant 5 is nonzero and the two roots in $\mathbb{F}_{r^2}$ are distinct. If $\chi_5(r) = 1$, both roots belong to $\mathbb{F}_r$, so their nonzero quotient has order dividing $r - 1$. If $\chi_5(r) = -1$, the Frobenius automorphism interchanges the two roots. For $t = u/v$ this gives $t^r = v/u = t^{-1}$, hence $t^{r+1} = 1$. In either case the least positive exponent giving $t^n = 1$ is exactly the least index with $r \mid F_n$.

Modulo 2, the successive pairs in the Fibonacci recurrence cycle through $(0, 1), (1, 1), (1, 0)$, yielding precisely the indices divisible by 3. Modulo 5 the roots coalesce at 3; direct substitution in the recurrence gives $F_n = n\, 3^{n-1}$ for $n \geq 1$, with $n = 0$ handled separately. Since 3 is invertible, vanishing is equivalent to $5 \mid n$. The selected carriers follow from $F_3 = 2$ and $F_5 = 5$. □

The use of finite fields is through the integral quadratic identity, not a reduction map from the real field. For the selected carrier and any odd prime label $\ell \neq 5$, the appropriate distinct-root field therefore gives $\operatorname{ord}(-u^2) = \ell$. At $P_3 = 2$ one may use a root of $T^2 + T + 1$ in $\mathbb{F}_4$; its quotient has order 3. At $P_5 = 5$ the repeated-root formula supplies the clock instead. This explicitly identifies the two exceptional labels.

For $\ell > 5$ the carrier is odd and different from 5. The order bound consequently gives

$$\ell \mid P_\ell - \chi_5(P_\ell), \qquad 2\ell - 1 \leq P_\ell \leq F_\ell < \varphi^\ell.$$

Indeed, $P_\ell \pm 1$ is even, so its positive multiple of the odd prime ℓ is a multiple of 2ℓ. The upper bound uses $P_\ell \mid F_\ell$ and Binet's formula. These inequalities connect the real growth scale and the modular clock while keeping label order and carrier magnitude distinct.

Theorem 21.58 (The marked Fibonacci law and an inherited RH criterion)**.** *For a finite set S of odd primes put $M_S = \prod_{\ell \in S} \ell$. Uniform index sampling over $0 \leq n < M_S$ gives, for every binary pattern $b : S \to \{0, 1\}$,*

$$\mathbb{P}\big(\mathbf{1}_{P_\ell \mid F_n} = b_\ell \text{ for all } \ell \in S\big) = \prod_{\ell \in S} \left(\frac{1}{\ell}\right)^{b_\ell} \left(1 - \frac{1}{\ell}\right)^{1 - b_\ell}.$$

The empty support is allowed, with period 1. For $N \geq 1$ and sampling over $1 \leq n \leq N$, the probability of any fixed pattern differs from this product by at most M_S/N.

For real $X \geq 3$ let S_X be the odd primes at most X, $M_X = M_{S_X}$, and

$$A(X) = \lim_{N \to \infty} \frac{1}{N} \#\{1 \leq n \leq N : P_\ell \nmid F_n \text{ for every } \ell \in S_X\}.$$

Then

$$A(X) = \prod_{\ell \in S_X} (1 - 1/\ell),$$

and the classical Nicolas criterion gives the exact equivalence

$$\text{RH} \quad \Longleftrightarrow \quad A(X) < \frac{2e^{-\gamma}}{\log\log(2M_X)} \quad \text{for all sufficiently large } X.$$

Proof. The carrier theorem replaces every Fibonacci event by $\ell \mid n$. The Chinese remainder theorem identifies a uniform residue modulo M_S with

independent uniform residues modulo the distinct labels. There is one successful residue at each label and $\ell - 1$ unsuccessful residues, proving every pattern probability. For arbitrary $N = kM_S + t$, $0 \le t < M_S$, there are k complete periods and one remainder. If a residues in a full period give the pattern, its total count is $ka + r$ with $0 \le r \le t$. The difference from Na/M_S is $r - ta/M_S$, of absolute value at most $t < M_S$. This proves the stated error and the fixed-cutoff limit. Divisibility at $F_0 = 0$ is well-defined; this sampling argument does not apply a reciprocal-divisor measure to zero.

For $X \ge 3$, the full primorial is $2M_X$, so $\theta(X) = \log(2M_X) > 1$ and

$$f(X) = e^{\gamma} \log \theta(X) \prod_{p \le X} (1 - 1/p) = \frac{e^{\gamma}}{2} \log\log(2M_X)\, A(X).$$

Every factor outside $A(X)$ is positive. Thus the displayed probability inequality is exactly $f(X) < 1$, or $F(X) < 0$, with f, F as in (18). The Nicolas equivalence already used above is precisely eventual $F(X) < 0$ [7, 8]. □

The order of limits is essential: S_X is fixed while $N \to \infty$, and only afterward does $X \to \infty$. The explicit sampling error M_X/N gives no unproved uniform growing-cutoff assertion. The finite generating polynomial is

$$G_X(z) = \prod_{\ell \in S_X} (1 + (z - 1)/\ell).$$

After canonical compensation the profiles

$$\mathcal{F}_X(z) = \prod_{\ell \in S_X} (1 + (z - 1)/\ell) e^{-(z-1)/\ell}$$

converge locally uniformly to an entire function, since $\sum_{\ell} \ell^{-2} < \infty$. The existing Bernoulli profile zero theorem gives zeros exactly $1 - \ell$ at the odd prime labels. These zeros encode index labels; the naturally ordered, unmarked prime factors of F_n form a different observable.

The maintained candidate owners

```
Mathlib/NumberTheory/Primes/StrongDivisibilityCarrier.lean
Mathlib/NumberTheory/Primes/FibonacciGoldenCarrier.lean
```

check twenty-two declarations: the generic carrier and its exact finite count/product transport, the ring Binet identity, the field power tests, prime-index orders, and the two exceptional integer clocks. Six more checked declarations in

reuse the existing CRT equivalence to identify the actual marked events and prove their full finite-period joint law. All twenty-eight axiom audits contain only the standard foundational axioms. The finite-field existence and Frobenius classification, interval remainder estimate, and displayed Nicolas equivalence are ordinary proofs here. Entry-point arithmetic and finite-field torus descriptions are classical; see Cubre–Rouse [15]. The contribution here is their exact use as a marked probability carrier and the explicit transfer to this existing RH criterion. No infinitude of Fibonacci primes is assumed, and no improved zero-free exponent follows without a new independent estimate.

21.30 Actual Fermat primes: finite or infinite

Here the prime family is exactly

$$T_j = 2^{2^j} + 1, \qquad J = \{j \geq 0 : T_j \text{ is prime}\}, \qquad \mathcal{P}_F = \{T_j : j \in J\}.$$

No assertion about the finiteness of J is assumed. This family differs from choosing one prime factor of every T_j: a composite Fermat number contributes nothing to $\mathcal{P}_F$. The distinction matters for the RH criterion. In the joint Fibonacci–Fermat construction below, the RH information comes from all odd prime labels in the Fibonacci block; the nested Fermat block contributes a fixed vacancy factor $1/2$. That construction is not an RH equivalence for actual Fermat primes by themselves.

Theorem 21.59 (Actual Fermat-prime avoidance and deficiency)**.** *Whether J is finite or infinite, the product*

$$b_F = \prod_{j \in J} (1 - 1/T_j)$$

converges and is positive. It is the natural density of positive integers divisible by no actual Fermat prime. More precisely,

$$\frac{2^{32}+1}{2^{33}} \;\leq\; b_F \;\leq\; \frac{2^{31}}{2^{32}-1}, \qquad \frac{2^{31}}{2^{32}-1} - \frac{2^{32}+1}{2^{33}} = \frac{1}{2^{33}(2^{32}-1)}.$$

Every positive integer n whose prime divisors all belong to $\mathcal{P}_F$ satisfies

$$\frac{\sigma(n)}{n} < 2, \qquad P_n(0) > \frac{1}{2}.$$

Consequently restricting Robin's inequality to these integers $n \geq 5041$ gives a statement already true unconditionally, rather than an RH equivalence.

Proof. The finite Fermat product telescopes:

$$\prod_{j<k} T_j = 2^{2^k} - 1, \qquad \prod_{j<k} (1 - 1/T_j) = \frac{2^{2^k-1}}{2^{2^k}-1}, \qquad k \geq 0.$$

The empty product at $k = 0$ is 1. Thus the product over *all* Fermat numbers tends to $1/2$, and deleting factors in $(0, 1)$ can only increase it. Also $\sum_j T_j^{-1} < \infty$, so the subproduct defining b_F converges positively. The first five Fermat numbers are prime, whereas $T_5 = 641 \cdot 6700417$ is composite. Retaining the first five factors gives the upper bound. Retaining every factor except $j = 5$ gives the lower bound $\frac{1}{2}(1 - 1/T_5)^{-1} = (2^{32} + 1)/2^{33}$. Subtracting gives the displayed interval width. These elementary facts about Fermat numbers are recalled in the introduction to [17]; no later primality decisions are used.

For any fixed finite prime cutoff, the Chinese remainder theorem gives the avoidance density as its finite product. The upper density of positive integers divisible by any omitted Fermat prime is at most the reciprocal tail: for each $N \geq 1$,

$$\frac{1}{N}\#\{1 \leq n \leq N : \exists p \in \mathcal{P}_F,\ p > X,\ p \mid n\} \leq \sum_{\substack{p \in \mathcal{P}_F \\ X < p \leq N}} \frac{1}{p} \leq \sum_{\substack{p \in \mathcal{P}_F \\ p > X}} \frac{1}{p}.$$

The final tail tends to zero, proving the full natural density without exchanging two unproved sampling limits.

Finally write $n = \prod_{j \in S} T_j^{a_j}$, where S is a finite subset of J and $a_j \geq 1$. Choose k with $S \subseteq \{0, \ldots, k-1\}$. The divisor-sum product gives

$$\frac{\sigma(n)}{n} = \prod_{j \in S} \frac{1 - T_j^{-(a_j+1)}}{1 - T_j^{-1}} \leq \prod_{j<k} (1 - 1/T_j)^{-1} = 2 - 2^{1-2^k} < 2.$$

This also covers $n = 1$ with $k = 0$. Since $P_n(0) = n/\sigma(n)$, its vacancy is greater than $1/2$. On $n \geq 5041$, $e^{-\gamma}/\log\log n < 1/2$, as used in the perfect-number discussion above, so Robin's probability inequality follows with no RH hypothesis. □

This proves that even allowing arbitrary powers and arbitrarily many actual Fermat primes does not reach the large-abundancy regime responsible for Robin counterexamples. It does not prove that J is finite, infinite, or independent of RH.

Theorem 21.60 (The Fermat Euler factor preserves the zeta zeros)**.** *For* $\Re s > 0$ *define*

$$H_F(s) = \prod_{p \in \mathcal{P}_F} (1 - p^{-s}).$$

This is holomorphic and nowhere zero on that half-plane, with $H_F(1) = b_F$*. The meromorphic function*

$$\zeta_{\overline{F}}(s) = \zeta(s) H_F(s)$$

has exactly the same zeros, with the same multiplicities, as ζ *in* $\Re s > 0$*, and still has a simple pole at* 1*. In* $\Re s > 1$ *it is the Euler product over primes that are not Fermat primes. In particular,*

$$\text{RH} \quad \Longleftrightarrow \quad \zeta_{\overline{F}}(s) \neq 0 \quad (\Re s > 1/2,\ s \neq 1).$$

This holds in both the finite and infinite cases for J*.*

Proof. For every $\sigma > 0$,

$$\sum_{j \in J} T_j^{-\sigma} \leq \sum_{j \geq 0} 2^{-\sigma 2^j} \leq \sum_{j \geq 0} 2^{-\sigma(j+1)} < \infty.$$

On a compact subset of $\Re s > 0$ choose a lower bound $\Re s \geq \sigma > 0$. The factors p^{-s} have modulus at most $p^{-\sigma} < 1$, and the displayed sum gives locally uniform absolute convergence of $\sum_p \log(1 - p^{-s})$ using the branch defined by its power series in the unit disk. Its exponential is H_F, proving holomorphy and nonvanishing. This argument does not distinguish a finite subproduct from an infinite one.

The ordinary Euler product of ζ in $\Re s > 1$ gives the asserted complementary product. Multiplication by a holomorphic unit preserves every zero and its order, and $H_F(1) = b_F > 0$ preserves the pole. The final equivalence is the usual zero-free half-plane formulation of RH. □

Thus the Fermat-prime Euler product H_F^{-1} itself is already zero-free throughout $\Re s > 0$. The displayed RH equivalence retains the complementary primes; it is not a criterion involving only the number or spacing of Fermat primes.

Theorem 21.61 (An infinitude-free complementary probability criterion)**.** *For* $X \geq 3$ *put*

$$B_F(X) = \prod_{\substack{p \leq X \\ p \in \mathcal{P}_F}} (1 - 1/p), \quad A_{\overline{F}}(X) = \prod_{\substack{p \leq X \\ p \notin \mathcal{P}_F}} (1 - 1/p), \quad \theta_{\overline{F}}(X) = \sum_{\substack{p \leq X \\ p \notin \mathcal{P}_F}} \log p,$$

where every p in these finite products and sums is prime. Then, without an assumption on the size of J,

$$\text{RH} \quad \Longleftrightarrow \quad A_{\overline{F}}(X) < \frac{e^{-\gamma}}{b_F \log \theta_{\overline{F}}(X)} \quad \textit{for all sufficiently large } X.$$

The product $A_{\overline{F}}(X)$ is the fixed-cutoff limiting probability of avoiding all non-Fermat primes at most X in uniform positive-integer sampling. If $f(X)$ is the original Nicolas function and

$$f_{\overline{F}}(X) = e^{\gamma} b_F \log \theta_{\overline{F}}(X) A_{\overline{F}}(X),$$

then an absolute eventual error bound is

$$0 \le \log f(X) - \log f_{\overline{F}}(X) \le \frac{19}{X}.$$

The constant 19 is explicit; the starting threshold here is only asserted to exist, not numerically specified.

Proof. Let m be the least label for which $T_m > X$. Telescoping the full Fermat-number product gives

$$1 - \frac{1}{T_m - 1} = \prod_{j \ge m} (1 - 1/T_j) \le \frac{b_F}{B_F(X)} \le 1.$$

Consequently, for $X \ge 3$,

$$0 \le \log B_F(X) - \log b_F \le -\log(1 - 1/(X-1)) \le \frac{3}{X}.$$

For $r_F(X) = \theta(X) - \theta_{\overline{F}}(X)$, the other finite product identity gives

$$0 \le r_F(X) \le \sum_{j<m} \log T_j = \log(T_m - 2) < 2 \log T_{m-1} \le 2 \log X.$$

Here $m \ge 1$ because $X \ge 3$. The unconditional prime number theorem implies that, eventually, $\theta(X) \ge X/2$, $2 \log X \le X/4$, and $\log(X/4) \ge \frac{1}{2} \log X$. Therefore $\theta_{\overline{F}}(X) \ge X/4 > 1$ and

$$0 \le \log\log \theta(X) - \log\log \theta_{\overline{F}}(X) \le \frac{r_F(X)}{\theta_{\overline{F}}(X) \log \theta_{\overline{F}}(X)} \le \frac{16}{X}.$$

All denominators used in the criterion are positive in this eventual range. Since the full prime product is $B_F(X) A_{\overline{F}}(X)$, exact subtraction gives

$$\log f - \log f_{\overline{F}} = \log B_F - \log b_F + \log\log \theta - \log\log \theta_{\overline{F}}.$$

The preceding estimates prove the claimed signed error bound.

Under RH, Nicolas's criterion gives $\log f(X) < 0$ eventually; the non-negative subtraction then gives $\log f_{\overline{F}}(X) < 0$. If RH fails, the same source gives positive excursions $\log f(X) \geq cX^{-b}$ at arbitrarily large X, for some constants $c > 0$ and $0 < b < 1/2$ [8, equation (1.10)]. For sufficiently large points on that excursion sequence, $cX^{-b} > 19/X$, hence $\log f_{\overline{F}}(X) > 0$. The eventual inequality is therefore false. This proves both directions without assuming finitely or infinitely many Fermat primes, and without using an RH-dependent reconstruction in the non-RH direction. □

These are ordinary proofs in this section, not new Lean declarations or claims of a new RH estimate. The last criterion is an application of the existing Nicolas excursion theorem with a proved $O(X^{-1})$ deletion error. If J is finite, $B_F(X)$ eventually equals the finite constant b_F; if J is infinite it approaches the same well-defined positive constant without stabilizing. Neither case changes the equivalence. No implication from RH to finiteness or infinitude of actual Fermat primes has been established here, and the certified zero-free boundary remains $7/8$.

21.31 Which prime classes may be removed?

The preceding argument depends on the size of the removed prime class, rather than its being a Fermat family. The following gives a sufficient condition, not a classification of all possible zero-preserving restrictions.

Theorem 21.62 (Critical-exponent prime deletion)**.** *Let S be a fixed set of primes. Suppose that for every $\epsilon > 0$ there are constants $C_\epsilon > 0$ and $X_\epsilon \geq 2$ such that*

$$\pi_S(X) = \#\{p \in S : p \leq X\} \leq C_\epsilon X^{1/2+\epsilon} \qquad (X \geq X_\epsilon).$$

Then

$$H_S(s) = \prod_{p \in S} (1 - p^{-s})$$

is holomorphic and nowhere zero for $\Re s > 1/2$. Thus $\zeta(s)H_S(s)$ retains exactly the zeros of ζ, with their multiplicities, in that half-plane, as well as its pole at 1.

Let C denote the complementary prime set, and define

$$b_S = \prod_{p \in S} (1 - 1/p) > 0, \quad A_C(X) = \prod_{\substack{p \in C \\ p \leq X}} (1 - 1/p), \quad \theta_C(X) = \sum_{\substack{p \in C \\ p \leq X}} \log p.$$

There is an exact probability version of the equivalence:

$$\text{RH} \quad \Longleftrightarrow \quad A_C(X) < \frac{e^{-\gamma}}{b_S \log \theta_C(X)} \quad \textit{for all sufficiently large } X.$$

Here $A_C(X)$ is the limiting avoidance probability for uniform sampling of positive integers at each fixed cutoff X. No existence or infinitude assumption on the class S is needed.

Proof. For every $\sigma > 1/2$, choose $\epsilon > 0$ with $1/2 + \epsilon < \min(\sigma, 1)$. Partial summation gives $\sum_{p \in S} p^{-\sigma} < \infty$. The logarithmic product argument in Theorem 21.60 therefore gives the claimed holomorphic unit. In particular, the product b_S converges positively.

We record the complete signed error for the probability criterion. Write

$$B_S(X) = \prod_{\substack{p \in S \\ p \le X}} (1 - 1/p), \qquad r_S(X) = \sum_{\substack{p \in S \\ p \le X}} \log p,$$
$$f_C(X) = e^{\gamma} b_S \log \theta_C(X) A_C(X).$$

Whenever $\theta_C(X) > 1$, the exact identity is

$$\begin{aligned} \Delta_S(X) &= \log f(X) - \log f_C(X) \\ &= \sum_{\substack{p \in S \\ p > X}} -\log(1 - 1/p) + \log\log\theta(X) - \log\log(\theta(X) - r_S(X)) \ge 0. \end{aligned}$$

Both the reciprocal tail and the change of the theta normalizer are retained. For a quantitative bound, suppose $\pi_S(t) \le C_0 t^{\alpha}$ on $t \ge X_0$, where $C_0 > 0$ and $0 \le \alpha < 1$. For $X \ge \max(2, X_0)$,

$$\sum_{\substack{p \in S \\ p > X}} -\log(1 - 1/p) \le 2 \sum_{\substack{p \in S \\ p > X}} \frac{1}{p} \le \frac{2C_0}{1 - \alpha} X^{\alpha - 1}, \qquad r_S(X) \le C_0 X^{\alpha} \log X.$$

Eventually the unconditional prime number theorem and $r_S(X) = o(X)$ give $\theta_C(X) \ge X/4$ and $\log \theta_C(X) \ge \frac{1}{2} \log X$. Hence

$$0 \le \Delta_S(X) \le C_0 \left(8 + \frac{2}{1 - \alpha} \right) X^{\alpha - 1} \quad \text{eventually.}$$

Every denominator is positive in this range. The numerical coefficient is explicit; the threshold depends on the stated counting bound and is not specified numerically here.

Under RH, $\log f < 0$ eventually and $\Delta_S \geq 0$, so $\log f_C < 0$. If RH fails, the existing Nicolas theorem supplies some $c > 0$ and $0 < b < 1/2$ with $\log f(X) \geq cX^{-b}$ at arbitrarily large X. Choose ϵ with $0 < \epsilon < 1/2 - b$ and apply the last bound with $\alpha = 1/2 + \epsilon$. Then $\Delta_S(X) = o(X^{-b})$, so $\log f_C(X) > 0$ at arbitrarily large points of the same sequence. This contradicts the eventual inequality and proves the converse. The CRT proves the fixed-cutoff probability interpretation. □

Finite unions of classes satisfying the hypothesis can be deleted simultaneously. Examples include actual Fermat primes, Mersenne primes and Fibonacci primes: their counting functions are at most logarithmic, with no assumed infinitude. For a fixed integer polynomial P of degree $d \geq 2$ and positive leading coefficient, the primes among $P(n)$, $n \geq 0$, have counting function $O_P(X^{1/d})$ and can also be deleted. These are prime *values*, not all prime divisors of those values. Infinite unions or varying polynomial parameters require additional uniform estimates. The theorem alone does not cover a positive-density residue class, nor does it supply the needed bound for Sophie Germain primes. Failure of this sufficient counting condition is not an impossibility theorem for those classes.

Theorem 21.63 (All Fermat factors form a deletable class)**.** *Let*

$$\mathcal{D}_F = \{p \text{ prime} : p \mid 2^{2^j} + 1 \text{ for some } j \geq 0\}.$$

This is an infinite set. For every real $X \geq 3$,

$$\pi_{\mathcal{D}_F}(X) \leq (1 + 6\log 2)\sqrt{\frac{X}{\log X}}.$$

Consequently every prime in $\mathcal{D}_F$ can be deleted simultaneously in Theorem 21.62. The result concerns all prime factors of all Fermat numbers, regardless of whether there are finitely or infinitely many actual Fermat primes.

Proof. Let m_j be the number of distinct prime factors of T_j, and $W_k = \sum_{j<k} m_j$. The proved factor-count and cumulative weight estimates give

$$(j+1)m_j \leq 2^j, \qquad (k+1)W_k \leq 3 \cdot 2^k.$$

Their maintained providers are `Nat.fermatFactorCount_mul_le_pow` and `Nat.dyadicPrefixWeight_mul_le_three_pow`. Every factor has multiplicative order 2^{j+1} for base 2. Lagrange's theorem in the nonzero residue group

therefore gives $2^{j+1} \mid p-1$. In particular, all factors from labels $j \geq k$ lie in the *single* progression $p \equiv 1 \pmod{2^{k+1}}$. Since $p > 1$, there are at most $X/2^{k+1}$ such primes up to X. Thus the finite covering bound, including the low-label factors, is

$$\pi_{\mathcal{D}_F}(X) \leq W_k + \frac{X}{2^{k+1}} \qquad (k \geq 0).$$

This bound does not sum overlapping progression counts.

Set $Y = \sqrt{X \log X}$ and choose $k \geq 0$ with $2^k \leq Y < 2^{k+1}$. Since $X \geq 3$,

$$k + 1 > \frac{\log Y}{\log 2} \geq \frac{\log X}{2 \log 2}.$$

The low part is at most $6 \log 2 \sqrt{X/\log X}$, while the high part is at most $X/Y = \sqrt{X/\log X}$. Adding proves the stated bound. Every Fermat number has a prime factor, and distinct labels have disjoint prime-factor supports, so $\mathcal{D}_F$ is infinite. The counting bound implies the hypothesis of Theorem 21.62. □

The finite covering and its optimized real bound are ordinary consequences here of the named checked weight estimates and the exact order theorem. They are not yet separately compiled declarations. No improved zero-free exponent is inferred.

Theorem 21.64 (Deletion by smooth multiplicative order)**.** *Fix an integer $a \geq 2$ and a finite set B of primes. Let $S_{a,B}$ be the primes $p \nmid a$ for which every prime factor of $\operatorname{ord}_p(a)$ belongs to B; order 1 is allowed. For $0 < \eta < 1$ set*

$$C_B(\eta) = \prod_{q \in B} (1 - q^{-\eta})^{-1}.$$

Then, for every $X \geq 2$,

$$\pi_{S_{a,B}}(X) \leq \left(1 + \frac{\log a}{\log 2}\right) C_B(\eta) X^{(1+\eta)/2}.$$

In particular $S_{a,B}$ satisfies the critical-exponent deletion criterion. This gives an RH equivalence after deleting all primes whose multiplicative orders use any specified finite set of prime factors.

Proof. Let E_B be the positive integers all of whose prime factors are in B, including 1. Unique factorization and finite geometric products give

$$\sum_{m \in E_B} m^{-\eta} = C_B(\eta).$$

Partition the primes being counted by whether their actual order m is at most $Y \geq 1$. Every low-order prime divides

$$\prod_{\substack{m \in E_B \\ m \leq Y}} (a^m - 1).$$

The distinct product of these primes is at least 2 to the power of their number and divides the displayed integer. Consequently their number is at most

$$\frac{\log a}{\log 2} \sum_{\substack{m \in E_B \\ m \leq Y}} m \leq \frac{\log a}{\log 2} C_B(\eta) Y^{1+\eta}.$$

For the high-order primes, $m \mid p-1$ and $p \leq X$ give at most X/m possibilities at any fixed m. Thus their total number is at most

$$X \sum_{\substack{m \in E_B \\ m > Y}} \frac{1}{m} \leq X C_B(\eta) Y^{-1+\eta}.$$

These are bounds on the two parts of the actual order partition; they do not assume distinctness of the integers $a^m - 1$. Set $Y = \sqrt{X}$ and add the bounds. Since η can be arbitrarily small and positive, the critical-exponent hypothesis follows. For $a = 2$ and $B = \{2\}$, a nontrivial power-of-two order is equivalent to divisibility by a Fermat number, giving $\mathcal{D}_F$. □

The explicit factor $C_B(\eta)$ records the cost of enlarging the removed order class. The theorem is uniform in X for fixed a, B, η, and does not exchange an unbounded B with the cutoff limit. These deletions still leave a set of primes of relative density 1. To minimize the retained *proportion* requires a different argument; identifying safe sparse deletions alone does not accomplish that.

21.32 Sparse positive prime carriers for all Dirichlet zeros

Safe deletion of thin classes leaves almost all primes. Reducing the retained proportion is a different problem. Fixed-rank thinning, retaining every kth prime for a fixed k, already occurs in partial-Euler-product RH criteria [18]. The next construction instead allows positive integer multiplicities and retains a single set with only polynomially many primes in $\log X$. The multiplicities are part of the data and cannot be erased.

Theorem 21.65 (A simultaneous positive integer prime compression)**.** *There are a fixed set Q of ordinary primes and positive integer weights w_p, $p \in Q$, with*

$$\#\{p \in Q : p \le X\} \le (\lfloor \log_2 X \rfloor + 2)^6 \qquad (X \ge 2),$$

such that the following holds for every *Dirichlet character χ, of any fixed modulus $q \ge 1$. The weighted Euler product*

$$L_w(s,\chi) = \prod_{p \in Q} (1 - \chi(p)p^{-s})^{-w_p}, \qquad \Re s > 1,$$

converges absolutely and has a meromorphic continuation to $\Re s > 0$ of the form

$$L_w(s,\chi) = L(s,\chi) \exp G_\chi(s), \qquad G_\chi \text{ holomorphic on } \Re s > 0.$$

Consequently all zeros and their multiplicities in that half-plane, and any principal-character pole at 1, are preserved exactly. In particular, with $\mathbf{1}$ the character modulo 1,

$$\mathrm{RH} \iff L_w(s,\mathbf{1}) \ne 0 \quad (\Re s > 1/2,\ s \ne 1).$$

The weights can be chosen with $w_p \le 2^j$ for $2^j \le p < 2^{j+1}$. The same Q and the same weights work for all characters; no uniform bound for their compensators over unbounded conductors or heights is asserted.

Proof. For $j \ge 1$ let P_j be the primes in $[2^j, 2^{j+1})$, and $N_j = \#P_j \le 2^j$. We first choose nonnegative integer weights $w_{j,p}$ on this finite set, preserving all moments

$$\sum_{\substack{p \in P_j \\ p \equiv a \pmod m}} w_{j,p} p^r = \sum_{\substack{p \in P_j \\ p \equiv a \pmod m}} p^r, \quad 1 \le m \le j, \quad 0 \le a < m, \quad 0 \le r \le j. \tag{66}$$

The all-one weights are feasible. The coordinate $m = 1, r = 0$ forces $\sum_p w_{j,p} = N_j$, so every feasible weight is at most N_j and the feasible collection is finite. Choose a feasible weight vector with minimum positive support S_j. For an empty block take the empty vector.

The map taking a subset $U \subseteq S_j$ to its vector of *unweighted* moment sums is injective. Otherwise two distinct subsets with equal vectors give, after deleting their intersection, disjoint nonempty subsets A, B with equal vectors. They are both nonempty because the zeroth moment also gives

equal cardinalities. Subtract $t = \min_{p\in A} w_{j,p} \geq 1$ from the weights on A and add t to the weights on B. All weights remain nonnegative integers, every moment is preserved, and at least one support point disappears without adding a new one. This contradicts minimality.

Write $K_j = \#S_j$. Each coordinate of a subset moment vector at degree r is an integer between 0 and $2^{j+(j+1)r}$. Its number of possible values is at most $2^{(j+1)(r+1)}$. Counting all coordinates of the injective map therefore gives

$$2^{K_j} \leq 2^{\sum_{m=1}^{j}\sum_{a=0}^{m-1}\sum_{r=0}^{j}(j+1)(r+1)}, \qquad K_j \leq \frac{j(j+1)^3(j+2)}{4} \leq (j+2)^5.$$

Set $Q = \bigcup_{j\geq 1} S_j$ and retain the positive weights on each block. For $J = \lfloor \log_2 X \rfloor \geq 1$,

$$\#\{p \in Q : p \leq X\} \leq \sum_{j=1}^{J} K_j \leq J(J+2)^5 \leq (J+2)^6.$$

It remains to prove that these exact finite cancellations preserve the full Euler factors, including all prime powers. For $|c| \leq 1$, $x > 1$ and $\Re s > 0$ put

$$f_{s,c}(x) = -\log(1 - cx^{-s}) = \sum_{\ell\geq 1} \frac{c^\ell x^{-\ell s}}{\ell},$$

using the power-series branch. Every finite block difference

$$D_{j,\chi}(s) = \sum_{p\in P_j} (w_{j,p} - 1) f_{s,\chi(p)}(p)$$

is holomorphic on $\Re s > 0$. Fix a compact subset K of that half-plane, and bounds $\Re s \geq \sigma_0 > 0$ and $|\Im s| \leq T$ on K. Let $u = 2^j$ and $c_j = 3u/2$. On the complex disk $|z - c_j| \leq u$, we have $\Re z \geq u/2$ and, for sufficiently large j,

$$|z^{-s}| \leq 2^{\sigma_0} e^{T\pi/2} u^{-\sigma_0} \leq \frac{1}{2}.$$

Thus $f_{s,c}(z)$ is holomorphic there and is bounded uniformly for $s \in K$ and $|c| \leq 1$ by $M_K u^{-\sigma_0}$, where one may take $M_K = 2^{1+\sigma_0} e^{T\pi/2}$ in this eventual range.

The Taylor polynomial of degree j about c_j has remainder at most

$$M_K u^{-\sigma_0} 2^{-j} \quad \text{on } u \leq x \leq 2u,$$

by Cauchy's estimate and $|x-c_j|/u \le 1/2$. If $j \ge q$, apply this separately in each residue class modulo q. The value $\chi(p)$ is constant on that class. Equation (66) cancels its entire Taylor polynomial, including the constant term. Only after this exact signed cancellation do we bound the two remainder measures, whose total mass is $2N_j \le 2u$. It follows that

$$\sup_{s\in K} |D_{j,\chi}(s)| \le 2M_K 2^{-j\sigma_0} \qquad (j \ge \max(q, J_K)),$$

where J_K makes the preceding disk estimate valid. Hence $G_\chi = \sum_{j\ge1} D_{j,\chi}$ converges locally uniformly on $\Re s > 0$ and is holomorphic there. More explicitly,

$$\sup_{s\in K} \left| \sum_{j\ge J} D_{j,\chi}(s) \right| \le \frac{2M_K 2^{-J\sigma_0}}{1-2^{-\sigma_0}} \quad (J \ge \max(q, J_K)).$$

This also makes the conductor and height restrictions on the tail estimate explicit.

For $\Re s > 1$, absolute convergence follows from

$$\sum_{p\in Q} w_p p^{-\sigma} \le \sum_{j\ge1} N_j 2^{-j\sigma} \le \sum_{j\ge1} 2^{-j(\sigma-1)} < \infty.$$

The usual Dirichlet Euler product then gives, in that domain, $\log L_w - \log L = G_\chi$. Exponentiation and analytic continuation prove the displayed factorization. The multiplier is a holomorphic unit, so it preserves zeros, multiplicities and poles. Taking the character modulo 1 proves the RH equivalence. □

The finite support-reduction argument is the classical integer moment-elimination principle, related to the integer-cone bounds of Eisenbrand–Shmonin [20]. Grosswald–Schnitzer [19] preserve zeta and Dirichlet zeros by perturbing Euler bases. Jurzak's fixed-rank thinning retains a positive proportion of the prime list. The theorem above specifies a single actual prime support of polynomial logarithmic size, positive integer multiplicities, and simultaneous preservation for every fixed Dirichlet character. Its proof is given here; priority for this combination has not been established.

The associated positive probability law. The weights give a genuine probability model, without an assumed prime-clock sampling theorem. For $\sigma > 1$, expand

$$Z_w(s) = L_w(s, \mathbf{1}) = \sum_{n\ge1} \frac{a_w(n)}{n^s}, \qquad a_w(p^r) = \binom{w_p + r - 1}{r} \quad (p \in Q),$$

with a_w multiplicative and zero on integers having a prime factor outside Q. All coefficients are nonnegative integers. The distribution

$$\mathbb{P}_{w,\sigma}(n) = \frac{a_w(n)n^{-\sigma}}{Z_w(\sigma)}$$

is normalized on positive integers. Its prime exponents are independent negative-binomial variables with integer shape w_p and parameter $p^{-\sigma}$. For every Dirichlet character and real t,

$$\mathbb{E}_{w,\sigma}[\chi(n)n^{-it}] = \frac{L_w(\sigma+it,\chi)}{Z_w(\sigma)}.$$

The sampled variable is the integer n under this explicitly weighted law; $\sigma > 1$ is fixed. This is not an assertion about uniform integer sampling, nor a transfer of a prime-clock estimate to a new family without an adapter.

What compression does and does not give. The main terms vanish through the exact moment identities; the proof bounds only their retained analytic remainder. It preserves the complete zero signal, rather than showing that the signal cannot occur. The integer weights encode moments of all primes in each block and can grow with the prime size. A useful attack would need an independent estimate on the resulting weighted arithmetic sums, with that growth charged explicitly. No such zero-excluding estimate follows from support size alone.

Indeed, deleting the multiplicities would destroy this construction's signal: the unweighted Euler product on Q is already holomorphic and nowhere zero on $\Re s > 0$, since its prime count is only polynomial in $\log X$. The sparse weighted product need not admit a direct pointwise Nicolas comparison at every cutoff: weights are chosen on complete dyadic blocks, and no uniform small error for partial blocks has been proved. Nor does holomorphic continuation alone supply a square-root bound for its signed inverse coefficients without the necessary vertical growth estimates.

Lemma 21.66 (Exact finite integer moment compression)**.** *Let I, K be finite sets and $v : I \times K \to \mathbb{N}$. There are weights $w_i \in \mathbb{N}$ such that, writing $S = \{i : w_i > 0\}$ and $M_k = \sum_{i\in I} v(i,k)$,*

$$\sum_{i\in I} w_i v(i,k) = M_k \quad (k \in K), \qquad 2^{|S|} \le \prod_{k\in K} (M_k + 1).$$

They can be chosen so that distinct subsets of S have distinct vectors of unweighted coordinate sums. Empty index or coordinate sets are allowed.

Proof. The all-one vector is feasible. Minimize its support cardinality among feasible natural weight vectors. If two distinct subsets of the support have equal signatures, delete their intersection and orient the resulting pair so that the first subset is nonempty. Subtract its minimum positive weight there and add it on the second subset. Disjointness preserves nonnegativity and every coordinate sum, while strictly decreasing support. This contradiction proves injectivity, including when the second subset is empty. For each k, every unweighted subset sum is an integer between 0 and M_k, because the support is contained in I and v is nonnegative. Counting signatures gives the bound. No nonemptiness hypothesis was used. □

The finite lemma, including its injectivity and exact support bound, is compiled in the maintained Lean candidate `PrimeFactorOscillations/Mathlib/Combinatorics/IntegerMomentCompression.lean`. Its quantitative declaration is `Finset.exists_nat_weights_card_pow_le`. The axiom audit contains only `propext`, `Classical.choice` and `Quot.sound`. The prime-class deletion results and the analytic continuation and zero preservation in Theorem 21.65 retain their ordinary proofs; those analytic conclusions are not claimed as compiled Lean declarations. The certified 7/8 zero-free boundary is unchanged.

21.33 Controlled weights and a square-root prime carrier

The preceding construction minimizes support aggressively, but its large integer weights encode many moments. A different choice gives a prescribed small weight cap and an explicit height bound. It can simultaneously avoid every prime in a fixed safely deletable class.

Theorem 21.67 (A weight-controlled carrier at the RH threshold)**.** *Let S be a fixed prime set with $\pi_S(X) \ll_\epsilon X^{1/2+\epsilon}$ for every $\epsilon > 0$, and put $C = \mathcal{P} \setminus S$. There are a fixed set $Q \subseteq C$ and positive integer weights $w_p \le \sqrt{p}$, $p \in Q$, such that, for $J = \lfloor \log_2 X \rfloor$,*

$$\pi_Q(X) \le 6\sqrt{X} + \frac{J(J+1)}{2} \qquad (X \ge 2).$$

The prime number theorem sharpens the order to $\pi_Q(X) \asymp \sqrt{X}/\log X$ as $X \to \infty$. For every fixed Dirichlet character χ, the product

$$L_Q(s,\chi) = \prod_{p \in Q} (1 - \chi(p)p^{-s})^{-w_p} \qquad (\Re s > 1)$$

has a meromorphic continuation to $\Re s > 1/2$ of the form

$$L_Q(s,\chi) = L(s,\chi)H_{S,\chi}(s)\exp G_\chi(s), \qquad H_{S,\chi}(s) = \prod_{p\in S}(1-\chi(p)p^{-s}),$$

where both multipliers are holomorphic and nowhere zero there. Thus every zero and multiplicity in that half-plane is preserved, as is the principal pole at 1. In particular, RH is equivalent to $L_Q(s,\mathbf{1}) \neq 0$ for $\Re s > 1/2$, $s \neq 1$. The same Q and weights work for all fixed characters. For every fixed modulus q and $\sigma_0 > 1/2$,

$$|G_\chi(s)| \le B_{q,\sigma_0} + A_{\sigma_0}|s| \quad (\Re s \ge \sigma_0,\ \chi \bmod q),$$

with constants specified in the proof. There is no uncharged uniformity in a growing modulus.

Proof. For $j \ge 1$ let $C_j = C \cap [2^j, 2^{j+1})$ and $N_j = |C_j|$. Let k_j be the largest integer $k \le j$ with $k! \le j$, and set $M_j = k_j!$. Then $1 \le M_j \le j$, and $q \mid M_j$ whenever $j \ge q!$. Set $h_j = 2^{\lfloor j/2 \rfloor}$. Within each residue class modulo M_j, list the elements of C_j in increasing order and divide this list into successive groups of length h_j, allowing a shorter last group. Retain the largest prime of each nonempty group and give it weight equal to the group length.

Writing $K_j = |Q \cap C_j|$, this gives the exact identities and bounds

$$\sum_{\substack{p\in Q\cap C_j \\ p\equiv a \pmod{M_j}}} w_p = \#\{p \in C_j : p \equiv a \pmod{M_j}\}, \qquad \frac{N_j}{h_j} \le K_j \le \frac{N_j}{h_j} + M_j.$$

All weights are positive integers at most $h_j \le \sqrt{p}$. In particular,

$$\pi_Q(X) \le \sum_{j=1}^{J} 2^{\lceil j/2 \rceil} + \sum_{j=1}^{J} j \le 6\sqrt{X} + \frac{J(J+1)}{2}.$$

For the first sum, pair $j = 2r-1, 2r$; the possible final unpaired term is also covered by the constant 6. The deletion hypothesis implies $\pi_S(X) = o(X/\log X)$. Consequently the prime number theorem gives $N_j \asymp 2^j/j$. The displayed two bounds on K_j, followed by a geometric sum for the upper bound and the last complete block for the lower bound, prove the sharper order for $\pi_Q(X)$. Its comparison constants and starting point are eventual; no numerical starting point is asserted.

Here is the signed cancellation preserving the full Euler factors. For $u = 2^j$, $u \le x \le 2u$, and $0 \le a < M_j$, put

$$E_{j,a}(x) = \sum_{\substack{p \in C_j,\ p \le x \\ p \equiv a \pmod{M_j}}} \big(w_p \mathbf{1}_Q(p) - 1\big),$$

where $w_p \mathbf{1}_Q(p)$ means zero off Q. A completed group contributes zero. Inside an unfinished group there are fewer than h_j omitted primes. The shorter final group also has total contribution zero. Hence, including both endpoints,

$$-(h_j - 1) \le E_{j,a}(x) \le 0, \qquad E_{j,a}(2u) = 0.$$

For $|c| \le 1$ and $\Re s > 0$, retain the entire Euler function

$$f_{s,c}(x) = -\log(1 - cx^{-s}) = \sum_{\ell \ge 1} \frac{c^\ell x^{-\ell s}}{\ell}, \qquad f'_{s,c}(x) = -\frac{scx^{-s-1}}{1 - cx^{-s}}.$$

No prime powers have been discarded. If $j \ge q!$, the character has a constant value c_a on each of these residue classes. Finite Abel summation gives the exact coupled identity

$$D_{j,\chi}(s) := \sum_{p \in C_j} (w_p \mathbf{1}_Q(p) - 1) f_{s,\chi(p)}(p) = -\sum_{a=0}^{M_j - 1} \int_u^{2u} E_{j,a}(x) f'_{s,c_a}(x)\, dx.$$

The endpoint term vanishes because each class has exactly matched total mass. Only after this cancellation do we estimate the retained derivative integral. For $\sigma = \Re s > 1/2$,

$$|D_{j,\chi}(s)| \le \frac{M_j (h_j - 1)|s|\, 2^{-j\sigma}(1 - 2^{-\sigma})}{\sigma(1 - 2^{-j\sigma})}.$$

In particular, for $\sigma \ge \sigma_0 > 1/2$, put $\rho = 2^{-(\sigma_0 - 1/2)} < 1$ and $C_{\sigma_0} = 1/[\sigma_0(1 - 2^{-\sigma_0})]$. For every $R \ge q!$ this yields the explicit tail

$$\left| \sum_{j \ge R} D_{j,\chi}(s) \right| \le C_{\sigma_0} |s| \rho^R \left(\frac{R}{1 - \rho} + \frac{\rho}{(1 - \rho)^2} \right).$$

This proves normal convergence on $\Re s > 1/2$ of $G_\chi(s) = \sum_{j \ge 1} D_{j,\chi}(s)$; the finitely many blocks $j < q!$ are individually holomorphic on $\Re s > 0$. For the stated height bound one may take

$$B_{q,\sigma_0} = 2 \sum_{1 \le j < q!} \frac{2^{j(1-\sigma_0)}}{1 - 2^{-j\sigma_0}}, \qquad A_{\sigma_0} = C_{\sigma_0} \frac{\rho}{(1 - \rho)^2}.$$

Indeed, the absolute coefficient mass in an initial block is at most $2N_j$, and $|f_{s,c}(p)| \le p^{-\sigma_0}/(1-p^{-\sigma_0})$. These are explicit constants, although the dependence on $q!$ is too large to ignore in a growing-conductor application.

Exact mass preservation gives $\sum_{p\in Q\cap C_j} w_p = N_j \le 2^j$. Thus the weighted Euler product converges absolutely on $\Re s > 1$. In that domain the full logarithms satisfy

$$\log L_Q(s,\chi) = \sum_{p\in C} f_{s,\chi(p)}(p) + G_\chi(s).$$

The deletion hypothesis makes $H_{S,\chi}$ a holomorphic unit on $\Re s > 1/2$, by the same absolute-convergence proof as Theorem 21.62. Its power-series logarithm also satisfies, uniformly in χ and $\Re s \ge \sigma_0 > 1/2$,

$$|\log H_{S,\chi}(s)| \le B_{S,\sigma_0} := \sum_{p\in S} \frac{p^{-\sigma_0}}{1-p^{-\sigma_0}} < \infty.$$

Thus this deleted-prime factor has also been charged in the full multiplier bound; its constant depends on S and σ_0. Exponentiating proves the factorization and continuation. The multiplier being a unit, all stated zeros, multiplicities and poles are preserved. The ordinary RH criterion follows by taking the character modulo 1. □

What the weight cap costs. The estimate $K_j \ge N_j/h_j$ is necessary for *any* nonnegative weight vector with weight cap h_j and the same zeroth block moment: $N_j = \sum w_p \le h_j K_j$. The construction is therefore within $M_j \le j$ support points of the minimum permitted by this mass constraint. This is an optimality statement for that specified constraint, not a lower bound on the support of every possible RH detector.

The full signed identity also gives an untwisted real-axis comparison. When $\chi = \mathbf{1}$ and $s = \sigma > 1/2$ is real, $E_{j,a} \le 0$ and $f'_{\sigma,1} \le 0$ show $D_{j,\mathbf{1}}(\sigma) \le 0$, so $G_{\mathbf{1}}(\sigma) \le 0$. On $\sigma > 1$, the weighted product is at most the Euler product on C. This sign is not a zero-excluding inequality at complex s.

More generally, the same construction with any integer caps $h_j \ge 1$ has the exact same signed identity. It works on every half-plane where $\sum_j j h_j 2^{-j\sigma}$ converges, together with the continuation domain of the deleted-prime multiplier. For example, $h_j = (j+1)^r$, with fixed integer $r \ge 1$ and $S = \varnothing$, gives weights bounded by $(1+\log_2 p)^r$, support $\pi_Q(X) \asymp X/(\log X)^{r+1}$, and preservation of all Dirichlet zeros on $\Re s > 0$. For every fixed $\sigma_0 > 0$ its compensator again has a bound $B_{q,r,\sigma_0} + A_{r,\sigma_0}|s|$. Thus the

price of support reduction can be chosen explicitly: polylogarithmic weights permit logarithmic thinning while retaining the entire open positive half-plane; square-root weights permit a square-root prime count while retaining the open RH half-plane.

The positive negative-binomial probability law from the preceding subsection applies to these integer weights as well. It is still a weighted sampling law. The finite rank-clock step is compiled in `PrimeFactorOscillations/Mathlib/Combinatorics/IntegerBlockGrouping.lean`. For $n \geq 0$, $h \geq 1$, define

$$P_{n,h}(m) = \begin{cases} h\lfloor m/h \rfloor, & m < n, \\ n, & m \geq n, \end{cases} \qquad v_i = P_{n,h}(i+1) - P_{n,h}(i).$$

The verified statements give $0 \leq v_i \leq h$, $\sum_{i<n} v_i = n$, and, for every $0 \leq m \leq n$,

$$-(h-1) \leq \sum_{i<m} v_i - m \leq 0.$$

The signed declaration is `Nat.groupedPrefix_signed_balance`; the exact mass declaration is `Nat.groupedTerminalWeight_total`. The final shorter group is included in both statements.

The ordinary proof here does not establish a bound on its signed inverse coefficients or improve the certified 7/8 boundary. The analytic Abel-summation and normal-convergence steps are not yet formalized in Lean. This controlled-weight construction is stated with its proof; no established novelty priority is claimed.

21.34 Power sparsity, weight caps and height growth

The finite integer compression lemma can be used locally, rather than on an entire dyadic block. This retains much smaller weights and controls the full height dependence with a fixed number of moments.

Theorem 21.68 (A tunable prime-carrier tradeoff)**.** *Fix an integer $k \geq 2$ and put*

$$C_k = k(k+1)\frac{2^{1/k}}{2^{1/k}-1}.$$

There are a fixed prime set Q_k and positive integer weights $w_p \leq p^{1-1/k}$ such that

$$\pi_{Q_k}(X) \leq C_k J(J+1) X^{1/k}, \qquad J = \lfloor \log_2 X \rfloor, \quad X \geq 2.$$

For every fixed Dirichlet character χ the absolutely convergent Euler product

$$\mathcal{L}_k(s,\chi) = \prod_{p\in Q_k} (1-\chi(p)p^{-s})^{-w_p}, \qquad \Re s > 1,$$

continues meromorphically to $\Re s > 0$ as

$$\mathcal{L}_k(s,\chi) = L(s,\chi)\exp G_{k,\chi}(s), \qquad G_{k,\chi} \text{ holomorphic on } \Re s > 0.$$

All zeros, multiplicities and poles in that half-plane are therefore preserved. The same set and weights work for every fixed character. In particular this gives an RH-equivalent zero criterion on each Q_k. For every fixed modulus q and $\sigma_0 > 0$,

$$|G_{k,\chi}(s)| \le B_{q,\sigma_0} + \frac{2}{k}\prod_{r=0}^{k-1}(|s|+r)\frac{2^{-\sigma_0}}{(1-2^{-\sigma_0})^{k+1}} \quad (\Re s \ge \sigma_0,\ \chi \bmod q),$$

where, for every $\sigma_0 > 0$,

$$B_{q,\sigma_0} = 2\sum_{1\le j<q!} \frac{2^{j(1-\sigma_0)}}{1-2^{-j\sigma_0}}.$$

This is the same finite initial-block expression as before, now used and justified throughout $\sigma_0 > 0$. Thus it is the logarithm of the multiplier *that has polynomial height growth of degree k. All assertions keep k fixed; no uniformity as $k \to \infty$ or the conductor grows is implicit.*

Proof. Use the moduli $M_j \le j$ from the preceding construction, so $q \mid M_j$ for $j \ge q!$. For $u = 2^j$, put

$$\ell_j = \left\lfloor \frac{(k-1)j}{k} \right\rfloor, \qquad L_j = 2^{\ell_j}, \qquad b_j = 2^{j-\ell_j}.$$

Divide $[u, 2u)$ into its b_j consecutive integer intervals of length L_j. In each such interval, and each residue class modulo M_j, let I be the finite set of primes in that cell and class. Apply Lemma 21.66 to the natural vectors $(1, p, \dots, p^{k-1})$, $p \in I$. It supplies nonnegative integer weights satisfying

$$\sum_{p\in I} w_p p^r = \sum_{p\in I} p^r \quad (0 \le r < k), \qquad 2^{|\{p:w_p>0\}|} \le \prod_{r=0}^{k-1}\left(1+\sum_{p\in I} p^r\right).$$

This is exactly the already compiled finite compression input, instantiated with coordinate type $\mathrm{Fin}(k)$. The subsequent analytic argument is an ordinary proof.

Write $n = |I| \le L_j$. The zeroth moment forces $\sum_{p\in I} w_p = n$, hence every weight is at most $n \le L_j \le u^{1-1/k} \le p^{1-1/k}$. Discard the zero weights. Empty cells contribute no primes. For $0 \le r < k$,

$$1 + \sum_{p\in I} p^r \le 2^{\ell_j + r(j+1)+1}.$$

Consequently the retained support in one cell and class has at most

$$k(\ell_j + 1) + \frac{k(k-1)}{2}(j+1) \le \frac{k(k+1)}{2}(j+1)$$

points. There are at most j residue classes and $b_j \le 2 \cdot 2^{j/k}$ cells. The total block support is therefore at most $k(k+1)j(j+1)2^{j/k}$. Summing this bound for $1 \le j \le J$, using

$$\sum_{j=1}^{J} 2^{j/k} \le \frac{2^{1/k}}{2^{1/k} - 1}\, 2^{J/k},$$

proves the claimed bound for all $X \ge 2$. The denominator is positive because $k \ge 2$.

We now estimate the exact full Euler-block difference

$$D_{j,k,\chi}(s) = \sum_{\substack{p \text{ prime} \\ u \le p < 2u}} (w_p \mathbf{1}_{Q_k}(p) - 1) f_{s,\chi(p)}(p), \qquad f_{s,c}(x) = \sum_{m\ge 1} \frac{c^m x^{-ms}}{m}.$$

The series retains every prime power. If $j \ge q!$, the character value is constant on each cell and residue class. The moment identities annihilate the entire Taylor polynomial of degree $k-1$ of $f_{s,c}$ about the left endpoint of the cell. In particular,

$$D_{j,k,\chi}(s) = \sum_{\text{cells and classes}} \sum_{p\in I} (w_p - 1)\big(f_{s,c}(p) - T_{k-1,s,c,I}(p)\big).$$

This signed identity, with all moments removed exactly, precedes any bound on the Taylor remainder.

For $\sigma = \Re s > 0$, termwise differentiation gives

$$|f_{s,c}^{(k)}(x)| \le \prod_{r=0}^{k-1} (|s| + r)\, x^{-k} \sum_{m\ge 1} m^{k-1} x^{-m\sigma} \le (k-1)! \prod_{r=0}^{k-1} (|s| + r) \frac{x^{-\sigma-k}}{(1 - x^{-\sigma})^k}.$$

For the last inequality, use $m^{k-1} \le (k-1)!\binom{m+k-2}{k-1}$ and its geometric generating series. The differentiation is justified locally uniformly on $x > 1$, $\Re s > 0$, $|c| \le 1$. Taylor's integral remainder on an interval of length L_j is bounded by $L_j^k/k!$ times this derivative bound. The residual coefficient mass in a cell is at most $2n$. Summing the remainders over cells and classes uses $\sum n \le u$ and $L_j^k \le u^{k-1}$, and yields

$$|D_{j,k,\chi}(s)| \le \frac{2}{k} \prod_{r=0}^{k-1} (|s| + r) \frac{2^{-j\sigma}}{(1 - 2^{-j\sigma})^k} \qquad (j \ge q!).$$

In particular, for $\Re s \ge \sigma_0 > 0$ and $R \ge q!$,

$$\left| \sum_{j \ge R} D_{j,k,\chi}(s) \right| \le \frac{2}{k} \prod_{r=0}^{k-1} (|s| + r) \frac{2^{-R\sigma_0}}{(1 - 2^{-\sigma_0})^{k+1}}.$$

This proves normal convergence of $G_{k,\chi} = \sum_{j \ge 1} D_{j,k,\chi}$ on $\Re s > 0$. The initial blocks $j < q!$ are bounded by B_{q,σ_0}, because they have total residual mass at most twice their original prime count. Bounding the remaining tail by the sum over all $j \ge 1$ proves the stated height estimate.

Each block still has total weight equal to its original prime count. Thus the Euler product is absolutely convergent for $\Re s > 1$, and the complete logarithms in that domain differ by $G_{k,\chi}$. Exponentiation gives $\mathcal{L}_k = L(s,\chi) \exp G_{k,\chi}$ and hence the meromorphic continuation. The exponential is a holomorphic unit, so the full zero signal and multiplicities survive. □

Deletion, probability and the unavoidable mass cost. The construction can instead be carried out on primes outside any fixed class S from Theorem 21.62. All cell bounds and the analytic estimate for $G_{k,\chi}$ remain valid. The factorization becomes

$$\mathcal{L}_{k,S}(s,\chi) = L(s,\chi) H_{S,\chi}(s) \exp G_{k,S,\chi}(s).$$

For a critical-thin class S this preserves zeros on $\Re s > 1/2$, the domain where the deleted-prime factor is proved to be a unit; it does not assert preservation on all of $\Re s > 0$ for that class. The full logarithmic multiplier bound additionally includes B_{S,σ_0} from the preceding proof. Thus all Fermat-number prime divisors can be excluded while the support is further compressed to $O_k(X^{1/k}(\log X)^2)$. The positive integer weights again define the stated negative-binomial probability law.

The mass cost is explicit. In a dyadic block, preserving zeroth mass N_j under the cap $w_p \le p^{1-1/k}$ requires at least $N_j/(2u)^{1-1/k}$ support points. The prime number theorem makes this $\gg_k u^{1/k}/\log u$. Our construction is within logarithmic factors of the smallest support power permitted by that cap and mass constraint. It is not a lower bound on all possible detectors.

For example, $k = 2$ gives square-root weights and support $O(\sqrt{X}(\log X)^2)$, with a quadratic height bound for the logarithmic compensator and preservation on $\Re s > 0$. The preceding terminal-group construction has the sharper support order $\sqrt{X}/\log X$, a linear height bound and a real-axis sign, but only preserves $\Re s > 1/2$. These are distinct quantitative choices. Taking larger fixed k makes the prime support sparser, increases the allowed weight exponent, and raises the degree of the logarithmic height bound. It does not remove the mass or height costs.

The finite integer moment theorem used here is compiled. The short-interval prime construction, Taylor cancellation, analytic continuation and resulting RH criterion have the complete ordinary proof above; they are not claimed as compiled Lean declarations. No independent bound on the weighted signed inverse coefficients has been proved, and the certified zero-free boundary remains 7/8. This explicit tradeoff is proved here without an established novelty-priority claim.

21.35 The exact scope of deletion by an Euler unit

The sufficient deletion condition also has a converse for the precise mechanism used here. Positivity prevents a larger deleted prime class from having a holomorphic-unit Euler factor throughout the same half-plane.

Theorem 21.69 (Classification of pure Euler-unit deletions)**.** *Let S be any fixed prime set, and let $\alpha \ge 0$. The following are equivalent:*

1. *For every $\epsilon > 0$, $\pi_S(X) \ll_\epsilon X^{\alpha+\epsilon}$.*
2. *The Euler product $E_S(s) = \prod_{p\in S}(1-p^{-s})^{-1}$, initially defined on $\Re s > 1$, has a holomorphic continuation to $\Re s > \alpha$.*
3. *The product $H_S(s) = \prod_{p\in S}(1-p^{-s})$, initially defined on $\Re s > 1$, continues to a holomorphic nowhere-zero function on $\Re s > \alpha$.*

In these cases both Euler products actually converge absolutely on $\Re s > \alpha$. In particular, at $\alpha = 1/2$, the critical-thin counting condition is necessary and sufficient for deletion by an Euler unit on the entire open RH half-plane.

Proof. Assume (1), and fix $\sigma > \alpha$. Choose $0 < \epsilon < \sigma - \alpha$. Dyadic summation of the counting bound gives $\sum_{p \in S} p^{-\sigma} < \infty$. Since $\sigma > 0$, the full logarithmic sums

$$\sum_{p \in S} \sum_{m \geq 1} \frac{p^{-ms}}{m}$$

converge normally on $\Re s > \alpha$. Exponentiation and its reciprocal give absolutely convergent Euler products that are holomorphic and nowhere zero there. This proves (3), and (3) implies (2) by taking a reciprocal.

For (2) implies (1), write the initial Euler product as

$$E_S(s) = \sum_{n \geq 1} \frac{a_S(n)}{n^s}, \qquad a_S(n) = \mathbf{1}_{\{\text{every prime factor of } n \text{ belongs to } S\}}.$$

Here $a_S(1) = 1$, and all coefficients are nonnegative and at most 1. The expansion and its differentiated series are absolutely convergent on $\Re s > 1$.

Fix a real $\sigma > \alpha$ and choose $c > \max(1, \sigma)$. The continued function is holomorphic on the disk $|s - c| < c - \alpha$. Set $t = c - \sigma$; then $0 < t < c - \alpha$. Termwise differentiation at c, followed by its convergent Taylor expansion at $c - t$, gives

$$E_S(\sigma) = \sum_{r \geq 0} \frac{t^r}{r!} \sum_{n \geq 1} a_S(n) (\log n)^r n^{-c}.$$

Every term is nonnegative. Tonelli's theorem therefore allows the two sums to be interchanged, yielding

$$E_S(\sigma) = \sum_{n \geq 1} a_S(n) n^{-c} \sum_{r \geq 0} \frac{(t \log n)^r}{r!} = \sum_{n \geq 1} a_S(n) n^{-\sigma} < \infty.$$

This is the classical positive-Taylor argument behind Landau's theorem, included here so that the continuation hypothesis and the use of positivity are explicit. In particular, $\sum_{p \in S} p^{-\sigma} \leq E_S(\sigma) < \infty$. For $\sigma = \alpha + \epsilon$ and every $X \geq 2$,

$$\pi_S(X) \leq X^{\alpha + \epsilon} \sum_{p \in S} p^{-(\alpha + \epsilon)}.$$

This convergent prime sum gives a finite bound constant, depending on S and ϵ; no numerical value is asserted. This proves (1), and the already proved implication from (1) also establishes absolute convergence of both products. □

The classification applies to an Euler multiplier required to be a unit at *every* point of $\Re s > 1/2$, including $s = 1$, as needed to preserve the original pole. It is not a classification of every RH-equivalent observable. At this threshold the deleted class has $\pi_S(X) = o(\pi(X))$, so its unweighted complement retains relative prime density 1. An entire positive proportion of primes therefore cannot be removed by this exact unit mechanism. Finite and many infinite special-prime classes both qualify; infinitude itself is not the criterion.

The smaller weighted carriers proved above achieve something different: they preserve zeroth mass in their blocks and move that mass onto fewer locations. Their support can have zero relative density because the weights or normalization carry the missing information. The classification explains precisely why the cost must be tracked; it does not rule out a different compensated observable or prove a zero-excluding estimate. The classification has the complete ordinary proof above and uses the classical Landau positivity principle. No separate Lean compilation or novelty claim is made. The weighted carriers retain the pole at 1, so they do not satisfy the holomorphy hypothesis in (2). Applying this positivity argument after compensating that pole would require a separately proved holomorphic function with nonnegative Dirichlet coefficients; no such input is supplied by compression alone.

21.36 A reverse construction with capped prime capacities

We now prescribe a favorable analytic model first and ask whether actual primes can supply it. The construction separates an exact normalization identity from the independent size estimate needed to exclude zeros. All prime powers and overlaps between congruence classes remain in the identity.

We use the following positive-series principle. Let $a_n \geq 0$, let $\sum_{n\geq 1} a_n n^{-c} < \infty$, and suppose its sum to the right of c agrees with a real-analytic function on (α, ∞). Then

$$\sum_{n\geq 1} a_n n^{-\sigma} < \infty \qquad (\sigma > \alpha). \tag{67}$$

Indeed put $w_n = a_n n^{-c}$ and $x_n = \log n$. The finite positive atomic measure $\sum_n w_n \delta_n$ has exponential transform $\sum_n w_n e^{t x_n}$. Its analytic continuation to $t < c - \alpha$ forces exponential integrability there by the positive-measure Landau frontier theorem. Taking $t = c - \sigma$ proves the claim. This argument has now been compiled in `PositiveDirichletContinuation.lean`, including the atomic measure identities and the final theorem `summable_`

`positiveDirichlet_of_analyticContinuation`. The five declarations use only the standard Lean axioms. The Euler-product adapters and the reverse construction below have ordinary proofs here; they are not yet Lean declarations.

Lemma 21.70 (A positive continuous model with a prescribed pole)**.** *For $x > 1$, define the positive measure*

$$d\nu(x) = \frac{1}{x \log x} \sum_{r \geq 1} \frac{(\log x)^r}{r!\, \zeta(r+1)}\, dx.$$

Its density is at most $1/\log x$, is bounded near 1, and its full Euler transform is

$$M(s) := \exp\left(\int_1^\infty -\log(1 - x^{-s})\, d\nu(x) \right) = \frac{s}{s-1} \qquad (\Re s > 1). \tag{68}$$

Thus its continuation is zero-free on $\Re s > 0$, with one simple pole at 1. The logarithm on the initial half-plane is the full series $\sum_{m \geq 1} x^{-ms}/m$.

Proof. Since $\zeta(r+1) > 1$, the density is bounded above by $(1 - x^{-1})/\log x$, which has a finite limit at 1. For real $s > 1$, substitute $x = e^t$ and use Tonelli:

$$P_0(s) := \int_1^\infty x^{-s}\, d\nu(x) = \sum_{r \geq 1} \frac{1}{r\, \zeta(r+1) s^r}.$$

Another nonnegative interchange gives

$$\begin{aligned} \log M(s) &= \sum_{m \geq 1} \frac{P_0(ms)}{m} \\ &= \sum_{r \geq 1} \frac{1}{r s^r \zeta(r+1)} \sum_{m \geq 1} m^{-r-1} \\ &= \sum_{r \geq 1} \frac{1}{r s^r} = \log \frac{s}{s-1}. \end{aligned}$$

The same sums and integrals converge absolutely and normally on $\Re s > 1$, so the identity holds there for complex s as well. The rational expression gives the stated continuation. No assertion about the zeros of ζ was used. □

Fix a nonempty finite set $\mathcal{Q}$ of positive integer moduli. Write $R = |\mathcal{Q}|$, $d_q = \varphi(q)$, and

$$\mathcal{Z}_{\mathcal{Q}}(s) = \prod_{q \in \mathcal{Q}} \prod_{\chi \bmod q} L(s, \chi), \qquad A(p) = \sum_{\substack{q \in \mathcal{Q} \\ p \equiv 1 \pmod q}} d_q. \tag{69}$$

These are fixed moduli throughout; no conductor-uniform assertion is implicit. For $p \nmid q$, let $f_q(p)$ be its multiplicative order modulo q. The local factor of the product over characters is

$$\prod_{\chi \bmod q} (1 - \chi(p)p^{-s})^{-1} = (1 - p^{-f_q(p)s})^{-d_q/f_q(p)}. \tag{70}$$

For $p \mid q$ it is 1. In particular,

$$\mathcal{Z}_{\mathcal{Q}}(s) = \prod_p (1 - p^{-s})^{-A(p)}\, U_{\mathcal{Q}}(s), \tag{71}$$

where $U_{\mathcal{Q}}$ retains every nonsplit local factor, with $f_q(p) \geq 2$. It is an absolutely convergent holomorphic unit on $\Re s > 1/2$. The order and the exponent are integers since $f_q(p) \mid d_q$. Formula (70) follows by multiplying over the characters of the finite unit group. It includes the trivial groups for $q = 1, 2$. Overlapping split classes contribute their summed capacity $A(p)$ exactly once.

The usual continuations of the Dirichlet L-functions show that $\mathcal{Z}_{\mathcal{Q}}$ has a pole of order R at 1. Every zero ρ of ζ with $\Re\rho > 1/2$ is a zero of this reference of multiplicity at least $R\, m_\zeta(\rho)$: each principal factor is $\zeta(s) \prod_{p|q}(1 - p^{-s})$, the finite product is a unit on $\Re s > 0$, and all other factors are holomorphic there. Additional nonprincipal zeros may increase the multiplicity; they cannot cancel it.

Choose $0 < \theta < 1$. For $j \geq 1$ put

$$u_j = 2^j, \quad L_j = 2^{\lfloor \theta j \rfloor}, \quad B_j = 2^{j - \lfloor \theta j \rfloor}, \quad \tau = \max(\theta, 1 - \theta).$$

Partition $[u_j, 2u_j)$ into its B_j half-open intervals I of integer length L_j, and denote their integer left endpoints by t_I. Define the target, available capacity, and deficit by

$$\begin{aligned} H_I &= \lceil R\nu(I) \rceil, \qquad C_I = \sum_{\substack{p \in I \\ p \text{ prime}}} A(p), \\ \delta_I &= (H_I - C_I)_+, \quad \Delta_j = \sum_{I \subset [u_j, 2u_j)} \delta_I. \end{aligned} \tag{72}$$

All these sums are over the stated finite sets. Choose integer weights on the actual primes of each interval satisfying

$$0 \leq v_p \leq A(p), \qquad \sum_{\substack{p \in I \\ p \text{ prime}}} v_p = \min(H_I, C_I) = H_I - \delta_I. \tag{73}$$

They exist by filling the finite integer capacities in any fixed order until the target is met or all capacities are exhausted. This finite existence and exact-deficit assertion is compiled as `exists_capped_selection` in `CapacitySelection.lean`. This constructs a set $Q = \{p : v_p > 0\}$ without assuming that every interval contains a prime. At fixed total weight, using larger capacities first minimizes the number of selected primes in that interval. The zero-capacity primes are never selected.

Theorem 21.71 (Reverse model and retained-zero criterion)**.** *For the construction above, set initially on* $\Re s > 1$

$$F(s) = \prod_p (1 - p^{-s})^{-v_p}, \qquad D(s) = \sum_{j \geq 1} \sum_{I \subset [u_j, 2u_j)} \delta_I [-\log(1 - t_I^{-s})].$$

There is a holomorphic G *on* $\Re s > \tau$*, defined independently of any zero-free hypothesis, such that*

$$F(s) e^{D(s)} = M(s)^R e^{G(s)} \qquad (\Re s > 1). \tag{74}$$

Consequently the completed model has a zero-free meromorphic continuation to $\Re s > \tau$*, with a pole of order* R *at* 1*.*

If $\tau \leq \eta < 1$ *and, independently,*

$$\forall \epsilon > 0 \quad \Delta_j \ll_\epsilon 2^{j(\eta+\epsilon)} \qquad (j \geq 1), \tag{75}$$

then every $L(s, \chi)$ *for* $\chi \bmod q$*,* $q \in \mathcal{Q}$*, and* $\zeta(s)$ *have no zeros on* $\Re s > \eta$*. In particular, with* $\theta = 1/2$*, a deficit bound of exponent* $3/4$ *gives the requested boundary* $3/4$*, and exponent* $1/2$ *would imply RH as well as the corresponding assertions for the included Dirichlet L-functions.*

More quantitatively, put

$$\gamma = \limsup_{j \to \infty} \frac{\log(1 + \Delta_j)}{j \log 2} \in [0, 1].$$

Every zero ρ *of the reference with* $\Re\rho > \tau$ *satisfies*

$$\Re\rho \leq \gamma. \tag{76}$$

Thus a hypothetical off-line zero has an unconditional nonzero response in the capacity deficits. No estimate such as (75) *is assumed in this last readout.*

Proof. Write $f_s(x) = -\log(1-x^{-s})$, with its full power series. The initial Euler products and logarithms converge absolutely on $\Re s > 1$, since

$$\sum_{I \subset [u,2u)} H_I \le R\nu([u,2u)) + B_j \le Ru/\log 2 + B_j \ll_R u.$$

In particular this controls both $\sum v_p$ and Δ_j.

The exact identity $\sum_{p\in I} v_p + \delta_I = H_I$ gives

$$\log F(s) + D(s) - R\log M(s) = G_{\mathrm{init}}(s) + \sum_{j\ge 1} G_j(s), \tag{77}$$

where the initial interval, which is not part of the dyadic partition, is retained explicitly:

$$G_{\mathrm{init}}(s) = -R\int_1^2 f_s(x)\,d\nu(x).$$

The complete dyadic summand is

$$\begin{aligned} G_j(s) = &\sum_{I\subset[u,2u)} (H_I - R\nu(I)) f_s(t_I) \\ &+ R\sum_{I\subset[u,2u)} \int_I [f_s(t_I) - f_s(x)]\,d\nu(x) \\ &+ \sum_{I\subset[u,2u)} \sum_{\substack{p\in I \\ p \text{ prime}}} v_p[f_s(p) - f_s(t_I)]. \end{aligned}$$

This is an exact signed mass cancellation. In particular the deficit term has not been discarded or included in an error norm.

For $\sigma = \Re s > 0$ and $x \ge u$,

$$|f_s(x)| \le \frac{x^{-\sigma}}{1-x^{-\sigma}}, \qquad f_s'(x) = \frac{-sx^{-s-1}}{1-x^{-s}}, \qquad |f_s'(x)| \le \frac{|s|x^{-\sigma-1}}{1-x^{-\sigma}}.$$

The rounding mass in each interval is less than 1. Moreover $L_j \le u^\theta$, $B_j \le 2u^{1-\theta}$, $\nu([u,2u)) \le u/\log 2$, and $B_j \le u$. After the displayed mass cancellation, these inequalities give the explicit remainder bound

$$|G_j(s)| \le \frac{2u^{1-\theta-\sigma} + (1+2R/\log 2)|s|u^{\theta-\sigma}}{1-u^{-\sigma}}. \tag{78}$$

The first term is rounding; the other two transports together have mass at most $(1+2R/\log 2)u$ and displacement at most L_j. The bound is summable,

uniformly on compact subsets of $\Re s > \tau$. The initial integral is holomorphic on $\Re s > 0$: its density is bounded near 1, and on every closed subhalf-plane $|f_s(x)| \le -\log(1 - x^{-\sigma_0})$, an integrable logarithmic singularity at 1. This proves normal convergence of G and (74). It also gives linear height growth for the local error on each closed subhalf-plane. The rational function M^R has exactly the asserted pole and no zeros on this domain.

Now suppose (75). If $\sigma > \eta$, choose $\epsilon < \sigma - \eta$. Since $t_I \ge 2^j$, the full deficit logarithm is absolutely normally convergent on $\Re s > \eta$: on closed subhalf-planes it is majorized by

$$\sum_{j \ge 1} \frac{\Delta_j \, 2^{-j\sigma}}{1 - 2^{-j\sigma}} < \infty.$$

Thus e^D is a holomorphic unit there and the continuation

$$F = M^R e^{G-D}$$

is zero-free with pole order R at 1.

The cap $v_p \le A(p)$ now supplies the independent positivity input. On the initial half-plane the full quotient

$$B(s) := \frac{\mathcal{Z}_{\mathcal{Q}}(s)}{F(s)} = \prod_p (1 - p^{-s})^{-(A(p) - v_p)} \, U_{\mathcal{Q}}(s) = \sum_{n \ge 1} b_n n^{-s} \tag{79}$$

has nonnegative integer coefficients and $b_1 = 1$. All nonsplit prime powers and every overlap multiplicity are still present. The meromorphic continuations of the numerator and F make B holomorphic on $\Re s > \eta$: F has no zeros, and their equal pole orders cancel at 1. The positive-series principle above therefore forces actual convergence of $\sum b_n n^{-\sigma}$ for every $\sigma > \eta$.

For completeness, this convergence gives a unit, not just a positive value on the real axis. Every finite subproduct of the positive Euler factors in (79) is bounded by $\sum b_n n^{-\sigma}$ for real $\sigma > \eta$. Its nonnegative logarithm is therefore uniformly bounded. Increasing the finite set of factors proves convergence of the full logarithmic Euler sum at σ, and hence absolute normal convergence at every complex point to its right. Varying σ shows that B is a holomorphic unit on $\Re s > \eta$. Thus $\mathcal{Z}_{\mathcal{Q}} = FB$ has no zeros there. Since none of its factors has a pole away from 1, each factor is zero-free there as well, including every principal factor and hence ζ. No RH-dependent recovery theorem is used.

Finally $\Delta_j \ll_R 2^j$, so $0 \le \gamma \le 1$. If a reference zero had $\Re\rho > \max(\tau, \gamma)$, choose $\max(\tau, \gamma) < \eta < \Re\rho < 1$. The definition of the limsup implies (75), contradicting the result just proved. The standard zero-free line $\Re s = 1$ handles the endpoint if necessary. This proves (76). □

The size requirement is an Euler-capacity requirement. Under (75), positivity of the prime coefficients in (79) also gives

$$\begin{aligned} \sum_{p}(A(p)-v_p)p^{-\sigma} &< \infty \quad (\sigma>\eta), \\ \sum_{p\le X}(A(p)-v_p) &\le X^{\sigma}\sum_{p}(A(p)-v_p)p^{-\sigma}. \end{aligned} \tag{80}$$

In particular the number of eligible but unselected primes is $O_\sigma(X^\sigma)$. For $\eta<\sigma<1$, the fixed-modulus prime number theorem in progressions implies that Q retains relative density 1 within the union of the eligible split classes. With $\mathcal{Q}=\{1\}$ this means almost all primes; with $\mathcal{Q}=\{q\}$ the ambient class is $p\equiv 1 \pmod q$, whose primes have capacity $\varphi(q)$. Taking a large fixed q can make this class a small positive proportion of all primes, but does not give a zero-density carrier. Growing moduli would require new uniform estimates. The lighter carriers of the preceding subsections use a different weight budget and cannot inherit the positive quotient by assumption.

For $\sigma>1$ the selected weights also give an honest probability law:

$$a_v(p^r)=\binom{v_p+r-1}{r} \quad (v_p>0), \qquad \mathbb{P}_\sigma(n)=\frac{a_v(n)n^{-\sigma}}{F(\sigma)}.$$

Here a_v is multiplicative and is zero on positive powers of unselected primes. The quotient has its analogous positive Dirichlet law. These are explicitly weighted integer laws, not an unstated extension of a uniform prime-clock theorem. Their positive-measure structure is exactly what the continuation argument consumes.

Exact joint compensation between congruence classes. The advantage of combining moduli can be stated without discarding signs. In one fixed interval I, let

$$\begin{aligned} h_I=\lceil \nu(I)\rceil, \qquad a_{q,I}&=h_I-C_{q,I}, \\ C_{q,I}=d_q\,\#\{p\in I: p \text{ prime},\ p&\equiv 1 \pmod q\}. \end{aligned}$$

These last quantities are signed integers. Set

$$\begin{aligned} P_I=\sum_{q\in\mathcal{Q}}(a_{q,I})_+, \qquad N_I&=\sum_{q\in\mathcal{Q}}(-a_{q,I})_+, \\ r_I=Rh_I-\lceil R\nu(I)\rceil &\in\{0,\dots,R-1\}. \end{aligned}$$

Then the combined deficit obeys the exact identity

$$\delta_I = (P_I - N_I - r_I)_+ = P_I - \min(P_I, N_I + r_I). \tag{81}$$

Indeed $\sum_q a_{q,I} = P_I - N_I$ and $H_I - \sum_q C_{q,I} = \sum_q a_{q,I} - r_I$. Thus capacity surplus from one split class cancels another class's deficit in the same interval, including the precise rounding correction. Overlaps are already counted correctly in $C_I = \sum_q C_{q,I}$. This identity identifies an actual joint saving; it is not an estimate for its size. For an exponent improvement one must prove the full retained sum

$$\sum_{I \subset [2^j, 2^{j+1})} \big[P_I - \min(P_I, N_I + r_I)\big] \ll_\epsilon 2^{j(\eta+\epsilon)}. \tag{82}$$

A centered variance, a finite sample, or the existing 7/8 boundary does not establish this inequality at a smaller η.

The reverse theorem supplies a sufficient arithmetic criterion, not an equivalence between RH and these fine-interval deficit bounds. RH alone has not been shown to imply their asserted size, and including nonprincipal characters adds their own zero-free requirements. Unconditionally the construction only gives $\Delta_j \ll_R 2^j$. The certified boundary remains 7/8. The new conclusion is the exact model, its noncancelling zero readout, and the explicit capped-capacity estimate that would change that boundary. No literature priority is asserted for this construction.

21.37 Capped moment transport and longer prime intervals

Higher moment accuracy changes the interval scale in the reverse construction. The following selection theorem supplies the missing finite feasibility statement while preserving nonnegative weights and the Euler-capacity cap. Its deficit is controlled by an explicit one-sided interval discrepancy, rather than by an unsigned moment matrix norm.

Theorem 21.72 (Capped moment selection with positive defects)**.** *Let $I = [a, b]$ with $a < b$, let μ be a finite positive measure on I of mass H, and let*

$$\alpha = \sum_{i \in S} A_i \delta_{x_i}, \qquad x_i \in I, \quad A_i \geq 0,$$

where S is finite. Repeated locations are allowed and their capacities add. For an integer $k \geq 1$ define

$$d_I = \sup_{J \subseteq I} \big(\mu(J) - \alpha(J)\big)_+, \qquad m_k = \lceil k/2 \rceil. \tag{83}$$

Here J ranges over all subintervals, with every choice of included or excluded endpoints, and includes the empty interval.

There exist $0 \le v_i \le A_i$ and a positive measure λ on I, supported on at most k points, such that

$$\begin{gathered}\sum_{i\in S} v_i x_i^r + \int_I x^r \, d\lambda(x) = \int_I x^r \, d\mu(x) \quad (0 \le r < k), \\ 0 \le \lambda(I) \le \min(H, m_k d_I).\end{gathered} \tag{84}$$

Thus the actual selected mass is at least $H - m_k d_I$. If every capacity is at most $A_* > 0$, its selected support has cardinality at least $(H - m_k d_I)_+ / A_*$.

Moreover the smallest possible defect mass δ_* satisfies the exact finite-dimensional dual identity

$$\delta_* = \sup_{\substack{P\in\mathbb{R}[x],\ \deg P<k \\ P(x)\le 1\ (x\in I)}} \left(\int_I P \, d\mu - \sum_{i\in S} A_i (P(x_i))_+ \right). \tag{85}$$

The zero polynomial is allowed, so the supremum is nonnegative.

Proof. The case $H = 0$ follows by taking all weights and λ zero. Write $z(x) = (1, x, \dots, x^{k-1}) \in \mathbb{R}^k$ and $c = \int_I z \, d\mu$. For $\delta \ge 0$ the compact convex set

$$K_\delta = \left\{ \sum_{i\in S} v_i z(x_i) : 0 \le v_i \le A_i \right\} + \delta \ \mathrm{conv}(\{0\} \cup z(I))$$

is exactly the set of moments obtainable with these capacities and a positive defect measure of mass at most δ. Indeed a convex combination gives such a measure immediately. Conversely the normalized moment vector of any positive measure lies in the closed convex hull of the compact moment curve; a separating linear functional would otherwise contradict its own pointwise upper bound after integration. In finite dimensions this hull is compact, so finite convex combinations suffice.

Suppose $c \notin K_\delta$. Strict separation of a point from a compact convex set gives a polynomial P of degree less than k with

$$\int_I P \, d\mu > \sum_{i\in S} A_i (P(x_i))_+ + \delta \max(0, \max_{x\in I} P(x)). \tag{86}$$

The maximum on the right is positive: otherwise $P \le 0$ on I, the left side is nonpositive, and the right side is zero. Divide by this maximum. We obtain

a polynomial $Q \le 1$ on I such that

$$\int_I Q\,d\mu - \sum_i A_i (Q(x_i))_+ > \delta. \tag{87}$$

For $0 < t < 1$, the relative open set $\{x \in I : Q(x) > t\}$ is a union of at most m_k intervals. To verify the count, a polynomial of degree d has at most d distinct real roots unless it is identically zero; the greatest possible number of positive components on a compact interval is $\lfloor d/2 \rfloor + 1$. Between successive positive components, either two sign-changing roots or an even-multiplicity root are required; counting multiplicities proves this bound. Here $d \le k-1$, which gives $m_k = \lceil k/2 \rceil$. Endpoint conventions are covered by the definition of d_I. The constant and identically-zero cases have at most one component. Layer-cake integration now gives

$$\begin{aligned}\int_I Q\,d\mu - \sum_i A_i (Q(x_i))_+ &\le \int_I Q_+\,d(\mu - \alpha) \\ &= \int_0^1 [\mu(\{Q > t\}) - \alpha(\{Q > t\})]\,dt \le m_k d_I.\end{aligned}$$

Take $\delta = m_k d_I$. This contradicts (87), so $c \in K_\delta$. Also $c \in K_H$ by taking no actual atoms and the original measure as defect. This proves the claimed bound with $\min(H, m_k d_I)$.

A representation of the defect moments can be reduced to at most k atoms: whenever more than k positive weights occur, their vectors $z(x)$ are linearly dependent. Move the weights along a nonzero dependence until one first reaches zero, choosing the direction that decreases some weight. The first coordinate preserves their total mass, and all coordinates preserve the moments. Repeat finitely. This proves the support assertion. The support lower bound for the actual atoms follows from the zeroth moment and $v_i \le A_*$.

For the dual identity, any feasible choice with defect mass δ and any polynomial $P \le 1$ on I obey

$$\int_I P\,d\mu - \sum_i A_i (P(x_i))_+ = \sum_i [v_i P(x_i) - A_i (P(x_i))_+] + \int_I P\,d\lambda \le \delta.$$

This proves that the dual supremum is at most δ_*. A feasible optimum exists: restrict to defect mass at most H; the capped atom box and the compact truncated moment hull are compact, and the zeroth moment expresses the defect mass continuously as $H - \sum_i v_i$. For any $0 \le \delta < \delta_*$, strict separation

gives the normalized polynomial in (87), so the supremum is greater than δ. Letting δ increase to δ_* proves equality. If $\delta_* = 0$, the zero polynomial and the already proved upper inequality give the same conclusion. □

Apply this theorem in each half-open dyadic cell of the previous construction, using its closed hull for the compact moment problem. Take

$$\mu_I = R\,\nu|_I, \qquad \alpha_I = \sum_{\substack{p\in I\\ p \text{ prime}}} A(p)\delta_p.$$

Actual prime atoms remain assigned to exactly one half-open cell. Only the synthetic defect measures may share an endpoint; their masses add in the transform. There is no rounding: $H_I = R\nu(I)$ is now real, and the selected weights may be real. For $u = 2^j$ and $L_j = 2^{\lfloor \theta j \rfloor}$ define

$$\mathcal{D}_{j,k,\theta} = \sum_{I\subset[u,2u)} d_I, \qquad \widetilde{\Delta}_j = \sum_{I\subset[u,2u)} \lambda_I(I) \le m_k \mathcal{D}_{j,k,\theta}. \tag{88}$$

Choose one of the feasible moment selections in every cell. This defines the actual prime set and its capped weights independently of a zero-free hypothesis.

Theorem 21.73 (Moment accuracy extends the admissible interval length)**.** *For the preceding construction and any fixed $k \ge 1$, put*

$$\tau_k = \max\bigl(1/2,\ 1 - k(1-\theta)\bigr).$$

If $\tau_k \le \eta < 1$ and

$$\forall \epsilon > 0 \qquad \mathcal{D}_{j,k,\theta} \ll_\epsilon 2^{j(\eta+\epsilon)} \quad (j \ge 1), \tag{89}$$

then ζ and every included Dirichlet L-function have no zeros on $\Re s > \eta$. In particular, choosing

$$\theta = 1 - \frac{1}{2k} \tag{90}$$

keeps $\tau_k = 1/2$ while allowing intervals of length $X^{1-1/(2k)}$. The sufficient arithmetic exponent remains $3/4$ for boundary $3/4$, or $1/2$ for RH; the intervals on which it must be proved are longer as k increases.

Unconditionally, a reference zero with $\beta = \Re\rho > \tau_k$ forces

$$\limsup_{j\to\infty} \frac{\log(1+\mathcal{D}_{j,k,\theta})}{j\log 2} \ge \beta. \tag{91}$$

Proof. Use all the full Euler factors, now setting

$$\widetilde{D}(s) = \sum_{j\geq 1} \sum_{I\subset[u,2u)} \int_I f_s(x)\, d\lambda_I(x).$$

Initial absolute convergence on $\Re s > 1$ follows because the combined mass of actual and synthetic atoms in a block is exactly $R\nu([u,2u)) \ll_R u$. The complete identity is

$$\log F(s) + \widetilde{D}(s) - R\log M(s) = -R\int_1^2 f_s(x)\, d\nu(x) + \sum_{j\geq 1} \widetilde{G}_j(s).$$

For each cell, the degree less than k Taylor polynomial of f_s at its left endpoint cancels exactly by (84). For $x \geq u$ and $\sigma = \Re s > 0$, the full-series derivative bound from the earlier moment compression proof is

$$|f_s^{(k)}(x)| \leq (k-1)! \prod_{r=0}^{k-1} (|s|+r)\, \frac{x^{-\sigma-k}}{(1-x^{-\sigma})^k}.$$

It also follows directly by differentiating $\sum_{m\geq 1} x^{-ms}/m$: use $|(ms)(ms+1)\cdots(ms+k-1)| \leq m^k \prod_{r<k}(|s|+r)$ and $\sum_{m\geq 1} m^{k-1}z^m \leq (k-1)!z/(1-z)^k$.

The two compared positive measures in a cell both have mass H_I, so Taylor's remainder, after the exact moment cancellation, gives

$$|\widetilde{G}_j(s)| \leq \frac{2R}{k\log 2} \prod_{r=0}^{k-1} (|s|+r)\, \frac{u^{1-k(1-\theta)-\sigma}}{(1-u^{-\sigma})^k}. \tag{92}$$

This is normally summable when $\sigma > \max(0, 1-k(1-\theta))$. The initial integral has already been proved holomorphic on $\Re s > 0$. Thus the completed model is zero-free there, apart from its prescribed pole, with polynomial height cost of degree k. Both k and the moduli are fixed.

By (88) and (89), the synthetic logarithm converges normally on $\Re s > \eta$, with majorant

$$\sum_{j\geq 1} \frac{\widetilde{\Delta}_j\, 2^{-j\sigma}}{1-2^{-j\sigma}}.$$

It follows that $F = M^R \exp(\widetilde{G} - \widetilde{D})$ is zero-free there, with the exact same pole order R. Although the v_p are now real, the quotient is still positive: for every real $a \geq 0$,

$$(1-z)^{-a} = \sum_{r\geq 0} \frac{a(a+1)\cdots(a+r-1)}{r!}\, z^r, \qquad |z| < 1,$$

where the constant coefficient is 1. Thus $a = A(p) - v_p \geq 0$ gives nonnegative real Dirichlet coefficients in the full quotient $\mathcal{Z}_Q/F$. The compiled positive-series core explicitly allows real coefficients, so the same positive continuation and Euler-logarithm argument from Theorem 21.71 makes that quotient a unit on $\Re s > \eta$. The zero-free conclusion follows.

Finally $d_I \leq H_I$, so $\mathcal{D}_{j,k,\theta} \leq R\nu([u, 2u)) \ll_R u$. If its displayed limsup were smaller than a reference zero's real part $\beta > \tau_k$, choose η strictly between their maximum and β. The resulting power bound would contradict the zero-free conclusion. This proves the unconditional readout. □

This is a quantitative use of higher accuracy: $k = 1, 2, 3, 4$ permit cell-length exponents $1/2, 3/4, 5/6, 7/8$, respectively, with the same RH-side analytic threshold. The new arithmetic quantity is a sum of maximal *one-sided subinterval capacity deficits*; it is not automatically controlled by the deficit of each whole cell. The capped moment theorem proves the transfer to that quantity, including its factor m_k, and its dual identity preserves the positive and negative contributions before layer-cake comparison. It does not prove the required power bound for primes. In particular, longer intervals, finite numerical diagnostics, or a larger character portfolio alone do not certify an exponent gain. The global certified boundary is still 7/8.

The two theorems in this subsection have full ordinary proofs. Finite moment compression and positive Dirichlet continuation have compiled providers, but the compact separation argument, real-capacity moment selection and analytic consumers above have not yet been formalized in Lean. No such formalization or literature priority is claimed.

21.38 The two-moment defect as a finite signed prime test

For two moments the polynomial dual can be evaluated exactly from a finite list of integrated prime-capacity sums. This makes the reverse construction explicit without relaxing its signed main terms.

Theorem 21.74 (Exact finite formula for the two-moment deficit)**.** *Let μ be a positive finite measure on $[a, b]$, where $a < b$, with mass $H > 0$ and first moment M, and assume $m = M/H \in (a, b)$. Let $\alpha = \sum_{i \in S} A_i \delta_{x_i}$, where S is finite, $x_i \in [a, b]$, $A_i \geq 0$, and $C = \sum_i A_i$. Define*

$$L(c) = \frac{cH - M - \sum_i A_i (c - x_i)_+}{c - a} \qquad (c > m),$$

$$R(c) = \frac{M - cH - \sum_i A_i (x_i - c)_+}{b - c} \qquad (c < m).$$

Set

$$T_L = \{b\} \cup \{x_i : m < x_i < b\}, \qquad T_R = \{a\} \cup \{x_i : a < x_i < m\}.$$

These are finite nonempty sets of distinct locations; repeated atoms contribute their summed capacities in the formulas. The least defect mass preserving zeroth and first moments is

$$\delta_* = \max\left\{0,\ H - C,\ \max_{c \in T_L} L(c),\ \max_{c \in T_R} R(c)\right\}. \tag{93}$$

In particular every retained denominator is strictly positive. The formula includes both boundary tests and the raw total mass deficit, even when that last term is negative.

Proof. Use the exact dual in Theorem 21.72 for $k = 2$. A dual affine polynomial whose maximum on $[a, b]$ is nonpositive has nonpositive objective. If its maximum is $0 < t < 1$ and its objective is positive, division by t increases that objective. Thus it suffices to consider the zero polynomial and affine polynomials with maximum 1.

The constant polynomial 1 gives $H - C$. A nonconstant normalized affine polynomial has one of the forms

$$1 - \lambda(x - a) \quad \text{or} \quad 1 - \lambda(b - x), \qquad \lambda > 0.$$

For the first form put $c = a + 1/\lambda$. Its exact objective is $L(c)$, since $(1 - \lambda(x - a))_+ = (c - x)_+/(c - a)$. When $c \le m$, its model integral $(cH - M)/(c - a)$ is nonpositive, so its objective cannot improve on 0. This leaves $c > m$. On each interval between consecutive atom locations, the numerator of L is affine in c. A quotient of an affine function by $c - a > 0$ is monotone or constant on that interval: the sign of its derivative is constant. Consequently its supremum for $m < c \le b$ is attained at a location in T_L, or is bounded by its limiting value at m, which is nonpositive. For $c \ge b$ all atoms are active, and

$$L(c) = \frac{c(H - C) - M + \sum_i A_i x_i}{c - a}.$$

Its supremum is bounded by the larger of $L(b)$ and its limit $H - C$ at infinity. Thus the entire decreasing-affine branch reduces to $0, H - C$, and the T_L tests.

For the increasing-affine form set $c = b - 1/\lambda$. The identical argument gives $R(c)$, discards $c \ge m$ because its model integral is nonpositive, and reduces the remaining branch to T_R and the limit $H - C$ at negative infinity.

Conversely every retained test is the objective of an admissible affine polynomial; 0 and $H - C$ come from admissible constants. The dual equality proves (93). □

The principal compensation is explicit. For $a \le t \le b$ write

$$E_L(t) = \mu([a,t]) - \alpha([a,t]), \qquad E_R(t) = \mu([t,b]) - \alpha([t,b]).$$

Fubini's theorem, with all endpoint atoms retained, gives

$$L(c) = \frac{\displaystyle\int_a^c E_L(t)\,dt - \int_{(c,b]} (x-c)\,d\mu(x)}{c-a} \qquad (m < c \le b), \tag{94}$$

$$R(c) = \frac{\displaystyle\int_c^b E_R(t)\,dt - \int_{[a,c)} (c-x)\,d\mu(x)}{b-c} \qquad (a \le c < m). \tag{95}$$

For example, $\int_a^c \alpha([a,t])\,dt = \sum_i A_i(c-x_i)_+$. The analogous identity for μ supplies its left truncated moment; subtracting the right tail restores the complete signed moment $cH - M$. The right formula is proved the same way. These are exact identities, including compensation across the two sides of the strike c.

For the actual prime model $\mu = R\nu|_{[a,b]}$ in a cell $[a,b] \subset [u,2u]$, $u \ge 2$, the coefficient inequalities

$$\frac{1}{\zeta(2)} \le \frac{1}{\zeta(r+1)} \le 1 \qquad (r \ge 1)$$

give the unconditional pointwise lower density

$$\nu'(x) \ge \frac{1-x^{-1}}{\zeta(2)\log x} \ge \ell_u := \frac{1-u^{-1}}{\zeta(2)\log(2u)} > 0.$$

Hence the retained negative model tails have explicit reserves:

$$L(c) \le \frac{\displaystyle\int_a^c E_L(t)\,dt - R\ell_u(b-c)^2/2}{c-a}, \tag{96}$$

$$R(c) \le \frac{\displaystyle\int_c^b E_R(t)\,dt - R\ell_u(c-a)^2/2}{b-c}. \tag{97}$$

The domains are respectively $c \in T_L$ and $c \in T_R$. No absolute value has replaced the signed integrated prime-capacity discrepancy. At $c = b$ in the left test and $c = a$ in the right test, the corresponding reserve is exactly zero. Those endpoint tests and $H - C$ remain in (93); the interior reserve does not estimate them.

The atomic support calculation used in the dual now has its own compiled owner, `CappedSupportFunction.lean`. It proves the signed capped-dot upper bound and an actual supported real maximizer attaining $\sum_i A_i(P(x_i))_+$, for arbitrary real coefficients $P(x_i)$, including negative ones. This checks the exact box support term used in the separation proof. The finite strike reduction, continuous moment selection, and analytic zero readout retain their ordinary proofs here; they are not reported as compiled.

For the prime application, choose $k = 2$ and $\theta = 3/4$. The analytic error in (92) is then normally summable for $\Re s > 1/2$, while the cell length is $X^{3/4}$. The complete arithmetic consumer is

$$\sum_{I \subset [2^j, 2^{j+1})} \max\left\{0, H_I - C_I, \max_{c \in T_{L,I}} L_I(c), \max_{c \in T_{R,I}} R_I(c)\right\} \ll_\epsilon 2^{j(\eta+\epsilon)}. \quad (98)$$

Here $H_I = R\nu(I)$ is real, with no rounding, and the actual prime capacity is $C_I = \sum_{p \in I,\ p \text{ prime}} A(p)$. The preceding theorem constructs capped weights with exactly this optimal synthetic defect in every cell. If (98) is proved for $\eta = 3/4$, the positive quotient argument proves the zero-free boundary $3/4$; $\eta = 1/2$ would imply RH. One can use this exact optimal defect in place of the sufficient maximal interval deficit, since only its positive mass enters the convergence majorant.

This is a universal finite formula and an independently bounded model compensation, not a finite numerical experiment or a proved prime-discrepancy saving. The missing estimates are the full signed integrated-capacity terms including their endpoint tests and total mass deficit. The only unconditional aggregate bound installed here is $O_R(2^j)$, so the certified zero-free boundary remains $7/8$. No literature priority is asserted for this reduction.

21.39 One-sided overweight control for a favorable prime model

The reverse order can also be expressed directly for an actual weighted prime Euler product. It is enough to control its overweight relative to the available reference capacities; pointwise domination is sufficient but is not necessary.

Theorem 21.75 (One-sided favorable-model transfer)**.** *Fix the finite character portfolio and its capacities $A(p)$, and let $1/2 \le \eta < 1$. The following are equivalent.*

1. *The reference $\mathcal{Z}_{\mathcal{Q}}$ is zero-free on $\Re s > \eta$.*
2. *There exist real weights $w_p \ge 0$, with $\sum_p w_p p^{-\sigma} < \infty$ for every $\sigma > 1$, whose initial full Euler product*

$$F_w(s) = \prod_p (1 - p^{-s})^{-w_p}$$

has a holomorphic nowhere-zero continuation on $\{\Re s > \eta\} \setminus \{1\}$, a pole of order R at 1, and satisfies the one-sided size bound

$$\forall \epsilon > 0 \qquad E(X) := \sum_{p \le X} (w_p - A(p))_+ \ll_\epsilon X^{\eta+\epsilon} \quad (X \ge 2). \tag{99}$$

For $\mathcal{Q} = \{1\}$ and $\eta = 1/2$, this is an RH equivalence. For a general portfolio at $\eta = 1/2$ it is the zero-free assertion for all of its Dirichlet L-functions, including ζ.

More generally, if a particular F_w has the stated favorable continuation on $\Re s > \tau$ with $1/2 \le \tau < 1$, put

$$E_j = \sum_{2^j \le p < 2^{j+1}} (w_p - A(p))_+.$$

Any reference zero of real part $\beta > \tau$ forces

$$\limsup_{j \to \infty} \frac{\log(1 + E_j)}{j \log 2} \ge \beta. \tag{100}$$

Thus a favorable prime model cannot erase a hypothetical bad-zero response while keeping its positive overweight smaller than that response.

Proof. Set $e_p = (w_p - A(p))_+$. Dyadic summation shows that (99) is equivalent to $\sum_p e_p p^{-\sigma} < \infty$ for every $\sigma > \eta$. For the converse direction of this elementary equivalence, use $E(X) \le X^\sigma \sum_p e_p p^{-\sigma}$. Accordingly

$$E_+(s) = \prod_p (1 - p^{-s})^{-e_p}$$

is an absolutely convergent holomorphic unit on $\Re s > \eta$, with every prime power retained.

Under (2), the complete quotient is initially

$$\frac{\mathcal{Z}_{\mathcal{Q}}(s)E_+(s)}{F_w(s)} = \prod_p (1-p^{-s})^{-(A(p)-w_p)_+}\, U_{\mathcal{Q}}(s). \tag{101}$$

Its coefficients are nonnegative real numbers. Its continuation is holomorphic on $\Re s > \eta$: F_w is zero-free and its pole of order R cancels the reference pole, while E_+ is a unit. The positive Dirichlet continuation principle therefore gives actual series convergence throughout this half-plane. The finite-positive-product and logarithm argument in Theorem 21.71 makes the full quotient a unit there. Equation (101) then proves that $\mathcal{Z}_{\mathcal{Q}}$ is zero-free. This proves (2) implies (1).

Conversely, under (1) choose $w_p = A(p)$. Then $F_w = \mathcal{Z}_{\mathcal{Q}}/U_{\mathcal{Q}}$ has the required favorable continuation and pole order, and $E(X) = 0$. For $\mathcal{Q} = \{1\}$ the reference is ζ; the usual functional-equation symmetry identifies its zero-free open right half of the critical strip with RH. For other portfolios, the reference's factors have no poles away from 1, so its zero-free assertion is exactly the corresponding assertion for every included factor.

For the last statement, if the displayed limsup were smaller than $\beta > \tau$, choose an exponent strictly between their maximum and β. Dyadic summation makes E_+ a unit above that exponent, and the same quotient proof excludes the zero, a contradiction. □

The necessary size of the selected set. Let $Q_w = \{p : w_p > 0\}$ and $A_* = \sum_{q\in\mathcal{Q}} \varphi(q) > 0$. Write $W_j = \sum_{2^j \le p < 2^{j+1}} w_p$. Since $A(p) \le A_*$, the exact one-sided comparison gives

$$\#(Q_w \cap [2^j, 2^{j+1})) \ge \frac{(W_j - E_j)_+}{A_*}. \tag{102}$$

Indeed $w_p \le A(p) + e_p$ on each selected prime, and both w_p and e_p vanish outside the selected set. For a prime quadrature with $W_j = R\nu([u, 2u))$, $u = 2^j$, the explicit density lower bound above gives $W_j \ge R\ell_u u$. If $E_j \ll_\epsilon u^{\eta+\epsilon}$ with $\eta < 1$, this forces selected support of order at least $u/\log u$, with constants depending only on the fixed portfolio. A small support with large weights must pay for those weights in E_j; its cardinality alone is not the signal-retention test.

The converse in the equivalence uses the zero-free assertion itself. The next theorem supplies an independently favorable model on actual primes, completing the construction part of the proposed reverse order. Its one-sided overweight still requires a separate arithmetic estimate. The transfer

theorem identifies the exact sufficient size condition and its noncancelling zero response. Its proof is ordinary, not yet a separate Lean declaration, and the certified unconditional reference boundary remains 7/8.

21.40 An unconditional favorable model on actual primes

The favorable model can be built on actual primes before estimating the size of its overweight. Only a weak classical prime-gap input is needed to guarantee that the finite moment problems have solutions: there are constants $C_g > 0$ and p_0 such that consecutive primes $p < p'$ with $p \geq p_0$ satisfy

$$p' - p \leq C_g p^{2/3}. \tag{103}$$

This follows, for example, from the stronger short-interval theorem of Baker–Harman–Pintz [21, Theorem 1]. It is used only for the availability of prime locations. It supplies no estimate for the one-sided overweight below.

Theorem 21.76 (Actual prime model with optimal local overweight)**.** *Fix the finite character portfolio. There is a fixed integer J_0 such that the following construction works for every $j \geq J_0$. Partition $[u, 2u)$, $u = 2^j$, into half-open cells of length $h = 2^{\lfloor 3j/4 \rfloor}$. For a cell $I = [a, b)$ put*

$$H_I = R\nu(I), \qquad M_I = R\int_I x\, d\nu(x), \qquad m_I = M_I/H_I,$$

and let S_I be its actual primes, with extreme primes p_-, p_+. They satisfy $p_- < m_I < p_+$. Choose a nonnegative weight vector solving the finite problem

$$\min \sum_{p\in S_I} (w_p - A(p))_+, \qquad \sum_{p\in S_I} w_p = H_I, \qquad \sum_{p\in S_I} p\, w_p = M_I. \tag{104}$$

A minimizer exists. Fix it, for example, by lexicographic minimization of the ordered coordinates among all minimizers. Set $w_p = 0$ for primes below 2^{J_0}.

The resulting actual prime Euler product has an unconditional continuation

$$F_w(s) = \left(\frac{s}{s-1}\right)^R e^{G(s)} \qquad (\Re s > 1/2), \tag{105}$$

where G is holomorphic. Thus F_w is zero-free on this half-plane and has precisely a pole of order R at 1.

Let E_I denote the optimal value in (104)*, and let δ_I^* be the ideal two-moment defect from Theorem 21.74 on the original cell hull $[a,b]$, with the original mass H_I, moment M_I and actual prime capacities. Then*

$$\delta_I^* \le E_I \le K_I \delta_I^*, \qquad K_I = \max\left\{1, \frac{m_I - a}{m_I - p_-}, \frac{b - m_I}{p_+ - m_I}\right\}. \tag{106}$$

All these denominators are positive. Uniformly over the cells,

$$1 \le K_I \le 1 + C_1 u^{-1/12}, \qquad C_1 = \frac{4C_g\, 2^{2/3}}{c_0}, \quad c_0 = \frac{1}{8\zeta(2)}. \tag{107}$$

In particular passing from the ideal defect to actual prime weights loses neither a power nor an asymptotic leading coefficient. The model is independently favorable; excluding reference zeros still requires the separate overweight estimate.

Proof. On $[u, 2u]$ the model density lies between ℓ_u and $r_u = 1/\log u$. Therefore the two distances of its cell mean from the endpoints obey

$$m_I - a,\ b - m_I \ge \frac{\ell_u}{2r_u} h \ge c_0 h.$$

The last inequality uses $u \ge 2$, $1 - u^{-1} \ge 1/2$, and $\log u/\log(2u) \ge 1/2$.

For all sufficiently large u, the prime-gap bound gives

$$0 \le p_- - a \le g_u, \qquad 0 < b - p_+ \le g_u, \qquad g_u = C_g(2u)^{2/3}.$$

For the left edge use the prime just preceding a, unless a itself is prime; for the right edge use the prime preceding b, since b is excluded. Bertrand's theorem ensures that these preceding primes are at least $u/2$ when u is large, so the starting threshold p_0 is respected. Because $h \ge u^{3/4}/2$, choose $J_0 \ge 2$ with

$$2^{J_0} > 2p_0 + 2, \qquad 2C_g 2^{2/3} 2^{-J_0/12} \le c_0/2.$$

These monotone conditions hold at every later block and give $g_u \le c_0 h/2$. In particular there are actual primes on both sides of m_I. No numerical value for the classical starting threshold is asserted.

Nonnegative weights on just p_-, p_+ can already match H_I, M_I, by the barycentric formulas

$$w_- = \frac{H_I(p_+ - m_I)}{p_+ - p_-}, \qquad w_+ = \frac{H_I(m_I - p_-)}{p_+ - p_-}.$$

The feasible set of all weight vectors is nonempty and compact: it is closed and every coordinate lies in $[0, H_I]$. The overweight objective is continuous, so it attains its minimum. Successive coordinate minima on its compact minimizer set justify the stated tie rule.

To identify the minimum, consider the capped moment-defect problem on the smaller hull $[p_-, p_+]$, with model measure $H_I\delta_{m_I}$. This is only a device for this finite moment problem: its mass and first moment are exactly the original H_I, M_I. The continuous model used in the analytic comparison remains $R\nu|_I$. Let its optimal defect be δ'_I. Any feasible actual w can be clipped to $v_p = \min(w_p, A(p))$; its removed weights give a positive actual-prime defect of mass $\sum_p (w_p - A(p))_+$. Thus $\delta'_I \le E_I$. Conversely, redistribute any optimal positive defect on $[p_-, p_+]$ to the two actual extreme primes by barycentric weights, preserving its mass and first moment. Add those weights to the capped vector. The resulting actual vector is feasible for (104), and its positive overweight is at most the added mass. Hence $E_I \le \delta'_I$, proving equality.

The finite strike formula for δ'_I has exactly the same numerators as the original ideal formula. Its left strikes are the actual primes above m_I, including p_+, and its right strikes are the actual primes below m_I, including p_-. Its denominators are respectively $c - p_-$ and $p_+ - c$. The ideal formula tests these same locations with denominators $c - a$ and $b - c$. For $c > m_I$,

$$\frac{c-a}{c-p_-} \le \frac{m_I - a}{m_I - p_-};$$

the symmetric right ratio is bounded by $(b - m_I)/(p_+ - m_I)$. If a numerator is negative it is already dominated by the zero test; otherwise these ratios bound the corresponding objectives. The common tests $0, H_I - C_I$ require no factor. This proves $E_I \le K_I \delta^*_I$. Every smaller-hull defect is feasible on the larger cell hull, so $\delta^*_I \le \delta'_I = E_I$.

Finally

$$K_I \le 1 + \frac{g_u}{c_0 h - g_u} \le 1 + \frac{2g_u}{c_0 h} \le 1 + C_1 u^{-1/12},$$

which proves the asserted uniform loss.

It remains to verify favorability independently of the overweight size. Every block has exact total weight $R\nu([u, 2u)) \ll_R u$, so the full prime product converges absolutely for $\Re s > 1$. The complete signed logarithmic

identity there is

$$\log F_w(s) - R\log M(s) = -R\int_1^{2^{J_0}} f_s(x)\,d\nu(x) + \sum_{j\ge J_0} G_j(s),$$

$$G_j(s) = \sum_{I\subset[2^j,2^{j+1})} \left[\sum_{p\in S_I} w_p f_s(p) - R\int_I f_s(x)\,d\nu(x)\right].$$

Both the constant and linear Taylor terms cancel in every cell. The $k=2$ remainder calculation already proved above gives the full block bound

$$|G_j(s)| \le \frac{R}{\log 2}|s|(|s|+1)\frac{u^{1/2-\Re s}}{(1-u^{-\Re s})^2}. \tag{108}$$

It converges normally on $\Re s > 1/2$. The retained finite initial integral is holomorphic on $\Re s > 0$, including its integrable endpoint at 1. Consequently the right side defines a holomorphic G on $\Re s > 1/2$, and exponentiation proves (105). This proves the favorable continuation without assuming any zero-free assertion for the reference. □

For this explicit optimized model, put $E_j = \sum_{I\subset[u,2u)} E_I$ and $\Delta_j^* = \sum_{I\subset[u,2u)} \delta_I^*$. The complete comparison is

$$\Delta_j^* \le E_j \le (1 + C_1 u^{-1/12})\Delta_j^*. \tag{109}$$

An independent estimate $E_j \ll_\epsilon u^{3/4+\epsilon}$ therefore gives boundary 3/4 by Theorem 21.75; exponent 1/2 would imply RH and the corresponding zero-free assertions for the included characters. Every hypothetical reference zero with real part $\beta > 1/2$ forces the dyadic overweight exponent to be at least β. The ideal and actual exponents agree by (109). This completes the construction and signal-retention parts of the reverse strategy.

The remaining arithmetic problem is the full finite-strike sum from (98), including its total mass deficit and both boundary tests. Unconditionally $E_I \le H_I$, so $E_j \le R\nu([u,2u)) \ll_R u/\log u$. That logarithmic bound supplies no new fixed power saving. The favorable model is not ζ; its positive overweight still carries the potential zeros. The certified reference boundary remains 7/8. This actual-prime construction and comparison have ordinary proofs here and consume the cited classical prime-gap theorem; they have not yet been formalized in Lean. No literature priority is asserted.

21.40.1 Higher moments and sparse repair primes

The actual construction extends to every fixed moment order. Its overweight can moreover be confined to a sparse set of actual primes. This is sparsity of the repair, not an estimate for its weighted mass.

Lemma 21.77 (Positive-objective polynomial norming)**.** *Let $I=[a,b]$, $h=b-a>0$, and let μ have density between ℓ and r, where $0<\ell\le r$. Let a finite set $S\subset I$ have covering radius at most ϵh on I. For an integer $d\ge 1$ put*

$$B_d=\frac{2^d d^{d+1}}{d!},\qquad C_d=4B_d^2 r/\ell.$$

If $C_d\epsilon<1$, every real polynomial P of degree at most d with $\int_I P\,d\mu>0$ satisfies

$$0<\max_I P\le\frac{\max_{x\in S}P(x)}{1-C_d\epsilon}. \tag{110}$$

In particular the moment vector of $\mu/\mu(I)$ through degree d belongs to the convex hull of the corresponding moment vectors of the actual nodes.

Proof. Write $L=\max_I|P|$. Interpolation at the $d+1$ nodes $a+ih/d$, $0\le i\le d$, gives the elementary derivative bound

$$\max_I|P'|\le\frac{B_d}{h}L.$$

Indeed, the denominator of the ith Lagrange basis polynomial has absolute value $(h/d)^d i!(d-i)!$. Its derivative numerator is a sum of d products, each bounded by h^{d-1} on I. Summing these bounds and using $\sum_{i=0}^{d}1/(i!(d-i)!)=2^d/d!$ proves the assertion. This argument also covers constant polynomials represented on this interpolation grid.

Let $M=\max_I P>0$. If $L=M$, then $L\le 4B_d(r/\ell)M$. Otherwise P attains $-L$. On the intersection of I with a radius $h/(2B_d)$ interval about such a point it is at most $-L/2$. This intersection has length at least $h/(2B_d)$; here $B_d\ge 2$. Consequently

$$\int_I P_-\,d\mu\ge\frac{\ell hL}{4B_d},\qquad\int_I P_+\,d\mu\le rhM.$$

The positive integral gives $L\le 4B_d(r/\ell)M$ in this case as well. A node within ϵh of a positive maximum therefore has value at least $M-B_d\epsilon L\ge(1-C_d\epsilon)M$. This proves (110) with all signs retained.

If the normalized moment vector were outside the finite convex hull, strict separation would give a polynomial whose μ-average is larger than its

maximum on S. Subtract that node maximum. The resulting polynomial has positive integral and nonpositive values at every node, contradicting (110). □

Lemma 21.78 (Actual overweight dual and sparse attainment)**.** *Let $k \geq 2$, let μ be a positive measure of mass $H > 0$ on a compact interval I, and let $S \subset I$ be finite. Give its nodes capacities $A_x \geq 0$. Suppose nonnegative weights on S can match the moments of μ through degree $k-1$. Let E be the minimum positive overweight among such weights, and let δ^* be the ideal capped defect on the whole interval from Theorem 21.72. Then*

$$E = \sup_{\substack{\deg P < k \\ P(x) \leq 1 \ (x \in S)}} \left(\int_I P \, d\mu - \sum_{x \in S} A_x (P(x))_+ \right). \tag{111}$$

A minimizer exists with $(w_x - A_x)_+ > 0$ on at most k nodes. If the norming conclusion (110) *holds with $C_d \epsilon = t < 1$, $d = k-1$, then*

$$\delta^* \leq E \leq (1-t)^{-1} \delta^*. \tag{112}$$

Proof. The feasible weights form a nonempty compact set: their zeroth moment is H, so each coordinate is in $[0, H]$. The continuous overweight objective therefore attains its minimum.

Put $z(x) = (1, x, \ldots, x^{k-1})$ and $c = \int_I z \, d\mu$. For $\delta \geq 0$ the capped weights with actual-node repair mass at most δ have the compact convex moment set

$$K_\delta = \left\{ \sum_{x \in S} v_x z(x) : 0 \leq v_x \leq A_x \right\} + \delta \operatorname{conv}\big(\{0\} \cup \{z(x) : x \in S\}\big).$$

Clipping w to $\min(w, A)$ shows $c \in K_E$. Conversely, $c \in K_\delta$ gives feasible actual weights with overweight at most δ. Hence $c \notin K_\delta$ when $0 \leq \delta < E$.

Strict separation in that case gives a polynomial P with

$$\int_I P \, d\mu > \sum_{x \in S} A_x (P(x))_+ + \delta \max\{0, \max_S P\}.$$

Its node maximum is positive: otherwise any already feasible nonnegative actual quadrature would make its integral nonpositive, contrary to this strict inequality. Normalize by that maximum. The resulting dual objective is larger than δ. Weak duality follows directly, node by node, from $P(x) \leq 1$:

$$w_x P(x) - A_x (P(x))_+ \leq (w_x - A_x)_+.$$

Letting δ increase to E proves the dual equality; if $E = 0$, the zero polynomial supplies equality.

For sparse attainment start with an optimal w and write $v = \min(w, A)$, $e = (w - A)_+$. A nonnegative combination $\sum e_x z(x)$ can be reduced to at most k nodes while preserving that entire vector. If more than k vectors remain, take a linear dependence among them. Its coefficients sum to zero by the first coordinate, so they have both signs. Move the positive weights along that dependence until one vanishes, and repeat. Denote the resulting weights by e'. They have the same mass E. Then $w' = v + e'$ matches all moments and has overweight at most E, hence exactly E by optimality. Its positive overweight is supported among the at most k nodes supporting e'.

Clipping also makes every actual minimizer an admissible ideal capped-defect solution, proving $\delta^* \le E$. For the other inequality take a polynomial in the actual dual with positive objective. Its integral is positive, so norming gives $M = \max_I P \le (1-t)^{-1}$. The polynomial P/M is at most 1 on all of I. The exact ideal dual already proved above, and positive homogeneity of its objective, give an actual objective at most $M\delta^* \le (1-t)^{-1}\delta^*$. Nonpositive objectives require no bound beyond $0 \le \delta^*$. Taking the supremum proves (112). □

Theorem 21.79 (Actual favorable models with sparse repair)**.** *Fix the finite character portfolio and an integer $k \ge 2$. Put*

$$\theta_k = 1 - \frac{1}{2k}, \qquad d = k - 1, \qquad C_k = 16\zeta(2)B_d^2.$$

There is a fixed $J_0(k, \mathcal{Q})$ with the following properties. For each $j \ge J_0$, $u = 2^j$, partition $[u, 2u)$ into cells of length $h = 2^{\lfloor \theta_k j \rfloor}$. On every cell choose nonnegative actual prime weights minimizing

$$\begin{aligned} E_{I,k} &= \min \sum_{p \in S_I} (w_{p,k} - A(p))_+, \\ \sum_{p \in S_I} p^r w_{p,k} &= R \int_I x^r \, d\nu(x) \quad (0 \le r < k). \end{aligned} \tag{113}$$

A minimizer with at most k positive-overweight coordinates exists. Choose the lexicographically least such minimizer in increasing prime order, and put $w_{p,k} = 0$ below 2^{J_0}. This defines the model before any RH assumption.

Its full actual prime Euler product has an unconditional continuation

$$F_{k,\mathcal{Q}}(s) = \prod_p (1 - p^{-s})^{-w_{p,k}} = \left(\frac{s}{s-1}\right)^R e^{G_k(s)}, \qquad \Re s > 1/2, \tag{114}$$

where G_k is holomorphic. The product initially means its full Euler logarithm on $\Re s > 1$.

Let $e_{p,k} = (w_{p,k} - A(p))_+$, $\mathcal{B}_k = \{p : e_{p,k} > 0\}$, and $E_{j,k} = \sum_{u \le p < 2u} e_{p,k}$. Then, unconditionally,

$$\#(\mathcal{B}_k \cap [u, 2u)) \le k\, u/h \le 2k\, u^{1/(2k)}. \tag{115}$$

For the ideal degree-$(k-1)$ defects on the original cells, write $\Delta^*_{j,k} = \sum_I \delta^*_{I,k}$. With the same fixed prime-gap constants as above, put

$$\epsilon_{j,k} = C_g (2u)^{2/3}/h.$$

The starting threshold can be chosen so $C_k \epsilon_{j,k} \le 1/2$, and then

$$\begin{aligned} \Delta^*_{j,k} \le E_{j,k} \le (1 - C_k\epsilon_{j,k})^{-1} \Delta^*_{j,k}, \\ E_{j,k} \le (1 + 2C_k\epsilon_{j,k})\Delta^*_{j,k}, \qquad \epsilon_{j,k} \ll_k u^{2/3-\theta_k} = o(1). \end{aligned} \tag{116}$$

Every reference zero with real part $\beta > 1/2$ forces

$$\limsup_{j\to\infty} \frac{\log(1 + E_{j,k})}{j \log 2} \ge \beta. \tag{117}$$

In particular, for every $\epsilon > 0$, infinitely many repair primes satisfy

$$e_{p,k} \ge p^{\beta - 1/(2k) - \epsilon}. \tag{118}$$

All constants, thresholds and choices here may depend on the fixed k and the fixed portfolio; k and conductors do not grow with j.

Proof. On a cell in $[u, 2u]$, the density of $R\nu$ lies between $R\ell_u$ and $R/\log u$, with ratio at most $4\zeta(2)$. The prime-gap argument in Theorem 21.76 gives covering radius at most $g_u = C_g(2u)^{2/3}$ using actual primes inside the cell. This includes both end strips. Since

$$\epsilon_{j,k} \le 2C_g 2^{2/3} u^{2/3-\theta_k} \longrightarrow 0,$$

choose $J_0 \ge 3$ with $2^{J_0} > 2p_0 + 2$ and $2C_kC_g 2^{2/3} 2^{J_0(2/3-\theta_k)} \le 1/2$. These inequalities hold for all later blocks; they ensure the grid exists inside each cell and the norming hypothesis holds. The cell endpoints are even integers greater than 2, so they carry no prime atoms. The original half-open partition and its closed hulls therefore use identical prime nodes and continuous moments.

The norming lemma places the normalized target moment vector in the convex hull of actual prime moment vectors. It gives feasibility of (113). The second lemma gives attainment, sparse attainment and the ideal comparison, uniformly on all cells. The set of optimal weights with at most k positive-overweight coordinates is a finite union of closed subsets of the compact optimal set: outside a chosen set of at most k coordinates require $w_p \le A(p)$. Successive coordinate minima therefore define the announced tie rule. Summing the cell comparisons and counting the cells proves (115) and (116).

Every block has total weight $W_j = R\nu([u,2u)) \le Ru/\log u$, ensuring initial absolute convergence on $\Re s > 1$. Retain the complete normalization:

$$G_k(s) = -R\int_1^{2^{J_0}} f_s(x)\,d\nu(x) + \sum_{j\ge J_0} G_{j,k}(s),$$

$$G_{j,k}(s) = \sum_{I\subset[u,2u)} \left[\sum_{p\in S_I} w_{p,k} f_s(p) - R\int_I f_s(x)\,d\nu(x)\right], \qquad u = 2^j. \quad (119)$$

Here $f_s(x) = -\log(1-x^{-s})$ retains every prime power. The moments cancel every Taylor term of degree less than k. The full remainder estimate from Theorem 21.73 gives

$$|G_{j,k}(s)| \le \frac{2R}{k\log 2}\left(\prod_{r=0}^{k-1}(|s|+r)\right)\frac{u^{1/2-\Re s}}{(1-u^{-\Re s})^k}. \quad (120)$$

Indeed $h^k \le u^{k-1/2}$, and the model and the actual quadrature have equal positive mass. This bound converges normally on $\Re s > 1/2$. The retained finite initial integral is holomorphic on $\Re s > 0$, with an integrable logarithmic endpoint at 1. Exponentiating the initial exact identity proves (114).

Apply Theorem 21.75 to this independently favorable model. If the limsup in (117) were below β, choose η strictly between their maximum with $1/2$ and β. The dyadic bounds would give cumulative overweight $O(X^\eta)$, excluding that reference zero. This proves the signal statement without assuming RH. For the last claim choose arbitrarily late blocks with $E_{j,k} > u^{\beta-\epsilon/3}$ and use (115). Some repair weight is at least $(2k)^{-1}u^{\beta-1/(2k)-\epsilon/3}$. For large blocks this is at least $p^{\beta-1/(2k)-\epsilon}$, since $u \le p < 2u$. The chosen primes tend to infinity. □

Corollary 21.80 (An RH equivalence for predefined sparse repairs)**.** *Take $\mathcal{Q} = \{1\}$ and construct all the above models unconditionally. Let $\mathcal{K}$ be any unbounded subset of $\{2,3,\dots\}$. The following are equivalent:*

1. *RH.*

2. *For every fixed $k \in \mathcal{K}$ and $\epsilon > 0$,*

$$E_{j,k} \ll_{k,\epsilon} u^{1/2+1/(2k)+\epsilon}, \qquad u = 2^j. \tag{121}$$

3. *For every fixed $k \in \mathcal{K}$ and $\epsilon > 0$,*

$$\max_{\substack{u \le p < 2u \\ p \text{ prime}}} e_{p,k} \ll_{k,\epsilon} u^{1/2+1/(2k)+\epsilon}. \tag{122}$$

An empty maximum is defined as zero. The ideal defects $\Delta^*_{j,k}$ may replace $E_{j,k}$ in the second condition. The quantifiers fix k before taking $j \to \infty$.

Proof. First compare the reference counting model with the usual logarithmic integral $\mathrm{Li}_2(X) = \int_2^X dt/\log t$. For $r \ge 1$, the integral test gives

$$0 \le 1 - \zeta(r+1)^{-1} \le \zeta(r+1) - 1 \le 2^{-r}(1/2 + 1/r) \le (3/2)2^{-r}.$$

In the defining series of ν, sum this bound with $t = \log x$. For $x \ge 2$ it gives

$$0 \le \frac{1}{\log x} - \nu'(x) \le \frac{x^{-1} + (3/2)x^{-1/2}}{\log x} \le \frac{5x^{-1/2}}{2\log x}. \tag{123}$$

Integration shows

$$0 \le \mathrm{Li}_2(X) - \nu([2, X]) \le \frac{5}{\log 2}(\sqrt{X} - \sqrt{2}).$$

Thus the model-count error is $O(\sqrt{X})$, unconditionally.

Assume RH. Its classical consequence $\pi(X) - \mathrm{li}(X) = O(\sqrt{X}\log X)$ is used only in this forward direction; see [22, Eq. 6.16.5]. The preceding model estimate shows, uniformly over every interval $J \subset [u, 2u]$ and every endpoint convention,

$$\big(\nu(J) - \#\{p \in J\}\big)_+ \ll \sqrt{u}\log u.$$

One-sided limits of the same cumulative estimate handle prime endpoints; no boundary prime is silently discarded. There are $u/h \le 2u^{1/(2k)}$ cells. The ideal defect bound $\delta^*_{I,k} \le \lceil k/2 \rceil d_I$, followed by the actual comparison (116), gives the stronger estimate

$$E_{j,k} \ll_k u^{1/2+1/(2k)} \log u. \tag{124}$$

This proves the second condition and hence the third, since every individual overweight is at most its block sum.

Conversely, a failure of RH gives a zero with real part $\beta > 1/2$, by the functional equation. Under the second condition choose one fixed $k \in \mathcal{K}$ with $1/2 + 1/(2k) < \beta$, and then a sufficiently small fixed $\epsilon > 0$. Its rate contradicts (117). Under the third condition use the sparse repair count to obtain

$$E_{j,k} \ll_{k,\epsilon} u^{1/2+1/k+\epsilon}.$$

Choose instead one fixed $k \in \mathcal{K}$ with $1/2 + 1/k < \beta$ and then small ϵ; the same unconditional signal statement gives a contradiction. Finally (116) compares the actual and ideal defects by a factor tending to 1, proving the stated replacement. □

For clarity, the full positively weighted model and its repair set have different sizes. If $\mathcal{S}_k = \{p : w_{p,k} > 0\}$ and $A_* = \sum_{q\in\mathcal{Q}} \varphi(q) > 0$, the exact identity and its cardinality consequence are

$$\sum_{u\le p<2u} \min\{w_{p,k}, A(p)\} = W_j - E_{j,k}, \qquad \#(\mathcal{S}_k \cap [u, 2u)) \ge \frac{(W_j - E_{j,k})_+}{A_*}. \tag{125}$$

The unconditional density comparison just proved gives $W_j = (R + o(1))u/\log u$. Therefore an independent overweight bound with any fixed exponent less than 1 forces at least $(R/A_* + o(1))u/\log u$ selected primes per block. For $\mathcal{Q} = \{1\}$ the classical prime number theorem then makes the full carrier density one among primes. Meanwhile the repair set has only $O_k(u^{1/(2k)})$ primes per block. A small unweighted repair count therefore does not exclude a zero: its weights can be large, as (118) quantifies.

These are ordinary proofs for independently defined actual prime models, not yet Lean formalizations of the polynomial norming, finite moment duality or Euler continuation. The already compiled positive-Dirichlet continuation and capped support-function lemmas remain inputs to that work. No unconditional power estimate for $E_{j,k}$ has been proved here; in particular the certified reference boundary remains 7/8. No literature priority is asserted.

Exact contact geometry of the optimized carrier. The finite optimum has an explicit polynomial description, retaining the mass deficit and every signed capacity term.

Theorem 21.81 (Binary interpolation and the contact-prime profile)**.** *Fix a cell in the all-moment construction. Write $\mu = R\nu|_I$, $S = S_I$, $H = \mu(I)$,*

$C = \sum_{p \in S} A(p)$, *and* $E = E_{I,k}$. *For* $\deg P < k$ *put*

$$\mathcal{D}(P) = \int_I P \, d\mu - \sum_{p \in S} A(p)(P(p))_+.$$

There are at least k *actual prime nodes in* S. *For each* k*-element subset* $T \subset S$ *and map* $b : T \to \{0,1\}$, *form the unique degree-less-than-*k *Lagrange interpolant* $P_{T,b}$. *Retain those satisfying* $P_{T,b}(p) \le 1$ *at every node in* S, *and call the resulting finite nonempty set* $\mathfrak{P}_{k,S}$. *Then*

$$\begin{aligned} E &= \max_{P \in \mathfrak{P}_{k,S}} \mathcal{D}(P) \\ &= \max\left(\{0, H - C\} \cup \{\mathcal{D}(P) : P \in \mathfrak{P}_{k,S},\ P \text{ nonconstant}\}\right). \end{aligned} \tag{126}$$

Let P *be any maximizing feasible dual polynomial and* w *any optimal actual weight vector. The complete contact rules are*

dual value	*optimal actual weight*
$P(p) < 0$	$w_p = 0$
$P(p) = 0$	$0 \le w_p \le A(p)$
$0 < P(p) < 1$	$w_p = A(p)$
$P(p) = 1$	$w_p \ge A(p)$.

(127)

If $E > \max\{0, H - C\}$, *every maximizing dual polynomial is nonconstant. There are then at most* $k-1$ *repair primes* $w_p > A(p)$, *at most* $k-1$ *partly used nodes* $0 < w_p < A(p)$, *and at most* $2(k-1)$ *nodes with weight different from both* 0 *and* $A(p)$. *If* $E = H - C > 0$, *the constant polynomial* $P = 1$ *is optimal and every optimal vector has* $w_p \ge A(p)$ *at all nodes. If* $E = 0$, *every optimal vector has* $w_p \le A(p)$.

Proof. We first justify dual attainment. If $E = 0$, the zero polynomial attains it. If $E > 0$, take a maximizing sequence with $\mathcal{D}(P) \ge E/2$. Its polynomials have positive μ-integral and node values at most 1. The proof of Lemma 21.77, with $t = C_k \epsilon_{j,k} < 1$, gives

$$\max_I P \le (1-t)^{-1}, \qquad \max_I |P| \le 4B_{k-1}(r/\ell)(1-t)^{-1}.$$

The constants here concern this fixed cell. Interpolation on the fixed equispaced grid bounds every coefficient. The feasible inequalities and $\mathcal{D}(P) \ge E/2$ are closed. Compactness in the finite-dimensional coefficient space therefore supplies a limit attaining E.

The actual nodes are unisolvent for degree less than k. If Q of that degree vanishes at every node, a feasible actual quadrature gives $\int_I Q\,d\mu = 0$. For every real v, the polynomial $1 + vQ$ has integral $H > 0$ and node values 1. Norming bounds its maximum on I by $(1-t)^{-1}$, independently of v. Varying both signs of v forces $Q = 0$ on I, hence identically. If $|S| < k$, the nonzero polynomial $\prod_{p\in S}(x-p)$ contradicts this fact. Thus $|S| \geq k$.

Start with an optimal dual polynomial. If fewer than k distinct nodes have value 0 or 1, choose a nonzero polynomial Q of degree less than k vanishing at all those contacts. For sufficiently small positive and negative v, $P + vQ$ stays feasible and keeps every noncontact node's sign. The objective is affine on that sign region; maximality in both directions makes its slope zero. Unisolvence supplies a noncontact node where $Q \neq 0$. Orient v so its value increases, and move to the first time a noncontact node reaches 0 or 1. That time is positive and finite: the selected node must reach 0 if it started negative or 1 if positive. Until then all signs and node upper bounds are preserved. The objective stays constant, all old contacts stay fixed, and a new contact appears. Repeat until at least k distinct contacts are present. Their binary values determine P by Lagrange interpolation. This proves the finite maximum. Constant interpolants 0 and 1 supply the explicit tests 0 and $H - C$.

For the weight classification put $P_+ = \max(P, 0)$, $P_- = \max(-P, 0)$ and $e = (w - A)_+$ at a node. The exact signed slack identity is

$$e - (wP - AP_+) = (1 - P_+)e + P_+(A - w)_+ + P_-w. \tag{128}$$

All three terms on the right are nonnegative for $w \geq 0$, $P \leq 1$. Moment matching and primal-dual equality give

$$\begin{aligned}\sum_{p\in S}\big[(1 - P(p)_+)e_p + P(p)_+(A(p) - w_p)_+ + P(p)_-w_p\big] \\ = E - \mathcal{D}(P) = 0.\end{aligned} \tag{129}$$

Thus every node gap vanishes. If $P < 0$, its last term forces $w = 0$; if $P = 0$, its first forces $w \leq A$. For $0 < P < 1$, the first two force both positive parts of $w - A$ to vanish, so $w = A$. For $P = 1$, the second forces $w \geq A$. No absolute-value estimate of the signed arithmetic main terms enters these conclusions.

A constant feasible polynomial has objective at most $\max\{0, H - C\}$: for $c \leq 0$ it is $cH \leq 0$, and for $0 \leq c \leq 1$ it is $c(H - C)$. The strict non-mass-deficit case therefore has only nonconstant optimizers. Each of $P = 0$ and $P = 1$ has at most $k - 1$ distinct roots; the contact rules give all three

exceptional-node counts. When $E = H - C > 0$, the constant optimizer 1 instead forces $w \geq A$ at every node. When $E = 0$, all nonnegative excesses vanish. □

Apart from contact primes, the optimized carrier therefore uses either the full available capacity or none of it. For $k = 2$, the nonconstant interpolation candidates are affine, recovering the actual-hull finite-strike description. For larger fixed k the candidate list is still finite and uses the actual prime locations. Estimating its signed objectives, including $H - C$, remains the separate arithmetic task.

The exact signed decomposition, its nonnegativity, the summed primal-dual gap, vanishing of every optimal node gap, and the four contact implications have maintained Lean proofs in `Proof/PrimeClassDeletion/OverweightSlack.lean`. They use only `propext`, `Classical.choice` and `Quot.sound`, and are exported by the prime-class-deletion assembly. Polynomial-dual attainment, finite interpolation and the root-count consequences above retain their full ordinary proofs here. This classification supplies no new weight estimate or certified improvement of 7/8.

21.40.2 A computable detector with dyadic rational weights

The preceding real optimization can be replaced by a finite algorithm using rational data. The algorithm needs no unknown prime-gap starting threshold, and keeps the sparse repair support.

We first record the quantitative interpolation input now checked in Lean. For $N \geq 1$ distinct nodes $v_i \in [0, 1]$, a polynomial $\deg P < N$ with $|P(v_i)| \leq L$, $L \geq 0$, and node separation at least $\delta > 0$, the maintained proof gives

$$\begin{aligned} |P'(x)| &\leq L \sum_{i=1}^{N} \frac{N-1}{\left|\prod_{j \neq i}(v_i - v_j)\right|} \\ &\leq L \frac{N(N-1)}{\delta^{N-1}} \qquad (0 \leq x \leq 1). \end{aligned} \tag{130}$$

Every denominator is nonzero. The same owner proves exact interpolation and its differentiated finite-product formula, using cached polynomial-root and derivative APIs. Its fifteen theorem declarations have standard-axiom audits. The owner is `Proof/PrimeClassDeletion/PolynomialPrimeNorming.lean`. The integral norming argument and the rational detector theorem below remain ordinary proofs.

Theorem 21.82 (A computable sparse rational repair detector)**.** *Fix $k \ge 2$ and the finite character portfolio. Put $d = k-1$, $B_d = 2^d d^{d+1}/d!$ and*

$$C_k^{\mathrm{alg}} = 32B_d^2, \qquad K_k = 64B_d\, k(2d)^d.$$

The following finite algorithm defines nonnegative dyadic rational weights $\widehat{w}_{p,k}$ on actual primes.

Start at $j = 3$. On $[u, 2u)$, $u = 2^j$, use the previous cells of length $h = 2^{\lfloor (1-1/(2k))j \rfloor}$. For each cell $I = [a, b)$ enumerate its actual primes S. If S is empty, declare it bad. Otherwise compute its closed-hull covering radius

$$r_I = \max\left\{ p_{\min} - a,\ b - p_{\max},\ \frac{1}{2} \max_{\text{consecutive } p,p' \in S} (p' - p) \right\},$$

where the last maximum is zero for a singleton. Declare the cell good exactly when $C_k^{\mathrm{alg}} r_I / h \le 1/2$. On bad cells put all weights zero.

On a good cell, write $y_p = (p-a)/h$, $\tau = u^{-3}$, $M = u^4$, and compute rational numbers $\widetilde{m}_r$ with certified errors

$$\left| \widetilde{m}_r - R \int_I \left(\frac{x-a}{h} \right)^r d\nu(x) \right| \le \tau \qquad (0 \le r < k). \tag{131}$$

Search the finite set of vectors $v_p \in M^{-1}\mathbb{Z} \cap [0, 2Ru]$ satisfying

$$\begin{gathered} \left| \sum_{p \in S} v_p y_p^r - \widetilde{m}_r \right| \le 2\tau \quad (0 \le r < k), \\ \#\{ p \in S : v_p > A(p) \} \le k. \end{gathered} \tag{132}$$

This set is nonempty. Minimize the rational objective $\sum_{p \in S} (v_p - A(p))_+$ and break ties lexicographically in increasing prime order. Its unique output defines $\widehat{w}_{p,k}$. Put all weights below 8 equal to zero.

This algorithm terminates for each prime and uses only finite prime enumeration, rational comparisons and certified evaluation of the positive model moments. Its full Euler product satisfies

$$\widehat{F}_{k,\mathcal{Q}}(s) = \prod_p (1 - p^{-s})^{-\widehat{w}_{p,k}} = \left(\frac{s}{s-1} \right)^R e^{\widehat{G}_k(s)} \qquad (\Re s > 1/2), \tag{133}$$

with $\widehat{G}_k$ holomorphic, unconditionally.

Set $\widehat{e}_{p,k} = (\widehat{w}_{p,k} - A(p))_+$, $\widehat{\mathcal{B}}_k = \{p : \widehat{e}_{p,k} > 0\}$, and $\widehat{E}_{j,k} = \sum_{u \le p < 2u} \widehat{e}_{p,k}$. For every $j \ge 3$,

$$\#(\widehat{\mathcal{B}}_k \cap [u, 2u)) \le 2k\, u^{1/(2k)}. \tag{134}$$

For all sufficiently large j, the exact real optimum from Theorem 21.79 obeys

$$0 \le E_{j,k} - \widehat{E}_{j,k} \le 6K_k\, u^{-3+1/(2k)}. \tag{135}$$

Thus rationalization preserves its dyadic growth exponent. For $\mathcal{Q} = \{1\}$, and any unbounded set $\mathcal{K}$ of fixed integers $k \ge 2$,

$$\text{RH} \quad \Longleftrightarrow \quad \widehat{E}_{j,k} \ll_{k,\epsilon} u^{1/2+1/(2k)+\epsilon} \quad \textit{for every } k \in \mathcal{K},\ \epsilon > 0. \tag{136}$$

The analogous individual-overweight criterion from Corollary 21.80 holds as well. Every reference zero with real part $\beta > 1/2$ forces the rational repair mass to have dyadic limsup exponent at least β, and forces infinitely many $\widehat{e}_{p,k} \ge p^{\beta-1/(2k)-\epsilon}$ for every $\epsilon > 0$. These are fixed-k, fixed-portfolio assertions.

Proof. All grid tests are exact rational tests. Here B_d is rational, the endpoints and prime locations are integers, and r_I is rational. On the cell the density ratio of $R\nu$ is at most $4\zeta(2) < 8$. Hence the good-cell test implies

$$4B_d^2(r/\ell)(r_I/h) \le 1/2.$$

The positive-objective norming and actual quadrature results therefore apply on every good cell, even at a scale where a nonexplicit prime-gap threshold is unavailable.

For completeness the target moments admit certified rational approximations uniformly in their finite input data. The series defining ν' has coefficients $1/\zeta(n+1)$, $n \ge 1$. The defining Dirichlet series for $\zeta(n+1)$ has a rational partial sum and remainder at most T^{-n}/n after the integer cutoff T; its reciprocal can be enclosed because $\zeta(n+1) \ge 1$. On a fixed cell choose an integer $U > \log b$. For $N \ge 2U$, the exponential-series tail is bounded by

$$\sum_{n>N} \frac{U^n}{n!} \le \frac{2U^{N+1}}{(N+1)!}.$$

It controls the defining density series uniformly on the cell. Finite partial sums have computable derivative bounds there, since $a \ge 8$. Rational interval Riemann sums, with certified elementary logarithms and the displayed tail, therefore enclose each normalized moment to arbitrarily small error. Taking a rational midpoint once its enclosing interval has width at most 2τ gives (131). No decision about an equality of unknown real numbers enters this procedure.

Let w be an exact sparse optimal quadrature on a good cell. Its total mass is $H = R\nu(I) < 2Ru$, so every coordinate is within the finite search bound. Round it down:

$$v_p = M^{-1}\lfloor Mw_p \rfloor, \qquad 0 \le w_p - v_p < M^{-1}.$$

There are at most u prime nodes in the cell and $0 \le y_p \le 1$. Thus each normalized moment changes by at most $u/M = \tau$. Combining this with the target enclosure makes v feasible for (132). Rounding down creates no new repair primes and never increases the positive-overweight objective. This proves both feasibility and the cellwise bound $\widehat{E}_{I,k} \le E_{I,k}$. The finite minimization and tie rule are consequently effective. Every chosen output, not just this rounded witness, satisfies

$$\left| \sum_{p \in S} \widehat{w}_{p,k} y_p^r - m_r \right| \le 3\tau \qquad (0 \le r < k), \tag{137}$$

where m_r denotes the true normalized moment. Its mass is in particular at most $H + 3\tau$.

We next bound the optimization loss. If $E_{I,k} = 0$, its rational optimum is also zero. Otherwise take an attaining optimal dual polynomial, expressed in the normalized coordinate as $Q(y) = \sum_{r<k} c_r y^r$. The positive-objective norming proof and the good-cell test give

$$\max_{[0,1]} |Q| \le 4B_d(r/\ell)(1 - 1/2)^{-1} \le 64B_d.$$

Interpolate at $0, 1/d, \ldots, 1$. Each denominator product has absolute value at least d^{-d}, and each numerator product has coefficient absolute sum at most 2^d: the coefficient sum of a product is at most the product of coefficient sums, and each factor $y - j/d$ contributes at most 2. Summing the $d+1 = k$ basis terms gives the explicit bound

$$\sum_{r<k} |c_r| \le 64B_d\, k(2d)^d = K_k. \tag{138}$$

This is an auxiliary finite-dimensional conditioning bound; the signed prime-capacity objective itself is retained.

The checked node weak-duality inequality, summed over the actual finite

support, now gives

$$\begin{aligned}\widehat{E}_{I,k} &\ge \sum_{p\in S} \widehat{w}_{p,k} Q(y_p) - \sum_{p\in S} A(p) Q(y_p)_+ \\ &= E_{I,k} + \sum_{r<k} c_r \left(\sum_{p\in S} \widehat{w}_{p,k} y_p^r - m_r \right) \\ &\ge E_{I,k} - 3K_k\tau.\end{aligned}$$

All constant and nonconstant dual candidates, including the mass-deficit row, are covered. Together with the preceding upper bound this proves the cell comparison.

The classical weak prime-gap estimate makes all cells good after some finite scale: their covering radii are $O(u^{2/3})$ and $h \ge u^{1-1/(2k)}/2$. The algorithm does not need to know that scale. There are $u/h \le 2u^{1/(2k)}$ cells, so summing the comparison after that scale proves (135). The imposed repair-count constraint, including zero weights on bad cells, proves (134) at every scale where the algorithm runs.

It remains to prove favorability for the chosen algorithmic output. Small moment residuals do not imply proximity in total variation to a particular exact optimizer; we instead compare its full Euler logarithm directly with the continuous model. Let $\mathcal{H}_k$ be the union of $(1, 8)$ and all bad cells. There are only finitely many bad cells, so this is a bounded union, with the initial endpoint at 1 retained. Every block has actual mass at most $R\nu([u, 2u)) + 3\tau u/h \ll_R u/\log u$. Thus the initial full Euler logarithm converges absolutely on $\Re s > 1$, where the exact identity is

$$\begin{aligned}\log \widehat{F}_{k,\mathcal{Q}}(s) - R\log M(s) &= -R\int_{\mathcal{H}_k} f_s(x)\, d\nu(x) + \sum_{j\ge 3} \widehat{G}_{j,k}(s), \\ \widehat{G}_{j,k}(s) &= \sum_{\substack{I\subset[u,2u)\\ I \text{ good}}} \left[\sum_{p\in S_I} \widehat{w}_{p,k} f_s(p) - R\int_I f_s(x)\, d\nu(x) \right].\end{aligned} \tag{139}$$

Every prime power is retained in f_s.

Fix a compact set in $\Re s > 1/2$, and let $\sigma_0 = \inf \Re s > 1/2$ on that set. Expand $f_s(a + hy)$ through degree $k - 1$. Its coefficient of y^r is $h^r f_s^{(r)}(a)/r! = O_{k,\mathrm{compact}}(u^{-\sigma_0})$, including $r = 0$. The full Euler-derivative estimate used above proves this uniformly, since $h/u \le 1$. By (137), the surviving Taylor polynomial contributes at most $O_{k,\mathrm{compact}}(\tau u^{1/(2k)-\sigma_0})$ per

block. The model and actual positive masses in the remainder sum to at most $2R\nu([u,2u)) + 3\tau u/h$. The same order-$k$ Taylor remainder, using $h^k \le u^{k-1/2}$, consequently gives

$$|\widehat{G}_{j,k}(s)| \ll_{k,R,\mathrm{compact}} \frac{u^{1/2-\sigma_0}}{\log u} + u^{-3+1/(2k)-\sigma_0}. \tag{140}$$

Both terms are summable over dyadic blocks. The finite integral over $\mathcal{H}_k$ is holomorphic on $\Re s > 0$, with the same integrable logarithmic endpoint at 1 as before. Thus (139) defines $\widehat{G}_k$ holomorphically on $\Re s > 1/2$. Exponentiation proves (133), including its unchanged pole order R.

Finally the exact/rational cost comparison transfers the previous RH equivalence. Alternatively, the reverse implication and bad-zero readout follow directly from the independently favorable rational model and the one-sided positive quotient. Its sparse count gives the individual-weight criterion and the heavy-atom consequence exactly as before. RH is used only for the forward arithmetic estimate. □

This supplies an actual computable prime set $\widehat{\mathcal{B}}_k$ with computable dyadic rational weights. Its weighted signal reaches every $\beta > 1/2$. The finite algorithm defines the observable at every scale; running it on a finite collection of scales is not a proof of its eventual RH-equivalent bound. The real and rational mass exponents agree, but neither construction supplies the independent power estimate.

The relation to modified Euler products is structural: Grosswald–Schnitzer [19] modify Euler bases, whereas the construction here fixes the actual primes and chooses effective rational exponents with a controlled one-sided capacity cost. No literature priority is asserted. The rational construction, coefficient budget and Euler continuation above are ordinary full proofs. The fifteen interpolation/derivative theorems and the earlier signed weak-duality/contact core are separately Lean checked. The certified reference boundary remains 7/8.

Corollary 21.83 (Rational linear programming gives certified repair intervals)**.** *On a good cell of Theorem 21.82, solve the bounded rational linear program*

$$E_I^{\mathrm{LP}} = \min \sum_{p\in S} e_p, \qquad 0 \le v_p \le 2Ru, \quad 0 \le e_p \le 2Ru, \quad e_p \ge v_p - A(p),$$
$$\left| \sum_{p\in S} v_p y_p^r - \widetilde{m}_r \right| \le \tau \quad (0 \le r < k). \tag{141}$$

It has a rational optimum obtainable by finite exact linear algebra. Rational conic compression followed by rounding down to multiples of M^{-1} produces weights $\widehat{w}_{p,k}^{\mathrm{LP}}$ with at most k repair primes, the same moment tolerances as (132)*, and costs satisfying*

$$\begin{aligned} E_I^{\mathrm{LP}} &\le E_{I,k} \le E_I^{\mathrm{LP}} + 2K_k\tau, \\ E_I^{\mathrm{LP}} - k/M &\le \widehat{E}_{I,k}^{\mathrm{LP}} \le E_I^{\mathrm{LP}}. \end{aligned} \tag{142}$$

Thus this procedure gives both a rational interval containing the exact real optimum and an actual sparse dyadic model. Its full favorable continuation, zero readout and RH criterion are those of the preceding theorem. For all sufficiently large blocks its complete error is bounded by

$$0 \le E_{j,k} - \widehat{E}_{j,k}^{\mathrm{LP}} \le 4K_k u^{-3+1/(2k)} + 2ku^{-4+1/(2k)}. \tag{143}$$

The LP output need not minimize the original dyadic grid problem; the displayed certificates are sufficient for every conclusion.

Proof. An exact real quadrature w and its excess $(w - A)_+$ give a feasible point of (141), since the target error is at most τ. Both variable boxes are valid because $w_p \le H < 2Ru$ and $A(p) \ge 0$. The feasible polytope is nonempty and compact, with rational coefficients. It has an optimal vertex. At a vertex the active constraint normals span the ambient space; otherwise a nonzero common null direction would allow small feasible moves in both directions. A full-rank active subsystem therefore determines that vertex by rational Gaussian elimination. Enumerating such subsystems is one terminating exact solver. This proves rational attainment without comparing unknown real moments.

At an optimum $e_p = (v_p - A(p))_+$: any strict excess slack can be decreased without changing the moment constraints. Clip v to $a_p = \min(v_p, A(p))$. Its rational excess vector e has moment vector

$$\sum_{p \in S} e_p(1, y_p, \dots, y_p^{k-1}).$$

While more than k excess coordinates are positive, rational Gaussian elimination gives a nonzero rational dependence among their moment columns. The first coordinate makes its coefficients sum to zero, so they have both signs. Move along the dependence by the least positive ratio needed to make one weight vanish. Nonnegativity, every moment and the total excess mass are preserved. The support shrinks, so the process terminates with a rational e' on at most k actual nodes.

The vector $v' = a + e'$ has exactly the original LP moments. Its total mass is at most $H + 2\tau < 2Ru$: use $u \geq 8$, $H \leq Ru/\log 8 < Ru$ and $2\tau < Ru$. Thus it respects every coordinate box. Its actual overweight is at most $\sum e' = E_I^{\mathrm{LP}}$. It is itself LP feasible, so optimality forces equality. In particular its repairs are supported on the at most k nodes of e'.

The true normalized moment errors of this compressed optimizer are at most 2τ. The attaining exact polynomial dual, with coefficient budget K_k, therefore gives $E_I^{\mathrm{LP}} \geq E_{I,k} - 2K_k\tau$ by the same complete signed weak-duality calculation used above. The reverse inequality follows from the feasible exact quadrature. This proves the first certificate.

Round v' down on the M^{-1} grid. Every normalized moment changes by at most $u/M = \tau$, so its discrepancy from the rational targets is at most 2τ, and from the true moments at most 3τ. No repair coordinate is created. Only the at most k existing repairs can decrease the positive-overweight cost, each by less than M^{-1}. This proves the second certificate. Summing these cell bounds, with $u/h \leq 2u^{1/(2k)}$, proves (143) after all cells are good. On bad cells use zero weights as before.

Finally the analytic proof above uses only nonnegative weights, these normalized moment tolerances, the mass bound, and the sparse repair count; it does not require a dyadic grid minimizer. All its continuation and zero-signal conclusions therefore apply to this LP construction. This is an ordinary proof of the rational optimization and compression procedure; those steps have not yet been formalized in Lean. □

21.41 Sparse repairs as a prime-Poisson overlap criterion

There is a probability interpretation of the computable repair weights which retains their unconditional zero signal. The probability space here is an auxiliary product of Poisson coordinates indexed by prime powers. It is constructed from the weights, rather than asserted to be the sampling law of a polynomial or affine family. The positive floor below is needed: an unregularized zero weight would give a point mass at a coordinate where the reference law has full support.

Lemma 21.84 (The signed repair–energy comparison)**.** *For $a, w, \delta \geq 0$, put*

$$h_\delta(a, w) = \left(\sqrt{w + \delta} - \sqrt{a + \delta}\right)^2, \qquad e = (w - a)_+.$$

Then

$$\begin{aligned} 0 \leq h_\delta(a, w) &\leq |w - a|, \\ e &\leq 2h_\delta(a, w) + (a + \delta)\mathbf{1}_{\{w > a\}}, \\ |w - a| &= 2e + a - w. \end{aligned} \tag{144}$$

In particular, for any finite index set S, nonnegative a_p, w_p, δ_p on S, and $B = \{p \in S : w_p > a_p\}$, define

$$E = \sum_{p\in S}(w_p - a_p)_+, \quad H = \sum_{p\in S} h_{\delta_p}(a_p, w_p), \quad C = \sum_{p\in S} a_p, \quad W = \sum_{p\in S} w_p.$$

The complete finite comparison is

$$\begin{aligned} H &\le 2E + C - W, \\ E &\le 2H + \sum_{p\in B}(a_p + \delta_p) \\ &\le 2H + c\,\#B \quad \text{if } a_p + \delta_p \le c \ (p \in S). \end{aligned} \tag{145}$$

The signed mass correction $C - W$ has not been discarded.

Proof. Write $x = \sqrt{w + \delta}$ and $y = \sqrt{a + \delta}$. Then $x, y \ge 0$ and $x^2 - y^2 = w - a$. Since $x + y \ge |x - y|$, their squared difference is at most $|w - a|$. If $w > a$, the exact square identity

$$2(x - y)^2 + y^2 - (x^2 - y^2) = (x - 2y)^2$$

gives the second inequality. If $w \le a$, its left side is zero and the energy is nonnegative. Splitting into the same two cases proves $|w - a| = 2(w - a)_+ + a - w$. Sum these statements on the specified finite sets. The final comparison uses only the capacity bound on B. □

The maintained module `RepairHellinger.lean` proves these scalar and finite-sum statements, including common-floor positivity and the explicit repair-cardinality consumer, in ten standard-axiom-audited declarations. The infinite probability products and analytic consequences in the following theorem are ordinary mathematics.

Theorem 21.85 (A computable prime-Poisson overlap criterion for RH)**.** *Fix the finite character portfolio $\mathcal{Q}$, an integer $k \ge 2$, and either computable rational construction of Theorem 21.82 and Corollary 21.83. Write $w_{p,k}$ for its output, $A(p)$ for the portfolio capacity, and*

$$\delta_p = p^{-4}, \quad h_{p,k} = h_{\delta_p}(A(p), w_{p,k}), \quad H_{j,k} = \sum_{\substack{u\le p<2u \\ p \text{ prime}}} h_{p,k}, \qquad u = 2^j.$$

The repair support and repair mass below always refer to the original weights $w_{p,k}$, before adding δ_p.

For each fixed real $\sigma > 0$, define independent Poisson coordinates (p, m), where p is prime and $m \geq 1$, with means

$$\lambda^0_{p,m}(\sigma) = (A(p) + \delta_p)\frac{p^{-m\sigma}}{m}, \qquad \lambda^k_{p,m}(\sigma) = (w_{p,k} + \delta_p)\frac{p^{-m\sigma}}{m}. \tag{146}$$

Let $\mathsf{P}^0_{\sigma,\mathcal{Q}}$ and $\mathsf{P}^k_{\sigma,\mathcal{Q}}$ be their product probability measures on $\mathbb{N}_0^{\{(p,m)\}}$ with its product sigma algebra. Every coordinate mean is strictly positive.

Put $f_\sigma(p) = -\log(1 - p^{-\sigma})$. For each integer cutoff $X \geq 2$, the limiting affinity of the complete prime-power laws at primes $p \leq X$ is

$$\rho_{X,k}(\sigma) = \exp\left(-\frac{1}{2}\sum_{\substack{p \leq X \\ p \text{ prime}}} h_{p,k} f_\sigma(p)\right). \tag{147}$$

Its decreasing limit $\rho_{\infty,k}(\sigma)$ satisfies

$$\begin{aligned} \rho_{\infty,k}(\sigma) > 0 \quad &\Longleftrightarrow \quad \sum_p h_{p,k} p^{-\sigma} < \infty \\ &\Longleftrightarrow \quad \mathsf{P}^k_{\sigma,\mathcal{Q}} \text{ and } \mathsf{P}^0_{\sigma,\mathcal{Q}} \text{ are mutually absolutely continuous.} \end{aligned} \tag{148}$$

If the affinity is zero, the two product measures are mutually singular.

Unconditionally, a reference zero with real part $\beta > 1/2$ has the following response: for every fixed k, $1/2 < \sigma < \beta$, and $0 < \epsilon < \beta - \sigma$, infinitely many dyadic $u = 2^j$ satisfy

$$\rho_{2u,k}(\sigma) \leq \exp\left(-2^{-\sigma-3} u^{\beta-\sigma-\epsilon}\right). \tag{149}$$

Thus the product laws are mutually singular at every such σ. No RH assumption is used for this response.

For $\mathcal{Q} = \{1\}$ and any prescribed unbounded set $\mathcal{K}$ of integers $k \geq 2$,

$$\begin{aligned} \text{RH} \quad \Longleftrightarrow \quad &\text{for every } \sigma \in \mathbb{Q} \cap (1/2, 1) \text{ there exists a fixed } k \in \mathcal{K} \\ &\text{with } \mathsf{P}^k_{\sigma,\{1\}} \text{ mutually absolutely continuous with } \mathsf{P}^0_{\sigma,\{1\}}. \end{aligned} \tag{150}$$

In the RH direction, all sufficiently large $k \in \mathcal{K}$ work for a given σ. All finite affinities in (147) are computable for rational $\sigma > 0$. This assertion does not decide positivity of their infinite limit.

Proof. For positive means λ, μ, the one-coordinate affinity is obtained directly from the exponential series:

$$\sum_{n=0}^{\infty} \sqrt{e^{-\lambda}\frac{\lambda^n}{n!} e^{-\mu}\frac{\mu^n}{n!}} = e^{-(\lambda+\mu)/2} e^{\sqrt{\lambda\mu}} = e^{-(\sqrt{\lambda}-\sqrt{\mu})^2/2}.$$

Independence gives the product formula for finitely many (p, m). At a fixed prime, summing every $m \geq 1$ gives

$$\sum_{m \geq 1} \frac{p^{-m\sigma}}{m} = -\log(1 - p^{-\sigma}) = f_\sigma(p).$$

The series is finite for $\sigma > 0$; passing to its increasing partial sums proves (147). No prime-power correction has been omitted.

The nonnegative exponent increases with X. Moreover

$$p^{-\sigma} \leq f_\sigma(p) \leq \frac{p^{-\sigma}}{1 - 2^{-\sigma}} \qquad (p \text{ prime},\ \sigma > 0). \tag{151}$$

The lower estimate is the $m = 1$ term, and the upper estimate bounds $1/m$ by 1 in the geometric series. This proves the first equivalence in (148). All coordinate Poisson laws assign positive probability to every nonnegative integer. They are therefore mutually absolutely continuous coordinate by coordinate. Kakutani's product-measure dichotomy, in the form proved by Kudō [24], now gives the second equivalence and the mutually singular alternative. Its local equivalence hypothesis is why the same strictly positive floor was added to both means, including at primes with $A(p) = 0$.

We next retain the zero signal. Put $c_{\mathcal{Q}} = 1 + \sum_{q \in \mathcal{Q}} \varphi(q)$, so that $A(p) + \delta_p \leq c_{\mathcal{Q}}$ for every prime. On the dyadic block the finite comparison gives

$$\widehat{E}_{j,k} \leq 2H_{j,k} + c_{\mathcal{Q}} \#(\widehat{\mathcal{B}}_k \cap [u, 2u)) \leq 2H_{j,k} + 2kc_{\mathcal{Q}} u^{1/(2k)}. \tag{152}$$

The rational repair detector already proves, for a reference zero at real part $\beta > 1/2$, that $\limsup_j \log(1 + \widehat{E}_{j,k})/\log u \geq \beta$. Consequently $\widehat{E}_{j,k} \geq u^{\beta - \epsilon}$ at infinitely many sufficiently large dyadic scales. One can use the limsup with $\epsilon/2$ first to absorb the added 1 without changing this conclusion. Since $\beta - \epsilon > \sigma > 1/2$ and $1/(2k) \leq 1/4$, the last support term in (152) is at most $\widehat{E}_{j,k}/2$ on those scales. Thus $H_{j,k} \geq u^{\beta - \epsilon}/4$. For $p \in [u, 2u)$, $f_\sigma(p) \geq (2u)^{-\sigma}$; keeping just this block in the nonnegative affinity exponent gives (149). It forces $\rho_{\infty,k}(\sigma) = 0$.

For the RH direction specialize to $\mathcal{Q} = \{1\}$, so $A(p) = 1$. Write $C_j = \#\{p \text{ prime} : u \leq p < 2u\}$ and $W_{j,k} = \sum_{u \leq p < 2u} w_{p,k}$. All sufficiently late cells are good, and their zeroth moment errors give

$$|W_{j,k} - \nu([u, 2u))| \ll_k u^{-3+1/(2k)}.$$

Under RH the previously used prime-counting estimate and the unconditional comparison of ν with the logarithmic integral give

$$|C_j - W_{j,k}| \ll_k u^{1/2} \log u.$$

For every fixed $\epsilon > 0$, the existing rational repair bound and the full signed energy comparison therefore imply

$$0 \le H_{j,k} \le 2\widehat{E}_{j,k} + C_j - W_{j,k} \ll_{k,\epsilon} u^{1/2+1/(2k)+\epsilon}. \tag{153}$$

Fix $\sigma > 1/2$. Choose $k \in \mathcal{K}$ large enough that $1/(2k) < \sigma - 1/2$, and then choose $0 < \epsilon < \sigma - 1/2 - 1/(2k)$. The finitely many initial blocks have finite energy, and the remaining series obeys

$$\sum_p h_{p,k} p^{-\sigma} \le \text{a finite initial contribution} + \sum_{j \text{ sufficiently large}} u^{-\sigma} H_{j,k} < \infty,$$

by geometric summation. This proves the asserted probability equivalence under RH, for every sufficiently large fixed k.

Conversely, if RH is false, reflection of the nontrivial zeros supplies a zero with real part $\beta > 1/2$. Choose rational σ with $1/2 < \sigma < \beta$. The unconditional bad-zero response makes the two product laws mutually singular for every fixed k, contradicting the right side of (150). This proves the reverse implication. Finally, the rational weights and common floors are computable, and at rational σ the finite expression uses only algebraic powers, square roots, logarithms and an exponential of computable real numbers. This proves the finite computability assertion. □

The new criterion interprets a hypothetical bad zero as asymptotic statistical separation of explicit probability laws. The sparse repair set retains the signal through (152); a positive density of repair primes is unnecessary. An independent upper bound on this energy or on the original repair mass would still be needed to exclude the zero. The theorem does not supply such an unconditional bound or change the certified 7/8 boundary. Kakutani's dichotomy and the one-coordinate Poisson calculation are classical inputs; the construction used here is the computable actual-prime repair model with its quantitative capacity and zero signal.

21.42 A computable deleted-prime class retaining the RH signal

The weighted construction admits a criterion involving the cardinality of a fixed, computable class of actual primes. Two facts are needed. The positive-series continuation principle can be applied to the opposite one-sided Euler quotient, provided the reference has no real zero on the ray of continuation. Also, an optimal LP vertex has few coordinates between its zero and full-capacity states, including the constant-dual cases.

Theorem 21.86 (Underweight transfer along the real axis)**.** *Fix a nonempty finite character portfolio and $1/2 \le \tau \le \eta < 1$. Let $w_p \ge 0$, with $\sum_p w_p p^{-\sigma} < \infty$ for every $\sigma > 1$. Assume its full Euler product F_w has a nowhere-zero holomorphic continuation on $\{\Re s > \tau\} \setminus \{1\}$ and a pole of order R at 1. Set*

$$d_p = (A(p) - w_p)_+, \quad e_p = (w_p - A(p))_+, \quad D_w(X) = \sum_{\substack{p \le X \\ p \text{ prime}}} d_p.$$

Suppose

$$\mathcal{Z}_{\mathcal{Q}}(\sigma) \ne 0 \quad (\eta < \sigma < 1), \qquad D_w(X) \ll_\epsilon X^{\eta+\epsilon} \quad \text{for every } \epsilon > 0.$$

Then $\mathcal{Z}_{\mathcal{Q}}$ has no zeros on $\Re s > \eta$.

For the zeta reference $\mathcal{Q} = \{1\}$, the real-axis nonvanishing condition is unconditional. In particular any particular independently favorable F_w as above has this unconditional response to a zeta zero with real part $\beta > \tau$:

$$\limsup_{j \to \infty} \frac{\log(1 + D_{j,w})}{j \log 2} \ge \beta, \qquad D_{j,w} = \sum_{\substack{2^j \le p < 2^{j+1} \\ p \text{ prime}}} (1 - w_p)_+. \tag{154}$$

For general Dirichlet portfolios the displayed real-axis hypothesis is retained; a possible exceptional real zero cannot be discarded.

Proof. Define the complete products

$$E_-(s) = \prod_p (1 - p^{-s})^{-d_p}, \qquad E_+(s) = \prod_p (1 - p^{-s})^{-e_p}.$$

The assumed growth bound and dyadic summation make E_- an absolutely normally convergent holomorphic unit on $\Re s > \eta$. This includes every prime-power term in its logarithm. The exact exponent identity $w_p + d_p - A(p) = e_p$ gives, initially on $\Re s > 1$,

$$\frac{F_w(s) E_-(s) U_{\mathcal{Q}}(s)}{\mathcal{Z}_{\mathcal{Q}}(s)} = E_+(s). \tag{155}$$

The nonsplit Euler factor $U_{\mathcal{Q}}$ is present. The right side has nonnegative real Dirichlet coefficients a_n, with $a_1 = 1$ and $a_p = e_p$ for each prime p. Indeed the coefficient of z^m in $(1-z)^{-e_p}$ is $e_p(e_p+1)\cdots(e_p+m-1)/m! \ge 0$; finite products and monotone summation give the stated Dirichlet expansion. It converges absolutely for $\Re s > 1$ because $e_p \le w_p$.

Treat the left side of (155) as a meromorphic quotient, without assigning a value at a reference zero. At every real $\sigma > \eta$, $\sigma \neq 1$, it is holomorphic in a neighborhood of σ: there are no reference zeros on $(\eta, 1)$ by hypothesis, and none on $(1, \infty)$ by the initial Euler product. At 1, the two poles of order R cancel and give a removable, nonzero quotient. The factors E_- and $U_{\mathcal{Q}}$ are holomorphic units there. Thus the real part of this quotient, with its removable value at 1, is a real analytic function on the entire ray $\sigma > \eta$. For real $\sigma > 1$ it equals $\sum_{n\geq 1} a_n n^{-\sigma}$.

Choose an absolutely summable real anchor $c > 1$. The positive Dirichlet continuation principle proved in `PositiveDirichletContinuation.lean` applies to these coefficients and this real analytic function. It gives actual summability of $\sum_{n\geq 1} a_n n^{-\sigma}$ for every $\sigma > \eta$. The prime subseries therefore gives

$$\sum_p e_p p^{-\sigma} = \sum_{p \text{ prime}} a_p p^{-\sigma} \leq \sum_{n\geq 1} a_n n^{-\sigma} < \infty.$$

Consequently the full logarithm of E_+ converges normally on $\Re s > \eta$, since at a real $\sigma > \eta$

$$\sum_p \sum_{m\geq 1} \frac{e_p p^{-m\sigma}}{m} \leq \frac{1}{1 - 2^{-\sigma}} \sum_p e_p p^{-\sigma} < \infty.$$

It defines a holomorphic unit there.

Cross multiplication now gives the meromorphic identity

$$\mathcal{Z}_{\mathcal{Q}}(s) E_+(s) = F_w(s) E_-(s) U_{\mathcal{Q}}(s) \qquad (\Re s > \eta). \tag{156}$$

It holds first on $\Re s > 1$ and then throughout the connected half-plane by meromorphic uniqueness. The right side is nonzero away from the pole at 1, and E_+ is a unit. This excludes every reference zero on $\Re s > \eta$.

For $0 < \sigma < 1$, the alternating Dirichlet series gives

$$\zeta(\sigma) = \frac{\sum_{m\geq 1} \big((2m-1)^{-\sigma} - (2m)^{-\sigma}\big)}{1 - 2^{1-\sigma}} < 0.$$

The paired numerator converges and is strictly positive. Thus the real-axis hypothesis for zeta does not assume RH. Finally, if the limsup in (154) were smaller than a zero real part $\beta > \tau$, choose $\max(\tau, \limsup) < \eta < \beta$. Dyadic summation would give the required underweight bound above η, and the result just proved would exclude that zero. This contradiction proves the readout. □

Lemma 21.87 (Few nonbinary coordinates at an optimal rational vertex)**.** *On any good cell of the nonempty-portfolio rational program* (141)*, every optimal vertex* (v, e) *satisfies*

$$\#\{p \in S : v_p \notin \{0, A(p)\}\} \le k. \tag{157}$$

This includes all constant-dual and mass-deficit cases. Choosing a deterministic minimizing rational vertex and rounding its v_p *down to multiples of* M^{-1}*, without the previous conic compression, gives weights* $w^{\mathrm{v}}_{p,k}$ *with*

$$\begin{aligned} &\#\{p \in S : w^{\mathrm{v}}_{p,k} \notin \{0, A(p)\}\} \le k, \\ &\left|\sum_{p\in S} w^{\mathrm{v}}_{p,k} y_p^r - m_r\right| \le 3\tau \quad (0 \le r < k), \\ &\sum_{p\in S} (w^{\mathrm{v}}_{p,k} - A(p))_+ \le E_I^{\mathrm{LP}} \le E_{I,k}. \end{aligned} \tag{158}$$

Here m_r *are the true normalized model moments,* $\tau = u^{-3}$ *and* $M = u^4$ *as before.*

Proof. At an optimum $e_p = (v_p - A(p))_+$, since otherwise decreasing that e_p alone would lower the objective while remaining feasible. The zeroth moment tolerance gives

$$\sum_{p\in S} v_p \le \widetilde{m}_0 + \tau \le H + 2\tau < 2Ru, \qquad H = R\nu(I).$$

For the last strict inequality use $u \ge 8$, $R \ge 1$, $H \le Ru/\log 8 < Ru$ and $2\tau < Ru$. Thus every upper coordinate box for v is strict. Every upper box for e is strict as well, since $0 \le e_p \le v_p$.

Suppose $T = \{p \in S : v_p \notin \{0, A(p)\}\}$ has more than k elements. Its rational moment columns $(1, y_p, \dots, y_p^{k-1})$ are linearly dependent over $\mathbb{Q}$. Choose rational c_p, not all zero, supported on T, with

$$\sum_{p\in S} c_p y_p^r = 0 \quad (0 \le r < k).$$

For a sufficiently small rational $t > 0$, perturb in both signs:

$$v_p^{\pm} = v_p \pm t c_p, \qquad e_p^{\pm} = e_p \pm t c_p \mathbf{1}_{\{v_p > A(p)\}}.$$

If $0 < v_p < A(p)$, keep $e_p = 0$ and choose t small enough to remain in that strict interval. If $v_p > A(p)$, keep $e_p^{\pm} = v_p^{\pm} - A(p) > 0$. Coordinates with

$v_p = 0$ or $v_p = A(p)$ are unchanged. There are finitely many strict margins, including the upper boxes, so one positive rational t preserves all of them in both directions. All normalized moments are unchanged, hence all their two-sided tolerance constraints are preserved. The two feasible points are distinct and have (v, e) as their midpoint, contradicting the vertex property. This proves (157).

The earlier finite rational active-basis construction supplies an optimal rational vertex. Fix an ordering of its finite candidate bases and choose the first minimizing feasible vertex; all comparisons are rational and decidable. For $w^{\mathrm{v}}_{p,k} = \lfloor M v_p \rfloor / M$, the coordinates 0 and $A(p)$ are unchanged, because $A(p)$ and M are integers. Rounding therefore creates no nonbinary coordinate. It decreases each coordinate by less than M^{-1}, so each normalized moment changes by at most $|S|/M \le u/M = \tau$. The LP tolerance and target approximation contribute the other two copies of τ. The positive overweight cannot increase. Finally an exact optimal real quadrature is feasible for the rational LP, since its exact moments differ from the rational targets by at most τ. Hence $E^{\mathrm{LP}}_I \le E_{I,k}$. □

Lemma 21.88 (The exact deleted-count ledger)**.** *Let S be a finite set and $w_p \ge 0$ on S. Define*

$$Z = \{p \in S : w_p = 0\}, \quad T = \{p \in S : 0 < w_p < 1\}, \quad D = \sum_{p \in S} (1 - w_p)_+.$$

Then

$$D = \#Z + \sum_{p \in T} (1 - w_p), \qquad 0 \le D - \#Z \le \#T. \tag{159}$$

Proof. A zero coordinate contributes exactly 1. A coordinate in T contributes exactly $1 - w_p \in (0, 1)$. Every remaining coordinate has $w_p \ge 1$ and contributes zero. Partition the specified finite support into these three cases. The maintained module `PartialCapacityCounts.lean` proves this identity and its two count inequalities, including their difference form, in seven standard-axiom-audited declarations. The real-axis Euler transfer and the LP vertex geometry remain ordinary proofs. □

Theorem 21.89 (A predefined deleted-prime RH criterion)**.** *For each fixed integer $k \ge 2$, specialize the rational model to $\mathcal{Q} = \{1\}$ and use the deterministic vertex and rounding procedure of Lemma 21.87. Use zero weights at primes below 8 and on bad cells, as in the previous construction. Define actual prime classes*

$$\mathcal{D}_k = \{p \text{ prime} : w^{\mathrm{v}}_{p,k} = 0\}, \qquad \mathcal{T}_k = \{p \text{ prime} : 0 < w^{\mathrm{v}}_{p,k} < 1\}.$$

Membership in both classes is decidable by a terminating finite computation from the integer inputs p, k. Their definition uses no RH or zero-location assumption.

The full product F_k^{v} is independently favorable: it has the same unconditional nowhere-zero continuation on $\Re s > 1/2$ and simple pole at 1 as before. For $j \geq 3$, write $u = 2^j$, $S_j = \{p \text{ prime} : u \leq p < 2u\}$ and

$$N_{j,k} = \#(\mathcal{D}_k \cap S_j), \qquad D_{j,k} = \sum_{p \in S_j} (1 - w_{p,k}^{\mathrm{v}})_+.$$

At every such dyadic block,

$$\#(\mathcal{T}_k \cap S_j) \leq 2ku^{1/(2k)}, \qquad 0 \leq D_{j,k} - N_{j,k} \leq 2ku^{1/(2k)}. \tag{160}$$

Every hypothetical zeta zero with real part $\beta > 1/2$ forces, for every fixed k,

$$\limsup_{j \to \infty} \frac{\log(1 + N_{j,k})}{j \log 2} \geq \beta. \tag{161}$$

In particular for every $\epsilon > 0$ there are infinitely many j with $N_{j,k} \geq u^{\beta - \epsilon}$.

For every prescribed unbounded set $\mathcal{K}$ of integers $k \geq 2$, the exact equivalence is

$$\mathrm{RH} \quad \Longleftrightarrow \quad N_{j,k} \ll_{k,\epsilon} u^{1/2 + 1/(2k) + \epsilon} \quad \text{for every } k \in \mathcal{K},\ \epsilon > 0. \tag{162}$$

There is also a one-model sufficient condition. For any fixed $k \geq 2$ and $1/2 \leq \eta < 1$,

$$\left[\forall \epsilon > 0, \quad N_{j,k} \ll_{\epsilon} u^{\eta + \epsilon}\right] \quad \Longrightarrow \quad \zeta(s) \neq 0 \quad (\Re s > \eta). \tag{163}$$

Thus finiteness of even one of these constructed deleted-prime classes would imply RH.

Proof. The earlier good-cell test is a finite rational test after enumerating the actual primes in the cell. The target moments admit certified rational approximations. The vertex is selected from finitely many rational active bases, feasibility and objective comparisons are exact, and rounding produces dyadic rational weights. Given a prime $p \geq 8$, its dyadic block and its unique half-open cell are obtained by integer arithmetic. Run this finite procedure on that cell and compare the output at p with 0 and 1. The finitely many smaller primes have their specified zero weights. This proves the claimed membership computation.

By Lemma 21.87, there are at most k nonbinary coordinates on each good cell. On a bad cell all coordinates are zero, so there are no partially weighted coordinates. There are at most $2u^{1/(2k)}$ cells per block. This proves the first inequality in (160). Apply Lemma 21.88 on the actual finite support S_j for the second inequality.

For each fixed k, all sufficiently late cells are good by the same unconditional prime-mesh argument as before. The rounded vertex model has nonnegative weights, true normalized moment errors at most $3u^{-3}$ and the previous block mass bound. Those are precisely the inputs of the complete Euler-logarithm argument (139)–(140). The initial interval and all bad cells are retained there in the finite head factor. It follows unconditionally that

$$F_k^{\mathrm{v}}(s) = \frac{s}{s-1}\exp(G_k^{\mathrm{v}}(s)), \qquad \Re s > 1/2,$$

with G_k^{v} holomorphic. This proves favorability independently of the reference zeros.

Apply Theorem 21.86 with this model and $\tau = 1/2$. A hypothetical zero at real part $\beta > 1/2$ forces the dyadic deficit $D_{j,k}$ to have limsup exponent at least β. If the limsup exponent of $N_{j,k}$ were smaller than β, choose

$$\max\left(1/(2k), \limsup_j \frac{\log(1+N_{j,k})}{j\log 2}\right) < \alpha < \beta.$$

The exact finite comparison $D_{j,k} \le N_{j,k} + 2ku^{1/(2k)}$ would give $D_{j,k} \ll u^{\alpha}$, a contradiction. This proves (161). Using the limsup with $\epsilon/2$ and absorbing the added 1 proves the stated infinitely-often power lower bound.

For the RH direction put

$$E_{j,k}^{\mathrm{v}} = \sum_{p\in S_j} (w_{p,k}^{\mathrm{v}} - 1)_+, \quad C_j = \#S_j, \quad W_{j,k}^{\mathrm{v}} = \sum_{p\in S_j} w_{p,k}^{\mathrm{v}}.$$

The complete signed balance is

$$D_{j,k} = E_{j,k}^{\mathrm{v}} + C_j - W_{j,k}^{\mathrm{v}}. \tag{164}$$

For sufficiently large j, the vertex-model overweight is at most the exact real optimum $E_{j,k}$, by (158). Under RH the existing optimal-repair estimate gives $E_{j,k}^{\mathrm{v}} \ll_{k,\epsilon} u^{1/2+1/(2k)+\epsilon}$. Its zeroth moment error and the prime-counting estimate give $|C_j - W_{j,k}^{\mathrm{v}}| \ll_k u^{1/2}\log u$, exactly as in the preceding Poisson proof. Combining the signed balance with $N_{j,k} \le D_{j,k}$ proves the

forward half of (162). The finitely many initial blocks are absorbed in its constant.

If RH is false, choose a zero with real part $\beta > 1/2$, then choose a fixed $k \in \mathcal{K}$ with $1/(2k) < \beta - 1/2$, and choose $0 < \epsilon < \beta - 1/2 - 1/(2k)$. The claimed count upper bound would contradict (161). This proves the reverse half.

Finally suppose the premise of (163) holds. Since $1/(2k) \le 1/4 < \eta$, the finite comparison in (160) gives $D_{j,k} \ll_\epsilon u^{\eta+\epsilon}$. Dyadic summation, including the finite initial prime factors, gives the cumulative underweight hypothesis of Theorem 21.86. Its zeta specialization excludes zeros on $\Re s > \eta$. If $\mathcal{D}_k$ is finite, this premise holds with $\eta = 1/2$, which yields RH by reflection. □

This criterion places the hypothetical bad-zero response in the cardinality of a predefined actual prime class. The nonbinary weights contribute only the explicitly bounded partial-capacity term in (160). An independent arithmetic estimate for $N_{j,k}$ would still be needed for a new unconditional zero-free boundary; the installed boundary remains 7/8. Jurzak [18] studies analytic-continuation criteria for partial products on prime-index arithmetic lists. Here the classes are selected by finite rational moment programs, and the response is an unconditional lower exponent for their counting functions. The finite vertex geometry and positive-series continuation principle are classical inputs; no established literature priority is asserted for this application.

21.43 Critical-line retention and the zero supremum in prime counts

A second scale choice continues the favorable model through the critical line. It keeps the same rational vertex algorithm, but defines new prime sets; the preceding $\mathcal{D}_k$ retain their original definitions.

Lemma 21.90 (Prime-count error to the right of the zero supremum)**.** *Put*

$$\Theta_\zeta = \sup\{\Re\rho : \zeta(\rho) = 0,\ 0 < \Re\rho < 1\}.$$

Then $1/2 \le \Theta_\zeta \le 1$. *For every fixed* $\Theta_\zeta < b < 1$,

$$\pi(X) - \mathrm{Li}_2(X) = O_b(X^b \log^2 X) \quad (X \ge 2). \tag{165}$$

The constant may depend on $b - \Theta_\zeta$ *and is not asserted uniform as that distance tends to zero.*

Proof. The nontrivial zeros are nonempty and symmetric about $\Re s = 1/2$, giving the bounds for their supremum; see [23, §25.10(i)]. We supply the analytic estimate used below, keeping a finite contour rather than an unsummed infinite explicit formula.

The classical Hadamard product for the completed zeta function and the local zero count $\#\{\rho : |\Im\rho - t| \le 1\} = O(\log(|t|+2))$ give

$$\left|\frac{\zeta'}{\zeta}(\sigma+it)\right| \ll_b \log(|t|+2) \quad (b \le \sigma \le 2,\ |t| \ge 2). \tag{166}$$

Indeed, subtract the completed logarithmic derivative at $2+it$ from that at $\sigma+it$. The zero sum becomes

$$\sum_\rho \left(\frac{1}{\sigma+it-\rho} - \frac{1}{2+it-\rho}\right).$$

Zeros with $|\Im\rho - t| \le 1$ contribute $O_b(\log(|t|+2))$, since every first denominator has modulus at least $b - \Theta_\zeta > 0$. For the other zeros each difference is $O_b((\Im\rho - t)^{-2})$; summing unit height intervals with the local zero count gives the same bound. Stirling's formula controls the gamma and rational factors. The logarithmic derivative at $2+it$ converges absolutely. This proves (166).

For a half-integer $x \ge 3$, put $c = 1 + 1/\log x$ and $T = x^2$. Truncated Perron inversion gives

$$\psi(x) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} -\frac{\zeta'}{\zeta}(s)\frac{x^s}{s}\,ds + O(x\log^2 x/T).$$

Its error is bounded by

$$\sum_{n\ge 2} \Lambda(n)(x/n)^c \min\{1, (T|\log(x/n)|)^{-1}\}.$$

On $x/2 < n < 2x$, use $|\log(x/n)| \gg |n-x|/x$, $|n-x| \ge 1/2$, and the harmonic sum over these distances. Outside that interval use $\Lambda(n) \le \log n$ and $\sum_{n\ge 2}(\log n)n^{-c} \ll (c-1)^{-2}$. These give the stated error. The truncated kernel bound follows by integrating y^s/s and bounding the omitted tails.

Move the finite contour to $\Re s = b$. No zero pole is crossed; the pole at 1 contributes x. Equation (166) bounds the horizontal segments by $O_b(x\log T/T)$ and the new vertical segment by $O_b(x^b\log^2 T)$. The bounded portion of that line has a finite bound since it misses 1 and every zero. Thus $\psi(x) - x = O_b(x^b\log^2 x)$. The half-integer $\lfloor X\rfloor + 1/2$ preserves $\psi(X)$ for arbitrary real X and changes the main term by at most $1/2$. Higher prime

powers contribute $O(\sqrt{X}\log^2 X)$, absorbed because $b > 1/2$. Partial summation proves (165). These classical inputs give no new upper bound for Θ_ζ. □

Theorem 21.91 (Computable prime classes retaining critical-line zeros)**.** *Fix an integer $k \geq 4$. In each block $u = 2^j$, $j \geq 3$, use the zeta portfolio, $A(p) = R = 1$, and set*

$$\theta_k^0 = 1 - \frac{1}{k}, \quad h_{j,k} = 2^{\lfloor (k-1)j/k \rfloor}, \quad \tau = u^{-3}, \quad M = u^4. \tag{167}$$

Use the same rational good-cell test, deterministic minimizing LP vertex and downward rounding as Theorem 21.89, with this cell length. Keep weight zero at primes below 8 and on bad cells. Write the output as $w_{p,k}^0$ and define

$$\mathcal{D}_k^0 = \{p \text{ prime} : w_{p,k}^0 = 0\}, \qquad \mathcal{T}_k^0 = \{p \text{ prime} : 0 < w_{p,k}^0 < 1\}.$$

These fixed prime classes are computable from integer inputs p, k. For $S_j = \{p \text{ prime} : u \leq p < 2u\}$, put

$$N_{j,k}^0 = \#(\mathcal{D}_k^0 \cap S_j), \qquad D_{j,k}^0 = \sum_{p \in S_j} (1 - w_{p,k}^0)_+.$$

The full Euler model has the unconditional representation

$$F_k^0(s) = \prod_p (1 - p^{-s})^{-w_{p,k}^0} = \frac{s}{s-1} \exp(G_k^0(s)), \qquad \Re s > 0, \tag{168}$$

where G_k^0 is holomorphic. The product initially means its absolutely convergent full Euler logarithm on $\Re s > 1$. For every $j \geq 3$,

$$0 \leq D_{j,k}^0 - N_{j,k}^0 \leq \#(\mathcal{T}_k^0 \cap S_j) \leq 2k\, u^{1/k}. \tag{169}$$

Every nontrivial zeta zero with real part $\beta > 1/k$ forces

$$\limsup_{j \to \infty} \frac{\log(1 + N_{j,k}^0)}{j \log 2} \geq \beta. \tag{170}$$

In particular, unconditionally, for every fixed $k \geq 4$ and $\epsilon > 0$,

$$N_{j,k}^0 \geq u^{1/2-\epsilon} \quad \text{for infinitely many } j. \tag{171}$$

Thus every $\mathcal{D}_k^0$ is infinite, independently of RH.

Proof. The earlier classes used $1-1/(2k)$; these use $1-1/k$. Here $2/3 < \theta_k^0 < 1$ and $u/h_{j,k} \le 2u^{1/k}$. The unconditional prime-mesh bound therefore makes all sufficiently late cells good, for each fixed k. The finite algorithm tests goodness itself and needs no knowledge of that eventual threshold. Its grid and LP data are rational, so the membership decision terminates exactly as before. The vertex theorem applies unchanged on every good cell. Rounding preserves zero and unit coordinates; bad cells have no partial coordinates. Count cells and apply Lemma 21.88 to obtain (169).

We check the full new continuation region. Fix a compact $K \subset \{\Re s > 0\}$ and $\sigma_0 = \inf_{s\in K} \Re s > 0$. The complete Euler kernel and differentiated series are

$$f_s(x) = \sum_{m\ge 1} x^{-ms}/m = -\log(1-x^{-s}),$$

$$f_s^{(k)}(x) = (-1)^k \sum_{m\ge 1} \frac{(ms)(ms+1)\cdots(ms+k-1)}{m}\, x^{-ms-k}.$$

For $x \ge u$, the full derivative bound is

$$|f_s^{(k)}(x)| \le (k-1)! \prod_{r=0}^{k-1} (|s|+r)\, \frac{x^{-\Re s-k}}{(1-x^{-\Re s})^k}.$$

No prime-power term has been removed. Now $h_{j,k}^k \le u^{k-1}$, and the exact exponent identity is

$$1 - \sigma - k(1-\theta_k^0) = -\sigma. \tag{172}$$

The rounded normalized moment errors are at most $3u^{-3}$. Every Taylor coefficient of $f_s(a+hy)$ through degree $k-1$ is $O_{k,K}(u^{-\sigma_0})$. The block's model plus actual positive mass is $O_k(u/\log u)$, by the zeroth moment tolerance and the zero weights on bad cells. The complete good-cell discrepancy therefore satisfies

$$|G_{j,k}^0(s)| \ll_{k,K} \frac{u^{-\sigma_0}}{\log u} + u^{-3+1/k-\sigma_0} \qquad (s \in K). \tag{173}$$

The second term retains the Taylor-polynomial residual from all cells; moments are not claimed to cancel exactly. Both terms are summable over dyadic blocks. The exponent identity, mesh range and negativity of both powers have separate checked Lean proofs in `CriticalCellScale.lean`; the Euler analysis is ordinary.

Let $\mathcal{H}_k^0$ be $(1,8)$ together with all bad cells, a bounded union. The full initial normalization is

$$\log F_k^0(s) - \log\frac{s}{s-1} = -\int_{\mathcal{H}_k^0} f_s(x)\, d\nu(x) + \sum_{j\ge 3} G_{j,k}^0(s) \quad (\Re s > 1). \tag{174}$$

The head integral is holomorphic on $\Re s > 0$. Near $x = 1$, uniformly on K, its kernel is $O_K(1 + |\log(x-1)|)$ and ν' is bounded. Together with (173), this defines G_k^0 throughout $\Re s > 0$ and proves (168). Only zeta is used: its nonsplit factor is $U_{\{1\}} = 1$. No general portfolio's nonsplit unit is extended below its proved half-plane.

The proof of Theorem 21.86 now applies with any $0 < \eta < 1$. All its factors are units on $\Re s > \eta$, zeta has no real zero on $(\eta, 1)$, and the positive Dirichlet continuation theorem has no $1/2$ restriction on its endpoint. Explicitly, set

$$d_p = (1 - w_{p,k}^0)_+, \qquad e_p = (w_{p,k}^0 - 1)_+,$$
$$E_- = \prod_p (1 - p^{-s})^{-d_p}, \qquad E_+ = \prod_p (1 - p^{-s})^{-e_p}.$$

The complete identity is $\zeta E_+ = F_k^0 E_-$. First assume a deficit bound above η, making E_- a convergent unit. The meromorphic quotient $F_k^0 E_-/\zeta$ is real analytic on the real ray after canceling the pole at 1. Its positive Dirichlet series really converges above η, making E_+ a unit too. Only then continue the cross-multiplied identity. It excludes a zeta zero in that half-plane.

Thus every zero at real part $\beta > 0$ forces the deficit $D_{j,k}^0$ to have limsup exponent at least β. For $\beta > 1/k$ the partial allowance $2ku^{1/k}$ has strictly smaller exponent and cannot absorb this signal. This proves (170). Critical-line zeros exist unconditionally [23, §25.10(i)]. Take $\beta = 1/2 > 1/k$ and use the limsup with $\epsilon/2$ to absorb the added 1. This proves (171). □

Theorem 21.92 (The zero supremum recovered from literal prime counts)**.** *For these fixed computable classes define*

$$\lambda_k = \limsup_{j\to\infty} \frac{\log(1 + N_{j,k}^0)}{j \log 2} \qquad (k \geq 4).$$

Then, unconditionally,

$$\Theta_\zeta \leq \lambda_k \leq \min\{1, \Theta_\zeta + 1/k\}, \qquad \lim_{k\to\infty} \lambda_k = \inf_{k\geq 4} \lambda_k = \Theta_\zeta. \tag{175}$$

In particular,

$$\begin{aligned} \mathrm{RH} \quad &\iff \quad \lim_{k\to\infty} \lambda_k = \tfrac{1}{2} \\ &\iff \quad \left[\forall k \geq 4, \epsilon > 0, \quad N_{j,k}^0 \ll_{k,\epsilon} u^{1/2+1/k+\epsilon}\right]. \end{aligned} \tag{176}$$

Each k is fixed before the dyadic limsup; no uniformity in growing k is asserted.

Proof. The zero readout gives $\lambda_k \geq \beta$ for every zero with $\beta > 1/k$. Since $\Theta_\zeta \geq 1/2 > 1/k$, taking the supremum gives $\lambda_k \geq \Theta_\zeta$, without assuming that a rightmost zero is attained. There are at most u integer nodes in S_j, so $\lambda_k \leq 1$. If $\Theta_\zeta = 1$, these two bounds already prove the sandwich and its limit.

Suppose $\Theta_\zeta < 1$ and fix $\Theta_\zeta < b < 1$. The unconditional comparison (123) and Lemma 21.90 give

$$F(t) := \nu([1,t]) - \pi(t) = O_b(t^b \log^2 t) \quad (t \geq 2),$$

also for one-sided endpoint limits. Take a late block where all cells are good. For each exact real optimum $E_{I,k}$, choose an attaining degree-less-than-k dual polynomial P_I. Use $P_I = 0$ if the optimum is zero. Otherwise positive-objective norming on this very good cell gives $\|P_I\|_I \leq 64 B_{k-1}$, as in (138). Writing π also for its counting measure, retain

$$E_{I,k} = \int_I P_I \, d(\nu - \pi) - \sum_{p \in S_I} (P_I(p))_-. \tag{177}$$

The negative dual values remain in the exact objective.

A nonconstant polynomial of degree less than k has at most $k-1$ monotonicity intervals, so $\mathrm{Var}_I(P_I) \leq 2(k-1)\|P_I\|_I$; the constant case has variation zero. Stieltjes integration by parts on $I = [a, b_I)$ uses $F(a^-), F(b_I^-)$ and the integral of $F(t)P_I'(t)$. Its endpoint and derivative contributions are bounded by

$$(|P_I(a)| + |P_I(b_I)| + \mathrm{Var}_I(P_I)) \sup_{u \leq t \leq 2u} \max\{|F(t)|, |F(t^-)|\}.$$

Hence $E_{I,k} \ll_{k,b} u^b \log^2 u$. This uses the classical prime error to the right of Θ_ζ, and proves no new zero-free strip.

There are at most $2u^{1/k}$ cells. The rounded vertex overweight is at most the exact optimum cellwise, so $E_{j,k}^0 \ll_{k,b} u^{b+1/k} \log^2 u$. The zeroth moment tolerance gives $W_{j,k}^0 = \nu([u,2u)) + O_k(u^{-3+1/k})$. With $C_j = \#S_j$, the full signed consumer is

$$\begin{aligned} N_{j,k}^0 \leq D_{j,k}^0 &= E_{j,k}^0 + C_j - W_{j,k}^0 \\ &\ll_{k,b} u^{b+1/k} \log^2 u + u^b \log^2 u + u^{-3+1/k}. \end{aligned} \tag{178}$$

Thus $\lambda_k \leq b + 1/k$. Let $b \downarrow \Theta_\zeta$ and retain the independent upper bound $\lambda_k \leq 1$. The sandwich proves both the limit and the infimum. Zero symmetry gives RH $\iff \Theta_\zeta = 1/2$. Under RH take $b > 1/2$ arbitrarily close in the same bound to obtain the count criterion. Conversely that criterion gives $\lambda_k \leq 1/2 + 1/k$ for every fixed k; let $k \to \infty$ and use $\lambda_k \geq \Theta_\zeta$. □

The size needed to retain the signal is now explicit: the fixed $\mathcal{D}_k^0$ cannot have counting exponent below Θ_ζ, and approach it within $1/k$. They are infinite even if RH holds, unlike the preceding classes whose finiteness was only a sufficient RH condition. The construction consumes completed moment, vertex and positive-series inputs. Its new analytic consumers have the full ordinary proofs above; only the three scalar cell-scale lemmas are newly Lean checked. No independent arithmetic bound on these cardinalities has been proved, and the installed boundary remains 7/8. No established literature priority is asserted.

Theorem 21.93 (One computable prime class with the exact zero exponent)**.** *There is one fixed computable class $\mathcal{D}_*$ of actual primes, defined without any zero-location assumption, such that*

$$\limsup_{X\to\infty} \frac{\log(1+\pi_{\mathcal{D}_*}(X))}{\log X} = \Theta_\zeta, \qquad \pi_{\mathcal{D}_*}(X) = \#\{p \le X : p \in \mathcal{D}_*\}. \quad (179)$$

For every $\eta \in [1/2, 1)$ the exact equivalence is

$$\begin{aligned} &\zeta(s) \neq 0 \quad (\Re s > \eta) \\ \iff &\left[\forall \epsilon > 0, \quad \pi_{\mathcal{D}_*}(X) \ll_\epsilon X^{\eta+\epsilon}\right]. \end{aligned} \quad (180)$$

In particular the square-root count criterion is equivalent to RH. Unconditionally $\mathcal{D}_*$ is infinite, and its dyadic count is at least $u^{1/2-\epsilon}$ at infinitely many scales, for every $\epsilon > 0$.

Proof. We give an effective diagonal construction with a uniform analytic budget. Merely substituting a growing k into the preceding fixed-k big-O statements would not suffice.

For integers $k \ge 4$, use the explicit rational norming constant $B_{k-1} = 2^{k-1}(k-1)^k/(k-1)!$ and put

$$C_k = (2k)^{2k}, \qquad Q_k = 1 + 16kC_k + 128kB_{k-1}. \quad (181)$$

For $j \ge 3$ put $u = 2^j$ and $m_j = \lfloor \sqrt{j} \rfloor$, computed as an integer square root. An integer k is admissible at this block if

$$4 \le k \le m_j, \qquad Q_k \le 2^{m_j}, \quad \text{and every length-}h_{j,k}\text{ cell is good.} \quad (182)$$

The last condition is the previous finite rational good-cell test on the actual primes. All these are decidable finite tests. If any order is admissible, take the largest, k_j, and run the corresponding rounded minimizing-vertex

algorithm on every cell of this block. If there is none, assign weight zero throughout the block. Also keep weight zero below 8. Write the resulting rational weight as w_p^* and define

$$\mathcal{D}_* = \{p \text{ prime} : w_p^* = 0\}.$$

Given an integer prime p, its block, every candidate order, all prime enumerations, rational moment approximations and LP vertex comparisons require only terminating finite computations. This proves computability, without asserting that the algorithm is efficient.

For each fixed $k \geq 4$, all its cells are good at every sufficiently late block. Also $m_j \to \infty$ and $2^{m_j} \to \infty$, so this fixed k is eventually admissible at every block. Consequently only finitely many blocks are skipped, and

$$k_j \longrightarrow \infty, \qquad k_j \leq m_j, \qquad Q_{k_j} \leq 2^{m_j}, \qquad m_j/j \longrightarrow 0 \tag{183}$$

on all sufficiently late blocks. No monotonicity of k_j is needed, and no unknown prime-gap threshold is used by the algorithm.

We now verify the uniform budget in (181). If $k \geq 4$, $u \geq 8$, $\Re s = \sigma \geq 1/k$ and $|s| \leq k$, then

$$1 - u^{-\sigma} \geq 1 - 8^{-1/k} \geq \frac{1}{2k}.$$

The last inequality follows from Bernoulli: $(1 - 1/(2k))^k \geq 1/2 > 1/8$. The full differentiated Euler series gives, for $1 \leq r \leq k$,

$$\frac{h^r}{r!}|f_s^{(r)}(a)| \leq \frac{1}{r}(2k)^{2r}u^{-\sigma}(h/u)^r, \qquad a \geq u.$$

For $r = 0$, $|f_s(a)| \leq 2ku^{-\sigma}$. Thus every retained Taylor coefficient is at most $C_k u^{-\sigma}$. There are at most $2u^{1/k}$ cells, each with every normalized moment discrepancy at most $3u^{-3}$. Their polynomial residual is at most $6kC_k u^{-3+1/k-\sigma}$. The combined model and actual positive mass is at most $3u$: $\nu([u, 2u)) \leq u/\log u$ and the extra mass is at most $3u^{-3}$ times the number of cells, which is at most u. Using $h^k \leq u^{k-1}$, the full Taylor remainder is at most $3C_k u^{-\sigma}$. We have therefore proved, uniformly in the indicated k, s, u,

$$|G_{j,k}^0(s)| \leq Q_k\big(u^{-\sigma} + u^{-3+1/k-\sigma}\big). \tag{184}$$

All prime powers, all moment residuals and the mass in every cell have been included.

Fix an arbitrary compact $K \subset \{\Re s > 0\}$ and $\sigma_0 = \inf_K \Re s > 0$. Since $k_j \to \infty$, eventually $1/k_j \le \sigma_0$ and $\sup_K |s| \le k_j$. Equations (183) and (184) bound the tail uniformly on K by

$$2^{m_j - \sigma_0 j} + 2^{m_j - (3-1/4+\sigma_0)j}.$$

Both series converge, because $m_j/j \to 0$. The finite head consists of $(1,8)$ and the finitely many skipped blocks, with every initial integral retained. Exactly the complete normalization from (174) now proves

$$F_*(s) = \prod_p (1-p^{-s})^{-w_p^*} = \frac{s}{s-1}\exp(G_*(s)) \quad (\Re s > 0), \tag{185}$$

with G_* holomorphic. Initial absolute convergence on $\Re s > 1$ follows from the same block mass bound.

For a late block let $N_j^* = \#(\mathcal{D}_* \cap S_j)$ and $D_j^* = \sum_{p \in S_j} (1 - w_p^*)_+$. The vertex and finite ledger give

$$0 \le D_j^* - N_j^* \le 2k_j u^{1/k_j} = u^{o(1)}. \tag{186}$$

Indeed $1/k_j \to 0$ and $\log(2k_j)/(j \log 2) \le \log(2m_j)/(j \log 2) \to 0$. The zeta-only underweight argument above, applied to (185), implies that every zero of real part $\beta > 0$ forces D_j^* to have dyadic limsup exponent at least β. The subpower allowance in (186) cannot absorb any such positive-power signal. Therefore

$$\lambda_* := \limsup_{j \to \infty} \frac{\log(1 + N_j^*)}{j \log 2} \ge \Theta_\zeta. \tag{187}$$

A critical-line zero gives the stated infinitely-often square-root lower exponent, so $\mathcal{D}_*$ is infinite.

For the opposite inequality suppose first $\Theta_\zeta < 1$ and fix $\Theta_\zeta < b < 1$. The cumulative discrepancy $F(t) = \nu([1,t]) - \pi(t)$ is bounded by $C_b t^b \log^2 t$ as above, with C_b independent of k. On any good cell, retain the signed dual identity (177). The endpoint-plus-variation factor used there is at most

$$2k\, 64 B_{k-1} \le Q_k.$$

There is therefore no untracked fixed-k constant: summing cells gives $E_j^* \ll_b Q_{k_j} u^{b+1/k_j} \log^2 u$. The zeroth moment tolerance and the same cumulative discrepancy control the remaining signed mass balance. Explicitly, with $C_j = \#S_j$ and $W_j^* = \sum_{p \in S_j} w_p^*$,

$$\begin{aligned} N_j^* \le D_j^* &= E_j^* + C_j - W_j^* \\ &\ll_b 2^{m_j} u^{b+1/k_j} \log^2 u + u^b \log^2 u + u^{-3+1/k_j}. \end{aligned} \tag{188}$$

Its exponent is at most b, since $m_j/j \to 0$ and $1/k_j \to 0$. Let $b \downarrow \Theta_\zeta$ in this conclusion to obtain $\lambda_* \le \Theta_\zeta$. If $\Theta_\zeta = 1$, the elementary bound $N_j^* \le u$ already gives the same conclusion. Together with (187), this proves $\lambda_* = \Theta_\zeta$.

Dyadic summation, using $\Theta_\zeta \ge 1/2 > 0$, transfers this exponent unchanged to the cumulative count. For example every bound $N_j^* \ll u^{b+\epsilon}$ with $b > 0$ sums geometrically to $\pi_{\mathcal{D}_*}(X) \ll X^{b+\epsilon}$; the reverse exponent inequality follows from $N_j^* \le \pi_{\mathcal{D}_*}(2u)$. This proves (179). Finally $\Theta_\zeta \le \eta$ is exactly the absence of zeros on $\Re s > \eta$. Equation (179) gives (180); zero symmetry specializes $\eta = 1/2$ to RH. □

This single-class construction removes the order parameter from the final arithmetic question. Its weights and its zero-weight prime set are fixed in advance. The uniform budgets above are ordinary proofs; they do not follow merely by exchanging the earlier fixed-order quantifiers. An independent estimate of this one class's count would move the zero-free boundary by exactly the same exponent. No such improved estimate is asserted here, and the certified 7/8 boundary is unchanged.

21.44 Endpoint divergence and the arithmetic cost of a zero

The exponent identity leaves an endpoint question: a positive series can converge at its abscissa of convergence. The full Euler quotients rule out that possibility here when the endpoint is the real part of an actual zero.

We use a boundary fact about positive series. If $f_j(n) \ge 0$, every $\sum_n f_j(n)$ converges and is at most M, and $f_j(n) \to g(n) \ge 0$ for every n, then $\sum_n g(n)$ converges. Indeed every finite sum of g is the limit of the corresponding finite sums of f_j, hence is at most M. Bounded nonnegative finite sums imply summability. The maintained `PositiveDirichletBoundary.lean` proves this fact and its explicit specialization to $a_n n^{-\sigma_j}$, $\sigma_j \to \alpha$, in two standard-axiom-audited declarations. No zero theorem is asserted to be Lean formalized by those lemmas.

Theorem 21.94 (Both one-sided Euler masses diverge at a zero)**.** *Let $w_p \ge 0$, with $\sum_p w_p p^{-\sigma} < \infty$ for every $\sigma > 1$. Suppose its complete Euler product F_w has a nowhere-zero holomorphic continuation on $\{\Re s > \tau\} \setminus \{1\}$ and a simple pole at 1, where $0 \le \tau < 1$. If $\rho = \beta + i\gamma$ is a nontrivial zeta zero with $\beta > \tau$, put*

$$d_p = (1 - w_p)_+, \qquad e_p = (w_p - 1)_+,$$
$$\mathcal{P}_d(\sigma) = \sum_p d_p p^{-\sigma}, \qquad \mathcal{P}_e(\sigma) = \sum_p e_p p^{-\sigma}.$$

Then both nonnegative series diverge at that endpoint:

$$\mathcal{P}_d(\beta) = \mathcal{P}_e(\beta) = \infty. \tag{189}$$

The statement is unconditional and makes no rightmost-zero or RH assumption.

Proof. Retain the full products

$$E_- = \prod_p (1 - p^{-s})^{-d_p}, \qquad E_+ = \prod_p (1 - p^{-s})^{-e_p}, \qquad \zeta E_+ = F_w E_-.$$

The identity holds initially on $\Re s > 1$. Assume first $\mathcal{P}_d(\beta) < \infty$. Because $\beta > 0$,

$$\sum_p \sum_{r \geq 1} \frac{d_p p^{-r\beta}}{r} \leq \frac{\mathcal{P}_d(\beta)}{1 - 2^{-\beta}} < \infty.$$

Thus the logarithm of E_- converges uniformly on the closed half-plane $\Re s \geq \beta$. Its exponential is continuous and nonzero there, and holomorphic in the interior.

On the real ray $\sigma > \beta$, the meromorphic quotient $F_w E_-/\zeta$ is real analytic after cancellation of the two simple poles at 1. Zeta has no real zero in $(0, 1)$ or $(1, \infty)$. This quotient agrees initially with the positive Dirichlet series of E_+. The positive-continuation principle therefore forces actual Dirichlet summability for E_+ at every real $\sigma > \beta$, and its prime coefficients give $\mathcal{P}_e(\sigma) < \infty$ there. This argument uses real-axis neighborhoods, not an RH-dependent complex zero-free region.

As $\sigma \downarrow \beta$, the same real quotient remains bounded: $F_w(\beta)$ and $E_-(\beta)$ are finite, whereas $\zeta(\beta) \neq 0$. Choose a sequence of such σ tending to β. The positive-series boundary fact just proved gives actual Dirichlet summability of E_+ at β. Its prime subseries then gives $\mathcal{P}_e(\beta) < \infty$. Consequently E_+ also has a continuous nonzero boundary value at ρ, with every prime power retained. Continue the cross-multiplied identity on $\Re s > \beta$, then let $s = \rho + h$, $h \downarrow 0$. It would give

$$0 \cdot E_+(\rho) = F_w(\rho) E_-(\rho) \neq 0,$$

a contradiction.

If instead $\mathcal{P}_e(\beta) < \infty$, interchange the roles of the products and use the positive quotient $E_- = \zeta E_+/F_w$. Its real continuation, cancellation at 1, and bounded boundary limit give $\mathcal{P}_d(\beta) < \infty$ by the same argument, and the same contradiction follows. All products and boundary limits above are justified by absolute convergence before a value at ρ is used. This proves (189). □

Theorem 21.95 (Logarithmic one-sided cost of a boundary zero)**.** *In the preceding setting let the order of the zero at ρ be $m \geq 1$. Assume in addition*

$$\mathcal{P}_d(\sigma) < \infty \quad \text{for every } \sigma > \beta. \tag{190}$$

Then $\mathcal{P}_e$ also converges on that real ray, and as $h \downarrow 0$,

$$\begin{aligned} \mathcal{P}_d(\beta + h) &\geq \frac{m}{2} \log(1/h) - O_{\rho,w}(1), \\ \mathcal{P}_e(\beta + h) &\geq \frac{m}{2} \log(1/h) - O_{\rho,w}(1). \end{aligned} \tag{191}$$

The constants are for this fixed zero and fixed model; no height-uniform bound is asserted.

Proof. The same positive-continuation argument makes both complete Euler products convergent units on $\Re s > \beta$. Write

$$H_{\pm}(\sigma) = \log E_{\pm}(\sigma), \quad K(s) = F_w(s)/\zeta(s), \quad B(\sigma) = \log |F_w(\sigma)/\zeta(\sigma)|.$$

The real function B stays bounded as $\sigma \downarrow \beta$, since its numerator and denominator are both nonzero at that real point. The exact signed real balance is

$$H_+(\sigma) - H_-(\sigma) = B(\sigma).$$

For $s = \sigma + it$ the full Euler factors give

$$|E_+(s)| \leq E_+(\sigma), \qquad |E_-(s)| \geq \frac{E_-(2\sigma)}{E_-(\sigma)}. \tag{192}$$

For the second inequality use $|1 - p^{-s}| \leq 1 + p^{-\sigma}$ and the exact identity $1 + p^{-\sigma} = (1 - p^{-2\sigma})/(1 - p^{-\sigma})$. Take the nonnegative powers d_p and pass from finite products by the already established absolute convergence. Thus the scalar factor $E_-(2\sigma)$ is retained.

Since F_w is nonzero at ρ and ζ has order m there, $|K(\rho + h)| \geq c_{\rho,w} h^{-m}$ for all sufficiently small $h > 0$, where $c_{\rho,w} > 0$. With $\sigma = \beta + h$, the complete quotient and real compensation therefore give

$$\begin{aligned} m \log(1/h) + \log c_{\rho,w} &\leq \log |K(\rho + h)| \\ &\leq H_+(\sigma) + H_-(\sigma) - H_-(2\sigma) \\ &= 2H_-(\sigma) + B(\sigma) - H_-(2\sigma). \end{aligned} \tag{193}$$

Here $H_-(2\sigma) \geq 0$, and B is bounded. It follows that $H_-(\beta + h) \geq (m/2) \log(1/h) - O(1)$. The signed real balance gives the same bound for H_+.

Finally the retained higher prime powers have bounded mass at the endpoint:

$$0 \le H_-(\sigma) - \mathcal{P}_d(\sigma) \le \frac{\mathcal{P}_d(2\beta)}{1-2^{-\beta}} \quad (\sigma \ge \beta,\ \sigma > \beta).$$

The numerator is finite by (190), since $2\beta > \beta$. The analogous bound for H_+ uses $\mathcal{P}_e(2\beta) < \infty$, already supplied by positive continuation. Subtracting these bounded tails proves (191). □

Corollary 21.96 (Endpoint size of the computable prime class)**.** *The single class $\mathcal{D}_*$ satisfies, unconditionally,*

$$\sum_{p \in \mathcal{D}_*} p^{-1/2} = \infty. \tag{194}$$

If a zero ρ of order m attains $\Re\rho = \beta = \Theta_\zeta < 1$, then

$$\begin{gathered} \sum_{p \in \mathcal{D}_*} p^{-\beta-h} \ge \frac{m}{2}\log(1/h) - O_\rho(1), \qquad h \downarrow 0, \\ \limsup_{X\to\infty} \frac{\pi_{\mathcal{D}_*}(X)\log X}{X^\beta} \ge \frac{m}{2\beta}. \end{gathered} \tag{195}$$

For every critical-line zero of order m, regardless of RH,

$$\limsup_{X\to\infty} \frac{\pi_{\mathcal{D}_*}(X)\log X}{\sqrt{X}} \ge m. \tag{196}$$

In particular the last limsup is at least 1. If RH is false, that limsup is infinite.

Proof. The strict-partial coordinates of the diagonal model have subpower counting function. For every fixed $b > 0$ they therefore have finite sum $\sum p^{-b}$, by dyadic summation. Their deficits are between 0 and 1. At every prime, its deficit is 1 on $\mathcal{D}_*$, is strictly between 0 and 1 on those partial coordinates, and is zero elsewhere. Consequently

$$\mathcal{P}_d(\sigma) = \sum_{p \in \mathcal{D}_*} p^{-\sigma} + \sum_{\substack{p \text{ prime} \\ 0<w_p^*<1}} (1-w_p^*)p^{-\sigma}, \tag{197}$$

where the second sum is finite for every fixed $\sigma > 0$. The model F_* is independently favorable on $\Re s > 0$. Apply Theorem 21.94 to an unconditional critical-line zero. The second sum cannot supply its divergent deficit, proving (194).

When $\beta = \Theta_\zeta$, the exact counting exponent proved above makes $\mathcal{P}_d(\sigma)$ finite for all $\sigma > \beta$. The logarithmic cost theorem applies. The partial-coordinate sum in (197) is bounded uniformly for $\sigma \geq \beta$ near that endpoint, so it gives the first inequality of (195).

We check the count normalization explicitly. If the second limsup in (195) is finite, call it L. For any $A > L$, $\pi_{\mathcal{D}_*}(X) \leq AX^\beta/\log X$ after some finite X_A. Positive partial summation gives

$$\begin{aligned}\sum_{p\in\mathcal{D}_*} p^{-\beta-h} &= (\beta+h)\int_2^\infty \pi_{\mathcal{D}_*}(x)x^{-\beta-h-1}\,dx \\ &\leq A(\beta+h)\int_{X_A}^\infty \frac{x^{-1-h}}{\log x}\,dx + O_A(1) \\ &= A\beta\log(1/h) + O_A(1).\end{aligned}$$

For the last equality set $t = \log x$, split the integral at $t = 1/h$, and use $\int_{\log X_A}^\infty e^{-ht}\,dt/t = \log(1/h)+O_A(1)$. The pole lower bound forces $A\beta \geq m/2$. Let $A \downarrow L$ to obtain $L \geq m/(2\beta)$. An infinite limsup already satisfies that conclusion.

If RH holds, every critical-line zero attains $\beta = \Theta_\zeta = 1/2$, so the factor $m/(2\beta)$ is exactly m. If RH is false, $\Theta_\zeta > 1/2$ and the exact counting exponent of $\mathcal{D}_*$ makes the square-root-normalized limsup infinite. This proves (196) in both cases. □

Corollary 21.97 (Root-summable exclusions preserve the signal)**.** *Let $\mathcal{A}$ be any prime class with $\sum_{p\in\mathcal{A}} p^{-1/2} < \infty$, and set $\mathcal{D}' = \mathcal{D}_* \setminus \mathcal{A}$. Then*

$$\sum_{p\in\mathcal{D}'} p^{-1/2} = \infty, \qquad \limsup_{X\to\infty} \frac{\log(1+\pi_{\mathcal{D}'}(X))}{\log X} = \Theta_\zeta. \tag{198}$$

The zero-free-strip and RH count equivalences hold for $\mathcal{D}'$ as well, and so does (196)*. If $\mathcal{A}$ is computable, $\mathcal{D}'$ is computable.*

In particular one may exclude simultaneously all actual Fermat primes and all actual Fibonacci primes:

$$\begin{aligned}\mathcal{A}_{\mathrm{Fer}} &= \{2^{2^n}+1 : n \geq 0,\ 2^{2^n}+1 \text{ prime}\},\\ \mathcal{A}_{\mathrm{Fib}} &= \{F_n : n \geq 3,\ F_n \text{ prime}\}.\end{aligned}$$

These are sets of prime values; overlaps in their union are counted once. Neither class is assumed infinite.

Proof. Subtracting a finite positive subseries cannot remove the divergence in (194). Also $\pi_{\mathcal{A}}(X) \le \sqrt{X}\sum_{p\in\mathcal{A}} p^{-1/2}$. If $\Theta_\zeta > 1/2$, this smaller-exponent count cannot absorb the Θ_ζ signal of $\mathcal{D}_*$. If $\Theta_\zeta = 1/2$, the divergent root sum of $\mathcal{D}'$ forces its counting exponent to be at least $1/2$: a smaller exponent would make that sum convergent by dyadic summation. Its exponent is at most that of $\mathcal{D}_*$, since it is a subset. This proves (198) and hence the same strip and RH equivalences.

Under RH, the removed prime Dirichlet sum stays bounded as $\sigma \downarrow 1/2$, so subtracting it preserves the logarithmic lower bound $(m/2)\log(1/h) - O(1)$. The partial-summation argument therefore preserves the constant m in the square-root-normalized count limsup. If RH is false, the retained exponent is greater than $1/2$ and that limsup is infinite. Computability follows by testing membership in the two specified computable sets.

Finally the two named classes have finite root sums even before primality filtering:

$$\sum_{n\ge 0}(2^{2^n}+1)^{-1/2} < \infty, \qquad \sum_{n\ge 3} F_n^{-1/2} \le \sum_{n\ge 3} \varphi^{-(n-2)/2} < \infty, \quad \varphi = \frac{1+\sqrt{5}}{2}.$$

The Fibonacci bound $F_n \ge \varphi^{n-2}$ for $n \ge 2$ follows from the recurrence and $\varphi^2 = \varphi + 1$, or the earlier Binet formula. Both prime-value membership tests terminate by monotone integer sequence enumeration. Their union is root-summable even when the two sets overlap. This proves the stated actual-prime exclusions. □

These results give an unconditional endpoint size obstruction for the constructed prime classes, stronger than their earlier infinitely-often $X^{1/2-\epsilon}$ lower bounds. The multiplicity parameter supplies a lower count cost, not an independently proved upper bound on zero multiplicity. The Euler and prime-class conclusions are ordinary full proofs; only the two positive boundary-summability statements are newly Lean checked. No new independent count upper estimate is installed, and the certified zero-free boundary remains 7/8. No established literature priority is asserted.

21.45 The independent arithmetic count problem

Computability decides membership at each integer cutoff; it does not give an asymptotic count bound. We now retain the signed optimizer in an arithmetic form, without inserting the Θ_ζ-dependent estimate used in (188).

Take a sufficiently late admitted block, with $u = 2^j$, $k = k_j$, $h = h_{j,k}$ and cells $I = [a, a+h)$. Here h divides u, so there is no unaccounted terminal cell.

For the exact-moment problem on I write $H_I = \nu(I)$, $C_I = \#S_I$, and let E_I be the minimum overweight. Its minimum deficit is $D_I^{\mathrm{ex}} = E_I + C_I - H_I$. The rounded algorithm has

$$N_j^* \le \sum_I D_I^{\mathrm{ex}} + O(u^{-3+1/k}). \tag{199}$$

Indeed its overweight is no larger than the exact optimum, and its zeroth moment has the already proved total tolerance. The exact dual, after substituting $Q = 1 - P$, is

$$D_I^{\mathrm{ex}} = \max_{\substack{\deg Q<k \\ Q(p)\ge 0\ (p\in S_I)}} \left(\sum_{p\in S_I} \min\{Q(p),1\} - \int_I Q\,d\nu \right). \tag{200}$$

This is the preceding finite LP duality, since $(1-Q)_+ = 1 - \min\{Q,1\}$ at the prime nodes. An attaining polynomial can be chosen with $\|Q - 1\|_I \le 64 B_{k-1}$, using the proved positive-objective norming and $P = 0$ when $E_I = 0$.

For any such tuple put $T(x) = \min\{1, \max\{0, Q_I(x)\}\}$ on its cell, and $T = 0$ outside $[u, 2u)$. Use the right cell at every shared endpoint. Define the complete signed terms

$$\begin{aligned} \mathcal{A}_j(T) &= \sum_{p\in S_j} T(p) - \int_u^{2u} T\,d\nu, \\ \mathcal{B}_j(Q) &= \sum_I \int_I (Q_I - 1)_+\,d\nu, \qquad \mathcal{N}_j(Q) = \sum_I \int_I (Q_I)_-\,d\nu. \end{aligned} \tag{201}$$

The objective in (200), summed over cells, is exactly $\mathcal{A}_j(T) - \mathcal{B}_j(Q) + \mathcal{N}_j(Q)$. In particular $Q_I = 1$ retains $\mathcal{A}_j(1) = \#S_j - \nu([u,2u))$ and both penalties vanish. That constant row cannot be removed by centering the other polynomials.

Here the continuum negativity really is a smaller error. Let G bound the first and last prime's distances to the cell endpoints and all successive prime gaps in the block. The proved prime-mesh input gives $G \ll u^{2/3}$ for all sufficiently large u. If $Q(a+hy) = \sum_{r=0}^{d} c_r y^r$, $d < k$, set $A(Q) = \sum_r |c_r|$. Then $|Q'| \le dA(Q)/h$ and $|Q''| \le d^2 A(Q)/h^2$. Between successive prime nodes b, c, where $Q(b), Q(c) \ge 0$, linear interpolation gives

$$Q(x) \ge -\tfrac{1}{2}\|Q''\|_I (x-b)(c-x).$$

Its negative integral is at most $\|Q''\|_I(c-b)^3/12$. At each end, the derivative bound gives at most $\|Q'\|_I G^2/2$. Since $\nu'(x) \le 1/\log u$, summing these pieces, including both endpoints, proves

$$\int_I (Q)_- \, d\nu \le \frac{A(Q)(d + d^2/12)G^2}{h \log u}. \tag{202}$$

A constant polynomial has no negative part here. For $d \ge 1$, Lagrange interpolation at $0, 1/d, \dots, 1$ bounds its coefficient sum by

$$A(Q) \le \|Q\|_I \, 4^d d^d / d!.$$

Indeed each numerator has coefficient sum at most 2^d, and the sum of reciprocal absolute denominators is $d^d 2^d / d!$. The norming bound, $k \le \sqrt{j}$, and this explicit inequality give $A(Q)k^2 = \exp(O(k \log(k+1))) = u^{o(1)}$ uniformly for all the retained tuples. There are u/h cells and $h \ge \frac{1}{2} u^{1-1/k}$. Consequently the full, uniform correction is

$$\mathcal{N}_j(Q) \ll u^{1/3 + o(1)}. \tag{203}$$

Node positivity has not been confused with continuum positivity.

We next expand the remaining main term with every divisor sign. For integer $n \ge 2$, $\Lambda(n) = \sum_{d \mid n} \mu(d) \log(n/d)$. Also the defining density of ν gives the absolutely convergent identity

$$\nu'(x) = \frac{1}{x \log x} \sum_{r \ge 1} \frac{\mu(r)}{r} (x^{1/r} - 1), \qquad x > 1.$$

To verify it, insert $1/\zeta(\ell+1) = \sum_{r \ge 1} \mu(r) r^{-\ell - 1}$ in the defining exponential series and sum over $\ell \ge 1$. Absolute convergence follows from $(x^{1/r} - 1)/r = O_x(r^{-2})$. Thus, with all endpoints and prime powers retained,

$$\begin{aligned} \mathcal{A}_j(T) = & \sum_{\substack{d, m \ge 1 \\ u \le dm < 2u}} \mu(d) \frac{\log m}{\log(dm)} T(dm) \\ & - \sum_{\substack{r \ge 2, \ p \text{ prime} \\ u \le p^r < 2u}} \frac{T(p^r)}{r} \\ & - \sum_{r \ge 1} \frac{\mu(r)}{r} \int_u^{2u} \frac{x^{1/r} - 1}{x \log x} T(x) \, dx. \end{aligned} \tag{204}$$

The first two sums are finite. For integer $R \geq 1$ the last sum's tail has absolute value at most $(2u)^{1/R}\log(2)/R$, since $0 \leq T \leq 1$. In particular $R = u$ makes its tail $O(u^{-1})$ uniformly in every adaptive polynomial tuple. This is a universal truncation estimate, not a finite numerical test. The reciprocal zeta identity above is used only at the absolutely convergent integers $\ell+1 > 1$.

One actual cancellation test shows why the high divisor rows must stay coupled. Put $T = 1$ and take $n = 2q \in [u, 2u)$ with $q > \sqrt{2u}$ an odd prime. Keeping only $d \leq \sqrt{2u}$ leaves $\log(2q) - \log q = \log 2$ in the numerator. The retained full row $d = q$ contributes $-\log 2$; $d = 2q$ contributes zero. Thus these composite terms cancel exactly. The positive small-divisor subpacket has size

$$\sum_{\substack{u/2 \leq q < u \\ q \text{ prime}}} \frac{\log 2}{\log(2q)} \sim \frac{\log 2}{2}\frac{u}{(\log u)^2}. \tag{205}$$

Here the omitted finitely many small u are immaterial. This follows from the ordinary prime number theorem on $[u/2, u)$ and $\log(2q) = \log u + O(1)$. It rules out discarding that paired subpacket as a square-root remainder. It is not a lower bound for the entire high-divisor sum, which has further signs and cross terms.

The independent RH estimate being attacked is therefore the uniform signed bound

$$\sup_{Q}\{\mathcal{A}_j(T_Q) - \mathcal{B}_j(Q)\} \ll_\epsilon u^{1/2+\epsilon}, \tag{206}$$

over the explicitly bounded, prime-dependent dual tuples above. Equations (199) and (203) show that this estimate would imply the required literal count bound and RH. All mesh, clipping, rounding and model-series tails now have displayed costs compatible with that endpoint. The estimate (206) itself has not been proved. The signed expansion and the evaluated semiprime cancellation do not supply its upper bound. These are ordinary calculations; no new Lean verification or zero-free exponent is claimed in this subsection. The certified boundary remains 7/8.

What the reverse construction has and has not achieved. The independently favorable weighted model has been constructed; its signal-retention size estimate has not been proved. Large support by itself cannot replace that weighted estimate. For example the computable rational perturbation

$$\widetilde{w}_p = w_p^* + 2^{-p}\mathbf{1}_{\{w_p^*=0\}}$$

has support equal to every prime. Its additional Euler factor is

$$U(s) = \exp\left(\sum_{p\in\mathcal{D}_*} 2^{-p}[-\log(1-p^{-s})]\right).$$

The logarithm converges normally on every compact subset of $\Re s > 0$, so U is a holomorphic unit there. Thus $F_{\widetilde{w}} = F_* U$ has the same favorable region and simple pole, without an RH assumption. But exactly

$$(\widetilde{w}_p - 1)_+ = (w_p^* - 1)_+, \qquad (1-\widetilde{w}_p)_+ = (1-w_p^*)_+ - 2^{-p}\mathbf{1}_{\{w_p^*=0\}}.$$

The overweight condition is unchanged, and the deficit changes by a summable mass at every positive exponent. Full support has therefore been obtained without reducing the unresolved arithmetic burden. Although the new zero-weight set is empty, its strict-partial coordinates now contain all of $\mathcal{D}_*$; the subpower partial-coordinate hypothesis needed for literal count recovery is no longer available. The deleted-prime theorem cannot be applied to this modified model by ignoring that hypothesis.

The original vertex construction does make the deficit comparable to a literal count, but that comparison is not an independent upper estimate. For dyadic endpoints $X = 2^J$, complete block mass matching, the finite initial head and the summable moment errors give $\sum_{p<X} w_p^* = \nu([1,X]) + O(1)$. The strict-partial support below X is subpower. Since total overweight is nonnegative, the exact signed ledger therefore implies, for every $\epsilon > 0$,

$$\pi_{\mathcal{D}_*}(X) \geq \pi(X) - \nu([1,X]) - O_\epsilon(X^\epsilon), \qquad X = 2^J.$$

Thus even its constant row retains an ordinary one-sided prime-count discrepancy. The other rows add adaptive polynomial tests and their compensating penalties; their uniform estimate has not been established. Compared with the earlier overweight and prime-Poisson energy criteria, the single class improves the arithmetic formulation and removes the final order parameter. No better independent arithmetic bound, and hence no advantage for proving RH or a smaller zero-free boundary, has yet been demonstrated.

21.46 What selection by the theta sign can establish

Here selection of prime factors and selection of prime cutoffs must be distinguished. The following statements concern the original full-prime Nicolas quantity

$$F(p) = \gamma + \log\log\theta(p) + \sum_{q\leq p,\ q \text{ prime}} \log(1-1/q).$$

They do not replace this product by one over the selected class.

For consecutive primes $q < p$, $p \geq 5$, put $T = \theta(q) > 1$. The exact increment is

$$F(p) - F(q) = \int_T^{T+\log p} \frac{dt}{t \log t} + \log(1 - 1/p). \tag{207}$$

If $\theta(p) \leq p$, this increment is strictly positive. Indeed, writing $L = \log p > 1$, its integral is at least $-\log(1 - L/p)/L$. The convergent logarithmic series gives

$$\frac{-\log(1 - L/p)}{L} = \sum_{m \geq 1} \frac{L^{m-1}}{mp^m} > \sum_{m \geq 1} \frac{1}{mp^m} = -\log(1 - 1/p).$$

If instead $\theta(q) \geq p$, the integral in (207) is at most $1/p$, whereas $\log(1 - 1/p) < -1/p$, so the increment is negative. These are directional statements about changes in F; neither gives the absolute inequality $F(p) < 0$.

Define the actual prime classes

$$\begin{aligned} S_+ &= \{p \text{ prime} : \theta(p) > p\}, \\ U &= \{p \text{ prime} : \theta(p) - \log p < p < \theta(p)\} \subset S_+. \end{aligned} \tag{208}$$

The class U consists of upward crossings at the prime jump. Its membership is computable: both theta values are logarithms of explicitly computable integer primorials, and neither equals the integer p. Otherwise e would satisfy an integer polynomial $X^p - N = 0$, contrary to its transcendence. Certified approximations therefore decide both strict signs.

Theorem 21.98 (Theta-upcrossing cutoffs retain Nicolas violations)**.** *RH is equivalent to $F(p) < 0$ for every sufficiently large $p \in U$. It is also equivalent to that inequality for every sufficiently large $p \in S_+$. If RH is false, there are infinitely many $p \in U$ with $F(p) > 0$. No cardinality or density lower bound for U is assumed in this statement.*

Proof. Under RH the established Nicolas inequality holds at all sufficiently large cutoffs, so it holds on both selected classes.

Suppose RH is false. The proved two-sided excursion theorem for K gives positive values arbitrarily far out and negative values arbitrarily far out. Consequently $\{x : K(x) > 0\}$ has infinitely many bounded positive components tending to infinity. On the closure of each component, continuity gives an interior positive maximum. Such a maximum cannot lie strictly between primes: there $\theta(t) = c$ is constant and $K'(t) = (t - c)\kappa(t)$

has either a constant sign or changes from negative to positive, giving a minimum. Thus a maximum occurs at a prime p. Its one-sided derivative signs imply $\theta(p) - \log p \le p \le \theta(p)$. The non-equality just proved makes both inequalities strict, so $p \in U$ and $K(p) > 0$.

On U the overshoot obeys $0 < \theta(p) - p < \log p$. The exact nonlinear correction therefore has the unconditional Taylor bound

$$0 \le D(p) \le \frac{\log p + 1}{2p^2}.$$

The positive prime-power tail is larger than this correction. Indeed PNT gives at least $p/(3\log p)$ primes in $(p, 2p]$ for all sufficiently large p. Each contributes at least $1/(2q^2) \ge 1/(8p^2)$ to $A(p)$, whence

$$A(p) \ge \frac{1}{24p\log p} > D(p)$$

eventually. The exact identity $F = K - D + A$ now yields $F(p) > K(p) > 0$ at the infinitely many maximizing primes. This contradicts the proposed eventual inequality on U, and also contradicts it on S_+ since $U \subset S_+$. $\square$

This is a qualitative cutoff-retention theorem, not an independent proof of the inequality on those cutoffs. In particular the theta-positive class has not been shown to satisfy the RH condition unconditionally. The earlier quantitative excursion proof uses maxima of $K(x) - A_0 x^{-b}$ and a wider theta-error band; the present unweighted component maxima give the exact logarithmic band but do not retain a claimed power-size amplitude. No restricted moving-rank coefficient criterion follows here without checking its comparison remainder against that amplitude. The inputs are the existing Nicolas identities and excursions; no established literature priority or new Lean check is claimed.

For selection inside an Euler product there is a further hypothesis to keep visible. If an unweighted prime subset S had a zero-free continuation of $\prod_{p\in S}(1-p^{-s})^{-1}$ on $\Re s > 1/2$, apart from a simple pole of order one at 1, the existing one-sided transfer theorem would already imply RH: its weights are bounded by the reference capacity 1, so its overweight is identically zero. No separate cardinality lemma is needed under those full hypotheses. Theta-sign selection has not proved them. The weighted model avoids this conclusion precisely by leaving an unestimated overweight; its prescribed zero-free continuation does not itself provide an arithmetic saving.

21.47 A greedy prime carrier with its inequality built in

A direct selection inside the prime measure avoids testing the desired Nicolas sign at each member. List all primes increasingly as $p_1, p_2, \ldots$. Starting with $A_0 = M_0 = 0$, retain p_j in $\mathcal{S}$ exactly when $A_{j-1} + \log p_j \le p_j$. On retention put $A_j = A_{j-1} + \log p_j$, $M_j = M_{j-1}$; on rejection put $A_j = A_{j-1}$, $M_j = M_{j-1} + \log p_j$. Let $\mathcal{D}$ be the rejected prime class and write

$$\theta_{\mathcal{S}}(x) = A_{\pi(x)}, \qquad M(x) = \theta_{\mathcal{D}}(x) = M_{\pi(x)}, \qquad N_{\mathcal{D}}(x) = \#(\mathcal{D} \cap [2, x]).$$

This is a fixed class, not a new choice at each cutoff. Every step is decidable by certified logarithms: equality $\log N = p_j$ with an integer primorial subproduct N is impossible by transcendence of e.

Theorem 21.99 (Greedy cap and optimal deletion cost)**.** *The selected class satisfies, unconditionally, $\theta_{\mathcal{S}}(x) \le x$ for every real $x \ge 2$. Put*

$$R(x) = \max\big(\{0\} \cup \{\theta(p) - p : \ p \le x, \ p \text{ prime}\}\big). \tag{209}$$

Then

$$R(x) \le M(x) \le R(x) + \log x. \tag{210}$$

For any other fixed prime class $\mathcal{T}$ satisfying $\theta_{\mathcal{T}}(x) \le x$ for every $x \ge 2$, its deleted class $\mathcal{T}^c$ (complement inside the primes) obeys

$$\begin{aligned} M(x) &\le \theta_{\mathcal{T}^c}(x) + \log x, \\ N_{\mathcal{D}}(x) &\le \#(\mathcal{T}^c \cap [2, x]) \qquad (x \ge 2). \end{aligned} \tag{211}$$

Thus the fixed greedy carrier minimizes the number of deletions exactly at every cutoff among every class with this cap. Moreover $\theta_{\mathcal{S}}(x) \sim x$ and $\#(\mathcal{S} \cap [2, x]) \sim x/\log x$ unconditionally.

Proof. The invariant $A_j \le p_j$ follows immediately by induction: retention enforces it, while rejection preserves $A_{j-1} \le p_{j-1} < p_j$. It holds between primes as well, proving the selected inequality.

Put $E_j = \theta(p_j) - p_j$ and $R_j = \max(0, E_1, \ldots, E_j)$. The selection rule is exactly

$$M_j = \begin{cases} M_{j-1}, & E_j \le M_{j-1}, \\ M_{j-1} + \log p_j, & E_j > M_{j-1}. \end{cases} \tag{212}$$

At a rejected prime, the cap invariant gives $M_j \ge E_j$, while the rejection test gives $M_j < E_j + \log p_j$. At a retained prime, $E_j \le M_j = M_{j-1}$. Induction, using the increasing prime weights $\log p_j$, therefore gives $R_j \le$

$M_j \le R_j + \log p_j$ in both cases. This proves (210) at primes and hence at all real cutoffs.

For any other capped class, $\theta_{\mathcal{T}^c}(t) = \theta(t) - \theta_{\mathcal{T}}(t) \ge \theta(t) - t$. Its left side is nondecreasing and nonnegative, so $\theta_{\mathcal{T}^c}(x) \ge R(x)$. This proves the first optimality inequality. For the sharper count comparison, apply Theorem 21.125 to the actual ordered prime stream, with weight $w_j = \log p_j$ and budget p_j. The weights are nonnegative and increasing. Membership in any competing capped class is a feasible selection at every prime prefix, because its retained logarithmic sum is at most p_j. The theorem therefore gives $\#(\mathcal{T} \cap [2, x]) \le \#(\mathcal{S} \cap [2, x])$. Taking complements inside the same finite prime prefix proves the exact deletion-count inequality. No logarithmic overhead is needed.

Finally PNT gives $R(x) = o(x)$: split the defining maximum into a fixed initial segment and a tail on which $|\theta(t) - t| \le \epsilon t$. Thus $M(x) = o(x)$ and $\theta_{\mathcal{S}}(x) = \theta(x) - M(x) \sim x$. For the retained count, use the direct squeeze

$$\theta_{\mathcal{S}}(x) \le \#(\mathcal{S} \cap [2, x]) \log x \le \pi(x) \log x.$$

The lower bound follows term by term from $\log p \le \log x$, and the upper bound from inclusion in the full prime prefix. Divide by x and use the theta and prime-count forms of PNT. This proves the retained-prime count asymptotic. □

The complete statement of Theorem 21.99, including both density limits at real cutoffs and comparison with every competing capped class, is now formalized by

`greedyPrimeClass_complete_cap`.

The exact identification of the regulator with the displayed finite maximum is checked by

`greedyPrimeRegulatorReal_le_iff`.

The proof transfers actual prime indices to the literal filtered prime sums and counts, including natural cutoffs zero and one, and uses the established unconditional theta and count PNT. No RH assumption is used. The preceding certified-logarithm membership algorithm remains a separate formalization obligation; the maintained class here is defined by the exact real comparison.

The independent favorable condition can be written in the exact integral form of the earlier RH criterion:

$$K_{\mathcal{S}}(x) = \int_x^\infty (\theta_{\mathcal{S}}(t) - t)\kappa(t)\, dt \le 0, \qquad K(x) = K_{\mathcal{S}}(x) + \int_x^\infty M(t)\kappa(t)\, dt. \tag{213}$$

Both integrals converge unconditionally. For example the quantitative PNT error $O(t\exp(-c\sqrt{\log t}))$ bounds its running maximum by $O(t\exp(-c'\sqrt{\log t}))$ for a possibly smaller $c' > 0$. The integrand on the selected class is nonpositive pointwise. No RH hypothesis supplies this inequality.

The same selected class satisfies an actual calibrated Nicolas product inequality, not only the integral sign. Define its finite missing-product constant and its error by

$$\begin{aligned} C_{\mathcal{D}} &= \sum_{p\in\mathcal{D}} -\log(1-1/p), \\ F_{\mathcal{S}}(x) &= \gamma - C_{\mathcal{D}} + \log\log\theta_{\mathcal{S}}(x) + \sum_{\substack{p\le x\\ p\in\mathcal{S}}} \log(1-1/p). \end{aligned} \tag{214}$$

The constant is finite by the quantitative PNT regulator bound; the error is defined once $\theta_{\mathcal{S}}(x) > 1$. The subtraction of $C_{\mathcal{D}}$ is essential.

Theorem 21.100 (The selected class already has the product sign)**.** *There are constants $c_0 > 0$ and X_0 such that, unconditionally,*

$$F_{\mathcal{S}}(x) \le -\frac{c_0}{x} < 0 \qquad (x \ge X_0). \tag{215}$$

The exact difference from the original all-prime error is

$$F(x) - F_{\mathcal{S}}(x) = \log\log\theta(x) - \log\log\theta_{\mathcal{S}}(x) + \sum_{\substack{p>x\\ p\in\mathcal{D}}} -\log(1-1/p). \tag{216}$$

Both terms on the right are nonnegative, and both must be estimated in a zero-signal transfer.

Proof. Write $D_{\mathcal{S}}$ and $A_{\mathcal{S}}$ for the earlier nonlinear correction and prime tail with $\theta_{\mathcal{S}}$ and the selected primes in place of θ and all primes. The calibrated exact identity is

$$F_{\mathcal{S}} = K_{\mathcal{S}} - D_{\mathcal{S}} + A_{\mathcal{S}}.$$

Indeed subtract the original identity, and use

$$\int_x^\infty M(t)\kappa(t)\,dt = \frac{M(x)}{x\log x} + \sum_{\substack{p>x\\ p\in\mathcal{D}}} \frac{1}{p},$$

which is partial summation against $1/(t\log t)$. The nonlinear corrections supply the difference of the two $\log\log$ terms, and the remaining prime tail

supplies exactly the quadratic part of the missing logarithmic product. This verifies the normalization as well as (216). Concavity gives $D_{\mathcal{S}} \geq 0$, while PNT gives $0 \leq A_{\mathcal{S}}(x) \leq A(x) = O(1/(x \log x))$.

There is also a definite negative area in $K_{\mathcal{S}}$. Choose a constant $C \geq 1$ with $\pi(2x) \leq Cx/\log x$ eventually, and put $h = \log x/(8C)$. Inside $[x, 2x]$, the union of intervals $[p, p+h)$ over all primes $p \leq 2x$ has measure at most $x/8$. At every remaining point t, its last full prime is at most $t - h$. Since the selected theta function is bounded by the last selected prime, the cap gives $t - \theta_{\mathcal{S}}(t) \geq h$ there. Also $\kappa(t) \geq 1/(8x^2 \log x)$ on $[x, 2x]$ eventually. Consequently

$$-K_{\mathcal{S}}(x) \geq \int_x^{2x} (t - \theta_{\mathcal{S}}(t))\kappa(t)\, dt \geq \frac{7}{512Cx}.$$

The pointwise cap makes the omitted part of the integral nonnegative. The $O(1/(x \log x))$ upper bound on $A_{\mathcal{S}}$ is eventually smaller than half this supply. The calibrated identity proves (215), with a smaller $c_0 > 0$. □

Theorem 21.101 (The size condition retains every off-line zero)**.** *For every fixed $1/2 \leq \eta < 1$, the following are equivalent:*

$$\zeta(s) \neq 0 \quad (\Re s > \eta), \qquad N_{\mathcal{D}}(X) \ll_\epsilon X^{\eta+\epsilon} \quad \textit{for every } \epsilon > 0.$$

Here the zero statement concerns nontrivial zeros; the pole at 1 is not a zero. In particular the same explicit selected class has an independently proved favorable inequality, and

$$\text{RH} \quad \Longleftrightarrow \quad \#\{p \leq X : p \text{ prime},\ p \notin \mathcal{S}\} \ll_\epsilon X^{1/2+\epsilon} \quad \textit{for every } \epsilon > 0. \tag{217}$$

Under RH the stronger count $N_{\mathcal{D}}(X) = O(\sqrt{X} \log X)$ follows from the usual Chebyshev error bound.

Proof. Suppose first that $\Theta_\zeta \leq \eta$. For $\eta < \kappa < 1$, the proof of Lemma 21.90 gives $|\theta(t) - t| = O_\kappa(t^\kappa \log^2 t)$. The regulator bound and splitting the rejected primes below and above $\sqrt{X}$ give

$$M(X) = O_\kappa(X^\kappa \log^2 X), \qquad N_{\mathcal{D}}(X) = O_\kappa(X^\kappa \log X).$$

Taking κ strictly between η and $\eta + \epsilon$ proves the claimed bound; when $\eta + \epsilon \geq 1$, the elementary count bound suffices. Under RH insert instead $|\theta(t) - t| = O(\sqrt{t} \log^2 t)$ to obtain the stronger count.

Conversely suppose the stated count holds, and fix $0 < \epsilon < 1 - \eta$. Then $M(t) \le \log t\, N_{\mathcal{D}}(t) = O_\epsilon(t^{\eta+\epsilon} \log t)$. The complete positive correction in (213) consequently satisfies

$$0 \le K(x) - K_{\mathcal{S}}(x) \le C_\epsilon \frac{x^{\eta+\epsilon-1}}{1-\eta-\epsilon}. \tag{218}$$

The denominator is positive on the stated parameter domain. The same count controls both terms in (216): since $\theta_{\mathcal{S}}(x) \sim \theta(x) \sim x$, its clock term is $O(M(x)/(x \log x))$, and partial summation bounds the entire missing-product tail by $O_\epsilon(x^{\eta+\epsilon-1}/(1-\eta-\epsilon))$. Consequently $0 \le F(x) - F_{\mathcal{S}}(x)$ has this same upper order. Positive excursions of the original Nicolas error are therefore preserved as well, including the full normalization and both deletion corrections. If there were a zero of real part $\beta > \eta$, choose also $\epsilon < \beta - \eta$ and $1 - \beta < b < 1 - \eta - \epsilon \le 1/2$. The unconditional positive excursion theorem gives $\limsup x^b K(x) = +\infty$. But $K_{\mathcal{S}} \le 0$ and the displayed correction give $\limsup x^b K(x) \le 0$, since $b + \eta + \epsilon - 1 < 0$. This is a contradiction. The zero-free implication is therefore established without using an RH-dependent wave under the bad-zero hypothesis. □

This carrier supplies the two requested obligations separately. The cap and $K_{\mathcal{S}} \le 0$ are unconditional properties of the constructed class. The root-size deletion estimate would make it large enough to retain every off-line zero signal. The independent bound proved so far is only $N_{\mathcal{D}}(X) \ll_\epsilon X^{7/8+\epsilon}$, from the installed boundary and the preceding theorem. An improvement to $X^{3/4+\epsilon}$ would give the boundary $3/4$ by the same complete consumer; the square-root target gives RH.

The actual size attack is now the concrete positive-record problem

$$\max_{\substack{p \le X \\ p \text{ prime}}} (\theta(p) - p)_+ \ll_\epsilon X^{\eta+\epsilon}.$$

The recursion and optimality theorem show exactly why changing the capped selection cannot by itself improve its exponent: every other capped prime class must delete at least this running positive mass, and the greedy algorithm pays at most one prime weight more. Every rejected prime is a strict new positive record of $\theta(p) - p$, because its rejection requires $\theta(p) - p > M(p^-) \ge R(p^-)$. In particular, if $q < p$ are consecutive full primes and p is rejected, then $p - q < \log p$. This necessary short-gap condition is insufficient for the desired count: it discards the record-height comparison, and supplies no power-saving upper bound on the number of

rejections. The retained exact target is the quantized record process (212), not a count of all short prime gaps. No independent improvement of the positive-record bound has been proved here. The construction simplifies the arithmetic object and proves optimal selection; it is not claimed to make its remaining size bound easier than the earlier RH criterion. The cap, regulator, exact count optimality, and density claims in Theorem 21.99 have the Lean proofs identified above. The product-sign and zero-signal transfer theorems in this subsection remain ordinary proofs with separate formalization obligations. No established originality priority is asserted.

21.48 The signed energy of the positive-record process

The count attack must retain the stopped prime weights, rather than assume an independent prime-gap law. Put $E(t) = \theta(t) - t$, retain the running positive record R from (209), and let $\Delta R(p) = R(p) - R(p^-) \geq 0$. For every real $X \geq 2$ define the finite signed quantity

$$\mathcal{C}_R(X) = \sum_{\substack{p \leq X \\ p \text{ prime}}} R(p^-) \log p - \int_2^X R(t)\, dt, \qquad Q_R(X) = \sum_{\substack{p \leq X \\ p \text{ prime}}} (\Delta R(p))^2.$$

Then the exact energy identity is

$$R(X)^2 = 2\mathcal{C}_R(X) + 2R(X)(R(X) - E(X)) + Q_R(X). \tag{219}$$

To prove it, integrate the finite step function product RE. The continuous part of dE is $-dt$, its prime jump is $\log p$, and R has only prime jumps. The right-endpoint form of integration by parts gives

$$R(X)E(X) = \mathcal{C}_R(X) + \sum_{\substack{p \leq X \\ p \text{ prime}}} R(p)\Delta R(p).$$

Here $E(p) = R(p)$ whenever $\Delta R(p) > 0$, and $R(2^-) = 0$ removes the initial endpoint. Finally $2\sum_{\substack{p \leq X \\ p \text{ prime}}} R(p)\Delta R(p) = R(X)^2 + Q_R(X)$. All sums retain their finite prime support.

At a positive record prime X, the terminal correction vanishes:

$$R(X)^2 = 2\mathcal{C}_R(X) + Q_R(X), \qquad 0 \leq Q_R(X) \leq R(X) \log X. \tag{220}$$

The last bound uses $0 \leq \Delta R(p) \leq \log p$ and $\sum_{\substack{p \leq X \\ p \text{ prime}}} \Delta R(p) = R(X)$. At these record endpoints the complete signed bounds are

$$\frac{R(X)^2 - R(X) \log X}{2} \leq \mathcal{C}_R(X) \leq \frac{R(X)^2}{2}.$$

For $A \geq 0$, an independent bound $\mathcal{C}_R(X) \leq A$ implies

$$R(X) \leq \log X + \sqrt{2A}.$$

More generally, suppose $a > 0$, $K \geq 0$, $X_0 \geq 2$, and $\mathcal{C}_R(p) \leq Kp^{2a}$ at every strict record prime $p \geq X_0$, where strict means $\Delta R(p) > 0$. Then the full bound, including all nonrecord cutoffs and the initial range, is

$$R(X) \leq \left(R(X_0) + \frac{1}{a} + \sqrt{2K} \right) X^a \qquad (X \geq 2).$$

Indeed $\log t \leq t^a/a$ for $t > 0$, and the record-endpoint root bound gives $R(p) \leq (a^{-1} + \sqrt{2K})p^a$. Every positive running maximum is attained at a strict record jump. If that prime is below X_0, its height is bounded by $R(X_0)$; otherwise the preceding estimate applies. Monotonicity of t^a and $X^a \geq 1$ give the displayed uniform constant. The independent signed-energy hypothesis is retained throughout.

Consequently an independent estimate $\mathcal{C}_R(X) \ll_\epsilon X^{3/2+\epsilon}$ at all sufficiently large positive record primes would give $R(X) \ll_\epsilon X^{3/4+\epsilon}$, hence the boundary 3/4 through the complete deletion consumer. The corresponding root target is $\mathcal{C}_R(X) \ll_\epsilon X^{1+\epsilon}$. The installed input supplies only $R(X) \ll_\epsilon X^{7/8+\epsilon}$ and thus the record-endpoint bound $\mathcal{C}_R(X) \ll_\epsilon X^{7/4+\epsilon}$. Constants are independent of the record cutoff, but may depend on ϵ and the stated source estimate.

This is a signed calculation and a checked failed application, not an improved estimate. The prime-divisibility probability law samples integers at a fixed prime cutoff; it does not make the successive prime gaps, stopped at their positive records, a mean-zero independent process. Thus its existing independence hypothesis does not apply to $\mathcal{C}_R$. Outside record endpoints, the terminal term in (219) must also remain present. No new zero-free exponent follows from this identity.

The entire signed energy calculation and its quantitative record-to-cutoff consumer are now Lean-checked. `primeRecordEnergy_identity` uses the actual finite prime sum and ordinary real interval integral for every $X \geq 2$. `primeRecordJump_left_limit` identifies each prime jump with its one-sided limit; the first-prime left limit is zero. `greedyPrimeRegulatorReal_integral` evaluates the actual record integral by the finite sum of its right-continuous jumps. The two finite product identities and the record-endpoint variation bound are compiled with their full support. `primeRecordEnergy_root_bound` and `primeRecordEnergy_power_bound_explicit` retain the independent energy estimate. `primeRecordEnergy_all_cutoff_bound` proves

the displayed uniform constant, using strict record primes and the finite initial record. All proof owners are maintained, exported, and registered by exact name for recurring foundational-axiom checks. The separate zero-response theorems are required for the zero-free interpretation; no independent power-saving estimate for $\mathcal{C}_R$ is supplied by these identities.

21.49 An extremal probability law among capped prime classes

The count comparison in Theorem 21.99 has an exact strengthening, with consequences for every finite decreasing local probability law.

Theorem 21.102 (Rank domination and exact optimality)**.** *Fix a real cutoff $x \geq 2$, and let $\mathcal{T}$ be any prime class with $\theta_{\mathcal{T}}(t) \leq t$ for $2 \leq t \leq x$. List the selected primes through x as $s_1 < \cdots < s_a$ for the fixed greedy class $\mathcal{S}$, and $t_1 < \cdots < t_b$ for $\mathcal{T}$. Then*

$$a \geq b, \qquad s_i \leq t_i \quad (1 \leq i \leq b). \tag{221}$$

In particular the greedy class minimizes the number of deleted primes exactly at every cutoff. For every nonnegative nonincreasing function w on primes,

$$\sum_{\substack{p \leq x \\ p \in \mathcal{S}}} w(p) \geq \sum_{\substack{p \leq x \\ p \in \mathcal{T}}} w(p).$$

If w is positive and strictly decreasing, equality is possible only when the two selected prime sets through x coincide.

Proof. Induct on the competitor's rank i. Suppose $s_j \leq t_j$ for $j < i$ and the greedy class has not made its ith selection by t_i. It has then selected exactly $i - 1$ primes before t_i. Since log is increasing,

$$A_{\pi(t_i)-1} + \log t_i = \sum_{j<i} \log s_j + \log t_i \leq \sum_{j \leq i} \log t_j = \theta_{\mathcal{T}}(t_i) \leq t_i.$$

The greedy rule therefore retains t_i, a contradiction. For $i = 1$ the same proof has empty preceding sums. This proves every required rank and hence $a \geq b$. The weighted comparison follows from $w(s_i) \geq w(t_i)$ on the matched ranks and the nonnegative unmatched terms. For a positive strictly decreasing w, distinct selected sets give either an unmatched positive term or a strict inequality on at least one matched rank. □

The entire statement of Theorem 21.102 is now formalized, including its local cap hypothesis $2 \le t \le x$, arbitrary decreasing weights, and the strict equality criterion. Its complete certificate is

`greedyPrimeClass_local_rank_complete`.

The rank of a prime in a class is the number of class primes strictly below it. The direct comparison at any specified ordinal rank is checked by

`greedyPrimeClass_local_rank_at`.

The formal proof truncates only the competitor, so the greedy class remains fixed at every cutoff. The weighted comparison retains the exact signed prefix ledger; equality with a positive strictly decreasing weight forces all prefix count surpluses to vanish and hence the two finite prime sets to coincide.

Theorem 21.103 (Stochastic and coefficient extremality)**.** *Let $\eta(p) \in [0,1]$ be nonincreasing on primes. At a fixed cutoff x, let $K_{\mathcal{V},x}$ denote the sum of independent Bernoulli variables of success probabilities $\eta(p)$ over the finite set $p \le x$, $p \in \mathcal{V}$. For every capped competitor $\mathcal{T}$ as above and every integer rank $r \ge 0$,*

$$\mathbb{P}(K_{\mathcal{S},x} \ge r) \ge \mathbb{P}(K_{\mathcal{T},x} \ge r). \tag{222}$$

The same comparison holds for expectations of every increasing function on $\{0, \dots, a\}$. It implies

$$\mathbb{E}z^{K_{\mathcal{S},x}} \ge \mathbb{E}z^{K_{\mathcal{T},x}} \quad (z \ge 1), \qquad \mathbb{E}z^{K_{\mathcal{S},x}} \le \mathbb{E}z^{K_{\mathcal{T},x}} \quad (0 \le z \le 1).$$

Separately, for every nonnegative nonincreasing finite odds function $v(p)$, one has coefficientwise domination

$$[z^r] \prod_{\substack{p \le x \\ p \in \mathcal{S}}} (1 + v(p)z) \ge [z^r] \prod_{\substack{p \le x \\ p \in \mathcal{T}}} (1 + v(p)z) \qquad (r \ge 0). \tag{223}$$

No comparison of individual normalized point masses or of peak heights is asserted.

Proof. The rank theorem gives $\eta(s_i) \ge \eta(t_i)$ for $i \le b$. Pad the competitor's array by $a - b$ zeros. Increasing one coordinate from u to $v \ge u$ changes the tail at rank $r + 1$ by exactly

$$(v - u)\mathbb{P}(K = r) \ge 0,$$

where K is the sum over the other coordinates. This follows by conditioning on that coordinate: the new tail is $(1-v)\mathbb{P}(K \geq r+1) + v\mathbb{P}(K \geq r)$. The rank-zero tail is always one. Iterating the replacement over the padded arrays proves (222), including coordinates equal to zero or one. For either finite law, use the common support bound a. The exact finite Abel identity is

$$\mathbb{E}\phi(K) = \phi(0) + \sum_{k=0}^{a-1} \big(\phi(k+1) - \phi(k)\big)\mathbb{P}(K \geq k+1).$$

The boundary term vanishes because $K \leq a$; for the competitor, all tails above its smaller support b vanish as well. Every difference in this sum is nonnegative, whereas $\phi(0)$ may have either sign and cancels between the two laws. This proves the expectation assertion. The two moment comparisons follow from the monotonicity of $k \mapsto z^k$ in their stated domains, retaining $z = 0, 1$.

For the coefficient assertion, each matched factor has $v(s_i) \geq v(t_i) \geq 0$, so replacing the matched factors increases every coefficient. The remaining factors have nonnegative coefficients and constant term one; appending them cannot decrease a coefficient. This proves the assertion also for ranks above the competitor's support and at rank zero. □

For the ordinary integer-divisibility law $\eta(p) = 1/p$, the fixed-cutoff Bernoulli law is the finite prime-divisibility density. More explicitly, put $V_x = \{p \leq x : p \text{ prime},\ p \in \mathcal{V}\}$ and $m = |V_x|$. For every integer $N > 0$ and every $r \geq 0$,

$$\left| \frac{1}{N} \#\{1 \leq n \leq N : \#\{p \in V_x : p \mid n\} \geq r\} - \mathbb{P}(K_{\mathcal{V},x} \geq r) \right| \leq \frac{3^m}{N}.$$

The bound follows from the proved coefficient error at parameter zero, followed by summation over the finite support of the tail indicator. Thus the greedy capped carrier maximizes every limiting upper-tail density. The limit is over sampled integers with x fixed; no uniform growing-cutoff sampling assertion is used. The coefficient conclusion applies directly to $v(p) = 1/(p-1)$. For affine odds $v(p) = \rho/(p+\beta)$, it applies when $\rho > 0$ and $\beta > -2$. The associated probability law $\eta(p) = \rho/(p+\beta+\rho)$ obeys the full stochastic comparison for every $\beta \geq -2$. These are prime-coordinate laws; no sampling theorem for an affine polynomial prime-pair family is asserted.

At $\beta = -2$, only $p = 2$ is forced. Write $V_x^\circ = V_x \setminus \{2\}$ and

$$A_{\mathcal{V},x} = \prod_{p \in V_x^\circ} (1 - \eta(p)) > 0, \qquad Q_{\mathcal{V},x}(z) = \prod_{p \in V_x^\circ} \left(1 + \frac{\rho}{p-2} z\right).$$

If $2 \in V_x$, then $K_{\mathcal{V},x} = 1 + K^{\circ}_{\mathcal{V},x}$ in distribution and, for every $r \geq 0$,

$$\begin{aligned} \mathbb{P}(K_{\mathcal{V},x} = 0) &= 0, \\ \mathbb{P}(K_{\mathcal{V},x} = r+1) &= A_{\mathcal{V},x}[z^r]Q_{\mathcal{V},x}(z), \\ \mathbb{P}(K_{\mathcal{V},x} \geq r+1) &= \mathbb{P}(K^{\circ}_{\mathcal{V},x} \geq r). \end{aligned}$$

If $2 \notin V_x$, no shift occurs and $\mathbb{P}(K_{\mathcal{V},x} = r) = A_{\mathcal{V},x}[z^r]Q_{\mathcal{V},x}(z)$. The fixed greedy class contains 2, so its shifted identities apply at every real cutoff $x \geq 2$. These identities retain the survival factor, including the empty remaining product. They do not extend the separate odds-coefficient order to $\beta = -2$.

Applying the decreasing weight $w(p) = -\log(1 - 1/p)$ shows, at each finite cutoff, that the greedy class minimizes the missing logarithmic Euler-product cost among all capped competitors. The passage to infinity also proves convergence, rather than assuming it for the greedy class. For each finite set A of primes outside $\mathcal{S}$, choose a cutoff X containing A. Nonnegativity and the exact finite complement identity give

$$\sum_{p \in A} w(p) \leq \sum_{\substack{p \leq X,\ p \text{ prime} \\ p \notin \mathcal{S}}} w(p) \leq \sum_{\substack{p \leq X,\ p \text{ prime} \\ p \notin \mathcal{T}}} w(p) \leq C_{\mathcal{T}^c}.$$

If $C_{\mathcal{T}^c}$ is finite, these bounds force convergence of the nonnegative greedy missing-cost series and give $C_{\mathcal{D}} \leq C_{\mathcal{T}^c}$. The same proof works for every nonnegative decreasing prime weight whose missing total for the competitor converges. This sharpens the earlier additive count comparison to exact optimality for counts and for all these probability observables. It does not estimate the optimal count's growth: the independent 3/4 and square-root size bounds remain open.

The whole stochastic and coefficient theorem is now Lean-checked. The actual finite product probability space is constructed from bit tuples; each tuple has the product of its coordinate success and failure weights. In `Proof/PrimeClassDeletion/BernoulliFiniteLaw.lean`, these weights are proved nonnegative and normalized, and their success-count masses equal the existing Bernoulli coefficients. `List.bernoulliOutcomeTail_eq` and `List.bernoulliOutcomeExpectation_eq` identify its actual tail probabilities and expectations. The finite generating polynomial is identified with the literal prime product by `primeClassBernoulliMass_eq_product_coeff`. The assembled certificate `greedyPrimeClass_local_probability_complete` retains every rank, every increasing test on the actual finite support, both moment ranges, and only the local competing cap $2 \leq t \leq x$.

Separately, `greedyPrimeClass_local_odds_coefficient` proves the displayed coefficient inequality for the literal prime products and arbitrary finite nonnegative decreasing odds. All owners are exported through the prime-deletion assembly.

The whole adjacent ordinary-divisibility and affine application block is also Lean-checked. `greedyPrimeClass_local_divisibility_limits` proves both actual integer-density limits and their comparison, using `ordinaryPrimeTail_density_error`'s explicit $3^m/N$ error. `greedyPrimeClass_local_affine_probability` and `greedyPrimeClass_local_affine_odds_coefficient` retain the distinct closed and open parameter domains above.

The exceptional-coordinate certificate `affinePrimeEndpoint_rank_shift` treats both prime cases. Its shifted and unshifted odds identities retain the full survival factor. The actual greedy endpoint is discharged by `greedyPrimeClass_two` and its endpoint consumers. All these owners are exported and included by exact declaration name in the recurring foundational-axiom audit.

The infinite missing-product comparison is now Lean-checked in `Proof/PrimeClassDeletion/CappedPrimeMissingCost.lean`. The full general consumer `greedyPrimeClass_missing_weight_summable` proves both convergence and the infinite comparison. Its logarithmic specialization `greedyPrimeClass_missing_logEulerCost` has the exact prime filter and class complement in every summand; cutoffs zero and one are included in the finite proof. The complete owner is exported and registered for recurring axiom checks. The signed record-energy identity, actual integral and left limits, and the complete record-to-cutoff power implication are now Lean-checked as described above. The independent signed-energy estimate and the separate zero-response applications remain open obligations. No established originality priority or new zero-free boundary is asserted.

21.50 How a prime-counting wave becomes a persistent deletion signal

The exact regulator permits a more precise statement than a comparison of growth exponents. It describes what happens to the whole wave, including its negative phases. The next theorem is conditional on a specified leading asymptotic for the *entire* prime-counting error; it does not assert that a single zeta zero supplies that asymptotic.

Theorem 21.104 (The deletion envelope of a periodic leading error)**.** *Let $0 < \beta < 1$, let $T > 0$, and let $f : \mathbb{R} \to \mathbb{R}$ be continuous and T-periodic, with*

$f(v_0) > 0$ *for some* v_0*. Suppose, as* $x \to \infty$ *through real cutoffs, that*

$$\theta(x) - x = x^\beta f(\log x) + o(x^\beta). \tag{224}$$

Define

$$\Phi_{\beta,f}(v) = \max\left\{0,\ \sup_{u\geq 0} e^{-\beta u} f(v-u)\right\}. \tag{225}$$

Then $\Phi_{\beta,f}$ *is continuous and* T*-periodic, and*

$$\begin{aligned} R(x) &= x^\beta \Phi_{\beta,f}(\log x) + o(x^\beta), \\ M(x) &= x^\beta \Phi_{\beta,f}(\log x) + o(x^\beta), \\ N_{\mathcal{D}}(x) &= \frac{x^\beta}{\log x}\Phi_{\beta,f}(\log x) + o\left(\frac{x^\beta}{\log x}\right). \end{aligned} \tag{226}$$

In particular

$$0 < f(v_0)e^{-\beta T} \leq \Phi_{\beta,f}(v) \leq \|f\|_\infty \qquad (v \in \mathbb{R}),$$

so $N_{\mathcal{D}}(x) \asymp x^\beta/\log x$ *at* all *sufficiently large cutoffs, including those on a negative phase of* f*. Constants may depend on* β, T, f *and the error in* (224)*, but not on* x*.*

Proof. Between primes the function $\theta(t) - t$ decreases. Consequently its positive running maximum over all $2 \leq t \leq x$ is exactly $R(x)$. For each $\epsilon > 0$ choose Y_ϵ such that the error in (224) is at most ϵt^β for $t \geq Y_\epsilon$. On the remaining bounded initial interval the error is bounded by a constant C_ϵ. The maximum of two functions differs by at most their uniform difference. Replacing $\theta(t) - t$ by $t^\beta f(\log t)$ therefore changes $R(x)$ by at most $\epsilon x^\beta + C_\epsilon$. After division by x^β this is $o(1)$.

Put $v = \log x$ and $u = \log(x/t)$. The model maximum divided by x^β is

$$\max\left\{0,\ \sup_{0\leq u\leq \log(x/2)} e^{-\beta u} f(v-u)\right\}.$$

Extending this domain to all $u \geq 0$ changes the value by at most $\|f\|_\infty (2/x)^\beta$. This proves the first line of (226). The second follows from $0 \leq M(x) - R(x) \leq \log x$.

Partial summation on the *actual deleted primes* gives

$$N_{\mathcal{D}}(x) = \frac{M(x)}{\log x} + \int_2^x \frac{M(t)}{t\log^2 t}\,dt.$$

Here $M(t) = O(t^\beta)$, and the integral is $O(x^\beta/\log^2 x)$. For completeness, split at $\sqrt{x}$: the initial part is $O(x^{\beta/2})$, while the rest is bounded by a constant times $\log^{-2} x \int_{\sqrt{x}}^x t^{\beta-1}\,dt$. This proves the count asymptotic, without identifying a mass estimate with a count estimate.

Continuity follows from the uniform continuity of the periodic function f: the modulus of continuity of every $v \mapsto e^{-\beta u} f(v-u)$ is bounded by that of f, uniformly in $u \geq 0$. Taking a supremum and the positive part preserves this bound. Periodicity is immediate. For each v, choose $u \in [0,T]$ with $v - u \equiv v_0 \pmod T$. The corresponding term is at least $e^{-\beta T} f(v_0) > 0$, which proves the uniform lower bound. □

Theorem 21.105 (Joint almost-periodic waves)**.** *Let $0 < \beta < 1$. The conclusions* (226) *also hold when f is bounded, uniformly continuous and uniformly almost periodic, provided $f(v_0) > 0$ somewhere and* (224) *holds. Here uniformly almost periodic means that for every $\epsilon > 0$ there is $L > 0$ such that every interval of length L contains a real τ with $\sup_v |f(v+\tau) - f(v)| < \epsilon$. The resulting $\Phi_{\beta,f}$ is uniformly almost periodic, continuous and bounded away from zero. In particular this applies to any finite joint wave $f(v) = \sum_{j=1}^m A_j \cos(\tau_j v + \varphi_j)$ that is positive somewhere; no common period of the frequencies is required.*

Proof. The running-maximum and partial-summation parts of the preceding proof use boundedness of f, not its periodicity. They therefore give all three asymptotics once $\Phi_{\beta,f}$ is defined by (225). Uniform continuity again passes to this envelope. Every ϵ-almost period of f is an ϵ-almost period of $\Phi_{\beta,f}$: each translated term $e^{-\beta u} f(v-u)$ changes by at most ϵ, uniformly for $u \geq 0$, and the same is true after taking the supremum and the positive part. Thus the envelope is uniformly almost periodic.

Put $a = f(v_0)/2 > 0$, and choose an almost-period length L for tolerance a. For any v, choose an a-almost period τ in $[v - v_0 - L, v - v_0]$ and put $u = v - v_0 - \tau$. Then $0 \leq u \leq L$ and $f(v-u) = f(v_0 + \tau) > a$. Hence $\Phi_{\beta,f}(v) \geq a e^{-\beta L} > 0$ uniformly in v.

For completeness the finite joint wave satisfies the stated almost-period condition even with incommensurable frequencies. Put $h(t) = (e^{i\tau_1 t}, \dots, e^{i\tau_m t})$. Compactness of the product of the m unit circles supplies, for any $\delta > 0$, a finite δ-net for this orbit with centres $h(t_1), \dots, h(t_q)$ in the orbit itself. Let $c = \max_j t_j$ and $L = \max(1, c - \min_j t_j)$. For any real b, some j satisfies $\max_k |e^{i\tau_k(b+c)} - e^{i\tau_k t_j}| < \delta$. Then $\tau = b + c - t_j \in [b, b+L]$ and $\max_k |e^{i\tau_k \tau} - 1| < \delta$. It follows, uniformly in v, that $|f(v+\tau) - f(v)| \leq \delta \sum_j |A_j|$. Choosing δ sufficiently small proves

the required almost-period condition. Finite joint waves are also bounded and uniformly continuous, so all hypotheses have been checked. □

Theorem 21.106 (Explicit response to one leading harmonic)**.** *Under Theorem 21.104's hypotheses, suppose* $f(v) = A\cos(\tau v + \varphi)$ *with* $A > 0$ *and* $\tau > 0$. *Put* $\alpha = \arctan(\beta/\tau)$, $w = \tau v + \varphi$, *and let* $d(v)$ *be the representative of* $w - \alpha$ *modulo* 2π *in* $[0, 2\pi)$. *Then*

$$\Phi_{\beta,f}(v) = A\max\left\{\cos w,\ \cos\alpha\exp\left(-\frac{\beta}{\tau}d(v)\right)\right\}, \tag{227}$$
$$A\cos\alpha\, e^{-2\pi\beta/\tau} \le \Phi_{\beta,f}(v) \le A.$$

Proof. Put $k = \beta/\tau > 0$, $a = \arctan k$ and $h(r) = e^{kr}\cos(a + r)$. Then $\cos a > 0$, $\sin a = k\cos a$, and

$$h'(r) = -e^{kr}\cos a\,(1 + k^2)\sin r.$$

Consequently h decreases on $[0, \pi]$ and increases on $[\pi, 2\pi]$. For $0 \le r \le d < 2\pi$ this gives $h(r) \le \max\{\cos a, h(d)\}$.

The entire earlier-wave tail satisfies the same bound. For $t \ge 0$, the elementary inequalities $\sin t \le t$, $\cos t \le 1$ and $1 + kt \le e^{kt}$ give

$$\cos(a - t) = \cos a\,(\cos t + k\sin t) \le \cos a\,(1 + kt) \le \cos a\, e^{kt}.$$

Taking $t = -r$ proves $h(r) \le \cos a$ for every $r \le 0$. Thus

$$h(r) \le \max\{\cos a, h(d)\} \qquad (r \le d).$$

The representative is implemented explicitly as

$$d = w - a - 2\pi\left\lfloor\frac{w - a}{2\pi}\right\rfloor, \qquad 0 \le d < 2\pi.$$

For each $u \ge 0$, put $r = d - \tau u \le d$. Periodicity and $k\tau = \beta$ give

$$e^{-\beta u}\cos(w - \tau u) = e^{-kd}h(r) \le \max\{\cos w, \cos a\, e^{-kd}\}.$$

Both candidate values are attained: use $u = 0$ and $u = d/\tau$, respectively. The second is strictly positive, so adjoining zero to the supremum changes nothing. Multiplication by A proves the exact formula. The lower bound follows from $d < 2\pi$ and $\cos a > 0$; the upper bound follows from $e^{-\beta u} \le 1$ and $\cos \le 1$. Continuity at the modular endpoint also follows from the preceding theorem. □

This is a precise memory effect of the *same* selected class: a negative phase of the original error cannot undo primes already deleted. It is not an independent upper bound on that class's size. For actual zeta applications, a putative zero alone does not justify (224); other frequencies and their remainders must first be controlled. No assumption of an attained largest zero real part, a unique leading zero pair, or an RH-dependent Fourier limit is used or inferred here. The certified unconditional boundary remains 7/8.

In the prime-deletion assembly, `GreedyRegulator.lean` contains the finite regulator core, and `GreedyPrimeRegulator.lean` identifies its clock with the actual greedy prime class. Theorem 21.104, including all three real-cutoff asymptotics, is now proved by the maintained declaration `greedyPrime_periodic_wave` in `PrimeWavePeriodic.lean`. Its proof retains the full leading-error hypothesis (224); it does not infer that hypothesis from the existence of a zero. The formal theorem permits every $\beta > 0$, so it covers the stated range $0 < \beta < 1$.

The maintained proof includes the literal positive-supremum formula, the finite sup norm, continuity and periodicity of the envelope, and the lower bound $f(v_0)e^{-\beta T}$. `PrimeWaveRecordComparison.lean` proves the uniform finite-prefix comparison with its initial-range and logarithmic overshoot terms; `PrimeWaveLimits.lean` proves the actual regulator and deleted-mass limits. `PrimeClassAbel.lean` proves the exact partial-summation formula for every actual prime class, with the first prime 2 retained. `PrimeClassAbelBigO.lean` proves the displayed $O(x^\beta/\log^2 x)$ remainder from a uniform class-mass power bound. `PrimeWaveCount.lean` then proves the actual deleted-count asymptotic. The companion declaration `greedyPrime_periodic_wave_count_bounds` gives, for every sufficiently large real x, the explicit comparison constants $f(v_0)e^{-\beta T}/2$ and $\|f\|_\infty + 1$.

Theorem 21.105 is now proved in `PrimeWaveJoint.lean`. The declaration `greedyPrime_almost_periodic_wave` contains the complete response for bounded uniformly continuous almost-periodic profiles. The declaration `greedyPrime_finite_joint_wave` discharges those hypotheses for every finite trigonometric wave with arbitrary real amplitudes, frequencies and phases. Both conclusions retain all three real-cutoff limits, uniform continuity and almost periodicity of the envelope, its finite sup norm and a strictly positive uniform lower bound, and count bounds at every sufficiently large real cutoff.

`FiniteUnitCharacterRecurrence.lean` proves the relatively dense simultaneous returns by compactness of the finite phase orbit. `FiniteJointWave.lean` proves boundedness, uniform continuity and almost periodicity without a common-period assumption.

`UniformWaveSupremum.lean` identifies the pointwise epsilon convention with the strict full-line supremum convention stated here. The envelope contraction, almost-period inheritance and positive recurrence bound are in `DampedWaveRecurrence.lean`. All these declarations are exported and included by exact name in the recurring foundational-axiom audit. Theorem 21.106 is now proved in `CosineEnvelope.lean` by `dampedWaveEnvelope_cosine` and `dampedWaveEnvelope_cosine_bounds`. The literal floor phase and its full range are proved in `DiscountedCosinePhase.lean`. `GrowingCosineKernel.lean` proves the exact derivative, both monotonicity intervals and the entire historical-tail estimate; the supremum proof exhibits both attaining lookbacks. These owners are exported and every named proof is registered for the recurring foundational-axiom audit. The independent zero-size estimates and zero-response applications remain separate obligations.

21.51 An infinitely-often size criterion from positive Gaussian probes

The leading-wave hypothesis in the preceding subsection can be removed for the purpose of detecting an off-critical zero in the deletion count. The class $\mathcal{S}$, its complement $\mathcal{D}$, and the functions $R, M, N_{\mathcal{D}}$ are exactly the same fixed objects as above. They do not depend on a hypothetical zero or on any probe parameter.

Theorem 21.107 (A forbidden zero forces deletions at every large cutoff)**.** *Suppose $\zeta(\rho_0) = 0$, where $\rho_0 = \beta_0 + i\gamma_0$ is nontrivial and $\beta_0 > 1/2$. For every $\epsilon > 0$ there are constants $d_\epsilon > 0$ and X_ϵ such that*

$$N_{\mathcal{D}}(X) \geq d_\epsilon X^{\beta_0 - \epsilon} \qquad (X \geq X_\epsilon). \tag{228}$$

The constants may depend on the chosen zero and ϵ, but not on the real cutoff X. No largest zero real part, single leading frequency or RH hypothesis is assumed.

We first establish the exact smoothed identity and its errors. Write $E_\theta(t) = \theta(t) - t$ and $E_\psi(t) = \psi(t) - t$, with $\theta(t) = \psi(t) = 0$ for $0 < t < 2$. For $X > 1$, $V > 0$ and real τ, put

$$G_V(u) = \frac{e^{-u^2/(2V)}}{\sqrt{2\pi V}}, \qquad H_\tau^\theta(X, V) = \int_{\mathbb{R}} E_\theta(Xe^u) G_V(u) e^{-i\tau u}\, du, \tag{229}$$

and define H_τ^ψ in the same way. These integrals converge absolutely: the elementary bound $\theta(t), \psi(t) = O(t \log(2t))$ for $t \geq 2$ suffices.

Lemma 21.108 (Signed Gaussian explicit formula with the prime powers retained)**.** *Fix $0 < c < 1$ and a real τ, and put $V = c \log X$. Let $\mathcal{Z}$ be the set of distinct nontrivial zeta zeros, and let m_ρ denote their multiplicities. For $V \geq 1$,*

$$\begin{aligned} H_\tau^\psi(X, V) = &- \sum_{\rho \in \mathcal{Z}} \frac{m_\rho}{\rho} X^\rho \exp\left(\frac{V}{2}(\rho - i\tau)^2\right) \\ &- \log(2\pi) e^{-V\tau^2/2} + O_\tau(X^{-1+c/2}). \end{aligned} \tag{230}$$

The zero sum is absolutely convergent. Uniformly in real τ, the complete prime-power correction satisfies

$$\begin{aligned} |H_\tau^\psi - H_\tau^\theta| &\leq 2X^{1/2} e^{V/8} \left(\log X + \frac{V}{2} + \sqrt{\frac{2V}{\pi}}\right) \\ &= O_c\left(X^{1/2+c/8} \log X\right). \end{aligned} \tag{231}$$

Proof. We use the classical unconditional meromorphic continuation, functional equation and zero-count estimates for ζ. In particular $0 < \Re\rho < 1$ for $\rho \in \mathcal{Z}$, the zero set is locally finite, $N(T) = O(T \log(2 + T))$, and $\zeta'/\zeta(-1 + it) = O(\log(2 + |t|))$. The last estimate follows directly from the functional equation and absolute convergence of ζ'/ζ on $\Re s = 2$. None of these facts assumes a zero-free substrip of the critical strip.

Gaussian-smoothed Perron inversion on $\Re s = 2$ gives

$$\int_{\mathbb{R}} \psi(Xe^u) G_V(u) e^{-i\tau u}\, du = \frac{1}{2\pi i} \int_{(2)} -\frac{\zeta'(s)}{\zeta(s)} \frac{X^s}{s} \exp\left(\frac{V}{2}(s - i\tau)^2\right) ds.$$

For example apply Mellin inversion to $\int_{\log(n/X)}^\infty G_V(u) e^{-i\tau u}\, du$ and then sum against $\Lambda(n)$. The Dirichlet series is absolutely convergent on this line, and the vertical Gaussian gives absolute convergence of the resulting contour integral. The Gaussian moment is exactly $\int G_V(u) e^{(s-i\tau)u}\, du = \exp(V(s - i\tau)^2/2)$.

Move the line only to $\Re s = -1$. On horizontal contours tending to infinity through heights avoiding zeros, the usual logarithmic-derivative estimate is $O(\log^2(2 + |t|))$ on the bounded strip $-1 \leq \Re s \leq 2$. Such heights exist by the local zero-count bound; the factor $e^{-V(t-\tau)^2/2}$ makes their integrals vanish. The pole at $s = 1$ contributes $X \exp(V(1 - i\tau)^2/2)$. This cancels exactly the Gaussian integral of Xe^u in E_ψ. A zero ρ contributes the signed residue $-m_\rho X^\rho \exp(V(\rho - i\tau)^2/2)/\rho$. At $s = 0$ the residue is

$-(\zeta'/\zeta)(0)e^{-V\tau^2/2} = -\log(2\pi)e^{-V\tau^2/2}$. There is no trivial zero in this contour strip.

On the new line the modulus of the exponential factor is $e^{V/2}e^{-V(t-\tau)^2/2}$. For $V \geq 1$ the remaining vertical integral, after extracting $X^{-1}e^{V/2}$, is bounded by a constant depending on τ. This proves the remainder in (230). The zero-count bound and Gaussian decay in $\Im\rho$ prove absolute convergence of its zero sum. It would be invalid to continue this shift to $-\infty$ and sum all trivial zeros: the Gaussian grows quadratically in the real direction.

Finally $Q(t) = \psi(t) - \theta(t)$ is nonnegative and satisfies the existing bound $Q(t) \leq 2\sqrt{t}\log t$ for $t \geq 2$. Below the first prime power 4, $Q = 0$, so there is no omitted small-input correction. The same bound is the input used in the maintained Nicolas prime-power-tail estimates. Since $|e^{-i\tau u}| = 1$, its complete Gaussian contribution is at most

$$2X^{1/2}\int_{\mathbb{R}} e^{u/2}(\log X + |u|)G_V(u)\,du.$$

Use $e^{u/2}G_V(u) = e^{V/8}G_V(u - V/2)$ and $\int |u|G_V(u)\,du = \sqrt{2V/\pi}$. This gives (231). Only this explicitly separated prime-power remainder has been estimated in modulus; the full signed zero sum remains in (230). □

Lean formalization status. Lemma 21.108 is formalized in full. Its maintained statement retains the actual physical probes, distinct nontrivial zeta zeros with their analytic multiplicities, absolute convergence, the explicit uniform remainder and the complete prime-power correction. Lemma 21.109 is also formalized in full. Theorem 21.107 is also formalized in full. The exact analytic proof chain and its complete counting consumer are described below.

For $X > 0$, $V > 0$ and every real frequency τ, put

$$T_{X,V,\tau}(s) = \frac{X^s \exp\bigl(V(s - i\tau)^2/2\bigr)}{s}.$$

`LogGaussianMellinKernel.lean` proves its exact vertical norm. For every $\sigma > 0$, `LogGaussianIntegrability.lean` proves integrability on that vertical line, and `LogGaussianMajorant.lean` proves

$$|T_{X,V,\tau}(\sigma + it)| \leq \frac{C_{X,V,\tau,\sigma}}{|\sigma + it|^2},$$
$$C_{X,V,\tau,\sigma} = X^\sigma e^{V\sigma^2/2}\left(|\sigma| + |\tau| + 1 + \frac{1}{V/2}\right).$$

The reciprocal s factor, variance and shifted frequency are retained throughout. The constant is explicit and the norm denominator is strictly positive on the stated domain.

`XiZeroStrip.lean` proves $0 < \Re\rho < 1$ for actual xi zeros without RH. `LogGaussianXiInterchange.lean` consumes the existing upper-strip interchange theorem on those actual zeros, with the actual Gaussian kernel and its proved majorant. It retains every canonical divisor-zero index, so multiplicities are included. Absolute integrated summability and the complete sum–integral interchange are proved.

`LogGaussianSignedIdentity.lean` proves integrability and the full signed decomposition on $\Re s = 3/2$, with the constant, paired xi-zero series, reference pole and digamma term all retained. Its declaration is `logGaussian_safeLine_signed_identity`. The physical Perron identification is now also proved. For every $X, V > 0$, every real τ and every $\sigma > 1$, `LogGaussianPhysicalProbe.lean` establishes

$$H_\tau^\psi(X,V) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{-\zeta'(\sigma+it)}{\zeta(\sigma+it)} T_{X,V,\tau}(\sigma+it)\,dt - X\exp\left(\frac{V}{2}(1-i\tau)^2\right).$$

Here the left side is the actual integral of $\psi(Xe^u) - Xe^u$ against $G_V(u)e^{-i\tau u}$. Absolute integrability is proved for this physical integrand. The proof includes the normalized complex Gaussian moment, exact inversion on every real vertical line, the cumulative Gaussian tail, every von Mangoldt threshold, absolute arithmetic summability, and absolute sum–integral interchange. No additional RH or zero-location hypothesis is used.

At $\sigma = 3/2$, the same maintained owner proves the full signed identity for this physical probe, retaining the constant, every multiplicity-counted paired xi term, the reference pole, the gamma contribution and the explicitly evaluated Gaussian reference term. Its declaration is `logGaussianPsiProbe_eq_full_signed_xi_identity`. These proofs are exported and registered by exact name.

The full physical residue expansion and its remainder are now also proved in `LogGaussianExplicitFormula.lean`. For every $X > 0$, $V \geq 1$ and real τ, let ρ_p range over the canonical xi divisor-zero indices, which retain analytic

multiplicity, and put

$$B_V = \frac{e^{V/2}}{\sqrt{2\pi V}}, \qquad S_\xi = \sum_p |\rho_p|^{-2},$$
$$S_\Gamma = \sum_{k=0}^{\infty} \frac{1}{2(k+1)(2(k+1)-1)},$$
$$K_\tau = (5+|\tau|)S_\xi + |\log(2\pi)| + \frac{1}{2} + S_\Gamma.$$

Both series defining the constants converge. The maintained declarations prove the actual physical identity and uniform remainder bound

$$H_\tau^\psi(X,V) = -\sum_p \frac{X^{\rho_p}}{\rho_p} \exp\left(\frac{V}{2}(\rho_p - i\tau)^2\right) - \log(2\pi)e^{-V\tau^2/2} + R_\tau(X,V),$$
$$|R_\tau(X,V)| \le K_\tau \frac{B_V}{X}.$$

The full residue series is absolutely convergent. The declarations are

`logGaussianPsiProbe_eq_explicit_formula`,
`logGaussianExplicitRemainder_norm_bound`,
`logGaussianExplicitRemainder_log_scale_bound`.

The last declaration proves $|R_\tau(X, c\log X)| \le K_\tau X^{-1+c/2}$ whenever $X > 0$ and $c\log X \ge 1$.

The proof starts from the complete signed safe-line identity. For each zero, it retains the full Gaussian moment and separates only the omitted physical left tail E_{ρ_p}. Integration by parts gives the independently summable estimate

$$|E_{\rho_p}| \le (5+|\tau|)\frac{B_V}{X|\rho_p|^2}.$$

The actual zero sum, its full left-tail error, the constant and reference pole, and every negative-even gamma atom remain coupled until the exact identity is assembled. The complete gamma remainder is bounded by $B_V S_\Gamma/X$; this uses the physical tails of all gamma atoms, rather than a divergent full Gaussian residue sum at the trivial zeros. The constant and reference-pole tails contribute the remaining $(|\log(2\pi)| + 1/2)B_V/X$. No principal zero contribution is estimated away.

The distinct-zero formula is proved in `LogGaussianZetaExplicit.lean`. It reuses the existing `XiZetaMultiplicity.lean` bridge, which identifies the

xi and zeta analytic orders unconditionally on the positive half-plane away from one. Exact finite-fiber regrouping gives the coefficient m_ρ/ρ at each distinct nontrivial zeta zero. The complete assembled declaration is

`logGaussian_signed_zeta_explicit_with_prime_powers`.

It proves the signed formula, absolute convergence, explicit $K_\tau X^{-1+c/2}$ remainder and the first inequality in (231) together, for all $X > 1$, $c > 0$, $c \log X \geq 1$ and real τ. This includes the lemma's full $0 < c < 1$ domain.

The prime-power proof uses the exact normalized Gaussian mass and absolute first moment, the shift $u \mapsto u - V/2$, and the full $\psi - \theta$ contribution, including small inputs. `logGaussian_prime_power_uniform_log_scale` additionally proves the displayed O_c estimate with the explicit positive constant

$$C_c = 2\left(2 + \frac{c}{2} + \frac{2c}{\pi}\right)$$

for every real τ and every $X \geq e$. All these declarations are exported and registered by exact name. The every-large-cutoff zero response is now formalized as well. The full proof and its exact maintained declaration are described below. The certified unconditional zero-free boundary remains 7/8.

Lemma 21.109 (A Gaussian shift exposes a strictly dominant zero)**.** *For every fixed $c > 0$ define*

$$A_c(\rho, \tau) = \beta + \frac{c}{2}\big(\beta^2 - (\gamma - \tau)^2\big), \qquad B_c(\tau) = \max_{\rho \in \mathcal{Z}} A_c(\rho, \tau), \quad \rho = \beta + i\gamma. \tag{232}$$

If a zero ρ_0 has $\beta_0 > 1/2$, one can choose $\tau > 0$ such that the maximum at τ is attained at a unique distinct zero ρ_* and

$$B_c(\tau) > \max\left\{B_c(0),\ \frac{1}{2} + \frac{c}{8},\ \beta_0\right\}. \tag{233}$$

The maximum is over distinct zeros; multiplicity at ρ_* is retained as the nonzero coefficient m_{ρ_*}/ρ_*.

Proof. The maximum exists: uniformly for τ in a compact interval,

$$A_c(\rho, \tau) \leq 1 + \frac{c}{2} - \frac{c}{2}(\gamma - \tau)^2 \longrightarrow -\infty$$

as $|\gamma| \to \infty$, and there are only finitely many zeros in any bounded part of the critical strip. There are no real nontrivial zeros; conjugation permits us to take $\gamma_0 > 0$.

Put $b = \beta_0 + c\beta_0^2/2$. Then $b > \max(\beta_0, 1/2 + c/8)$. If $B_c(0) < b$, the choice $\tau = \gamma_0$ gives $B_c(\tau) \geq b > B_c(0)$ and the required strict inequalities. If $B_c(0) \geq b$, choose a maximizer at 0 with $\gamma > 0$, using its conjugate if necessary. For every $0 < \tau < 2\gamma$,

$$A_c(\rho, \tau) - A_c(\rho, 0) = \frac{c}{2}\,\tau(2\gamma - \tau) > 0.$$

Thus in this case too there is a positive τ with all the strict inequalities in (233).

It remains to remove ties without changing these inequalities. Locally the maximum is over finitely many candidate zeros, by the uniform tail bound above. For two different heights, the difference of their exponents is an affine function of τ with nonzero slope $c(\gamma - \gamma')$. At the same height the exponent strictly increases with β, since the difference for $\beta' < \beta$ is $(\beta - \beta')(1 + c(\beta + \beta')/2) > 0$. Consequently only finitely many crossing parameters need be avoided in a sufficiently small interval about the chosen τ. Perturbing within that interval gives a unique maximizing zero, while continuity of the finite maximum preserves every strict gap. □

Lean formalization status: complete. `GaussianExposure.lean` proves

`gaussianZetaEnvelope_unique_exposure`.

For every $c > 0$ and actual nontrivial zero with $\beta_0 > 1/2$, its conclusion gives a positive shift, an attained unique maximum over the entire distinct-zero set, every strict inequality in (233), and the nonzero analytic multiplicity coefficient m_{ρ_*}/ρ_*. The proof reuses the existing real-axis nonvanishing and compact zero-set results, and retains conjugation and analytic multiplicity.

`GaussianFiniteEnvelope.lean` proves the scalar height cutoff for $0 \leq \beta \leq 1$: if $A_c(\beta + i\gamma, \tau) \geq L$, then

$$|\gamma| \leq |\tau| + 1 + \frac{2(1 + c/2 + |L|)}{c}.$$

It proves uniform finiteness of the actual candidate set on bounded shift intervals, and attainment of the actual supremum. `GaussianShiftGain.lean` proves the simultaneous strict gain at a positive shift.

`GaussianCrossings.lean` proves that unequal heights can tie only at their explicit affine crossing, while equal-height scores are injective in the real part. A small open interval preserving the strict gain contains a shift outside the finite crossing set. All these real proofs are maintained, exported and registered by exact name. The every-large-cutoff response theorem below is now formalized with its complete real-cutoff and deleted-count conclusion.

Proof of Theorem 21.107. It suffices to treat $0 < \epsilon < 1$. Choose $0 < a < 1/4$ so small that $\kappa = a^2 + 2a < \epsilon/2$, and set $c = a^2$. Fix τ and the unique ρ_* given by the exposure lemma, and abbreviate $B = B_c(\tau)$ and $q = 1/2 + c/8$. The uniqueness supplies a positive gap between B and every other zero exponent: only finitely many zeros have exponent near B.

For $V = c \log X$, the signed term at ρ_* in (230) has modulus $(m_{\rho_*}/|\rho_*|)X^B$. The sum of all other zero terms is $O(X^{B-\delta})$ for some $\delta > 0$. To justify this also for the infinite tail, fix $X_1 = e^{1/c}$ and use the summable majorant $\sum_\rho (m_\rho/|\rho|) X_1^{A_c(\rho,\tau)-B+\delta}$. The prime-power error is $O(X^q \log X)$, strictly smaller since $q < B$. The $s = 0$ term and the left contour remainder are smaller as well. It follows that, for all sufficiently large X,

$$|H_\tau^\theta(X, V)| \geq dX^B, \qquad H_0^\theta(X, V) = o(X^B)$$

for some $d > 0$. The second estimate uses the full unshifted zero sum and the strict gap $B_c(0) < B$; its prime-power error is again retained. The oscillation of the leading residue cannot cancel its modulus, because there is a unique dominant distinct zero.

Now retain the *coupled signed* expression

$$\begin{aligned} P_\phi(X) &= H_0^\theta(X, V) + \Re\big(e^{i\phi} H_\tau^\theta(X, V)\big) \\ &= \int_{\mathbb{R}} E_\theta(Xe^u) G_V(u) \big(1 + \cos(\tau u - \phi)\big)\, du. \end{aligned} \tag{234}$$

Its kernel is nonnegative for every phase. Choosing the phase to maximize the first line gives, eventually,

$$\max_\phi P_\phi(X) = H_0^\theta(X, V) + |H_\tau^\theta(X, V)| \geq \frac{d}{2} X^B.$$

In particular the unshifted term has not been discarded. It is lower order by the independent exponent gap, rather than by a main-term triangle or Gram estimate.

Put $Y = X^{1+\kappa}$. For $u \leq \kappa \log X$ one has $E_\theta(Xe^u) \leq R(Y)$. This includes $Xe^u < 2$, where $E_\theta(Xe^u) = -Xe^u < 0$. The kernel in (234) is at most $2G_V$. For the remaining tail use only the elementary $\theta(t) \leq t \log t$ for $t \geq 2$. Writing $L = \log X$, $z = (\kappa - c)L/\sqrt{V}$ and $\overline{\Phi}(z) = (2\pi)^{-1/2} \int_z^\infty e^{-v^2/2}\, dv$, one has

$$\begin{aligned} \int_{\kappa L}^\infty e^u (L + u) G_V(u)\, du &= e^{V/2} \left((L + V)\overline{\Phi}(z) + \sqrt{\frac{V}{2\pi}} e^{-z^2/2} \right) \\ &\ll_c L X^{c/2-2}. \end{aligned} \tag{235}$$

Indeed $\kappa - c = 2a$ and $V = a^2 L$, so $z = 2\sqrt{L}$; use $\overline{\Phi}(z) \le e^{-z^2/2}$. Consequently, uniformly in the phase,

$$P_\phi(X) \le 2R(Y) + O_c(X^{-1+c/2} \log X).$$

Combining this with the positive lower bound proves $R(Y) \ge d' X^B$ for every sufficiently large X.

The substitution $Y = X^{1+\kappa}$ covers every sufficiently large real Y, not just a subsequence. Since $B > \beta_0$ and $\beta_0 < 1$,

$$\frac{B}{1+\kappa} > \frac{\beta_0}{1+\kappa} \ge \beta_0 - \kappa > \beta_0 - \epsilon/2.$$

Every denominator is positive. Finally $N_{\mathcal{D}}(Y) \log Y \ge M(Y) \ge R(Y)$. Absorbing this single logarithm into the remaining $\epsilon/2$ margin proves (228). All probe parameters were fixed before allowing X to grow. □

Lean formalization of the full theorem. The maintained declaration

`greedyDeletedPrimeClass_uniform_zero_response`

proves Theorem 21.107 with its full quantifiers: every actual nontrivial zero with real part greater than $1/2$, every $\epsilon > 0$, and every real cutoff beyond one fixed threshold. Its conclusion uses the literal cardinality of the deleted prime prefix at $\lfloor X \rfloor$. No RH, largest-real-part, single-frequency or assumed oscillation-asymptotic hypothesis is added.

`GaussianSpectralGap.lean` proves a positive gap from the uniquely exposed zero to the entire remaining zero set. `GaussianSpectralWeight.lean` identifies its exact residue norm as $(m_\rho/|\rho|)X^{A_c(\rho,\tau)}$. `GaussianSpectralMajorant.lean` controls every subset of the remaining zeros by its absolutely summable series at $X_1 = e^{1/c}$. The exact signed split and the resulting full infinite-series lower bound are proved in `GaussianSpectralDominance.lean`.

The actual theta identity in `GaussianThetaBounds.lean` retains the reference term, physical remainder and complete prime powers. With $b = m_{\rho_*}/|\rho_*|$, its signed consumer `GaussianSignedProbeBound.lean` proves

$$\begin{aligned}\Re H_0^\theta + |H_\tau^\theta| \ge\ & bX^B - C_1 X^{B-\delta} - C_0 X^{B_c(0)} - 2|\log(2\pi)| \\ & - (K_\tau + K_0) X^{-1+c/2} - 4(2 + c/2 + 2c/\pi) X^{1/2+c/8} \log X.\end{aligned}$$

`GaussianPositiveSignal.lean` absorbs all five allowances using their strict exponent gaps and proves a lower bound $(b/2)X^B$ at every sufficiently large real scale.

`GaussianPositiveKernel.lean` constructs a unit complex rotation realizing the shifted probe's full norm, proves the exact weighted-integral identity, and proves $0 \le K(u) \le 2G_V(u)$. For the full complementary tail, the formal proof uses the real tilt

$$e^{tu}G_V(u) = e^{Vt^2/2}G_V(u - Vt)$$

and the existing shifted absolute moment. `GaussianChernoffTail.lean` proves the supported exponential tail inequality, and `GaussianLocalizationTail.lean` specializes it with $\lambda = 2/a$. Including the physical factor $2X$, the complete tail $T_a(X)$ satisfies

$$T_a(X) \le 2(2 + a^2 + 2a + 2a^2/\pi)X^{-1+a^2/2}\log X \qquad (X \ge e).$$

This bounds the same tail directly, without requiring the normal-tail identity as a formal input.

`GaussianRecordLocalization.lean` reuses the existing unconditional prime number theorem for the large-input bound. It retains the native record allowance and proves

$$\Re H_0^\theta + |H_\tau^\theta| \le 2\big(R(Y) + \log Y\big) + T_a(X), \qquad Y = X^{1+a^2+2a}.$$

The extra logarithm comes from the already formalized record comparison and is retained until it is absorbed. `GaussianRegulatorLower.lean` then proves, for every sufficiently large real X,

$$R\Big(X^{1+a^2+2a}\Big) \ge \frac{b}{8}X^B.$$

`GaussianRealCutoffTransfer.lean` proves that the power substitution covers every sufficiently large real cutoff and applies the actual deleted-mass and deleted-count comparisons. Finally `GaussianUniformZeroResponse.lean` selects the localizer and exposure and proves the complete theorem, with $d_\epsilon = b/8 > 0$ for those fixed choices. All thirty-four new proof declarations are maintained, exported and registered by exact name. This closes the every-cutoff response formalization; it supplies no improved zero-free boundary. The certified unconditional boundary remains 7/8.

Theorem 21.110 (Infinitely-often deletion-size criterion)**.** *For every fixed* $1/2 \le \eta < 1$*, the following are equivalent:*

$$\Re\rho \le \eta \qquad (\rho \in \mathcal{Z}),$$
$$\liminf_{\substack{n\to\infty \\ n\in\mathbb{N},\ n\ge 2}} \frac{\log(1 + N_{\mathcal{D}}(n))}{\log n} \le \eta. \tag{236}$$

Equivalently the second condition says that for every $\epsilon > 0$ and every integer N there exists an integer $n \geq \max(N, 2)$ with $N_{\mathcal{D}}(n) \leq n^{\eta+\epsilon}$. In particular $\eta = 1/2$ gives an RH equivalence for this fixed computable prime class. If RH is false and Θ_ζ is the supremum of the nontrivial zero real parts, then the counting exponent has an actual limit:

$$\lim_{\substack{n\to\infty \\ n\in\mathbb{N},\ n\geq 2}} \frac{\log(1 + N_{\mathcal{D}}(n))}{\log n} = \Theta_\zeta.$$

Proof. The zero-free condition and Theorem 21.101 give the stronger all-cutoff upper bound $N_{\mathcal{D}}(X) \ll_\epsilon X^{\eta+\epsilon}$. It implies the stated lower-limit condition and the equivalent infinitely-often formulation, absorbing its constant into $\epsilon/2$ of exponent margin. Conversely a zero with $\beta_0 > \eta$ has $\beta_0 > 1/2$. The uniform lower bound (228), for every positive ϵ, implies that the displayed lower limit is at least $\beta_0 > \eta$, a contradiction. The elementary equivalence between the lower-limit formulation and the integer subsequence formulation follows by taking logarithms; the extra 1 changes the quotient by $o(1)$ on any such subsequence.

If RH is false, $\Theta_\zeta > 1/2$. Taking the supremum over all off-critical zeros in the uniform lower bound gives a lower limit at least Θ_ζ. For $\Theta_\zeta < 1$ the existing zero-strip-to-count estimate gives an upper limit at most Θ_ζ. If $\Theta_\zeta = 1$, the elementary $N_{\mathcal{D}}(n) \leq n$ gives that same upper limit. This proves the last assertion. □

Maintained Lean proofs of the cofinal and logarithmic RH criteria. For $\eta = 1/2$, the cofinal integer-count formulation has a complete direct proof in `GreedyPrimeRHCriterion.lean`. The maintained theorem `riemannHypothesis_iff_cofinal_greedyDeletedPrimeClass_count` states exactly

$$\mathrm{RH} \quad \Longleftrightarrow \quad \left[\forall \epsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq \max(N, 2), \quad N_{\mathcal{D}}(n) \leq n^{1/2+\epsilon}\right].$$

Its forward implication proves the bound at every sufficiently large integer cutoff. The direct formal proof first establishes constants $A \geq 0$ and $C > 0$ such that, for every integer $n \geq 2$,

$$N_{\mathcal{D}}(n) \log 2 \leq A + C\sqrt{n}\,(\log n)^2 + \log n.$$

The constant A retains the entire initial prime range. Division by $n^{1/2+\epsilon}$ makes each of the three allowances tend to zero; the proof carries all of them through the count comparison. The reverse implication consumes the

unconditional uniform zero-response theorem at the same cofinal integer cutoffs. The supporting maintained owners are `GreedyPrimeRHUpper.lean` and `GreedyPrimeSparseCriterion.lean`.

The maintained owner `GreedyPrimeCountLiminf.lean` proves the literal logarithmic lower-limit reformulation as well. Writing $g_{\mathcal{P}}(n) = \log(1 + N_{\mathcal{P}}(n))/\log n$, its proof first establishes $0 \le g_{\mathcal{P}}(n) \le 2$ for every prime class $\mathcal{P}$ and integer n, with the formal real division convention giving $g_{\mathcal{P}}(0) = g_{\mathcal{P}}(1) = 0$. For $n \ge 2$ and $a \ge 0$, it retains the entire logarithmic allowance:

$$N_{\mathcal{P}}(n) \le n^a \quad \Longrightarrow \quad g_{\mathcal{P}}(n) \le a + \frac{\log 2}{\log n}.$$

Conversely, $g_{\mathcal{P}}(n) < a$ implies $1 + N_{\mathcal{P}}(n) < n^a$ by multiplying by $\log n > 0$ and exponentiating. Given $y > \eta \ge 0$, choose $a = \eta + (y - \eta)/2$. At all sufficiently large cutoffs the displayed allowance is less than $(y - \eta)/2$. The cofinal power bound therefore gives $g_{\mathcal{P}}(n) < y$ arbitrarily far out. The reverse follows by taking $y = \eta + \epsilon$ in the bounded lower-limit characterization. In particular the exact maintained theorem `riemannHypothesis_iff_greedyDeletedPrimeClass_log_count_liminf` states RH if and only if $\liminf_{n\to\infty} g_{\mathcal{D}}(n) \le 1/2$. Both new declarations and the preceding six are registered in the recurring foundational audit.

This verifies both RH formulations, including their exact cofinal transfer for every $\eta \ge 0$. The upper zero-strip implication for general $1/2 \le \eta < 1$ and the final zero-supremum limit in Theorem 21.110 remain separate formalization obligations; the larger labelled claim is not closed by this specialization.

The independent size burden has changed quantifiers: an unbounded sequence of favorable integer cutoffs is sufficient; an estimate at every cutoff is no longer required. No such independent sequence estimate at exponent $3/4$ or $1/2$ is proved here, and no claim that finding it is arithmetically easier is made. The certified unconditional zero-free boundary remains $7/8$. At $\beta_0 = 1/2$ the strict separation from the prime-power exponent fails; no all-scale square-root lower bound under RH is asserted.

Theorem 21.111 (Stabilizing probability-profile rank test)**.** *For an integer $n \geq 2$, form the finite Bernoulli probability profile*

$$P_{\mathcal{S},n}(z) = \prod_{\substack{p \leq n \\ p \in \mathcal{S}}} \left(1 - \frac{1}{p} + \frac{z}{p}\right).$$

Define the computable bit

$$b(n) = \mathbf{1}_{\{\pi(n) - \deg P_{\mathcal{S},n} \leq \sqrt{n}(\log n)^2\}}. \tag{237}$$

This binary sequence stabilizes unconditionally, and its limit is

$$\lim_{n\to\infty} b(n) = \begin{cases} 1, & \text{if RH holds,} \\ 0, & \text{if RH fails.} \end{cases} \tag{238}$$

Consequently, RH is equivalent both to $b(n) = 1$ for infinitely many integers n and to $b(n) = 1$ for all sufficiently large integers n.

Proof. Every factor has positive leading coefficient $1/p$. Thus, also for the empty product,

$$\deg P_{\mathcal{S},n} = \#\{p \leq n : p \in \mathcal{S}\}, \qquad \pi(n) - \deg P_{\mathcal{S},n} = N_{\mathcal{D}}(n).$$

These are finite independent Bernoulli variables with success probabilities $1/p$ at a fixed cutoff. No simultaneous growing-cutoff sampling limit is used.

Under RH, Theorem 21.101 gives $N_{\mathcal{D}}(n) \ll \sqrt{n}\log n$. Its fixed implied constant is absorbed by the additional factor $\log n$, so $b(n) = 1$ eventually. If RH fails, functional reflection gives a zero with $\beta > 1/2$. Put $\delta = (\beta - 1/2)/2 > 0$ and apply Theorem 21.107 with $\varepsilon = \delta$. It gives $N_{\mathcal{D}}(n) \geq cn^{1/2+\delta}$ at every sufficiently large integer n, for some fixed $c > 0$. Since $cn^{\delta} > (\log n)^2$ eventually, $b(n) = 0$ eventually.

For completeness, each finite bit is computable. The greedy membership test already has effective comparisons at integer cutoffs. The threshold $\sqrt{n}(\log n)^2$ cannot be an integer: otherwise $(\log n)^2$ and hence $\log n$ would be algebraic, contradicting Lindemann's theorem since $\log n \neq 0$ and $e^{\log n} = n$ is algebraic. Certified rational intervals for the logarithm therefore eventually separate the threshold from the integer $N_{\mathcal{D}}(n)$. This proves computability of each bit, not an effective unconditional bound on its last change. No finite run of zeros or ones certifies the limiting bit, and the theorem supplies no decision algorithm for RH. □

The nine spectral-exponent identities, strict comparisons and localization-parameter statements in `Proof/PrimeClassDeletion/GaussianRecordExponents.lean` are Lean-checked. They include the exact signed shift, strict dominance over the prime-power exponent, the positive-denominator exponent comparison and the cross-multiplied Gaussian tail identity. The complex contour formula, its infinite zero-sum audit, phase-positive integral argument and the resulting prime-count/RH theorems have full ordinary proofs here. They are not claimed as separately compiled Lean declarations. The required analytic formalization still includes the Gaussian-smoothed Perron contour shift and its zero-sum limit. No established literature priority is asserted.

21.52 A critical-line record floor and a genuine counting-exponent limit

The preceding Gaussian argument separates an off-critical zero from the square-prime term by a strict exponent gap. At the critical line that gap vanishes. A compact window and coherent use of many zero frequencies supply a different mechanism. Throughout this subsection the literal classes $\mathcal{S}, \mathcal{D}$ are unchanged.

Lemma 21.112 (Compact positive window with the square-prime term retained)**.** *For $h > 0$, define*

$$W_h(u) = \frac{1}{h}\left(1 - \frac{|u|}{h}\right)_+, \qquad K_h(u) = e^{-u/2} W_h(u),$$
$$w(y) = \left(\frac{\sin(y/2)}{y/2}\right)^2, \qquad w(0) = 1.$$

Here $r_+ = \max(r, 0)$. Assume RH and count positive zero heights γ with multiplicities m_γ. Then, for this fixed h,

$$\begin{aligned} J_h(X) &= \int_{\mathbb{R}} (\theta(Xe^u) - Xe^u) K_h(u)\, du, \\ \frac{J_h(X)}{\sqrt{X}} &= f_h(\log X) + o_h(1), \\ f_h(v) &= -1 - 2\sum_{\gamma > 0} m_\gamma w(h\gamma) \frac{\frac{1}{2}\cos(\gamma v) + \gamma \sin(\gamma v)}{\frac{1}{4} + \gamma^2}. \end{aligned} \tag{239}$$

The series converges absolutely and uniformly in real v. In particular f_h is uniformly almost periodic. The constant -1 is the full leading square-prime contribution.

Proof. The triangular Fourier pair is

$$\int_{\mathbb{R}} W_h(u)e^{itu}\,du = w(ht).$$

Its entire Mellin multiplier is

$$\Phi_h(s) = \int_{\mathbb{R}} K_h(u)e^{su}\,du = \left(\frac{\sinh((s-\frac{1}{2})h/2)}{(s-\frac{1}{2})h/2}\right)^2,$$

with its removable value at $s = 1/2$. On every fixed vertical strip, $\Phi_h(\sigma + it) = O_h((1+|t|)^{-2})$. The same Perron argument as in Lemma 21.108, now with this compact kernel, gives the unconditional identity

$$\begin{aligned}\int_{\mathbb{R}} (\psi(Xe^u) - Xe^u)K_h(u)\,du = &-\sum_{\rho\in\mathcal{Z}} \frac{m_\rho}{\rho} X^\rho \Phi_h(\rho) \\ &- \log(2\pi)\Phi_h(0) + O_h(X^{-1}).\end{aligned} \tag{240}$$

For clarity, its initial integrand on $\Re s = 2$ is the complete

$$-(\zeta'/\zeta)(s)X^s\Phi_h(s)/s.$$

Shift only to $\Re s = -1$. The pole at 1 cancels the integral of Xe^u, the pole at 0 gives the displayed constant, and no trivial zero is crossed. On zero-avoiding horizontal heights the integrand is $O_{h,X}(\log^2(2+|t|)/(1+|t|)^3)$. On the left vertical line the integral is absolutely convergent and $O_h(X^{-1})$. The same decay and the zero count make the zero sum absolutely convergent. This also justifies the contour limit for the piecewise linear window; differentiability at its corners is not needed.

The prime number theorem, together with $\theta(t) = O(t)$, gives

$$Q(t) := \psi(t) - \theta(t) = \theta(\sqrt{t}) + O(t^{1/3}\log t) = \sqrt{t} + o(\sqrt{t}).$$

Indeed the terms with prime-power exponent at least 3 are at most $O(\log t)$ terms, each $O(t^{1/3})$. Since $|u| \le h$ on the kernel support, this asymptotic is uniform at $t = Xe^u$ as $X \to \infty$, and

$$\int Q(Xe^u)K_h(u)\,du = \sqrt{X}\int W_h(u)\,du + o_h(\sqrt{X}) = \sqrt{X} + o_h(\sqrt{X}).$$

Under RH, $\Phi_h(1/2 + i\gamma) = w(h\gamma)$. Divide (240) by $\sqrt{X}$ and pair conjugate zeros to obtain (239). For fixed h, its coefficients are $O_h(m_\gamma/\gamma^3)$ at large height. Their sum converges by $N(T) = O(T\log(2+T))$. Finite trigonometric sums are uniformly almost periodic by the compact-torus argument used above, and their uniform limit has the same property. No independence of zero ordinates is assumed. □

Lemma 21.113 (Coherent critical frequencies exceed the square-prime bias)**.** *Under RH, the profiles in* (239) *satisfy*

$$f_h(-2h) = \frac{1}{2}\log(1/h) + O(1) \qquad (h \downarrow 0). \tag{241}$$

The implied constant is uniform for sufficiently small $h > 0$. In particular, some fixed h has $f_h(-2h) > 0$.

Proof. Keep the complete signed expression

$$f_h(-2h) = -1 - \sum_{\gamma>0} \frac{m_\gamma w(h\gamma)\cos(2h\gamma)}{\frac{1}{4}+\gamma^2} + 2\sum_{\gamma>0} \frac{m_\gamma w(h\gamma)\gamma\sin(2h\gamma)}{\frac{1}{4}+\gamma^2}.$$

The cosine sum is $O(1)$ uniformly in h, since $0 \le w \le 1$ and $\sum m_\gamma/(1/4+\gamma^2) < \infty$. Write the last sum without its factor 2 as S_h. Replacing $\gamma/(1/4+\gamma^2)$ by $1/\gamma$ changes S_h by $O(h)$: use $|\sin(2h\gamma)| \le 2h\gamma$ and that same convergent reciprocal-square sum.

We give the signed evaluation, rather than an unsigned zero bound. The Riemann–von Mangoldt formula, with multiplicities, is

$$N(t) = \frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + O(\log(2+t)).$$

Stieltjes integration by parts consequently gives

$$S_h = \frac{1}{2\pi}\int_2^\infty w(ht)\frac{\sin(2ht)}{t}\log\frac{t}{2\pi}\,dt + O(1). \tag{242}$$

Here the bounded initial interval contributes $O(h)$. To verify the uniform remainder, set $g(y) = w(y)\sin(2y)/y$ with its removable value at 0. Then

$$\int_0^\infty |g'(y)|(1+\log(2+y))\,dy < \infty, \qquad w(ht)\sin(2ht)/t = hg(ht).$$

Thus the error integral against $O(\log(2+t))$ is $O(h\log(2/h))$, and the endpoint errors are bounded. This proves the claimed uniformity in (242).

Substitute $y = ht$ there. The integral with the additional $\log(y/(2\pi))$ is absolutely convergent, including at 0. It follows that

$$S_h = \frac{\log(1/h)}{2\pi}\,I + O(1), \qquad I = \int_0^\infty w(y)\frac{\sin(2y)}{y}\,dy.$$

This signed integral is positive and has the exact value $\pi/2$. Indeed, if $I(a) = \int_0^\infty w(y)\sin(ay)/y\,dy$ for $a \ge 0$, Fourier inversion for $W_1(u) = (1-|u|)_+$ gives

$$I'(a) = \int_0^\infty w(y)\cos(ay)\,dy = \pi(1-a)_+.$$

Differentiation is justified by integrability of w; $I(0) = 0$ therefore gives $I(2) = \pi/2$. Consequently $2S_h = \frac{1}{2}\log(1/h) + O(1)$. Together with the retained constant -1 and the bounded cosine contribution this proves (241). □

Theorem 21.114 (An unconditional square-root floor for deleted primes)**.** *There are constants $c > 0$ and X_0 such that, for every real $X \geq X_0$,*

$$R(X) \geq c\sqrt{X}, \qquad N_{\mathcal{D}}(X) \geq \frac{c\sqrt{X}}{\log X}. \tag{243}$$

In particular the same fixed computable deleted-prime class $\mathcal{D}$ is infinite unconditionally.

Proof. First assume RH. Choose a fixed small h for which $d = f_h(-2h) > 0$. Uniform almost periodicity supplies $L > 0$ such that every real interval of length L contains a v with $f_h(v) \geq 3d/4$: translate the relatively dense set of $d/4$-almost periods by $-2h$. The error in (239) tends to zero for all $X \to \infty$, so at such sufficiently large v, with $Y = e^v$, one has

$$J_h(Y) \geq (d/2)\sqrt{Y}.$$

Given any sufficiently large real cutoff X, choose this v in $[\log X - h - L, \log X - h]$. Every point Ye^u in the compact kernel support is then at most X. Since $\theta(t) - t \leq R(X)$ for every $0 < t \leq X$, including $t < 2$, and $K_h \geq 0$,

$$J_h(Y) \leq C_h R(X), \qquad C_h = \int_{\mathbb{R}} K_h(u)\,du = \Phi_h(0) > 0.$$

Also $\sqrt{Y} \geq e^{-(h+L)/2}\sqrt{X}$. This proves the first inequality in (243) under RH. The regulator comparison $M(X) \geq R(X)$ and $N_{\mathcal{D}}(X)\log X \geq M(X)$ give its second inequality.

If RH fails, take a zero with real part $\beta > 1/2$ and put $\epsilon = (\beta - 1/2)/2$. Theorem 21.107 gives $N_{\mathcal{D}}(X) \geq d_\epsilon X^{1/2+\epsilon}$ at every sufficiently large cutoff. Now use both sides of the regulator estimate:

$$R(X) \geq M(X) - \log X \geq (\log 2)N_{\mathcal{D}}(X) - \log X \geq \sqrt{X}$$

eventually. The last inequality holds because $\epsilon > 0$. This proves the same first inequality in the other case; the same regulator argument again gives the second. The two cases establish the unconditional assertion. No RH-dependent Fourier identity was used under the bad-zero hypothesis. □

Theorem 21.115 (An exact all-scale exponent and its inverse prime sequence)**.** *Unconditionally,*

$$\lim_{X\to\infty} \frac{\log(1+N_{\mathcal{D}}(X))}{\log X} = \Theta_\zeta. \tag{244}$$

If d_j is the jth prime in $\mathcal{D}$, then

$$\lim_{j\to\infty} \frac{\log d_j}{\log j} = \frac{1}{\Theta_\zeta}, \qquad d_j \ll j^2(\log j)^2. \tag{245}$$

In particular RH is equivalent to $d_j = j^{2+o(1)}$. Under RH there are constants $a, A > 0$ such that, eventually,

$$\frac{aj^2}{(\log j)^2} \le d_j \le Aj^2(\log j)^2. \tag{246}$$

Proof. Under false RH, the exponent limit was proved in Theorem 21.110. Under RH the unconditional lower floor just proved and the upper bound $N_{\mathcal{D}}(X) \ll \sqrt{X}\log X$ from Theorem 21.101 squeeze the exponent to $1/2 = \Theta_\zeta$. These cases prove (244) for all real cutoffs.

The class is infinite by Theorem 21.114. Its prime enumeration is strictly increasing and $N_{\mathcal{D}}(d_j) = j$ exactly. Since $1/2 \le \Theta_\zeta \le 1$, substituting $X = d_j$ into (244) and inverting its positive limit gives the first formula in (245). For its quantitative upper bound, put $X = Aj^2(\log j)^2$ in the unconditional count lower floor. For a sufficiently large fixed A this gives $N_{\mathcal{D}}(X) \ge j$ eventually, hence $d_j \le X$. In particular $\log d_j \le 3\log j$ eventually after increasing the cutoff. Under RH, the count upper bound at d_j then gives $j \le C\sqrt{d_j}\log d_j \le 3C\sqrt{d_j}\log j$, which proves the lower inequality in (246). Finally zero symmetry gives $\Theta_\zeta = 1/2$ if and only if RH, proving the stated inverse-sequence equivalence. □

Theorem 21.116 (Exact record-to-count calibration and logarithmic concentration)**.** *Unconditionally,*

$$N_{\mathcal{D}}(X) \sim \frac{M(X)}{\log X} \sim \frac{R(X)}{\log X} \qquad (X\to\infty). \tag{247}$$

For every fixed $0 \le a < 1$,

$$\frac{N_{\mathcal{D}}(X^a)}{N_{\mathcal{D}}(X)} \longrightarrow 0. \tag{248}$$

Thus if P_n is sampled uniformly from the nonempty finite set $\{p \in \mathcal{D} : p \le n\}$ at a sufficiently large integer cutoff, then $\log P_n/\log n$ tends to 1 in probability. This is a statement about that specified uniform finite sample.

Proof. The denominators are positive for all sufficiently large X, by Theorem 21.114. Fix $0 < a < 1$ and write $\Theta = \Theta_\zeta \geq 1/2$. The actual exponent limit in Theorem 21.115, applied at both X and X^a, gives

$$\frac{N_{\mathcal{D}}(X^a)}{N_{\mathcal{D}}(X)} = X^{-(1-a)\Theta+o(1)} \longrightarrow 0.$$

For $a = 0$ the numerator is zero. Now retain the entire finite weighted mass

$$M(X) = \sum_{\substack{p \leq X \\ p \in \mathcal{D}}} \log p.$$

Its terms from $X^a < p \leq X$ each have weight at least $a \log X$, and every term has weight at most $\log X$. Hence

$$a\left(1 - \frac{N_{\mathcal{D}}(X^a)}{N_{\mathcal{D}}(X)}\right) \leq \frac{M(X)}{N_{\mathcal{D}}(X)\log X} \leq 1.$$

First let $X \to \infty$ for each fixed a, then let $a \uparrow 1$. The middle ratio tends to 1. The regulator sandwich $0 \leq M(X) - R(X) \leq \log X$ and the lower floor $R(X) \geq c\sqrt{X}$ also give $M(X)/R(X) \to 1$. This proves (247). Finally $P_n \leq n$, and for each fixed $0 < a < 1$ the probability that $\log P_n / \log n \leq a$ is exactly $N_{\mathcal{D}}(n^a)/N_{\mathcal{D}}(n)$. Equation (248) proves the asserted convergence in probability. □

These results concern the very same automatically favorable capped class, rather than a new model or a selected subsequence of cutoffs. The compact window retains the square-prime bias and obtains positivity from its signed interaction with all critical zero frequencies. They strengthen the earlier limsup readout for moment-rounded classes: for this greedy class the exponent is a genuine limit, including when RH holds. The Bernoulli profile above consequently has a degree deficit at least $c\sqrt{n}/\log n$ at every sufficiently large integer cutoff.

This is a lower-count theorem. The independent upper-count estimate at exponent $3/4$ or $1/2$ is still unproved, and the certified unconditional zero-free boundary remains $7/8$. All the compact-window contour, zero-count Stieltjes estimate, almost-periodic recurrence and arithmetic conclusions in this subsection have ordinary proofs above. They are not claimed as Lean declarations. Their analytic Lean blockers are the compact-window Perron shift, the Riemann–von Mangoldt Stieltjes asymptotic and the uniform almost-periodic passage for the full zero series. The classical explicit formula and zero-count law are inputs; no established originality priority is asserted.

21.53 Optimal prime-component sampling and its moment spectrum

The fixed deleted class also defines a statistic of ordinary integers, with every prime power retained. Let $\mathcal{H}_{\mathcal{D}}$ be the positive integers all of whose prime factors belong to $\mathcal{D}$, and put

$$A_{\mathcal{D}}(m) = \prod_{\substack{p \in \mathcal{D} \\ p \mid m}} p^{v_p(m)}.$$

The empty product is 1. The sampled variable is the integer $m \in \{1, \dots, N\}$; $\mathcal{D}$ is fixed before sampling, and N is the sample endpoint. This is neither a polynomial input nor a substitution into a two-input family. Every finite $A_{\mathcal{D}}(m)$ is computable by factorization and the already certified greedy membership test.

Write $\Theta = \Theta_\zeta$. The installed unconditional boundary gives $\Theta \le 7/8 < 1$. In particular $\sum_{p \in \mathcal{D}} 1/p < \infty$. The results below use that installed input; they do not improve it.

Lemma 21.117 (Every reciprocal tail has an exact exponent)**.** *For every fixed $s > \Theta$,*

$$T_s(X) := \sum_{\substack{p \in \mathcal{D} \\ p > X}} p^{-s} = X^{\Theta - s + o(1)} \qquad (X \to \infty). \tag{249}$$

The implied asymptotics are for each fixed s; no uniformity as $s \downarrow \Theta$ is asserted.

Proof. By Theorem 21.115, $N_{\mathcal{D}}(X) = X^{\Theta + o(1)}$. For $0 < \epsilon < s - \Theta$, partial summation retains the exact endpoint correction

$$\begin{aligned} T_s(X) &= -X^{-s} N_{\mathcal{D}}(X) + s \int_X^\infty N_{\mathcal{D}}(t) t^{-s-1}\, dt \\ &\ll_{s,\epsilon} X^{\Theta + \epsilon - s}. \end{aligned}$$

The upper constant can contain $(s - \Theta - \epsilon)^{-1}$. For a fixed $b > 1$, logarithmic concentration from Theorem 21.116 gives $N_{\mathcal{D}}(X)/N_{\mathcal{D}}(X^b) \to 0$. The independently positive interval contribution therefore gives

$$T_s(X) \ge \frac{N_{\mathcal{D}}(X^b) - N_{\mathcal{D}}(X)}{X^{bs}} = X^{b(\Theta - s) + o(1)}.$$

Take logarithms, then let $\epsilon \downarrow 0$ in the upper bound and $b \downarrow 1$ in the lower bound. This proves (249). □

Theorem 21.118 (A full prime-power law with a uniform sampling error)**.** *Let*

$$c_{\mathcal{D}} = \prod_{p\in\mathcal{D}}(1-1/p) > 0, \qquad \nu(a) = \frac{c_{\mathcal{D}}}{a} \quad (a \in \mathcal{H}_{\mathcal{D}}).$$

Then ν is a probability law on $\mathcal{H}_{\mathcal{D}}$. Equivalently, its valuations at primes of $\mathcal{D}$ are independent, with probabilities $(1-1/p)p^{-k}$ at $k \geq 0$. Only finitely many of these valuations are nonzero almost surely.

For an integer $N \geq 1$ let

$$\mu_N(a) = \frac{1}{N}\#\{1 \leq m \leq N : A_{\mathcal{D}}(m) = a\}.$$

For every fixed $\Theta < \sigma < 1$, the complete Euler constant

$$B_\sigma = \prod_{p\in\mathcal{D}} \frac{1+p^{-\sigma}}{1-p^{-\sigma}} < \infty \tag{250}$$

gives, with total variation equal to half the sum of absolute point-mass differences,

$$d_{\mathrm{TV}}(\mu_N, \nu) \leq \min\{1, \tfrac{1}{2}B_\sigma N^{\sigma-1}\} \qquad (N \geq 1). \tag{251}$$

The constant does not depend on a prime cutoff or on a statistic subsequently applied to the component.

Proof. The reciprocal prime sum converges by the preceding lemma with $s = 1$. Thus the product defining $c_{\mathcal{D}}$ is positive. The nonnegative Euler expansion, including every prime power, gives

$$\sum_{a\in\mathcal{H}_{\mathcal{D}}} \frac{1}{a} = \prod_{p\in\mathcal{D}}(1-1/p)^{-1} = c_{\mathcal{D}}^{-1}.$$

This proves normalization. For the equivalent product measure, the probability that the valuation at p is nonzero is $1/p$. The summable union bound shows that the probability of any nonzero valuation beyond a prime cutoff tends to zero. Hence the product law is concentrated on finite-support configurations, and its mass at each a is exactly $c_{\mathcal{D}}/a$.

Write μ_{Mob} for the ordinary Möbius function, to distinguish it from μ_N. For each $a \in \mathcal{H}_{\mathcal{D}}$ the full signed finite identity is

$$\mu_N(a) = \sum_{\substack{e\in\mathcal{H}_{\mathcal{D}} \\ e \text{ squarefree} \\ ae\leq N}} \mu_{\mathrm{Mob}}(e)\frac{\lfloor N/(ae)\rfloor}{N}. \tag{252}$$

Indeed $A_{\mathcal{D}}(m) = a$ means $a \mid m$ and m/a has no prime factor in $\mathcal{D}$; inclusion-exclusion on those actual prime divisors gives the formula. There is no coprimality restriction between a and e: a prime dividing a must also be excluded from m/a. Thus all higher valuations and their cross terms remain present. Absolute convergence at exponent 1 also gives

$$\nu(a) = \sum_{\substack{e \in \mathcal{H}_{\mathcal{D}} \\ e \text{ squarefree}}} \frac{\mu_{\mathrm{Mob}}(e)}{ae}.$$

Subtract these identities, giving zero floor terms when $ae > N$. For any integer $q \geq 1$ and $0 < \sigma < 1$,

$$0 \leq \frac{1}{q} - \frac{\lfloor N/q \rfloor}{N} \leq N^{\sigma-1} q^{-\sigma}.$$

For $q \leq N$ use the rounding bound $1/N$; for $q > N$ use $1/q$. These two domains form the entire supported sum.

Only this explicit finite-sampling remainder is now bounded in absolute value. Summing it over the whole state space gives

$$\begin{aligned} \sum_{a \in \mathcal{H}_{\mathcal{D}}} |\mu_N(a) - \nu(a)| &\leq N^{\sigma-1} \sum_{a \in \mathcal{H}_{\mathcal{D}}} \sum_{\substack{e \in \mathcal{H}_{\mathcal{D}} \\ e \text{ squarefree}}} (ae)^{-\sigma} \\ &= N^{\sigma-1} \prod_{p \in \mathcal{D}} \frac{1 + p^{-\sigma}}{1 - p^{-\sigma}}. \end{aligned}$$

The prime-power geometric denominator and the squarefree numerator are both charged in this product. Its convergence follows from $\sum_{p \in \mathcal{D}} p^{-\sigma} < \infty$ and $1 - p^{-\sigma} \geq 1 - 2^{-\sigma} > 0$. Tonelli applies to the displayed nonnegative majorant. Dividing by 2 and using that total variation is at most 1 proves (251). This is a uniform law for the full component; a fixed-cutoff sampling limit has not been substituted for a growing-cutoff estimate. □

Theorem 21.119 (The exact statistical convergence exponent)**.** *Put $\Delta_N = d_{\mathrm{TV}}(\mu_N, \nu)$. If $L(a)$ is the largest prime factor of a, with $L(1) = 1$, let $\Delta_N^{\max}$ be the total variation distance of the two L-laws and let $K_N^{\max}$ be their Kolmogorov distance. Then all three are positive and*

$$\begin{aligned} \lim_{N \to \infty} \frac{\log \Delta_N}{\log N} &= \Theta - 1, \\ \lim_{N \to \infty} \frac{\log \Delta_N^{\max}}{\log N} &= \Theta - 1, \\ \lim_{N \to \infty} \frac{\log K_N^{\max}}{\log N} &= \Theta - 1. \end{aligned} \tag{253}$$

Consequently, for every fixed $1/2 \le \eta < 1$,

$$\begin{aligned} &\Re\rho \le \eta \quad (\rho \in \mathcal{Z}) \\ \iff\ &\Delta_N \ll_\epsilon N^{\eta-1+\epsilon} \quad \textit{for every } \epsilon > 0. \end{aligned} \tag{254}$$

In particular RH is equivalent to the square-root convergence exponent $\Delta_N = N^{-1/2+o(1)}$.

Proof. Under ν the exact tail of the largest deleted prime is

$$q(X) = \nu\{a : L(a) > X\} = 1 - \prod_{\substack{p\in\mathcal{D}\\ p>X}} (1 - 1/p) \sim T_1(X) = X^{\Theta-1+o(1)}.$$

For the comparison, the logarithm of the reciprocal tail product is $T_1(X) + O(T_1(X)/X)$, and $T_1(X) \to 0$. Every prime divisor of $A_{\mathcal{D}}(m)$ is at most N when $m \le N$. The empirical tail above N is therefore exactly zero. At that moving cutoff the limiting tail is $q(N) > 0$, since $\mathcal{D}$ is infinite. The uniform full-law theorem and contraction under deterministic maps consequently give

$$\begin{aligned} &0 < q(N) \le K_N^{\max} \le \Delta_N^{\max} \le \Delta_N, \\ &\Delta_N \le \tfrac{1}{2} B_\sigma N^{\sigma-1} \qquad (\Theta < \sigma < 1). \end{aligned}$$

Take logarithms and then $\sigma \downarrow \Theta$. This proves (253). It also proves both directions of (254). Zero symmetry gives the RH specialization. The phrase square-root exponent allows every positive exponent margin; no bound $O(N^{-1/2})$ is asserted. □

Theorem 21.120 (A positive divisor identity gives the full moment spectrum)**.** *For every fixed real* $t > 0$, *put*

$$U_t(N) = \frac{1}{N} \sum_{m=1}^{N} A_{\mathcal{D}}(m)^t.$$

Then unconditionally

$$\lim_{N\to\infty} \frac{\log U_t(N)}{\log N} = \max\{0, \Theta + t - 1\}. \tag{255}$$

For $0 < t < 1 - \Theta$ *its finite limiting moment is*

$$U_t(\infty) = c_{\mathcal{D}} \prod_{p\in\mathcal{D}} (1 - p^{t-1})^{-1},$$

and the deficit has a strict sign and an exact decay exponent:

$$0 < U_t(\infty) - U_t(N) = N^{\Theta+t-1+o(1)}. \tag{256}$$

The square-root statistic always diverges, with

$$U_{1/2}(N) \gg \log\log N, \qquad \lim_{N\to\infty} \frac{\log U_{1/2}(N)}{\log N} = \Theta - \frac{1}{2}. \tag{257}$$

In particular RH is equivalent to $U_{1/2}(N) \ll_\epsilon N^\epsilon$ for every $\epsilon > 0$; boundedness of this square-root mean would be false even under RH.

Proof. For $d \in \mathcal{H}_\mathcal{D}$ define the positive Jordan weight

$$J_t(1) = 1, \qquad J_t(d) = d^t \prod_{p\mid d}(1 - p^{-t}) \quad (d > 1).$$

On a prime power, $J_t(p^k) = p^{kt} - p^{(k-1)t}$; telescoping and multiplication give $\sum_{d\mid a} J_t(d) = a^t$. Since the $\mathcal{D}$-smooth divisors of m are exactly the divisors of $A_\mathcal{D}(m)$, the complete finite consumer is

$$U_t(N) = \sum_{\substack{d\in\mathcal{H}_\mathcal{D}\\ d\le N}} J_t(d)\frac{\lfloor N/d\rfloor}{N}. \tag{258}$$

Every weight is nonnegative. This identity includes all prime powers; it is not a bound obtained by discarding signed main terms.

For every $r > \Theta + t$ the full positive Euler budget is

$$Z_t(r) := \sum_{d\in\mathcal{H}_\mathcal{D}} J_t(d)d^{-r} = \prod_{p\in\mathcal{D}} \frac{1 - p^{-r}}{1 - p^{t-r}} < \infty.$$

The local identity follows by summing every $k \ge 1$; convergence follows from $r - t > \Theta$. All denominators are positive, since $r > t$.

If $\Theta + t < 1$, $Z_t(1) < \infty$ and dominated convergence in (258) gives $U_t(N) \to Z_t(1) = U_t(\infty)$. Also $U_t(N) \ge 1$, so its exponent is zero. If $\Theta + t \ge 1$, choose any $r > \Theta + t$, hence $r > 1$. The same exact identity gives

$$U_t(N) \le \sum_{\substack{d\in\mathcal{H}_\mathcal{D}\\ d\le N}} \frac{J_t(d)}{d} \le N^{r-1}Z_t(r).$$

Together with $U_t(N) \ge 1$, this proves the upper exponent, including equality $\Theta + t = 1$.

For the remaining lower exponent, fix $0 < a < 1$. There are $N^{\Theta+o(1)}$ deleted primes in $(N^a, N]$ by logarithmic concentration. At each such prime, eventually $J_t(p) \ge p^t/2$ and $\lfloor N/p \rfloor / N \ge 1/(2p)$. Their positive contribution to the same consumer is at least

$$\frac{1}{4} N^{\min\{t-1, a(t-1)\}} \big(N_{\mathcal{D}}(N) - N_{\mathcal{D}}(N^a) \big).$$

Its lower exponent tends to $\Theta + t - 1$ as $a \uparrow 1$. This proves (255).

In the subcritical case $0 < t < 1 - \Theta$, subtraction retains the exact nonnegative rounding and tail deficit

$$U_t(\infty) - U_t(N) = \sum_{d \in \mathcal{H}_{\mathcal{D}}} J_t(d) \left(\frac{1}{d} - \frac{\lfloor N/d \rfloor}{N} \right).$$

Choose $\Theta + t < r < 1$ and use the same two-domain rounding inequality as before. The deficit is at most $N^{r-1} Z_t(r)$. Its terms at deleted primes $p > N$ are at least $\frac{1}{2} T_{1-t}(N)$ eventually. Lemma 21.117 therefore supplies a strictly positive lower bound of exponent $\Theta + t - 1$. Let $r \downarrow \Theta + t$ to obtain (256).

Finally use the unconditional count floor $N_{\mathcal{D}}(x) \ge c\sqrt{x}/\log x$. Partial summation gives

$$\sum_{\substack{p \le N \\ p \in \mathcal{D}}} p^{-1/2} \ge \frac{1}{2} \int_{X_0}^{N} N_{\mathcal{D}}(x) x^{-3/2} \, dx \ge \frac{c}{2} \log\log N - O(1).$$

The prime terms of the positive Jordan consumer at $t = 1/2$ give

$$U_{1/2}(N) \ge \frac{1}{2} \sum_{\substack{p \le N \\ p \in \mathcal{D}}} (p^{-1/2} - p^{-1}).$$

The reciprocal-prime sum is finite, so this proves the asserted logarithmic divergence. Since $\Theta \ge 1/2$, the moment-spectrum formula at $t = 1/2$ gives exactly $\Theta - 1/2$. Reflection again proves the RH equivalence. □

The same uniform law also controls the previously studied probability profiles. If $K_{\mathcal{D}}(m)$ counts distinct deleted prime divisors, its limiting probability generating function is the entire product

$$P_{\mathcal{D}}(z) = \prod_{p \in \mathcal{D}} (1 - 1/p + z/p).$$

Uniform convergence on every compact z-set follows from $\sum_{p\in\mathcal{D}} 1/p < \infty$. For its coefficients b_k and the empirical probabilities $b_{N,k} = N^{-1}\#\{m \le N : K_{\mathcal{D}}(m) = k\}$, contraction of total variation gives the full profile bound

$$\sum_{k\ge 0} |b_{N,k} - b_k| \le B_\sigma N^{\sigma-1}.$$

This is an upper estimate for the count-profile error; its exponent is not claimed to be sharp for that particular statistic. In contrast the full component, largest-prime total variation and largest-prime Kolmogorov error all have the exact exponent above.

The elementary fixed-prime CRT law and the earlier Fermat avoidance density are completed inputs for the sampling interpretation. The present conclusion concerns all valuations in one fixed infinite greedy class, a uniform error for the entire law, and its exact zero-supremum exponent. It differs from the earlier prime-Poisson repair criterion, whose measures use fractional repair weights and whose zero response was a dyadic limsup. The positive Jordan identity supplies a genuine one-sided moment deficit, with no omitted prime-power or cutoff correction.

For the RH programme the exact numerical consumer is now

$$\Theta \le \eta \quad\Longleftrightarrow\quad \sum_{m=1}^{N} \sqrt{A_{\mathcal{D}}(m)} \ll_\epsilon N^{\eta+1/2+\epsilon}$$
$$\text{for every } \epsilon > 0, \qquad \frac{1}{2} \le \eta < 1.$$

The installed $7/8$ gives mean exponent at most $3/8$; boundary $3/4$ requires at most $1/4$; RH requires exponent zero. No independent improvement of that mean estimate is proved here. The certified zero-free boundary remains $7/8$.

The pointwise rounding bound for all $N, q \ge 1$ and $0 \le \sigma \le 1$, together with the exact finite signed residue-sum identity and its weighted error estimate, is Lean-verified in `PrimeClassSamplingFloor`. The declaration `floorSamplingError_le_rpow` proves the rounding estimate used above. The finite sampling argument is also Lean-verified directly for the actual integer counts, rather than under a sampling-error hypothesis. For a class $\mathcal{P}$ of primes put $S_Y = \{p \in \mathcal{P} : p \le Y\}$ and

$$B_{S_Y}(z) = \prod_{p\in S_Y} (1 - 1/p + z/p), \qquad C_{N,k} = \#\{1 \le m \le N : K_{\mathcal{P}}(m) = k\}.$$

For every $1 \le N \le Y$, $0 \le \sigma \le 1$, and $S_\sigma = \sum_{p \in \mathcal{P}} p^{-\sigma} < \infty$, the maintained declaration

`actualSample_coefficient_error_le_exp`

proves

$$\sum_{k=0}^{|S_Y|} \left| \frac{C_{N,k}}{N} - [z^k] B_{S_Y}(z) \right| \le N^{\sigma-1} \prod_{p \in S_Y} (1 + 2p^{-\sigma}) \le e^{2S_\sigma} N^{\sigma-1}.$$

The proof uses the exact subset expansion with coefficient $\lfloor N / \prod_{p \in T} p \rfloor / N$, the signed floor remainder, and the identity that the coefficient norm of $(z-1)^r$ is 2^r. Thus the estimate is uniform in the independently varying cutoff Y. The infinite count-profile consumer is now also Lean-verified. For $0 \le \sigma < 1$ with the same summability hypothesis, the declaration

`classEulerProfile_count_sampling_law`

proves that the literal product $P_{\mathcal{P}}(z) = \prod_{p \in \mathcal{P}} (1 - 1/p + z/p)$ is entire, that its Taylor coefficients b_k satisfy

$$b_k \ge 0, \qquad \sum_{k \ge 0} b_k = 1, \qquad \sum_{k \ge 0} \left| \frac{C_{N,k}}{N} - b_k \right| \le e^{2S_\sigma} N^{\sigma-1} \quad (N \ge 1).$$

The formal proof constructs the coefficient limits from the uniform finite sampling estimate, passes every finite coefficient sum to the limit, and then proves the full infinite sum bound and normalization. Uniform convergence of the finite Euler products and Cauchy's coefficient estimate identify this same limiting law with the entire product's Taylor coefficients. The separate declaration

`classEulerProfile_eq_zero_iff`

proves that its zeros are exactly $1 - p$ for $p \in \mathcal{P}$. All these statements include finite and empty classes and the smallest prime 2; no infinitude or additional sampling-law hypothesis is used.

Normalization of the full prime-power component law, its absolutely summable Möbius floor expansion, the full total-variation estimate retaining all valuations, and the positive Jordan moment consumer remain to be formalized. The verified count-profile law does not establish their exact convergence and moment spectra. No originality priority over the literature is established.

21.54 Universal optimality of capped prime-component laws

The exact rank theorem has consequences for entire prime-power laws. Here a capped class $\mathcal{T}$ always means a subset of the primes satisfying $\theta_{\mathcal{T}}(x) \le x$ for every $x \ge 2$. Its complement inside the primes is $\mathcal{E}$. For a prime class $\mathcal{V}$, let $A_{\mathcal{V}}(m)$ be the component of m supported on $\mathcal{V}$, with all valuations retained. The previous subsection defines this statistic for $\mathcal{D}$; the same definition applies to $\mathcal{E}$.

Lemma 21.121 (Geometric prime factors are ordered by every concave loss)**.** *For a real $p > 1$, let G_p have probabilities $\mathbb{P}(G_p = k) = (1 - p^{-1})p^{-k}$ at $k \ge 0$, and put $Z_p = p^{G_p}$. For $x \ge 1$, with $k = \lfloor \log x / \log p \rfloor$,*

$$S_p(x) := \mathbb{E}\min(Z_p, x) = (k+1)(1-p^{-1}) + xp^{-k-1}. \tag{259}$$

For every $x > 1$, $S_p(x)$ is strictly decreasing in $p > 1$. Consequently, for $1 < p \le q$ and every nonnegative nondecreasing concave function $g : [1, \infty) \to [0, \infty)$,

$$\mathbb{E}g(Z_q) \le \mathbb{E}g(Z_p),$$

allowing infinite expectations.

Proof. The probabilities at $j \le k$, multiplied by p^j, each contribute $1 - p^{-1}$, while the entire remaining tail has probability p^{-k-1}. This proves (259), including $x = p^k$. For fixed $x > 1$, on a cell where k is constant its derivative is

$$\frac{\partial S_p(x)}{\partial p} = \frac{k+1}{p^2}\left(1 - \frac{x}{p^k}\right) \le 0. \tag{260}$$

All denominators are positive. The derivative is strictly negative inside each cell. At a boundary $x = p^k$, $k \ge 1$, the formulas for k and $k-1$ both equal $k + 1 - k/p$. There are only finitely many such boundaries on any compact interval $1 < p_0 \le p \le p_1$. Thus the function is continuous and strictly decreasing on that interval. When $x = 1$ it is identically 1.

We supply the test-function passage. For a continuous nonnegative nondecreasing concave g, interpolate its values on nested finite dyadic grids in $[1, 2^n]$ and make the interpolant constant past 2^n. These nonnegative concave interpolants increase to g. Each is a constant plus a nonnegative linear combination of $\min(z, x_i) - 1$, since its successive nonnegative slopes decrease and its last slope is zero. The capped-mean inequality therefore compares its expectations. Monotone convergence gives the comparison for g.

A finite concave nondecreasing function can only have an upward right jump at the endpoint 1. Writing $b = g(1+) - g(1) \geq 0$, it is the sum of a continuous nonnegative nondecreasing concave function and $b\mathbf{1}_{\{z>1\}}$. The latter indicator is the increasing limit, as $\epsilon \downarrow 0$, of $(\min(z, 1+\epsilon) - 1)/\epsilon$. Its expectation has the same order. This covers the endpoint case and all stated test functions. □

Theorem 21.122 (The greedy deletion law is the least concave-cost law)**.** *Let $\mathcal{T}$ be any capped class and let $\mathcal{E}$ be its deleted complement. Take independent geometric valuations at the primes of $\mathcal{E}$, and let $Y_{\mathcal{E}}$ be their product, allowing the value $+\infty$. Let $Y_{\mathcal{D}}$ have the proper integer-valued law ν from Theorem 21.118. For every nonnegative nondecreasing concave g as above, put $g(+\infty) = \lim_{z\to\infty} g(z)$. Then*

$$\mathbb{E}g(Y_{\mathcal{D}}) \leq \mathbb{E}g(Y_{\mathcal{E}}). \tag{261}$$

If $\mathcal{T} \neq \mathcal{S}$, the capped-mean comparison is strict at every threshold $x > 1$:

$$\mathbb{E}\min(Y_{\mathcal{D}}, x) < \mathbb{E}\min(Y_{\mathcal{E}}, x). \tag{262}$$

Thus the greedy law is uniquely least in this concave-cost order. The comparison concerns limiting laws, rather than exact ordering of arbitrary finite-sample component distributions.

Proof. Theorem 21.102 gives $N_{\mathcal{D}}(x) \leq N_{\mathcal{E}}(x)$ at every cutoff. The unconditional lower floor shows that both deleted classes are infinite. List their primes as $d_1 < d_2 < \cdots$ and $e_1 < e_2 < \cdots$. The count comparison gives

$$d_j \geq e_j \qquad (j \geq 1).$$

For any independent nonnegative multiplier $B \geq 1$, the function $z \mapsto g(Bz)$ is again nondecreasing and concave. Conditioning on B therefore preserves the preceding one-prime comparison, with no integrability hypothesis needed for a nonnegative expectation. Replace the coordinates one at a time in a finite product to obtain the comparison for its first n geometric factors. The products increase to $Y_{\mathcal{D}}$ and $Y_{\mathcal{E}}$. Monotone convergence gives (261). For $\mathcal{D}$ only finitely many valuations are nonzero almost surely, so its limit really has the already proved law ν.

If $\sum_{p\in\mathcal{E}} 1/p = \infty$, then $Y_{\mathcal{E}} = +\infty$ almost surely. Indeed, for each finite L, the event $Y_{\mathcal{E}} \leq L$ requires all valuations at $p > L$ to vanish; their finite avoidance products tend to zero. In this case the right side of (262) is x, whereas the left side is strictly below x because $\mathbb{P}(Y_{\mathcal{D}} = 1) = c_{\mathcal{D}} > 0$.

In the remaining case both products are finite almost surely. If the classes differ, some rank j has $d_j > e_j$. Form an auxiliary independent product by replacing only the jth greedy coordinate by the factor with base e_j. Such an auxiliary coordinate product need not itself be a prime-component law; only its independence is used. All other greedy factors equal 1 with positive probability

$$\prod_{i \neq j} (1 - 1/d_i) > 0.$$

On that event the strict one-prime capped-mean difference applies; on every other event its conditional difference is nonnegative. The auxiliary law therefore has strictly greater capped mean than the greedy law at each $x > 1$. Changing its remaining coordinates to the e_i coordinates can only increase that mean, by the non-strict product comparison already proved. This proves strictness. □

Theorem 21.123 (Maximal retained density and minimal component entropy)**.** *For every capped $\mathcal{T}$, the set of positive integers all of whose prime factors belong to $\mathcal{T}$ has a natural density*

$$c_{\mathcal{E}} = \prod_{p \in \mathcal{E}} (1 - 1/p) \leq c_{\mathcal{D}}. \tag{263}$$

The inequality is strict unless $\mathcal{T} = \mathcal{S}$.

Among capped competitors with $\sum_{p \in \mathcal{E}} 1/p < \infty$, let $\nu_{\mathcal{E}}(a) = c_{\mathcal{E}}/a$ on $\mathcal{E}$-smooth positive integers. Their Shannon entropies, in natural logarithms, satisfy

$$H(\nu_{\mathcal{D}}) \leq H(\nu_{\mathcal{E}}), \qquad H(\nu_{\mathcal{V}}) = \sum_{p \in \mathcal{V}} \left[-\log(1 - 1/p) + \frac{\log p}{p - 1} \right]. \tag{264}$$

Here $H(\nu_{\mathcal{D}}) < \infty$; a competitor's entropy may be infinite. Equality holds only for $\mathcal{T} = \mathcal{S}$.

Proof. If the reciprocal sum over $\mathcal{E}$ converges, the finite CRT avoidance law and its uniform reciprocal-tail bound prove the density formula, exactly as in the earlier Fermat avoidance argument. If it diverges, upper density is at most every finite avoidance product, and those products tend to zero. Thus the stated density exists in both cases.

The weight $w(p) = -\log(1 - 1/p)$ is positive and strictly decreasing. The deleted-prime rank comparison $d_j \geq e_j$ therefore gives $\sum_j w(d_j) \leq \sum_j w(e_j)$. If the latter sum diverges, the density inequality is strict. If it

converges and the classes differ, one strict rank gives a positive strict gap between the sums. Exponentiating proves (263) and its equality characterization.

For a proper component law, normalization and Tonelli give

$$\begin{aligned} H(\nu_{\mathcal{V}}) &= -\log c_{\mathcal{V}} + \mathbb{E}\log Y_{\mathcal{V}} \\ &= -\log c_{\mathcal{V}} + \sum_{p\in\mathcal{V}} \frac{\log p}{p-1}. \end{aligned}$$

This identity allows infinite entropy. The single-prime entropy $h(p) = -\log(1-1/p) + \log p/(p-1)$ obeys

$$h'(p) = -\frac{\log p}{(p-1)^2} < 0 \qquad (p > 1).$$

It is positive and strictly decreasing. The same rank comparison proves the entropy inequality and strictness when the classes differ. For the greedy class, $h(p) \ll (1+\log p)/p$ is summable by the proved count exponent $\Theta < 1$. This proves finiteness of its entropy. Entropy is asserted here only for the proper countable integer-valued laws, namely competitors with summable deleted reciprocal primes. □

Theorem 21.124 (Adaptive finite caps cannot reduce the moment exponent)**.** *For an integer $N \geq 2$, let $\mathcal{P}_N = \{p \leq N : p \text{ prime}\}$. A subset $\mathcal{T} \subseteq \mathcal{P}_N$ is admissible if*

$$\sum_{\substack{p\in\mathcal{T}\\ p\leq q}} \log p \leq q \qquad (q \in \mathcal{P}_N).$$

For $0 < t \leq 1$ define the finite optimization

$$\begin{aligned} V_t(N) &= \min_{\mathcal{T} \text{ admissible}} U_t(\mathcal{T}; N), \\ U_t(\mathcal{T}; N) &= \frac{1}{N}\sum_{m=1}^{N} \left(\prod_{\substack{p\in\mathcal{P}_N\setminus\mathcal{T}\\ p\mid m}} p^{v_p(m)} \right)^t. \end{aligned} \tag{265}$$

With $c_t = (1-2^{-t})/2 > 0$, every admissible choice satisfies

$$U_t(\mathcal{T}; N) \geq c_t \sum_{\substack{p\leq N\\ p\in\mathcal{D}}} p^{t-1} \geq c_t N^{t-1} N_{\mathcal{D}}(N). \tag{266}$$

Consequently

$$\lim_{N\to\infty}\frac{\log V_t(N)}{\log N}=\max\{0,\Theta+t-1\}. \tag{267}$$

In particular there are $c>0, N_0$, independent of the admissible selection, such that

$$V_{1/2}(N)\ge c\log\log N\quad(N\ge N_0),\qquad \lim_{N\to\infty}\frac{\log V_{1/2}(N)}{\log N}=\Theta-\frac{1}{2}. \tag{268}$$

For each fixed $1/2\le\eta<1$,

$$\Theta\le\eta\quad\Longleftrightarrow\quad V_{1/2}(N)\ll_\epsilon N^{\eta-1/2+\epsilon}\quad\text{for every }\epsilon>0. \tag{269}$$

The value and an attaining subset for $V_{1/2}(N)$ are computable from the integer N, without an efficiency claim.

Proof. There are finitely many subsets of $\mathcal{P}_N$, and the empty subset is admissible. Thus the minimum exists. The finite prime inequalities are equivalent to the cap throughout $2\le x\le N$: the theta function is constant between prime nodes. Theorem 21.102 therefore applies at every prefix of this same finite problem. If $e_1<\cdots<e_b$ are the primes in $\mathcal{E}_N=\mathcal{P}_N\setminus\mathcal{T}$ and $d_1<\cdots<d_a$ are those in $\mathcal{D}\cap[2,N]$, it gives $b\ge a$ and $e_j\le d_j$ for $j\le a$. Since p^{t-1} is nonnegative and nonincreasing for $0<t\le1$,

$$\sum_{p\in\mathcal{E}_N}p^{t-1}\ge\sum_{\substack{p\le N\\ p\in\mathcal{D}}}p^{t-1}.$$

Apply the exact positive Jordan divisor identity to $\mathcal{E}_N$; it does not require a fixed infinite class. Retaining its nonnegative prime terms gives

$$U_t(\mathcal{T};N)\ge\sum_{p\in\mathcal{E}_N}(p^t-1)\frac{\lfloor N/p\rfloor}{N}.$$

For every such prime, $p^t-1\ge(1-2^{-t})p^t$ and $\lfloor N/p\rfloor/N\ge1/(2p)$. These inequalities prove the first bound in (266). The second uses $p^{t-1}\ge N^{t-1}$ for $p\le N$. All constants are independent of $\mathcal{T}$ and N.

The fixed greedy choice restricted to $\mathcal{P}_N$ is admissible, so $V_t(N)\le U_t(N)$ for its previously proved moment mean. That mean has exponent $\max(0,\Theta+t-1)$. On the other side the new uniform floor and $N_{\mathcal{D}}(N)=N^{\Theta+o(1)}$ give the lower exponent $\Theta+t-1$, while the elementary $V_t(N)\ge1$ supplies zero when that exponent is nonpositive. This proves (267). For

$t = 1/2$, partial summation of the unconditional deletion count floor gives $\sum_{p \le N, p \in \mathcal{D}} p^{-1/2} \gg \log\log N$. Substitute this into the same uniform inequality to obtain (268). Since $\Theta \ge 1/2$, its exponent yields (269), including RH at $\eta = 1/2$.

Finally enumerate the finite prime set and all its subsets. Every admissibility comparison is $\log M \le q$ for an integer prime subproduct M and a positive integer prime q. Equality is impossible by transcendence of e^q, so certified logarithm intervals decide it. At $t = 1/2$ each cost is a finite sum of square roots of integers divided by N, hence is algebraic. Exact algebraic comparisons, including equality, select a minimum and an attaining subset. This proves the computability assertion on its stated integer domain. □

The limiting-law order and the finite optimization have different quantifiers. The former compares full geometric laws for fixed global capped classes. The latter is uniform over every admissible subset chosen separately at each sample size. The selection must be made before the integer m is sampled. Randomizing that selection independently of m preserves the lower bound by averaging. Thus a cutoff-dependent or independently randomized family cannot conceal the same moment exponent.

The one-prime result is a concave-cost order. It does not imply ordinary stochastic order: for the factors with bases 2 and 3, their tails at $3/2$ are $1/2$ and $1/3$, whereas at $5/2$ they are $1/4$ and $1/3$. These exact crossing tails concern the one-prime comparison; no claim about failure of stochastic order for a particular whole greedy-versus-competitor pair is made.

The new conclusions use the completed rank theorem and full prime-power law. They extend the earlier Bernoulli count-profile extremality to geometric components, concave losses, actual integer density, entropy, and a finite optimization with a uniform exponent. They do not supply an independent upper estimate for that exponent. The certified boundary is still $7/8$; $3/4$ requires $V_{1/2}(N) \ll_\epsilon N^{1/4+\epsilon}$, and RH requires $V_{1/2}(N) \ll_\epsilon N^\epsilon$ for every $\epsilon > 0$. The universal logarithmic lower floor is compatible with RH.

All proofs in this subsection are ordinary proofs. No new Lean verification is claimed for the geometric capped mean, its concave-test extension, the infinite-product comparison or the adaptive finite optimization. Those are the concrete formalization consumers; the earlier finite Bernoulli tail comparisons remain separate compiled inputs. No established originality priority is asserted.

21.55 A signed certificate for greedy rank optimality

The exact rank comparison has a finite mechanism that does not depend on primes. It is useful to retain this mechanism rather than just its cardinality consequence. Let $w_0, w_1, \dots$ be nondecreasing real weights and let $b_0, b_1, \dots$ be arbitrary real budgets. For a selection $a \subseteq \mathbb{N}_0$, set

$$\begin{aligned} C_a(n) &= \#(a \cap \{0, \dots, n-1\}), \\ S_a(n) &= \sum_{\substack{0 \le j < n \\ j \in a}} w_j, \\ n \in g &\iff S_g(n) + w_n \le b_n. \end{aligned} \tag{270}$$

The last line defines the greedy selection recursively, starting with $S_g(0) = 0$. A competitor t is admissible when $S_t(n) + w_n \le b_n$ at every index $n \in t$. The budget test is imposed at selected indices. If budgets decrease, this definition does not assert a budget inequality at an index where nothing is selected.

Theorem 21.125 (Greedy rank optimality with arbitrary budgets)**.** *For every such admissible competitor and every $n \ge 0$,*

$$C_t(n) \le C_g(n), \qquad S_g(n) - S_t(n) \le (C_g(n) - C_t(n))w_n.$$

In particular, equal counts imply $S_g(n) \le S_t(n)$, and $n - C_g(n) \le n - C_t(n)$. Define the signed count–cost slack by $Q_n = (C_g(n) - C_t(n))w_n - (S_g(n) - S_t(n))$. For arbitrary selections g, t, before imposing either greedy admissibility or monotonicity, its exact identities are

$$Q_{n+1} - Q_n = (C_g(n+1) - C_t(n+1))(w_{n+1} - w_n) \tag{271}$$

and

$$Q_n = \sum_{j=0}^{n-1} (C_g(j+1) - C_t(j+1))(w_{j+1} - w_j). \tag{272}$$

For the greedy selection and an admissible competitor, $Q_n \ge 0$. More precisely, every selected competitor index i has a greedy index of the same zero-based rank at or before it:

$$i \in t \implies \exists j \le i : \quad j \in g, \qquad C_g(j) = C_t(i). \tag{273}$$

No monotonicity of b_n or nonnegativity of w_n is required.

Proof. Write $\Delta C_n = C_g(n) - C_t(n)$ and $\Delta S_n = S_g(n) - S_t(n)$. The two exact recursions give

$$\Delta S_{n+1} - \Delta S_n = (\Delta C_{n+1} - \Delta C_n) w_n.$$

Substitution proves (271); telescoping from $Q_0 = 0$ proves (272). All endpoints and signs are retained.

We prove $\Delta C_n \geq 0$ and $Q_n \geq 0$ simultaneously. Both vanish at zero. Assume both inequalities at n. The count gap can become negative at $n+1$ only when it is zero at n, the competitor selects n, and greedy does not. But when $\Delta C_n = 0$, the inequality $Q_n \geq 0$ says $S_g(n) \leq S_t(n)$. Competitor admissibility would give

$$S_g(n) + w_n \leq S_t(n) + w_n \leq b_n,$$

so greedy must select n, a contradiction. Therefore $\Delta C_{n+1} \geq 0$. Since $w_{n+1} - w_n \geq 0$, the exact slack recursion now implies $Q_{n+1} \geq Q_n \geq 0$. This closes the joint induction and gives all count and cost conclusions.

Finally fix $i \in t$ and put $k = C_t(i)$. Then $C_t(i+1) = k+1$, so $C_g(i+1) \geq k+1$. The greedy count starts at zero and changes only by zero or one. Consequently some $j \leq i$ is selected with $C_g(j) = k$, proving the rank assertion. □

For the prime construction take the stream to be the increasing primes, $w_n = \log p_n$ and $b_n = p_n$. This gives the rank mechanism used in Theorem 21.102, which in turn feeds the full-law and adaptive finite-cap results above. The signed slack is a finite certificate for that ordering; it supplies no independent upper estimate for the square-root moment and changes no zero-free boundary.

Theorem 21.125 is now Lean-verified in its full finite ordered-stream form. The module `PrimeClassDeletion.CapacitySelection` implements the actual greedy recursion and proves its signed invariant. The maintained module `CapacitySelectionFinite` identifies the recursive counts and costs with the literal finite-set cardinalities and sums used here. Its declarations

`prefixSlack_step`, `prefixSlack_sum`,
`greedySelection_finset_prefix_optimality`,
`greedySelection_finset_rank_le`

verify the exact difference and Abel identities, every count and cost conclusion, nonnegative slack, and the actual selected same-rank witness. They include $n = 0$, arbitrary real budgets, and signed nondecreasing weights. The analytic prime enumeration and the infinite geometric probability-law consumers remain separate formalization obligations.

21.56 Rare profiles, arithmetic records, and spectral dimension

The full sampling law also controls ranks growing with the sample size. The following statements apply first to an arbitrary fixed infinite prime class $\mathcal{P} = \{p_1 < p_2 < \cdots\}$ with

$$0 < \alpha < 1, \qquad \lim_{j\to\infty} \frac{\log p_j}{\log j} = \frac{1}{\alpha}. \tag{274}$$

Equivalently $N_{\mathcal{P}}(X) = X^{\alpha+o(1)}$; bracketing X between successive p_j proves the equivalence. For positive integers m define $K_{\mathcal{P}}(m) = \#\{p \in \mathcal{P} : p \mid m\}$. The prime class is fixed before sampling m uniformly from $\{1, \dots, N\}$. The integer N is the sample endpoint, k is a factor-count level, and r below is a generating-function argument.

For every $\sigma > \alpha$ the prime sum $\sum_{p\in\mathcal{P}} p^{-\sigma}$ converges. Indeed, choose $\epsilon > 0$ with $\sigma(1/\alpha - \epsilon) > 1$ and compare $p_j^{-\sigma}$ with $j^{-\sigma(1/\alpha-\epsilon)}$. In particular $\sum_{\mathcal{P}} 1/p < \infty$. There is therefore a proper independent count law $K_{\mathcal{P}} = \sum_{p\in\mathcal{P}} B_p$, where $\mathbb{P}(B_p = 1) = 1/p$ and $\mathbb{P}(B_p = 0) = 1 - 1/p$. Only finitely many B_p are nonzero almost surely. Put

$$\begin{aligned} c_{\mathcal{P}} &= \prod_{p\in\mathcal{P}} (1 - 1/p) > 0, \\ P_{\mathcal{P}}(z) &= \mathbb{E}z^{K_{\mathcal{P}}} = \prod_{p\in\mathcal{P}} (1 - 1/p + z/p) = c_{\mathcal{P}} \prod_{p\in\mathcal{P}} (1 + z/(p-1)), \\ b_k &= \mathbb{P}(K_{\mathcal{P}} = k), \qquad q_k = \mathbb{P}(K_{\mathcal{P}} \ge k). \end{aligned} \tag{275}$$

All denominators $p - 1$ are positive. The products converge uniformly on compact sets, so $P_{\mathcal{P}}$ is entire, with nonnegative coefficients b_k. Its mean $\sum_{\mathcal{P}} 1/p$ is finite.

Theorem 21.126 (Reciprocal indices of growth and rare probabilities)**.** *For every class satisfying* (274)*, let* $M_{\mathcal{P}}(r) = \max_{|z|\le r} |P_{\mathcal{P}}(z)|$*. Then* $M_{\mathcal{P}}(r) = P_{\mathcal{P}}(r)$ *and*

$$\lim_{r\to\infty} \frac{\log\log M_{\mathcal{P}}(r)}{\log r} = \alpha. \tag{276}$$

Every b_k *and* q_k *is positive, including* $b_0 = c_{\mathcal{P}}$ *and* $q_0 = 1$*. At large ranks the point mass and the tail have the same exact logarithmic decay:*

$$\lim_{k\to\infty} \frac{-\log b_k}{k\log k} = \lim_{k\to\infty} \frac{-\log q_k}{k\log k} = \frac{1}{\alpha}. \tag{277}$$

More explicitly, for each fixed $\alpha < \sigma < 1$, put $C_\sigma = \sigma^{-1} \sum_{\mathcal{P}} p^{-\sigma} < \infty$. For every integer $k \geq 2$,

$$\frac{c_{\mathcal{P}}}{\prod_{j=1}^{k}(p_j - 1)} \leq b_k \leq q_k \leq \exp\left(C_\sigma k - \frac{k}{\sigma} \log k \right). \tag{278}$$

No uniform constant as $\sigma \downarrow \alpha$ is asserted.

Proof. For $r \geq 1$, independence and nonnegative product expansion give

$$\log P_{\mathcal{P}}(r) = \sum_{p \in \mathcal{P}} \log(1 + (r-1)/p) \leq C_\sigma r^\sigma.$$

Here $\log(1+y) \leq y^\sigma/\sigma$ for $y \geq 0$ and $0 < \sigma < 1$: for $t > 0$, $t^{1-\sigma} \leq 1 + t$, and integration of $(1+t)^{-1} \leq t^{\sigma-1}$ proves the inequality. For $r \geq 2$, each factor with $p \leq r$ is at least $3/2$, whereas all other factors are at least one. Thus

$$\log P_{\mathcal{P}}(r) \geq (\log(3/2)) N_{\mathcal{P}}(r) = r^{\alpha + o(1)}.$$

Nonnegative Taylor coefficients imply $|P_{\mathcal{P}}(z)| \leq P_{\mathcal{P}}(|z|)$ and equality in the maximum at $z = r$. The last two bounds show that the lower growth index is at least α and its upper index is at most every fixed $\sigma > \alpha$. Let $\sigma \downarrow \alpha$ after taking the limit in r. This proves (276).

The event in which exactly the first k Bernoulli variables are one has probability

$$\left(\prod_{j=1}^{k} p_j^{-1} \right) \prod_{j>k} (1 - 1/p_j) = \frac{c_{\mathcal{P}}}{\prod_{j=1}^{k}(p_j - 1)} > 0.$$

It supplies the independent positive lower bound for b_k. For $r \geq 1$, exponential Markov gives $q_k \leq r^{-k} P_{\mathcal{P}}(r)$. Taking $r = k^{1/\sigma}$ proves the upper bound in (278).

Finally, (274) and elementary integral bounds for $\sum_{j \leq k} \log j$ give

$$\sum_{j=1}^{k} \log(p_j - 1) = \left(\frac{1}{\alpha} + o(1) \right) k \log k.$$

Replacing $\log p_j$ by $\log(p_j - 1)$ changes this sum by a bounded amount, since $\sum_j 1/p_j < \infty$. The positive configuration bound therefore makes the upper decay index of b_k at most $1/\alpha$. The exponential upper bound makes the lower decay index of q_k at least $1/\sigma$ for every $\sigma > \alpha$. Since $b_k \leq q_k$, letting $\sigma \downarrow \alpha$ proves both limits in (277). □

Theorem 21.127 (Arithmetic and independent records agree)**.** *For integer $N \geq 1$ put $H_{\mathcal{P}}(N) = \max_{1 \leq m \leq N} K_{\mathcal{P}}(m)$. On one probability space let $K_1, K_2, \ldots$ be independent copies of $K_{\mathcal{P}}$ and put $W_{\mathcal{P}}(N) = \max_{1 \leq i \leq N} K_i$. Then*

$$\begin{aligned} H_{\mathcal{P}}(N) &\sim \alpha \frac{\log N}{\log \log N}, \\ W_{\mathcal{P}}(N) &\sim \alpha \frac{\log N}{\log \log N} \quad \textit{almost surely,} \\ \frac{W_{\mathcal{P}}(N)}{H_{\mathcal{P}}(N)} &\longrightarrow 1 \quad \textit{almost surely.} \end{aligned} \tag{279}$$

The last ratio is asserted for all sufficiently large N, when its denominator is positive.

Proof. Let $Q_0 = 1$ and $Q_k = \prod_{j=1}^{k} p_j$. An integer with k distinct prime divisors in $\mathcal{P}$ is divisible by their product, which is at least Q_k. Conversely Q_k has exactly k such prime divisors. Consequently, including $k = 0$ and both integer endpoints,

$$H_{\mathcal{P}}(N) = \max\{k \geq 0 : Q_k \leq N\}, \qquad \log Q_k = \left(\frac{1}{\alpha} + o(1)\right) k \log k. \tag{280}$$

The maximum is finite because $Q_k \geq 2^k$, and tends to infinity with N. If $h = H_{\mathcal{P}}(N)$, then $Q_h \leq N < Q_{h+1}$, so $\log N \sim h \log h/\alpha$ and $\log \log N \sim \log h$. This proves the deterministic limit.

Write $\ell_N = \log N / \log \log N$ for $N \geq 3$. For fixed $0 < \epsilon < \alpha$, the tail limit in (277) gives, for $k_N^{\pm} = \lceil (\alpha \pm \epsilon) \ell_N \rceil$,

$$N q_{k_N^+} = N^{-\epsilon/\alpha + o(1)}, \qquad N q_{k_N^-} = N^{\epsilon/\alpha + o(1)}.$$

A union bound and independence, respectively, give

$$\mathbb{P}(W_{\mathcal{P}}(N) \geq k_N^+) \leq N q_{k_N^+}, \qquad \mathbb{P}(W_{\mathcal{P}}(N) < k_N^-) \leq \exp(-N q_{k_N^-}).$$

Both bounds are summable along $N = 2^j$. Borel–Cantelli, followed by an intersection over a countable sequence $\epsilon \downarrow 0$, proves the almost-sure limit along these dyadic samples. The record $W_{\mathcal{P}}(N)$ is nondecreasing, ℓ_N is eventually increasing, and $\ell_{2^{j+1}}/\ell_{2^j} \to 1$. Bracketing each integer N between successive powers of two therefore proves the full almost-sure limit. Combining with the deterministic limit proves the ratio. No independence of the arithmetic values $K_{\mathcal{P}}(1), K_{\mathcal{P}}(2), \ldots$ is assumed. □

Theorem 21.128 (Uniform relative recovery at growing ranks)**.** *For integers $N \geq 1$ and $k \geq 0$ define the actual empirical mass and tail by*

$$b_{N,k} = \frac{1}{N}\#\{m \in \{1,\dots,N\} : K_{\mathcal{P}}(m) = k\},$$
$$q_{N,k} = \frac{1}{N}\#\{m \in \{1,\dots,N\} : K_{\mathcal{P}}(m) \geq k\}.$$

Fix

$$0 < c < \alpha(1-\alpha), \qquad 0 < \delta < 1-\alpha-\frac{c}{\alpha}, \qquad L_N = \left\lfloor c\frac{\log N}{\log\log N}\right\rfloor \quad (N \geq 3). \tag{281}$$

Then, for all sufficiently large integer N,

$$\max_{\substack{k\in\mathbb{N}_0\\ 0\leq k\leq L_N}} \left(\left|\frac{b_{N,k}}{b_k} - 1\right| + \left|\frac{q_{N,k}}{q_k} - 1\right|\right) \ll_{\mathcal{P},c,\delta} N^{-\delta}. \tag{282}$$

The constant and the starting point are independent of k. All denominators are positive; in this range the empirical masses are also positive eventually. For any integer sequence $k_N \sim c\log N/\log\log N$ with $0 < c < \alpha(1-\alpha)$, one moreover has

$$b_{N,k_N} = N^{-c/\alpha+o(1)}, \qquad q_{N,k_N} = N^{-c/\alpha+o(1)}. \tag{283}$$

Proof. For each $\alpha < \sigma < 1$ define the finite constant

$$B_{\mathcal{P},\sigma} = \prod_{p\in\mathcal{P}} \frac{1+p^{-\sigma}}{1-p^{-\sigma}}.$$

The normalization and full sampling proof of Theorem 21.118 require only these summable Euler products, and therefore apply to this fixed class $\mathcal{P}$. For clarity, on its positive smooth state space $\mathcal{H}_{\mathcal{P}}$ the exact finite identity is

$$\mu_{\mathcal{P},N}(a) = \sum_{\substack{e\in\mathcal{H}_{\mathcal{P}}\\ e \text{ squarefree}\\ ae\leq N}} \mu_{\mathrm{Mob}}(e)\frac{\lfloor N/(ae)\rfloor}{N}, \qquad \nu_{\mathcal{P}}(a) = \frac{c_{\mathcal{P}}}{a}.$$

Here $\mu_{\mathcal{P},N}$ is the law of the full $\mathcal{P}$-component of a uniform $m \in \{1,\dots,N\}$. There is no coprimality restriction between a and e. The infinite counterpart for $\nu_{\mathcal{P}}$ is absolutely convergent. After subtraction, only the rounding and tail remainder is estimated, using

$$0 \leq q^{-1} - N^{-1}\lfloor N/q\rfloor \leq N^{\sigma-1}q^{-\sigma} \qquad (q \geq 1).$$

This includes the domain $q > N$, where the floor is zero. Summing over all a and all squarefree e charges both the prime-power denominator and the squarefree numerator. Contraction to $K_{\mathcal{P}}$ consequently gives

$$\sum_{k\geq 0} |b_{N,k} - b_k| \leq B_{\mathcal{P},\sigma} N^{\sigma-1}, \qquad |q_{N,k} - q_k| \leq B_{\mathcal{P},\sigma} N^{\sigma-1}. \tag{284}$$

This is uniform in the statistic level k.

The positive coefficient lower bound in Theorem 21.126 implies, for each $\epsilon > 0$, a constant $A_\epsilon > 0$ such that

$$b_k \geq A_\epsilon \exp\!\left(-\left(\frac{1}{\alpha} + \epsilon\right) k \log(\max\{2, k\})\right) \qquad (k \geq 0).$$

The finitely many initial ranks, including zero and one, are absorbed using $b_k > 0$. For $0 \leq k \leq L_N$, eventually $k \log(\max\{2, k\}) \leq c \log N$. Thus the independent positive supply and its negative sampling allowance remain explicit:

$$b_k \geq A_\epsilon N^{-c(1/\alpha+\epsilon)},$$
$$b_{N,k} \geq A_\epsilon N^{-c(1/\alpha+\epsilon)} - B_{\mathcal{P},\sigma} N^{\sigma-1}.$$

Choose fixed $\sigma > \alpha$ and $\epsilon > 0$ sufficiently close to α and zero, respectively, that

$$\sigma - 1 + c(1/\alpha + \epsilon) < -\delta.$$

Such choices exist precisely because of the strict parameter inequalities in (281). Divide the two absolute sampling errors by $b_k \geq A_\epsilon N^{-c(1/\alpha+\epsilon)}$ and $q_k \geq b_k$. This proves (282), with a bound uniform in the entire displayed finite rank range. It also proves eventual positivity of the empirical masses.

Finally, if $k_N \sim c\ell_N$, choose $c < c' < \alpha(1-\alpha)$. Then $k_N \leq \lfloor c'\ell_N \rfloor$ eventually and the relative estimate with c' applies. Since $k_N \log k_N / \log N \to c$, the exact coefficient and tail limits prove (283). □

Theorem 21.129 (Exact adaptive minimax for the factor-count record)**.** *Take the fixed greedy retained and deleted classes $\mathcal{S}, \mathcal{D}$ of the preceding subsections, and write $\Theta = \Theta_\zeta$. For each integer $N \geq 2$, let $\mathcal{P}_N = \{p \leq N : p \text{ prime}\}$ and let $\mathcal{T} \subseteq \mathcal{P}_N$ range over all choices satisfying the logarithmic cap at every prime through N. Put $\mathcal{E}_N = \mathcal{P}_N \setminus \mathcal{T}$. Then*

$$\min_{\mathcal{T} \text{ admissible}} \max_{1\leq m\leq N} K_{\mathcal{E}_N}(m) = H_{\mathcal{D}}(N),$$
$$H_{\mathcal{D}}(N) \sim \Theta \frac{\log N}{\log\log N}. \tag{285}$$

The first equality is exact at every N and is attained by $\mathcal{T} = \mathcal{S} \cap \mathcal{P}_N$. A competitor may be chosen separately at every N, before taking its worst case over m. Among fixed global capped classes, $\mathcal{S}$ is the unique class whose record is no larger than every competitor's record at every integer cutoff.

Proof. Theorem 21.115 supplies (274) for $\mathcal{D}$ with $\alpha = \Theta$. Here $1/2 \le \Theta \le 7/8 < 1$ unconditionally. The record asymptotic follows from Theorem 21.127.

For exact finite optimality put $h = H_{\mathcal{D}}(N)$. If $h = 0$, every competing maximum is nonnegative. If $h > 0$, let $d_1, \dots, d_h$ be the first h deleted primes. Their product is at most N, so all lie in $\mathcal{P}_N$. The exact rank comparison of Theorem 21.102 gives at least h primes $e_1 < \cdots < e_h$ in $\mathcal{E}_N$, with $e_j \le d_j$ for $1 \le j \le h$. Their product is an integer at most N with exactly h prime divisors in $\mathcal{E}_N$. Thus the competing record is at least h. The greedy prefix is admissible and gives exactly the original statistic on every $m \le N$, proving equality.

For a fixed different global cap, its complement has ranks $e_j \le d_j$ for every j, with at least one strict inequality. At such a rank h, $\prod_{j \le h} e_j < \prod_{j \le h} d_j$. At the integer cutoff $N = \prod_{j \le h} e_j$, its record is at least h, whereas the greedy record is below h. This proves the asserted uniqueness across all cutoffs. No uniqueness of a finite optimizer at each separate N is asserted. □

Corollary 21.130 (A fully discrete rare-count readout of the zero supremum)**.** *Fix a rational number $0 < c < 7/64$. For integer $N \ge 4$ define the entirely discrete ranks*

$$u_N = \lfloor \log_2 N \rfloor, \qquad v_N = \lfloor \log_2 u_N \rfloor, \qquad \kappa_c(N) = \left\lfloor \frac{c u_N}{v_N} \right\rfloor. \tag{286}$$

Here $u_N \ge 2$ and $v_N \ge 1$. Both logarithmic floors are integer bit-length operations. Define actual integer counts

$$C_c^{=}(N) = \#\{m \in \{1, \dots, N\} : K_{\mathcal{D}}(m) = \kappa_c(N)\},$$
$$C_c^{\ge}(N) = \#\{m \in \{1, \dots, N\} : K_{\mathcal{D}}(m) \ge \kappa_c(N)\}.$$

Then, unconditionally, for either $\star \in \{=, \ge\}$,

$$C_c^{\star}(N) = N^{1 - c/\Theta + o(1)}, \qquad \Theta = \frac{c}{1 - \lim\limits_{N \to \infty} \dfrac{\log C_c^{\star}(N)}{\log N}}. \tag{287}$$

The counts are positive for every sufficiently large N. For each fixed $1/2 \le \eta < 1$,

$$\Theta \le \eta \quad \Longleftrightarrow \quad C_c^\star(N) \ll_\epsilon N^{1-c/\eta+\epsilon} \quad \text{for every } \epsilon > 0. \tag{288}$$

At $\eta = 1/2$ this is equivalent to RH. The limiting entire-function order is Θ, the rare-probability decay index is $1/\Theta$, and both the arithmetic record and its independent-law counterpart have record constant Θ.

In particular, for $\mathcal{P} = \mathcal{D}$, the uniform relative estimate holds for every $0 < c < 7/64$ and every $0 < \delta < 1/8 - 8c/7$. One explicit unconditional specialization is

$$\max_{\substack{k\in\mathbb{N}_0 \\ 0\le k\le \lfloor (\log N)/(10\log\log N)\rfloor}} \left(\left| \frac{b_{N,k}}{b_k} - 1 \right| + \left| \frac{q_{N,k}}{q_k} - 1 \right| \right) \ll N^{-1/100}. \tag{289}$$

The same $O(N^{-1/100})$ relative estimate holds at the single binary rank $\kappa_{1/10}(N)$. All these discrete counts are computable from integer N, without an efficiency assertion.

Proof. For $1/2 \le \Theta \le \eta < 1$ and $0 < c < \eta(1-\eta)$ one has

$$1 - \Theta - \frac{c}{\Theta} - \left(1 - \eta - \frac{c}{\eta} \right) = (\eta - \Theta) \left(1 - \frac{c}{\eta\Theta} \right) \ge 0.$$

Indeed $c < \eta(1-\eta) \le \eta\Theta$. Taking the installed $\eta = 7/8$ proves that $c < 7/64$ is inside the actual growing-rank range, and every $\delta < 1/8 - 8c/7$ is allowed.

The bit-length definitions give $u_N = (\log N)/\log 2 + O(1)$ and $v_N = (\log\log N)/\log 2 + O(1)$. Hence $\kappa_c(N) \sim c \log N/\log\log N$. Apply (283) to this sequence, for the same fixed class $\mathcal{D}$. Multiplying each probability by N proves both count exponents in (287). The denominator in its recovery formula is $c/\Theta > 0$.

The exact count exponent is at most $1 - c/\eta$ if and only if $\Theta \le \eta$. A sequence with an existing logarithmic exponent has that upper exponent precisely when it satisfies every displayed ϵ-upper bound; this proves (288). Zero symmetry gives the RH specialization. The remaining growth and record indices follow from the two preceding general theorems.

For $c = 1/10$ the available margin is $1/8 - 8c/7 = 3/280 > 1/100$, proving (289). For the binary rank choose a fixed $c' > 1/10$ sufficiently close to $1/10$ that $1/100 < 1/8 - 8c'/7$. Eventually $\kappa_{1/10}(N) \le \lfloor c' \log N/\log\log N \rfloor$, so the same uniform theorem proves its pointwise relative estimate. Finally the already proved finite decidability of the greedy class determines its prime membership through N. Bit lengths, rational floors and integer factorization then compute the two displayed finite counts. □

For $c = 1/10$, the installed boundary supplies count upper exponent $31/35$; a boundary $3/4$ would supply $13/15$, and RH supplies $4/5$, each with an arbitrary positive ϵ allowance. No independent arithmetic estimate reducing that count exponent is proved here. The certified zero-free boundary remains $7/8$. The growing-rank interval is a proved sufficient range; no failure of relative approximation outside it is claimed.

The Bernoulli-product and ordinary prime-factor-count background is classical; compare [9, Proposition 3.1 and §4]. That proposition treats a divergent sum of Bernoulli parameters. The present class has a finite parameter sum, a fixed entire limiting law, and rare levels of order $\log N/\log\log N$ tied to its all-scale prime exponent. The independent-record argument is also elementary. The specific conclusions here are the spectral interpretation for the greedy class, its exact adaptive record minimum, and the uniform relative arithmetic recovery. No literature-priority claim is made.

These are full ordinary proofs. No new Lean verification is claimed for the countable Euler law, the coefficient asymptotics, the growing-rank transfer or the independent record limit. The maintained finite greedy count–cost and rank comparisons are compiled inputs to the cap mechanism. Concrete remaining formalization tasks are countable Euler-law normalization, its elementary-symmetric coefficient limits, the uniform asymptotic division, and the dyadic Borel–Cantelli consumer.

21.57 Affine samples, optimal prime filters, and a disappearing law

Fix a prime class $\mathcal{P}$ satisfying (274), with exponent α. Write $\mathcal{H}_{\mathcal{P}}$ for its positive smooth integers and $A_{\mathcal{P}}(m) = \prod_{p\in\mathcal{P}} p^{v_p(m)}$. All samples in this subsection are *integer* samples, different from the earlier affine prime-pair ensembles. The local congruence count is classical. The conclusions concern uniform moving filters, their optimal product budget, and rare events whose exponent survives convergence of the whole count law to zero. The comparison with the divergent-mean framework of [9] made above still applies; no literature priority claim is required.

Theorem 21.131 (Full affine law with its forced component)**.** *Let A, q, M be arbitrary positive integers, and sample $F(n) = A + qn$ uniformly over $0 \le n < M$. Put*

$$Y = A + q(M-1), \quad g = \gcd(A,q), \quad A_0 = A/g, \quad q_0 = q/g, \quad R = Y/g,$$
$$\mathcal{V} = \mathcal{P} \setminus \{p : p \mid q_0\}, \quad \mathcal{Q} = \mathcal{P} \setminus \{p : p \mid q\}, \quad r = K_{\mathcal{P}}(g). \tag{290}$$

Let $Z_{\mathcal{V}}$ have the full component law $\mathbb{P}(Z_{\mathcal{V}} = a) = c_{\mathcal{V}}/a$ on $\mathcal{H}_{\mathcal{V}}$. For each fixed $\alpha < \sigma < 1$ the actual full component law $\mu_{A,q,M}$ satisfies

$$\begin{aligned} &\|\mu_{A,q,M} - \mathrm{Law}(A_{\mathcal{P}}(g)Z_{\mathcal{V}})\|_1 \le B_{\mathcal{P},\sigma}R^{\sigma}/M, \\ &d_{\mathrm{TV}}(\mu_{A,q,M}, \mathrm{Law}(A_{\mathcal{P}}(g)Z_{\mathcal{V}})) \le \min\{1, B_{\mathcal{P},\sigma}R^{\sigma}/(2M)\}. \end{aligned} \tag{291}$$

The induced model for the distinct count is exactly $r + K_{\mathcal{Q}}$, with independent Bernoulli probabilities $1/p$ for $p \in \mathcal{Q}$. The valuation mask uses q_0, while the count mask uses q.

Every small-prime exception is included. At a prime $p \mid q$, write $a = v_p(A)$ and $b = v_p(q)$. If $a < b$, its valuation is forced to equal a. If $a \ge b$, its model is $b + G_p$, where $\mathbb{P}(G_p = j) = (1 - 1/p)p^{-j}$ for $j \ge 0$. At $p \nmid q$ it is G_p. These model coordinates are independent. This includes $p = 2$, $M = 1$, and $g > 1$.

Proof. Reduction by g gives $F_0(n) = A_0 + q_0 n$, with $\gcd(A_0, q_0) = 1$ and $1 \le F_0(n) \le R$. In particular $R = A_0 + q_0(M - 1) \ge M$. No prime dividing q_0 divides any $F_0(n)$. For $d \in \mathcal{H}_{\mathcal{V}}$ let $C_d = \#\{0 \le n < M : d \mid F_0(n)\}$. Since $\gcd(d, q_0) = 1$, its congruence is one residue class modulo d, so $|C_d/M - 1/d| \le 1/M$. If $d > R$, then $C_d = 0$. Hence for every such d,

$$|C_d/M - 1/d| \le (R^{\sigma}/M)d^{-\sigma}. \tag{292}$$

For $d \le R$ use $1 \le (R/d)^{\sigma}$. For $d > R$ use $M \le R \le R^{\sigma}d^{1-\sigma}$.

For each $a \in \mathcal{H}_{\mathcal{V}}$ the exact identity is

$$\mathbb{P}(A_{\mathcal{V}}(F_0(n)) = a) = \sum_{\substack{e \in \mathcal{H}_{\mathcal{V}} \\ e \text{ squarefree} \\ ae \le R}} \mu_{\mathrm{Mob}}(e)C_{ae}/M.$$

There is no restriction $\gcd(a, e) = 1$. Its model counterpart is the absolutely convergent sum $c_{\mathcal{V}}/a = \sum_e \mu_{\mathrm{Mob}}(e)/(ae)$ over all squarefree $e \in \mathcal{H}_{\mathcal{V}}$. Subtract these complete signed identities. Only their rounding and omitted-tail remainder is bounded by (292). Summing that bound over all a and squarefree e gives

$$\frac{R^{\sigma}}{M} \sum_{a \in \mathcal{H}_{\mathcal{V}}} a^{-\sigma} \sum_{\substack{e \in \mathcal{H}_{\mathcal{V}} \\ e \text{ squarefree}}} e^{-\sigma} = \frac{R^{\sigma}}{M} \prod_{p \in \mathcal{V}} \frac{1 + p^{-\sigma}}{1 - p^{-\sigma}} \le \frac{R^{\sigma}}{M} B_{\mathcal{P},\sigma}.$$

All series converge since $\sigma > \alpha$. Multiplication by $A_{\mathcal{P}}(g)$ is injective and preserves both distances. The identity $A_{\mathcal{P}}(F(n)) = A_{\mathcal{P}}(g)A_{\mathcal{V}}(F_0(n))$ proves (291).

For the distinct count, primes dividing g are already present. A prime dividing q but not g never occurs. All other coordinates are the geometric presence indicators, with probability $1/p$. Thus both the arithmetic statistic and its model equal r plus the count from $\mathcal{Q}$. Comparing $v_p(A)$ and $v_p(q)$ after division by g gives the full local classification, with no omitted exceptional prime. □

Lemma 21.132 (Uniform positive supply after a moving finite filter)**.** *For finite $S \subset \mathcal{P}$ put $b_k^S = \mathbb{P}(K_{\mathcal{P}\setminus S} = k)$ and $q_k^S = \mathbb{P}(K_{\mathcal{P}\setminus S} \geq k)$. If $|S| \leq J$, then*

$$b_k^S \geq c_{\mathcal{P}} p_{J+k}^{-k} \ (k \geq 1), \quad b_0^S \geq c_{\mathcal{P}}, \quad b_k^S \leq q_k^S \leq q_k \ (k \geq 0). \tag{293}$$

For $S = \mathcal{P} \cap \{p : p \mid q\}$, integer $1 \leq q \leq Y$, fixed $c > 0$ and $\epsilon > 0$, a constant $C_\epsilon > 0$ gives

$$\min_{0 \leq k \leq \lfloor c \log Y / \log\log Y \rfloor} b_k^S \geq C_\epsilon Y^{-c/\alpha - \epsilon} \tag{294}$$

for all sufficiently large Y, uniformly in q. Individual point masses need not be monotone under filtering.

Proof. The ith surviving prime t_i is at most p_{J+i}. The event in which exactly its first k coordinates are one has probability $c_{\mathcal{P}\setminus S}/\prod_{i \leq k}(t_i - 1)$, and $c_{\mathcal{P}\setminus S} \geq c_{\mathcal{P}} > 0$. This proves the lower bounds. Bernoulli coupling with the original law proves the tail upper bound.

For the second assertion, $J = K_{\mathcal{P}}(q) \leq H_{\mathcal{P}}(Y) = O(\log Y / \log\log Y)$ by (280). Put $L = \lfloor c \log Y / \log\log Y \rfloor$. Bounding $J + L$ by a fixed multiple of $\log Y / \log\log Y$ and using (274) gives $L \log p_{J+L} \leq (c/\alpha + o(1)) \log Y$, uniformly. For $1 \leq k \leq L$, $p_{J+k}^k \leq p_{J+L}^L$; the zero-rank bound is independent of J. This proves (294). □

Theorem 21.133 (Uniform affine recovery of rare profiles)**.** *Fix $0 < h \leq 1$, $0 \leq \gamma < 1$, $h + \gamma \leq 1$, and $0 < \rho \leq 1$, with $h > (1-\gamma)\alpha$. Use* (290) *and suppose*

$$\begin{gathered} M \geq \rho Y^h, \quad g \geq Y^\gamma, \quad 0 < c < \alpha\{h - (1-\gamma)\alpha\}, \\ 0 < \delta < h - (1-\gamma)\alpha - c/\alpha. \end{gathered} \tag{295}$$

Write $b_{A,q,M}(j), q_{A,q,M}(j)$ for the empirical point mass and tail of $K_{\mathcal{P}}(F(n))$. With $S = \mathcal{P} \cap \{p : p \mid q\}$ one has

$$\max_{0 \leq k \leq \lfloor c \log Y / \log\log Y \rfloor} \left(\left| \frac{b_{A,q,M}(r+k)}{b_k^S} - 1 \right| + \left| \frac{q_{A,q,M}(r+k)}{q_k^S} - 1 \right| \right) \ll Y^{-\delta}. \tag{296}$$

The constant and starting point depend only on $\mathcal{P}, h, \gamma, \rho, c, \delta$. For every rank sequence $k_Y \sim c \log Y / \log\log Y$ in this strict range, both empirical probabilities at $r + k_Y$ equal $Y^{-c/\alpha+o(1)}$, uniformly over these families.

Proof. For large Y, $M \geq 2$ and hence $q \leq Y$. The full-law error contracts to every count event with allowance $B_{\mathcal{P},\sigma} R^\sigma / M$. The independent positive supply and full negative allowance are

$$\begin{aligned} b_k^S &\geq C_\epsilon Y^{-c/\alpha-\epsilon}, \\ b_{A,q,M}(r+k) &\geq C_\epsilon Y^{-c/\alpha-\epsilon} - B_{\mathcal{P},\sigma} R^\sigma / M. \end{aligned} \tag{297}$$

Since $R \leq Y^{1-\gamma}$ and $M \geq \rho Y^h$, the relative error is at most a constant times $Y^{(1-\gamma)\sigma-h+c/\alpha+\epsilon}$. Choose fixed $\alpha < \sigma < 1$ and $\epsilon > 0$ close enough to α and zero that this exponent is below $-\delta$. The strict range permits this choice. The same division works for tails since $q_k^S \geq b_k^S$.

At $k_Y \sim c \log Y / \log\log Y$, the positive supply and the original tail upper bound (278) give $b_{k_Y}^S, q_{k_Y}^S = Y^{-c/\alpha+o(1)}$ uniformly in $q \leq Y$. In the upper bound take the limit first and then $\sigma \downarrow \alpha$. Choose a slightly larger permitted c' to contain these ranks, and apply the relative estimate. □

Theorem 21.134 (Exact optimal filter under a product budget)**.** *For integer $B \geq 1$ put $J = H_{\mathcal{P}}(B)$ and $S_* = \{p_1, \ldots, p_J\}$, empty when $J = 0$. Among all finite $S \subset \mathcal{P}$ with $\prod_{p \in S} p \leq B$, the set S_* minimizes $\mathbb{P}(K_{\mathcal{P}\setminus S} \geq k)$ for every integer $k \geq 0$ simultaneously. For each $k \geq 1$ it is the unique minimizer. This is a tail-order assertion, not minimization of each point mass.*

Proof. Every feasible S has $|S| \leq J$, since any j primes of $\mathcal{P}$ have product at least $\prod_{i \leq j} p_i$. Its successive surviving primes satisfy $t_i \leq p_{J+i}$. Using independent uniforms U_i, couple the indicators $\mathbf{1}_{\{U_i \leq 1/t_i\}}$ and $\mathbf{1}_{\{U_i \leq 1/p_{J+i}\}}$. Their sums are finite almost surely and ordered. The prefix S_* is feasible by the exact primorial inverse, so this proves every tail inequality.

If $S \neq S_*$, at least one coordinate satisfies $t_i < p_{J+i}$; otherwise the survivor sets and hence the deleted sets coincide. At that coordinate there is a positive probability of selecting only the larger indicator. Independently, the other coordinates of the smaller law have positive probability of summing to $k-1$: select any $k-1$ of them and set all others to zero. The infinite product of their failure probabilities is positive. On this event the smaller total is $k-1$ and the larger at least k. Thus the tail inequality is strict for every $k \geq 1$. For $k = 0$ all tails are one. □

Theorem 21.135 (Sharp budget threshold for erasing a rare-event exponent)**.** *Let $X \to \infty$ through integers, let $B_X \geq 3$ be integers, and suppose*

$$\frac{\log\log B_X}{\log\log X} \longrightarrow s, \qquad 0 \leq s < \infty.$$

Let $k_X \sim c \log X / \log\log X$ be positive integers, where $c > 0$ is fixed. Take minima over the finite nonempty family $\mathfrak{S}_X = \{S \subset \mathcal{P} : \prod_{p\in S} p \leq B_X\}$. Then

$$\begin{aligned} \frac{-\log \min_{S\in\mathfrak{S}_X} b_{k_X}^S}{\log X} &\longrightarrow c\Phi_\alpha(s), \\ \frac{-\log \min_{S\in\mathfrak{S}_X} q_{k_X}^S}{\log X} &\longrightarrow c\Phi_\alpha(s), \\ \Phi_\alpha(s) &= 1 + (1/\alpha - 1)\max\{1, s\}. \end{aligned} \tag{298}$$

The prefix filter $S_* = \{p_1, \dots, p_{H_{\mathcal{P}}(B_X)}\}$ attains both limiting exponents. For $s \leq 1$, every feasible filter has point mass and tail $X^{-c/\alpha+o(1)}$, uniformly over all filters. In particular every polynomial product budget preserves this exponent. Some larger budgets also have $s = 1$; polynomial growth is not claimed to be a sharp boundary within that plateau.

Proof. Every prime in a feasible nonempty set is at most B_X. Thus the family is finite; it contains the empty set, and all of its count point masses are positive. Put $J = H_{\mathcal{P}}(B_X)$ and $k = k_X$. The primorial inverse gives

$$\frac{\log(1+J)}{\log\log X} \longrightarrow s. \tag{299}$$

For $s > 0$, use $H_{\mathcal{P}}(B) \sim \alpha \log B / \log\log B$. For $s = 0$, the bound $0 \leq J \leq \log B_X / \log 2$ also covers bounded budgets.

If $s \leq 1$, then $\log(J+k)/\log\log X \to 1$. The uniform bound $b_k^S \geq c_{\mathcal{P}} p_{J+k}^{-k}$ gives probability at least $X^{-c/\alpha+o(1)}$. The unfiltered bound $q_k^S \leq q_k = X^{-c/\alpha+o(1)}$ gives the matching upper probability. Both bounds are independent of S, proving all plateau assertions.

If $s > 1$, then $J/k \to \infty$. For every feasible filter, its first J surviving primes are at most p_{2J}. Retain all ways of selecting k of these J coordinates in the positive coefficient expansion:

$$b_k^S \geq c_{\mathcal{P}} \binom{J}{k} p_{2J}^{-k} \geq c_{\mathcal{P}} (J/k)^k p_{2J}^{-k}. \tag{300}$$

Indeed, each factor $(J-i)/(k-i)$ in the binomial product is at least J/k for $0 \leq i < k$. Since $\log p_{2J} = (1/\alpha + o(1)) \log J$, $\log J = (s+o(1)) \log\log X$, and

$\log k = (1+o(1))\log\log X$, this gives upper decay exponent $c\{1+s(1/\alpha-1)\}$ uniformly over all filters.

For the prefix filter the remaining mean satisfies

$$\lambda_J = \sum_{j>J} p_j^{-1} = J^{\,1-1/\alpha+o(1)}.$$

For the upper bound compare with $\sum_{j>J} j^{-1/\alpha+\epsilon}$ for fixed sufficiently small $\epsilon > 0$. For the lower bound retain $J < j \le 2J$ and use J/p_{2J}. Letting $\epsilon \downarrow 0$ proves the exponent. Its factorial moment gives

$$q_k^{S_*} \le \mathbb{E}\binom{K_{\mathcal{P}\setminus S_*}}{k} \le \lambda_J^k/k!.$$

Monotone convergence justifies the infinite nonnegative expansion. The estimate $\log k! = k\log k - k + O(\log k)$ now makes the lower decay exponent of this tail at least $c\{1+s(1/\alpha-1)\}$. Since $b_k^{S_*} \le q_k^{S_*}$, the prefix supplies the matching upper bounds for both minimum probabilities. Together with the uniform positive supply this proves both limits. Its exact tail minimization was proved above. □

Corollary 21.136 (An optimal disappearing law still carries the zero index)**.** *Let $\mathcal{P}$ satisfy* (274) *and fix $0 < b < 1-\alpha$. For sufficiently large integer X put*

$$B_X = \lfloor X^b \rfloor, \quad J_X = H_{\mathcal{P}}(B_X), \quad Q_X = \prod_{j\le J_X} p_j, \quad M_X = 1 + \left\lfloor \frac{X-1}{Q_X} \right\rfloor,$$
$$e_X = \frac{1}{M_X}\#\{0 \le n < M_X : K_{\mathcal{P}}(1+Q_X n) \ge 1\}. \tag{301}$$

Let E_X be the minimum nonzero-count probability over all unit residue classes $a \bmod q$ with $1 \le q \le B_X$, sampled uniformly over $\{1 \le m \le X : m \equiv a \pmod q\}$. These finite sample sets are nonempty. Then

$$e_X,\ E_X = \left(\frac{\log X}{\log\log X}\right)^{1-1/\alpha+o(1)}, \tag{302}$$
$$\mathrm{Law}(K_{\mathcal{P}}(1+Q_X n)) \longrightarrow \delta_0 \quad \textit{in total variation.}$$

Thus the explicit progression is asymptotically optimal for suppressing nonzero counts at this modulus budget, but does not remove their logarithmic exponent.

For the computable greedy deleted class $\mathcal{D}$, take $b = 1/16$ and $\Theta = \Theta_\zeta$. One has

$$\begin{aligned} \lim_{X\to\infty} \frac{\log e_X}{\log\log X} &= \lim_{X\to\infty} \frac{\log E_X}{\log\log X} = 1 - \frac{1}{\Theta}, \\ \text{RH} \quad &\iff \quad \text{for every } \epsilon > 0,\ \ e_X \ll_\epsilon (\log X)^{-1+\epsilon}. \end{aligned} \tag{303}$$

The same equivalence holds with E_X. For the binary rank selector $\kappa_{1/20}(X)$ of (286)*, the actual progression point mass and tail moreover satisfy*

$$\begin{aligned} \mathbb{P}_X(K_{\mathcal{D}} = \kappa_{1/20}(X)) &= X^{-1/(20\Theta)+o(1)}, \\ \mathbb{P}_X(K_{\mathcal{D}} \geq \kappa_{1/20}(X)) &= X^{-1/(20\Theta)+o(1)}. \end{aligned} \tag{304}$$

Both have relative error $O(X^{-1/200})$ from the model with its first J_X deleted-prime coordinates removed. In this specialization, integer X and the integer root $B_X = \lfloor X^{1/16} \rfloor$ give computable finite cutoffs, progressions and counts.

Proof. Use representatives $1 \leq a \leq q$, $\gcd(a, q) = 1$. For $q \leq X^b$, the sample size is $M_a = X/q + O(1)$, hence at least $X/(2q)$ eventually, uniformly in a, q. The largest sample is at most X and the gcd shift is zero. For any fixed $\alpha < \sigma < 1 - b$, the full affine law gives the uniform event-probability error

$$O_{\mathcal{P},\sigma}(X^{\sigma+b-1}) \tag{305}$$

from each masked Bernoulli model.

Its erased set is $S = \mathcal{P} \cap \{p : p \mid q\}$, with prime product at most $q \leq B_X$. The optimality theorem minimizes the model probability of a nonzero count at $S_* = \{p_1, \ldots, p_{J_X}\}$. This filter is realized by the allowed modulus $Q_X \leq B_X$ and residue one. Thus both E_X and e_X differ from this optimal model probability by (305). Taking a minimum changes an additive uniform error by at most that error; no exact finite empirical optimizer is asserted.

The primorial inverse gives $J_X \sim \alpha b \log X / \log\log X$. Writing $\lambda_J = \sum_{j>J} p_j^{-1}$, one has

$$1 - \prod_{j>J} (1 - 1/p_j) \sim \lambda_J = J^{1-1/\alpha+o(1)}.$$

Indeed, $\lambda_J \to 0$ and $\sum_{j>J} p_j^{-2} \leq p_{J+1}^{-1} \lambda_J = o(\lambda_J)$. Expand the logarithm of the product and then its exponential. This positive logarithmic-scale mass dominates the polynomial error (305), proving the two asymptotics in

(302). The total variation distance of a nonnegative integer law from δ_0 is its probability of being nonzero, which also proves the asserted convergence.

For $\mathcal{D}$, the proved range $1/2 \le \Theta \le 7/8$ makes $b = 1/16 < 1-\Theta$ admissible. The logarithmic limits follow because $\log(\log X/\log\log X)/\log\log X \to 1$. Their common value is at least -1, with equality precisely when $\Theta = 1/2$, equivalently RH. The upper bounds for every $\epsilon > 0$ force the limit to be at most -1, and equality implies all those bounds. This proves both equivalences.

The model filter has polynomial budget, so the phase theorem preserves its rare exponent. For the relative error apply the affine rare-recovery theorem: the largest sample satisfies $X - Q_X < Y_X \le X$, so $Y_X \sim X$, and $M_X \ge \frac{1}{2} Y_X^{15/16}$ eventually. Take $h = 15/16$, $\gamma = 0$, $\rho = 1/2$. For $c = 1/20$ the available margin is

$$\frac{15}{16} - \Theta - \frac{1}{20\Theta} \ge \frac{15}{16} - \frac{7}{8} - \frac{1/20}{7/8} = \frac{3}{560} > \frac{1}{200}.$$

The first inequality holds since $t \mapsto 15/16 - t - 1/(20t)$ is decreasing on $[1/2, 7/8]$. Choose $c' > 1/20$ close enough that this margin still exceeds $1/200$. The binary rank is asymptotic to $(1/20)\log X/\log\log X$ and lies in the c' range. The relative error and both rare exponents follow. The integer sixteenth root and finite tests for the computable class implement the sample without a decision about a real cutoff. □

These full affine and optimization results have ordinary proofs; they are not advertised as Lean declarations. The RH equivalence is not an independent arithmetic upper estimate. The installed $\Theta \le 7/8$ gives the upper exponent $-1/7$ for e_X, E_X on the logarithmic scale; RH requires -1. No improvement of the certified zero-free boundary is claimed.

21.58 Conditional prime localization and reconstruction from the zero bin

Continue with the explicit progression of (301), for a fixed $0 < b < 1 - \alpha$. Let $\mathbb{P}_X$ be uniform measure on its M_X integers and put

$$C_X = A_{\mathcal{P}}(1 + Q_X n), \qquad K_X = K_{\mathcal{P}}(1 + Q_X n), \qquad b_{X,k} = \mathbb{P}_X(K_X = k).$$

Thus $C_X > 1$ is exactly the event $K_X \geq 1$. Abbreviate $J = J_X$ and write

$$\begin{gathered}\lambda_J = \sum_{j>J} p_j^{-1}, \quad c_J = \prod_{j>J}(1-1/p_j), \quad W_J = \sum_{j>J}(p_j-1)^{-1}, \\ S_{2,J} = \sum_{j>J}(p_j-1)^{-2}, \qquad v = 1/\alpha - 1 > 0.\end{gathered} \tag{306}$$

All sums converge and all displayed normalizations are positive. The previous proofs give

$$\begin{gathered}J \sim \alpha b \frac{\log X}{\log\log X}, \quad p_{J+1} = J^{1/\alpha+o(1)}, \quad \lambda_J = J^{-v+o(1)}, \quad 1 - c_J \sim \lambda_J, \\ d_{\mathrm{TV}}(\mathrm{Law}(C_X), \mathrm{Law}(Z_J)) \leq \Delta_X, \qquad \Delta_X \ll X^{-\xi}, \quad \xi = 1 - b - \sigma > 0,\end{gathered} \tag{307}$$

where $\alpha < \sigma < 1 - b$ is fixed and $Z_J = \prod_{j>J} p_j^{G_{p_j}}$ has the full independent geometric law. The last estimate is the full-component estimate of Theorem 21.131, not just convergence of count distributions. Since $\Delta_X = o(\lambda_J)$, both the zero and nonzero empirical bins are positive eventually. Classical Poisson approximation is background; the relative growing-rank statement and its arithmetic normalization below are proved directly.

Theorem 21.137 (A growing rare profile determined by the zero bin)**.** *Define the parameter from the actual finite sample alone by*

$$\widehat{\lambda}_X = -\log b_{X,0}. \tag{308}$$

It is finite and positive for large X, and $\widehat{\lambda}_X \sim \lambda_J$. For every fixed $0 < \epsilon < 1/2$,

$$\max_{0 \leq k \leq \lfloor J^{1/2-\epsilon} \rfloor} \left| \frac{b_{X,k}}{e^{-\widehat{\lambda}_X} \widehat{\lambda}_X^k / k!} - 1 \right| \leq J^{-2\epsilon+o(1)}. \tag{309}$$

In particular, the whole displayed growing collection of rare point masses is recoverable to relative accuracy from the zero bin. This asserts a relative point-mass estimate, beyond the unconditional convergence of both laws to δ_0.

Proof. For the model let $\beta_{J,k} = \mathbb{P}(K_{\mathcal{P}\setminus\{p_1,\dots,p_J\}} = k)$. With $w_j = (p_j - 1)^{-1}$ for $j > J$ its generating function is $c_J \prod_{j>J}(1 + w_j z)$. Normalize the positive weights w_j to a probability law by division by W_J. In k independent draws from that law, the probability that all labels are distinct is exactly

$k!\beta_{J,k}/(c_J W_J^k)$. The union bound on collisions of pairs therefore proves the finite quantitative inequality

$$0 \le 1 - \frac{k!\beta_{J,k}}{c_J W_J^k} \le \binom{k}{2} \frac{S_{2,J}}{W_J^2} \qquad (k \ge 0). \tag{310}$$

For $k = 0$ the middle expression is zero. The countable formula follows either by the independent-draw construction or by monotone convergence from finite supports.

For large J, $p_{J+1} \to \infty$, so

$$W_J = \lambda_J(1 + O(p_{J+1}^{-1})), \qquad S_{2,J} \ll p_{J+1}^{-1}\lambda_J.$$

Put $t_J = -\log c_J$. Expansion of $-\log(1-x)$, with $0 \le x \le 1/p_{J+1}$, also gives

$$t_J = \lambda_J + O(p_{J+1}^{-1}\lambda_J), \qquad |W_J - t_J| \ll p_{J+1}^{-1}\lambda_J.$$

The full sampling bound gives $|b_{X,0} - c_J| \le \Delta_X$. As both terms tend to one, the logarithm is Lipschitz on their common eventual interval $[1/2, 1]$, whence

$$|\widehat{\lambda}_X - t_J| \ll \Delta_X, \quad \widehat{\lambda}_X \sim \lambda_J, \quad \left|\frac{W_J}{\widehat{\lambda}_X} - 1\right| \ll p_{J+1}^{-1} + \Delta_X/\lambda_J.$$

Furthermore $c_J/b_{X,0} = 1 + O(\Delta_X)$.

Set $L = \lfloor J^{1/2-\epsilon} \rfloor$. The collision allowance in (310) is at most

$$L^2 S_{2,J}/W_J^2 \le J^{1-2\epsilon} J^{-1/\alpha+v+o(1)} = J^{-2\epsilon+o(1)}.$$

The normalization error after raising to any $k \le L$ is bounded by a constant times $L(p_{J+1}^{-1} + \Delta_X/\lambda_J)$. Its first part is $J^{1/2-\epsilon-1/\alpha+o(1)} = o(J^{-2\epsilon})$, since $\epsilon < 1/2$ and $\alpha < 1$; its second part decays faster than every fixed power of J. The elementary estimate $|(1+u)^k - 1| \le e^{2k|u|} - 1$ for $|u| \le 1/2$ justifies that normalization bound. Using $e^{-\widehat{\lambda}_X} = b_{X,0}$, these estimates prove the claimed relative comparison first for the model masses $\beta_{J,k}$.

It remains to transfer relative, rather than absolute, accuracy. Let $\pi_{X,k} = e^{-\widehat{\lambda}_X}\widehat{\lambda}_X^k/k!$. Since $\widehat{\lambda}_X = J^{-v+o(1)} < 1$ eventually, its minimum on $0 \le k \le L$ is at L, and

$$-\log \pi_{X,L} = O(L \log J) = o(\log X).$$

Thus $\min_{k \le L} \pi_{X,k} = X^{-o(1)}$. The full sampling error $|b_{X,k} - \beta_{J,k}| \le \Delta_X$ divided by this minimum is at most $X^{-\xi+o(1)}$, again smaller than every fixed negative power of J. This completes (309). At $k = 0$ the empirical ratio is exactly one. □

Theorem 21.138 (The conditional surviving component is one localized prime)**.** *Under the same hypotheses,*

$$\mathbb{P}_X(C_X \text{ is prime} \mid C_X > 1) = 1 - O(J^{-v+o(1)}). \tag{311}$$

Moreover, for every fixed $\eta > 0$, put $t = \min\{1, \alpha\eta/2\} > 0$. For all sufficiently large integer X,

$$\mathbb{P}_X\left(\left|\frac{\log C_X}{\log\log X} - \frac{1}{\alpha}\right| > \eta \,\middle|\, C_X > 1\right) \le J^{-vt+o(1)}. \tag{312}$$

Consequently $\log C_X/\log\log X \to 1/\alpha$ in conditional probability, even though $\mathbb{P}_X(C_X > 1) \to 0$. Prime powers and multiple distinct factors are both included in the error.

Proof. For the full model Z_J, the probability of at least two distinct active prime coordinates is at most $\lambda_J^2/2$, by its second factorial moment. The probability of a repeated prime factor is at most $\sum_{j>J} p_j^{-2} \le p_{J+1}^{-1}\lambda_J$, because $\mathbb{P}(G_p \ge 2) = p^{-2}$ exactly. These exhaust the ways a nontrivial component can fail to be a prime. Since $\mathbb{P}(Z_J > 1) = 1 - c_J \sim \lambda_J$,

$$\mathbb{P}(Z_J \text{ composite} \mid Z_J > 1) \ll \lambda_J + p_{J+1}^{-1} = J^{-v+o(1)}. \tag{313}$$

Here conditioning must be justified quantitatively. If two probability measures μ, ν have total variation at most Δ and $\nu(E) > \Delta$, then

$$d_{\mathrm{TV}}(\mu(\cdot \mid E), \nu(\cdot \mid E)) \le \frac{2\Delta}{\nu(E) - \Delta}. \tag{314}$$

Indeed, for each measurable $B \subseteq E$, subtract $\mu(B)/\mu(E)$ and $\nu(B)/\nu(E)$, keeping the numerator difference and normalization difference separately. Each contributes at most $\Delta/\mu(E)$, and $\mu(E) \ge \nu(E) - \Delta$. Apply this with the full component laws and $E = \{a > 1\}$. The conditional error is $O(\Delta_X/\lambda_J) = X^{-\xi+o(1)}$, smaller than every fixed negative power of J. This proves (311) for the actual arithmetic samples.

Every nontrivial model component is at least p_{J+1}. The same is exactly true for every actual sample, since $1 + Q_X n$ is coprime to all of $p_1, \dots, p_J$. As $\log p_{J+1}/\log\log X \to 1/\alpha$, the lower size exception in (312) is eventually empty.

For the upper exception retain the full composite allowance and consider the remaining case of a single prime. Let $T = \lfloor J^{1+t} \rfloor$. The unconditional

probability that this prime has label $j > T$ is at most λ_T. The already proved mean exponent gives

$$\frac{\lambda_T}{1 - c_J} = J^{-vt+o(1)}.$$

If $J < j \le T$, then

$$\frac{\log p_j}{\log\log X} \le \frac{1+t}{\alpha} + o(1) \le \frac{1}{\alpha} + \frac{\eta}{2} + o(1) < \frac{1}{\alpha} + \eta$$

eventually, by the choice of t. Thus the model upper exception is bounded by the sum of $J^{-v+o(1)}$ and $J^{-vt+o(1)}$. Since $t \le 1$, the latter controls both. Transfer once more with (314); its polynomially decaying error is negligible. This proves the full conditional bound with all multiplicities retained. □

Corollary 21.139 (Computable conditional quantiles recover the zero supremum)**.** *For fixed $0 < \tau < 1$ define the lower conditional quantile*

$$m_\tau(X) = \min\{a \in \{1, \dots, X\} : \mathbb{P}_X(C_X \le a \mid C_X > 1) \ge \tau\}. \tag{315}$$

It exists for all large integer X and satisfies

$$\frac{\log m_\tau(X)}{\log\log X} \longrightarrow \frac{1}{\alpha}, \tag{316}$$

uniformly for τ in any fixed compact subinterval of $(0,1)$.

For $\mathcal{P} = \mathcal{D}$, $b = 1/16$ and each rational $\tau \in (0,1)$, this quantile is a computable integer. Its limit is $1/\Theta_\zeta$. In particular, for the conditional median $m_{1/2}(X)$,

$$\begin{aligned} \text{RH} &\iff m_{1/2}(X) = (\log X)^{2+o(1)} \\ &\iff \text{for every } \epsilon > 0,\ \ m_{1/2}(X) \gg_\epsilon (\log X)^{2-\epsilon}. \end{aligned} \tag{317}$$

Similarly $\Theta_\zeta \le 3/4$ is equivalent to the lower bounds $(\log X)^{4/3-\epsilon}$ for every $\epsilon > 0$.

Proof. The conditioning event has positive probability eventually, and the actual component always lies in $\{1, \dots, X\}$. Hence the finite set in the quantile definition is nonempty. For every $\eta > 0$ the localization theorem puts probability $1 - o(1)$ inside the interval $[(\log X)^{1/\alpha-\eta}, (\log X)^{1/\alpha+\eta}]$. For any fixed $0 < \tau_0 \le 1/2$, this brackets every quantile with $\tau_0 \le \tau \le 1 - \tau_0$ eventually. Using slightly smaller interior windows removes any endpoint rounding issue. Take logarithms and let $\eta \downarrow 0$ to prove (316) uniformly.

For rational τ, compare finite integer counts by cross-multiplication and choose the first component value meeting the threshold. The progression uses an integer sixteenth root and the proved computable class $\mathcal{D}$, so no real-cutoff decision is needed. For this class $\alpha = \Theta_\zeta \geq 1/2$, so its quantile exponent is at most two, with equality exactly under RH. The family of median lower bounds forces its exponent to be at least two, and equality implies all those bounds. The same argument with 4/3 proves the 3/4 equivalence. □

These are ordinary proofs of relative probability and conditional localization results, not new Lean declarations. The classical independent-coordinate and collision arguments are explicit inputs, rather than a claim that Poisson approximation itself is new. The new arithmetic conclusions use the full affine sampling budget and its positive rare-event denominator. The installed bound gives median exponent at least 8/7; the 3/4 target requires 4/3 and RH requires two. No independent improvement of that median lower bound is proved, and the certified zero-free boundary remains 7/8.

21.59 Three-bin reconstruction beyond the Poisson collision scale

Retain the fixed thin prime class and the actual integer progression of (301). Use $J, c_J, W_J, \lambda_J, \Delta_X, \xi$ from (307). The square-root scale in the preceding Poisson comparison comes from collisions of labels in the coefficient expansion. We now retain that term rather than requiring it to tend to zero. Nonuniform birthday probabilities and logarithmic expansions are classical; see Mase [25, §§3–5]. The lemma below supplies an explicit uniform remainder for our moving weights. The arithmetic conclusion reconstructs actual sample probabilities from three rational empirical bins. No priority claim for the classical collision expansion is made.

Lemma 21.140 (A uniform pair-collision correction)**.** *Let (a_i) be a finite or countable probability vector, with $a_i \geq 0$, $\sum_i a_i = 1$, and $m = \sup_i a_i \leq 1/4$. Write $\theta_r = \sum_i a_i^r$ for $r \geq 2$. Let D_n be the probability that n independent draws have distinct labels, with $D_0 = 1$. If $n \geq 0$ is an integer and $nm \leq 1/4$,*

then $D_n > 0$ and

$$\begin{aligned} \left|\log D_n + \binom{n}{2}\theta_2\right| &\le 8n^3(\theta_3 + \theta_2^2), \\ \left|\log \frac{D_n}{(1-\theta_2)^{\binom{n}{2}}}\right| &\le 10n^3(\theta_3 + \theta_2^2). \end{aligned} \tag{318}$$

Every logarithm has a positive argument; $1 - \theta_2 \ge 3/4$.

Proof. Let e_j be the elementary symmetric sum of degree j in the a_i, with $e_0 = 1$. Then $D_j = j!e_j$. On the event of j distinct draws, let h_j be the conditional expected total weight of their occupied labels. The next draw avoids those labels with probability $1 - h_j$, so the exact recurrence and pointwise bound are

$$\frac{D_{j+1}}{D_j} = 1 - h_j, \qquad 0 \le h_j \le jm, \qquad h_0 = 0. \tag{319}$$

These statements inductively show $D_j > 0$ through $j = n + 1$: at every required step $jm \le 1/4$. There is no division by a zero coefficient.

Conditional on j distinct labels, an unordered set S of size j has probability $\prod_{i\in S} a_i/e_j$. Writing $e_j^{(i)}$ for the sum omitting label i gives

$$h_j = \frac{\sum_i a_i^2 e_{j-1}^{(i)}}{e_j} \quad (j \ge 1).$$

For $j \ge 2$ put $u_{j-1,i} = \mathbb{P}(i \in S \mid |S| = j-1)$ and $V_{j-1} = \sum_i a_i^2 u_{j-1,i}$. The elementary identity $e_{j-1}^{(i)} = e_{j-1}(1 - u_{j-1,i})$ and the preceding exact recurrence give

$$\begin{aligned} h_j &= \frac{j(\theta_2 - V_{j-1})}{1 - h_{j-1}}, \\ 0 \le u_{j-1,i} &= \frac{a_i e_{j-2}^{(i)}}{e_{j-1}} \le \frac{(j-1)a_i}{1 - h_{j-2}}, \\ 0 \le V_{j-1} &\le \frac{(j-1)\theta_3}{1 - h_{j-2}}. \end{aligned} \tag{320}$$

All sums converge absolutely by nonnegativity and $\sum_i a_i = 1$. The same formulas follow on finite supports first and then by monotone convergence if the support is infinite. For $j = 1$, directly $h_1 = \theta_2$.

Every denominator in the required range is at least $3/4$. The recurrence, with its negative term retained, implies $h_j \le (4/3)j\theta_2$. Subtracting the leading term before estimating gives, for $j \ge 2$,

$$h_j - j\theta_2 = \frac{j(\theta_2 h_{j-1} - V_{j-1})}{1 - h_{j-1}}, \qquad |h_j - j\theta_2| \le 2j^2(\theta_2^2 + \theta_3).$$

The latter bound also holds at $j = 0, 1$. For $0 \le h_j \le 1/4$, $|\log(1 - h_j) + h_j| \le h_j^2$, and $h_j^2 \le 2j^2\theta_2^2$. Summing these two remainders in $\log D_n = \sum_{j=0}^{n-1} \log(1 - h_j)$ and using $\sum_{j<n} j = \binom{n}{2}$ proves the first bound, with the displayed constant eight (and room to spare). Finally $|-\log(1-\theta_2) - \theta_2| \le \theta_2^2/[2(1-\theta_2)] \le \theta_2^2$. Its extra contribution is at most $n^2\theta_2^2/2$; for $n \ge 1$ this is absorbed by the second displayed constant. For $n = 0$ both assertions are exact. □

Theorem 21.141 (Rational reconstruction from the first three empirical bins)**.** *For the actual progression probabilities $b_{X,k}$ define*

$$\widehat{W}_X = \frac{b_{X,1}}{b_{X,0}}, \qquad \widehat{q}_X = \frac{2b_{X,0}b_{X,2}}{b_{X,1}^2}, \qquad \mathcal{R}_{X,k} = \frac{b_{X,0}\widehat{W}_X^k}{k!}\widehat{q}_X^{\binom{k}{2}}. \tag{321}$$

For all sufficiently large integer X, these quantities are positive and $0 < \widehat{q}_X < 1$. For every fixed $0 < \epsilon < 2/3$,

$$\max_{0 \le k \le \lfloor J^{2/3-\epsilon} \rfloor} \left| \frac{b_{X,k}}{\mathcal{R}_{X,k}} - 1 \right| \le J^{-3\epsilon + o(1)}. \tag{322}$$

The reference matches $b_{X,0}, b_{X,1}, b_{X,2}$ exactly. For a computable $\mathcal{P}$ and a computable integer budget, it is rational and computable from the first three finite counts and the integer input k. In particular this applies to $\mathcal{D}$ with $b = 1/16$. The reference need not sum to one over all k; the conclusion is the stated uniform relative estimate on its finite rank range.

Proof. For $j > J$ set $a_j = (p_j - 1)^{-1}/W_J$ and $\theta_{r,J} = \sum_{j>J} a_j^r$. Their maximum and second moment have the exact logarithmic indices

$$m_J = J^{-1+o(1)}, \qquad \theta_{2,J} = J^{-1+o(1)}, \qquad \theta_{3,J} + \theta_{2,J}^2 \le J^{-2+o(1)}. \tag{323}$$

Indeed $m_J = (p_{J+1} - 1)^{-1}/W_J = J^{-1+o(1)}$. The upper bounds follow from $\theta_{2,J} \le m_J$ and $\theta_{3,J} \le m_J^2$. For the matching second-moment lower bound retain $J < j \le 2J$:

$$\theta_{2,J} \ge \frac{J(p_{2J} - 1)^{-2}}{W_J^2} = J^{-1+o(1)}.$$

Here $W_J \sim \lambda_J = J^{1-1/\alpha+o(1)}$, as proved in the preceding subsection.

Write $\beta_{J,k}$ for the model count masses. The exact distinct-label coefficient formula gives $\beta_{J,k} = c_J W_J^k D_k/k!$. Put $q_J = 1 - \theta_{2,J}$ and

$$R^0_{J,k} = c_J W_J^k q_J^{\binom{k}{2}}/k!.$$

For $L = \lfloor J^{2/3-\epsilon} \rfloor$, $Lm_J \to 0$. The collision lemma applies uniformly to $0 \le k \le L$:

$$\beta_{J,k} = R^0_{J,k} e^{E_{J,k}}, \qquad \max_{k\le L} |E_{J,k}| \le 10L^3(\theta_{3,J} + \theta_{2,J}^2) \le J^{-3\epsilon+o(1)}. \tag{324}$$

The exact first three model coefficients are

$$\beta_{J,0} = c_J, \quad \beta_{J,1} = c_J W_J, \quad \beta_{J,2} = \tfrac{1}{2} c_J W_J^2 q_J.$$

Each empirical coefficient differs from its model by at most Δ_X. Since $c_J \to 1$ and $W_J \sim \lambda_J$, these identities prove

$$\begin{gathered} b_{X,0} = c_J(1 + O(\Delta_X)), \qquad \widehat{W}_X/W_J = 1 + O(\Delta_X/\lambda_J), \\ \widehat{q}_X - q_J = O(\Delta_X/\lambda_J^2). \end{gathered} \tag{325}$$

In particular $b_{X,0}, b_{X,1}, b_{X,2} > 0$ eventually. The error in $\widehat{q}_X$ is polynomially small in X, whereas $1 - q_J = \theta_{2,J} = J^{-1+o(1)}$. Thus $0 < \widehat{q}_X < 1$ eventually, including the strict upper inequality.

All factors in the two references are positive. Taking their logarithmic ratio and charging both growing powers in (321) gives

$$\max_{k\le L} \left| \log \frac{\mathcal{R}_{X,k}}{R^0_{J,k}} \right| \ll \Delta_X + L\Delta_X/\lambda_J + L^2\Delta_X/\lambda_J^2 = X^{-\xi+o(1)}.$$

Also uniformly on this range,

$$-\log R^0_{J,k} \le O(1 + L\log J + L^2\theta_{2,J}) = o(\log X).$$

Use $J \asymp \log X/\log\log X$ and $2/3 - \epsilon < 1$ for the first growing term; the second is at most $J^{1/3-2\epsilon+o(1)}$. Consequently both reference minima are at least $X^{-o(1)}$.

For clarity the complete relative consumer, including the signed arithmetic discrepancy, is

$$\begin{aligned} \frac{b_{X,k}}{\mathcal{R}_{X,k}} - 1 = \frac{R^0_{J,k}}{\mathcal{R}_{X,k}}(e^{E_{J,k}} - 1) + \left(\frac{R^0_{J,k}}{\mathcal{R}_{X,k}} - 1 \right) \\ + \frac{b_{X,k} - \beta_{J,k}}{\mathcal{R}_{X,k}}. \end{aligned} \tag{326}$$

The first term is bounded by $J^{-3\epsilon+o(1)}$ using the uniform logarithmic remainder. The other two are $X^{-\xi+o(1)}$, since $|b_{X,k} - \beta_{J,k}| \le \Delta_X$ and the reference minimum is $X^{-o(1)}$. This proves (322). Equivalently the positive model supply is $R^0_{J,k} e^{-|E_{J,k}|}$ and the negative sampling allowance is Δ_X; all empirical masses on the range are therefore positive eventually. Substitution at $k = 0, 1, 2$ proves exact matching of the three bins. Finally all displayed empirical parameters and reference values use only rational operations, integer powers and factorials. Finite enumeration proves the computability assertion on the stated decidable integer domains. □

Corollary 21.142 (A genuine transition for the uncorrected Poisson law)**.** *Let* $\widehat{\lambda}_X = -\log b_{X,0}$ *and* $\pi_{X,k} = e^{-\widehat{\lambda}_X} \widehat{\lambda}_X^k / k!$. *For* $k_X = \lfloor J^u \rfloor$ *one has*

$$\begin{cases} b_{X,k_X}/\pi_{X,k_X} \longrightarrow 1, & 0 < u < 1/2, \\ b_{X,k_X}/\pi_{X,k_X} = \exp[-J^{\,2u-1+o(1)}] \longrightarrow 0, & 1/2 < u < 2/3. \end{cases} \tag{327}$$

In the second range the three-bin reference still has relative error tending to zero. Thus the correction extends the logarithmic rank exponent from $1/2$ *to* $2/3$ *for a directly reconstructed reference. No limit at* $u = 1/2$ *is claimed under the present hypotheses.*

Proof. Below $1/2$ use Theorem 21.137 with $\epsilon = 1/2 - u$. For $1/2 < u < 2/3$, the three-bin theorem applies with $\epsilon = 2/3 - u$ and gives $b_{X,k_X}/\mathcal{R}_{X,k_X} \to 1$. Since $e^{-\widehat{\lambda}_X} = b_{X,0}$,

$$\log \frac{\mathcal{R}_{X,k_X}}{\pi_{X,k_X}} = k_X \log \frac{\widehat{W}_X}{\widehat{\lambda}_X} + \binom{k_X}{2} \log \widehat{q}_X.$$

The already proved parameter bounds give $\widehat{W}_X/\widehat{\lambda}_X = 1 + O(p_{J+1}^{-1} + \Delta_X/\lambda_J)$. Its growing-power contribution tends to zero, since $u < 2/3 < 1/\alpha$. On the other hand, $-\log \widehat{q}_X = \theta_{2,J}(1 + o(1)) = J^{-1+o(1)}$. The second term is therefore $-J^{2u-1+o(1)}$ and tends to minus infinity. The logarithm of the remaining empirical relative error is $o(1)$. This proves the second formula with its sign intact. □

The collision lemma is a quantitative version of classical nonuniform birthday analysis. The arithmetic reconstruction and transition above are full ordinary proofs, not Lean declarations. They improve the simple reconstructed profile, rather than the earlier Euler-model sampling range, which already allowed ranks of order J. They supply no independent estimate changing Θ_ζ; the certified zero-free boundary remains $7/8$.

21.60 A finite-bin hierarchy for every fixed sublinear rank power

The three-bin theorem extends to every fixed interpolation order. The underlying polynomial-degree phenomenon belongs to classical nonuniform birthday analysis; compare Mase [25, Proposition 5.2]. We give the analytic remainder and the full empirical transfer, including all powers of the measured parameters. The conclusion concerns the actual integer samples already fixed in (301).

Lemma 21.143 (Uniform interpolation of collision logarithms)**.** *Let $(a_i)_{i\in I}$ be a probability vector on a finite or countable set, put $m = \sup_{i\in I} a_i > 0$, and let D_n be the probability that n independent draws have distinct labels. Write $F(n) = \log D_n$ when $D_n > 0$. For integers $r \geq 1$ and $n \geq 0$ such that $m \max\{n, r\} \leq 1/16$, every required D_i is positive and*

$$\left| F(n) - \sum_{j=0}^{r} \binom{n}{j} \Delta^j F(0) \right| \leq C_r n^{r+1} m^r, \tag{328}$$

$$C_r = 8^r \left(1 + \frac{2^r r^{r+1}}{r!}\right), \qquad \Delta^j F(0) = \sum_{i=0}^{j} (-1)^{j-i} \binom{j}{i} F(i).$$

Here $F(0) = F(1) = 0$, so the sum may start at $j = 2$. The constant is explicit and depends only on r.

Proof. Put $\theta_\ell = \sum_{i\in I} a_i^\ell$, including $\theta_1 = 1$. Then $0 \leq \theta_\ell \leq m^{\ell-1}$. Use the falling factorial $(x)_s = x(x-1)\cdots(x-s+1)$, with $(x)_0 = 1$. Define polynomials $B_n(z)$ by

$$\begin{aligned} B_0(z) &= 1, \\ B_n(z) &= \sum_{\ell=1}^{n} (-1)^{\ell-1} (n-1)_{\ell-1} \theta_\ell z^{\ell-1} B_{n-\ell}(z) \quad (n \geq 1). \end{aligned} \tag{329}$$

Their formal exponential generating function is

$$\sum_{n\geq 0} B_n(z) \frac{t^n}{n!} = \exp\left(t + \sum_{\ell\geq 2} \frac{(-1)^{\ell-1}}{\ell} \theta_\ell z^{\ell-1} t^\ell \right).$$

The recurrence follows by differentiation in t, or verifies this identity coefficient by coefficient. At $z = 1$ the right side is $\prod_{i\in I}(1 + a_i t)$, so $B_n(1) = D_n$.

The countable product converges locally uniformly since $\sum_i a_i = 1$, and the coefficient identity follows by its finite-product limits. Also $B_n(0) = 1$ for every n.

We first establish a uniform analytic logarithm. Fix an integer $N \geq 1$ and set $\rho_N = (8Nm)^{-1}$. On $|z| \leq \rho_N$, induction in $1 \leq s \leq N$ gives

$$B_s(z) \neq 0, \qquad R_s(z) = B_s(z)/B_{s-1}(z), \qquad |R_s(z) - 1| \leq 1/3. \tag{330}$$

Indeed $R_1 = 1$. After division by B_{s-1}, the terms of the recurrence with $\ell \geq 2$ contain $\ell - 1$ reciprocal earlier ratios. If those ratios have distance at most $1/3$ from one, each reciprocal has modulus at most two. Thus

$$|R_s(z) - 1| \leq \sum_{\ell=2}^{s} (2sm|z|)^{\ell-1} \leq \frac{2sm|z|}{1 - 2sm|z|} \leq \frac{1}{3}.$$

This proves nonvanishing and completes the induction on the closed disc. In particular the logarithms $\log R_s(z) = \sum_{j\geq 1} (-1)^{j-1}(R_s(z) - 1)^j/j$ are analytic there and normalized to zero at $z = 0$. Define $H_N(z) = \sum_{s=1}^{N} \log R_s(z)$. It satisfies $e^{H_N(z)} = B_N(z)$, $H_N(0) = 0$, and

$$\begin{aligned} |\log R_s(z)| &\leq \tfrac{3}{2}|R_s(z) - 1| \leq 4sm|z|, \\ |H_N(z)| &\leq 4N^2 m|z| \leq N/2 \quad (|z| \leq \rho_N). \end{aligned}$$

Nonvanishing on the closed disc also gives analyticity in a neighborhood of it, as needed for Cauchy's estimate. Write $f_\ell(N) = [z^\ell]H_N(z)$. If $Nm \leq 1/16$, then $\rho_N \geq 2$, and Cauchy's estimate and a geometric tail give, for every $r \geq 1$,

$$\left| H_N(1) - \sum_{\ell=1}^{r-1} f_\ell(N) \right| \leq \frac{N}{2} \frac{(8Nm)^r}{1 - 8Nm} \leq 8^r N^{r+1} m^r. \tag{331}$$

On the real interval from zero to one the ratios are positive, so the chosen logarithm has $H_N(1) = \log D_N$. This also proves all positivity assertions in the lemma.

We next prove that each $f_\ell(N)$, initially defined for nonnegative integers, is a polynomial in N of degree at most $\ell + 1$, with $f_\ell(0) = 0$. This step is formal near $z = 0$, where every $B_N(0) = 1$, and therefore does not require a bound on Nm. Write $H_N = \log B_N$ for this normalized formal series. The recurrence is equivalently

$$e^{H_{N+1} - H_N} = 1 + \sum_{s=1}^{N} (-1)^s (N)_s \theta_{s+1} z^s e^{H_{N-s} - H_N}. \tag{332}$$

Induct on ℓ. In the coefficient of z^ℓ on the right, only $s \le \ell$ occurs and every exponential coefficient uses f_j with $j \le \ell - s < \ell$. By induction, $f_j(N-s) - f_j(N)$ has degree at most j. Consequently that exponential coefficient has degree at most $\ell - s$; multiplication by $(N)_s$ gives degree at most ℓ. When $N < s$, extend the coefficient term by zero using $(N)_s = 0$; only the already constructed lower-order polynomials are used in this extension. On the left the coefficient is $f_\ell(N+1) - f_\ell(N)$ plus a polynomial of degree at most ℓ built from lower differences. Thus $f_\ell(N+1) - f_\ell(N)$ is a polynomial of degree at most ℓ for all $N \ge 0$. A discrete antiderivative in the binomial polynomial basis, with value zero at zero, has degree at most $\ell + 1$. This proves the induction, including the first case, where the right coefficient is $-N\theta_2$.

For the desired order r, the polynomial $Q_r(N) = \sum_{\ell=1}^{r-1} f_\ell(N)$ has degree at most r. Let I_r denote polynomial interpolation on $0, 1, \dots, r$; in Newton form, $(I_r F)(n) = \sum_{j=0}^{r} \binom{n}{j} \Delta^j F(0)$. If $n \le r$, it equals $F(n)$ exactly. Otherwise $n \ge r \ge 1$. For every node $i \le r$, as well as at n, (331) bounds $T(i) = F(i) - Q_r(i)$ by $8^r i^{r+1} m^r$, with $T(0) = 0$. The absolute sum of the Lagrange interpolation weights at an integer $n \ge r$ is at most

$$n^r \sum_{i=0}^{r} \frac{1}{i!(r-i)!} = \frac{2^r n^r}{r!}.$$

Since $I_r Q_r = Q_r$, it follows that

$$\begin{aligned} |F(n) - (I_r F)(n)| &\le |T(n)| + |(I_r T)(n)| \\ &\le 8^r n^{r+1} m^r + 8^r r^{r+1} m^r \, \frac{2^r n^r}{r!} \\ &\le C_r n^{r+1} m^r. \end{aligned}$$

The case $n = 0$ is exact. This proves the full stated bound. □

Theorem 21.144 (The empirical finite-bin hierarchy)**.** *Fix an integer $r \ge 1$. For the progression counts $b_{X,i}$ put*

$$A_{X,i} = i!\, b_{X,i}, \qquad \Gamma_{X,j} = \prod_{i=0}^{j} A_{X,i}^{(-1)^{j-i}\binom{j}{i}} \quad (0 \le j \le r), \tag{333}$$

and define

$$\mathcal{R}_{X,k}^{(r)} = \frac{1}{k!} \prod_{j=0}^{r} \Gamma_{X,j}^{\binom{k}{j}} \qquad (k \ge 0). \tag{334}$$

For large X, every measured bin used here is positive, so even the negative integer powers are defined. For each fixed $0 < \epsilon < r/(r+1)$,

$$\max_{0\le k\le \lfloor J^{\,r/(r+1)-\epsilon}\rfloor} \left| \frac{b_{X,k}}{\mathcal{R}^{(r)}_{X,k}} - 1 \right| \le J^{-(r+1)\epsilon+o(1)}. \tag{335}$$

The reference exactly matches bins $0,\dots,r$. It is rational and computable from those $r+1$ finite counts on the decidable integer domains of the preceding theorems. For $r=2$ it is exactly the three-bin reference (321)*. No normalization over all ranks is asserted. All orders and constants in this arithmetic statement are fixed before letting $X\to\infty$; no uniformity in growing $r(X)$ is claimed.*

Proof. Use the normalized prime weights $a_j = (p_j-1)^{-1}/W_J$ for $j > J$. Their maximum is $m_J = J^{-1+o(1)}$ by (323). Let D_k denote their distinct-label probability and $F_J(k) = \log D_k$. The exact model coefficients satisfy $\beta_{J,k} = c_J W_J^k D_k/k!$. For $i \le r$ define the model parameters and reference by replacing every $b_{X,i}$ in (333)–(334) with $\beta_{J,i}$. Denote the reference by $R^{0,(r)}_{J,k}$. The logarithm of $i!\beta_{J,i}$ is $\log c_J + i\log W_J + F_J(i)$. Newton interpolation reproduces its constant and linear parts. Consequently the interpolation lemma gives

$$\begin{gathered} \beta_{J,k} = R^{0,(r)}_{J,k} e^{E_{J,k}}, \\ \max_{0\le k\le L} |E_{J,k}| \le C_r L^{r+1} m_J^r \le J^{-(r+1)\epsilon+o(1)}, \qquad L = \lfloor J^{r/(r+1)-\epsilon}\rfloor. \end{gathered} \tag{336}$$

Indeed $\max\{L,r\}m_J \to 0$.

For each fixed $i \le r$, $D_i \to 1$, either by the pair-collision bound or by a finite union bound. Thus $\beta_{J,i} \sim \lambda_J^i/i!$, including the zero-bin limit $\beta_{J,0} = c_J \to 1$. Since $|b_{X,i} - \beta_{J,i}| \le \Delta_X$, all measured bins are positive eventually and

$$\max_{i\le r} \left| \log \frac{i! b_{X,i}}{i!\beta_{J,i}} \right| \ll_r \Delta_X/\lambda_J^r.$$

The finite-difference coefficients have absolute sum at most 2^j at order j. Charging the power $\binom{k}{j}$ of every parameter therefore gives

$$\max_{k\le L} \left| \log \frac{\mathcal{R}^{(r)}_{X,k}}{R^{0,(r)}_{J,k}} \right| \ll_r (1+L)^r \Delta_X/\lambda_J^r = X^{-\xi+o(1)}. \tag{337}$$

In particular no order-r parameter error is charged as though it appeared only to its first power.

The analytic bound in the lemma, evaluated at $z = 1$, also gives $|\log D_k| \le 4k^2 m_J$ on this range. As $W_J = J^{1-1/\alpha+o(1)}$, we obtain

$$-\log \beta_{J,k} \le O(1 + L\log J + L^2 m_J) = o(\log X) \qquad (0 \le k \le L).$$

Here $L = J^u$ up to a floor with fixed $u < 1$, so $L\log J = o(\log X)$ and $L^2 m_J = J^{2u-1+o(1)} = o(\log X)$. Both references therefore have minimum at least $X^{-o(1)}$ on the full range, by (336) and (337).

Retain the complete relative identity

$$\begin{aligned}\frac{b_{X,k}}{\mathcal{R}^{(r)}_{X,k}} - 1 = \frac{R^{0,(r)}_{J,k}}{\mathcal{R}^{(r)}_{X,k}}(e^{E_{J,k}} - 1) + \left(\frac{R^{0,(r)}_{J,k}}{\mathcal{R}^{(r)}_{X,k}} - 1\right) \\ + \frac{b_{X,k} - \beta_{J,k}}{\mathcal{R}^{(r)}_{X,k}}.\end{aligned} \tag{338}$$

Its first error is $J^{-(r+1)\epsilon+o(1)}$. Its second and third errors are $X^{-\xi+o(1)}$: use the parameter bound and divide the full sampling allowance Δ_X by the proved reference minimum. This proves (335), including eventual positivity of all empirical masses on the range. The positive supply is the model reference times $e^{-|E_{J,k}|}$; its negative allowance is precisely Δ_X.

Newton interpolation at each node $k \le r$ reproduces $\log(k! b_{X,k})$ exactly, proving exact bin matching. The definitions use integer powers of positive rational numbers, so all finite references are rational and computable as asserted. At $r = 2$, $\Gamma_{X,0} = b_{X,0}$, $\Gamma_{X,1} = b_{X,1}/b_{X,0}$ and $\Gamma_{X,2} = 2b_{X,0}b_{X,2}/b^2_{X,1}$, which recovers the preceding formula. □

Corollary 21.145 (Finite data recover every fixed sublinear power)**.** *Fix $0 < u < 1$ and any integer $r > u/(1-u)$. Then the first $r+1$ empirical bins give*

$$\max_{0 \le k \le \lfloor J^u \rfloor} \left| \frac{b_{X,k}}{\mathcal{R}^{(r)}_{X,k}} - 1 \right| \le J^{-\{r-(r+1)u\}+o(1)} \longrightarrow 0. \tag{339}$$

The number of bins is independent of X. In particular four bins suffice for every fixed rank exponent $u < 3/4$, and five suffice for every fixed $u < 4/5$. These are exponents of the rank range, not zero-free boundaries. The statement does not include all sequences $k = o(J)$, nor the linear endpoint $k \asymp J$.

Proof. Set $\epsilon = r/(r+1) - u > 0$ in Theorem 21.144. The exponent $(r+1)\epsilon = r - (r+1)u$ is positive. Taking $r = 3$ and $r = 4$ proves the two displayed examples. □

Corollary 21.146 (Sharp scale for the canonical interpolation formula)**.** *Let $D_{N,n}$ be the no-collision probability for n draws from the uniform distribution on N labels. For fixed $r \geq 1$ and $N > r$ let*

$$T_{N,n}^{(r)} = \exp\left(\sum_{j=0}^{r} \binom{n}{j} \Delta^j \log D_{N,0}\right),$$

where the finite differences use the values $\log D_{N,i}$ for $0 \leq i \leq r$. If $n = n_N \to \infty$ and $n = o(N)$, then

$$\log \frac{D_{N,n}}{T_{N,n}^{(r)}} = -\frac{n^{r+1}}{r(r+1)N^r}(1 + o(1)). \tag{340}$$

In particular, for every fixed $t > 0$ and $n_N \sim tN^{r/(r+1)}$,

$$\frac{D_{N,n_N}}{T_{N,n_N}^{(r)}} \longrightarrow \exp\left(-\frac{t^{r+1}}{r(r+1)}\right) < 1. \tag{341}$$

Thus the exponent $r/(r+1)$ cannot be enlarged for this canonical interpolation scheme without another correction. This is not an information-theoretic lower bound for every procedure using finitely many bins: in the uniform model, its parameter N can itself be recovered from a low-order collision probability.

Proof. For $n \leq N$ the exact probability is $D_{N,n} = \prod_{j=0}^{n-1}(1 - j/N)$. For $n = o(N)$, expand each logarithm through order r:

$$\log D_{N,n} = -\sum_{\ell=1}^{r} \frac{1}{\ell N^\ell} \sum_{j=0}^{n-1} j^\ell + O_r(n^{r+2}/N^{r+1}).$$

The same expansion at the finitely many nodes $0, \ldots, r$ has remainder $O_r(N^{-r-1})$. Their interpolation weights are $O_r(n^r)$. The polynomial $\sum_{j=0}^{n-1} j^\ell$ has degree $\ell + 1$ and leading coefficient $1/(\ell+1)$; this follows by discrete integration of n^ℓ. For $\ell < r$ it is reproduced exactly by interpolation of degree r. For $\ell = r$, the difference from its interpolant vanishes at $0, \ldots, r$

and has leading coefficient $1/(r+1)$. It is therefore exactly $(n)_{r+1}/(r+1)$. Subtracting the interpolated node expansion yields

$$\log \frac{D_{N,n}}{T^{(r)}_{N,n}} = -\frac{(n)_{r+1}}{r(r+1)N^r} + O_r\left(\frac{n^{r+2}+n^r}{N^{r+1}}\right).$$

Since $n \to \infty$ and $n/N \to 0$, the remainder divided by n^{r+1}/N^r tends to zero, and $(n)_{r+1}/n^{r+1} \to 1$. This proves both limits. At a larger fixed power $n = \lfloor N^u \rfloor$ with $r/(r+1) < u < 1$, the logarithm tends to minus infinity, which proves the claimed obstruction for this formula. □

The hierarchy and its arithmetic consumer are full ordinary proofs; no additional Lean declarations are asserted. It replaces an infinite Euler reference by finitely many actual integer-count frequencies on every fixed sublinear power range. The earlier Euler-law sampling theorem already reached ranks of order J and is not weakened or superseded by this change of reference. For $\mathcal{D}$ all statements apply unconditionally with $b = 1/16$, but they do not independently bound Θ_ζ. The certified zero-free boundary is still $7/8$.

21.61 Adaptive reconstruction with a growing number of empirical bins

The preceding arithmetic theorem fixes the interpolation order. We now keep its explicit analytic constant and prove a new empirical transfer with an order growing with the sample. Only $O(\log J) = O(\log\log X)$ measured bins will be needed. The cutoff is chosen from rational measured data, rather than from an unavailable exact value of the maximal model weight. The classical collision ingredients and their relation to Mase [25] remain as described above; the growing-order arithmetic transfer is proved here.

Lemma 21.147 (Stability with every interpolation power charged)**.** *Let (a_i) be a probability vector as in Lemma 21.143, with maximum $m > 0$ and distinct-label probabilities D_k. Fix $c_0, W > 0$ and put $\beta_k = c_0 W^k D_k/k!$. Let $1 \le r \le L$ be integers with $Lm \le 1/16$. Suppose nonnegative observations b_k, $0 \le k \le L$, satisfy $|b_k - \beta_k| \le \Delta$. Define*

$$\begin{aligned} u &= \min_{0\le i\le r} \beta_i, \qquad v = \min_{0\le k\le L} \beta_k, \qquad S(L,r) = \sum_{j=0}^{r} 2^j \binom{L}{j}, \\ E &= C_r L^{r+1} m^r, \qquad \eta = \frac{2\Delta}{u} S(L,r), \qquad 0 \le \Delta \le u/2. \end{aligned} \tag{342}$$

Then $u, v > 0$, every observed node b_i, $i \le r$, is positive, and its reference

$$R_k = \frac{1}{k!} \prod_{j=0}^{r} \left\{ \prod_{i=0}^{j} (i! b_i)^{(-1)^{j-i}\binom{j}{i}} \right\}^{\binom{k}{j}} \qquad (0 \le k \le L)$$

obeys the completely explicit bound

$$\max_{0 \le k \le L} \left| \frac{b_k}{R_k} - 1 \right| \le e^{E+\eta} - 1 + \frac{\Delta}{v} e^{E+\eta}. \tag{343}$$

Moreover $R_k = b_k$ for $0 \le k \le r$ and $S(L, r) \le (r+1)(1+2L)^r$. No constant hidden in this statement depends on the order.

Proof. All model nodes are positive by the interpolation lemma. The assumption on Δ gives $b_i \ge \beta_i/2 > 0$ at the measured nodes and $|\varepsilon_i| \le 2\Delta/u$, where $\varepsilon_i = \log(b_i/\beta_i)$. Let R_k^0 be the reference with model nodes β_i. Interpolation reproduces the constant and linear terms in $\log(k!\beta_k) = \log c_0 + k \log W + \log D_k$. Thus $\beta_k = R_k^0 e^{E_k}$ with $|E_k| \le E$. Exactly, including all powers of the empirical parameters,

$$\Lambda_k = \log(R_k/R_k^0) = \sum_{j=0}^{r} \binom{k}{j} \sum_{i=0}^{j} (-1)^{j-i} \binom{j}{i} \varepsilon_i.$$

For integer $0 \le k \le L$, $\binom{k}{j} \le \binom{L}{j}$, with the usual zero convention when $j > k$. It follows that $|\Lambda_k| \le \eta$. Retain the signed identity

$$\frac{b_k}{R_k} - 1 = e^{E_k - \Lambda_k} - 1 + \frac{b_k - \beta_k}{R_k}, \qquad R_k = \beta_k e^{\Lambda_k - E_k} \ge v e^{-E-\eta}. \tag{344}$$

Estimating only these proved remainders gives (343). The positive arithmetic supply is β_k; the competing negative allowance is Δ. Newton interpolation is exact at its nodes, proving bin matching. Finally $\binom{L}{j} \le L^j$ and each $j \le r$ contributes at most $(1 + 2L)^r$, which proves the stated bound on S. □

Theorem 21.148 (An empirical reconstruction at almost linear rank)**.** *Use the same fixed thin prime class $p_j = j^{1/\alpha + o(1)}$, $0 < \alpha < 1$, and the actual progression sample of* (301)*, with $0 < b < 1 - \alpha$. Fix a positive rational c and an integer $A \ge 1$ such that*

$$0 < c < \alpha(1 - b - \alpha). \tag{345}$$

For every integer $X \geq 4$ and every positive rational c, use the same discrete formula as in (286)*:*

$$U_X = \lfloor \log_2 X \rfloor, \qquad V_X = \lfloor \log_2 U_X \rfloor, \qquad \kappa_c(X) = \lfloor cU_X/V_X \rfloor .$$

Here $U_X \geq 2$ and $V_X \geq 1$. This defines the selector on the full parameter domain above; the earlier rare-count corollary used its smaller interval $0 < c < 7/64$. For sufficiently large integer X, define

$$\begin{gathered} \ell_J = 1 + \lfloor \log_2 J \rfloor, \qquad r_X = (A+3)\ell_J, \\ \widehat{W}_X = b_{X,1}/b_{X,0}, \qquad \widehat{m}_X = \frac{2}{(p_{J+1}-1)\widehat{W}_X}, \\ L_X = \min\left\{ \left\lfloor \frac{1}{256\widehat{m}_X} \right\rfloor, \kappa_c(X) \right\}. \end{gathered} \tag{346}$$

Here ℓ_J is the binary length of the positive integer J. Use (333)–(334) *at the variable order r_X, and denote its value by $\mathcal{R}_{X,k}^{(r_X)}$. All its input bins are positive eventually, and*

$$\begin{gathered} L_X = J^{1+o(1)}, \qquad L_X = O(J), \qquad r_X = O(\log J) = O(\log\log X), \\ \max_{0 \leq k \leq L_X} \left| \frac{b_{X,k}}{\mathcal{R}_{X,k}^{(r_X)}} - 1 \right| \ll_{\mathcal{P},b,c,A,\delta} J^{-A} + X^{-\delta} \ll_{\mathcal{P},b,c,A,\delta} J^{-A}, \\ 0 < \delta < 1 - b - \alpha - c/\alpha. \end{gathered} \tag{347}$$

The number of measured bins is $r_X + 1$. Each reference is positive and rational and matches those measured bins exactly. For decidable $\mathcal{P}$ and a computable integer budget B_X, every cutoff and reference here is computable using finite integer counts and rational arithmetic. In particular rational b supplies such a budget by integer root extraction. No value of α is needed as an input to the algorithm once the fixed choices b, c, A are admissible.

Proof. First check the selector on its complete stated domain. The bit lengths satisfy $U_X = \log_2 X + O(1)$ and $V_X = \log_2 \log_2 X + O(1)$. As $V_X \geq 1$, the ratio is defined for every $X \geq 4$. For any fixed positive rational c, these estimates and the outer floor give $\kappa_c(X) \sim c \log X / \log\log X$. This argument does not use the parameter restriction of the earlier corollary.

Fix the displayed δ and choose a fixed $\sigma > \alpha$ sufficiently close to α that

$$\xi = 1 - b - \sigma > c/\alpha + \delta.$$

The full sampling theorem supplies $|b_{X,k} - \beta_{J,k}| \le \Delta_X \ll X^{-\xi}$ for every k, uniformly. It also gives $J \sim \alpha b \log X / \log\log X$. Write $m = m_J$ for the largest normalized prime weight. The exact first two model bins are $\beta_{J,0} = c_J$ and $\beta_{J,1} = c_J W_J$. Since $c_J \to 1$, $W_J = J^{1-1/\alpha+o(1)}$ and $\Delta_X/W_J \to 0$, division of these two bins gives

$$\frac{\widehat{W}_X}{W_J} = 1 + O(\Delta_X/W_J), \qquad \frac{\widehat{m}_X}{m_J} = \frac{2W_J}{\widehat{W}_X} \longrightarrow 2, \\ m_J \le \widehat{m}_X \le 3m_J \quad \text{eventually}. \tag{348}$$

Both measured bins are positive before making either division. Here $m_J = ((p_{J+1} - 1)W_J)^{-1} = J^{-1+o(1)}$ is an exact maximum, not just an upper estimate.

Both entries of the minimum defining L_X have logarithmic index one in J: the first is $J^{1+o(1)}$, and

$$\frac{\kappa_c(X)}{J} \longrightarrow \frac{c}{\alpha b}.$$

Taking floors does not change these statements, since both entries tend to infinity. Thus $L_X = J^{1+o(1)}$ and $L_X = O(J)$. In particular $L_X \ge r_X$ eventually. Also $L_X m_J \le 1/256$ and $r_X m_J \to 0$. Every use of the explicit interpolation lemma is therefore within its stated domain.

We first bound the analytic error without hiding dependence on r_X. The elementary integral bound on $\log(r!)$ gives $r! \ge (r/e)^r$ for every integer $r \ge 1$, and hence

$$C_r \le (r+1)(16e)^r, \qquad C_r L_X^{r+1} m_J^r \le (r+1) L_X \, 4^{-r} \quad (r = r_X). \tag{349}$$

Indeed $16eL_X m_J \le e/16 < 1/4$. Since $\ell_J > \log_2 J$, we have $4^{-r_X} \le J^{-2(A+3)}$. With $L_X = O(J)$, the last bound in (349) is $O_A((\log J) J^{-2A-5}) = O_A(J^{-A})$. All constants used to obtain this geometric decrease are valid for every order, not only for fixed r.

We next control the observed nodes, including their factorials. For $0 \le i \le r_X$, the analytic logarithm estimate in the interpolation lemma gives

$$D_i \ge e^{-4i^2 m_J} \ge \tfrac{1}{2} \quad \text{eventually}, \qquad \beta_{J,i} = \frac{c_J W_J^i D_i}{i!}.$$

Since $r_X = O_A(\log J)$, $c_J \ge 1/2$ eventually and $-\log W_J = O(\log J)$, it follows, uniformly over the nodes, that

$$u_X = \min_{0 \le i \le r_X} \beta_{J,i} \ge \exp(-C_A(\log J)^2), \qquad \Delta_X/u_X \le X^{-\xi+o(1)}. \tag{350}$$

For example, when $W_J \le 1$ one may use the explicit lower bound $u_X \ge W_J^{r_X}/(4r_X^{r_X})$. Thus $\Delta_X \le u_X/2$ eventually and all measured nodes are positive. The complete parameter amplification is bounded by

$$\eta_X = \frac{2\Delta_X}{u_X} S(L_X, r_X) \le \Delta_X \exp(O_A((\log J)^2)) \le X^{-\xi+o(1)}. \tag{351}$$

Here we used $L_X = O(J)$, $S(L_X, r_X) \le (r_X+1)(1+2L_X)^{r_X}$ and $(\log J)^2 = o(\log X)$. This is the required fresh growing-order transfer; it is not a substitution into a fixed-order implied constant.

The remaining allowance is the sampling discrepancy divided by the full positive model mass at the largest ranks. For every $0 \le k \le L_X$,

$$\begin{aligned} -\log \beta_{J,k} &\le O(1) + k\log(1/W_J) + \log(k!) + 4k^2 m_J \\ &\le (1/\alpha + o(1))k \log J + O(k) \le (c/\alpha + o(1)) \log X. \end{aligned}$$

For the second line use $\log(1/W_J) = (1/\alpha - 1 + o(1)) \log J$, $\log(k!) \le k \log(1+L_X)$, $L_X = O(J)$ and $k^2 m_J \le k/256$. For the final line use $k \le \kappa_c(X) \sim c \log X / \log\log X$ and $\log J \sim \log\log X$. Consequently

$$v_X = \min_{0 \le k \le L_X} \beta_{J,k} \ge X^{-c/\alpha - o(1)}, \qquad \Delta_X / v_X \le X^{-\xi + c/\alpha + o(1)} = O(X^{-\delta}). \tag{352}$$

In particular this supply dominates the arithmetic allowance, so the empirical bins throughout the output range are positive.

Apply Lemma 21.147 with these u_X, v_X, η_X, and with $E = O_A(J^{-A})$. Its full signed consumer yields $O_A(J^{-A}) + O(X^{-\delta})$, as asserted. The model mass has not been replaced by zero or by a centered variance in this step. Exact matching and rationality follow from the interpolation formula. Computability uses binary lengths, the rational selector κ_c, integer prime cutoffs and integer powers of positive rational counts. For finitely many small inputs a direct zero test can report that the required measured nodes are not yet positive; the theorem asserts eventual positivity, not a claimed uniform effective threshold in its asymptotic hypotheses. □

Corollary 21.149 (A linear window under power asymptotics)**.** *In addition to the preceding hypotheses, suppose $p_j \sim Cj^{1/\alpha}$ for some constant $C > 0$. Put $v = 1/\alpha - 1$. Then the same data-driven cutoff has the definite positive limit*

$$\frac{L_X}{J} \longrightarrow \min\left\{\frac{1}{512v}, \frac{c}{\alpha b}\right\} > 0. \tag{353}$$

Thus $O(\log J)$ actual empirical bins reconstruct a fixed positive fraction of the linear rank range with error $O(J^{-A})$, for each prescribed fixed integer $A \ge 1$.

Proof. For every fixed $\varepsilon > 0$, the bounds $(1-\varepsilon)Cj^{1/\alpha} \le p_j \le (1+\varepsilon)Cj^{1/\alpha}$ hold at all sufficiently large indices. Integral comparison of the convergent power tail, followed by $\varepsilon \downarrow 0$, gives

$$W_J \sim \frac{J^{1-1/\alpha}}{Cv}, \qquad m_J = \frac{1}{(p_{J+1}-1)W_J} \sim \frac{v}{J}.$$

The subtraction of one changes the tail by a relative $o(1)$, since its smallest prime tends to infinity. By (348), $\widehat{m}_X \sim 2v/J$. The two integer entries defining L_X, divided by J, therefore tend to $1/(512v)$ and $c/(\alpha b)$. Floors change either ratio by at most $1/J$. Their minimum has the stated limit, and the error estimate is that of the theorem. □

Corollary 21.150 (An unconditional adaptive profile for the greedy class)**.** *For $\mathcal{P} = \mathcal{D}$, take $b = 1/16$, $c = 1/20$ and any fixed integer $A \ge 1$. The construction uses no value of Θ_ζ as input and, unconditionally, satisfies*

$$\max_{0\le k\le L_X} \left| \frac{b_{X,k}}{\mathcal{R}^{(r_X)}_{X,k}} - 1 \right| \ll_A J^{-A} + X^{-1/200}, \qquad L_X = J^{1+o(1)}. \tag{354}$$

It uses only $O_A(\log\log X)$ measured bins. No stronger regularity of the deleted primes is assumed.

Proof. The proved range $1/2 \le \Theta_\zeta \le 7/8$ implies

$$1 - \frac{1}{16} - \Theta_\zeta - \frac{1}{20\Theta_\zeta} \ge \frac{3}{560} > \frac{1}{200}.$$

Indeed the left side is decreasing on $[1/2, 7/8]$: its derivative is $-1 + 1/(20\Theta_\zeta^2) < 0$. Thus the parameter condition holds with $\alpha = \Theta_\zeta$, and the theorem allows $\delta = 1/200$. All finite decisions use the proved decidable class $\mathcal{D}$, the integer sixteenth root and rational counts; neither the choices nor their computation require knowing Θ_ζ. □

These are full ordinary proofs and do not add Lean declarations. The constants in the new stability lemma are uniform in the order; the subsequent arithmetic constants depend on the fixed class and the displayed parameters. The earlier relative sampling theorem already compared the integer law with an infinite Euler reference on a linear range. The new conclusion constructs its reference from $O(\log\log X)$ actual empirical bins and one surviving prime. For the general index hypothesis, $L_X = J^{1+o(1)}$ does not assert coverage of every possible sequence $k = o(J)$. The stronger

linear limit requires the extra power asymptotic and is not asserted for $\mathcal{D}$. These profile reconstruction results supply no independent bound on Θ_ζ; the certified boundary remains 7/8.

21.62 The largest coordinate and the cutoff from empirical bins alone

The remaining prime input in the preceding cutoff can also be removed. Newton's identities recover power sums from the first coefficients, while a growing power sum locates the largest coordinate. Both facts are classical algebraic and norm mechanisms. The point here is a uniform error estimate for their signed composition, followed by the actual integer-sampling consumer. No individual surviving prime is needed by the new cutoff once X, J and the indicated empirical bins are supplied. This does not eliminate the class-membership computation needed to form those finite counts.

Lemma 21.151 (Uniform stability of collision-moment recovery)**.** *Let (a_i) be a finite or countable probability vector, $m = \sup_i a_i > 0$, and let e_j be its elementary symmetric coefficients, with $e_0 = 1$. For an integer $t \geq 1$, suppose real numbers $\widehat{e}_0, \dots, \widehat{e}_t$ satisfy*

$$\widehat{e}_0 = 1, \qquad \max_{1 \leq j \leq t} |\widehat{e}_j - e_j| \leq \varepsilon, \qquad 0 \leq \varepsilon \leq \tfrac{1}{2}. \tag{355}$$

Define the recovered moments by the signed triangular identities

$$s\widehat{e}_s = \sum_{j=1}^{s} (-1)^{j-1} \widehat{\theta}_j \widehat{e}_{s-j} \qquad (1 \leq s \leq t). \tag{356}$$

For $\theta_s = \sum_i a_i^s$ and every $1 \leq s \leq t$,

$$\begin{gathered} |\widehat{\theta}_s - \theta_s| \leq s4^s\varepsilon, \qquad m^s \leq \theta_s \leq m^{s-1}, \\ m \leq (\widehat{\theta}_s + s4^s\varepsilon)^{1/s}. \end{gathered} \tag{357}$$

The final root has a strictly positive argument. The constants are uniform in t, in the probability vector, and in the signs of the recovered moments.

Proof. The nonnegative expansion of $(\sum_i a_i)^j$ gives $0 \leq e_j \leq 1/j!$. The full product $E(z) = \prod_i (1 + a_i z)$ converges locally uniformly, and near zero

$$\log E(z) = \sum_{s \geq 1} \frac{(-1)^{s-1}}{s} \theta_s z^s.$$

Indeed the coordinate logarithms converge absolutely on every closed disc of radius less than one, since $\sum_i a_i = 1$. Let $E_t(z) = \sum_{j=0}^{t} e_j z^j$ and $\widehat{E}_t(z) = \sum_{j=0}^{t} \widehat{e}_j z^j$. The normalized logarithm of E_t has the same coefficients as $\log E$ through degree t. For the observed polynomial, its own recovered moments satisfy

$$[z^s] \log \widehat{E}_t(z) = (-1)^{s-1} \widehat{\theta}_s / s.$$

It follows directly by comparing coefficients in $\widehat{E}_t' = \widehat{E}_t (\log \widehat{E}_t)'$; the coefficient of degree $s-1$ is exactly (356). Thus the signed recursion, including every lower-order cross term, precedes the error estimate.

On $|z| \le 1/4$ one has

$$|E_t(z) - 1| \le e^{1/4} - 1 < 1/3, \qquad |\widehat{E}_t(z) - E_t(z)| \le \varepsilon/3 \le 1/6.$$

Every point on the segment between the two polynomials stays within distance less than $1/2$ of one. Their normalized logarithms are therefore analytic on a neighborhood of this disc, and integration along that segment gives

$$|\log \widehat{E}_t(z) - \log E_t(z)| \le 2 |\widehat{E}_t(z) - E_t(z)| \le 2\varepsilon/3.$$

Cauchy's estimate at radius $1/4$ now proves the first inequality of (357), with room in its constant. No tail of a truncated Newton recursion is discarded. The two power-sum bounds follow from $a_i \le m$ and $\sum_i a_i = 1$, taking a supremum for the lower bound. In particular $\theta_s > 0$. Finally $\widehat{\theta}_s + s 4^s \varepsilon \ge \theta_s \ge m^s > 0$, which proves the remaining assertion. □

We will use rational upper roots without an oracle for real roots. For an integer $s \ge 1$ and a positive rational y, choose the unique integer h with $2^{hs} \le y < 2^{(h+1)s}$. Among $q = s, s+1, \dots, 2s$, take the least one for which $(q 2^h / s)^s \ge y$, and define

$$\mathsf{U}_s(y) = q 2^h / s, \qquad y^{1/s} \le \mathsf{U}_s(y) \le (1 + 1/s) y^{1/s}. \tag{358}$$

These are finite integer or rational comparisons. The choice exists because $q = 2s$ works. If $q > s$, minimality shows that the upward rounding error is less than $2^h/s \le y^{1/s}/s$; if $q = s$, the root is exact. This proves both bounds, including exact dyadic endpoints. The search for h terminates in either direction since $y > 0$.

Theorem 21.152 (An almost optimal cutoff measured from the bins)**.** *Use the fixed prime class, progression, and parameter domain of Theorem 21.148:*

$p_j = j^{1/\alpha+o(1)}$, $0 < b < 1-\alpha$, *and rational* $0 < c < \alpha(1-b-\alpha)$. *For large integer* X, *write* $\ell_J = 1 + \lfloor \log_2 J \rfloor$ *and define*

$$\begin{aligned} t_X &= \ell_J^2, \qquad \widehat{W}_X = b_{X,1}/b_{X,0}, \\ \widehat{e}_{X,i} &= \frac{b_{X,i}}{b_{X,0}\widehat{W}_X^{\,i}} \quad (0 \le i \le t_X). \end{aligned} \tag{359}$$

Use (356) *to compute* $\widehat{\theta}_{X,s}$ *through* $s = t_X$, *and put*

$$\begin{aligned} \widetilde{m}_X = \mathsf{U}_{t_X}(2\widehat{\theta}_{X,t_X}), \qquad \widetilde{L}_X &= \min\left\{ \left\lfloor \frac{1}{256\widetilde{m}_X} \right\rfloor, \kappa_c(X) \right\}, \\ L_X^\star &= \min\left\{ \left\lfloor \frac{1}{256 m_J} \right\rfloor, \kappa_c(X) \right\}. \end{aligned} \tag{360}$$

The last cutoff is a theoretical benchmark only. All observed nodes are positive eventually, and the recovered moment used in the root is positive eventually. With constants depending on the fixed class and the displayed parameters, one has

$$\begin{gathered} 1 \le \frac{\widetilde{m}_X}{m_J} \le 1 + O(1/\log J), \qquad 0 \le 1 - \frac{\widetilde{L}_X}{L_X^\star} = O(1/\log J), \\ \widetilde{L}_X = J^{1+o(1)}, \qquad \widetilde{L}_X = O(J), \\ \max_{0 \le k \le \widetilde{L}_X} \left| \frac{b_{X,k}}{\mathcal{R}_{X,k}^{(t_X)}} - 1 \right| \ll \exp(-(\log J)^2) + X^{-\delta}, \\ 0 < \delta < 1 - b - \alpha - c/\alpha. \end{gathered} \tag{361}$$

Here $\mathcal{R}^{(t_X)}$ *is the same rational empirical Newton reference as before, now using its first* t_X+1 *bins. It exactly matches those bins and needs no further surviving prime as input. This uses* $O((\log\log X)^2)$ *measured bins and a single construction works for every fixed power accuracy* J^{-A}. *For* $\mathcal{P} = \mathcal{D}$, $b = 1/16$, $c = 1/20$, *one may take* $\delta = 1/200$ *unconditionally, with no value of* Θ_ζ *supplied to the computation. All assertions are eventual; no uniform effective onset threshold is assumed in the index hypothesis.*

Proof. Fix δ and choose a fixed $\sigma > \alpha$ so that $\xi = 1 - b - \sigma > c/\alpha + \delta$. Use the full signed sampling theorem, which gives $|b_{X,k} - \beta_{J,k}| \le \Delta_X \ll X^{-\xi}$ for all k. Let $m = m_J = J^{-1+o(1)}$, and abbreviate $t = t_X$. The exact model identity is

$$\beta_{J,i} = c_J W_J^i e_i, \qquad e_i = D_i/i!.$$

In particular the observed coefficients in (359) have no extra factorial. They obey $\widehat{e}_{X,0} = \widehat{e}_{X,1} = 1$. The order-two normalization, for example, gives exactly

$$\widehat{\theta}_{X,1} = 1, \qquad \widehat{\theta}_{X,2} = 1 - \frac{2b_{X,0}b_{X,2}}{b_{X,1}^2},$$

consistent with the pair-collision correction above.

First retain the growing-node positive supply. Since $t = O((\log J)^2)$ and $m = J^{-1+o(1)}$, $t^2 m \to 0$. The collision logarithm bound gives $D_i \geq 1/2$ for all $0 \leq i \leq t$, eventually. As $c_J \geq 1/2$ and $W_J \leq 1$ eventually, one may use

$$\begin{gathered} u_X = \min_{0\leq i\leq t} \beta_{J,i} \geq \frac{W_J^t}{4t^t} \geq \exp(-C(\log J)^3), \\ \Delta_X/u_X \leq X^{-\xi+o(1)}. \end{gathered} \tag{362}$$

The last step uses $-\log W_J = O(\log J)$ and $(\log J)^3 = o(\log X)$. Thus every observed node is positive before any division.

Let $\zeta_i = \log(b_{X,i}/\beta_{J,i})$. For $\Delta_X \leq u_X/2$, $|\zeta_i| \leq 2\Delta_X/u_X$. Subtracting the empirical normalization exactly gives

$$\begin{gathered} \log \frac{\widehat{e}_{X,i}}{e_i} = \zeta_i - \zeta_0 - i(\zeta_1 - \zeta_0), \\ \left|\log \frac{\widehat{e}_{X,i}}{e_i}\right| \leq 4(i+1)\Delta_X/u_X. \end{gathered} \tag{363}$$

All empirical powers and both normalizing denominators remain in this identity. Since $e_i \leq 1$, the coefficient error consequently satisfies

$$\varepsilon_X = \max_{1\leq i\leq t} |\widehat{e}_{X,i} - e_i| \leq 8(t+1)\Delta_X/u_X \leq X^{-\xi+o(1)}$$

eventually, when the logarithmic bound is at most one. The recovery lemma applies. Moreover $\theta_t \geq m^t \geq \exp(-C(\log J)^3)$, so

$$\left|\frac{\widehat{\theta}_{X,t}}{\theta_t} - 1\right| \leq \frac{t4^t\varepsilon_X}{\theta_t} \leq \Delta_X \exp(O((\log J)^3)) \leq X^{-\xi+o(1)}. \tag{364}$$

This proves positivity of the recovered high moment without assuming that arbitrary empirical Newton moments are positive.

Eventually $\theta_t \leq 2\widehat{\theta}_{X,t} \leq 3\theta_t$. Combining both power-sum inequalities of the lemma with the rational upper-root bounds gives

$$\begin{gathered} m \leq \widetilde{m}_X \leq (1+1/t)\, 3^{1/t} m^{\,1-1/t}, \\ 0 \leq \log(\widetilde{m}_X/m) \leq \log(1+1/t) + \frac{\log 3 + \log(1/m)}{t} = O(1/\log J). \end{gathered} \tag{365}$$

This proves the asserted relative accuracy of the maximum. The unobserved vector is used in the proof, not in the formula for $\widetilde{m}_X$.

For completeness account for both integer cutoffs. Write $\widetilde{m}_X/m = 1+\rho_X$ with $0 \le \rho_X = O(1/\log J)$. Monotonicity and a floor allowance of one give

$$\frac{L_X^\star}{1+\rho_X} - 1 \le \widetilde{L}_X \le L_X^\star.$$

Indeed the unrounded capacity $1/(256m)$ is at least $L_X^\star$, while the other entry $\kappa_c(X)$ is an integer at least $L_X^\star$. Both entries defining $L_X^\star$ have logarithmic index one in J, and $\kappa_c(X) = O(J)$ by the independently checked selector formula in the preceding theorem. Thus $L_X^\star = J^{1+o(1)} = O(J)$ and its reciprocal is $o(1/\log J)$. Division by this positive cutoff proves the claimed ratio, $\widetilde{L}_X = J^{1+o(1)}$ and $\widetilde{L}_X = O(J)$. In particular $t \le \widetilde{L}_X$ eventually. Also $\widetilde{L}_X m \le 1/256$.

We can now apply the explicit interpolation and stability lemmas on the full output range. With $L = \widetilde{L}_X$, their analytic remainder satisfies

$$E = C_t L^{t+1} m^t \le (t+1) L\, 4^{-t} \le \exp(-(\log J)^2) \quad \text{eventually.} \tag{366}$$

To check the final constant, note that $t = (\log J)^2/(\log 2)^2 + O(\log J)$, so the logarithm of the middle expression is at most $-(2/\log 2 + o(1))(\log J)^2$. The complete interpolation-parameter allowance is

$$\eta_X = \frac{2\Delta_X}{u_X} S(L,t) \le \Delta_X \exp(O((\log J)^3)) \le X^{-\xi+o(1)}. \tag{367}$$

Here $S(L,t) \le (t+1)(1+2L)^t$, and (362) charges the noise in every measured bin.

Finally the full positive mass over $k \le L$ satisfies the same physical estimate, with all terms retained:

$$\begin{aligned} -\log \beta_{J,k} &\le O(1) + k\log(1/W_J) + \log(k!) + 4k^2 m \\ &\le (c/\alpha + o(1))\log X, \\ v_X = \min_{0\le k\le L} \beta_{J,k} &\ge X^{-c/\alpha - o(1)}, \qquad \Delta_X/v_X = O(X^{-\delta}). \end{aligned} \tag{368}$$

The second inequality uses $L = O(J)$, $k^2 m \le k/256$, and $k \le \kappa_c(X)$. The final estimate follows from $\xi > c/\alpha + \delta$.

The exact signed consumer (344), or equivalently its proved bound (343), gives

$$e^{E+\eta_X} - 1 + (\Delta_X/v_X) e^{E+\eta_X} \ll \exp(-(\log J)^2) + X^{-\delta}.$$

This includes both the error in the inferred cutoff and all compensation by the empirical normalizations. The arithmetic positive supply is still $\beta_{J,k}$; its negative allowance is Δ_X. Since both $\exp(-(\log J)^2)$ and $X^{-\delta}$ are $o(J^{-A})$ for every fixed A, the same construction gives every fixed power accuracy. All its operations, including the upper root, are finite rational or integer computations once the bins and J are given. For finitely many small inputs one may test for a zero node or a nonpositive recovered moment before performing the divisions and root; the theorem proves those tests eventually pass. The $\mathcal{D}$ specialization follows from the already checked margin $15/16 - \Theta_\zeta - 1/(20\Theta_\zeta) \geq 3/560 > 1/200$. □

Corollary 21.153 (The same calibration within a logarithmic bin budget)**.** *Fix an integer $A \geq 1$, and instead take $t_X = (A+3)\ell_J$ in all the empirical formulas of Theorem 21.152. Denote the resulting maximum and cutoff by $\widetilde{m}_X^{(A)}$ and $\widetilde{L}_X^{(A)}$. Then, eventually,*

$$\begin{gathered} m_J \leq \widetilde{m}_X^{(A)} \leq \{2^{1/(A+3)} + o(1)\} m_J, \\ \widetilde{L}_X^{(A)} = J^{1+o(1)} = O(J), \\ 2^{-1/(A+3)} \leq \liminf_{X\to\infty} \frac{\widetilde{L}_X^{(A)}}{L_X^\star} \leq \limsup_{X\to\infty} \frac{\widetilde{L}_X^{(A)}}{L_X^\star} \leq 1, \\ \max_{0\leq k\leq \widetilde{L}_X^{(A)}} \left| \frac{b_{X,k}}{\mathcal{R}_{X,k}^{(t_X)}} - 1 \right| \ll_A J^{-A} + X^{-\delta}. \end{gathered} \tag{369}$$

Only $O_A(\log\log X)$ empirical bins are used, with no additional surviving-prime input. Any prescribed fixed relative uncertainty above one for the maximum can be reached by choosing a sufficiently large fixed A; no uniformity as A varies with X is asserted.

Proof. The proof just given applies with the smaller order, keeping each dependence explicit. Now the node supply is at least $\exp(-C_A(\log J)^2)$, and both coefficient recovery and relative moment recovery have error at most $\Delta_X \exp(O_A((\log J)^2))$. The rational-root bound is unchanged. Since

$$m_J^{-1/t_X} \longrightarrow 2^{1/(A+3)},$$

it gives the stated one-sided bound on the maximum. The floor comparison with $L_X^\star$, divided by that diverging cutoff, gives the displayed liminf and limsup. Also $\widetilde{L}_X^{(A)} m_J \leq 1/256$. The explicit analytic remainder is now at most $(t_X+1)\widetilde{L}_X^{(A)} 4^{-t_X} = O_A((\log J) J^{-2A-5})$. The full parameter and

sampling allowances are bounded just as before, giving the asserted relative error. Finally $2^{1/(A+3)} \downarrow 1$, which proves the last fixed-accuracy assertion. □

Corollary 21.154 (Exact calibrated linear limits under power asymptotics)**.** *If $p_j \sim Cj^{1/\alpha}$ for some $C > 0$, put $v = 1/\alpha - 1$. The two fully empirical cutoffs above satisfy*

$$\begin{aligned} \frac{\widetilde{L}_X}{J} &\longrightarrow \min\left\{\frac{1}{256v}, \frac{c}{\alpha b}\right\}, \\ \frac{\widetilde{L}_X^{(A)}}{J} &\longrightarrow \min\left\{\frac{1}{256\,2^{1/(A+3)}v}, \frac{c}{\alpha b}\right\}. \end{aligned} \tag{370}$$

Both limits are strictly positive. The first reaches the full benchmark capacity of this interpolation argument; the earlier prime-assisted cutoff had capacity constant $1/(512v)$ before the same physical rank cap.

Proof. The preceding power-asymptotic corollary gives $m_J \sim v/J$. The quadratic-order maximum estimate therefore yields $\widetilde{m}_X \sim v/J$, proving the first limit.

For the logarithmic order $t = (A+3)\ell_J$, let $q_J = \lfloor J/t \rfloor$. Then $q_J \to \infty$, $q_J = o(J)$, and monotonicity of the prime weights gives

$$\theta_t \geq q_J a_{J+q_J}^t, \qquad \frac{a_{J+q_J}}{m_J} = \frac{p_{J+1} - 1}{p_{J+q_J} - 1} \longrightarrow 1.$$

The last limit follows directly from the common asymptotic $p_j \sim Cj^{1/\alpha}$ and $q_J/J \to 0$. Also $q_J^{1/t} \to 2^{1/(A+3)}$. Together with $\theta_t \leq m_J^{t-1}$, this proves

$$\frac{\theta_t^{1/t}}{m_J} \longrightarrow 2^{1/(A+3)}.$$

Relative empirical recovery and $1 \leq \mathsf{U}_t(y)/y^{1/t} \leq 1 + 1/t$ then show that $\widetilde{m}_X^{(A)}/m_J$ has that same limit. Taking the two capacity minima, with floors costing at most $1/J$ after division, proves the second displayed limit. The comparison constant $1/(512v)$ is (353). No such additional power asymptotic is asserted for $\mathcal{D}$. □

All results in this subsection have full ordinary proofs. They do not add Lean declarations. Power-sum recovery and approximation of a maximum by high moments are classical ingredients, not new zero detectors. The new

arithmetic consumer preserves their precision through the signed empirical normalization and through the rare-event denominator, and it no longer enumerates a surviving prime to choose the reference's range. The quadratic bin budget improves calibration to relative error $O(1/\log J)$ and gives superpolynomial accuracy in J; the logarithmic budget retains the preceding fixed-power accuracy with no extra prime input. Neither statement proves a new estimate on Θ_ζ; the certified zero-free boundary remains 7/8.

21.63 Finite integer signs recover a thin prime class

Exact profile zeros already identify prime labels in Theorem 21.50; Theorem 21.54 gives their separation. Those results do not bound the displacement caused by an empirical function error. We now control the entire error on a growing disc and preserve the signs at negative half-integers. The decoder will use integer arithmetic, without locating roots. The forward Bernoulli-product mechanism is classical; compare [9, Proposition 3.1]. The finite arithmetic sign transfer, its quantitative range, and the decoding capacity below are the conclusions proved here; no established literature priority is claimed.

Fix any set of primes $\mathcal{P}$, possibly finite or empty, for which

$$0 < \sigma < 1, \qquad S_{\mathcal{P},\sigma} = \sum_{p \in \mathcal{P}} p^{-\sigma} < \infty. \tag{371}$$

For positive integers m use the actual divisor count $K_{\mathcal{P}}(m) = \#\{p \in \mathcal{P} : p \mid m\}$. This count is finite at every m. For integer $N \geq 4$ put

$$C_{N,k} = \#\{1 \leq m \leq N : K_{\mathcal{P}}(m) = k\}, \qquad b_{N,k} = C_{N,k}/N. \tag{372}$$

The prime class is fixed before sampling. The independent reference has success probabilities $1/p$ and the proper entire generating function

$$P_{\mathcal{P}}(z) = \sum_{k \geq 0} \beta_k z^k = c_{\mathcal{P}} \prod_{p \in \mathcal{P}} (1 + z/(p-1)),$$
$$c_{\mathcal{P}} = \prod_{p \in \mathcal{P}} (1 - 1/p) > 0.$$

No asymptotic prime index or infinitude is assumed in this subsection.

Fix positive rational numbers h, c satisfying

$$h\sigma < 1, \qquad 0 < c < (1-\sigma)/h. \tag{373}$$

All parameters are fixed before N tends to infinity. Define the integer cutoffs and empirical polynomial by

$$\begin{gathered} U_N = \lfloor \log_2 N \rfloor, \quad V_N = \lfloor \log_2 U_N \rfloor, \\ M_N = \lfloor cU_N/V_N \rfloor, \quad R_N = \lfloor U_N^h \rfloor, \\ T_N(z) = \sum_{k=0}^{M_N} b_{N,k} z^k. \end{gathered} \tag{374}$$

Here $U_N \geq 2$, $V_N \geq 1$ and $R_N \geq 1$. The selector formula is valid for every positive rational c, as independently checked in Theorem 21.148. For $h = a/b > 0$ with positive integers a, b, R_N is the integer bth root of U_N^a. Consequently no comparison with an unspecified real cutoff is used.

Lemma 21.155 (Sampling on a growing complex disc)**.** *For every fixed*

$$0 < \eta < \min\{1 - \sigma - hc,\ c(1/\sigma - h)\}, \tag{375}$$

one has

$$\sup_{|z| \leq 2R_N} |T_N(z) - P_{\mathcal{P}}(z)| \ll_{\mathcal{P},\sigma,h,c,\eta} N^{-\eta}. \tag{376}$$

The constant and eventual threshold are independent of z. This includes finite and empty $\mathcal{P}$. If σ is rational, the choice $c = \sigma(1-\sigma)$ balances the two displayed losses and permits every $\eta < (1-\sigma)(1-h\sigma)$.

Proof. The complete count-profile law identifies the reference coefficients with the Taylor coefficients of the actual entire Euler product. Its signed finite-subset sampling identity and the rounding estimate

$$0 \leq q^{-1} - N^{-1}\lfloor N/q \rfloor \leq N^{\sigma-1} q^{-\sigma} \quad (q \geq 1)$$

give the uniform finite-reference bound. Passing every partial coefficient sum to the reference limit gives

$$\begin{gathered} \sum_{k \geq 0} |b_{N,k} - \beta_k| \leq B_{\mathcal{P},\sigma} N^{\sigma-1}, \\ B_{\mathcal{P},\sigma} = \exp(2S_{\mathcal{P},\sigma}) < \infty. \end{gathered} \tag{377}$$

This uses the count law alone and includes finite and empty prime classes.

The coefficient proof of (278) likewise requires only this summability. In detail, with $A_\sigma = \sigma^{-1} S_{\mathcal{P},\sigma}$,

$$\begin{gathered} \log P_{\mathcal{P}}(r) \leq A_\sigma r^\sigma \quad (r \geq 1), \\ \beta_k \leq \exp\{A_\sigma k - (k/\sigma)\log k\} \quad (k \geq 2). \end{gathered}$$

The first inequality follows from $\log(1+y) \le y^\sigma/\sigma$. The second follows by bounding a nonnegative coefficient with $r^{-k}P_{\mathcal{P}}(r)$ and taking $r = k^{1/\sigma}$. No positive lower bound for high coefficients is required; they may be zero in a finite class.

Write $M = M_N$ and $R = R_N$. Retain the complete signed error before bounding its two remainders:

$$T_N(z) - P_{\mathcal{P}}(z) = \sum_{k=0}^{M} (b_{N,k} - \beta_k) z^k - \sum_{k>M} \beta_k z^k. \tag{378}$$

On $|z| \le 2R$, the first remainder is at most

$$B_{\mathcal{P},\sigma} N^{\sigma-1} (2R)^M = N^{-(1-\sigma-hc)+o(1)}.$$

This charges the full growing test $(2R)^M$. For the second, $M \to \infty$ and, eventually, $M + 1 \ge 2$. Put

$$\begin{aligned} a_N &= \sigma^{-1} \log(M+1) - \log(2R) - A_\sigma \\ &= (1/\sigma - h + o(1)) \log\log N \longrightarrow \infty. \end{aligned}$$

For every $k > M$, the coefficient bound gives $\beta_k (2R)^k \le e^{-a_N k}$. Consequently

$$\sum_{k>M} \beta_k (2R)^k \le \frac{e^{-a_N(M+1)}}{1 - e^{-a_N}} = N^{-c(1/\sigma - h)+o(1)}.$$

Here $M \sim c \log N / \log\log N$ and $\log R = (h + o(1)) \log\log N$, by the integer selector formulas. Both exponents are positive by the parameter domain. For each stated strict η, the first remainder is eventually at most $B_{\mathcal{P},\sigma} N^{-\eta}$. The tail slope eventually satisfies $a_N \ge \log 2$, so its denominator is at least $1/2$, and the second remainder is at most $2N^{-\eta}$. Thus the constant in (376) can be taken as $\exp(2S_{\mathcal{P},\sigma}) + 2$, with an eventual threshold depending on the fixed parameters. Finally substituting $c = \sigma(1-\sigma)$ makes both exponents $(1-\sigma)(1-h\sigma)$. □

Formalization status. The complete statement of Lemma 21.155, including the balanced specialization, is Lean-verified by `finite_decoding_complex_error` and `finite_decoding_complex_error_balanced` in `Helpers/PrimeClassDecoderComplexError.lean`. The proof derives the entire-product growth, all coefficient and series tails, the full signed error decomposition, and both losses for the literal integer selectors U_N, V_N, M_N, R_N. Its threshold is uniform over the whole disc and its

constant is explicit. The signed half-integer supply and the mathematical decoding guarantees below are also Lean-verified. The computation-cost clauses are tracked separately and are not included in that verification claim.

Theorem 21.156 (Exact prime labels from finite integer sign tests)**.** *Use* (371)–(374)*. For every integer* $0 \le n \le R_N$ *compute the integer*

$$I_{N,n} = \sum_{k=0}^{M_N} (-1)^k C_{N,k} (2n+1)^k 2^{M_N - k}. \tag{379}$$

For all sufficiently large integer N*, every such integer is nonzero and, simultaneously for all those* n*,*

$$\operatorname{sgn} I_{N,n} = (-1)^{N_{\mathcal{P}}(n+1)}. \tag{380}$$

In particular the completely finite decoder

$$\begin{aligned} \widehat{\mathcal{P}}_{N;h,c} &= \{n+1 : 1 \le n \le R_N,\ I_{N,n-1} I_{N,n} < 0\} \\ &= \mathcal{P} \cap \{2, \dots, R_N + 1\}. \end{aligned} \tag{381}$$

is eventually exact. It also gives every prefix count through $R_N + 1$ *by summing the indicated sign changes. Given* N, h, c *and the* $M_N + 1$ *integer counts, it uses no additional class-membership or primality queries. There are* $O(R_N M_N)$ *integer arithmetic operations, on integers of* $O_{h,c}(\log N)$ *bits. The cost of obtaining the input counts is not included.*

Proof. We give an independent lower bound for the magnitude of the signed model at the actual evaluation points. For any integer $R \ge 1$ and $0 \le n \le R$, put $w_n = -n - 1/2$ and $S = S_{\mathcal{P},\sigma}$. Use the actual probability-normalized Euler factors:

$$P_{\mathcal{P}}(z) = \prod_{p \in \mathcal{P}} \left(1 + \frac{z-1}{p}\right), \qquad f_p(w_n) = \frac{p - n - 3/2}{p}.$$

Every numerator is a nonzero half-integer and has absolute value at least $1/2$. For $p \le 4R + 2$,

$$|f_p(w_n)| \ge \frac{1}{2p} \ge \frac{1}{12R}, \qquad N_{\mathcal{P}}(4R+2) \le 6^\sigma S R^\sigma.$$

For $p > 4R + 2$, put $t = (n + 3/2)/p$. Then $0 \le t < 1/2$, so $\log f_p(w_n) = \log(1 - t) \ge -2t \ge -5R/p$. The tail over the same class satisfies

$$\sum_{\substack{p \in \mathcal{P} \\ p > 4R+2}} \frac{1}{p} \le R^{\sigma-1} S.$$

Moreover $f_p(w_n) < 0$ if and only if $p \le n+1$. Thus for every finite class prefix with endpoint $Y \ge 4R{+}2$, multiplication by $(-1)^{N_{\mathcal{P}}(n+1)}$ gives exactly the product of the factor magnitudes. Bounding that signed finite product and passing to its proved Euler-product limit gives

$$0 < e^{-6^\sigma SR^\sigma \log(12R) - 5SR^\sigma} \le (-1)^{N_{\mathcal{P}}(n+1)} P_{\mathcal{P}}(w_n). \tag{382}$$

This bound includes every probability-normalization factor: it is proved directly for $1 + (z-1)/p$, without removing the constant factor from the product.

Now take $R = R_N$. Since $h\sigma < 1$, $R_N^\sigma \log(2R_N) = o(\log N)$. Thus the independent lower bound in (382) is $N^{-o(1)}$, uniformly at all evaluation points. Lemma 21.155 makes the arithmetic and truncation allowance $O(N^{-\eta})$ for a fixed $\eta > 0$. It is eventually strictly smaller than that magnitude. Both quantities are real at w_n, so their signs agree:

$$\operatorname{sgn} T_N(w_n) = \operatorname{sgn} P_{\mathcal{P}}(w_n).$$

This preserves the signed model signal rather than replacing it by an unsigned coefficient mass. The exact integer normalization is

$$I_{N,n} = N2^{M_N} T_N(-n - 1/2), \qquad N2^{M_N} > 0. \tag{383}$$

It proves nonvanishing and (380). Subtracting the two adjacent prime counts gives the class criterion, because $N_{\mathcal{P}}(n+1) - N_{\mathcal{P}}(n)$ is one precisely when the integer $n+1$ belongs to $\mathcal{P}$, and is zero otherwise.

At $n = 0$ the sign is positive. The step $n = 1$ detects 2, and $n = 2$ detects 3, with no exceptional-prime deletion. All later composite labels produce no sign change. The same proof includes a finite class without knowing its last prime. For the empty class all probes are the positive integer $N2^{M_N}$, so the decoded set is empty even before the eventual range is reached.

The algorithm is well defined for every $N \ge 4$, even if a small-input probe vanishes: the strict integer inequality still defines a finite set. Its guarantee is eventual, with a threshold depending on the fixed class and parameters. For the evaluation algorithm, write $M = M_N$, $R = R_N$ and $x_n = -2n - 1$. Initialize $A_0 = 0$, $V_0 = 1$ and, for $0 \le j \le M$, compute

$$A_{j+1} = x_n A_j + C_{N,M-j} V_j, \qquad V_{j+1} = 2V_j.$$

Induction gives $V_j = 2^j$ and

$$A_j = \sum_{i<j} C_{N,M-i} x_n^{j-1-i} 2^i, \qquad A_{M+1} = I_{N,n}.$$

Thus the evaluation uses only the supplied counts through rank M. The executable decoder is unchanged if any higher counts are changed. Five arithmetic operations per step include the coefficient-index subtraction, the two value multiplications, addition and power doubling. Charging four setup operations per point and multiplication and comparison at every adjacent test gives, eventually,

$$(R+1)\{5(M+1)+4\}+2R \le 30RM.$$

Here $R, M \ge 1$ eventually. The literal executable program may instead evaluate both endpoints separately at each of the R adjacent tests. Charging those repeated evaluations gives the valid bound

$$2R\{5(M+1)+4\}+2R \le 30RM$$

as well. Thus no caching assumption is needed for its complexity. Input-count acquisition is excluded.

The actual sample partition gives $\sum_{k=0}^{M} C_{N,k} \le N$. The output satisfies $|I_{N,n}| \le N(2R+1)^M 2^M$. Every partial Horner accumulator, its next multiplication, scaled coefficient and stored power is bounded by

$$B_N = N(2R+1)^M 4^M = N(4R+2)^M 2^M.$$

This follows by retaining the partial sum above, bounding each coefficient by N and using $j \le M+1 \le 2^M$. Products of adjacent probes are charged the full bound B_N^2. The literal selector limits give

$$\frac{M_N}{\log N} \longrightarrow 0, \qquad \frac{M_N \log(2R_N)}{\log N} \longrightarrow ch, \qquad \log B_N \le (2+ch)\log N$$

eventually. For the binary width $b(z) = 1 + \lfloor \log_2(1+|z|) \rfloor$, put $D_{h,c} = (2+ch)/\log 2 + 2 > 0$. Every value bounded by B_N has width at most $D_{h,c} \log N$ eventually; comparison products have width at most $2D_{h,c} \log N$. All thresholds are simultaneous across the evaluation range and all Horner steps. Each shorter prefix uses the same signs and gives its exact class count.

The rational parameters also admit an integer-only construction of both selectors. Write $h = a/d$ and $c = u/v$ with positive integers and $d, v \ge 1$. Compute U_N and V_N by successive divisions by 2. Their numbers of visited tests are exactly U_N+1 and V_N+1, including inputs 0 and 1, and $V_N \le U_N$. The rank is the integer quotient

$$M_N = \frac{uU_N}{vV_N} \text{ rounded down,}$$

with quotient zero when $V_N = 0$, agreeing with the original formula. For the radius, form $Y = U_N^a$ and start at $r = 0$; increase r while $(r+1)^d \le Y$. The returned integer is exactly R_N. This forward search visits at most $R_N + 1$ tests. An arbitrary a priori termination fuel does not enter that test count.

Charge five operations per binary-log test, a multiplications to form Y, $d + 6$ operations per root test and twelve fixed setup/rank operations. The resulting preprocessing count is at most

$$(a + d + 28)(U_N + R_N + 1).$$

The proved selector limits imply

$$\frac{\log\log N}{R_N} \longrightarrow 0, \qquad \frac{\log N}{R_N M_N} \longrightarrow 0, \qquad \frac{U_N + R_N + 1}{R_N M_N} \longrightarrow 0.$$

For the first limit, divide $\log R_N / R_N \to 0$ by the earlier normalized logarithmic radius limit. Combine that limit with the normalized rank limit to obtain the second; the binary-cutoff limit then gives the third. Consequently preprocessing is eventually at most $R_N M_N$. The complete arithmetic count, including the literal program's repeated evaluations, is therefore at most $31 R_N M_N$.

Finally put $E = (a+1)(d+1)$ and $K = a + d + u + v + 1$. For $U_N \ge 1$, every root-search power with base at most $U_N^a + 1$ and exponent at most $d+1$ is at most $(U_N + 1)^E$. Both rank multiplication inputs and their quotient are bounded by $K(U_N + 1)^E$. The same scalar bounds all fixed parameter integers. For every fixed K, E, the proved limit

$$\frac{\log\{K(U_N + 1)^E\}}{\log N} \longrightarrow 0$$

implies $K(U_N+1)^E \le N$ eventually. The binary-log values and termination fuel are at most $N+1$. Thus every retained preprocessing integer is bounded by $N + 1 + K(U_N + 1)^E \le 3N \le B_N$ eventually. This proves its claimed logarithmic binary width on the same range. □

Formalization status. The complete statement of Theorem 21.156 is Lean-verified by `finite_integer_prime_decoding_complete` in `Helpers/PrimeClassDecoderComplete.lean`. It derives the actual signed infinite-product supply and literal integer signs, nonvanishing, all prefix sets and counts, executable selection from rational parameters and observed counts alone, the full preprocessing and evaluation arithmetic budget, and all retained binary-width bounds. One eventual threshold covers all ranges and

all these conclusions. The executable program makes no class-membership or primality queries and no caching assumption enters its displayed cost. The theorem includes 2, composites and finite or empty classes. The capacity classification and RH corollaries below remain separate formalization obligations; this completion makes no new claim about the zero-free boundary.

21.64 Exact decoding capacity and an operational RH criterion

The degree budget supplies a converse to the preceding analytic construction. This gives a critical exponent for this particular polynomial sign decoder, not an impossibility theorem for every algorithm that might use the same counts.

For every positive rational h, c, the sign-test set in the first line of (381) is defined by the integer formula, without the analytic parameter restrictions. Its eventual equality with the class is the property to be required, rather than part of that unrestricted definition.

For a class satisfying (371) for at least one $0 < \sigma < 1$, define its convergence exponent and admissible decoding powers by

$$\tau_{\mathcal{P}} = \inf\{s > 0 : \sum_{p \in \mathcal{P}} p^{-s} < \infty\} \in [0, 1),$$

$$\mathcal{H}_{\mathrm{dec}}(\mathcal{P}) = \left\{ h \in \mathbb{Q}_{>0} : \begin{array}{l} \text{there is a fixed } c \in \mathbb{Q}_{>0} \text{ such that} \\ \widehat{\mathcal{P}}_{N;h,c} = \mathcal{P} \cap \{2, \dots, R_N + 1\} \\ \text{for every sufficiently large integer } N \end{array} \right\}, \tag{384}$$

$$\mathcal{C}_{\mathrm{dec}}(\mathcal{P}) = \sup \mathcal{H}_{\mathrm{dec}}(\mathcal{P}) \in [0, +\infty].$$

In this definition the counts and the decoder are always (372) and (379)–(381); $M_N = \lfloor cU_N/V_N \rfloor$ and $R_N = \lfloor U_N^h \rfloor$. The constant c may depend on h and the class, but never on N.

Theorem 21.157 (The exact critical decoding exponent)**.** *With the convention* $1/0 = +\infty$*,*

$$\mathcal{C}_{\mathrm{dec}}(\mathcal{P}) = 1/\tau_{\mathcal{P}}. \tag{385}$$

Every positive rational $h < 1/\tau_{\mathcal{P}}$ *is admissible. No positive rational* $h > 1/\tau_{\mathcal{P}}$ *is admissible. The theorem makes no assertion about attainment at equality.*

If additionally $N_{\mathcal{P}}(x) = x^{\alpha + o(1)}$*,* $0 < \alpha < 1$*, then* $\tau_{\mathcal{P}} = \alpha$*. For any fixed positive rational* $h > 1/\alpha$ *and* c*, every decoder based on the consecutive signs*

of a nonzero real polynomial of degree at most M_N misses a fraction $1-o(1)$ of the class primes in $\{2,\dots,R_N+1\}$. This last assertion allows arbitrary real coefficients, not only the actual empirical polynomial.

Proof. The set of convergent exponents is upward closed. Consequently every $s > \tau_{\mathcal{P}}$ is a convergent exponent: the definition of the infimum supplies a convergent exponent strictly below s, and comparison gives convergence at s. If $0 < h < 1/\tau_{\mathcal{P}}$, choose

$$\tau_{\mathcal{P}} < \sigma < \min\{1, 1/h\}, \qquad 0 < c < (1-\sigma)/h$$

with c rational. The preceding decoder theorem applies, proving admissibility. When $\tau_{\mathcal{P}} = 0$, this works for every fixed positive rational h and already proves infinite capacity.

For the converse let h be admissible, with its fixed c. For every N, T_N is a nonzero polynomial: its constant coefficient is at least $1/N$, because $K_{\mathcal{P}}(1) = 0$. A strict sign change between $T_N(-n+1/2)$ and $T_N(-n-1/2)$ forces a real zero in the open interval $(-n-1/2, -n+1/2)$. Those intervals are pairwise disjoint. A nonzero polynomial of degree at most M_N has at most M_N distinct real roots. Thus the following is an exact degree constraint, before any analytic relaxation:

$$\begin{gathered} |\widehat{\mathcal{P}}_{N;h,c}| \le M_N, \\ N_{\mathcal{P}}(R_N+1) \le M_N \quad \text{whenever decoding is exact.} \end{gathered} \tag{386}$$

Zeros at shared interval endpoints cannot create a strict sign change and cause no exception.

To obtain a bound at every large integer cutoff x, put $q = \lceil (x+1)^{1/h} \rceil$ and $N = 2^q$. Then $U_N = q$, $R_N \ge x+1$, $q \sim x^{1/h}$, and $M_N = O_c(q/\log q)$. The eventual exactness hypothesis therefore gives

$$N_{\mathcal{P}}(x) \ll_{h,c} x^{1/h}/\log x.$$

For any $s > 1/h$, summing over dyadic prime intervals now gives

$$\sum_{p\in\mathcal{P}} p^{-s} \le \text{a finite initial sum} + \sum_{j\ge j_0} 2^{-js} N_{\mathcal{P}}(2^{j+1}) < \infty.$$

It follows that $\tau_{\mathcal{P}} \le 1/h$. For $\tau_{\mathcal{P}} > 0$ this says $h \le 1/\tau_{\mathcal{P}}$, proving the required upper bound on the capacity. Rational density and the proved admissibility below that value give the exact supremum.

Under the additional exact counting index, convergence holds for every $s > \alpha$ by dyadic comparison. If $0 < s < \alpha$, a convergent sum would imply

$N_{\mathcal{P}}(x) \le x^s \sum_{\mathcal{P}} p^{-s}$, contradicting $N_{\mathcal{P}}(x) = x^{\alpha+o(1)}$. Thus $\tau_{\mathcal{P}} = \alpha$. Finally if $h\alpha > 1$,

$$N_{\mathcal{P}}(R_N+1) = (\log N)^{h\alpha+o(1)}, \qquad M_N = (\log N)^{1+o(1)},$$
$$\frac{M_N}{N_{\mathcal{P}}(R_N+1)} \longrightarrow 0.$$

The degree argument applies to any nonzero real polynomial and permits at most M_N reported labels. At least $N_{\mathcal{P}}(R_N+1) - M_N$ true labels must be missed, proving the asserted fraction. This is a restriction of the polynomial sign-code scheme; it is not an information-theoretic bound on arbitrary finite-data procedures. □

Corollary 21.158 (A finite integer-sign criterion for RH)**.** *For the fixed greedy deleted class $\mathcal{D}$,*

$$\mathcal{C}_{\mathrm{dec}}(\mathcal{D}) = 1/\Theta_\zeta, \qquad \frac{8}{7} \le \mathcal{C}_{\mathrm{dec}}(\mathcal{D}) \le 2. \tag{387}$$

In particular

$$\begin{aligned} \mathrm{RH} \quad &\Longleftrightarrow \quad \mathcal{C}_{\mathrm{dec}}(\mathcal{D}) = 2, \\ \Theta_\zeta \le 3/4 \quad &\Longleftrightarrow \quad \mathcal{C}_{\mathrm{dec}}(\mathcal{D}) \ge 4/3. \end{aligned} \tag{388}$$

An equivalent literal form of the first statement is: for every rational $0 < h < 2$ there is a fixed positive rational c such that the integer tests (379) *recover exactly $\mathcal{D} \cap \{2, \dots, \lfloor U_N^h \rfloor + 1\}$ for every sufficiently large integer N.*

Unconditionally, one explicit choice is

$$\begin{aligned} &h = 10/9, \qquad \sigma = 8/9, \qquad c = 8/81, \\ &\sup_{|z| \le 2R_N} |T_N(z) - P_{\mathcal{D}}(z)| = O(N^{-1/1000}). \end{aligned} \tag{389}$$

The corresponding integer signs eventually decode the full displayed prefix, including the labels 2 and 3 when present.

Proof. The proved all-scale counting index for $\mathcal{D}$ is $N_{\mathcal{D}}(x) = x^{\Theta_\zeta+o(1)}$, with $1/2 \le \Theta_\zeta \le 7/8$. The critical-exponent theorem therefore gives $\mathcal{C}_{\mathrm{dec}}(\mathcal{D}) = 1/\Theta_\zeta$ and the displayed bounds. The already established equivalence RH $\Longleftrightarrow \Theta_\zeta = 1/2$ proves the first criterion, and positivity of Θ_ζ gives the 3/4 criterion. Admissibility of every $h < 1/\Theta_\zeta$ proves the literal RH implication. Conversely admissibility of all rational $h < 2$ implies $\Theta_\zeta \le 1/h$ for all those h, hence $\Theta_\zeta \le 1/2$.

For the numerical specialization, $\sigma = 8/9 > 7/8$ makes $S_{\mathcal{D},\sigma}$ finite without RH. Also $h\sigma = 80/81 < 1$ and $c = \sigma(1-\sigma)$. Both errors of the complex-disc lemma have exponent $(1-\sigma)(1-h\sigma) = 1/729 > 1/1000$. The independent half-integer magnitude is $N^{-o(1)}$, so that error preserves every required sign eventually. This proof supplies no new independent upper estimate on Θ_ζ; it uses the installed 7/8 boundary. □

Corollary 21.159 (Prime values of exponential sequences have infinite capacity)**.** *Let each of the following denote the set of distinct* prime values *of the indicated sequence:*

$$\begin{aligned}\mathcal{P}_{\text{Fermat}} &= \{2^{2^j}+1 : j \geq 0,\ 2^{2^j}+1 \text{ prime}\},\\ \mathcal{P}_{\text{Mersenne}} &= \{2^m-1 : m \geq 2,\ 2^m-1 \text{ prime}\},\\ \mathcal{P}_{\text{Fib}} &= \{F_m : m \geq 0,\ F_m \text{ prime}\}, \qquad F_0 = 0,\ F_1 = 1.\end{aligned}$$

The Fibonacci recurrence is $F_{m+2} = F_{m+1} + F_m$. For each class, whether it is finite or infinite,

$$\tau_{\mathcal{P}} = 0, \qquad \mathcal{C}_{\text{dec}}(\mathcal{P}) = +\infty. \tag{390}$$

Thus every fixed rational logarithmic power of the prime cutoff can be recovered by the corresponding finite sign decoder, with an admissible fixed bin constant c. This does not decide infinitude of any of the three classes.

Proof. For every fixed $\sigma > 0$ the unfiltered sequence bounds give

$$\begin{aligned}\sum_{p \in \mathcal{P}_{\text{Fermat}}} p^{-\sigma} &\leq \sum_{j \geq 0} 2^{-\sigma 2^j} < \infty,\\ \sum_{p \in \mathcal{P}_{\text{Mersenne}}} p^{-\sigma} &\leq \sum_{m \geq 2} 2^{-\sigma(m-1)} < \infty.\end{aligned}$$

The composite sequence values occur only in these upper bounds; the observations and decoded classes retain actual prime values. For Fibonacci, Binet's identity with the golden ratio $\phi = (1+\sqrt{5})/2$ gives

$$F_m = \frac{\phi^m - (-\phi^{-1})^m}{\sqrt{5}} \geq \frac{\phi^m}{2\sqrt{5}} \quad (m \geq 2).$$

Consequently $\sum_{p \in \mathcal{P}_{\text{Fib}}} p^{-\sigma} \leq (2\sqrt{5})^\sigma \sum_{m \geq 3} \phi^{-m\sigma} < \infty$. The Fibonacci values at indices zero, one and two are nonprime; the prime values from index three onward are distinct. The other two sequences are strictly increasing. Each class is a set, so a value is counted once. They are separate

observations; their overlaps, for example the prime 3, introduce no joint-independence assertion.

Since all positive exponents converge, each convergence exponent is zero. The capacity theorem proves the claim, including the case of a finite class with an unknown final prime. The sequences and primality filters are decidable on integer domains, so their finite counts and sign tests are computable. □

Removing any prime class $\mathcal{Q}$ with $\tau_{\mathcal{Q}} = 0$ preserves the complete set of convergent positive exponents:

$$\left\{ s > 0 : \sum_{p \in \mathcal{P} \setminus \mathcal{Q}} p^{-s} < \infty \right\} = \left\{ s > 0 : \sum_{p \in \mathcal{P}} p^{-s} < \infty \right\}.$$

Consequently $\tau_{\mathcal{P} \setminus \mathcal{Q}} = \tau_{\mathcal{P}}$. For the capacity conclusion retain this section's hypothesis: $\sum_{p \in \mathcal{P}} p^{-s_0} < \infty$ for some $0 < s_0 < 1$. Then $\mathcal{C}_{\mathrm{dec}}(\mathcal{P} \setminus \mathcal{Q}) = \mathcal{C}_{\mathrm{dec}}(\mathcal{P})$. This includes removing any subclass of the union of the three displayed special prime classes, regardless of their overlaps.

Here is the complete summability argument. Every prime class has a convergent exponent, since its inverse-square series is bounded by the natural inverse-square series. Thus the infimum defining $\tau_{\mathcal{Q}}$ has a nonempty defining set, and $\tau_{\mathcal{Q}} = 0$ implies convergence at every positive exponent. For each such exponent and every finite cutoff X, use the exact partition of partial sums:

$$\sum_{\substack{p \le X \\ p \in \mathcal{P}}} p^{-s} = \sum_{\substack{p \le X \\ p \in \mathcal{P} \setminus \mathcal{Q}}} p^{-s} + \sum_{\substack{p \le X \\ p \in \mathcal{P} \cap \mathcal{Q}}} p^{-s}.$$

The last series converges by comparison with the series for $\mathcal{Q}$. This proves the equality of convergence sets; the capacity classification gives the stated capacity equality. Finite unions of the special classes converge at every positive exponent by the sum bound, as does every subclass. The general deletion argument is Lean-verified by

`primeClassConvergentExponents_diff_eq`,
`primeClassConvergenceExponent_diff_eq`,
`integerDecoderCapacity_diff_eq`.

The union and arbitrary-subclass applications are also checked. Applying this to $\mathcal{D}$ uses its class-specific summability input; the general invariance proof does not verify that input or prove the RH endpoint.

The complete finite decoder theorem is now formalized in Lean, including the growing-disc approximation, independent signed half-integer supply, exact recovery, executable rational selectors, and the full arithmetic and bit-cost clauses. The maintained declaration

`finite_integer_prime_decoding_complete`

verifies Theorem 21.156.

The critical-capacity theorem, including every clause of Theorem 21.157, is verified by

`integer_decoder_capacity_classification`.

Its converse uses the exact samples $N_j = 2^{2^{bj}}$ for $h = a/b$. They satisfy $U_{N_j} = 2^{bj}$ and $R_{N_j} = 2^{aj}$. The actual degree constraint then bounds each geometric prime shell, and a finite-subsums argument proves convergence for every $s > b/a$. The proof retains the infinite-capacity case $\tau_{\mathcal{P}} = 0$, identifies $\tau_{\mathcal{P}} = \alpha$ under the counting-index hypothesis, and proves the missed-fraction limit for arbitrary real polynomial coefficients. It makes no endpoint-attainment assertion. The sharper all-cutoff inverse estimate is also formalized:

`integerDecoderAdmissible_all_cutoff_inverse_bound`

proves that each admissible positive rational h has a fixed $K > 0$ such that $N_{\mathcal{P}}(x) \le K x^{1/h}/\log x$ at every sufficiently large integer cutoff. For $h = a/b$ the proof retains the full factor bj in $M_{N_j} \le c\,2^{bj}/(bj)$ and uses $j = \lfloor \log_{2^a} x \rfloor + 1$. It gives $K = c\,2^b a \log 2/b$, with the witness c from admissibility. Corollary 21.159 is now verified for the three actual prime-value classes by

`special_prime_value_capacity`.

Its certificate includes convergence at every positive exponent, zero convergence exponent, infinite capacity, admissibility of every positive rational power, and finite membership tests. The real Binet identity and the displayed golden-ratio lower bound are checked by `fibonacci_binet_real` and `fibonacci_golden_exponential_lower`. The membership tests search only indices $n \le p$ for Fermat and Mersenne values and $n \le p + 2$ for Fibonacci values, with the primality filter retained. The monotonicity and small-index claims in the proof are also checked. No infinitude hypothesis is used. The general sparse-deletion argument and the three special-class applications

above are now formalized. The RH corollaries and their class-specific inputs remain separate formalization obligations.

These exact decoders use $M_N+1 = O(\log N/\log\log N)$ bins and recover prime labels on a growing prefix, a distinct conclusion from the existing full-profile inverse theorem. For $\mathcal{D}$ the certified zero-free boundary remains $7/8$; the operational endpoint 2 is an equivalence, not a newly proved arithmetic estimate at that endpoint.

21.65 Selecting certified good primes before estimating their size

We now reverse the order of the two obligations. First construct a decidable class whose members satisfy the Nicolas inequality independently; then ask how large that same class must be to retain every forbidden zero signal. Here, as in the preceding subsection, the selected primes are cutoffs for the full-prime product.

Fix a deterministic certified rational approximation a_p, for each prime $p \geq 5$, such that $|a_p - F(p)| \leq p^{-3}$. Such an approximation is computable from the finite primorial and finite prime product, together with certified logarithms and Euler's constant. Define the actual prime classes and their counting functions by

$$\begin{aligned} \mathcal{G} &= \{p \geq 5 :\ p \text{ prime},\ a_p \leq -2p^{-3}\}, \\ \mathcal{B} &= \{p \geq 5 :\ p \text{ prime},\ a_p > -2p^{-3}\}, \\ G(X) &= \#(\mathcal{G} \cap [5, X]), \qquad B(X) = \#(\mathcal{B} \cap [5, X]). \end{aligned} \tag{391}$$

The rational comparison decides membership, including equality. There is no need to decide whether the real number $F(p)$ equals zero. For every selected prime, unconditionally,

$$p \in \mathcal{G} \implies F(p) \leq -p^{-3} < 0, \qquad p \in \mathcal{B} \implies F(p) > -3p^{-3}. \tag{392}$$

Thus the member inequality is proved before any size assertion.

Lemma 21.160 (Persistence through consecutive prime indices)**.** *Suppose $|\theta(t) - t| = O_\kappa(t^\kappa \log^2 t)$ for some $1/2 \leq \kappa \leq 1$. For consecutive primes $q < p$ with $X \leq p \leq 2X$, one has, uniformly for sufficiently large X,*

$$|F(p) - F(q)| \leq C_\kappa X^{\kappa-2} \log^2 X. \tag{393}$$

Here $C_\kappa > 0$ is independent of the primes and X. If $0 < b < 1/2$, $\kappa + b > 1$, and arbitrarily large primes P satisfy $F(P) > 2P^{-b}$, then arbitrarily large

P satisfy

$$B(2P) \geq c_\kappa P^{2-\kappa-b}/\log^2 P. \tag{394}$$

The same conclusion with G in place of B follows from arbitrarily large P with $F(P) < -2P^{-b}$. The thresholds may depend on b, κ and their input bounds.

Proof. Put $L = \log p$, $T = \theta(q) = \theta(p) - L$, and $h(t) = 1/(t \log t)$. PNT ensures that the segment $[T, T + L]$ lies in $[p/2, 2p]$ eventually, also when $\kappa = 1$. On this segment, $|h'(t)| \ll 1/(p^2 \log p)$. Since $Lh(p) = 1/p$, the exact increment (207) gives

$$F(p) - F(q) = \int_T^{T+L} (h(t) - h(p))\, dt + \log(1 - 1/p) + 1/p,$$
$$|F(p) - F(q)| \ll \frac{|\theta(p) - p| + \log p + 1}{p^2}.$$

This proves (393). The cancellation of $1/p$ has been performed exactly; a bound on either uncancelled summand would lose that cancellation.

Increase C_κ if necessary and put

$$M_P = \left\lfloor \frac{P^{2-\kappa-b}}{4C_\kappa \log^2 P} \right\rfloor.$$

Because $0 < 2 - \kappa - b < 1$, this tends to infinity and is $o(P/\log P)$. PNT therefore supplies at least M_P successor primes between P and $2P$. Telescoping the signed increment, followed by its proved local variation bound, changes F by at most $P^{-b}/4$ over these successors. A positive starting value $> 2P^{-b}$ consequently remains $> P^{-b} > 0$. Every one of these primes belongs to $\mathcal{B}$. For a negative starting value $< -2P^{-b}$, every successor satisfies $F(p) < -P^{-b} < -3p^{-3}$ eventually. Then $a_p \leq F(p) + p^{-3} < -2p^{-3}$, so these primes belong to $\mathcal{G}$. Taking a fixed smaller positive multiple of the displayed expression for M_P proves both bounds. □

Theorem 21.161 (Exact size burden for the certified good class)**.** *The class $\mathcal{G}$ satisfies $F(p) < 0$ on every member without assuming RH. Nevertheless its counting exponent alone does not retain the forbidden signal: unconditionally,*

$$\limsup_{X\to\infty} \frac{\log(1 + G(X))}{\log X} = 1.$$

More precisely the complementary count has the dichotomy

$$\limsup_{X\to\infty} \frac{\log(1 + B(X))}{\log X} = \begin{cases} 0, & \text{if RH holds,} \\ 1, & \text{if RH fails.} \end{cases} \tag{395}$$

In particular,

$$\text{RH} \quad \Longleftrightarrow \quad B(X) = O(X^{1-\delta}) \text{ for some fixed } 0 < \delta < 1. \tag{396}$$

The constant and eventual cutoff in this proposed size estimate must be independent of X. They are not supplied here.

Proof. Under RH, (20) and $\beta_0 < 2$ give $F(p) \le -c/(\sqrt{p}\log p)$ eventually for some fixed $c > 0$. Thus $a_p \le -2p^{-3}$ eventually: $\mathcal{B}$ is finite, and $\mathcal{G}$ contains every sufficiently large prime. This proves the RH branches.

Suppose RH is false. Fix a zero with real part $\beta > 1/2$ and an exponent $1-\beta < b < 1/2$. Theorem 11.2 supplies both signs of F with arbitrarily large multiples of x^{-b}. The function F is constant between consecutive primes. PNT makes the last prime below x asymptotic to x, so the two signs occur at arbitrarily large primes with the starting inequalities in Lemma 21.160.

Write $\Theta = \Theta_\zeta$ as in Lemma 21.90. If $\Theta < 1$, fix $\Theta < \kappa < 1$. The proof of that lemma gives $|\theta(x) - x| = O_\kappa(x^\kappa \log^2 x)$. The condition $\kappa + b > 1$ follows from $\kappa > \Theta \ge \beta$. The persistence lemma gives, for each sign along its own unbounded sequence, at least $c_\kappa P^{2-\kappa-b}/\log^2 P$ primes in that class below $2P$. Let β approach Θ from the actual zeros, κ decrease to Θ, and b decrease to $1-\beta$. The counting exponents $2-\kappa-b$ approach 1. No zero attaining Θ is assumed and no constant is claimed uniform in these limits. The elementary upper bound $G(X), B(X) \le X$ proves that both limsup exponents are exactly 1.

If $\Theta = 1$, use $\kappa = 1$ and the elementary PNT bound $|\theta(x) - x| = O(x)$ in the persistence lemma. Zeros with β arbitrarily close to 1 permit $b > 1-\beta$ arbitrarily close to zero, giving exponents $1-b$ tending to 1. This proves the remaining case and the dichotomy. A bound in (396) would give a complementary limsup exponent at most $1-\delta$, contradicting the false-RH branch. The converse follows from finiteness of $\mathcal{B}$ under RH. □

This construction meets the first reverse-selection obligation literally: its selected primes already satisfy the inequality. It also proves the retention threshold for that same class: omitting only a power-saving number of primes would force an off-line zero's persistent positive signal to hit $\mathcal{G}$, which is impossible by (392). Merely proving that $\mathcal{G}$ has exponent 1 is insufficient, since both classes have exponent 1 under false RH.

The attempted direct size argument from theta-sign monotonicity has a precise unresolved edge. Its exact input is

$$F(p) = F(3) + \sum_{\substack{5 \le r \le p \\ r \text{ prime}}} \left\{ \int_{\theta(r)-\log r}^{\theta(r)} \frac{dt}{t \log t} + \log(1 - 1/r) \right\}.$$

The theta sign controls an individual increment as proved above, whereas membership of $\mathcal{B}$ depends on this entire signed prefix sum. Thus no bound on the number of increments of either sign supplies a bound on $B(X)$. The jump estimate proves persistence and the required burden; it does not supply the independent upper count in (396). That upper bound remains unproved. These are full ordinary proofs, not new Lean-checked declarations or a claimed originality priority. The certified zero-free boundary remains 7/8.

21.66 Nested Fermat carriers and a joint Fibonacci profile

Fermat numbers need not be prime to supply exact clocks. Write $T_j = 2^{2^j}+1$ for $j \geq 0$ and choose its least prime factor R_j. Here the sampled value is $2^n - 1$, with scalar exponent $n \geq 0$: we test $R_j \mid 2^n - 1$. This is a different observable from testing divisibility directly on the Fermat sequence T_n. The cutoff k selects the labels $0 \leq j < k$, independently of the sizes of R_j and $2^n - 1$.

Theorem 21.162 (Fermat divisors give a nested dyadic clock)**.** *If $r > 2$ divides T_j, then*

$$\operatorname{ord}_r(2) = 2^{j+1}, \qquad r \mid 2^n - 1 \quad \Longleftrightarrow \quad 2^{j+1} \mid n.$$

In particular all prime factors of a fixed T_j have exactly the same event on this sampled sequence. The selected primes R_j are pairwise distinct, and $R_0 = 3$. For

$$C_k(n) = \sum_{j=0}^{k-1} \mathbf{1}_{R_j \mid 2^n-1} = \sum_{j=0}^{k-1} \mathbf{1}_{2^{j+1} \mid n},$$

uniform sampling over $0 \leq n < 2^k$ gives

$$\mathbb{P}(C_k = t) = 2^{-t-1} \quad (0 \leq t < k), \qquad \mathbb{P}(C_k = k) = 2^{-k}.$$

Thus, with $G_k(z) = \mathbb{E}z^{C_k}$,

$$G_k(z) = \frac{1}{2} \sum_{t=0}^{k-1} (z/2)^t + (z/2)^k, \tag{397}$$

$$(2 - z)G_k(z) = 1 + (1 - z)(z/2)^k, \qquad \mathbb{E}C_k = 1 - 2^{-k}. \tag{398}$$

The cross-multiplied identity holds for every complex z, including $z = 2$, where $G_k(2) = 1 + k/2$.

Proof. In $\mathbb{Z}/r\mathbb{Z}$ the hypothesis gives $2^{2^j} = -1 \neq 1$, while $2^{2^{j+1}} = 1$. In particular 2 is a unit. Its order divides 2^{j+1} but does not divide 2^j, so the order is exactly 2^{j+1}. The power test proves the claimed divisibility equivalence, including $n = 0$. A shared prime factor at two Fermat labels would consequently have two different exact orders, which is impossible. The first selected factor is the prime $T_0 = 3$.

For $t < k$, exactly 2^{k-t-1} residues in a complete period are divisible by 2^t and not by 2^{t+1}. Their count C_k is t. Only the zero residue is divisible by 2^k, and it has count k. This proves the distribution and (397), also when $k = 0$. The indicator definition handles the zero residue without assigning a finite 2-adic valuation to zero. Multiplying the finite geometric sum proves the cross identity without division by $2 - z$. Alternatively, splitting the sampled indices into even and odd parts gives $G_0 = 1$ and $G_{k+1} = (1 + zG_k)/2$. Finally, summing the threshold probabilities gives $\mathbb{E}C_k = \sum_{j<k} 2^{-j-1} = 1 - 2^{-k}$. □

These events are nested, not independent Bernoulli trials. Nor is an infinitude of Fermat primes assumed: R_j exists even when T_j is composite. Direct sampling of T_n would give a different law: each fixed R_j divides only T_j, so its density among Fermat indices is zero.

Theorem 21.163 (A common-index product law and its exact compensation)**.** *Let S be a finite set of odd prime labels, $M_S = \prod_{\ell\in S} \ell$, and $k \geq 0$. Sample the same index n uniformly over $0 \leq n < 2^k M_S$, evaluating the pair $(F_n, 2^n - 1)$. For every binary Fibonacci pattern b and every complex z,*

$$\mathbb{E}\Big[z^{C_k(n)}\mathbf{1}_{\mathbf{1}_{P_\ell|F_n}=b_\ell \text{ for every } \ell\in S}\Big] = G_k(z)\prod_{\ell\in S}\left(\frac{1}{\ell}\right)^{b_\ell}\left(1-\frac{1}{\ell}\right)^{1-b_\ell}.$$

Consequently, for arbitrary complex marks w_ℓ,

$$\mathbb{E}\left[z^{C_k(n)}\prod_{\ell\in S} w_\ell^{\mathbf{1}_{P_\ell|F_n}}\right] = G_k(z)\prod_{\ell\in S}(1 + (w_\ell - 1)/\ell).$$

The entire nested Fermat block is independent of the Fibonacci block. No independence is asserted within the Fermat block.

For S_X the odd primes at most X, put $\mu_X = \sum_{\ell\in S_X} 1/\ell$ and let

$$T_{X,k}(n) = C_k(n) + \sum_{\ell\in S_X} \mathbf{1}_{P_\ell|F_n}.$$

Compensation by the actual mean *gives*

$$H_{X,k}(z) = e^{-(z-1)(\mu_X+1-2^{-k})}\,\mathbb{E}z^{T_{X,k}}.$$

As $X, k \to \infty$, independently, these functions converge locally uniformly on $|z| < 2$ to

$$H(z) = \frac{e^{1-z}\mathcal{F}_{\mathrm{odd}}(z)}{2-z}, \qquad \mathcal{F}_{\mathrm{odd}}(z) = \prod_{\ell \geq 3 \text{ prime}} (1 + (z-1)/\ell)e^{-(z-1)/\ell}.$$

This limit extends meromorphically to the whole plane. It has a simple pole at 2, with residue $-e^{-1}\mathcal{F}_{\mathrm{odd}}(2) \neq 0$. The exact compensation

$$(2-z)e^{z-1}H(z) = \mathcal{F}_{\mathrm{odd}}(z)$$

extends to an entire function and recovers all the simple zeros $1-\ell$, for odd primes ℓ.

Proof. The previous carrier theorems replace the events by $2^{j+1} \mid n$ and $\ell \mid n$, respectively. Since 2^k and M_S are coprime, CRT identifies one uniform scalar residue with independent uniform residues modulo 2^k and modulo M_S. The first coordinate determines the whole Fermat count; the second determines the full Fibonacci pattern. This proves the first formula. Summing over all binary patterns, with the stated marks, gives the second formula and the mean $\mu_X + 1 - 2^{-k}$. Empty supports and $k = 0$ are included.

The exact finite identity gives, for $z \neq 2$,

$$G_k(z) - \frac{1}{2-z} = \frac{(1-z)(z/2)^k}{2-z}.$$

For every fixed $0 \leq R < 2$ this implies the uniform bound

$$\sup_{|z| \leq R} \left| G_k(z) - \frac{1}{2-z} \right| \leq \frac{1+R}{2-R}(R/2)^k.$$

The dyadic mean correction tends uniformly on compact sets to e^{1-z}. The previously proved canonical Bernoulli products converge locally uniformly to $\mathcal{F}_{\mathrm{odd}}$. Their product therefore gives the stated two-parameter limit.

The displayed quotient supplies the meromorphic extension. Every factor at $z = 2$ is positive, and the logarithms of the factors have an absolutely convergent sum, because their absolute values are bounded by a constant times ℓ^{-2}. Thus $\mathcal{F}_{\mathrm{odd}}(2) > 0$ and the residue is nonzero. Multiplying by the stated entire compensator removes this pole. Away from 2 the removed factors are nonzero, so all odd-prime profile zeros and their multiplicities are preserved. □

A shared physical prime does not mean a shared index event. The same prime 5 occurs as $P_5 = 5$ and $R_1 = 5$, yet

$$5 \mid F_n \iff 5 \mid n, \qquad 5 \mid 2^n - 1 \iff 4 \mid n.$$

These two events have probabilities $1/5$ and $1/4$ and joint probability $1/20$ under uniform sampling modulo 20. Their independence comes from their coprime index clocks. Likewise $P_3 = 2$ supplies the index-3 event, while $R_0 = 3$ supplies the index-2 event. Thus the arithmetic names of the carrier primes are not the parameters of the probability law.

The RH information and the order of limits. For $k \geq 1$ the event $C_k = 0$ is exactly n odd. Writing $A(X)$ as in Theorem 21.58, the limiting density of avoiding every selected carrier in both sampled sequences is therefore

$$B(X) = \frac{1}{2}A(X) = \prod_{p \leq X} (1 - 1/p).$$

The existing Nicolas criterion becomes

$$\text{RH} \quad \iff \quad B(X) < \frac{e^{-\gamma}}{\log\log(2M_X)} \quad \text{for all sufficiently large } X \geq 3.$$

The extra nested Fermat events beyond R_0 do not change this vacancy event. This exact translation supplies no new independent bound on the product or on zeta zeros.

All finite probability formulas above first use a complete period. For fixed X, k, interval sampling over $1 \leq n \leq N$, $N \geq 1$, has pattern error at most $2^k M_X/N$, by the same period-remainder argument as before. If a fixed finite weight has absolute value at most W, the corresponding expectation error is at most $2^{k+1}M_X W/N$: the incomplete period and its predicted average each have absolute contribution at most $2^k M_X W$ before dividing by N. Only after $N \to \infty$ at fixed cutoffs are X, k allowed to grow. The local uniform profile limit does not assert simultaneous growing-cutoff sampling accuracy.

Formalization scope. The maintained candidate owners

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Mathlib/NumberTheory/Primes/FermatCarrier.lean
Mathlib/NumberTheory/Primes/FermatCarrierProbability.lean
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check twenty-nine declarations: exact orders and integer power tests, distinct selected carriers, actual dyadic sample sums and their geometric recurrence, compensation at all z, the exceptional value at 2, and the exact mean. Four further checked declarations in

`Helpers/FermatFibonacciProbability.lean`

prove coprimality of the sample periods, pointwise compatibility of the actual scalar events with CRT, the generating-sum factorization over arbitrary commutative semirings, and the normalized real joint law. All thirty-three axiom audits contain only the standard foundational axioms. The complex normalized law follows by the same finite algebra; the interval estimate, locally uniform limit, residue, and RH translation above are ordinary proofs, with no separate Lean compilation claim. The elementary order and geometric-law mechanisms are classical. The purpose of this construction is their common-index realization alongside the Fibonacci golden-root clock and the exact preservation of the marked profile.

21.67 All Fermat prime factors: a uniform logarithmic tail

The preceding construction selected one factor R_j at each Fermat label. We now retain *every distinct physical prime factor*. For $T_j = 2^{2^j} + 1$ define

$$m_j = \omega(T_j), \qquad W_t = \sum_{0 \le j < t} m_j, \qquad \mathcal{U}_k = \bigcup_{0 \le j < k} \mathrm{PF}(T_j),$$

where PF is the finite set of distinct prime factors. The supported count is

$$C_k^{\mathrm{all}}(n) = \#\{r \in \mathcal{U}_k : r \mid 2^n - 1\} \qquad (n \ge 0).$$

At $n = 0$ this count is W_k; it is not defined using $\mathrm{PF}(0)$.

Theorem 21.164 (The full weighted Fermat law)**.** *The supports* $\mathrm{PF}(T_j)$ *are pairwise disjoint,* $m_j \ge 1$*, and, for every* $j, t \ge 0$*,*

$$(j+1)m_j \le 2^j, \qquad W_t < 2^t, \qquad (t+1)W_t \le 3 \cdot 2^t.$$

Every index has the exact count

$$C_k^{\mathrm{all}}(n) = \sum_{j<k} m_j \mathbf{1}_{2^{j+1} \mid n}.$$

Uniform sampling over $0 \le n < 2^k$ *gives the distribution*

$$\mathbb{P}(C_k^{\mathrm{all}} = W_t) = 2^{-t-1} \quad (0 \le t < k), \qquad \mathbb{P}(C_k^{\mathrm{all}} = W_k) = 2^{-k},$$

and hence

$$Q_k(z) = \mathbb{E} z^{C_k^{\mathrm{all}}} = \frac{1}{2} \sum_{t<k} \frac{z^{W_t}}{2^t} + \frac{z^{W_k}}{2^k}.$$

No independence among the factors inside one Fermat number is used.

Proof. A shared prime factor at two distinct labels would have two different orders by Theorem 21.162. The supports are therefore disjoint, and each is nonempty.

For a prime factor r of T_j, the order 2^{j+1} is at most the size r of $\mathbb{Z}/r\mathbb{Z}$. Equality is impossible: $r > 2$ is an odd prime, whereas 2^{j+1} is even. Thus $r \geq 2^{j+1} + 1$. The product of the distinct prime factors divides T_j, so

$$2^{(j+1)m_j} < \prod_{r \in \mathrm{PF}(T_j)} r \leq T_j = 2^{2^j} + 1.$$

All quantities are integers. Therefore $2^{(j+1)m_j} \leq 2^{2^j}$ and $(j+1)m_j \leq 2^j$.

For general natural weights satisfying this last inequality, induction in t gives the slightly stronger first bound $W_t + 1 \leq 2^t$: use $m_t \leq 2^t$ and $W_{t+1} = W_t + m_t$. The second bound has the immediate bases $t = 0, 1$. For its induction step with $t \geq 1$,

$$(t+1)W_{t+1} = (t+1)(W_t + m_t) \leq 4 \cdot 2^t.$$

Consequently

$$(t+2)W_{t+1} \leq 4\frac{t+2}{t+1}\, 2^t \leq 6 \cdot 2^t = 3 \cdot 2^{t+1},$$

where $t \geq 1$ gives $(t+2)/(t+1) \leq 3/2$.

Every factor in the jth block divides $2^n - 1$ exactly when $2^{j+1} \mid n$. Disjointness allows the full supported count to be summed block by block, proving its identity. For a nonzero residue with 2-adic valuation $t < k$ the count is W_t; the zero residue has count W_k. The same dyadic residue counts as before prove the distribution. The levels are distinct because all m_j are positive. The generating formula follows, including $k = 0$. □

For positive indices the finite count stabilizes as k grows. Define the full observable, with an explicitly finite ambient set, by

$$C^{\mathrm{all}}(n) = \#\{r \in \mathrm{PF}(2^n - 1) : r \mid T_j \text{ for some } j \geq 0\} = W_{v_2(n)} \qquad (n \geq 1).$$

This is the component of the Mersenne-value prime factors whose base-two order is 2^a for some $a \geq 1$. Indeed, in the odd prime field the half-order

power then has order two, so it is -1; the prime therefore divides T_{a-1}. The converse is the proved Fermat order test. The observable is finite even though all Fermat labels are allowed.

Theorem 21.165 (A tail bound uniform in sample size and cutoff)**.** *For every real $x \geq 1$, every integer $N \geq 1$, and every cutoff $k \geq 0$,*

$$\frac{1}{N}\#\{1 \leq n \leq N : C_k^{\mathrm{all}}(n) \geq x\} \leq \frac{3}{x(1+\log_2 x)},$$
$$\frac{1}{N}\#\{1 \leq n \leq N : C^{\mathrm{all}}(n) \geq x\} \leq \frac{3}{x(1+\log_2 x)}.$$

The same bound holds for the complete-period laws and their limiting weighted geometric law. For every $0 < \theta < 1$, all these laws have the uniform fractional-moment bound

$$\mathbb{E}C^{\theta} \leq \frac{1}{2(1-2^{\theta-1})}.$$

With $H_0 = 0$ and $H_a = \sum_{j=1}^{a} 1/j$ for $a \geq 1$, their truncated means satisfy

$$\mathbb{E}_{\mathrm{period}} C_k^{\mathrm{all}} = \sum_{j<k} \frac{m_j}{2^{j+1}} \leq \frac{1}{2} H_k.$$

For $L = \lfloor \log_2 N \rfloor$, positive-interval sampling has the stronger cutoff-dependent bound

$$\frac{1}{N}\sum_{n=1}^{N} C_k^{\mathrm{all}}(n) \leq \frac{1}{2} H_{\min(k,L)}, \qquad \frac{1}{N}\sum_{n=1}^{N} C^{\mathrm{all}}(n) \leq \frac{1}{2} H_L.$$

Thus the full positive-interval mean is $O(\log\log N)$.

Proof. Since $W_t \geq t$ and $W_0 = 0$, for each $x \geq 1$ there is a least $t = t(x) \geq 1$ with $W_t \geq x$. If $k < t$, the truncated threshold event is empty. If $k \geq t$, the nested clock gives the exact equivalence

$$C_k^{\mathrm{all}}(n) \geq x \quad \Longleftrightarrow \quad 2^t \mid n.$$

The full observable has the same equivalence for every positive n. Hence the positive-interval probability is exactly $\lfloor N/2^t \rfloor / N$, at most 2^{-t}. There is no added period-remainder error, and no restriction on how k may grow with N.

The compiled cumulative inequalities give

$$x(t+1) \le W_t(t+1) \le 3 \cdot 2^t, \qquad x \le W_t < 2^t.$$

In particular $\log_2 x < t$ and all denominators below are positive. Therefore

$$\frac{\lfloor N/2^t \rfloor}{N} \le 2^{-t} \le \frac{3}{x(t+1)} \le \frac{3}{x(1+\log_2 x)}.$$

For the reference random variable T with $\mathbb{P}(T = s) = 2^{-s-1}$, the limiting count is W_T and its threshold probability is exactly 2^{-t}. The complete-period law is $W_{\min(T,k)}$, so it obeys the same upper bound.

For U_N uniform on $1, \dots, N$,

$$\mathbb{P}(v_2(U_N) \ge t) = \frac{\lfloor N/2^t \rfloor}{N} \le 2^{-t} = \mathbb{P}(T \ge t).$$

Thus $v_2(U_N)$ is stochastically dominated by T. The cumulative weights are increasing, and truncation can only decrease them. Every indicated fractional moment is therefore at most that of W_T. Since $W_s < 2^s$,

$$\mathbb{E}W_T^\theta \le \frac{1}{2} \sum_{s \ge 0} 2^{-(1-\theta)s} = \frac{1}{2(1-2^{\theta-1})}.$$

Finally, linearity applied to the exact block indicators gives the complete-period mean. The arithmetic bound $m_j/2^{j+1} \le 1/(2(j+1))$ proves its harmonic upper bound. On $1, \dots, N$, the jth indicator has probability $\lfloor N/2^{j+1} \rfloor / N \le 2^{-j-1}$ and is zero when $j + 1 > L$. Summing only these active rows proves both interval mean inequalities. No independent-factor approximation enters any of these bounds. □

The theorem does not assert that the infinite reference mean $\sum_{j \ge 0} m_j 2^{-j-1}$ is finite. Its finite partial sums have the proved harmonic envelope; fractional moments below one are uniformly finite without that extra assertion.

The infinite PGF records factor complexity. For $|z| \le 1$ put

$$Q(z) = \sum_{t \ge 0} 2^{-t-1} z^{W_t}.$$

The sum is absolutely convergent, and the finite reference laws satisfy the explicit uniform estimate

$$\sup_{|z| \le 1} |Q_k(z) - Q(z)| \le 2^{1-k}.$$

Indeed, the terminal atom of Q_k and the omitted tail of Q each have total mass 2^{-k}.

More generally, let $\alpha = \limsup_{t\to\infty} W_t/t \in [1,\infty]$. The radius of convergence of Q is exactly

$$R_Q = 2^{1/\alpha} \in [1,2], \qquad 1/\infty = 0.$$

Because the W_t are strictly increasing, its nonzero Taylor coefficients are precisely $a_{W_t} = 2^{-t-1}$. The Cauchy–Hadamard formula consequently gives

$$R_Q^{-1} = \limsup_{t\to\infty} 2^{-(t+1)/W_t} = 2^{-1/\alpha}.$$

Thus the one-carrier pole at 2 cannot simply be transferred to the full factor count. Its analytic radius measures the average growth of the actual Fermat factor counts. No value of α beyond the stated range is assumed.

The Fibonacci connection is retained. For a finite odd-prime support S and a common scalar sampled modulo $2^k M_S$, the same CRT argument gives

$$\mathbb{E}\left[z^{C_k^{\mathrm{all}}(n)} \prod_{\ell\in S} w_\ell^{\mathbf{1}_{P_\ell \mid F_n}} \right] = Q_k(z) \prod_{\ell\in S} (1 + (w_\ell - 1)/\ell).$$

The whole weighted dyadic block depends on the first CRT coordinate, and the Fibonacci vector on the second; this is the complete independence argument. With the Fibonacci compensation alone, the limits on $|z| \le 1$ are therefore $\mathcal{F}_{\mathrm{odd}}(z)Q(z)$. Adding independent noise to a mod-Poisson limit is classical [9, Example 2.2]; here the noise and its tail have an explicit arithmetic realization. A neighborhood outside the unit disc requires the actual radius condition above. These sampling and factor-count results supply no new independent estimate for the earlier zeta-zero exclusion bound.

Formalization scope. The maintained candidate modules

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Mathlib/NumberTheory/Primes/FermatFactorCount.lean
Mathlib/Probability/Distributions/WeightedDyadic.lean
Mathlib/NumberTheory/Primes/FermatAllFactorProbability.lean
Mathlib/Probability/Distributions/WeightedDyadicBound.lean
```

check twenty-one declarations. They prove the physical prime-factor lower and cardinal bounds, disjointness, the arbitrary-weight generating law, its actual full-factor transport, and both cumulative weight estimates. Every axiom audit contains only standard foundational axioms. The threshold/floor consumer, logarithmic tail, fractional moments, interval means,

infinite PGF and radius statements, and the final weighted CRT specialization are the complete ordinary proofs above, without a separate Lean compilation claim for those combined analytic conclusions.

21.68 Exact binary sampling and its sharp leading constant

The weighted geometric law has a stronger finite-sampling statement than the generic complete-period remainder estimate. It does not depend on the unknown sizes of the positive Fermat block weights. We use total variation in the convention $d_{\mathrm{TV}}(\mu,\nu)=\frac{1}{2}\sum_x|\mu(x)-\nu(x)|$.

Theorem 21.166 (Exact binary defect and sharp sampling constant)**.** *Let w_j be positive integers, $W_t=\sum_{0\le j<t}w_j$, U_N uniform on $\{1,\dots,N\}$ with $N\ge 1$, and let T have $\mathbb{P}(T=t)=2^{-t-1}$ for $t\ge 0$. For $k\ge 0$ put*

$$X_{N,k}=W_{\min(v_2(U_N),k)},\qquad X_k=W_{\min(T,k)}.$$

Write

$$\begin{aligned} r_t &= \frac{N \bmod 2^t}{2^t},\\ b_t &= \left\lfloor \frac{N}{2^t}\right\rfloor - 2\left\lfloor \frac{N}{2^{t+1}}\right\rfloor \in\{0,1\},\\ h &= \min\{k,1+\lfloor \log_2 N\rfloor\}. \end{aligned}$$

Then

$$d_{\mathrm{TV}}(X_{N,k},X_k)=\frac{1}{2N}\sum_{\substack{0\le t<k\\ b_t=1}}(1-r_t)\le \frac{h}{6N}+\frac{1}{2N}.$$

The same formula, with the finite sum over all nonzero bits of N, and the bound with $h=1+\lfloor\log_2 N\rfloor$ hold without truncation. For the untruncated variables the leading constant is optimal:

$$\limsup_{N\to\infty}\frac{N}{\log_2 N}\,d_{\mathrm{TV}}(W_{v_2(U_N)},W_T)=\frac{1}{6}.$$

For nonnegative weights the displayed error remains an upper bound; strict positivity is needed for the equality of transported defects.

Proof. Set $q_t=\lfloor N/2^t\rfloor$, so $N/2^t=q_t+r_t$, $r_0=0$ and

$$q_t=2q_{t+1}+b_t,\qquad r_{t+1}=\frac{r_t+b_t}{2}.$$

For $t < k$, the empirical mass at level W_t is $(q_t - q_{t+1})/N$. Its difference from 2^{-t-1} is

$$\frac{q_t - q_{t+1}}{N} - \frac{1}{2^{t+1}} = \frac{b_t - r_t}{2N}.$$

The terminal mass difference is $q_k/N - 2^{-k} = -r_k/N \le 0$. Since $0 \le r_t < 1$, the positive differences occur exactly at the bits $b_t = 1$. The levels W_t are distinct, so total variation, the total positive difference between two probability laws, is the stated sum. All bits above $\lfloor \log_2 N \rfloor$ vanish. For the full law there are likewise only finitely many positive differences, and the same calculation applies. If levels merge because some weights vanish, summing masses in each fiber can only decrease total variation.

Here is the joint estimate on the retained digit sum. Define

$$R(b, r) = b(1 - r), \qquad B(r) = r - \frac{|r - \frac{1}{2}|}{3}.$$

For $0 \le r \le 1$ and $b \in \{0, 1\}$, put $y = (r + b)/2$. Then

$$R(b, r) \le \frac{1}{3} + B(y) - B(r). \tag{$*$}$$

This is a two-branch identity with a nonnegative slack. For $b = 0$ that slack is 0 when $r \ge 1/2$, and $(1 - 2r)/3$ when $r \le 1/2$. For $b = 1$ it is 0 when $r \le 1/2$, and $(2r - 1)/3$ when $r \ge 1/2$. Thus all four cases prove $(*)$ directly. We also have $B(0) = -1/6$ and $B(r) \le 5/6$ on $[0, 1]$. Summing $(*)$ over the actual binary orbit makes all intermediate compensator terms cancel:

$$\sum_{0 \le t < h} b_t(1 - r_t) \le \frac{h}{3} + B(r_h) - B(0) \le \frac{h}{3} + 1.$$

Division by $2N > 0$ proves the claimed bound.

To prove sharpness symbolically for every $m \ge 1$, take

$$N_m = \frac{4^m - 1}{3}.$$

Its binary expansion has ones exactly at $t = 2i$, $0 \le i < m$, and $r_{2i} = (1 - 4^{-i})/3$. Therefore

$$\sum_{0 \le i < m} (1 - r_{2i}) = \frac{2m}{3} + \frac{4}{9}(1 - 4^{-m}),$$

$$d_{\mathrm{TV}}(W_{v_2(U_{N_m})}, W_T) = \frac{m}{3N_m} + \frac{2(1 - 4^{-m})}{9N_m}.$$

Since $\log_2 N_m = 2m - \log_2 3 + \log_2(1 - 4^{-m})$, the normalized limit along this sequence is $1/6$. The universal upper bound gives the reverse limsup inequality. For $k = 0$ both laws are the point mass at $W_0 = 0$; for $N = 1$ and $k \geq 1$ the exact defect is $1/2$. These cases also verify the endpoint conventions. □

Taking $w_j = m_j = \omega(T_j)$ applies the theorem to the actual all-Fermat-factor count, by the exact order and disjoint-support theorems. The result is independent of how many prime factors any particular Fermat number has. In particular, the complete reference law approximates the actual positive-interval law in total variation at the displayed rate, simultaneously for every cutoff.

Theorem 21.167 (Uniform common-index sampling with odd clocks)**.** *Let $M \geq 1$ be odd, and let A be uniform on $\mathbb{Z}/M\mathbb{Z}$, independently of T. For every $N \geq 1$ and integer $a \geq 0$,*

$$d_{\mathrm{TV}}\big((U_N \bmod M, v_2(U_N)), (A, T)\big) \leq \frac{Ma}{2N} + 2^{-a}.$$

Consequently, if $N \geq M$, this distance is at most

$$\min\left\{1,\ \frac{M}{2N}\left(\log_2 \frac{N}{M} + 3\right)\right\}.$$

Every deterministic joint observation of the two coordinates obeys the same upper bound. In particular, for any finite set S of odd prime labels, taking $M = M_S$ gives it for the full or truncated weighted Fermat count together with $(\mathbf{1}_{P_\ell | F_n})_{\ell \in S}$, uniformly in the Fermat cutoff. Thus $M_S = o(N)$ suffices for convergence of these growing finite vectors to the independent reference law.

Proof. For each $0 \leq t < a$ and each residue $c \bmod M$, the event $n \equiv c \bmod M$, $v_2(n) = t$ is exactly one residue class modulo $M2^{t+1}$: combine $n \equiv c \bmod M$ and $n \equiv 2^t \bmod 2^{t+1}$ by CRT. Its count on $1, \ldots, N$ differs from $N/(M2^{t+1})$ by at most one. The contribution of these Ma cells to half the sum of absolute mass differences is therefore at most $Ma/(2N)$. On the remaining levels, the empirical total mass is $\lfloor N/2^a \rfloor / N \leq 2^{-a}$, and the reference total mass is 2^{-a}. Their contribution to half the absolute difference is at most 2^{-a}. This proves the first estimate, including $a = 0$. For $N \geq M$ choose $a = \lceil \log_2(N/M) \rceil$. Then $2^{-a} \leq M/N$ and $a \leq \log_2(N/M) + 1$. Total variation is also at most one, giving the displayed minimum. A deterministic map cannot increase total variation: sum the signed masses over each fiber

before applying the triangle inequality. For the Fibonacci labels use the odd modulus $M = M_S$, and send c to the vector $(\mathbf{1}_{\ell|c})_{\ell\in S}$. The maintained Fibonacci carrier theorem identifies this with the actual vector $(\mathbf{1}_{P_\ell|F_n})_{\ell\in S}$. Send the other coordinate to $W_{\min(t,k)}$, or to W_t for the full count. Finally, writing $q = N/M \to \infty$, the bound $(\log_2 q + 3)/(2q)$ tends to zero. □

This common-index statement uses the Fibonacci clocks obtained from Binet's quadratic identity and the golden-root quotient in Section 21.29. It gives a growing-cutoff probability comparison for bounded tests. For example, a bounded complex test of absolute value at most one has expectation error at most twice the displayed total variation. Tests outside that bounded class require their own integrability estimates; the statement supplies no new zeta-zero exclusion. Geometric approximation for valuation vectors is classical, for example [16, Proposition 1.9]; the exact single-index binary defect and its transport to these arithmetic counts are the assertions proved here. No literature priority is claimed.

Corollary 21.168 (A simultaneous compensated Fibonacci–Fermat limit)**.** *For real $X \geq 3$ put*

$$S_X = \{\ell \leq X : \ell \text{ is an odd prime}\}, \qquad s_X = |S_X|,$$
$$M_X = \prod_{\ell\in S_X} \ell, \qquad \lambda_X = \sum_{\ell\in S_X} \frac{1}{\ell}.$$

Define the actual Fibonacci observation and compensated joint transform by

$$B_X(n) = \sum_{\ell\in S_X} \mathbf{1}_{P_\ell|F_n},$$
$$\mathcal{G}_{N,X,k}(z,u) = e^{-\lambda_X(z-1)} \frac{1}{N} \sum_{n=1}^{N} z^{B_X(n)} u^{C_k^{\mathrm{all}}(n)}.$$

Let $\mathcal{F}_{\mathrm{odd}}$ be the entire compensated Fibonacci profile and Q the full weighted Fermat PGF above. For each $R \geq 1$ there is a constant C_R depending only on R such that, for every $X \geq 3$, $N \geq M_X$ and $k \geq 0$,

$$\sup_{\substack{|z|\leq R\\ |u|\leq 1}} |\mathcal{G}_{N,X,k}(z,u) - \mathcal{F}_{\mathrm{odd}}(z)Q(u)| \leq E_{N,X}(R) + C_R(X^{-1} + 2^{-k}),$$
$$E_{N,X}(R) = R^{s_X} e^{(R+1)\lambda_X} \frac{M_X}{N} \left(\log_2 \frac{N}{M_X} + 3\right).$$

In particular, fix $0 < c < 1$, and for all sufficiently large N set

$$X(N) = \left\lfloor \frac{c \log N}{\log \log N} \right\rfloor .$$

For every integer sequence $k(N) \to \infty$, the compensated transforms converge to $\mathcal{F}_{\mathrm{odd}}(z)Q(u)$, uniformly on $|z| \le R$, $|u| \le 1$. More precisely, for every $0 < \eta < 1 - c$, the uniform error is

$$O_{c,R,\eta}\left(N^{c-1+\eta} + \frac{\log \log N}{\log N} + 2^{-k(N)} \right).$$

No fixed-cutoff sampling limit or uniform first moment of the full Fermat law is assumed.

Proof. In the independent reference law from Theorem 21.167, the Fibonacci coordinates have probabilities $1/\ell$, and the other coordinate has PGF $Q_k(u)$. Consequently its compensated joint transform is exactly

$$\mathcal{F}_X(z)Q_k(u), \qquad \mathcal{F}_X(z) = \prod_{\ell \in S_X} (1 + (z-1)/\ell) e^{-(z-1)/\ell}.$$

For the unnormalized test, $|z^{B_X} u^{C_k}| \le R^{s_X}$. Also $|e^{-\lambda_X(z-1)}| \le e^{(R+1)\lambda_X}$. The expectation bound $2\|f\|_\infty d_{\mathrm{TV}}$ and the joint total-variation estimate therefore give

$$\sup_{\substack{|z| \le R \\ |u| \le 1}} |\mathcal{G}_{N,X,k}(z,u) - \mathcal{F}_X(z)Q_k(u)| \le E_{N,X}(R).$$

This explicitly charges the growing complex test and its compensator.

The canonical product satisfies

$$\sup_{|z| \le R} |\mathcal{F}_X(z) - \mathcal{F}_{\mathrm{odd}}(z)| \le C_R / X.$$

For completeness, with $a = z - 1$ a factor $(1 + a/\ell)e^{-a/\ell}$ differs from one by $O_R(\ell^{-2})$. This follows by its Taylor expansion, uniformly for $|a| \le R + 1$. The tail sum is bounded by $\sum_{n > X} n^{-2} = O(X^{-1})$. The finite-product inequality $|\prod_j (1 + e_j) - 1| \le \exp(\sum_j |e_j|) - 1$ bounds the remaining tail product, while the initial products are uniformly bounded on the compact disc. This proves the displayed estimate first for large X depending on R; enlarging C_R covers $3 \le X$ in the bounded remaining range. It uses the cross-multiplied tail factorization $\mathcal{F}_{\mathrm{odd}} = \mathcal{F}_X \cdot \mathrm{tail}_X$, so it remains valid at

every zero of $\mathcal{F}_X$. Since $|Q_k(u)| \le 1$ and $\sup_{|u|\le 1} |Q_k(u) - Q(u)| \le 2^{1-k}$, the first conclusion follows.

The diagonal claim needs only elementary bounds:

$$s_X \le X, \qquad \lambda_X \le 1 + \log X, \qquad M_X \le X^X.$$

Indeed, the first and third count and bound the labels, and the second bounds their reciprocal sum by the harmonic sum. For the stated $X(N)$, $X = o(\log N)$ and $X \log X = (c + o(1)) \log N$. Thus $M_X \le N^{c+o(1)} < N$ eventually, $R^{s_X} e^{(R+1)\lambda_X} = N^{o(1)}$, and $\log_2(N/M_X) + 3 = O(\log N)$. It follows that $E_{N,X}(R) \le N^{c-1+o(1)}$, hence it is $O_{c,R,\eta}(N^{c-1+\eta})$. Finally $X(N)^{-1} = O_c(\log\log N / \log N)$. Substitution proves the stated rate and simultaneous limit. □

Formalization scope. The maintained candidate owner

```
Mathlib/Probability/Distributions/DyadicDiscrepancy.lean
```

checks nine declarations: the common compensator and its endpoint bounds, preservation of the unit interval, the one-step signed inequality, its telescoped and normalized forms, and the exact alternating reward sum. Its axiom audit contains only the standard foundational axioms. The exact total-variation identification, the sharp limsup, transport to the physical Fermat count, and the joint odd-clock theorem and simultaneous compensated limit are the complete ordinary proofs above, without a separate Lean compilation claim for those assembled probability conclusions.

21.69 Rational residue-class laws: a stronger inverse theorem

The preceding inverse theorem extends beyond a common leading coefficient. Fix an integer $Q \ge 1$. Suppose that outside finitely many primes the laws have the form $\eta_1(p) = R_a(p)$ and $\eta_2(p) = S_a(p)$ for $p \equiv a \pmod{Q}$, where a runs over the units modulo Q. Each R_a, S_a is a rational function with rational coefficients. Assume each is either identically zero or has an expansion $\nu/x + \kappa/x^2 + O(x^{-3})$ with rational $\nu > 0$. The actual probabilities are in $[0, 1]$.

Theorem 21.169 (Residue-rational profile rigidity)**.** *Equality of the entire profiles implies $R_a = S_a$ as rational functions for every unit class a. The complete laws then differ by a finite permutation of prime coordinates. More precisely, if one active R_a differs from S_a, there is a unit progression*

$r \pmod M$, *refining* $a \pmod Q$, *such that for every sufficiently large prime* $p \equiv r \pmod M$ *the value* $R_a(p)$ *is absent from the entire second law. Consequently the first profile has infinitely many zeros absent from the second. Using the prime number theorem in a fixed progression, their number in* $[-T, 0]$ *is at least* $(\nu/\varphi(M) + o(1))T/\log T$. *If only the second law is active in a discrepant class, interchange the two laws.*

Proof. Write $R_a(x) = \nu/x + \kappa/x^2 + O(x^{-3})$. For each active target class b, write $S_b(x) = \mu_b/x + \lambda_b/x^2 + O(x^{-3})$. If $R_a(p) = S_b(q)$ and p is large, then q is large and comparable to p: finitely many small target primes have probabilities bounded away from the positive values tending to zero under consideration. Taking reciprocals gives

$$q = \frac{\mu_b}{\nu} p + \left(\frac{\lambda_b}{\mu_b} - \frac{\mu_b \kappa}{\nu^2} \right) + O(p^{-1}).$$

Both displayed coefficients are rational. The error belongs to a fixed rational lattice because p, q are integers, so it is eventually zero. Thus every large equality satisfies an exact affine equation $C_b q = A_b p + B_b$, with positive integers A_b, C_b and integer B_b.

If $B_b = 0$, primeness and sufficiently large p, q force $q = p$ and $A_b = C_b$. Then $b = a$, and $R_a(p) = S_a(p)$ has only finitely many solutions unless the two rational functions are identical. For each remaining equation, $B_b \neq 0$. Choose a distinct auxiliary prime ℓ_b not dividing $QA_bB_bC_b$, and impose $p \equiv -B_b A_b^{-1} \pmod{\ell_b}$. This is a nonzero residue. Together with $p \equiv a \pmod Q$, the Chinese remainder theorem gives one unit class modulo $M = Q \prod_b \ell_b$. Each affine equality on this class would make $\ell_b \mid q$, impossible once the prime $q > \ell_b$. This excludes every target class simultaneously; the finite exceptional target primes were already removed.

Dirichlet's theorem supplies infinitely many primes in the chosen unit progression. Their probabilities are eventually distinct, since $R_a(p)^{-1} = p/\nu - \kappa/\nu^2 + O(p^{-1})$ is strictly increasing on large prime arguments. They therefore give distinct exclusive zeros. Their cutoff is $p \leq \nu T + O(1)$, which yields the stated count from the fixed-progression prime number theorem. Equal profiles cannot have such exclusive zeros. Hence all rational functions agree, the labelled tails agree, and cancellation of the common tail gives the finite-permutation conclusion. □

This is an ordinary proof; its finite affine-exclusion and residue-rational classification have not yet been formalized in Lean. It applies only after a family's rational local laws and exceptional prime set have been established.

It does not assert such a transfer for every polynomial, marked family, or Dirichlet L-function.

21.70 The completed signed-packet calculus and its open consumer

For a finite $S \subset \{1, \dots, N\}$ and complex coefficients a, b, put $A_d = \sum_{n \in S,\ d \mid n} a_n$ and $B_d = \sum_{n \in S,\ d \mid n} b_n$. The maintained signed covariance identity is exactly

$$\sum_{\substack{n,m \in S \\ (n,m)=1}} (a_n - c b_n)\overline{(a_m - c b_m)} = \sum_{d=1}^{N} \mu(d) |A_d - c B_d|^2.$$

Indeed $\sum_{d \mid (n,m)} \mu(d)$ is the coprimality indicator, and finite interchange of sums proves the identity. Each divisor row retains the full expansion

$$|A_d - c B_d|^2 = |A_d|^2 - c B_d \overline{A_d} - \overline{c} A_d \overline{B_d} + |c|^2 |B_d|^2.$$

In particular, if b is supported on primes larger than D, then $B_d = 0$ for every $2 \le d \le D$. Such a prime-only compensator changes none of these low rows, regardless of the complex coefficient c. This rules out that specific compensation mechanism on those rows; it is not an obstruction to other coupled compensators.

There is also an exact partition using every prime allowed by the finite support, rather than a chosen finite set. For $v_1 = 0$, define, for $1 \le n \le N$ and prime $p \le N$,

$$v_p(n) = \begin{cases} \mu(n) v_n \dfrac{\log p}{\log n}, & p \mid n,\ p \text{ prime}, \\ 0, & \text{otherwise}. \end{cases}$$

Only $n > 1$ enters the first branch. Since $\mu(n) = 0$ off the squarefree numbers, $\sum_{p \le N} v_p(n) = \mu(n) v_n$ for $1 \le n \le N$. The coprime form of two blocks with the same prime index vanishes exactly: both supported arguments are divisible by that prime. Writing $\mathcal{C}(f, g) = \sum_{n,m \in S,\,(n,m)=1} f_n \overline{g_m}$, this gives the exact identity

$$\mathcal{C}(\mu v, \mu v) = \sum_{\substack{p,q \le N \\ p,q \text{ prime},\ p \ne q}} \mathcal{C}(v_p, v_q).$$

The coefficient v may itself retain a full compensation, so this identity does not erase its cross terms. It does not estimate their joint cancellation.

The reconciliation inventory `paper/completed-research-inventory.json` and the public theorem-audit inventory record the maintained owners for these covariance identities, full-prime partitions, Gaussian principal and Poisson inputs, Euler-continuation results, detector adapters, and quantitative tail lemmas. Their public export is `PrimeFactorOscillations/Assembly/CompletedDetectorPackets.lean`. A copied source is not certified by this list: the inventory separately records the fresh compilation and axiom-audit status. An earlier 20 October reconciliation checked 1,139 owned Lean modules and 4,843 selected theorem closures with standard foundations only; both the public and canonical QRH statement comparisons passed. That reconciliation identified 301 maintained owners and transferred all 132 proved private graph nodes to maintained sources. A later 1,150-module promotion covered the identity, finite-permutation and prime-coordinate inverse consumers. Subsequent maintained owners add the full adaptive cutoff inverse and exact optimal-distortion theorem, with separate build and axiom receipts. The residue-rational classification and continuum-cardinality assertion remain ordinary proofs. The signed arithmetic estimate for an improved RH boundary remains open.

The unfinished analytic consumer remains explicitly open. In the fixed-mask notation of that programme it asks, for each $\varepsilon > 0$ and each fixed admissible G, for constants $C_{\varepsilon,G}, D_{0,\varepsilon,G}$ such that

$$|I_{D,G}| \le C_{\varepsilon,G} D^{3/2+\varepsilon} \quad (D \ge D_{0,\varepsilon,G}).$$

No uniformity for a growing G is implicit. The identities above do not prove this estimate. Detector precision, zero-signal preservation and smaller separate remainders are recorded as structural results; the certified unconditional boundary remains 7/8. Failed absolute-value reductions and restricted-range attempts retain their failure records and are not counted as improvements of that boundary.

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